

SKILLS & KNOWLEDGE OF COST ENGINEERING

6th Edition

aace International

The Authority for Total Cost Management®

Skills and Knowledge of Cost Engineering

Sixth Edition

Dr. Makarand Hastak, PE CCP, Editor

2015

Skills and Knowledge of Cost Engineering
Sixth Edition
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Skills and Knowledge of Cost Engineering

Sixth Edition

Dr. Makarand Hastak, PE CCP, Editor

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PREFACE

This 6th edition of *Skills and Knowledge of Cost Engineering* (S&K6) has revised and updated chapters from the previous edition. Also, several new chapters have been included to bring the latest knowledge in cost engineering to the readers. The S&K6 is well aligned with the basic processes of Total Cost Management (TCM) and includes knowledge in areas that are essential for an individual to function effectively as a cost professional. AACE International defines TCM as:

> *"…the effective application of professional and technical expertise to plan and control resources, costs, profitability and risks. Simply stated, it is a systematic approach to managing cost throughout the life cycle of any enterprise, program, facility, project, product, or service. This is accomplished through the application of cost engineering and cost management principles, proven methodologies and the latest technology in support of the management process."* [http://www.aacei.org/resources/tcm/]

This edition has been organized into six sections comprising thirty-four chapters. The Preface from the 5th edition offered a good description of cost engineering and has been included in this edition as an introduction under the title *"Cost Engineers: Who Are They and What Do They Do?"* However, for the Canons of Ethics, another essential area of knowledge for a cost professional, please refer to the association website for the latest description.
www.aacei.org/aboutUs/govDocuments/CanonEthics.shtml

At the AACE Education Board, we continuously strive to enhance the value offered by this publication and would appreciate any suggestions for improvement. Also, we would like to hear from you if you would be interested in contributing to future editions of this publication. You can send your comments to the AACE HQ at: education@aacei.org.

Please note that the proceeds from the sale of this and other publications published by the AACE Education Board help in providing scholarships to deserving undergraduate and graduate students from all locations where AACE Sections exist.

A work of this magnitude and quality could not be completed without dedicated contributions from (i) the Education Board champions who identified subject matter experts and diligently followed up with them to write and/or revise the chapters, (ii) all the authors for their contribution to this body of knowledge, (iii) the reviewers who methodically evaluated each chapter for accuracy and depth of coverage, as well as, (iv) my graduate students - Sayanti Mukhopadhyay and Saumyang Patel and all the AACE staff who worked with me and contributed countless hours in making this project successful. I feel honored to have worked on this project with these dedicated individuals and their efforts and support are much appreciated. A special thanks to all the past editors for their significant contribution to this preface, particularly in documenting the history of this premier publication.

The History of This Publication

This AACE International publication had its beginning in 1985, when the Education Board started to work on the AACE Recommended Practice: *Required Skills & Knowledge of a Cost Engineer*. Board members included: Brian D. Dunfield (Chair), Dr. Brisbane H. Brown Jr., Frank J. Kelly Jr., CCP, James M. Neil, PE CCP, and Gord Zwaigenbaum, CCP, and they were supported in this effort by Barry G. McMillan, the AACE International Executive Director at the time. The document, "*Required Skills & Knowledge of a Cost Engineer*" was published in August 1986 in the *Cost Engineering* journal.

In 1987, a 13-session workshop on the "*Required Skills and Knowledge of a Cost Engineer*" was organized and presented at the AACE Annual Meeting in Atlanta. Similar sessions have been presented at the AACE Annual Meetings ever since that first effort in 1987. The instructional material presented in this workshop was edited by Dr. James M. Neil, PE CCP, for a new AACE publication called the, *Special Supplement-1987 Transactions*. This was the 1st edition of the present S&K. Later that year, the 2nd edition was published with its present title, *Skills & Knowledge of Cost Engineering*. Dr. James M. Neil, PE CCP, again served as the editor. The content was increased to 17 chapters and a consistent style was adopted.

Five years later (1992), the 3rd edition was published with Donald F. McDonald Jr., PE CCP, and Dr. James M. Neil, PE CCP, serving as co-editors. A large number of AACE International members became involved in the updating and revising of the published materials. The 18 chapters in this edition were compiled as a system for teaching the basic skills and knowledge that any cost engineer should possess.

The 4th edition was published in 1999, with Dr. Richard E. Larew, PE CCP, serving as editor. In this edition, new chapters, problems and solutions were added, while grouping closely-related chapters into eight parts. The AACE International Canons of Ethics was also included as an appendix.

The 5th edition, published in 2004, was edited by Dr. Scott Amos, PE, and included 31 chapters organized into seven sections. The addition of 13 new chapters to this edition was a result of the attempt to align the content of the educational materials within the Association to improve its value in the certification preparation process. Special care was taken to align the material presented in the *Skills & Knowledge of Cost Engineering* and the *AACE International Certification Study Guide*.

This Edition

The 6th edition of Skills and Knowledge of Cost Engineering presents further refinement to the subject matter content. The content of each chapter has been updated and consistent formatting changes have been used throughout the publication. In addition to the updates, two chapters have been completely re-written and eleven new chapters and an introduction have been added. Certified cost professionals and other subject matter experts independently reviewed the content of each chapter to ensure its accuracy and learning value. The following chapters have been re-written for this publication:

Chapter 2–Pricing and Costing
Chapter 12–Project Planning

The following chapters have been added to this publication:

Introduction – Cost Engineers: Who Are They And What Do They Do?
Chapter 14 – Earned Value Overview
Chapter 18 – Project Communications
Chapter 22 – Value Engineering
Chapter 25 – Change Management Practical Guide
Chapter 26 – Overview of Construction Claims and Disputes
Chapter 27 – Financial and Cash Flow Analysis
Chapter 28 – Practical Corporate Investment Decision - Making Guide
Chapter 30 – Optimization
Chapter 32 – Risk Management Practical Guide
Chapter 33 – Total Cost Management Overview
Chapter 34 – The International System of Units

How to Use This Publication

All topics covered in this publication relate to the skills and knowledge expected of a cost professional as covered in AACE Recommended Practice 11R-88, *Required Skills and Knowledge of Cost Engineering* and the *Total Cost Management Framework: An Integrated Approach to Portfolio, Program, and Project Management*. This publication has two specific objectives:

- To provide a significant coverage of important cost engineering topics such that the S&K6 becomes a desk reference for all cost professionals.
- To serve as a reference for all those who want to prepare for the Certified Cost Professional (CCP) as well as other AACE International certification examinations.

Readers should consider this publication as a reference. Most of the chapters include supporting references and recommended reading material at the end that could be used as a resource for additional information. Also, this publication can be used in conjunction with the Certified Cost Professional (CCP) Certification Study Guide (CSG) that includes problem sets related to the knowledge areas covered in S&K6 to better prepare for the Certified Cost Professional (CCP) certification examination.

We hope you find this publication to be of value.

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August 2015

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INTRODUCTION

Cost Engineers: Who Are They and What Do They Do?

John K. Hollmann, PE CCP CEP DRMP FAACE AACE Hon. Life [1]

What is cost engineering and who are the people we call cost engineers? The first place to seek an answer is the *AACE International Constitution and Bylaws* which states:

> **Article I-Section 2**—The Association is dedicated to the tenets of furthering the concepts of total cost management and cost engineering. Total Cost Management (TCM) is the effective application of professional and technical expertise to plan and control resources, costs, profitability and risk. Simply stated, it is a systematic approach to managing cost throughout the life cycle of any enterprise, program, facility, project, product or service. This is accomplished through the application of cost engineering and cost management principles, proven methodologies and the latest technology in support of the management process.
>
> **Article I-Section 3**—Total cost management is that area of engineering practice where engineering judgment and experience are used in the application of scientific principles and techniques to problems of business and program planning; cost estimating; economic and financial analysis; cost engineering; program and project management; planning and scheduling; and cost and schedule performance measurement and change control.

The full AACE International Constitution and Bylaws is available at: www.aacei.org/aboutUs/govDocuments/constitution

What this says is that the list of practice areas in Article I-Section 3 are collectively called *cost engineering*; while the "process" through which these practices are applied is called *total cost management* or TCM. Let's elaborate a bit more.

TCM and its sub-processes (strategic asset management and project control) can be summarized as management processes focused on coming up with ideas for strategic assets, analyzing and deciding upon the best idea, and finally planning and creating the selected projects in a controlled way (i.e., project control). While this is the process, who performs it?

Many people would say that "engineers" and "engineering" are most often responsible for creating functional projects or strategic assets; they are correct. However, there are multiple elements to engineering projects, e.g., a bridge project. Most look at engineering and see the element of physical design and the calculation and analysis tasks that are done to support the design of a bridge; again, they are correct. However, many people do not see that beyond the physical dimension of the design of the bridge structure, there are also dimensions of money, time, and other resources that are invested in the creation of the asset. These investments are collectively referred to as costs. Someone needs to estimate what the bridge might cost, determine the activities needed to design and build the bridge, estimate how long these activities will take, etc. Furthermore, someone needs to continually monitor and assess the progress of the bridge design and construction to ensure that the completed bridge meets the owner's objectives. This is a significant amount of work that requires special skills and knowledge.

The cost dimension requires calculation, analysis, planning, and control. No bridge has ever been built without dealing with both the physical and cost dimensions. However, the engineering skills and knowledge required to deal with costs are quite different from those required to deal with the physical design. From that difference, the field of cost engineering was born. Cost engineers work alongside other engineers or software analysts, play producers, architects, and other creative disciplines to handle the cost dimension. Returning to the AACE Constitution and Bylaws definition stated earlier, the skills and knowledge needed by the cost dimension are: business and program planning; cost estimating; economic and financial analysis; cost engineering; program and project management; planning and scheduling; and cost and schedule performance measurement and change control. All these functions are performed as part of the TCM process.

Cost engineers often specialize in one functional aspect of the project process. They may have titles such as cost estimator, parametric analyst, strategic planner, scheduler, cost/schedule engineer, project manager, or project control leader. They may work for the business that owns and operates the asset or they may work for the contractor that executes the projects. No matter what their job title or business environment, a general knowledge of, and skills in, all areas of cost engineering are required to perform their job effectively. This edition of the Skills and Knowledge of Cost Engineering provides a comprehensive overview of many TCM subject areas.

[1] John K. Hollmann authored the introduction, but granted permission for its use by Scott J. Amos, PhD PE, in Skills and Knowledge of Cost Engineering, 5th Edition (S&K5). This introduction appeared as part of the preface and has been edited for inclusion in this publication.

SECTION 1 - COST

Chapter 1: Cost Elements
Chapter 2: Pricing and Costing
Chapter 3: Materials
Chapter 4: Labor
Chapter 5: Engineering Role and Project Success
Chapter 6: Machinery, Equipment, and Tools
Chapter 7: Economic Cost
Chapter 8: Activity-Based Cost Management

Chapter 1—Cost Elements[2]

Franklin D. Postula, PE CCP FAACE AACE Hon. Life

Abstract
Cost is a basic "yard stick" by which activities and assets are measured and compared. The word *cost* is so commonly used and generally related to monetary value, we often lose sight of its true meaning and importance as a cost engineering concept. This chapter provides a basic overview of cost engineering concepts as they relate to different activities or assets; describes how cost elements are structured within the various categories: such as materials, labor, and other resources; states the purpose of cost accounting and its importance; and finally describes how these cost elements and cost structures can be used for the purpose of cost management.

Keywords
Assets, Cost Accounting, Cost Elements, Cost Management, Cost Structure, and Resources.

INTRODUCTION
Cost is one of the three fundamental attributes associated with performing an activity or the acquisition of an asset. These are (1) price (cost), (2) features (performance), and (3) availability (schedule). The need to understand and quantify the attributes of cost spawned the engineering discipline of cost engineering. Cost engineering is the application of scientific principles and techniques to problems of estimation, cost control, business planning and management science, profitability analysis, project management, as well as planning and scheduling [1]. While this definition seemingly addresses non-cost areas, it lists key engineering activities that either generates cost when they are performed, or defines plans and processes that cause (or influence) cost to be generated in other activities and/or assets, such as: a) What are the elements that make up cost? b) How are these cost elements categorized and how do they relate to one another? c) Why is it important to collect and account for costs as they relate to specific activities and assets? d) How do we apply these cost elements and categories to the insight for managing activities and assets?

This chapter provides a basic understanding of these cost fundamentals and gives you the insight and background you will need as you study subsequent chapters in this Skills and Knowledge of Cost Engineering (S&K) textbook.

Learning Objectives
After completing this chapter, the reader should be able to:

- Understand what makes up cost—i.e., the basic resources (material, labor, etc.) that are needed to perform an activity or create an asset.
- Understand the distinction between cost elements that are either directly or indirectly applied to an asset.
- Relate the cost elements to the life cycle of the asset: acquisition, use, and disposal.
- Use the understanding of cost elements to further understand how cost is measured, applied, and recorded to arrive at the total activity and/or asset cost.
- Apply the knowledge gained to solve problems related to cost element source and definition.

CONCEPTS
Cost is the value of an activity or asset. Generally, this value is determined by the cost of the resources that are expended to complete the activity or produce the asset. Resources used are categorized as equipment, labor, material, and "other." Although money and time are sometimes thought of as resources, they only implement and/or constrain the use of the physical resources previously listed. The path by which resources are converted (via a "project") to activities and assets is illustrated in Figure 1.1. The final activity or asset produced depends on what can be "afforded," given the money and time allocated to the project. Normally, you think of material as the physical composition of the asset. However, the value of the asset may also include the cost elements of scrap material or manufacturing spares, construction form work and expendable safety items, and the cost of transporting the material to the work site.

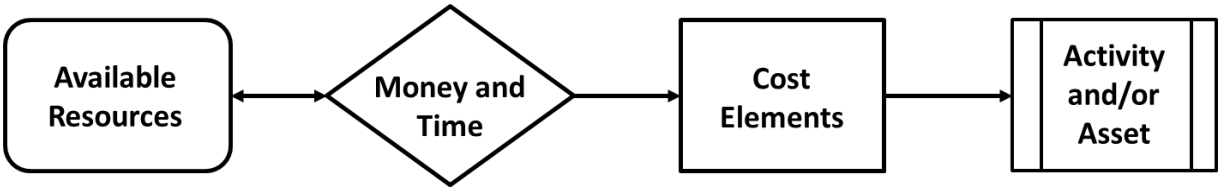


Figure 1.1—Conversion of Resources to Project Results

[2] Franklin D. Postula authored the original chapter in S&K5. Chris A. Boyd made revisions and minor updates to the manuscript for S&K 6.

Often, we think of labor as the value of the work needed to complete the activity or asset; i.e., the worker's labor in painting the building or soldering an electrical contact. Labor also includes the work of the engineer who prepares the design, the foreman supervising the field work, or the technician that maintains the wave soldering equipment. The "other" cost category consists of resources needed to support the activity and/or asset. An example would be the facilities needed to produce an activity or asset, which would include the tooling, miscellaneous equipment, electricity, taxes, and maintenance, etc., necessary to keep the facility available for use. Other costs might be office supplies, communication costs, travel costs, and security costs.

Another important aspect of cost relates to whether one is the producer or consumer of an activity or asset. The cost categories previously listed are those associated with the producer. The consumer has additional costs that add to the value of the activity or asset being acquired. A fundamental addition is profit for the producer. Generally, profit is established by market competition, although contract terms and conditions may limit profit on certain local, state, and US federal government procurements.

The value of an asset or activity also may be related to intangible costs. The value of a piece of art is often related to the name of the artist. When evaluating alternatives, the value of each alternative should be assessed in terms of the benefits that can be expected, if selected, and the consequences that might be suffered if not [2]. An intangible benefit might be an avoided cost if the alternative is chosen. For example, incorporating a spell checker into word processing software saves (avoids) the labor of manual checking and/or reworking the document. An intangible negative consequence might be the cost of missing an opportunity because resources were invested in another alternative instead of one more beneficial. The following example illustrates the resource cost categories:

Example 1— A homeowner decides to build a deck on the back of the house. The homeowner draws up plans for the project, gets the building permit from the city, buys the material, hauls it home, and constructs the deck. A representative sample of the cost elements and categories associated with building this asset are shown in Table 1.1. Notice that some of these cost elements are not part of the physical deck, but are necessary in order to complete the project. Furthermore, some of the cost elements are not part of the work activity needed to get the deck built, but are essential to support the project. In the next section, we will see how these cost categories are structured.

Table 1.1—Costs Associated with the Homeowner's Deck

Category	Cost Elements
Materials:	Draft paper/pencil, concrete, nails, lumber, deck screws, paint, brushes, turpentine, and drop cloths.
Labor:	Draw plans, get permit, get materials, construct footings, erect deck, paint deck, homeowner's support in making lunch or snacks.
Other:	Building permit fee, use of home to draw plans, shovel, power saw, power drill, electricity, pickup truck, and gasoline.

COST STRUCTURING

It is important to further structure the cost elements within the equipment, labor, material, and other resource categories, in order to understand how they influence the total cost of the activity or asset and to get a better understanding of how they can be controlled. This structuring sorts the cost elements into direct costs, indirect costs, fixed costs, and variable costs. In practice, some costs may fall in more than one of these groupings. This will be shown through an extension of the previous example.

Direct Costs

Direct costs are those resources that are expended solely to complete the activity or asset. In other words, "Any cost that is specifically identified with a particular final cost objective, but not necessarily limited to items that are incorporated in the end product as material, equipment or labor" [3]. Thus, the direct cost of a foundation for a house includes trenching for the footings, the wooden forms (if not reusable), the concrete, and the labor to place and finish the concrete. Direct costs for making a metal bowl would be the metal sheet stock and labor cost for the stamping machine operator. The material cost for manufacturing the bowl would include the scrap from the stamping process, less any salvage value.

Indirect Costs

Indirect costs are those resources that need to be expended to support the activity or asset, but are also associated with other activities and assets. In other words, "Any cost not directly identified with a single final cost objective, but identified with two or more final cost objectives…" [3]. Indirect costs are allocated to an activity or asset based upon some direct cost element, such as labor hours, materials, and equipment cost, or all three. Indirect costs also may be referred to as "overhead costs" or "burden costs." Indirect costs include general administrative activities associated with operating the business, costs for providing and maintaining field equipment or a manufacturing facility, and expenses for utilities, taxes, legal services, etc. In some industry circles the indirect and fixed cost are called "hotel costs", which represent the cost items necessary in order to conduct business.

Fixed Costs
Fixed costs are those cost elements that must be provided independent of the volume of work activity or asset production that they support. These can be either direct or indirect costs. The tool used to stamp the metal bowl is a direct fixed cost that is incurred whether 100 or 1,000 items are produced. The tools used to finish the concrete foundation are an indirect fixed cost, or consumable cost, since they can be reused on other concrete finishing work. The tools are an indirect fixed cost only if they are owned. If they are leased or provided as part of the labor charge, they would fall into a different classification. *AACE International Recommended Practice 10S-90, Cost Engineering Terminology* says that tools are not consumable unless they are not able to be used for additional work.

Variable Costs
Variable costs are those cost elements that must be provided and are dependent on the volume of work activity or asset production that they support. Again, these can be either direct or indirect costs. An example of a direct variable cost is the material used to form the metal bowl, since the amount varies with the quantity produced. An indirect variable cost could be the transportation/distribution used to operate the stamping machine, since it also varies with the quantity produced, but is considered to be an overhead cost.

In business practice, cost element information may be grouped in a variety of ways to provide the basis for management decisions. Some of these groupings are listed in Table 1.2. Bear in mind that this table only shows a representative list of cost groupings. Any set of cost groupings should be tailored to the individual company's method of doing business. It is important to categorize as many charges or costs as possible as "direct" in order to reduce the total overhead, general, and administrative cost as a percentage of total project cost. This maximizes the recognition of the actual costs of the project work. This more complete analysis will result in a more stable percentage allocation of overhead costs to project work.

Table 1.2 – Example Cost Element Groupings

Cost Center	Cost centers are groups of activities within a project that provide a convenient point for collecting and measuring costs. This could be a department in an engineering organization such as a structural design group, or it could be process related such as a metal stamping operation.
Labor Craft	It may be convenient to group types of labor such as electricians, plumbers, etc. on a construction site, or machinists, tool and die technicians, etc., in a manufacturing operation.
Material Type	These groups could be raw material, purchased parts, etc., in a manufacturing company, or could be concrete, 1.5-inch and smaller pipe, etc., for a construction project.
Inventory	This could be the value of purchased material and equipment waiting to be used in manufacturing or installed in a facility under construction. This could also be the cost of finished goods waiting to be sold.
Overhead	As discussed previously, these are indirect costs that are allocated to labor and material cost elements. Examples are the cost of maintaining a manufacturing facility or the cost of the home office of a construction company that has several projects underway.
Equipment	This is the value of all machines, tools, and other equipment needed to support a manufacturing operation or a construction project.
Subcontracts	It is important to separately collect and report work that is contracted out to others in support of the project. This could be labor or material, direct or indirect work.
Other Direct Costs	Sometimes it is convenient to directly charge a cost that may also be treated as an indirect cost. Examples are travel expenses, start-up costs, plant protection, etc.
Commitments	This is a group of future costs that are represented by obligations to obtain subcontracted/purchased material and services. It is extremely important to have this information available when changes to production or construction plans occur.

Example 2—The homeowner tries to better understand the costs associated with building the deck. The homeowner structures the costs as shown in Table 1.3.

Table 1.3—Costs Associated with the Homeowner's Deck

Direct	
Material:	Drafting paper, concrete, nails, lumber, deck screws, paint, and turpentine.
Labor:	Draw plans, get permit, get materials, construct footing, erect deck, paint deck.
Other:	Building permit fee.
Indirect	
Material:	Pencil, brushes, drop cloth.
Labor:	Homeowner's support in making lunch.
Other:	Use of house to draw plans, shovel, power saw, power drill, electricity, pickup truck, gasoline.
Fixed	
Direct:	Get permit.
Indirect:	Use of house to draw plans, shovel, power saw, power drill, pickup truck, pencil, brushes, drop cloth.
Variable	
Direct:	Drafting paper, building permit fee, concrete, nails, lumber, deck screws, paint, turpentine, draw plans, get materials, construct footings, erect deck, paint deck.
Indirect:	Homeowner's support in making lunch, electricity, gasoline.

As you consider these groupings of cost elements, you may realize that the variable direct cost elements are those that depend on the size of the deck. Although you might think of them as being "fixed" for this specific project, their value would change if the deck design were to change. Also, if a second identical deck were to be built, the value of these cost elements may go down as a result of efficiencies (learning) in performing the job. In the next section, we will investigate the purpose for collecting and reporting cost elements and how these elements might be organized to provide information for management decision making.

COST ACCOUNTING

What is the purpose of cost accounting and why is it important?

Cost accounting is defined as the historical reporting of disbursements, costs, and expenditures on a project. When used in conjunction with a current working estimate, cost accounting can assist in giving the precise status of the project to date [4]. Historical costs can also provide a sound basis for forecasting and budgeting costs of future activities and assets.

While detailed methods used for cost accounting may vary from business to business or from project to project, all accounting systems include the three basic steps of recording, classifying, and summarizing cost element data in terms of money expended with time [5]. The recording of cost information is nothing more than the mechanical gathering of data in a routine manner. There are a variety of methods to achieve this. Generally, it is accomplished with time sheets for labor, invoices for subcontracts, and Purchase Orders (PO) for procured items. Other costs are gathered from utility bills, expense reports, tax bills, etc. Every business enterprise has an established approach for classifying and summarizing costs that is organized around their business practices. This approach is called a "code of accounts," by which all recorded cost elements are classified. A code of accounts (sometimes referred to as a chart of accounts or as cost code of accounts) is a systematic numeric method of classifying various categories of costs incurred in the progress of a job; the segregation of engineering, procurement, fabrication, equipment utilization, construction, and associated project costs for accounting purposes [1]. A company's code of accounts is configured to support the recording of cost data in the general ledger. An example of summary-level accounts is shown in Table 1.4.

Table 1.4—Typical Code of Accounts (After Jelen [6])

2000 Assets	**6000 Expenses**
2100 Cash	6100 Cost of goods sold
2200 Accounts receivable	6200 Salaries and wages
2300 Notes receivable	6300 Heat, light, and power
2400 Inventory – materials and supplies	6400 Communications expense
2500 Inventory – finished products	6500 Reproduction expense
2600 Work-in-progress	6600 Insurance
2700 Equipment	6700 Taxes
2800 Building and fixtures	6800 Depreciation
2900 Land	6900 Interest expense
3000 Liabilities	**7000 Construction work in progress**
3100 Accounts payable	7100 Site preparation
3200 Notes payable	7200 Concrete work
3300 Taxes payable	7300 Structural steel
3400 Accrued liabilities	7400 Heavy equipment
3500 Reserve accounts	7500 Buildings
	7600 Electrical systems
4000 Equity	7700 Piping systems
4100 Capital stock issued and outstanding	
4200 Retained earnings	**8000 Manufactured goods in progress**
	8100 Direct materials
5000 Revenues	8200 Direct labor
5100 Sales of finished goods	8300 Overhead
5200 Other revenues	

While classifying costs in accordance with the general ledger breakout is a common practice, this approach does not generally provide the visibility needed to manage the work or to make informed forecasts of the cost of new jobs. An alternate method of cost element classification is called Activity-Based Costing (ABC). In the ABC approach, resources that are used are assigned to activities that are required to accomplish a cost objective [7]. ABC makes cost accounts understandable and logical, and much more useful for the cost engineer [8].

This method of collecting and summarizing cost elements reveals which resources and activities are the most significant contributors (drivers) to the cost of the cost objective. Another approach to classifying costs that is similar to ABC accounting is using a Work Breakdown Structure (WBS) to group cost elements. It has become a common practice for a WBS to be a required project management tool on most contracts. Not only does a WBS provide a framework for planning and controlling the resources needed to perform the technical objectives, but it facilitates a summary of project data regarding the cost and schedule performance. Table 1.5 shows an example WBS for a study project wherein the deliverable item is an intellectual product documented in technical publications. When used to classify and record costs, the WBS becomes the cost element structure (CES), as well.

Table 1.5—Typical WBS Format

Level 1	Level 2	Level 3
1.0 Research Project	1.1 Concept Study	1.1.1 State-of-the-Art Research
		1.1.2 Concept Definition
		1.1.3 Data Analysis
	1.2 Mathematical Model	1.2.1 Equation Formulation
		1.2.2 Computer Programming
		1.2.3 Test and Evaluation
	1.3 Deliverable Data	1.3.1 Technical Documents
		1.3.2 Engineering Data
		1.3.3 Management Data
	1.4 Project Support	1.4.1 Project Management
		1.4.2 Review Meetings

The general format, however, is applicable for the CES of a manufactured product or construction project [9]. While there is no universal WBS/CES standard, some have been developed for government acquisition. One comprehensive example is the U.S. Army Cost Element Structure [10]. This CES provides a definition of each cost element within the structure. It also provides structure for cost elements in all procurement phases: development, production, and operation and support. Sometimes the code of accounts to be used is determined by law. For example, the Federal Energy Regulatory Commission (FERC) has established a Uniform System of Accounts for the electrical power generation industry. The account numbering plan used consists of a system of three-digit whole numbers as shown in Table 1.6. Regardless of how cost elements are classified and grouped, it is important that this is done in a manner that is consistent with the way future work is estimated and budgeted. Historical cost records represent the way a company conducts business and can be analyzed to determine whether improvements have been made and how costs may trend in the future. Therefore, the integrity of the cost accounting system is essential to developing a project cost baseline.

Table 1.6—FERC Uniform System of Accounts

100-199	Assets and other debits
200-299	Liabilities and other credits
300-399	Plant accounts
400-432, 434, & 435	Income accounts
433, 436-439	Retained earnings accounts
440-459	Revenue accounts
500-599	Production, transmission and distribution expenses
900-949	Customer accounts, customer service and informational, sales, and general and administrative expenses

Example 3—The homeowner recognizes that the deck being built is an improvement to the home, and that it would be considered a capital investment. The homeowner decides that there needs to be a structure of cost accounts to provide data for future maintenance estimates. Table 1.7 shows the code of accounts the owner developed: Notice that the homeowner hired some of the labor and has cost records (invoices) for the work. For the labor the homeowner performed, fair market value might be used to account for the cost of these activities. Also, values of indirect cost elements are not shown on the list. For, example, the homeowner could have included the rental value of the tools used in the construction. This valuation approach would have allowed the homeowner to get a better understanding of the total value of the deck.

Table 1.7—Homeowner's Deck New Code of Accounts

1.0 Material	1.1 Footing form lumber, nails, and concrete 1.2 Deck lumber and screws 1.3 Paint and turpentine
2.0 Labor	2.1 Draw plans (self) 2.2 Get permit (self) 2.3 Get materials (self) 2.4 Construct footings – Sam's Handyman Service 2.5 Erect deck – Sam's Handyman Service 2.6 Pain deck – Sam's Handyman Service
3.0 Other	3.1 Permit fee

The homeowner's code of accounts allows for grouping material and labor costs to find the total cost of the footings, deck structure, and painting. Table 1.8 shows how the homeowner rearranges the cost elements to get this visibility.

Table 1.8—Cost Elements of the Homeowner's Deck Project

1.0 Footings	1.1 Allocation, in part, of: • Draw plans (self) • Get permit (self) • Permit fee • Get material (self) 1.2 Form lumber, nails, and concrete 1.3 Construct footings – Sam's Handyman Service
2.0 Deck Structure	2.1 Allocation, in part, of: • Draw plans (self) • Get permit (self) • Permit fee • Get material (self) 2.2 Lumber and screws 2.3 Erect deck – Sam's Handyman Service
3.0 Painting	3.1 Allocation, in part, of: • Draw plans • Get permit • Permit fee • Get material 3.2 Paint and turpentine 3.3 Paint deck – Sam's Handyman Service

Cost element allocation would be a ratio of the cost of the material and labor in each component of the asset. This arrangement of cost elements is similar to the ABC or WBS approach.

COST MANAGEMENT

There are many ways that cost elements and cost structure can be displayed to provide information for cost management. This chapter considers four of the most common methods of how cost information is applied to cost management. These are: cost estimating, cost trending, cost forecasting, and lifecycle costing. Although these methods will be discussed in more detail in later chapters, it is important to see how they relate to cost elements and structure.

Cost Estimating

Cost Estimating predicts the quantity and cost of resources needed to accomplish an activity or create an asset. The building blocks of a cost estimate are:

- A well-defined scope (what we are trying to estimate),
- A cost element structure (how we organize the information), and
- Historical cost data (data from cost accounting records and/or "experience" of subject matter experts (SMEs), commercially available books/databases).

Key questions to ask regarding a cost estimate always include: "What cost data was used?", "What is the Basis of Estimate?", "What are the lessons learned from similar projects?" and, "How can we reduce the cost of X using value engineering or other methods?" Cost element data and its structure are paramount ingredients of a sound cost estimate.

Cost Trending

Cost trends are established from historical cost accounting information. Cost management questions may focus on how expenditures are trending relative to physical accomplishments.

"How much are we spending for pipe fitters and how much piping has been installed during the last six months?" or "What has been our monthly cost for steel this last year and how many bowls have we produced?" Again, having access to cost history in the structure needed is key to providing the required information.

Cost Forecasting

Forecasts are much like estimates. An estimate is always for future activities and assets, and establishes the Budget at Completion (BAC) for performance baselines; whereas forecasts are predictions of the cost to complete or Estimate to Complete (ETC) for cost elements in progress, or that have not started. Therefore, a sound cost forecast will be based on cost element data from inception of the work to the date of the forecast, the cost trend of that data compared to accomplishments, and a cost estimate of the work

remaining to be completed. Cost element history in the proper activity structure is essential for realistic cost forecasts. The combinations of these elements produce the Estimate at Completion (EAC) for the project.

Life-Cycle Costing

Life-Cycle Costs (LCC) are associated with an asset, and they extend the cost management information beyond the acquisition (creation) of the asset to the use and disposal of the asset. Asset acquisition consists of the design/development phase and the production/construction phase. (Some industries/companies use phase reviews called "gates" to ensure all the required tasks for the phase have been completed, and/or to make project go/no-go decisions in the early phases. Common usage has occasionally morphed to refer to the entire phase as "Gate [name or number]"). Generally, cost elements are segregated into these phases or gates because design/development costs are often recovered over more than one asset. For example, design and development cost of a new airplane is amortized over the production. The design of a housing project is recovered through sales of the houses built. Once the asset is created, it enters the Operation and Support (O&S) phase, sometimes called Operations and Maintenance (O&M) phase. A new set of cost elements and CES is applicable to this phase or gate, and cost data must be collected to support the cost management efforts. The final phase is disposal of the asset with another unique set of cost elements. Refer to Table 1.9 to see how the homeowner might group cost elements of the deck project to reflect its LCC.

Table 1.9—Life-Cycle Cost Elements of Homeowner's Deck

1.0 Design	
	1.1 Draw plans
	1.2 Get permit
	1.3 Permit fee
2.0 Construction	
	2.1 Get materials
	2.2 Form lumber, nails, concrete
	2.3 Construct footings
	2.4 Deck lumber and screws
	2.5 Erect deck
	2.6 Paint and turpentine
	2.7 Paint deck
3.0 Maintenance	
	3.1 Repaint deck every 5 years
	3.2 Termite inspection every 2 years
4.0 Disposal	
	4.1 Tear down deck after 25 years
	4.2 Remove footings
	4.3 Haul demolition material to dump
	4.4 Replant grass

Example 4—The homeowner rearranges the project cost elements to understand what it cost to design and construct the deck and to get a perspective on what it might cost to maintain and eventually remove the deck. If desired, the homeowner can develop a cost estimate for the maintenance and disposal phases. As a first cut at the estimate, the homeowner can refer to the construction cost element history for most of the information and add estimates for termite inspection and replanting the grass.

CONCLUSION

In this chapter, we discussed some of the fundamental aspects of cost. We started with an understanding of resources and cost elements and how they relate to the performance of activities and the creation of an asset. Money and time were introduced as both enablers of and constraints on the execution of a project. Cost elements were illustrated by an example of building a wood deck for a home. Next, the chapter considered how cost elements are structured into direct, indirect, fixed, and variable cost groups. The purpose for organizing costs into groups is to determine which cost elements are used in performing the activity or creating the asset and which are in support of the work. Other possible groupings were introduced to illustrate how cost elements can be arranged to provide visibility on the cost of specific activities or resources. The deck example was extended by structuring the cost elements into cost groups. The importance of cost accounting in establishing a database for cost management was discussed. Several methods of classifying and summarizing cost elements were introduced: Code of Accounts, Activity-Based Costing (ABC), and Work Breakdown Structure (WBS). The cost elements of the deck example were organized into a code of accounts and WBS to illustrate these methods.

Finally, the topic of cost management was introduced. Four common methods for providing cost information were discussed as they apply to cost management. This illustrated the importance of having sound cost history that is structured in a usable format. The deck example was extended further to demonstrate the four phases or gates of life-cycle costing.

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RECOMMENDED READING

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Chapter 2—Pricing & Costing

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Abstract
This chapter highlights the difference between pricing and costing. It is very important to distinguish between the terms "price" and "cost." There is a very fine difference between them, which is why people often tend to use them interchangeably. This chapter discusses the concepts of pricing and costing of a project, illustrates the differences and helps the reader to identify the inputs, transforming mechanisms and outputs related to the cost and pricing process.

Keywords
Cash Flow , Competitive Advantage, Cost, Financial Management, Inputs, Opportunity Cost, Outputs, Price, Profit, Return on Assets (ROA), Return on Investments (ROI), and Transforming Mechanisms.

INTRODUCTION
Often price and cost are viewed interchangeably, but it is important to recognize the significant difference between the two. Remember the old adage, "Beware the cost of the lowest price?" Well, it is true. Just because a product is less expensive than its competitors', does not mean you or your business saves any money in the long run. Sometimes, perhaps most times, more spending occurs in part because of time and efficiency lost to a mediocre product. Or, potentially spending twice as much as originally hoped because you have to buy a replacement for the first purchase [3].

So what is the difference between "cost" and "price"? Price equals what you are selling an item for, or quoting for the performance of a project. However, if you are the customer in this transaction, this amount is not a price, but the cost of the item or service. What it comes down to for most major purchase decisions is Return on Investment (ROI). Where do you stand to gain the biggest bang for the buck [3]? In business, the price of a widget is what it can sell for, while the cost is how much is spent to make one. Profit is the difference between the two, determining the success or failure of a business [4].

Pricing and costing are terms which are not the same, but are commonly used interchangeably in economics, finance and project management, and even among cost engineering practitioners. Clarity of these terms is one of the objectives of this chapter. This is best achieved by its application to cost engineering in the Engineering, Procurement, and Construction (EPC) Industry for capital acquisitions and/or improvements, and by relating the terminologies to the cost-pricing process based on AACE International's TCM Framework. The basis of the cost and pricing process is the analysis of the inputs, transforming mechanisms, and the ensuing outputs or deliverables. The transforming mechanisms (tools and techniques) address both the cost engineering and business economic perspectives.

The cost engineering perspective which establishes the costing/pricing strategies, the business economic perspective can also be considered such as the financial ratios calculated from the financial data. Both perspectives are important to the transforming mechanisms (tools and techniques) for the costing and pricing process.

There are many reasons that cost engineers and cost controllers need to understand the financial ratio. The main reason is that the ratio establishes another set of tools and techniques for improving the cost decision process on projects at any phase of execution. Using financial ratios provides a quick method for project personnel to sort through financial information that is available in the financial reports.

Outputs are the deliverable of the process and can include items such as the completion of an estimate or the acquisition of a contract or project. Obtaining the best possible inputs, combined with optimizing the transforming mechanisms, is essential to these critical deliverables.

Learning Objectives
After completing this chapter, the reader should be able to:

- Differentiate between costing and pricing.
- Identify the inputs, transforming mechanisms and outputs related to the costing and pricing process.
- Understand the budgeting process.
- Calculate financial ratios related to the costing and pricing of projects.
- Understand the reasoning behind the costing and pricing process outputs.

COSTING AND PRICING
In researching definitions of costing and pricing from various sources, clear differences were found in business and economics, namely in terms of a buyer and a seller. Costing and pricing are presented here from the cost estimating perspective.

Pricing

Pricing can be defined as the determination of the amount to be charged to the client, so as to fully include everything necessary to provide a service or product, plus profit. Therefore, the cost includes all direct and indirect cost, plus any allowable contingency. In cost estimating, pricing a project follows cost. (See S&K6 Chapter 9, Cost Estimating) (See Table 2.1) [1].

Table 2.1: Project Cost versus Price-Example

Project SOW (Scope of Work) Cost	$1,000,000
Field Office Overheads (6%)	$60,000
Home Office Overheads (10%)	$100,000
Project Cost Subtotal	$1,160,000
Contingency (5%)	$58,000
Project Total Cost	$1,218,000
Profit (10%)	**$121,800**
Project Total Price	$1,339,800

Costing

Defining cost can become confusing for some, as they become entangled in accounting and business economics (e.g., COGS-cost of goods sold and fixed, variable and sunk costs) terminologies. Clarifying this terminology by relating it to the cost estimating process is recommended for cost engineering professionals.

In the cost estimating process, costing follows scope determination and quantification and *precedes* pricing and budgeting (see S&K6 Chapter 9, Cost Estimating) [1]. This concept is also substantiated by the TCM Framework. "You cost the scope contents and price the cost estimate" (see *TCM Framework*, p.42) [4]. Based on the cost estimating process, cost can be categorized into direct or indirect cost (see S&K6 Chapter 9, Cost Estimating) (Figure 2.1) [1].

Direct costs are those that are specific and tangible to the project, and include the costs of materials, labor, equipment, etc. Indirect costs are those costs not directly accountable or tangible to the project, such as business taxes, home office overhead, or transportation fleet distributed cost, but necessary for the business to remain solvent.

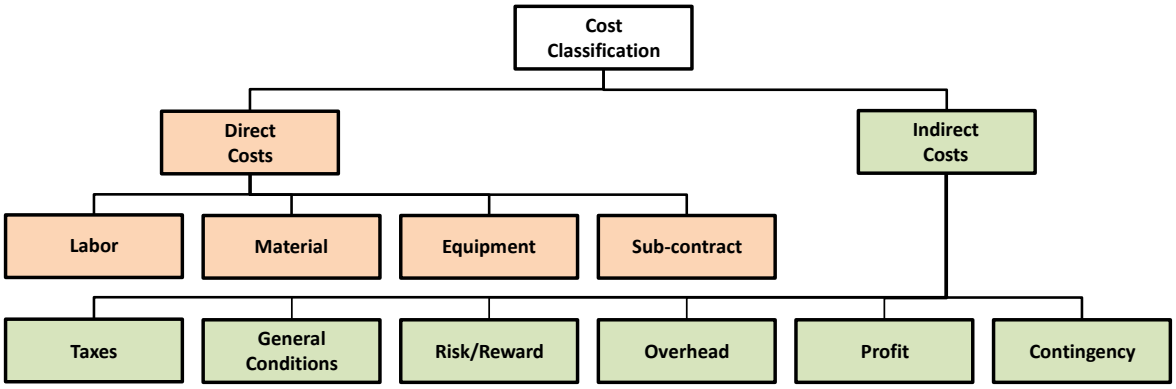


Figure 2.1—Cost Classification

THE COSTING-PRICING PROCESS

Keeping with the process mapping technique in the TCM Framework, inputs, transforming mechanism (tools and techniques), and outputs (see *TCM Framework***,** p.42) [2] use the mapping process for costing and pricing, as shown in Figure 2.2. They are as follows:

- The inputs are the documentable items that will be acted upon, and include, but are not limited to, the WBS/Scope (Work Breakdown Structure), historical records and cost estimation.
- Transforming mechanisms (tools and techniques) are applied to the inputs to produce the outputs.
- The outputs are the deliverables of the process and include items such as the project estimate, project acquisition and a business decision.

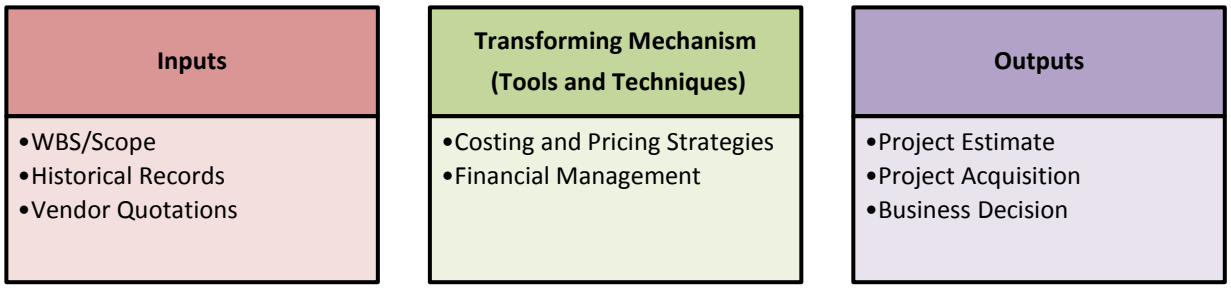


Figure 2.2—The Analytical Framework of the Costing-Pricing Process

Inputs
The main input to the costing-pricing process is the scope, such as equipment, for which the cost needs to be determined; or the systems (mechanical, electrical, and plumbing (MEP)) related to the project that has to be priced. In costing, the individual items that comprise the historical records of a system or process along with previous projects data will be beneficial for comparison purposes. Vendor quotations provide a more current value of the cost and final price of the equipment and system.

- WBS/scope
- Historical records
- Vendor quotations

WBS/Scope
The scope of work incorporates all the documents and relevant drawings and specifications that form the basis of the scope. The Work Breakdown Structure (WBS) provides a structure with increasing levels of functional decomposition of scope from top to bottom. At the bottom level of the WBS the elements of the scope are cost coded. It is at this level of the WBS that costing, and consequently, pricing occurs. Here is a great example where cost and price comingle in the estimate. The purchased equipment delivered to the job site represents a price in the estimate, as well as any subcontracted scope by other sub-vendors and contractors such as the MEP. While these items represent pricing, they are still cost in the general contractor's cost estimate.

Historical Records
According to the *TCM Framework*, project historical records referred to as, "historical database management," is a process for collecting, maintaining, and analyzing project historical information so that it is ready and available for use by other project control processes. Project historical records for costing and pricing a project includes, but is not limited to, the following documents from similar previous projects:

- Work Breakdown Structure
- Quantity take-offs or quantity surveys
- Schedules
- Cost reports, which would show the costing data
- Bid Documents, which would indicate the pricing information

The objective of using historical records is to enable continuous improvement and mitigate risk in the project installed systems based on lessons learned from previous projects.

Vendor Quotations
In coordination with the procurement and contracts department, those vendors that are being considered for major equipment purchases on the project are contacted for vendor quotations. The cost elements for major equipment quotations are generally for items such as: mechanical equipment inclusive of compressors, heat exchangers, condensers, or major electrical equipment. The cost elements are specific to the type of projects and the strategy of construction (modularization versus non-modularization). Vendors are contacted to provide a current estimate of these elements, which provides a current estimate of the cost of the major cost items. This is an important step that helps in qualifying the level of accuracy of the estimate. As stated earlier, a vendor estimate or quote is a price, not a cost. Hence, the vendor sells the equipment for a price and the installation contractor or facility owner receives the equipment as a cost to their project.

Transforming Mechanism (Tools and Techniques)
The tools and techniques are methodologies applied to costing and pricing to produce an optimal outcome, be it an estimate, a project, or a business decision. With the complexities involved, it should not come as a surprise that pricing is considered an art by many cost managers [5]. Establishing the right intelligence information on customer cost budgets and competitive pricing is an essential element in this part of pricing. A disciplined approach that is derived from combining the cost engineering and business valuation is very beneficial. This approach involves the following tools and techniques:

- Costing and pricing strategies
- Financial analysis

Costing and pricing strategies include:

- Cost estimating
- Budgeting
- Cost forecasting

Cost Estimating
Cost estimating involves the application of techniques that translate quantities into financial and resource information. (See *TCM Framework*, p.177) [2] The outputs of cost estimating are used as primary inputs, not only for business planning, but for the project cost and schedule control processes. The budgeting process is the technique used to transform the cost elements into a monitoring and controlling tool.

Budgeting
The inputs to the budget development process include:

- **Activity Cost Estimates**—The hours and dollars associated with each cost activity.
- **Scope Baseline**—The WBS and scope-based on drawings, specifications, previous documents and expert opinion.
- **Project Schedule**—An initial schedule that would provide an indication of cash flow.
- **Contracts**—Contract strategy information, indicating direct labor cost versus contract costs planned for the project.

From these inputs, a budget is established broken down to the activities level. The budget process is then set up and includes:

- Monthly review and evaluation of the budget cost items;
- A change control process that documents and controls changes to the budget; and,
- A monthly forecast that indicates predicted budget overrun/underrun.

Cost Forecasting
In order to forecast final cost, a forecasting technique called the earned value method is applied. This methodology establishes, on a periodic basis (monthly in most cases), the following (Refer to *S&K6 Chapter 14, Earned Value Overview*) [1]:

- The Estimate at Completion (EAC), which is based on the earned value Cost Performance Index (CPI), plus actual cost to date and the Estimate to Complete (ETC).
- Predicts an overrun in cost if the EAC cost value is higher than the Budget at Completion (BAC).
- The Cost Performance Index (CPI) is also provided for each work package, as indicated in the EAC value development.

Financial Management

- Return on Investment
- Return on Assets
- Net Profit Margin

Return on Investment (ROI) and Return on Assets (ROA) measures profitability of a company on its investment and assets respectively. Calculating the ROI provides a better understanding about the cost effectiveness of a project. ROA, when applied to projects, is typically used as a technique for project acceptance, or as a tool to evaluate overall performance. Companies having a choice of different projects would prefer those projects showing a higher ROA. Return on Assets is best used for comparing projects within the same industry.

Net Profit Margin is a measure of the effectiveness of a company at cost controlling. Depending on the manner in which the company's financial system is organized (i.e., costs are organized by function rather than by project), it may be difficult to ascertain the net profit for specific project(s). However, by judicious selection of appropriate accounts, a reasonable overall margin can be estimated.

Return on Investment (ROI)
Calculating simple ROI, for cash flow and capital investment analysis, measures the profitability in dollars invested. It is easy to understand and simple to use. It works well in situations where both the profits from the projects and the costs of the investment are easily known. Investments yielding a higher ROI will be the better investment; however, this calculation does not recognize any risk factor. Benefits are returns on the investment coming in as a result of the investment made into the project. (See Table 2.2)

$$Simple\ ROI\ (\%) = \frac{[(\text{Profit} = (\text{Project Outputs}) - \text{Project Costs (Inputs)}]}{\text{Project Costs (Inputs)}} = \text{x}\%$$

Table 2.2—Cost and Benefit Data for ROI Calculation

Project Sample						
		Year 1 Initial	Year 2	Year 3	Year 4	Cumulative Total
Total Costs	Labor Costs	$30,000	$25,000	$80,000	$30,000	$165,000
	Equipment Costs	$10,000	$25,000	$20,000	$5,000	$60,000
		$40,000	$50,000	$100,000	$35,000	$225,000
Total Benefits		$0	$75,000	$125,000	$100,000	$275,000
Net Benefits/Profit		($40,000)	$25,000	$25,000	$65,000	$75,000

$$Simple\ ROI\ (\%) \text{to the end of the project} = \frac{75{,}000}{225{,}000} = 33\%$$

Net Profit Margin

The net profit margin is equal to the net income after taxes and excluding extraordinary items divided by total revenues; and is a measure of the effectiveness of a company at cost control. (See Table 2.3)

Table 2.3—Sample Financial Data

Sample Data Income Statement	
Period Ending	**March 30, 2014**
Total Revenue	$1,010,000.00
Project Personnel Cost	$900,000.00
General/Administrative Expense	$25,000.00
Subcontractor Cost	$14,000.00
Net Income	$21,000.00
Sample Data Balance Sheet	
Period Ending	**March 30, 2014**
Property Plant/Equipment	$400,000.00
Other Assets	$100,000.00
Total Assets	$500,000.00

$$Net\ profit\ Margin = \frac{\text{Net Income}}{\text{Total Revenue}}$$

$$Net\ Profit\ Margin = \frac{21{,}000}{1{,}010{,}000} = 0.0208$$

Return on Assets (ROA)

This number tells you how effective your business has been at putting its assets to work. The ROA is a test of capital utilization - how much profit (before interest and income tax) a business earned on the total capital expended to achieve that profitability. ROA is recognized here to determine the anticipated return from a project. ROA is also applied to determine the viability of a project. Projects may not be considered viable in which the ROA is lower than the cost of funding the project. ROA is a ratio of the earnings generated from a project compared to the assets spent to generate that revenue. (See Table 2.3)

$$Return\ on\ Assets = \text{Net Profit Margin x Asset Turnover x } 100\%$$

$$Asset\ Turnover = \frac{\text{Total Revenue}}{\text{Total Assets}}$$

$$Asset\ Turnover = \frac{1{,}010{,}000}{500{,}000} = 2.02$$

$$Return\ on\ Assets = 0.0208 \text{ x } 2.02 = .042 \text{ or } 4.2\%$$

Outputs
The outputs of this process represent the documents that bring closure to the costing and pricing process and include:

- Project estimate
- Project acquisition
- Business decision

Project Estimate
This is a deliverable based on the culmination of the costing and pricing processes. The costing process involves developing the cost estimate and budget. The pricing process establishes the all-inclusive pricing of the project taking into account profitability.

Project Acquisition and Business Decision
Establishing a price for the project is only one of the parameters that need to be addressed in project acquisition and business decision, as the financial ratios need to be used to address the validity of the acquisition. These ratios are:

- Return on Assets (ROA)
- Return on Investment (ROI)
- Net Profit Margin

CONCLUSION
Differentiating between pricing and costing should not be a difficult concept to grasp.

- Just remember, in simple terms, the seller provides a price and the buyer receives a cost.
- Defining cost and price in some circles seems to be a confusing issue.

As a cost engineering professional, it is recommended that the definitions provided be consistently used in the cost estimating process to ensure clear communication. Cost, in this instance, is defined as a combination of the direct cost and indirect costs. These include the costs of materials, labor, equipment, contracts, and all indirect costs that are not directly accountable to the project. Indirect costs include business taxes, home office, overhead, etc. These are apportioned to the project on a pro rata basis. The direct cost is used for project controls monitoring on a monthly basis, in most part to validate field performance. The final price pulls together all costs, including contingency and profitability, and is used for billing purposes.

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Chapter 3—Materials

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Abstract
Selecting materials for a product, project, or project element, involves numerous considerations. Factors including cost, availability, durability, weight, wear resistance, and ease-of-fabrication are among relevant considerations regarding choice of materials. Other items for consideration include: quality assurance/quality control, materials handling, and materials hazards. The criticality of a material to the final product or project will dictate issues of quality assurance and control. The procurement of materials can range from raw materials to materials purchased at fabricated stages or already incorporated into select subassemblies. Make or buy decisions will strongly influence these decisions and the purchase of materials at various stages.

Keywords
Bulk Material, Competing Characteristics, Engineered Materials, Fabricated Material, Handling, Procurement, Raw Material, and Waste.

INTRODUCTION
Materials are a key resource in most projects and production endeavors. There may be isolated instances, where there is no requirement for materials; however, in most cases, materials and their related issues are important issues for those in responsible charge.

Materials have the quality of being purchased by those using them, rather than being manufactured by the subject entity. Thus, a tree log is material to the lumber mill that manufactures it into dimension lumber (product) which is material to the roof truss plant that fabricates it into a roof truss (product), which is material to the home builder that incorporates the roof trusses into a finished house.

Materials range from the simplest of raw materials to the most complex fabricated materials with a large range in between. The simplest of raw materials may be silica sand from a pit that is mined, in order to manufacture glass. On the other end of the spectrum, an electronic components manufacturer may require complex fabricated materials, such as a printed circuit board containing millions of transistors. While both the glass and the printed circuit boards rely on a form of silica, there are vast differences in the degree of complexity between the two. Besides the issues of materials type, materials must be procured in proper amounts, at the right time, and at the right cost, in order to lead to an efficient production process. In addition to materials, shaped in the particular production process, those machines and related equipment that perform the work also need maintenance materials again ranging from the simple to the complex. Simple materials may consist of grease and oil, whereas complex materials may again consist of fabricated items.

Learning Objectives
After completing this chapter, the reader should be able to:

- Identify types of project materials.
- Understand the issues involved in selecting and handling materials.
- Understand the principles of materials purchasing and management, including the proper amount of stock to save money and avoid waste or production delay.
- Understand possible safety hazards associated with materials and be aware of regulations governing worker and materials safety.

MATERIALS COMPETITION
Practical selection of materials for a given application must always take into account the competing characteristics of materials. Materials vary on a number of characteristics including cost, availability, service life, weight, corrosion/wear resistance, machinability, weldability, and other ease-of-fabrication criteria. A common phrase heard in industry is, "there are no bad materials, just bad applications of materials." The strong performance of materials in one application does not guarantee success in a differing application, and sometimes the differences may be difficult to ascertain. An ideal material may fail in a particular application because of fabrication difficulties. A strong, light-weight material, may pose fabrication/assembly problems of its own, as a result of machining costs or weldability issues.

The automobile industry is a prime example of continued competition between materials resulting from various factors, such as weight with its attendant impact on fuel economy. Steel has been replaced in numerous applications in automobiles by high performance plastics and aluminum. At first, targets were non-structural automobile applications, such as interior panels and interior trim items. Through further engineering, these plastic and aluminum components have migrated into structural automobile components, such as tanks and body parts. Steel automobile applications have also witnessed significant changes through the development and improvement of high-strength, low-alloy steels. Galvanized steels in automobile body fabrication have seen

increased usage through the advent of two-sided galvanized steels, with a mini-spangle spray application of zinc galvanizing that permits smooth-finish painting.

MATERIALS HANDLING

In the materials area, the materials handling issue is a significant concern regarding cost structure and system efficiencies. Materials handling is a requirement of the production process, but inefficiencies in this area can create plant-wide problems. Poor materials handling can result in damage to either raw materials or the finished product. An inefficient materials handling system can slow production operations, creating other excessive costs as a result of production delays [4].

Materials Handling Principles

Each materials manufacturing situation is somewhat different; however, there are some basic principles in this area that find wide application. These basic principles include the following:

- Material movement should be over the shortest distance possible;
- Terminal time should be in the shortest time possible, since the objective is to move materials;
- Eliminate manual material handling when mechanized methods are feasible;
- Avoid partial transport loads, since full loads are more economical; and,
- Materials should be readily identifiable and retrievable.

Some of these principles always apply, whereas others are situation specific. Mechanized material handling methods are most cost effective in high wage countries. In a lesser-developed country, the capital/labor tradeoff, given low wages, will tend to emphasize manual methods as compared to equipment-intensive operations. In virtually all situations, it is uneconomical to delay material handling equipment at terminal points for materials loading and unloading. Thus, materials can be loaded onto containers or pallets that are quickly transferred on and off materials handling platforms.

Materials Handling Decision Factors

There are four basic decision factors that affect materials handling. These four factors are:

- Material to be handled
- Production system type
- Facility type
- Materials handling system costs

The material to be handled will impact numerous other decisions. A brick manufacturing plant will be dependent upon a source of clay material. In part, this clay can be handled much as other earthen products are handled, with similar equipment, including front-end loaders and conveyors. A paint production facility will handle numerous raw materials from bulk tanker truck, rail car and/or through pipelines. These would require other criteria. Materials such as pipe or structural steel in a fabrication facility will require overhead cranes and forklifts for their movement.

The production system type will be divided into job shop or batch process, lean production, and continuous process types [2]. Continuous processes, such as seen in a petro-chemical plant or a steel mill, will find fixed-path equipment including conveyor lines and pipelines. Job shop, batch processes and lean production processes need more flexibility in their material handling systems. Many job shops perform unique jobs and large investments in single-application equipment, is often not economical.

The facility type will govern materials handling decisions. A facility with low-ceiling heights, and a barely adequate structural system, would not be a candidate for installation of overhead cranes. These constraints may instead dictate floor-based gantry cranes to accomplish the material handling requirements. Rectangular facilities versus other facility shapes, will govern production layout and material handling decisions.

The choice between job shop, batch process, and continuous process will impact capital investment. Material handling system costs and their economic feasibility should be analyzed as to labor costs and capital costs in the capital/labor tradeoff equation. Both labor and capital costs can vary depending on the country of manufacture. Manufacturers have gravitated toward countries with lower labor costs, but then material must be transported to these countries unless produced there as well. In some cases, the higher energy costs associated with shipping costs of materials to a certain country and return shipping costs back to markets, will eliminate or exceed labor-cost fabrication advantages.

Predicted demand for a facility can help in the economic and practical evaluation of alternatives. Facilities with high levels of demand over several years can better justify more expensive materials handling systems. Comparisons between alternative materials handling systems are difficult, and must include not only initial costs, but life-cycle costs as well. These include labor, maintenance, repair, energy, and disposal costs.

TYPES OF MATERIALS AND RELATED INFORMATION

For purposes of differentiation, materials can be segregated into four basic categories: raw materials, bulk materials, fabricated materials and engineered or designed materials. These categories are differentiated on the criteria of the amount of processing required for the material to be useful for its intended purpose.

Raw Materials

Raw materials are those materials used in a production or fabrication process that require a minimum amount of processing to be useful. The most basic example of this might be natural gravel from a river deposit that combined with some screening for size separation is then used as a sub base material for a roadway or foundation slab. The gravel can further be processed into materials for a concrete mix. In the steel-making process, certain raw materials, such as coal converted to coke, limestone, and iron ore, are mined for eventual combination and used for producing steel.

Bulk Materials

The steel product can be considered to be a bulk material in all of its various forms, including sheet steel, steel bars, steel pipe, and structural steel shapes, such as wide-flange beams and angles. The bulk materials category is distinguished by its availability. A customer desiring a bulk material, such as steel pipe, can call a distributor and achieve delivery of this pipe as soon as the next day, depending on transport distance. Bulk materials in common sizes are typically readily available with minimal lead times for order and delivery.

Fabricated Materials

Fabricated materials are bulk materials transformed into custom-fit items for a particular product or project. For example, the bulk material of steel pipe is transformed by fabrication operations into custom dimensions for a particular use, such as in a petroleum refinery. If the particular use is a welded piping system with flanged fittings, the pipe will be cut to dimension with welded flanges added where necessary, based on shop drawings. In the shop drawing phase, data from design drawings, which have been prepared by the project's or product's design professionals, is used to develop detailed shop drawings. These shop drawings need to convey all information necessary for the fabrication of the given item. In some cases, the shop drawings will also provide information for field assembly and erection of the fabricated items. The design professionals typically require a review of shop drawings prior to fabrication for acceptance. The design professional's review of shop drawings is undertaken to ensure conformance with original design intent. The historical separation between design drawings and shop drawings is a result of the fact that fabricators are more familiar with economies of fabrication. Therefore, as long as design intent is met, the fabricator has a significant degree of flexibility leading to lower costs in the fabrication process.

Engineered/Designed Materials

Engineered or designed materials constitute a category requiring substantial processing in order to attain their final form. Design or engineered materials are also based on shop drawings. These engineered materials may consist of many components and subcomponents that end with a completed product. These engineered materials consist of such diverse items as pumps, motors, boilers, chillers, fans, compressors, transformers, and motor control centers. The engineered materials producer, in some cases, will be producing these items as off-the-shelf products, and in other instances as custom products. An engineered item, such as a fan, will consist of such components as a fan housing, impeller, shaft, bearings, motor, and support base. Items such as the impeller, shaft, and fan housing may be manufactured directly by the subject vendor from various materials. The motor and bearings may come from other manufacturers to be added as components of the final product. The degree of capabilities possessed by the particular manufacturer will often be the deciding factor concerning what items in the final product are self-performed and which are subcontracted out to others.

The issue of custom or off-the-shelf products will have an availability impact to the potential customer. The manufacturer will often produce standard sizes and maintain these in inventory. Unusual sizes will require special designs and often-lengthy lead times. In materials procurement and selection, those responsible will want to analyze these issues for value-added benefits. In certain cases, the cost and use of these time effective alternatives may be an attempt to standardize as much as possible to allow purchase of off-the-shelf components rather than custom items.

PRODUCTION MATERIALS PURCHASE AND MANAGEMENT

Materials procurement is an important business function because it has a key role on the organization's ability to offer products at a competitive price. Purchasing can be defined as the acquisition of necessary materials of the correct quality, at the correct time, for a competitive price, from the selected vendor or supplier. Sufficient stocks of materials must be acquired to prevent delays in production operations. The concept of safety stocks is important, whereby ensuring there is a critical amount on hand when a replenishment quantity is received. The purpose of safety stock is to protect against the uncertainty in demand and in the length of the replenishment lead time. On the other hand, excessive materials supplies create additional costs and problems for the organization. Excess materials, beyond reasonable quantities, require an investment in operating capital to finance this inventory and require storage space. Moreover, excess material quantities can lead to damage and theft.

Materials Quality
Materials procurement must focus on the proper quality of the required materials. This implies the existence of predetermined standards and specifications to measure materials acceptance. Over-specifying on higher-quality materials in excess of requirements will lead to excessive costs and customers may not appreciate these benefits [1]. Similarly, the usage of poor quality materials can result in product defects leading to increased costs and potential litigation problems.

Materials Vendor Surveillance and Materials Traceability
In certain situations, the criticality of materials in a given application is such that vendor surveillance is an important requirement. Vendor surveillance may require periodic inspection by purchasers at the vendors' location(s) to ensure conformance with performance standards and specifications. Depending on volume and criticality, this periodic inspection may need to be conducted on a full-time basis at the subject vendor's facility. Materials traceability is accomplished by means of mill certifications. Before a material is used in a given application, mill certifications ensure that the material used meets the purchasing specifications. A mill certification will verify that a material, such as cross-linked polyethylene tubing, is in fact as it has been represented before incorporation into a finished product.

Materials traceability is a key issue because the improper substitution of one type of material for the specified material can lead to significant in-service problems and defects. A railroad car manufacturer that uses the wrong lower grade of steel in fabricating railroad car axles, as a result of an issue of mixed steel, will witness a significant number of service failures resulting in expensive call-backs and repairs. These mixed-steel issues can create accidents with tragic consequences.

Materials Quantity
Funds spent to acquire materials are a cost to the firm until these same materials can be sold as part of the completed product. Firms in industries where materials obsolescence is a factor, encounter special problems in holding excess material quantities. As an example, a large inventory of printed circuit boards may have to be discarded or drastically discounted as technology changes, creating obsolescence. Materials storage is a further burden that can sometimes exceed the value of the materials. The simple example of storing some bags of cement proves this point. If the inside storage space costs $100 per square meter per year and the storage of an excess of 20 bags of cement valued at $5 per bag takes up 1 square meter of space, any storage beyond one year therefore exceeds the actual value of the stored cement. On the other hand, maintaining insufficient materials inventories may create dangers of "stock-outs" interrupting the production process. Ordering materials in smaller quantities may create higher costs through missing the economies of volume discounts. The organization needs to balance these competing issues involving materials quantity.

Economic Order Quantity
As previously mentioned, there must be sufficient materials inventory to meet production requirements while still avoiding excessive inventory carrying costs and storage costs. In order to balance these competing demands, the firm must determine its Economic Order Quantity (EOQ) number. This EOQ number is determined based upon materials costs, storage costs, order costs, and annual demand. A garden tractor manufacturer has a requirement for 15,000 engines per year. The engines each cost $75. The order cost for a purchase order is $250. The storage costs for the engine are $12 per year, which includes space costs and financing costs. The standard formula used in industry for EOQ is:

EOQ = $[\sqrt{2xDxP/S}]$

Where:

D = annual demand
P = purchase order costs
S = storage/carrying costs

Thus, in this example:

EOQ = $\left[\sqrt{\left\{\frac{(2x15{,}000x\$250)}{\$12}\right\}}\right] = 791$

Therefore, in this case, the manufacturer should order 791 engines at a time which is the EOQ value.

The formula for computing Reorder Point (RP) is:

Reorder Point (RP) = (O x R) + I

Where:
RP = reorder point
O = order time

R = production rate
I = minimum inventory level or safety stock

Assume that the production process uses 60 engines per day for the 60 garden tractors produced per day. If the lead time for an order is 5 days, and the safety stock level is 180 engines (minimum level), then the reorder formula in this example yields:

RP = (5 days x 60 units/day) + 180 units = 480 units.

The EOQ value of 791 engines should be ordered whenever inventory drops to 480 units. Cycle stock levels maintained at very low levels can almost ensure the potential for production delays.

Just-In-Time Inventory Techniques
Recent years have seen the widespread introduction of just-in-time techniques for materials procurement across various industries. The just-in-time concept implies that the exact material quantities needed are delivered at the exact time needed. The goal is to reduce inventories and the related costs, such as storage space. Traditional practices that focus on safety stocks can mask unprofitable variations in the production process. By removing these safety stocks, the goal of just-in-time systems is a lean production process and an enhanced competitive position. The production facility therefore receives more frequent deliveries of smaller shipments on a periodic basis.

Individual Purchasing Orders and Systems Contracts
For non-standard and costly items, the purchasing function may procure these through a system of plans/specifications requirements and competitive bidding. For items that the organization uses on a continual basis, a systems contract may be the best solution. At the start of every year, the organization estimates potential quantities of required materials and places these out for bid on a systems contract. A fabricator during a year may require x thousand pounds of various types of welding wire and welding rod for its operations. The fabricator bids this out using a systems contract that calls for periodic deliveries throughout the year, as usage demands, based on actual production. If quantities vary significantly higher or lower, there may be additional provisions for price adjustments in the systems contract or for inflationary upstream price increases not controllable by the vendor. The advantage of the systems contract is reduced purchasing work load and improved pricing, based on economies of scale.

Materials Inspection
Prior to working on them in the facility, purchased materials must be inspected to assure they comply with specifications. Inspections can range from simple visual methods to machine-assisted inspections incorporating techniques such as X-ray and ultrasonic technology to detect flaws. The goal is to ensure that the correct grade and quality of materials are being received from suppliers. A fabricator of light poles needs to ensure that the correct steel grade has been received prior to the fabrication of these items. A light pole failure in a store parking lot, as a result of low-grade steel, could result in fatalities and/or property damage. The fabricator in turn may perform welding work on the steel in order to produce the light poles. There will first need to be the development of a welding procedure and welding process selection to ensure quality work in this area. Those performing the welding must be trained, and in some cases certified in the particular welding process and the chosen welding position(s). Once the light poles have been fabricated, inspection of the welding work must take place prior to shipment of the light poles. Inspection in critical cases may be required by a specification to be 100 percent of the items being produced; but more often, it is instead performed on a spot-check basis of sampling so many items out of the production run. Critical work, such as welded items like fabrications for a nuclear power facility, may require mill certifications as previously noted regarding traceability issues.

Expediting
Expediting involves the monitoring of all steps in the procurement cycle to ensure on-time delivery of the necessary materials. This monitoring includes checking design status, material status, production status, and shipping status. Analysis of potential delays is a key element in the expediting process. If shipping delays take place, alternative forms of delivery may be necessary to avoid production delays. By continually reviewing the status, the expediting process helps to avoid unpleasant interruptions of the production process. Expediting communication is conducted through telephone, fax and e-mail methods. Personal visits to vendors are part and parcel of vendor surveillance efforts, and can also be helpful. This writer remembers one site visit to a manufacturer of motor control centers for a wastewater treatment plant. The week before our surprise site visit, the manufacturer had promised over the telephone that the motor control centers were almost ready to ship. Instead, it was found upon a site visit, that this manufacturer had not even started fabrication of these units.

Global Materials Decisions
Materials fabrication decisions, as to locations and methods, are being made on a global basis. An example may assist in illustrating this concept. A large East Coast hotel in the U.S. wanted a series of simulated trees and canopies created inside the hotel areas with beads simulating the tree and canopy cover. These beads would be strung, such that there were 9,000 beads per square meter and there were almost 200 square meters of area to cover. A West Coast interior theme firm won the contract. The quantities meant that nearly 1.8 million beads would need to be strung on the canopies and trees at a competitive price. The trees and canopies were fabricated out of structural steel at the firm's West Coast fabrication shop. To be cost-competitive, the beads and the bead stringing

work were subcontracted to a firm in India. In India, skilled craftspeople strung the beads. The steel frames were shipped to India, strung with beads, and then trans-shipped to the U.S. East Coast hotel for final installation.

PLANT MATERIALS MANAGEMENT

Plant materials are a special category of materials. These are materials that are not associated with incorporation into any particular product or project. Instead, these are materials that assist the plant in completing materials fabrication and production operations. Examples range from oils, greases, solvents, and cutting bits, to parts such as spare motors or cylinders for a production machine.

These plant materials assist in the production, maintenance, and repair of the facilities. Commodity plant materials such as oils, greases, solvents, and cutting bits are typically low-cost items with predictable usage patterns. Moreover, sources of supply are readily available and order/procurement lead times are minimal.

Specialized Plant Materials

Specialized plant materials such as production equipment replacement parts, can pose more difficult problems. A particular replacement part may be available only from the Original Equipment Manufacturer (OEM) and require significant lead time. Those responsible for this area will want to maintain an inventory of critical replacement parts. Supplementing replacement parts inventory involves networking with other manufacturers or fabricators owning the same type of equipment that may be willing to "loan" out a replacement part item in an emergency. A secondary strategy is to have a backup plan in place to procure replacement parts through other manufacturers if there are difficulties with the OEM.

Plant Materials Benchmarking

Plant materials usage and longevity should be tracked through such measures as benchmarking. Benchmarking involves the examination of other organizations as to their practices. The benchmarking organizations do not need to be competitors but merely similar organizations. Therefore, a structural steel fabricator may examine the operations of a steel pipe fabricator for comparison of operational practices. Often it becomes easier to study the actions and practices of another organization rather than your own and learn from them because it is easier to be objective. Studying your own organizations' materials management practices brings with it a certain degree of defensiveness and rationalization of poor practices. Benchmarking can in turn invite others from various organizations into the subject organization for an evaluation and review of practices.

MATERIALS WASTE PRODUCT AND HAZARD ISSUES

Users and producers of materials, deemed as hazardous to humans, are required to comply with government regulations concerning hazard communication. The purpose of these regulations is to ensure that potential hazards are properly evaluated and hazard information is properly communicated to employers and employees. These requirements include labels, warnings, safety data sheets, information, and training.

A fabricator, for example, may be using a solvent for cleaning parts prior to a welding operation necessary to join the parts. The potential effects of the solvent on employees from fumes or direct contact must be properly evaluated prior to use. Moreover, some employees may have special allergic reactions to this particular cleaning solvent.

Safety Data Sheets and Hazard Communication

Materials safety is always an important issue in the safe handling, fabrication, and transport of materials. In the U.S., the Occupational Safety & Health Administration (OSHA), Department of Transportation (DOT), and other state agencies enforcing OSHA/DOT standards, regulate the vast majority of these interactions. Other countries may have similar or stricter regulations. The U.S., along with other countries, is now implementing what has been termed as, "the Globally Harmonized System (GHS) of Classification and Labeling of Chemicals," published by the United Nations [3]. The GHS is a system for standardizing and harmonizing the classification and labeling of chemicals. It addresses the following issues:

- Defining health, physical, and environmental hazards of chemicals;
- Creating classification processes that use available data on chemicals for comparison with the defined hazard criteria; and,
- Communicating hazard information, as well as protective measures, on labels and Safety Data Sheets (SDS).

While OSHA/DOT is still the standard, the U.S. has adopted GHS as their enforced standard unless GHS does not address other areas covered by OSHA. The previous OSHA and other regulatory references to Material Safety Data Sheets (MSDS) now are substituted by Safety Data Sheets (SDS). These SD sheets must be readily available and accessible to those dealing with the particular hazardous materials, as required by the applicable regulations. An organization with multiple work locations must have complete files on these SD sheets for review and inspection. The SDS requirements can force a re-examination of various issues in terms of the use and application of hazardous materials. The organization may find that they are using four different types of cleaning solvent for the same application, each with its own attendant SDS issues. Based on an evaluation, these four solvent types may be narrowed to one type promoting standardization and reducing potential problems. With the widespread advent of the Internet, web publishing of this SDS information may be a viable alternative. This web publishing ensures current up-to-date information, which is available anywhere in the organization through computer access.

These SDS sheets contain information on chemical and common names of ingredients including substances that may be carcinogenic along with physical and chemical material characteristics such as vapor pressure and flash point. Other SDS information includes physical hazards such as fire and explosion potentials. Health hazards listed in SDS information include primary routes of entry, exposure symptoms, and medical conditions which may be aggravated by exposure.

Environmental Regulations
Materials production operations involving potential hazards to the environment must pay attention to environmental regulations. As an example, in the U.S., solid waste is regulated from "cradle to grave" through the Resource Conservation and Recovery Act (RCRA). The RCRA regulates three categories of hazardous waste handlers:

- Generators
- Transporters
- Owners and operators of Treatment, Storage, and Disposal (TSD) facilities

Hazardous waste generators must keep accurate records, store waste in approved containers, label the waste, and use a manifest system to track the waste until delivery to a TSD facility. Violations of the RCRA can result in civil and criminal penalties, including fines of up to $25,000 per day assessed by the Environmental Protection Agency.

Material Life Cycle
All materials have a life cycle from initial development through to manufacture, incorporation in the product, and eventual disposal. Materials incorporating such elements as cadmium, lead, and asbestos, have created problems years after their manufacture because of environmental hazard issues. Radioactive waste from medical procedures and nuclear power production, have long half-lives measured in some cases at thousands of years. Materials such as these have created "cradle to grave" problems, and addressing these issues requires a systems engineering approach [5].

Waste Materials and Surplus Materials
Production operations on materials frequently results in the creation of significant quantities of waste materials. This production of waste materials or scrap is a significant cost in numerous production processes. This cost is comprised of the original materials' cost, plus waste material handling and disposal costs. Reduction in materials waste provides a cost reduction potential in three areas. Some waste materials can be reused in the production process. A grade of steel, not meeting requirements in one use, may be re-used for a product not requiring a premium steel grade. A steel manufacturer can take scrap steel from downstream production processes and transfer the scrap upstream to re-melt the scrap as a raw material component.

Surplus materials result from either mistakes in excessive ordering, changes in material requirements, and/or incorrect original quantity information. The production of an order for a job shop production process may have specified a given number of pieces of steel plate. The original take-off regarding material requirements was incorrect. Mistakes may have taken place in development of the bill of materials. If this is material for which there is a continuing use, it may make sense to return the materials to inventory with a credit to the job shop order. Purchasing may have incorrectly processed the order with an incorrect quantity requirement. With a fast-track fabrication job, the requirements for the steel plate may have been a late design change, where the materials procurement function could not act soon enough to cancel the order. In any case, the existence of surplus materials points to inefficiencies in the materials process. Returning the materials to the original vendor may be possible although, in most cases, a restocking fee of 15 to 20 percent or more may be charged to the organization. However, unless as noted above, the materials have a continuing use in fabrication operations, this may be the most economical course of action. The prevalence of surplus materials should be tracked with the goal of minimization of this expensive and wasteful practice.

Future Developments
The materials area has seen a revolution over the past few decades in terms of computerization/robotics/CAD/CAM for materials tracking, ordering, and fabrication or manufacturing. Computerized techniques for rapid prototyping/design and fatigue analysis, have allowed designers and others to significantly reduce costs and lead times in these areas. Developments in these areas are only expected to accelerate in the future.

A revolutionary development on the horizon in the materials area is that of 3D printing. 3D printers can take raw materials or bulk materials, such as hydraulic cement or plastic particles, and deposit them in layers on a surface. These printers are similar to ink-jet units, but instead of spraying ink, they actually spray material controlled through the computer to create the desired item. The end result is that these 3D printers create either physical models or actual physical items such as wall panels. The physical models enable design-production staffs to analyze the interaction of various material components in an assembly. The creation of actual physical components for a job could therefore be performed in remote areas, or on the Moon, without attendant transportation costs and other issues [6]. You still have transportation costs for the raw material(s) for the 3D printer and attendant software costs for driving the printer – either a license cost for the part or the purchase of the "print" document.

CONCLUSION

Users and producers of materials must recognize potential areas for practice improvement. The recognition of the impact and potential of materials issues will lead to more rational decision processes by those responsible. This will result in an improved cost structure for the organization better able to achieve competitive advantage. The general objective of materials procurement is to minimize total costs through reduction in purchase costs, material handling costs, storage costs, and shortage costs. Tools such as the Economic Order Quantity (EOQ) formula can lead to more rational decision processes in this area. In recent years, techniques such as just-in-time inventory methods have seen widespread introduction in materials-intensive applications.

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Chapter 4—Labor

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Abstract
Labor is one of the most important resources for a project. An owner, employer, or project manager from any industry needs to have a complete understanding of how the labor force works. This chapter provides an overview of the different classifications of labor, the different types of labor wages and benefits and also indirect and overhead labor and other costs. This chapter also illustrates the procedure to estimate the work hours for a given work scope at a given location; and how the labor hours can be used to monitor the progress of work.

Keywords
Direct Labor, Indirect Labor, Labor Rates, Labor Wage, Overhead Labor, and Performance Monitoring.

INTRODUCTION
As an owner, employer, project manager, estimator, etc., you will have work tasks that your workers need to complete. As such, you need to know how much this will cost. You also need to know how to set up and monitor the effort to ensure you are getting the desired work product, in the timeframe required, for a price you can afford. In order to do this, you need to understand the cost factors that affect this work and the techniques used to monitor progress.

Learning Objectives
After completing this chapter, the reader should be able to:

- Identify different classifications of labor and how each contributes to the completed project.
- Develop labor rates for estimating.
- Develop and use weighted average rates/composite crew rates.
- Include indirect and overhead labor and other costs.
- Estimate work hours for a given work scope at a given location.
- Use labor hours to monitor work progress.

LABOR CLASSIFICATIONS
The following definitions were taken or adapted from *AACE International's Cost Engineer's Notebook* [1].

- **Direct Labor**—The labor involved in the work activities that directly produce the product or complete the installation being built.
- **Indirect Labor**—The labor needed for activities which do not become part of the final installation, product, or goods produced but that are required to complete the project.
- **Overhead Labor**—The labor portion of costs inherent in the performing of a task (such as: engineering, construction, operating, or manufacturing), which cannot be charged to, or identified with a part of the work, and therefore must be allocated on some arbitrary basis believed to be equitable, or handled as a business expense independent of the volume of production.

Table 4.1 provides examples of different labor classifications and costs. The examples are not all-inclusive and only serve to illustrate the elements of each type of labor.

Table 4.1—Examples of Labor Costs

Cost Type	Direct Labor	Indirect Labor	Overhead labor
Construction	Carpenters, Electricians, Ironworkers, and Foremen working on the project	General Foremen, Construction Management, Field Purchasing, Field Warehouse Personnel, Payroll Personnel, Jobsite Computer Support, Project Cost Engineers and Schedulers, etc.	Home Office Support: Legal Assistance, Procurement, Human Resources, Senior Management Review, Corporate Computer Support, Estimating and Business Development, etc.
Manufacturing	Plant Equipment Operators, First Line Foreman, Supervisors, etc.	Plant Accountants, Maintenance Personnel, Purchasing Personnel, Security, Plant Supervision, Warehouse Personnel, Production Planning and Cost Personnel, On-site Computer Support	Corporate support: Legal, Human Resources, Computer Support, Corporate Finance and Accounting Support, Sales, etc.
Engineering	Civil, Mechanical, Electrical, Instrumentation and Controls Engineering and first line supervision	Documentation Support – duplicating and record keeping, Engineering Cost and Scheduling Personnel, Project Supervision	Corporate Support: Human Resources, Computer Support, Corporate Accounting Support, Estimating, Business Development, etc.

The difference between indirect and overhead labor appears to be somewhat vague. Depending upon the size of a project, plant, or office, and the location, some elements could shift from indirect labor to overhead labor, and there may be instances where direct labor moves to indirect labor and overhead labor as well.

For example, if the construction project is small, payroll and accounting may be located offsite and be composed of personnel who are splitting their time between several projects at different locations. In this instance, this function could be an indirect or an overhead. Therefore, it is imperative that the estimator and/or cost engineer understand where within his project, industry, and company, each of these costs are included, so that they can be correctly estimated and included in the estimate and budget.

DEVELOPING LABOR RATES

Base Wages

Base wages is the amount that goes directly to the employee. The source of these wage structures can be found in data bases from previous projects, labor contracts, unit rates supplied by contracting and engineering firms, local chamber of commerce data, government labor statistics, published labor data bases, and standardized estimating publications, such as Means and Richardson [4-7].

Base wages are usually calculated on a per hour basis; however, it can also be a breakdown of a weekly or monthly base salary prorated to a daily or hourly rate. The reason for an hourly breakdown is that estimates are usually developed based upon the amount of work hours to complete. Therefore, the labor cost rates need to be developed on a comparable basis.

If one is costing out craft labor, their pay rate is usually given in hourly increments. Supervision, support staff, engineering, etc., often are paid on a weekly, bi-weekly, or monthly rate. This rate can also be broken down to an hourly rate for estimating and payroll purposes.

The following are examples of base wages:

- Craft personnel—$27.00 per hour
- Supervision—$1,400 per week divided by 40 hours/week = $35.00 per hour and
- Engineering—$80,000 per year divided by 2,080 hours/year = $ 38.46 per hour

NOTE: 2,080 hours = 5 days per week @ 8 hours per day for 52 weeks

The examples above can be used to calculate the direct amount that each employee will earn and be paid for each hour worked.

Fringe Benefits

Paid Time-Off (PTO) — Most employees have additional benefits of time off for local and national holidays, vacation, sick time, etc. Therefore, in developing a unit cost for labor, a factor is added to increase the estimated and booked cost per hour worked each week to cover PTO. Most companies transfer this money to a special fund to be used when an employee takes paid time off. In the case of construction craft, that may work for many employers during a given period, the funds are usually paid into a fund managed by their union or trade organization and they pay out the salary for PTO.

Example:
An engineer gets 5 days of sick time, 10 days of vacation, and 10 holidays per year with a base salary of $38.46 per hour.

Adders are: Sick Time: 5 days @ 8 hours /day @ $38.46 =	$1,538 per year
Vacation: 10 days @ 8 hours /day @ $38.46 =	$3,077 per year
Holiday: 10 days @ 8 hours /day @ $38.46 =	$3,077 per year
Total	$7,692 per year

This engineer is now working 2,080 hours (52 weeks x 40 hours per week), less 25 days @ 8 hours or 200 hours for paid time off; so productive time is 1,880 hours.

The hourly cost is $38.46 base wage
+ $4.09 PTO adder ($7,692 divided by 1,880 hours)
Total = $42.55

Medical and Life Insurance Benefits—Some firms and labor contracts include contributions to a medical and life insurance program. These are usually calculated on an hourly, weekly, or monthly cost basis and added to the per hour work cost.

If the company that employs the engineer, in the example above, contributes $500 per month for medical and other insurances, the following should be added to the hourly costs:

$500/month x 12 months = $6,000/year divided by 1,880 hours = $3.19 per hour

Furthermore, if the company also contributes $400 per month to a 401k plan (a U.S. recognized company retirement plan) and/or any other retirement plans for the engineer, the following should be added:

$400/month x 12 months = $ 4800/year divided by 1880 hours = $2.55 per hour

Government Mandated Benefits:
These benefits include items such as: government retirement funds, unemployment insurance, retirement healthcare insurance, etc. In the U.S., these funds include Social Security, Medicare, state unemployment insurance, etc.

These costs are usually calculated on a straight percent of the worked hours. Continuing with our example engineer, we will add the following:

Retirement (6.2%)	= .062 x $38.46	=	$ 2.38
Retirement Medical (1.45%)	= .0145 x $38.46	=	$ 0.58
State Unemployment (1.0%)	= .01 x $38.46	=	$ 0.38
Total Government Mandated Benefits		=	$ 3.34

Summary of Engineer's Wages Example
The cost basis for an Engineer who makes $80,000 is summarized in Table 4.2:

Table 4.2—Example Labor Costs for an Engineer

		Per Hour
Base salary (1,880 hours/year)	=	$38.46
Fringe Benefits:		
Company Retirement Contributions	=	$2.55
Paid Time-Off (Holidays, vacation, sick time)	=	$4.09
Company Medical & Life Insurance	=	$3.19
Government Mandated Benefits (Retirement, etc.)	=	$3.34
Total Fringe Benefits (per hour)	=	$13.17
Total Costs (per hour)	=	$51.63
Benefits Adder	=	$13.17/$38.46 = 34.2%

Engineering/Contractors Overhead and Profit

The calculations in Table 4.2 will apply for the direct hire of individual workers. When hiring contract employees, or estimating an engineer's or contractor's costs, labor rates are usually broken down differently. The breakdown includes the following:

- Base wages, including fringes;
- Worker's compensation, (if applicable);
- Overhead; and,
- Profit, (if applicable for time and material situations).

In these instances, the vacation, sick time, retirement contributions, and medical contributions are included in the fringe benefits. Worker's compensation is a direct government rate. Overhead will apply to the home office cost of administration, payroll, billing, etc. Profit usually only applies to approved time and material charges. The negotiated contract determines each of these costs.

RS Means, for example, includes a table of average rates for various types of contractor personnel including overhead and profit in their manual, *Concrete & Masonry Cost Data 2007* [7]. These labor rates are based upon a survey of union rates in 30 cities. Similar information is available in the *Engineering News Record* (ENR) magazine and online.

Fully Loaded or Billing Rate

A fully loaded rate is the base wage, plus adders that will be paid for an hour's work on the job. An owner employing a contractor on a time and material basis only pays for the worker's time when he or she is on the job. If he or she is sick, on vacation, or it is a holiday, the contractor cannot bill the owner. However, the payment rate charged usually includes funds to cover this paid time off. The contractor either places the funds in a separate account for use when the worker is off; or, if the worker is in a union, the union may get the funds to disburse when the worker is off. All estimates relating cost of labor to work performed are usually calculated using the fully loaded rates.

Overtime Wages

There are many different overtime wage situations and there are several aspects that need to be evaluated in developing an overtime wage structure. Overtime can range from straight time pay for the additional hours, beyond the standard workweek of 40 hours, or 8 hours per day, to 1.5 and 2.0 times the regular pay.

When developing the overtime formula, the estimator needs to make sure that those benefits that are only charged on an 8 hour day, or 40 hours per week, are not added to the overtime pay rate. Benefits such as PTO, some insurance, and some governmental funding programs, may be included in this category. Government funded retirements, such as Social Security and Medicare, are calculated as a percentage of the wage and should be added to the overtime rate. The estimator needs to confirm what needs to be added for the specific work area where the project is located, in order to develop the correct rate. See the example in Table 4.3:

Table 4.3 – Overtime Wage Rate Calculation

		Per Hour
Craft worker earning	=	$27.00
Working overtime (1.5 times the base rate)	=	$40.50
(PTO, Company insurance, and State unemployment adders are not included)		
Federal retirement and medical (included at 7.6%)	=	$3.10
Total cost per hour (1.5 time OT)	=	**$43.60**

WEIGHTED AVERAGE RATES/CREW COMPOSITE RATES

Most estimates are for groups of workers who have a variety of backgrounds, years of experience, etc. In the craft area within each craft you could have a range of apprentices to journeymen at the top step of the wage ladder, based upon their training and years of experience. The same applies in the engineering ranks as your team mix will have beginning engineers right out of school, engineering aides, registered engineers, senior engineers, etc. Since estimators do not know who will be part of the actual team, they must make some assumptions in order to develop a comparable base wage rate to use. In most cases they will build a weighted average team, see Table 4.4.

Table 4.4—Weighted Average Examples

Civil Engineering Design Team			
Number	**Classification**	**Hourly Base Wage**	**Extension**
2	Engineering Aides	$18.00	$36.00
2	Junior Engineers	$25.00	$50.00
4	Engineers	$35.00	$140.00
2	Senior Engineers	$40.00	$80.00
1	Engineering Supervisor	$48.00	$48.00
11	**Total**		**$354.00**
Average cost for the group			$354.00/11 = $32.18/hour
With benefits adder of 34.2%			$43.19/hour

Composite Concrete Crew – Normal Time (40 hours/wk)			
Number	**Classification**	**Hourly Base Wage**	**Extension**
2	Laborers	$25.00	$50.00
4	Carpenters	$26.50	$106.00
2	Cement Masons	$26.00	$52.00
1	Ironworker	$26.50	$26.50
1	Foreman	$32.00	$32.00
10	**Total**		**$266.50**
Average cost for the group			$266.50/10 = 26.65/hour
With benefits adder of 30% (assumed)			$34.65/hour

Composite Concrete Crew – Overtime (1.5 times normal wages over 40 hours/wk)			
Number	**Classification**	**Hourly Base Wage**	**Extension**
2	Laborers	$25.00 x 1.5	$75.00
4	Carpenters	$26.50 x 1.5	$159.00
2	Cement Masons	$26.00 x 1.5	$78.00
1	Ironworker	$26.50 x 1.5	$39.75
1	Foreman	$32.00 x 1.5	$48.00
10	**Total**		**$399.75**
Average cost for the group			$399.75/10 = 39.98/hour
With benefits adder of 7.65%			$43.04/hour

Composite Concrete Crew – Double Time (2 times normal wages over 40 hours/wk)			
Number	**Classification**	**Hourly Base Wage**	**Extension**
2	Laborers	$25.00 x 2.0	$100.00
4	Carpenters	$26.50 x 2.0	$212.00
2	Cement Masons	$26.00 x 2.0	$104.00
1	Ironworker	$26.50 x 2.0	$53.00
1	Foreman	$32.00 x 2.0	$64.00
10	**Total**		**$533.00**
Average cost for the group			$533.00/10 = 53.30/hour
With benefits adder of 7.65%			$57.38/hour

Example Calculation:

A contractor needs to make up time in the schedule. If the contractor works the concrete crew, shown in Table 4.4—10 hours per day, Monday through Saturday for two weeks, how much extra will it cost the contractor?

Overtime is paid for all hours over eight Monday through Friday, and the first eight hours on Saturday.
Double-time is paid for hours greater than eight on Saturday and all Sunday work.

Monday through Friday @ 2 hours per day	= 10 hours of 1.5 time pay
Saturday	= 8 hours
Total 1.5 time pay	= 18 hours
Saturday double time pay	= 2 hours

Crew Cost (1.5 times) = $43.04 / hour x 10 workers x 18 hours x 2 weeks	=	$15,494
Crew Cost (double time) = $ 57.38/hour x 10 workers x 2 hours x 2 weeks	=	$2,295
Total	=	$17,789
Normal time cost if no OT worked		
$34.65/hour x 10 workers x 20 hours x 2 weeks	=	$13,860
Additional cost to work overtime	=	$3,929

Special Direct Labor Estimating Adjustments

Commercial estimating databases are composed of composite data that reflect costs and productivity for individual locations, industries, etc. However, some industries such as nuclear or individual plants or projects may require additional nonproductive or "non-wrench turning" activities, which cause their labor costs to increase. Examples of these situations include the need to respond to increased or unique regulatory requirements, special safety requirements, individual plant operational issues, etc. Example responses include: additional instructional meetings, supplemental training, and increased turnover times to comply with industry and plant or project unique situations. Therefore, when using a commercial database, you need to add an adjustment in order to properly align estimates and budgets with these unique requirements. The following is an example of how to determine adjustment factors to use in estimating work hour requirements in these situations:

Example:

In response to a need to improve project safety, Company A has instituted a weekly one hour safety meeting, and increased turnover meetings by 15 minutes at the beginning of each shift. A review of the commercial industry work hour database that Company A uses reveals that none of these activities are included in their commercial estimating database that they use.

The project is currently working a five day, 40 hour per week schedule.
Estimated work hours to set forms for Foundation B = 150 work hours

Additional meeting time:	Safety meeting	=	60 minutes
	Turnover time increase		
	@15 minutes x 5 days	=	75 minutes
	Total increase = 135 minutes/60 minutes/hour = 2.25 hours/week		
	Adjustment = 2.25/40 = 5.6% or 6% rounded		

Thus, six percent needs to be added to all required work hour calculations
The adjustment for setting forms for Foundation B = 150 hours x 1.06 = 159.0 hours required

Extended overtime and some shift work may lead to additional productivity reductions. Extended overtime is usually considered after two weeks, as fatigue can become a productivity factor. Also swing and night shifts may also have productivity impacts. Local and company historical data should be used to determine if additional adjustments need to be considered in developing labor estimates. Adjustments will usually result in adding a percentage productivity adjustment, similar to the example presented above.

INDIRECT AND OVERHEAD LABOR

The previous examples have shown how to determine wage rates and how to use them to develop an estimate for the direct portion of the work. However, a complete estimate needs to include indirect and overhead labor and other costs as well. Overhead and indirect positions were discussed at the beginning of this chapter and were illustrated in Table 4.1.

There are two methods of determining these costs that will be addressed here. The first method is to do a direct estimate of the indirect staff required and cost them out the same way as the direct work crews.

For example, if we were building a manufacturing facility that will take a year, the indirect support could consist of the personnel listed in Table 4.5, using wage rates determined by methods explained earlier:

Table 4.5— Example of Indirect Labor for a Facility Project

Indirect Positions	Duration on site (months)	No. of positions	Worker months	Monthly Rate*	Estimate
Warehouse workers	12	2	24	$3,800	$91,200
	6	2	12	$3,800	$45,600
Accounting clerks	12	1	12	$4,800	$57.60
	6	1	6	$4,800	$28,800
Payroll supervisor	8	1	8	$5,500	$44,000
First aid person	12	1	12	$6,000	$72,000
Safety engineer	10	1	10	$7,000	$70,000
Office manager	12	1	12	$7,500	$90,000
Clerical support	12	1	12	$4,000	$48,000
	6	1	6	$4,000	$24,000
On-site IT support	11	1	11	$7,000	$77,000
Project manager	10	1	10	$10,000	$100,000
Total Indirect Labor					**$748,200**
* Includes benefits					

As shown in Table 4.1, a manufacturing facility or power plant will have the same kinds of functions that will be included in their list of indirect labor positions.

The second method will be to use historical job percentages to determine an appropriate allowance for indirect labor. Typical examples would include applying a percentage of direct labor, based upon historical data, or applying a percentage of the total direct costs for indirect labor, material, and other costs.

Example:

	Estimated Direct Costs	=	$360,000
	Indirect Costs @ 25%	=	$90,000 (25% is from company historical data)
Or,			
	Estimated Direct Labor	=	$250,000
	Indirect Labor @ 30%	=	$75,000

In the second example, material and other indirect costs would have to be estimated separately. The choice of methods will depend upon how much detailed information is available for the estimator to use in developing the estimate.

Overhead Labor

While indirect costs are often located at the plant or jobsite, overhead personnel are more likely to be located at a corporate facility, which is physically separate from the manufacturing facility or construction site. These personnel usually work on many different projects for their company at the same time, or just spend short periods, days or weeks, on each project. Examples of these kinds of positions were included in Table 4.1.

For early estimates, the general methodology is to apply a percentage factor to either direct costs, direct labor costs, etc., as determined by corporate historical data, to develop the overhead estimate. This method can be used for more detailed estimates as well. A detailed estimate for overhead labor can also be developed using number of persons, similar to the indirect labor estimate already illustrated. It is up to management and the estimators to determine which method will supply them with the estimate accuracy they need.

ESTIMATING WORK HOURS TO COMPLETE A GIVEN WORK SCOPE

Estimating work hours is usually not done in detail until enough scope information is available to do at least a Class 3 estimate. A Class 3 estimate, as defined by AACE International[1], is a project on which the major equipment has been identified, layout drawings are available, and rough quantities can be estimated for many of the major elements (such as cubic yards of concrete, linear feet of pipe, etc.). Class 2 or Class 1 estimates would have more detailed unit information, as more of the final drawings would be available. With this kind of information available, the estimators will group these quantities into appropriate work packages, and collaborate with the schedulers to begin the development of the work package sequence. They can then estimate the labor costs by multiplying work hours, from their databases, with the identified quantities. It is critical to do this so that overall staffing requirements over the project duration can be used to confirm the sum of the individual work package requirements.

The AACE classification ranges from least detailed, a Class 5 estimate, to most detailed, a Class 1 estimate; in which all drawings and specifications are completed as outlined in AACE International Recommended Practice 18R-97, *Cost Estimate Classification System for Process Industries* [3].

Work Package Development (or Work Breakdown Structure—WBS)
The first step is to review the project and develop meaningful work packages. Often, several summary level estimates are completed before enough of the project is designed to provide the estimator with detailed quantities of material, so that a detailed labor estimate can be developed. An example of a WBS for construction of a small manufacturing facility consisting of a main building, warehouse and office building, and site work area including entrance roads and utilities, is shown in Table 4.6.

Table 4.6—Example WBS for a Small Facility

Level 1:	**ABC Manufacturing Plant**		
Level 2:	**Site Development**	**Main Building**	**Warehouse**
Level 3:	Roads	Foundation	Foundation
	Utilities	Excavation	Excavation
	Water	Base fill (gravel)	Base fill (gravel)
	Gas/electrical	Concrete	Concrete
	Sewerage	Structural steel	Structural steel
	Lighting	Building walls	Building walls
	Parking lot	Building roof	Building roof
		Building interior	Building interior
		Building lighting	Building lighting
		Building (utilities)	Building (utilities)
		Equipment (installation)	Office furniture
		Equipment (cost)	Warehouse equipment
		Electrical	
		Piping	

As seen in Table 4.6, the work hour estimates will be done at the Level 3. We will use the main building concrete foundation as an example.

- The building is 300 ft. (m) x 100 ft. (m);
- The concrete foundation consists of wall footings, foundation walls, and a slab;
- The slab will be our example, and it is 1 foot (meter) thick; and,
- The quantity of concrete in the slab is: 200 ft. (m) x 100 ft. (m) x 1 ft. (m) = 20,000 Cubic Ft (Cubic Meters) = 741 CY (CM).

Determination of the work hours required is done by consulting a reference database to determine how many hours it has taken historically to complete a foundation slab of this type. There are multiple places to obtain this data, including company historical data for projects on this site or in the area, a commercial estimating database, Means, Richardson, etc.

For example, the database you are using indicates that to place this slab it will take:

0.026 labor hours/m^2

For this example:
20,000 m^2 x .026 labor hours/m^2 = 520 labor hours

In this example, measurements are in feet or meters and are designated by ft (m) (feet/meters), sf (m^2) (square feet/meters squared), etc. Please note that this example uses 0.026 labor hours for both sf and m^2 solely so that the numbers stay the same. There was no attempt to convert hours for sf to hours for m^2.

Costs to Support the Worker
The costs above include only the direct work required to place the slab. The assumption is that all of the material is on hand, the site is prepared, and the crew is ready to start work. The reality is that the costs need to include many items and support personnel that allow the workers to perform their tasks. In addition to indirect and overhead labor, the following examples further illustrate other indirect and overhead costs.

Construction indirect costs and overhead costs can include items, such as storage and fabrication facilities, lunch and rest room facilities, tool rooms, etc. Manufacturing indirect costs could include warehouse space, administrative offices, rest rooms, lunchrooms, locker rooms, raw material loading and unloading facilities, etc. Engineering indirect costs could include duplicating facilities, computer facilities, administrative offices, and personnel, etc. When doing a conceptual, or Class 4 estimate*, these will normally be added as a percentage or allowance. When doing a budget, control, or Class 2 estimate*, these costs will be estimated in detail. When dealing with a less detailed estimate, often these support costs are estimated by adding a historical factor to the direct work estimate to provide for these necessary personnel and activities.

* **Note**: Estimate classifications are from *AACE International Recommended Practice 17R-97*, *Cost Estimate Classification System*, Revised November 29, 2011 [2].

Factors Affecting Productivity
Most estimates are developed from a common database that equates so much work done for so many work hours expended. In the petrochemical industry, common indices are based upon, "U.S. Houston-Gulf Coast" production. These rates are then adjusted for conditions at the jobsite under review. The following is a typical, but not all-inclusive, list of items that each estimator needs to review to determine if they affect the job and the estimate:

- Will union or non-union craft labor be used?
- Is sufficient labor available locally, or will workers have to come from a long distance away?
- If area is remote, do workers have to be bused in?
- What will the weather conditions be like? (hot, cold, rainy, etc.).
- Are there any local holidays?
- Are temporary living quarters needed?
- Is overtime necessary to attract workers?
- What are the standard work hours and work days?

The Richardson Estimating System suggests adjustments to their rates for the following (3, p. 1-2):

Jobsite Conditions	Good		+ 3% to 5%
		Average	+ 6% to 8%
		Poor	+ 9% to 15%
Worker Skill Level	High		+ 2% to 5%
		Average	+ 6% to 10%
		Poor	+ 11% to 20%
Temperature		Below 40 degrees F (4.4 degrees C) or above 85 degrees F (29.4 degrees C) add 1% per degree of variance	
Work Weeks in excess of 40 hours			
		41 to 48 hours	+ 5% to 10%
		49 to 50 hours	+ 11% to 15%
		51 to 54 hours	+ 16% to 20%
		55 to 59 hours	+ 21% to 25%
		60 to 65 hours	+ 26% to 30%
		66 to 72 hours	+ 31% to 40%

**Richardson Process Plant Construction Estimating Standards*, Volume 1, Page 1-2, Copyright 2007.

Example Calculation:
The standard labor cost for 100 LF (M) of footing 8 inches (cm) by 12 inches (cm) = $743.30*
*Richardson 2007, page 7, section 3-1.

The jobsite conditions are:

		Adders
Jobsite conditions	Good	4%
Worker Skill	Average	8%
Temperature	95° F	10%
Work week =	40 hours	0%
Total adders =		22%

Unit Rate [100 LF (M)] = $743.30 x 1.22 = $906.83

Productivity Improvements
The discussion above was meant to make the reader aware of various conditions that affected both the cost and schedule duration of the project.

Learning Curve—One of the most important items affecting the learning curve is the productivity improvement that results from a crew performing repetitive type operations. In a manufacturing environment, or a construction project where similar kinds of work are done, the more the crew does the work, the faster and more efficient they become as they gain familiarity of working together, using the tools, possibly fabricating special tooling to make the work easier and faster, etc. This needs to be encouraged and factored into any budget or estimate made.

Examples of Other Types of Productivity Improvements—While it is one thing to recognize existing factors, there is also the opportunity to put in place procedures or make changes to improve productivity and minimize the cost of some of these factors. For example, to shorten waiting time for a crew, material may be pre-staged at the work location, stored on trailers which can be easily moved to the work site, fabricated in sections, and assembled at the work site, etc. To minimize the impact of adverse weather, temporary shelters can be built which provide shelter from the elements. Using portable tool sheds, which can be moved around to various locations as the work progresses, can shorten the time required to pick up tools. In addition to these physical actions, there is a whole series of actions, including training and team building, which can be used to improve communications and working relations between the various crews and personnel on a site. A manufacturing plant has the added benefit of more permanent personnel and the same physical location at which all of these ideas can be used to improve the worker's efficiency. There are many books and programs dealing with ways to improve productivity and these should be consulted for a complete list of options available. The cost of these types of programs can be more than offset by the savings.

Using Commercially Available Data for Location Estimating and Comparisons
In addition to individual company internal databases, sources for comparison data include RS Means and Richardson Estimating Systems [4-7]. These firms, as well as others, now have online databases that are available for use.

RS Means publishes a U.S. city-by-city comparison that is broken down by material and installation costs and by cost division (concrete, masonry, etc.). Their system involves comparing the ratio of the different city indices to develop a multiplier to apply to your labor cost estimates. Since they also publish unit rates, they offer comparison factors by states, and by zip codes, which can then be adjusted to determine a factor to apply to their unit rate extensions.

An example of R.S. Means' comparison data for two cities is given in Table 4.7.

Table 4.7—Comparison Data for Two Cities per R.S. Means 2013 [4]

		Chicago, Illinois [1]			Los Angeles, California [2]		
		Matl	Inst	Total	Matl	Inst	Total
1	Contractor equipment		92.7	92.7		102.1	102.1
2	Site construction	106.6	94.2	97.9	99.4	107.6	105.2
3	Concrete (summary)	100.0	152.9	126.1	103.2	120.3	111.7
4	Masonry	100.7	151.0	131.4	89.9	119.7	108.1
5	Metals	95.4	133.1	107.2	98.3	104.8	100.3
6	Woods and plastics	99.5	157.6	132.6	102.9	120.2	112.8
7	Thermal/moisture protection	103.3	143.1	119.3	102.2	121.2	109.9
8	Openings (doors & windows)	99.4	158.4	113.7	99.6	120.7	104.7
9	Finishes	99.7	156.7	131.7	102.3	118.6	111.4
10 - 14	Equip & specialties	100.0	124.1	104.9	100.0	111.5	102.3
15	Mechanical (Plumbing, HVAC, etc.)	99.9	133.3	113.6	100.1	116.7	106.9
16	Electrical, communications, utilities	97.8	130.8	115.2	99.8	120.8	110.8
	Weighted average	**99.3**	**138.3**	**116.5**	**99.8**	**116.6**	**107.2**

[1] *RS Means 2013 Concrete & Masonry Cost Data*, page 1360 [7]
[2] *RS Means 2013 Concrete & Masonry Cost Data*, page 1354 [7]

The comparisons listed in Means are comparisons to a national average. Therefore, in order to compare one city to another, you need to calculate the RATIO difference, NOT the numerical difference.

$$\frac{Specific\ City\ Cost\ x\ 100}{National\ Average\ Cost} = The\ city\ index\ number$$

For example, a building in Chicago was erected for $1,540,000; estimate the cost of the same building in Los Angeles.

$$\frac{Index\ of\ LA\ x\ cost\ of\ Chicago}{Index\ of\ Chicago} = Cost\ in\ LA$$

$$\frac{107.2\ x\ \$1{,}540{,}000}{116.5} = \$1{,}420{,}000\ (rounded)$$

Note: explanation from *RS Means Concrete and Masonry Cost Data 2013*.

PERFORMANCE MONITORING

Work Packaging/Work Breakdown Structure (WBS)
An estimate is usually assembled in work packages. Work packages should be clearly identified. Items that determine what makes up a work package include portions of the job that complete a specific portion, building, or area, or are assigned to one subcontractor, can clearly be designed and scheduled, etc. Construction activities have been fairly standard for years.

However, manufacturing, power plants, and other portions of industry also use labor estimating and control. In their situations, work output and their organizations are more complicated than the labor craft vs. work activities on the construction site. In order to better address their situations, Activity Based Cost (ABC) methodology has been developed to aid in organizing, analyzing, and setting up labor monitoring systems within these industries. This methodology gives some guidelines and suggests procedures that can be used to clearly define appropriate work packages. While there is not sufficient time to discuss this methodology here, a discussion of ABC is elsewhere in this publication.

For purposes of this presentation, we will start with a simple construction related WBS for a project to install a new boiler at an existing manufacturing facility.

- Mobilization
- Excavation
- Sub Foundation
- Slab Placement
- Support Steel
- Piping
- Boiler Installation
- Utilities
- Startup
- Cleanup
- Demobilization

The labor estimate is usually the basis for project performance monitoring. The comparison of actual work hours expended versus the estimated work hours and the development of a relationship between milestone goals is a key tool in determining how much of the project is completed, how much effort has been expended to get to the current point, if there are problems, what has to be worked on to overcome these problems, and how this will affect the completion date and final costs. The estimate is shown in Table 4.8:

Table 4.8—Boiler Project Labor Estimate*

Work	Quantities	Units	Work Hours
Mobilization	1	Lot	50
Excavation	300	CF (CM)	100
Sub-foundation	100	CF (CM)	50
Slab placement	100	CY (CM)	270
Support steel	2	Tons	60
Piping	30	LF (M)	90
Boiler installation	1	Lot	50
Utilities	1	Lot	80
Startup	1	Lot	100
Cleanup	1	Lot	50
Demobilization	1	Lot	50
Total Work Hours			**950**

**Please note that this example uses work hours for CF, CY, and LF and CM, and M solely so that the numbers stay the same. There was no attempt to convert hours from English to Metric.*

Since each of these activities has different units of work, one cannot add the units for each piece together. However, work hours can be added together. So, if 100 CY (CM) of the slab is to be placed and is half or 50 percent completed, one has earned 135 work hours (50 percent x 270 estimated work hours). Adding the earned hours up for each of the work packages will then allow a composite percent complete for the entire project to be determined. A more detailed example is shown in Table 4.9:

Table 4.9—Detailed Boiler Project Labor Estimate*

Work	Quantities	Units	Work Hours	% Complete	Earned Hours
Mobilization	1	Lot	50	100%	50
Excavation	300	CF (CM)	100	100%	100
Sub-foundation	100	CF (CM)	50	100%	50
Slab placement	100	CY (CM)	270	50%	135
Support steel	2	Tons	60	10%	6
Piping	30	LF (M)	90	10%	9
Boiler installation	1	Lot	50	0%	0
Utilities	1	Lot	80	0%	0
Startup	1	Lot	100	0%	0
Cleanup	1	Lot	50	0%	0
Demobilization	1	Lot	50	0%	0
Total Work Hours			**950**	**35%**	**335**

**Please note that this example uses work hours for CF, CY, and LF and CM, and M solely so that the numbers stay the same. There was no attempt to convert hours from English to Metric.*

It is important when reporting percent complete that the milestones and the credit for each are as clearly defined as possible. This will build credibility with the system in that everyone reporting progress will do it the same way and the data collected will stand up to management scrutiny.

For some activities, such as cleanup or mobilization and demobilization, it may be very difficult to define the milestones. Progress for these activities may be monitored using a duration scale. In other words, if these are to take four days to complete, and two days have gone by, then this portion of the work is 50 percent complete. This is a subjective approximation and is one of many ways to obtain an earned value when a mathematical calculation is not practical. For these activities, it is imperative that management understand and agree to the methodology used prior to the start of the project.

Performance Monitoring Calculation Examples:

The example in Table 4.10 illustrates how the overall percent complete is calculated using work completed at any given time.

Table 4.10—Slab Placement Status Calculation Based Upon Physical Work Completed

Slab Placement	Estimated Work Hours	Activity % Complete	Work Hours Earned	Project % Complete
Formwork	67.5	100%	67.5	25%
Reinforcing steel	67.5	100%	67.5	25%
Concrete placed	108.0	0%	0.0	0%
Cured	27.0	0%	0.0	0%
Total	**270.0**		**135.0**	**50%**

In Table 4.11, each activity of the slab placement is estimated and the work hours are allocated to each of the five days in the plan.

Table 4.11—Example 1—Slab Placement Work Progress—Plan

Activity	Estimated Work Hours	Day 1	Day 2	Day 3	Day 4	Day 5
Formwork	67.5	67.5				
Reinforcing steel	67.5	12.5	55.0			
Concrete placed	108.0			54.0	54.0	
Cured	27.0					27.0
Total	270.0	80.0	55.0	54.0	54.0	27.0
Cumulative Hours (Plan)		80.0	135.0	189.0	243.0	270.0

In Table 4.12, the status through Day three is reported, for both the physical completion, and the work hours expended.

Table 4.12—Example 1—Slab Placement Work Package—Status

	Estimated Work Hours						
Activity	**Day 1**	**Day 2**	**Day 3**	**Day 4**	**Day 5**	**CPI**	**SPI**
Cumulative hours (plan)	80.0	135.0	189.0	243.0	270.0		
Cumulative actual hours expended	90.0	165.0	200.0			1.22	
Cumulative actual hours earned	80.0	205.0	243.0			1.29	

Graphic Presentation of Earned Value Data

The project status data for Example 1 is shown in Figure 4.1 in a graphic format. Reviewing this work package, it appears that this work will finish ahead of schedule and under budget. The differences on the graph between the plan and earned and expended lines provide a visual analysis of this example. As long as the curves are above the planned line the status is OK. The addition of the Cost Performance Index (CPI) and Schedule Performance Index (SPI) calculations provides a numerical calculation to further define the status.

CPI = hours earned/hours expended = 243/200 = 1.22
SPI = hours earned/hours planned = 243/189 = 1.29

If the indices are ≥ 1, then the project is generally on or ahead of schedule. If the project, or elements of the project, is < 1, then these activities need to be reviewed to determine if action is required to improve their performance, as they are not progressing as planned. (Note: If more work is completed on non-critical activities, it is mathematically possible for the indices to be > 1 and the project still to be in trouble. That will be because the critical path activities are behind schedule.)

Example 1 Work Package Status

In Example 1 (Figure 4.1, Table 4.12), the costs are higher than the plan. This is normally not good, except when the schedule is also ahead of the plan. Since the project is ahead of schedule and the CPI (Cost) is less than the SPI (schedule), this portion of the project will finish early and may finish less than planned.

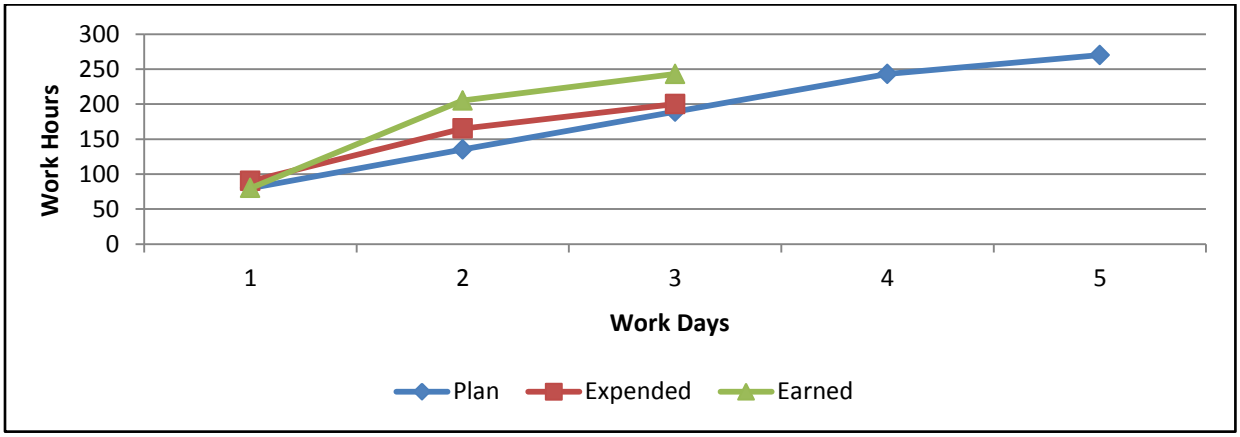


Figure 4.1—Earned Value Example 1

In Example 2, the costs (work hours expended) are higher than the plan. This is normally not good, and since the earned is less than the plan, it looks like there are some problems. Confirming that there are problems, the CPI (cost) and the SPI (schedule) are < 1.0. So, the work package appears to be over-running the budget, and behind schedule as well.

Table 4.13—Example 2 Work Package—Status

	Estimated Work Hours						
Activity	**Day 1**	**Day 2**	**Day 3**	**Day 4**	**Day 5**	**CPI**	**SPI**
Cumulative hours (plan)	135.0	189.0	243.0	270.0			
Cumulative actual hours expended	90.0	165.0	198.0			0.83	
Cumulative actual hours earned	80.0	135.0	164.0			0.87	

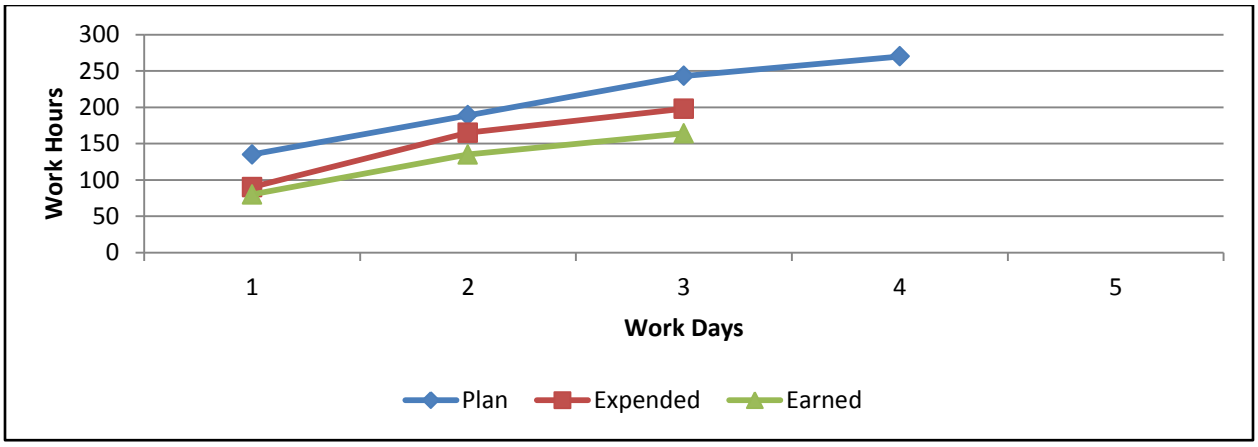


Figure 4.2—Earned Value Example 2

In Example 3 (Figure 4.3, Table 4.14), the costs are less than the plan. This would normally be OK. However, the earned is less than the plan; it looks like there are some schedule problems. Confirming that there are schedule problems, the CPI (cost) is 1.03 (> 1.0), and the SPI (schedule) is 0.93 (< 1.0). So, the work package appears to be under-running the plan at a greater rate than the work package is behind schedule. If the schedule slippage is acceptable, then the budget may still be OK.

Table 4.14—Example 3 Work Package—Status

	Estimated Work Hours						
Activity	**Day 1**	**Day 2**	**Day 3**	**Day 4**	**Day 5**	**CPI**	**SPI**
Cumulative hours (plan)	80.0	135.0	189.0	243.0	270.0		
Cumulative actual hours expended	90.0	150.0	170.0			1.03	
Cumulative actual hours earned		80.0	115.0	175.0		0.93	

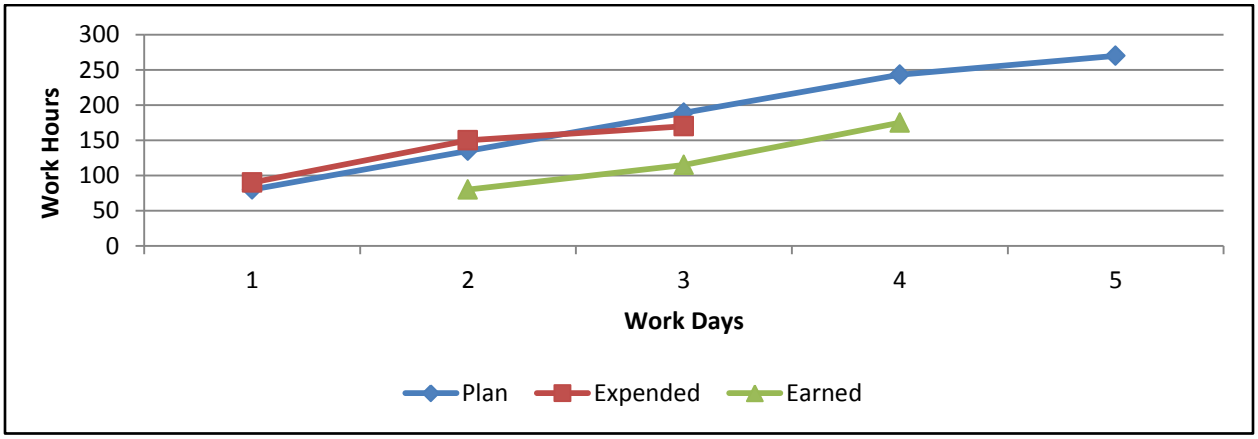


Figure 4.3—Earned Value Example 3

Work Sampling

Work sampling is a method that can be used to determine production or unit rates for specific work activities. These rates are to be used in setting up a company database, or determining the relationship between work at an individual site and labor standards, which have been or may be used for estimating projects in the future.

The process involves picking a sample work item or items and having personnel record all activities and labor hours associated with those activities so that unit rates per production measure can be determined. Personnel who witness and record the activities and the labor hours worked on each activity can collect data from timesheets, or create their own database based upon their observation of the work activities. Comparisons can then be made against existing experience or databases to determine the most reasonable data to use as the standard. Or, the data can be used to determine how the actual work is deviating from the standard.

CONCLUSION

In this chapter, the basics of labor cost development and progress monitoring have been discussed. In all instances, it is imperative that the estimator/cost engineer understand the basics in order to make sure that all of the areas impacting the estimate are thoroughly investigated and that the correct data is used in a consistent manner.

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The following is a list of companies, by no means all inclusive, that provide estimating data:

- ARES Corporation
- Building Systems Design, Inc.
- ICARUS/Richardson—Aspen Technology, Inc.
- RS Means
- Win Estimator

Chapter 5—Engineering Role & Project Success

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Abstract
Engineering is a key element toward the success of products and projects. Proper delivery of a service can also benefit through such techniques as flow chart analysis. Computerized techniques, such as CAD/CAM, BIM, simulation, and others, require significant upfront investments in software/training, but correctly applied have proven their worth in reducing overall costs toward the delivery of products and processes. Most of the costs of products/projects are determined in the engineering phase while the bulk of the expenditures are found downstream in production, construction, and operations. Techniques including constructability, manufacturability, and value analysis have found use in this area.

Keywords
Computer Aided Design (CAD), Computer Aided Engineering (CAE), Computer Aided Manufacturing (CAM), Constructability, Engineering Production/Construction, Prototype, and Standardization.

INTRODUCTION
The success of many products and projects is predicated upon an effective and efficient engineering effort. Improved engineering efforts are essential in many applications given globalized competition and product/project liability issues, among other concerns. Globalization means that countries around the world are now competitors. Adequate engineering is no longer sufficient, as previous trade barriers have fallen or seen substantial reductions. Labor costs, where not offset by advantages in materials and transportation costs, mean that labor-intensive tasks are exported around the world. Engineering tasks that are labor-intensive have in some instances followed the same pattern, as businesses out-source engineering tasks to lesser-developed countries where an educated engineering workforce in a lower cost environment yields competitive advantage. In some cases, engineering itself may be componentized with various elements of a design done in many countries and eventually integrated into a unified whole. Traditional product and project engineering has been revolutionized in many cases by the advent of CAD/CAM concepts which enable the rapid development and prototyping of design concepts. Businesses are realizing that they must automate many heretofore manual engineering practices in order to compete in a globalized environment. Practitioners in this area must understand the potential of CAD/CAE/CAM for competitive advantage. One must be able to relate engineering decisions on product selection to their impact on process selection.

Learning Objectives
After completing this chapter, the reader should be able to:

- Identify engineering issues involved in product, project, and process development, including research, the use of CAD/CAE/CAM, product liability, patents, trade secrets, and developing prototypes.
- Understand product and process design and production issues, including process selection, standardization, manufacturability, constructability, and "make" or "buy" issues.
- Identify production health and safety issues.
- Identify issues involved in planning facility layout.
- Design assembly and flow process charts.
- Understand other engineering production/construction concepts, such as reengineering, and relate engineering decisions on product selection to their impact on process selection.

PRODUCT, PROJECT AND PROCESS DEVELOPMENT
Development of products, projects, and processes ranges from the simple to the incredibly complex. Similarly, timeframes can range from days, weeks, and months to several years or more. In addition, development may depend on concurrent and predecessor development of other items. The advent of low-cost functional microprocessors has enabled control of processes on a real-time basis that was previously only imagined heretofore. In addition, the commercialization of computers has meant the automation of numerous design processes as a result of their ever-decreasing costs, bringing these advantages to even low-volume operations.

Pure and Applied Research
In the development of products, organizations conduct research. The research can be divided into two types; that of pure research and applied research. Pure or basic research involves work without a specific end product or use in mind. A common example might be a researcher in a lab examining the interactions of different chemical compounds. Pure research functions most often take place at universities or foundations. Pure research activities may be financed from government grants or private sector grants. In some progressive private entities they earmark a certain percentage of sales to be devoted to pure research, realizing that this is ultimately the seed corn for new products. On the other hand, applied research is the attempt to develop usable products or add new feature-sets to existing products. Applied research is more specific than pure research and is typically carried out by the

organization producing the product. These applied efforts can be directed at enhancing existing products or extending product applications to new areas.

Product, Project, and Process Life Cycles

The life cycle of a particular product/project will have a significant influence on all design decisions, including producing plant and equipment. Some products such as automobiles have long lives while seeing significant changes in feature sets throughout history. Civil infrastructure projects often have lives of many decades and should be designed for low-cost maintenance and upgrades. Other products may have a life cycle measured in a handful of years or less. With short-life-cycle items, time to market is essential requiring both rapid design and production to avoid missing windows of opportunity. There is no sense building a 30-year life factory with associated equipment for a product that will be obsolete in five years.

There is, of course, a prediction problem inherent in life cycles and forecasting demand. In the era of large mainframe computers in the 1960s, who could have foreseen the era when computers would be prevalent on desktops in both homes and offices? Phonograph records were a viable product for decades until supplanted by superior compact disc (CD) technology. Videotapes have been replaced by digital video discs (DVD) while other technologies such as large platter-type laser discs have fallen by the wayside. Similarly, existing DVD technology has been rapidly supplanted by improved technology. CRT-based televisions and computer monitors have been made obsolete by flat-panel LCD and plasma displays. These are being made in ever-larger sizes for movie-theatre-type experiences in the home and office conference rooms while their more-compact offspring are used in smart phones.

Computer-Aided Design (CAD)/Computer-Aided Engineering (CAE)

In product, project, and process development, designers have been aided significantly by the advent of computer-aided design (CAD) and computer-aided engineering (CAE) software. CAD/CAE software involves the use of computerized work stations and software, including data bases and computer graphics to quickly develop and analyze a product, project, or process design. Combined with the Internet revolution with high bandwidth connections, designers and engineers around the world can work on a single design. Electrical engineers in Germany may be working with mechanical engineers in India to develop a new product that will actually be produced in China. The Indian mechanical engineer can call up a part in a database, such as a motorcycle strut on a suspension system. The part then can be automatically generated as a finite model and run in a mechanical design optimizer package for a certain number of cycles. At the same time, the Indian mechanical engineer can be engaged in an on-line conversation with the German electrical engineer on a key mechanical/electrical interface issue for routing of a wiring harness. In a half-hour, the conversation ends and the results of the mechanical design optimizer software are now available. The optimizer software has continually refined the design through numerous iterations. The engineer can now review the results, including color graphics views of stress/strain diagrams. The mechanical engineer can then bring this strut into the complete design and perform analysis on this design as well.

Once designs are finalized, design data can then be ported to the manufacturing process through the interface between CAD/CAE software and CAM software. Concepts, such as 3D printing, first developed in the mid-1980s, are now being popularized as a result of enhanced technology and lower costs to allow rapid prototyping of design concepts.

Computer-Aided Manufacturing (CAM)

Computer-aided manufacturing (CAM) provides the counterpart advantages of CAD/CAE software to the factory floor. Design information from CAD/CAE software can be ported directly into CAM software. Design dimensional information then can directly be sent to machines to control their actions in producing parts and products. This degree of automation can range from the simple to the complex as noted above. A fabricator of wood roof trusses for residential and light commercial structures can take its designs and port them to the CAM software. The CAM software aggregates the design information for computer-controlled cutting of wood roof truss members. The software package calculates member's dimensional requirements against lumber piece length to optimize production and minimize waste. Truss designs are stored on computer diskette and when time comes to actually produce the truss, the truss layout is projected onto a laser layout table. Truss assembly personnel therefore do not have to measure truss chord and web locations with the computer laser table projection. The CAM process is thus faster with fewer mistakes. In more complex cases, computer-numerically controlled (CNC) machines are linked with design and receive tooling/machining instructions. Design information can be sent separately to various machines for production with final assembly to take place later. These CNC tools are typically multi-function for operations such as machining with numerous operations being accomplished automatically at one workstation.

Prototypes

For a variety of reasons, organizations will typically develop prototypes prior to large-scale production. Prototypes are developed to test designs and also test customer reaction. An equipment manufacturer planning a new type of equipment may place prototypes in the hands of customers for real-world testing and demonstration. Design concepts may be uncertain or user reaction may be a key element in downstream product success. Prototype development is expensive. However, it is less expensive than producing an unwanted item or an item with a key flaw that otherwise may only be discovered after numerous units are in customer hands. Prototype development may mean the actual construction of a small-scale pilot plant to test concepts, similar to what is done in the petro-chemical business.

The advent of CAD/CAE has led to the development of virtual prototypes on computer. Organizations at a fraction of the cost of physical prototypes can produce computer simulations of a prototype for testing by operators and designers. These computer simulations mean that operator mistakes will not cause real physical damage further reducing testing costs. Changes to a computer simulation can be readily accomplished far faster than with a physical prototype. This is important anywhere that time-based competition is an issue.

Whether the prototypes be actual physical models or computer simulations, the use and application provides significant benefits for the organization. Customers may be exposed to both physical and computer-simulation prototypes. This also provides for interaction of the design team with customers that can provide important feedback. This one-on-one interaction gathers information far better than simple survey form methods. Physical or computer prototypes represent a significant financial and time outlay by the organization, but eliminating this development step has typically proven to be a short-sighted measure.

Patents and Trade Secrets

Investments in new products and research are typically both expensive and time-consuming endeavors. This investment must receive protection. Typically, organizations can protect this investment through either patents or trade secrets. In the U.S., for patents starting June 8, 1995, a patent's duration is now 20 years from the date of patent filing. In return for publishing the information underlying the patent, protection is granted to the inventor(s) for an exclusive period of 20 years. Patents filed prior to June 8, 1995, in the U.S., have a protected period of 17 years from the date of patent issuance. The 20-year period has been standardized by the World Trade Organization (WTO). These WTO protections, with 20-year patent rights, have been adopted by virtually all industrialized nations around the world. Those organizations wishing to emulate the patent's provisions will either have to develop a creative approach different from the patented design, or pay royalties to the patent holder. Copying patented features before the 20-year patent expiration date will result in a case of patent infringement. This can lead to, in many instances, significant monetary damages along with potential cessation of producing the offending product.

Trade secrets also serve as protection for intellectual property. Trade secrets can be somewhat broader than patent protection in that they protect both commercial and technical information from disclosure. Organizations producing substances from a popular soft drink to a fried-chicken recipe, or services such as a particular business method, have all successfully protected these through the trade secret route. The advantage of trade secrets is their perpetual nature, as long as disclosure can be prevented for the trade secret. Organization employees are prevented from disclosure of trade secrets through employment contracts, as well as subdivision of processes to prevent the repository of total information in one or two employees. To avoid reverse engineering of a particular substance by competitors, items such as masking agents and inert chemicals are used as protection methods to prevent competitor discovery.

Product Liability

In today's increasingly litigious society, product liability is of importance in engineering design and production. In some cases, the product liability issue can act as a drag on engineering innovation retarding the advancement of design because of these concerns. Product liability provides a means by which those injured by a product can seek compensation for their damage. The tort law in this area has evolved over decades from a concept of buyer beware to a concept of seller beware. In part, this is a result of the increasing complexity of today's products. The old concept of buyer beware was more applicable in a time of peddlers selling a bolt of cloth. The buyer could reasonably examine the bolt of cloth as to quality. Compare this with the purchase of a rotary lawnmower at a store. The variety of pieces and parts on the lawnmower make it more difficult to examine and certain defects in design engineering may only become apparent after extended use of this unit. The law recognizing the increased complexity of products has evolved to placing increased responsibility upon the manufacturer. Defensive design in this area aims to attempt to make products much safer, even when used improperly by the end user. An example of defensive design is the switch on a consumer power tool that requires an additional switch activation to energize the tool. Defensive design on electrical power tools may also include the incorporation of a Ground-Fault Circuit Interrupter (GFCI) so that the tool shuts down if the cord or other apparatus has been damaged, preventing electric shocks to the user.

Product liability lawsuits can be won even if the manufacturer did nothing wrong. Therefore, some items may be over-engineered with more robust designs with high safety factors to counter the potential for future product liability lawsuits. A safety factor is a reserve amount incorporated into a product as protection against a failure. A cable-operated elevator system, for example, may have a safety factor of six to one. Examination of the elevator sheave-pulley system might find six cables being used to support the elevator cab in its vertical movements within a building. However, the weight could be supported by one cable, whereas six cables are used in this example to provide a significant safety factor. In addition, safeties such as governors would prevent an elevator cab from a free-fall event as they would activate to "grab" the elevator side rails. Steps such as these greatly increase product safety and mitigate the chances of a product liability issue.

PRODUCT, PROJECT, AND PROCESS DESIGN

Standardization

Engineering design must pay close attention to standardization concepts. Standardization attempts to base product designs, in whole or in part, on existing product items and tooling. The advantages of standardization are readily apparent in that by

incorporating existing elements into new products, overall product development costs will be lower and time to market will be shorter. Standardization also means that engineering development costs can be amortized over more units of product reducing selling costs. An example of design standardization may be through use of common parts, such as an automotive frame. One automotive frame system may provide the basis for several automobile types. The same principles would apply to engines, as one power plant is used in a number of applications. Engine horsepower may be varied by use of turbochargers versus naturally aspirated engines. Engine displacement can be varied by changing the stroke and holding the cylinder bore constant.

Not only are there economic benefits to standardization for the producer, but for the customer as well. Product standardization on equipment types can mean easier and better operation as users are already familiar with common controls along with other features. Operating personnel only have to learn one set or type of controls resulting in more proficient operation. Product standardization means less investment in spare parts inventory and lower general maintenance costs. Maintenance personnel are able to become more familiar with fewer equipment components, resulting in faster repairs and fewer mistakes.

Standardization can pose a problem concerning product defects. If there is a product flaw, the flaw will be spread over a wide variety of products. These general product flaws can be costly from a repair/recall standpoint, in addition to harming the organizations' overall image and reputation.

Process Selection

Part and parcel of product engineering design will be process selection. Process selection relates to the production methods chosen to produce the product. There are two basic types of production methods: continuous production and discrete production. Examples of continuous production methods would be petro-chemical plants, power plants and manufacturers with assembly-line methods, such as wire/cable manufacturers and automotive manufacturers. Examples of discrete production would be manufacturers of any type of custom product, such as a concrete pre-cast plant, structural steel fabrication shop, or machine shop. Some products will envelope both methods, sometimes by the same firm. A structural steel fabrication shop will take basic steel products from a steel manufacturer engaged in a continuous process method, such as continuous casting and turn them into customized "one-off" structural steel fabrications for a particular project.

Continuous production method systems are less expensive in the long run as the high fixed costs of extensive production machinery can be amortized over many units of production. The determining factor in deciding in favor of continuous production is whether demand is such that production volumes can justify the investments required for this method. Continuous systems result in lesser costs on a volume basis, but during economic downturns saddle the organization with significant fixed costs. In general, continuous production methods use specialized equipment including conveyor lines and specialized machinery. Discrete production methods use general purpose equipment. Production equipment such as forklifts, welders and fabrication tables and machining centers can be used for a wide variety of items. Discrete production methods will have a higher labor factor versus continuous methods. In regions of the world where labor costs are less expensive, relative to capital equipment, discrete methods may be favored because of their more favorable capital/labor tradeoff ratios. In addition, during economic downturns, the organization has a significantly less-expensive fixed-cost base to support.

Manufacturability

Engineering design methods must focus, not only on issues of product design, but also on issues of manufacturing the design [1]. A design may work perfectly in terms of function, but if it is unnecessarily complicated to produce, problems may ensue. Design tolerances may be specified that are unnecessary for product function, yet create substantial problems in production. Designs, where possible, should be:

- Forgiving of minor inaccuracies
- Easy to fabricate
- Based on efficient use of labor, materials, and equipment.

Designs done on the computer can prove problematic to manufacture on the factory floor. Slight changes or modifications in a design that does not affect product integrity, as to function or aesthetics, but instead promote ease of assembly of the product are referred to as manufacturability. Whether items, such as jigs and fixtures, are used to support components while being assembled (saving substantial production time) is often a key item for review in this area. During the design and development process, experienced manufacturing personnel should view products from the manufacturability perspective. These reviews can pinpoint problems before designs are developed to the point where changes create significant delays and associated costs.

Since manufacturing a product may involve the incorporation of numerous items from other subsidiary firms, these entities should be made part of the team evaluating manufacturability issues. The team approach is a necessity in this area to produce a product that meets key design intent concepts, while being feasible for the firm to manufacture. The advent of embossed patterns in sheet steel was marketed to consumers for products, such as refrigerators, as not showing fingerprints. This was true and a benefit to product users. The cost benefit for manufacturers was that the embossed steel "hid" minor nicks and weld spatter from the manufacturing process, thus eliminating this source of rework.

Constructability

Constructability is the counterpart of manufacturability applied to constructed projects and elements. The same issues apply to the realm of constructability. Designs can be developed on paper and in the computer that may make sense, from the designer's viewpoint, but create significant problems in their construction. Generally, work such as welded construction is better suited to a fabrication shop, whereas field structural steel construction prefers bolted construction designs.

Improved constructability of designs means, in many cases, that large elements of construction can be produced in a fabrication facility, yielding controlled costs in a controlled environment. This approach can then reduce field labor content where costs are more variable as a result of issues such as weather, field workforce skill levels, and labor availability.

The Construction Industry Institute (CII) has defined constructability as the optimum use of construction knowledge and experience in planning, design, procurement, and field operations to achieve overall project objectives [2]. CII was founded by large industrial owners in the US to enhance the construction process, including these issues. They have researched and published the results of this work.

The same precepts of manufacturability apply to constructability. Early pre-construction implementation of these techniques can pinpoint problems before designs are developed to the point where changes create significant delays and associated costs. Constructability analysis of designs by experienced construction site management can pinpoint key design implementation concerns. An increasingly-popular trend in constructability analysis is design evaluation through computerized Building Information Modeling (BIM) software. Evaluation of engineering designs through BIM analysis can point out routing conflicts between structural elements and mechanical/electrical/plumbing systems. BIM is increasingly being used to not only achieve constructability goals, but to reduce project delivery time.

Make or Buy Decisions

Product, project, and process development must concern decision-makers with "make" or "buy" decisions. That is to say, what items should be subcontracted out to others versus what items should be made in-house. How the organization's quality and cost on an item compare with outside suppliers are typical issues to be considered. If trade secrets are involved in an item, this decision will typically be a "make" decision, unless the trade secret is the result of certain combinations of widely-sourced ingredients. For example, if a commodity, such as sugar, is part of a fermentation trade secret process, the organization will not find it practical to produce its own sugar, unless it is already in that line of business.

The overarching goal with make or buy decisions is making the best selections that will enhance overall quality at a lower cost. If a manufacturer is producing 10,000 units of a piece of equipment on an annual basis, it may make far more sense to purchase motors from another manufacturer specializing in motors. The motor manufacturer may be making 1,000,000 of these motors for their own use and as OEM-sourced equipment for others. There is no practical way to achieve these kinds of economies-of-scale benefits at the relatively low 10,000 units per year. Therefore, the motor selection will be a buy decision.

The materials selection decision as to make or buy must consider actual product demand issues. Particularly with a new product, forecasted demand may be uncertain. It is far easier to subcontract with other organizations to buy needed components when potential demand is uncertain. Properly selected, these entities already have production facilities and experienced personnel making similar or the same components as needed for the production of an item. On the other hand, making an item means that plant, equipment, and personnel must be marshaled to produce the necessary components. The organization may not have the critical mass or capital to be able to afford the luxury of producing these items by themselves. By choosing the "buy decision," the subcontracting entity brings specialized labor, management, tools, equipment, plant, and capital into the equation. If the subcontracting entity has spare capacity, this can be a fast and cost-effective alternative to making the item themselves.

ENGINEERING PRODUCTION/CONSTRUCTION

Numerous decisions must be made with regards to engineering production/construction. These decisions will naturally occur based on decisions reached in other areas as outlined above.

Production Health and Safety

Personnel health and safety is paramount in any production situation; as well as from both a humanitarian and an economic standpoint. In many countries, organizations producing work through their own plants, or by subcontracting, have also encountered public relations backlash as a result of substandard health and safety conditions.

Advanced civilizations place a premium on human health and safety. From the economic viewpoint, health and safety lapses are expensive. An accident can result in the loss of a trained worker or an interruption in the process. Systems must be selected that reduce, and/or eliminate, the potential of accidents. Health issues are more difficult to ascertain, because unlike a safety-related accident, the effects may not be immediately known. Health issues include exposure to fumes, dust, noise, and heat. Fumes from a production process may cause long-term health problems. The exposure risk may necessitate a change to eliminate fumes or to include robotic operation, so that human exposure is minimized.

The first priority is to attempt to remove as many safety hazards from the process as is practicable. An example of this would be the design of a sound-proof enclosure substitution in the development of engineering specifications for a work process to lessen noise impact on the workforce. If the engineering solution of the noise enclosure is not practical, then work practice controls become the next priority. Engineering analysis may find sound decibel levels to be too high for a worker on an eight-hour Time-Weighted Average (TWA) exposure basis. Therefore, the solution is to rotate workers from a high-noise area to a low-noise area every two hours to meet TWA-noise safety standards. But, a low-noise area may not be available, or it may not be practical to rotate the workers; therefore, another solution must be found. The last alternative is to provide workers with Personnel Protective Equipment (PPE) such as ear plugs or ear muffs, in order to meet TWA-noise safety standards.

Facility Layout

Facility layout involves decisions as to arrangement, including transportation concerns, equipment location, labor location, and services location. The facility may be existing, requiring renovation, or one that is being built from the ground up on a "greenfield" basis. Layout decisions should always consider the potential impact of additional demand; thereby, considering future expansion and additions to the base layout. If the facility will be incorporated into an existing plant site, with other components, then the meshing of the new facility with existing production units is a key concern.

Assembly and Flow Process Charts

Assembly and Flow Process Charts assist in planning the facility layout and material flow. They help to analyze production operations in terms of operations sequences performed, distances between operations, and operation time requirements [3].

Process charts are developed based on standard symbols for studying operations. The symbol O represents an operation. An operation occurs when any change takes place whether of a physical nature or when new information is received. The symbol ∇ represents storage of an item. Storage is when any item is kept or held in the process. The symbol ⇒ represents transportation or movement in the process, as when items are moved from one location to another. The symbol □ represents inspection. An inspection occurs when an object is examined for conformance as to quality or quantity. The symbol *D* represents delay. Delays occur with interruptions to the process prohibit the next operation or item from taking place. With process chart development, operations can be timed to analyze potential production issues. On top of this, personnel balance charts can analyze work activity by each worker to optimize product flow and results.

A production operation might entail the cutting of steel beams to length and drilling holes in the beams. Flow charting of this operation is shown in Table 5.1.

Table 5.1—Flow Chart of Steel Beam Production

Symbol	Operation
∇	Beams in storage
⇒	Transport beam to saw
O	Cut beam to length at saw
∇	Place on pallet
⇒	Transport by forklift to vertical drill
O	Drill beam
□	Inspect beam to verify quality
D	Store on pallet until needed
⇒	Transport to paint booth
O	Prime beam

Flow charts offer the advantage of mapping product flow, which helps in spotting inefficiencies in the production process. Steps can be changed and rearranged to promote higher productivity and lower costs. Therefore, these charts can be a powerful tool in facility planning.

Quantitative Analysis in Facility Layout

Techniques such as linear programming and Monte Carlo techniques can significantly assist in facility layout and production decisions. Linear programming is a mathematical technique which is widely used to find optimal solutions to problems. Linear programming techniques are designed to either minimize or maximize an objective function. For example, material distances in a facility can be minimized or available space can be maximized. These linear programming decisions are also bound by limiting or restraining parameters. There may be a limited amount of space available or limited milling machines available, for a particular production operation. Detailed linear programming techniques are beyond the scope of this chapter since entire books have been written on these topics.

Monte Carlo techniques provide for simulation. Queuing problems, such as wait time for a crane in a plant, can be analyzed with Monte Carlo techniques. Data can be generated via computer programs with random number generators. Information such as wait time costs are factored into the equation and simulations are conducted on this basis. Again, these techniques are beyond the scope

of this chapter since entire books have been written on these topics. Practitioners in this area are encouraged to consult these books for further information.

Reengineering

Reengineering is the fundamental rethinking and radical redesign of business processes to achieve dramatic improvements in critical contemporary measures of performance, such as cost, quality, service, and speed. Some of these steps include combining two or more jobs into one and enabling workers to make decisions, with work being performed where it makes most sense. An example of reengineering might have your supplier monitoring your inventory of their supplied item, since they are better able to accomplish this task. This then frees the organization to focus on its own mission-critical work.

Reengineering focuses on the optimization of the total organization, rather than sub-optimization of individual departments or units. In the context of this section, disparate units such as design, manufacturing, sales/marketing, and customer service are brought together to deliver optimal solutions that benefit the entire organization, as opposed to an individual unit. Moreover, reengineering focuses on the "why" of an action or process as opposed to the "how." An appliance manufacturer may be concerned with welding spatter on new appliances and how to control this. However, a change in the steel purchased from a smooth pattern to an embossed pattern on steel sheet may make the welding spatter not noticeable in the finished product. Here, a focus on why the welding spatter needs to be controlled is more important than the how, and results in a simple purchasing specifications change. Reengineering focuses on these global issues.

CONCLUSION

The recognition of the impact and potential of engineering issues will lead to more rational decision processes by those responsible. Traditional product and project engineering has been revolutionized in many cases by the advent of CAD/CAE/CAM concepts, which enable the rapid development and prototyping of design concepts. Businesses are realizing that they must automate many heretofore manual engineering practices, in order to compete in a globalized environment. Individuals practicing in this area must understand the potential of CAD/CAM and business reengineering for competitive advantage. One must be able to relate engineering decisions on product selection to their impact on process selection.

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Chapter 6—Machinery, Equipment, & Tools

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Abstract
This chapter illustrates the management of machineries, equipment, and tools related to a project and their impact on the project's schedule and cost. An overview of how to establish an equipment valuation database and identify the different categories and subcategories of the equipment value is provided. This chapter also discusses equipment price and cost information as well as the current and residual values for new and used equipment.

Keywords
Fair Market Value-In-Exchange, Fair Market Value-In-Place, Fair Value, Forced Liquidation Value, Orderly Liquidation Value, Replacement Cost, Reproduction Cost, Salvage Value, and Scrap Value.

INTRODUCTION
Selecting, purchasing, tracking, storing, maintaining, and selling machinery, equipment, and tools is an important project management function that can greatly impact a project's cost and schedule. This chapter outlines current issues and industry practices regarding equipment from a cost engineer's perspective.

Learning Objectives
After completing this chapter, the reader should be able to:

- Establish an equipment valuation database and identify the different equipment value categories and subcategories.
- Research equipment price and cost information.
- Understand the factors that affect current and residual values for new and used equipment.

ESTABLISHING AN EQUIPMENT VALUATION DATABASE
In order to establish a reliable equipment valuation database, upon which to base estimates or appraisals, emphasis must be placed on the quantity, quality, and trade level (sales tier) of data being cataloged. This is so the values contained in the database are appropriate for their intended use. Such data can be used to establish reliable current value appraisals and residual value estimates (future values) for equipment needed on a project.

Equipment Value Categories
Equipment values can be divided into two major categories:

- Replacement cost (the cost new of an item having the same or similar utility)
- Market value (used equipment, secondary market value)

Within each category are subcategories. The first category, **Replacement Cost**, has the following subcategories:

- **Reproduction Cost**— is the cost new of an identical item.
- **Fair Value**— is the adjusted cost of a new item, giving consideration for the cost of similar items, and taking into account utility and all standard adjustments and discounts to list price.

The second category, **Market Value**, also contains several subcategories:

- **Fair Market Value-in-Place**— is the monetary amount that may reasonably be expected for an installed property in an exchange between a willing buyer and a willing seller, with equity to both, neither under any compulsion to buy or sell, and both fully aware of all relevant facts, including installation as of a specified date. This amount includes all normal direct and indirect costs, such as installation and other assemblage costs necessary to make the property fully operational.
- **Fair Market Value-in-Exchange**—is the estimated monetary amount that may be reasonably expected for a property in an exchange between a willing buyer, and a willing seller, with equity to both, neither under any compulsion to buy or sell, and both fully aware of all relevant facts, as of a specific date (also referred to as retail value).
- **Orderly Liquidation Value**— is the estimated gross monetary amount that could typically be realized from a liquidation sale, given a reasonable period of time to find a purchaser (or purchasers), with the seller being compelled to sell on an as-is, where-is basis, as of a specific date (also referred to as wholesale value).
- **Forced Liquidation Value**— is the estimated gross monetary amount that could typically be realized from a properly advertised and conducted public auction, with the seller being compelled to sell with a sense of immediacy on an as-is, where-is basis, as of a specific date (also referred to as "under the hammer" or "blow-out" value).
- **Salvage Value/Part-Out Value**—is the estimated monetary amount that may be expected for the whole property or a component of the whole property that is retired from service for possible use elsewhere, as of a specific date.

- **Scrap Value**—is the estimated monetary amount that could be realized for the property if it were sold for its material content, not for productive use, as of a specific date. For example, dollars per ton of steel or pound of copper.

Valuation Examples/Subcategories of Value

Table 6.1 is an example of market value subcategories for the sale of a certain four-year-old CNC machining center (original cost new $350,000). This example illustrates the relative meaning and weight of some of the valuation terms previously discussed. It is based on these terms and definitions that valuation data should be collected and cataloged.

Table 6.1—Market Value Subcategories for CNC Machining Center

Type of Value	Sales Price	% of FMV
Fair Market Value-in-Place (FMVIP)	$250,000	125%
Fair Market Value-in-Exchange (FMV)	$200,000	100%
Orderly Liquidation Value (OLV)	$160,000	80%
Forced Liquidation Value (FLV)	$120,000	60%
Salvage/Part-Out Value	$8,000	4%
Scrap Value (SV)	$2,000	1%

Replacement Cost, Sources of Data

Replacement cost is the highest value that can be attributable to a piece of equipment. Thus, it can be considered 100 percent of value (new). Several sources are available for collecting and monitoring equipment replacement cost (new) data. Among these are prices and data obtained from the following sources:

- Manufacturer's price lists.
- Data obtained verbally from sales representatives and new equipment dealers.
- Published prices from technical and trade journals.
- Published cost data in pricing guides.
- Literature obtained at trade shows.
- Invoices containing cost data relating to past transactions.
- Purchase orders from past transactions.
- Equipment quotations from manufacturers or dealers.
- Internet data.
- Appraised values obtained from replacement value (new) or insurance appraisals, which typically list replacement cost (new).

By cataloging valuation data obtained through the various sources listed above, the estimator/appraiser can establish a sizeable and meaningful database from which to plot value trends for a specific piece of equipment over time, or to validate the fair value of a similar item currently and/or in the future.

Market Value, Sources of Data

An equipment valuation database can also be established for market values. Sources of this data (for the various subcategories already mentioned) include the following:

- Trade publications listing sales advertisements for used equipment.
- Retail prices obtained from used equipment dealers or brokers.
- Equipment price quotations for the purchase of used equipment documented in previous transactions.
- Values from "market data publications" available for purchase.
- Auction "sales catalogs" available from auction companies at a nominal cost ($20 to $50), or through auction sales data reporting subscription.
- Regulatory filings.
- Past remarketing and sales results from one's own firm.

Market Valuation Example

To better illustrate the basis for some of the subcategories of market value (how equipment is bought and sold) review the example of a 10-year old metal turning lathe shown in Table 6.2. This is followed by Table 6.2, an example summary of the relationship of subcategories of market value to one another.

Table 6.2—Market Value Subcategories for Metal Lathe

Event	Cost/Value
a. Acquisition price: "as-is" at auction	$5,500 (forced liquidation value)
b. Sales tax	Exempt (equipment dealer)
c. Deinstallation: rigging, shipping, and delivery to warehouse (a+b+c) = purchase price	$600
d. Cost of money (estimated) time to sell: 90 days, 10% annual rate x 3 months x $6,100 purchase price	$154
e. Overhead: 20% of purchase price (includes some preparation and advertising)	$1,220
f. Profit (15% to 20%): use 20% of purchase price (including deinstallation)	$1,220
Subtotal: (a+b+c+d+e+f)	$8,694 (minimum desired selling price)
g. Ask: (advertise for sale)	$9,800 (retail asking)
h. Take: (sale to end user)	$8,800 (fair market value-in-exchange)
i. Buyer: end user pays sales tax (7%)	$616
j. Delivery	$600
k. Installation and debugging	$1,200
Total installed cost to end user (h+i+j+k)	$11,216 (fair market value-in-place)

Table 6.3—Metal Turning Lathe Example Summary

Type of Market Value		% of FMV
Fair Market Value-in-Place	($11,216)	127%
Fair Market Value-in-Exchange	($8,000)	100%
Forced Liquidation Value	($5,500)	63%

Trade Data/Cost Adjustments

Used equipment sales data is often inconsistent for the same or similar pieces of equipment. In order to adjust or "normalize" this data, the following points should be considered and value added or deducted as appropriate. These adjustments include the following:

- The same equipment, but with different years of manufacture (normally adding value for newer equipment).
- The same equipment, but with different attachments, drive motors, etc.
- The location of the sale (i.e., equipment sold in a prime geographic market area will usually achieve a higher sales price than equipment sold in a remote area).
- Use (amount of wear/use).
- Condition (one of the most important considerations).

In some instances, condition alone can cause considerable value swings for identical pieces of equipment. It is extremely important that the condition of a piece of equipment be noted at the time of sale, to ensure that equipment sold in excellent condition for a relatively high price will not later be used to anticipate the price of a piece of equipment expected to be in average or fair condition at the end of the lease. Once again, this is one of the most important parts of any database and should not be overlooked.

Equipment Condition Terms and Definitions

The following is a sample selection of condition terms and definitions used to describe equipment:

Example 1:

- **Very Good (VG)—**Describes equipment that is in excellent appearance and being used to its full design specifications without being modified and without requiring any repairs or abnormal maintenance at the time of inspection, or within the foreseeable future.
- **Good (G)—**Describes equipment that was modified or repaired and is being used at or near its fully specified use.
- **Fair (F)—**Describes equipment being used at some point below its fully specified use, because of the effects of age and/or application. This equipment will require general repairs and some replacement of minor elements in the foreseeable future, to raise the level of use to or near its original specifications.
- **Poor (P)—**Describes equipment being used at some point well below the fully specified use. This equipment will require extensive repairs and/or the replacement of major elements in the near future to realize full capacity.
- **Scrap (X)—**Describes equipment that is no longer serviceable and cannot be used to any practical degree regardless of the extent of the repairs or modifications that may be undertaken. This condition applies to equipment that has been used for 100 percent of its useful life, or that is 100 percent technologically or functionally obsolete.

Another variation is **Example 2:**

- **Excellent (E)—**Describes equipment that is new, or in practically new condition, with extremely low use, no defects, and that may still be under warranty.
- **Good (G)—**Describes equipment that has a good appearance, may have just recently been completely overhauled, or rebuilt with new materials, and/or has had such use and maintenance that no repairs or worn part replacement is necessary. This equipment shows no deferred maintenance.
- **Average (A)—**Describes equipment that is in 100 percent operating condition, with no known major mechanical defects; but, may have some worn parts that will need repair or replacement in the future. This equipment may have high use, but any defects are not obvious.
- **Fair (F)—**Describes equipment that shows high use; defects are obvious and will require repair or a general overhaul or rebuild soon. This equipment is operational, but questionable, and may exhibit deferred maintenance.
- **Poor (P)—**Describes equipment that has seen severe and long hours of service and is not operational. It requires rebuild, repair, or overhaul, before it can be profitably used.

In summary, the logging of raw data and condition, along with the cost adjustments previously mentioned, will permit the fine tuning of market data, yielding a more reliable and meaningful value.

Data Filing Systems

Once data is obtained, in sufficient quantities, a decision can be made on how to record it. Most valuation and research firms file data using one of four methods:

- The first is by Standard Industrial Classification (SIC) code, or by the more detailed North American Industry Classification System (NAICS). In this case, data is stored in broad industry category codes, such as SIC #34-machine tools, SIC #44-marine, etc. This method is quite effective when using an electronic database, since a numerical method of encoding each piece of data lends itself to an SIC type of listing.
- Another method of filing is to list data by equipment class and type, such as machine tools-lathes, aircraft-commuter, construction-crawler crane, trailers-dry van, barges-covered hopper, or railcar-center beam flat.
- A third method lists equipment by industry category, such as machine shop equipment, construction, mining equipment, aircraft, or marine vessels.
- Finally, another filing system is based strictly on the equipment manufacturer's name, such as Freuhauf-trailers, Mazak-machine tools, Caterpillar-construction equipment, Boeing-commercial aircraft, and IBM-computers, etc.

By using any one of the above methods of filing, a workable cataloging system can be created that will enable the user to obtain comparable valuation data in a minimal amount of time. The larger the database, however, the more complex the filing system must be for prompt sorting and retrievals.

Data Storage

Data can be stored via several methods. The system most widely used is Excel spreadsheets. Here, systems can be developed that file (log) everything from individual equipment specifications to a manufacturer's total equipment production run, listing all pertinent data and serial numbers for each piece of equipment studied. For larger, centralized (and more efficiently stored) databases, a database application, such as Oracle or Access may be used. When establishing an electronic data filing system, sufficient time should be devoted to organizing the data to be user-friendly and accessible, by providing for listings and searches on multiple identifying factors, such as equipment class and type, industry, and manufacturer. Also, numerous sites on the Internet have databases that can be easily accessed when needed. Use of third-party information can augment or support an in-house (proprietary) database.

Additionally, estimators/appraisers may wish to consider purchasing certain trade data available in the public domain. There is no substitute for actual trade data.

Equipment Valuations

One of an estimator/appraiser's most difficult tasks is the analysis and calculation of equipment residual values (future values) for leases or life cycle costing. Unfortunately, there are no magic formulas or short cuts, and no college to teach how to properly calculate residuals—the only way to gain this knowledge is through hard work, years of experience, and a good database.

Industry and business place such an importance on (future) residual values, the role of the estimator/appraiser has become vital. One key component of estimating residual values is the appraiser understands of the equipment and the industry in which it operates.

The analyst must have access to trade (sales) data. There is no substitute for data. Sometimes unqualified opinions can cause serious financial damage to a company. Although the determination of residual values, using trade data, does not guarantee the future value of equipment, it nonetheless provides a sound basis upon which to predicate future values. In addition, there are many variables the analyst should consider in arriving at a realistic residual value estimate.

Residual Value Curves

Residual Value is the expected future amount of money an owner/lessor is expected to realize from an asset, as of a specified future date. Included are such things as the end of a lease term or project, from any and all sources, including sale, re-lease, holdover rent, penalties, damage, litigation, and payments for noncompliance to the lease documentation (such as return and maintenance provisions).

Figures 6.1 and 6.2 illustrate that wholesale market value (Orderly Liquidation Value) of equipment. This does not follow predefined curves or formulas. In fact, the shape of a residual curve may change on an annual or even monthly basis, depending on circumstances. It is important to remember that the market is constantly changing. The variables that cause this change will be discussed later.

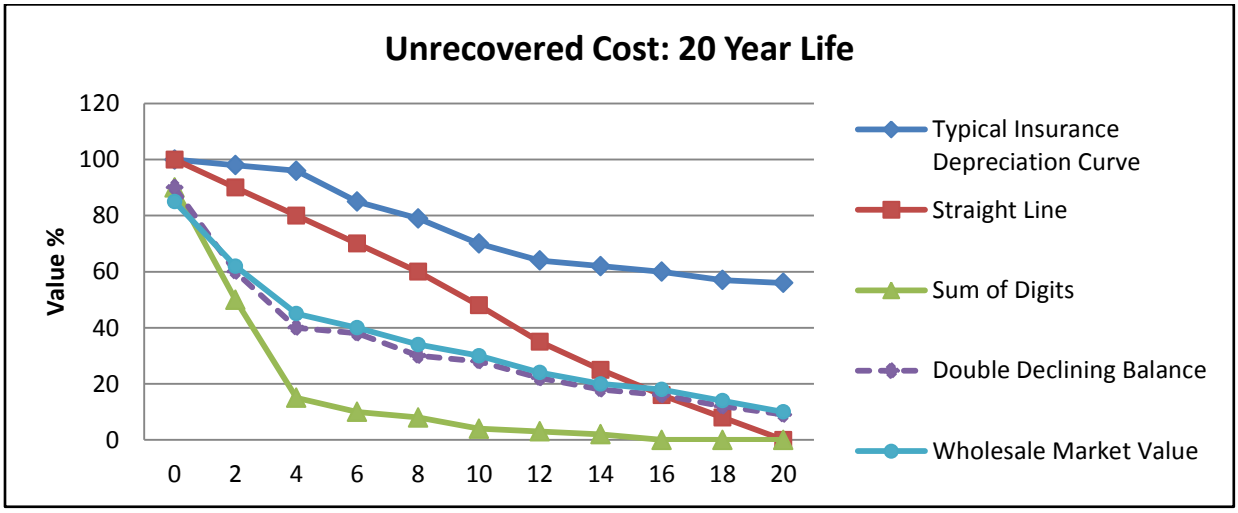


Figure 6.1—Executive Aircraft Methods of Depreciation

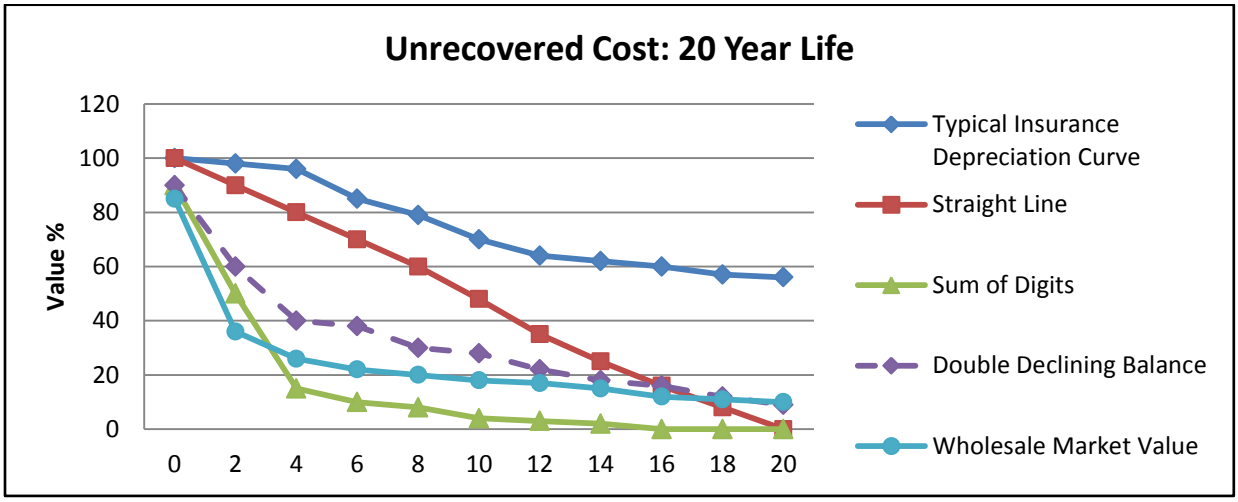


Figure 6.2—Machine Tools Methods of Depreciation

Figure 6.3 illustrates some of the classic, as well as unusual, residual value curves for used equipment.

The "normal" residual value curve of long-lived equipment usually follows an L-shaped curve that is illustrated in Figures 6.1–6.3. Another shape is the disrupted market curve, which is a deviation from the normal curve and is usually in the shape of a "U". This curve typically results from excess supply or regulatory pressures causing the market value for a piece of equipment to suddenly plunge or deviate from normal for a period of time. Classical examples include the "U" curves for covered hopper railcars and intercoastal barges during the mid-1980s.

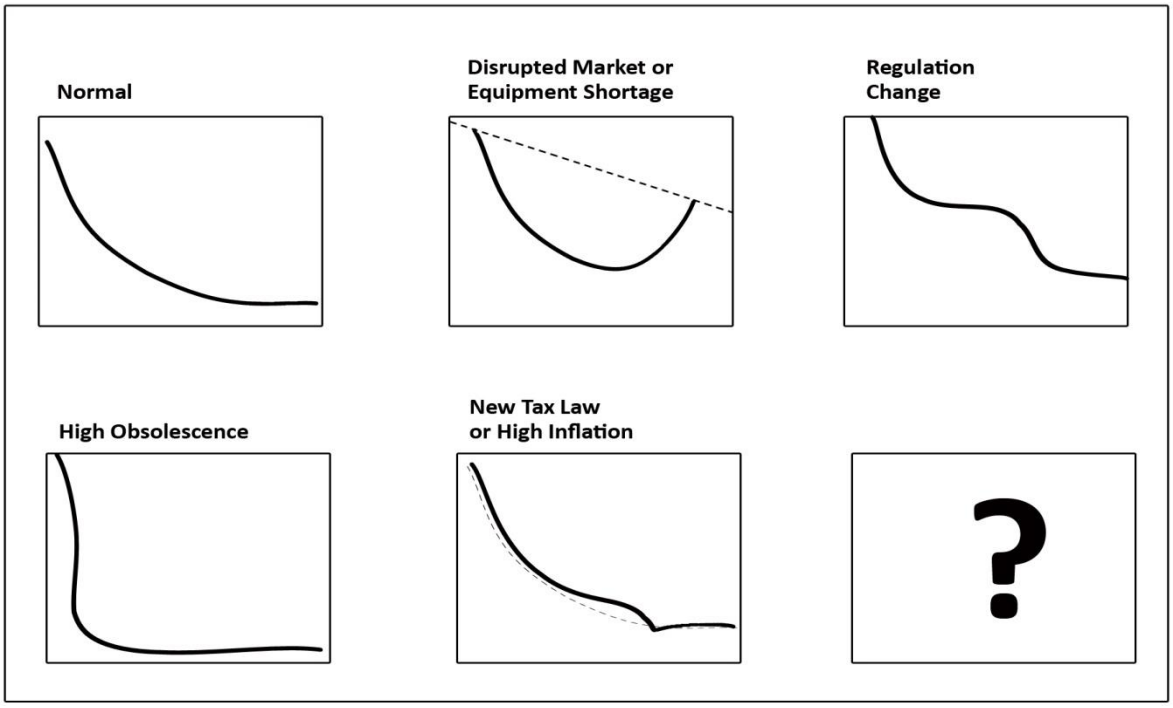


Figure 6.3—Residual Value Curves for Used Equipment

Another curve that can be studied is the regulatory change curve, which illustrates the sudden impact on market value that regulation can cause, such as the effect of FAA Part 36 noise and exhaust regulations on the values of some Boeing 737s, AirBus 319s, etc. Another example is the impact of the Environmental Protection Agency's Mercury and Air Toxics Standards (MATS), California's "Carbon Cap and Trade," and similar regulation, on the value of pulverized coal plants. Many such plants were or will be forced to close as a result of the high cost of pollution controls.

The high obsolescence curve is a truncated shape that illustrates the impact of technological obsolescence. This curve is particularly prevalent among items such as computers, data storage units, and certain other types of high-tech equipment.

Another deviation from the normal shape is the (new) tax law/high inflation residual curve. Tax laws and inflation can in some cases cause a normal residual curve to rise dramatically in a short time. Although this pattern is quite appealing to the owner/lessor when selling in a strong market, it should also be recognized that an elevated curve will sooner or later likely fall back to its "normal" shape.

Variables That Affect Residual Value
In general, there are 12 items that should be considered in estimating residual values:

- Initial Cost
- Maintenance
- Use/Wear and Tear
- Population
- Age
- Economy
- Changes in Technology
- Foreign Exchange
- Tax Laws
- Legislation/Regulation
- Location of Equipment
- Method of Sales

Initial Cost—one of the most frequently overlooked items in any transaction, is the initial cost of equipment. Many times, the owner/lender/lessor is presented with funding; containing indirect costs that he or she may not be aware are included in the price. For residual purposes, the estimator/appraiser should consider basing his or her residual estimate on 'hard costs' only for individual items. The hard cost of an asset includes the new cost of the basic machine and the cost of any additional items necessary

to make it operate, including drives, motors, electrical and controls. Items referred to as "soft costs" that should not be included are as follows:

- Foundations (which some large machines may require as a result of their size or overall sensitivity).
- Freight.
- Debugging.
- Taxes (which may include federal, state, and local taxes, as well as duty on foreign imports).
- Installation (which includes fastening the equipment to a foundation, piping, and electrical wiring from a main distribution system, and other items necessary to make the machine operable).

The following is an example of an actual situation involving a transfer stamping press line in an automotive facility. A leasing company was presented with a transaction valued at $2.1 million. Subsequent investigation found the following:

- The basic cost of the machine was $1.5 million.
- The cost of foundations was about $400,000.
- The cost of freight was $40,000.
- Taxes were $60,000.
- Installation was estimated at $100,000 (all totaling $2.1 million).

In this instance, if a leasing company's historical "residual curve" for this equipment indicated 30 percent of the new cost at the end of the term, the difference between calculating the residual on a value-in-exchange basis for the hard asset only, versus the value-in-place including soft costs, would equal $180,000 at lease term.

Total Cost: $2.1 million x 30% =$630,000
Hard Cost: $1.5 million x 30% =$450,000
Difference: =$180,000

This difference could present a future shortfall, if the asset were sold on a value-in-exchange basis. It should be understood that in some instances—such as a facility lease or financing or life-cycle costing—soft costs should be considered in determining residual values.

Maintenance—the next consideration is maintenance. The difference in value received from a well-maintained versus a poorly-maintained piece of equipment can be substantial. Maintenance can also affect the useful life of equipment. In calculating a residual value, estimators/appraisers must consider how the equipment will be maintained and/or the maintenance language in the lease. Maintenance provisions in leases can be relied upon by the lessor as a future condition statement, and also for future claims for damage, abuse, or deferred maintenance. If the lease contains strong maintenance provisions, a higher residual usually can be assumed strictly on the basis of the condition in which the equipment is expected to be returned at the end of the lease.

Use/Wear and Tear—use/wear and tear is another important consideration that should be understood by the estimator/appraiser in estimating residual values. The difference between equipment in harsh service, versus mild service, can be substantial. In some instances, this can affect the equipment's useful life. An example is covered hopper railcars. Used in grain service, they can have useful lives of approximately 40 to 50 years. However, if used in salt service, their useful lives can be as short as 15 years. Other examples include construction equipment used in sand pits, which subject the undercarriage to excessive wear. Thus, the analyst should be aware of all such conditions when estimating residual values.

Some types of equipment, such as aircraft, define use in hours of use and cycles (takeoffs and landings); other transportation equipment defines use in miles or kilometers. Mechanical equipment use is usually recorded by the hour. Typically, most mechanicals tend to wear out at around 10,000 to 20,000 hours. At these milestones, usually some form of rebuild or refurbishing is required.

For analysis purposes, 1,500 to 2,000 hours per year use is usually considered one shift; 2,500 to 3,500 hours is considered two shifts; and around 5,000+ hours per year is considered three shifts. Leasing terms for equipment that will be used 5,000+ hours per year (as frequently happens in the mining industry) usually contain strong return and maintenance provisions in order to ensure the future condition of the equipment. The appraiser may then base his or her residual value estimates on this assumption.

An example of the effects of high use and wear and tear is the partial disintegration, in flight, of a commercial jet that had approximately 80,000 cycles of use in addition to high time. Besides the high use rate in this example, the geographic area of use, which was tropical, could have also played a part in the incident, since it may have caused the aircraft to be subjected to further wear and tear from the corrosion caused by the moist, tropical, salty air.

Population—another important consideration is the overall population of the subject equipment. Typically, the larger the population of equipment, the more data can be obtained. This gives statistical significance to the residual value, because the value will be based on a large sample referred to as a commodity. Residual valuations are particularly difficult and oftentimes meaningless for

prototype equipment. When valuing a prototype, the equipment analyst can only attempt to compare the item being valued to another item having similar utility. Technology could be a large risk in such a transaction, as well as a very limited secondary market.

Many times estimators/appraisers are able to obtain information on annual production runs of certain equipment. A review of this data may reveal a sizeable increase in the manufacture of the product during a certain year, which may have an impact on the future residuals; e.g., if much of the equipment produced in a boom year comes off lease all at once.
Boom years for particular types of equipment are generally related to the overall health of the economy, or, in particular market segments, to regulatory mandates and/or special tax incentives. One classic illustration of this is the chassis industry, which had production rates during the 1950s and 1960s in the area of 4,000 units per year. In the 1970s, this average increased to about 10,000 units per year, while in 1984 the number jumped to about 24,000, then in 1985 to almost 29,000 units. In such instances, the estimator/appraiser should have considered the potential downward impact of boom years on future residual values (This, however, was mitigated by the explosion of the intermodal industry in the 1990's and 2000's.). These production spikes have also recently occurred in the truck and trailer markets, and in the blue water marine vessel market. What drives the primary market today may not drive the secondary market in five to eight years.

Age—another item that is often overlooked is the actual age of date of manufacture of a piece of equipment. Equipment presented as new in January 2012, could have a 2011 or 2010 build date. This is particularly important for transportation equipment. If, for example, the equipment comes off of a lease or project in 2017, which truck will sell for more: the 2011 or 2012 model? The answer is obvious. The estimator/appraiser should verify model years and serial numbers on each transaction.

The following example is given to illustrate the different sales tiers that should be considered when estimating residual values. An 8-year-old insulated trailer fleet was sold to an end-user. The equipment manager surveyed the market at all levels. The results of this survey are shown in Table 6.4.

Table 6.4—The Transaction: 195 Trailers, 8 Years Old, Original Cost $3,510,000

Sales Tier	% of Highest Sales Price	Comments
Retail sales level (FMV)	100%	Actual sales price $1,040,000
Wholesale (OLV)	76%	Offers received to purchase trailers by wholesalers
Auction (FLV)	63%	Estimated proceeds from the sale at auction received from several auctioneers
End of economic useful life (Salvage Value)	7%	Estimate of the value of trailers to be used as storage units only

The difference in the values above for the same equipment illustrates the importance of an appraiser recognizing at what level or sales tier, the equipment typically sells, and then adjusting its residuals to reflect this level.

Economy—the equipment appraiser must also consider the overall shape of the economy. For instance, the U.S. economy usually performs at four-to-seven-year business cycles. A used truck that sells for $35,000 in a robust economy, may sell for $18,000 in a recession. Further, the sale of equipment in a robust economy might take from 1 to 30 days, while in a recession it could take 60 to 90 days. The additional time it takes to sell should also be calculated in the overall cost of the sale (cost of money).

Changes in Technology—changes in technology can also affect residual values. These changes occur in every type of equipment. However, some changes have a more profound impact on value. An analysis of technological changes, occurring over the past 20 years, shows that future advances in technology were generally known at the time of lease origination. The astute analyst may choose to make an adjustment in the residual of a piece of equipment that is subject to such obsolescence.

Some recent changes can be found in the computer micro-processor and hard disk storage capacity. Big ticket medical imaging equipment, such as CT scanners, are currently using scanners that have reduced the time necessary to "fix" an image from minutes to one second.

Even the railcar industry now shows examples of obsolescence, such as the impact of modern 286,000-lb. gross rail weight capacity vs. the older 263,000-lb. capacity, or the impact the container and chassis has had on the 50-foot boxcar. Locomotives have also undergone significant changes in technology affecting the levels of tractive horsepower and mode of power; i.e., AC versus DC.

Trailers have also experienced changes in technology based on the "maximizing of cubes." Trailers with a length of 53 feet conform to state regulations for length, height, and weight, plus can carry more volume (cubic feet). Thus, in today's market, 53-foot trailers hold an advantage over 48-foot trailers, and 48-foot trailers hold an advantage over the older and more obsolete 45-foot trailers, and 45-foot trailers hold an advantage over the yet older and more obsolete 40-foot trailers.

Foreign Exchange—foreign exchange is also a consideration that may enter into a residual calculation. Factors that influence foreign exchange include international trade relations and the international political environment, which can cause trade wars, embargoes and higher or lower duties. Consider the following example (which actually occurred during the mid-1980s to 1990s):

A Japanese tractor manufacturer produces a tractor which sells for 10 million yen. At an exchange rate of 100 yen to the dollar, the tractor would cost U.S. $100,000; however, at 78 yen to the dollar, that same tractor would cost U.S. $128,000. This increase of 28 percent is strictly a result of currency.
Thus, the value of foreign exchange, such as the yen or euro, can increase or decrease quite rapidly. Such changes could put pressure on selling prices (and residual values), causing them to suddenly drop or increase. The analyst should attempt to compare the cost of foreign manufactured equipment to that of similar domestically manufactured equipment in order to calculate a realistic base value on which to predicate the residual analysis. Strong foreign currency may cause the price of foreign equipment to rise in the U.S., which, in turn, may pull residuals up. However, it should also be understood that the reverse can also occur.

Tax Law—Tax law is another concern that owners and leasing companies deal with regularly. Tax law can affect such things as build rates (when incentives are given), and thus the supply on the secondary market.

As a result of 1990 tax laws, depreciation rates have been extended. The net effect is that, in many instances, it is more appealing for an end-user to buy used equipment rather than new equipment.

It is imperative that the equipment appraiser consider the true economics of a transaction and not simply the tax ramifications. The following two classic examples illustrate the impact of certain tax legislation and regulation.

In the early 1980s, because of tax incentives, the cost of new hopper barges suddenly increased from $220,000 to about $330,000. Between 1978 and 1981, approximately 3,000 hopper barges were built, increasing the existing capacity by 33 percent. Further, because of certain tax incentives, two-thirds of the new barges were owned by private investors (limited partnerships, etc.). Many of these investors participated in leasing transactions for tax purposes. Most residuals were set without considering that the cost of the barges had increased almost 50 percent in only two to three years. In addition to the sudden jump in prices, the early 1980s also signaled the end of the U.S. coal export boom, and the end of the Polish and Australian coal strikes.

Investors found themselves in a situation where too many barges were built for artificial purposes (i.e., tax benefits). The supply/demand balance was drastically altered. Charter rates for hopper barges fell from $130 per day to $35 per day, and the value of the $330,000 barge plummeted to about $80,000 for a five-year-old asset with a 30+ year useful life (see Figure 6.4). After a few years, many barge manufacturers went out of business. No new barges were built, and since there were a great number of barges laid up, up to 25 percent of the older barges were scrapped over a five-year period. This soaked up some of the glut.

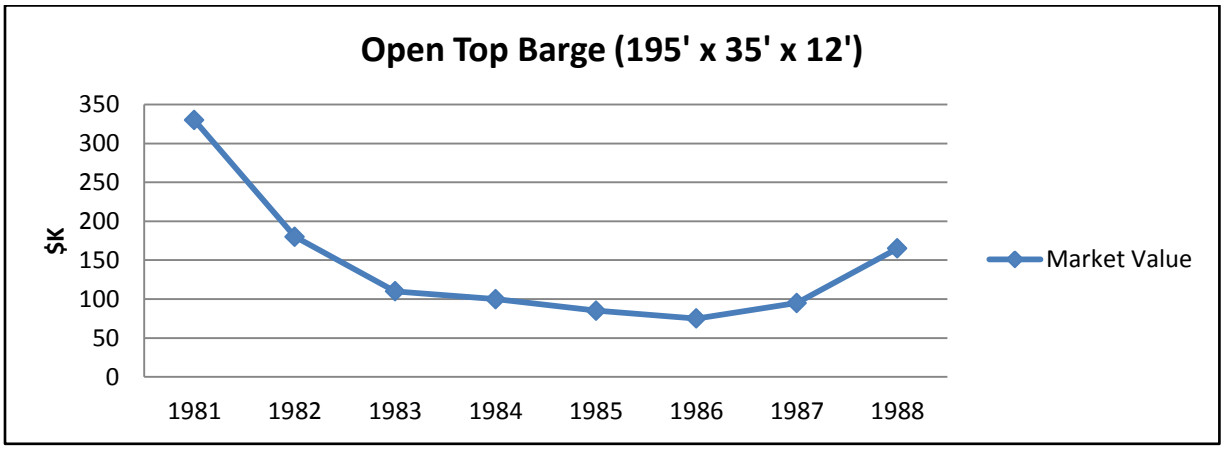


Figure 6.4—1981 Open-Top Barge Market Values

At that time, the dollar weakened 60 to 80 percent against foreign currencies, and exports once again started to flow from the US. Increased exports, coupled with a sharp decrease in the production of new hopper barges, and the ongoing scrapping of the excess fleet, caused a barge shortage. The price of barges suddenly escalated in 1988, to approximately $175,000 (for the same 1981-built equipment). This example illustrates the disrupted U curve.

Figure 6.5 illustrates an almost identical situation that occurred with covered hopper railcars, which were likewise over-built in the early 1980s because of tax incentives, and were owned to a great extent by private investors not familiar with the industry. An imbalance in the supply/demand curve resulted, causing the price of five-year-old equipment with a useful life of 50 years to plummet to the

point where covered hopper railcars manufactured new in 1981 for $44,000 were selling a few years later for about $14,000, if at all.

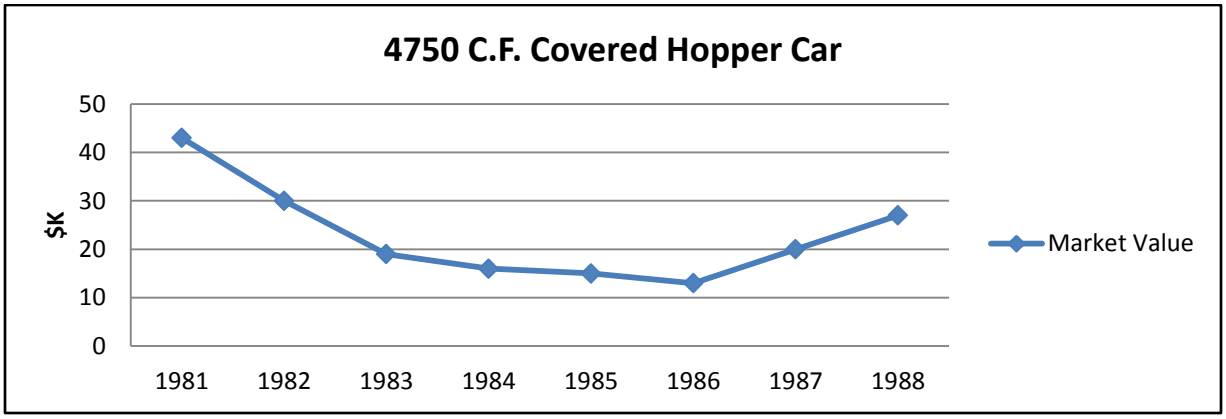


Figure 6.5—Covered Hopper Car Market Values

Once again, this scenario caused many railcar builders to cease operations. This situation was further aggravated by the Russian grain embargo and the Railroad Deregulation Act of 1980 (Stagger's Act), which deregulated the rail industry and caused downward pressure on shipping rates.

For five years, the rail industry had essentially not built any new railcars and had scrapped a significant portion of the older covered hopper car fleet. When the dollar weakened and exports started to flow, there was once again an equipment shortage that caused the price of the asset to suddenly rise, once again illustrating the disrupted U curve. Several operating lessors that followed this situation correctly predicted when the demand curve would pass above supply, and made tens of millions of dollars overnight.

Similar instances have recently occurred in the truck/trailer equipment segment, construction equipment, ISO and domestic containers and chassis, automobiles, machine tools, plastic injection molding machines, and many more. The 2008 financial crisis caused many industries to experience U curves that continue even today.

Legislation/Regulation—another important variable is legislation/regulation. Although it cannot be predicted, a sense of future legislation can be recognized, such as current talk about increased regulation related to commercial aircraft, or the impact of the CAFE or Clean Air Act on engine technology. Regulations sometimes impact values in positive ways, such as the effect of the Jones Act on U.S. hulled marine vessels. However, more often than not, the impact of legislation and regulation on the equipment market is a negative one.

Equipment Location—another factor to be considered is the location of equipment at the end of the lease. Does the lessee require that the equipment be delivered to a prime market location, or will it have to be sold in a remote area? To illustrate this point, consider the sale of seven off-highway rock trucks. Ideally, these trucks should have been marketed in prime mining areas of the country, such as Wyoming, Arizona, Minnesota, or the Appalachian area, where they had an estimated value of $160,000 each. However, since the owner could find no buyer in the immediate geographic area, each truck had to be physically cut in half, match marked, disassembled, and shipped on nine flatbeds to a buyer out of the area. The incremental cost of performing this work and reassembly amounted to a reduction in the sales price of approximately 22 percent. As a result, the owner/lessor was able to realize only $125,000 per truck. Thus, because the equipment was not delivered to a prime geographic marketing area, the owner/lessor lost $245,000 (7 x $35,000) in residual value for the trucks. This illustrates the importance of recognizing the location of prime geographic equipment markets.

Method of Sale—In general, if the equipment analyst considers all of these factors, along with the method of sale and current and historical trade data, the lessor should be well positioned to meet or exceed its estimated or booked residual (Future Value) at lease termination.

Calculating Residual Values

The calculation of residual values is more art than science. It has been said that economists only give major predictions on the economy every six months, the reason being that someone might remember. The same can be said for the calculation of residual values. Professional economists sometimes miss short-term predictions by 100 percent or more, so a lessor might wonder what degree of accuracy a residual value calculation might have in 10 years. However, if an analyst performs a thorough and proper analysis based on available market data, a high degree of accuracy is possible.

In fact, during the 1980s, when the residual value guarantee industry was in its heyday (because of regulation limiting the amount of residual value a bank may hold), such values were typically calculated for equipment with terms of three to eight years. Based on available information, it was found that guarantee companies annually experienced losses (on their guarantees) in the area of only 1.5 percent of the total values guaranteed. This is a somewhat startling statistic, but once again shows that if estimators/appraisers are thorough and perform a proper analysis, a high degree of accuracy is possible.

Residual Valuation Formats
Residual analysis formats differ from company to company, but should contain the same basic elements. Following are two of the more popular formats used in the industry:

Example Format 1:

- **Marketplace**—a general description of the marketplace in which the equipment is used.
- **Manufacturer**—background on the manufacturer and the part in which the equipment plays in the overall sales and marketing of the manufacturer.
- **Model Run**—the number of years the equipment has been manufactured.
- **US Market**—the number of units of the subject equipment in the US market.
- **Installation**—the estimated time to install a subject machine and estimated cost.
- **Software**—a brief discussion of software, such as used in controls.
- **Versatility**—the number of uses for the subject equipment.
- **Life Expectancy**—the estimated economic useful life of the equipment, allowing for proper maintenance.
- **Residual Value**—the analyst's estimate of residual values as of a given future date(s).
- **Manufacturer Name**—a listing of the complete manufacturer's name, address of corporate headquarters, and phone number.
- **Type of Equipment**—a complete description of the equipment being analyzed.
- **Manufacturing/Marketplace**—a general discussion of the manufacturer's future marketing plans and outlook.
- **Future Improvements**—any technological breakthroughs on the horizon.
- **Price Increases**—a listing of any price increases that are planned in the future, as well as an analysis of past increases.
- **De-installation**—the estimated time and cost to de-install the equipment.
- **Conclusion**—a narrative summary of the entire analysis.

Example Format 2:

- **Equipment Description**—a detailed narrative on the piece of equipment being analyzed.
- **Transaction Price**—the cost of equipment listed in the lease transaction (lessor's cost).
- **Fair Value**—the adjusted transaction price on which the residual is based, including only hard costs, those adjusted for discounts, and the prices of competitor's equipment of similar quality and utility.
- **Estimated Residuals**—the numerical estimate of residual value given in terms of dollars as of a specified future date(s).
- **Comments (Discussion)**—a narrative description of the analysis, including a brief description of the equipment being analyzed with comments on the equipment's desirability in the secondary market, what makes it better than a competitor's, the manufacturer's reputation in the industry, the amount of time the manufacturer has been in existence, the manufacturer's estimated current market share, a list of market competitors, the equipment's estimated useful life, obsolescence considerations (planned introductions of new models, etc.), and estimated time and cost to remarket the equipment, etc.

As stated previously, appraisers may develop their own style or format, but regardless of which format is used, all of the items shown in the two previous examples should be addressed somewhere within the analysis.

Residual Curves
Estimators/appraisers who have been studying equipment values over a number of years should save their work. Since these analyses are based on actual market data, it is felt that similar equipment being valued over a period of years will start to develop a standard shape or residual curve as a percentage of fair value (standard discounted price new).

To illustrate the fair value of an item, consider an over-the-road truck tractor. The truck carries a list price of $140,000, yet can be readily purchased on a fleet basis for $90,000. Thus, the fair value is $90,000, and not the $140,000 list price. Using fair value will help the analyst avoid the trap of applying a "standard curve" (for conceptual purposes) against a list price that is inflated for whatever reason, yielding a residual curve that is also inflated and most likely in error.

From time to time, equipment manufacturers raise prices for no apparent reason other than perhaps not to be caught in case the government someday applies price controls. The same manufacturers routinely apply large discounts to their equipment, showing the importance of using a fair value as a benchmark for a realistic discounted selling price.

"Standard curves" for indicative purposes can be developed through constant residual analysis over many years. Table 6.5 shows some residual curves that are used from time to time on a conceptual basis only.

A study of these curves can be made to determine which curves may be applicable to "rusty iron" and which may apply to high-tech equipment. The higher the obsolescence factor, the shorter is the equipment's life, and the lower the residual curve. In some cases, the curve becomes almost truncated after a short period of time. Unless absolutely necessary, residual curves should not be used to determine actual residual values; it should be only for the conceptual "order of magnitude" purposes. They can, however, be useful in an analysis method called curve fitting, when only several points of data are obtained and the appraiser is left with information gaps at certain years. (This concept will be explained later in this chapter.)

Methodology

Estimators/appraisers have various methods available to use in determining residual values, including the "Cost Approach" (depreciated replacement cost), the "Income Approach" (present value of future cash flows), and the "Market" (or trade data) approach to value. From a practical standpoint, the "Market Approach" is oftentimes believed to be the most accurate, because of its reliance on sales and trade data. A "Market Approach" analysis involves the following steps:

- A complete market analysis and technical review of the subject piece of equipment is undertaken. Once that is finished, the appraiser collects all available sales trade data relating to the same or similar equipment, noting manufacturer, model, capacity, equipment age at the time of trade, trade level, condition, and sales price. If different trade levels are found, notations should be made next to the price noting levels, such as auction (liquidation value), retail (fair market value-in-exchange), and retail asking (fair market value-in-exchange/asking).

- The current list price (new) is obtained for each piece of equipment analyzed and all standard discounts are applied. The discounted list price of the subject equipment is then compared to other equipment of similar quality and utility to arrive at a fair value. The fair value new represents a reference point of 100 percent that is the basis of the residual calculation. From this absolute dollar value, a residual curve in the form of a percentage (of fair value) can be determined.

- After the fair value is determined and all available trade data is collected, an equipment valuation history is compiled (see Figure 6.6). This form is used to compile available residual information (in raw form) characterized by the age of the equipment sale. The example notes the fair value (100 percent) and various trades (auction) for the same or similar piece of equipment at different ages up to six years. The trade data shown represents equipment sold in average condition with normal wear and tear.

Table 6.5—Sample Residual Curves

	Years (% of Fair Value)								
Curve #	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**
1	76%	63%	51%	43%	45%	30%	24%	17%	14%
2	60%	55%	50%	45%	40%	35%	30%	25%	20%
3	55%	50%	45%	42%	39%	36%	32%	30%	25%
4	50%	45%	35%	30%	25%	22%	20%	18%	15%
5	45%	33%	27%	22%	20%	18%	17%	16%	15%
6	45%	32%	20%	15%	10%	5%	4%	3%	2%
7	35%	30%	25%	20%	15%	10%	7%	5%	3%
8	31%	27%	23%	19%	16%	12%	9%	3%	1%
9	45%	25%	10%	5%	3%	0%	0%	0%	0%

	(All Amounts in $1,000s)						
	Fair Value Adjusted Price	**1 year**	**2 years**	**3 years**	**4 years**	**5 years**	**6 years**
Residual Value Actual, $	200.0	--	130.0	122.5	101.6	85.0	72.5
			130.0				
				120.0			
				125.0			
					110.0		
					110.0		
	TRADE DATA				95.0		
						90.0	
						80.0	
						85.0	
							75.0
							70.0

Figure 6.6—Equipment Valuation History

- After the trade data is documented, an average is determined and entered into the residual analysis line. When a wide discrepancy exists in the values, the high and low may be discarded, or simple common sense can be used to determine what looks like the average.

- The residual value in a percentage form is then calculated from the available data (see Figure 6.7). This data now represents the historical relationship between the current secondary market and the current fair value new. If this historical analysis is turned around, it can also be used in a conservative way to represent the residual value of equipment of the same age carried into the future. This type of analysis is based on zero inflation. For equipment where no trade data is available, data is obtained on the most comparable machine and applied.

  In instances where available trade data is limited, "curve fitting" (see Figure 6.8) will fill the voids. Analysts who routinely log residual curves will, over time, accumulate a large number of these curves. Each curve is based on actual equipment trades and represents how certain types of equipment tend to depreciate in trade value over a period of years.

  Thus, an analyst can complete the residual curve presented in Figure 6.8, by matching the known percentages with a standard curve having the same or similar percentage at a given age (see Figure 6.7), and then filling in the unknown "gaps" on the curve with the known corresponding percentages of the fitted curve.

- After the final curve is completed, it should be reviewed by the analyst and adjusted up or down accordingly for factors such as maintenance, use, location, regulation, technology, and other relevant elements that could affect residual values.

- After a final review and adjustment, the residual curve should be cross-checked with other curves of similar equipment. The analyst should notice a similarity in shape to the known curve. If not, then it should be re-analyzed to determine if an error was made. Additional research may be required to increase the sample of trade data, etc. An increase in sample size and additional research may lead to a change in the curve or confirm the original shape.

  In any event, the analyst should review the final results and base the findings on the body of knowledge as well as experience. This will lead the analyst to the selection of the final curve and appropriate residual values.

(All Amounts in $1,000s)							
	Fair Value Adjusted Price	**1 year**	**2 years**	**3 years**	**4 years**	**5 years**	**6 years**
Residual Value Actual, $	200.0	--	130.0	122.5	101.6	85.0	72.5
Residual Value Curve, %	100%	--	65%	61%	51%	43%	36%

Figure 6.7—Residual Value Curves

(All Amounts in $1,000s)							
	Fair Value Adjusted Price	**1 year**	**2 years**	**3 years**	**4 years**	**5 years**	**6 years**
Residual Value Actual, $	200.0	--	130.0	122.5	101.6	85.0	72.5
Subject Curve, %	100%	--	65%		51%		36%

Figure 6.8—Curve Fitting

In summary, the residual valuation method described above is basic. It is also non-inflated, and based on actual historical sales, trades, and in some instances the use of curve fitting to fill voids in certain curves. Finally, and most importantly, the analyst must review the curve and make subjective adjustments based on personal experience and variables that affect value.

Inflation Factors

The trade data method of residual value analysis described here does not use inflation factors. However, many owners/leasing companies routinely apply an inflation factor to residual values. Care must be taken in selecting an index. Several types of indices are published and are available for purchase by trade associations, insurance companies, appraisal companies, and/or the government.

If an inflation factor is used, it should be based on a "machine-specific" index, that is, an index devoted strictly to that type of equipment being valued (e.g., a "handy-size" bulk carrier index, a 6-axle, AC locomotive index, a Class 8 over-the-road truck/tractor index, a locomotive index, a plastic injection molding machine index, etc.)

The use of machine-specific indices for equipment of a similar type provides much greater reliability than the use of "industry-specific" indices. Industry-specific indices are based on a basket of goods believed to be representative of a typical manufacturing plant or operation in a certain industry. For example, a machine shop index, reflective of the typical machine shop, may include milling machines, lathes, drilling machines, machining centers, turning centers, grinder, welders, cranes, measuring machines, and other related items. In reality, each type of equipment will escalate at its own rate. It is not uncommon to see one type of equipment escalating at one percent per year, while another type may be escalating at five percent per year. Thus, a blending of weighting of the index can yield an answer that is half right (also half wrong). If the error is compounded over a number of years, the deviation from the actual inflation rate can be significant.

This principle also applies to leasing companies that use governmental indices, such as the Consumer Price Index (CPI). An analysis of the CPI would probably startle most leasing company executives. Not too many lessors would care to have the inflated value of their truck fleet or aircraft based on the cost of a loaf of bread, a pound of butter, or one month's home rent.

The analyst should take particular care and exercise caution when applying inflation factors to non-inflated residual values. Each owner/leasing company will have its own philosophy regarding the use of inflation factors, which the analyst should follow in a responsible way.

CONCLUSION

This chapter discussed the two categories of equipment values, “Replacement Cost” and “Market Value” that are used to establish an equipment valuation database. Establishing a reliable equipment valuation database is very important because estimates and appraisals for future projects are based on this database. The current and residual values of equipment are also determined using this database. Of all the methods available to the estimators/appraisers for determining the residual values, this chapter illustrated the trade data method of residual value analysis. This method provides inflated residual value as it does not take into account the associated inflation factors. However, the analysts should be careful in applying the inflation rate factors to get the non-inflated residual values, as each owner/leasing company has their own philosophies in using such inflation rate factors.

Chapter 7—Economic Costs

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Abstract
Facing the heightened global competition, cost professionals need to sharpen their knowledge and skills of economic costs. Making the right decision to pursue a project, product or service requires a sound understanding of the time value of money, tax impact, depreciation, and economic analysis techniques. Readers will also learn controllable factors impacting decisions and those outside the control of cost professionals.

Editor's note: This chapter works in tandem with Chapter 27, *Financial and Cash Flow Analysis,* to provide fundamentals in engineering economics.

Keywords
Depreciation, Discount Rate, Economic Analysis, Economic Life, Future Value, Opportunity Costs, Present Value, Price Index, Taxes, and Time Value of Money.

INTRODUCTION
The area of economic costs is a wide-ranging area of importance for cost engineers and cost managers. Ultimately, there is an owner or client for the project, public or private, that is supplying the funds for the project. The project must be economically feasible to construct. If instead of a project, the item is a product to be produced, the same criteria will apply to producing a product. The price of the product must be in a range that is affordable to the target market for the product.

With globalization, the area of economic costs is complicated by the production of various components for a product in a number of countries. These various countries, by definition, will have their own unique set of economic cost factors, including currency issues, inflation rates, and taxation rates. Cost engineers and cost managers need to be informed of economic cost concepts and potential applications to their work. Often, one is faced with the differences between alternatives and the cost implications may not be immediately apparent. Economic analysis techniques, properly applied, can pinpoint the alternative with the most favorable cost aspects.

Learning Objectives
After completing this chapter, the reader should be able to:

- Evaluate, on an economic analysis basis, the differences between two or more alternative courses of action.
- Understand such concepts and techniques as depreciation methods, net present value, annual cash flow analysis, rate-of-return analysis, benefit-cost analysis, and payback periods.

CONCEPTS
Cost engineering professionals must make sensible decisions in the arena of economic costs. Often, these decisions are complicated by the fact that elements of these decisions (such as costs, revenues, and benefits) occur at different times such as in the life cycle of a project. As an example, manufacturing processes that proved economical over a lengthy period of time may end up as a net negative cost when the downstream costs of hazardous waste cleanup are factored into the equation. While no one can peer unerringly into the future, the more effective the analysis of the totality of economic costs, the better will be the resulting decisions. The recognition of the impact of the various factors of economic costs will lead to more rational decision processes by cost engineering professionals.

Elements of economic costs may be outside the ability of the cost engineering professional to control. Taxation policies and the respective rates of taxation are set by political entities responding to their often diverse constituencies. Depreciation rules, again, are enacted by political entities. Currency variations may be outside even the control of the particular political entity, and instead are influenced by the actions and inactions of governments and currency traders/speculators around the world. Some aspects of inflation may be the result of government actions, as the government expands the money supply resulting in too much money chasing too few goods and services. While some elements of economic costs may be outside the control of any one decision maker, one still cannot ignore their potential impact on investment decisions.

TYPES OF COSTS

Opportunity Costs
Economic decisions need to consider the impact of opportunity costs. An opportunity cost represents the foregone benefit by choosing one alternative over another. A student upon graduation from high school has many choices including work, military service, and further education. If the student decides to pursue further education, the costs of this endeavor represent not only tuition costs, books, and living costs, but also wages or salary foregone from work opportunities. Thus, a total cost calculation will

include the lost salary or wages given up in favor of the further education decision. Similarly, other economic decisions need to consider foregone benefits for an accurate analysis.

Sunk Costs
Sunk costs represent funds already spent in by virtue of past decisions. Since these expenditures are in the past, by definition, they should not influence current decisions. While a past investment may still be yielding benefits from an income stream, these current and future benefits are relevant. The past expenditure is considered to be a sunk cost and is ignored in current and future decision-making.

Book Costs
Assets are carried on the firm's books at original cost less any depreciation. Book costs represent the value of an item as reflected in the firm's books. Book costs do not represent cash flows and thus are not taken into account for economic analysis decisions, except for potential depreciation impacts for tax consequences. Conservative accounting principles dictate that if the market price for financial assets, such as a stock, is lower than the original price, this asset will be carried at the lower cost or market value. The underlying land values may have significantly escalated over a period of years, however, the asset will still be carried at its original or book cost. For the firm, the sale of the land at a higher market price would not make sense unless this was part of a strategy to dispose of surplus assets.

Incremental Costs
In economic analysis decisions, focus must be on incremental costs or those cost differences between alternatives. As an example, in the comparison between two alternative pieces of processing equipment, if both units have the same yearly annual maintenance costs of $2,500 each, there is no incremental difference. Therefore, maintenance costs can be excluded from the analysis. On the other hand, if a third piece of processing equipment has annual maintenance costs of $1,500, then there is an obvious incremental cost difference of $1,000, which must be considered in the analysis between the three alternative units.

CHANGES IN COSTS
Changes in costs occur for a number of reasons in the economy. Cost professionals must be conversant with the potential for cost changes and their implications. The most common cost changes concepts are: inflation, deflation, escalation, and currency variation.

Cost changes are usually measured by price indexes which represent relative prices for either a single good or service, or a market basket of goods and services. Generally, indexes are used over time to measure the relative price changes in goods and services. Price indexes can be inaccurate barometers of price changes in certain contexts. If the quality of a good or service changes for better or worse, price indexes do not account for this change in quality. Price indexes do not account for changes in size, such as in recent years in the US whereby plywood sheet length and width dimensions have remained constant, but sheet thicknesses have gotten slightly thinner. Changes in packaging may be less costly for a producer of a product, but have some effect or no effect depending upon the end-user. They are not reflected in cost indices. Comparing a 1980-era car to a 2010-era car will find not only a change in relative price, but in quality as well. Price indexes lose relevancy when the items being compared are not the same. A factory built in one country in 2005, of a given size with its own price index, could be compared to a factory built in another country in 2012, only if adjustments are made for relevant differences. If the 2005 factory construction included several miles of additional utility lines for services or expensive soils problems, these differences must be considered in the analysis, along with currency translation issues.

Inflation
Inflation is a rise in the price level of a good or service, or market basket of goods and/or services. Inflation does not occur by itself but must have a driving force behind it. There are four effects that can result in inflation, either by themselves or in combination with other effects. These four effects are: money supply, exchange rates, demand-pull inflation, and cost-push inflation.

Money supply is influenced by the central bank of a country. Most countries' governments are able to operate by selling and buying bonds and setting certain internal interest rates in that country. These central bank operations will therefore have an impact on monetary policy, which can impact inflation. A loosening of monetary policy will increase the flow of money in the system, which means the increased money supply is chasing the same amount of goods and services. This increases the price of the goods and services, resulting in inflation.

Exchange rates can impact inflation by influencing the price of imported goods and services. If the currency of Country A falls in relation to the currency of Country B, imports from Country A are relatively less expensive. On the other hand, a rising Country A currency relative to Country B currency will make those same imports more expensive. If the import is a basic industrial commodity, such as copper used in a wide range of products, the rise in relative price will lead to inflationary price ripples throughout the economy of Country A.

Demand-pull inflation is when excessive quantities of money are chasing a limited amount of goods and services', resulting in what is essentially a "seller's market," as sellers receive premium prices. Examples of this can be anything from automobiles to real estate. An auto in short supply commands prices above list invoice. The same demand-pull inflation would apply to real estate in a popular location.

Cost-push inflation takes place when product producers encounter higher costs and then push these costs along to others in the production chain through higher prices. These higher costs may be for labor, material, or any other item with a significant cost element. A labor contract with workers may dictate 10 percent wage increases in a given year. If not offset by productivity gains in a product with 50 percent labor content, these wage increases will be passed on to purchasers of the product.

Deflation

Deflation is the opposite of inflation, with a fall in the general price level for goods and services or a representative market basket of goods and services. The same aforementioned factors of money supply, exchange rates, demand-pull, and cost-push factors operate, but in the opposite direction with a resultant decrease in prices. If costs for producers of goods and services fall in competitive industries, this cost decrease will be passed on to purchasers. Prices for personal computers have fallen across the board since their large-scale introduction in the 1980s, even while features have increased in the units. Complete personal computers can now be purchased, 20 years later, for a fraction of their original cost. A contracting money supply can result in price deflation. Exchange rates that rise in one country, giving that country a stronger currency relative to those countries from which it imports goods and services, will result in deflationary price decreases for those same imports.

Escalation

Escalation is a technique to accommodate price increases or decreases during the life of the contract. An escalation clause is incorporated into the contract. This provides that when prices increase, the purchaser for these relevant inflationary changes will compensate the supplier for the price increases. For instance, an aircraft maintenance firm may sign a five-year contract for the maintenance and fueling for a firm's aviation fleet. If aviation fuel increases in cost during the contract, the escalation will be paid for by the purchaser as an additional contract amount. Similarly, if the price of fuel declines or de-escalates during the contract, the supplier will either charge less or rebate a like amount to the purchaser. These escalation and de-escalation clauses help to shield both the supplier and the purchaser from unpredictable cost changes. Without such clauses, suppliers would include contingency amounts that might later be found to be totally unrealistic and on the high side. However, the supplier would gain from this windfall and the purchaser would be the loser. Similarly, excessive price increases in a commodity such as aircraft fuel might force a supplier into contractual default, depending on the magnitude of the increase. These escalation clauses may or may not be allowed in certain governmental sector contracts. For the firm facing escalation issues, and difficulty in forecasting inflation rates, a common solution is to pre-purchase items most subject to inflationary price increases. Another alternative is to perform hedging through futures contracts for various commodities, including oil on the commodity exchanges around the world.

Currency Variation

Currency changes can have a significant cost impact both on those inside the country, as well as those outside the country. Currency prices are set in markets around the world and change on a constant basis as the result of daily trading fluctuations and moves by central banks. Many organizations operate on a multi-national basis. Therefore, the currency fluctuations in one country or many countries can have an overall impact on earnings. A contract for work, or to supply products to one country, if set in that countries' currency can make the value of that contract go up or down when those earnings are repatriated to the home country of the firm. Financial assets held in one country can witness a significant decline in value if that country's currency is devalued by the central bank. Protecting against currency variation is complicated and can be accomplished through currency futures hedging or valuing contracts against very stable currencies to cite two examples.

GOVERNMENTAL COST IMPACTS

The actions of governmental units and jurisdictions can impose significant cost impacts on a firm. In some cases, the cost impacts of governmental units are direct, such as in the case of imposed taxes. In other cases, governmental cost impacts are more difficult to measure, such as in governmental regulations that require or prohibit certain actions. Whether direct or indirect, an accounting of costs must recognize these governmental actions.

Governmental Regulations

Regulations mandating certain actions by organizations may or may not have an impact on costs. An example might be safety laws mandating compliance by all organizations. However, certain organizations may have in-place safety requirements that already meet or exceed the new safety laws. Therefore, there is no additional cost for regulatory compliance. On the other hand, concurrent with these new governmental safety laws may be reporting requirements proving that the entity is complying with governmental strictures. In this case, there is then an additional cost impact. Issues such as importing equipment into a foreign country may be no problem in one country and take six months in a neighboring country as a result of customs regulations. Regulatory uncertainty is another burden placed on entities as they postpone certain actions because they do not know what burdens will be placed upon them. Then there may be further uncertainty as to enforcement. There are numerous laws on the books around the world that are not enforced by governmental units or the enforcement is non-uniform. Moreover, there may be uneven enforcement between nationally-based firms and foreign-based firms. Attempting to assess the cost of these situations can pose a high degree of difficulty for an organization. In a number of countries, other firms can assist in navigating through the myriad of laws and their enforcement levels.

Governmental Taxes

Governments are most often maintained by the taxes that they impose. These taxes take many forms, such as income taxes, property taxes, inventory taxes, employment taxes, gross receipts taxes, and sales taxes. With a sales tax the firm merely acts as the tax collector for the government, adding the sales tax and collecting it from customers. While the firm does not pay the sales tax itself, excepting sales tax on items it may purchase for its own use, there are costs involved for the firm in administering these tax types. Other taxes such as income taxes directly impact the firm in terms of profitability as they tax the net income of the firm. Some countries have a Value-Added Tax (VAT). The VAT is applied to the added value applied by the firm. Therefore, if a firm took $100 worth of raw materials and produced a product valued at $250, the Value-Added Tax (VAT) would be applied to the $150 difference or value added by the firm. Similarly, another firm taking the same product and incorporating it in an assembly would be assessed the VAT, based only on the added value.

Effective Tax Rates and Marginal Tax Rates

Effective tax rates also termed average tax rates are calculated for income taxation purposes by the percentage of total taxable income paid in taxes. The effective tax rate results from dividing the tax liability by the total taxable income. The marginal tax rate is the tax rate on the next dollar of taxable income. For financial decision-making, the marginal tax rate is a key element because the firm is concerned with the tax impact of additional income or income deductions.

Investment Tax Credits

Governmental entities, in order to encourage economic activity, may give firms tax credits based on their investments in a given location. Governments may want to encourage the location of new plants in economically depressed areas and therefore promote this through investment tax credits including abeyance on certain taxes, such as property taxes. In other cases, tax credits may be granted for certain types of investment in a plant and equipment. The investment tax credits may be tied to certain public policy goals. Thus, a firm installing more energy efficient equipment that reduces energy consumption may be able to take advantage of energy investment tax credits.

Depreciation and Depletion

In order to encourage firms to invest in new plants and equipment, governmental entities often allow these firms to depreciate their investments over time. This investment depreciation allows the firm to reduce its income by a set proportion per year with a depreciation write-off until the investment is fully depreciated by the firm. The limits on investment depreciation write-off are proscribed by the governmental tax code. It must be realized that depreciation itself is not a cash flow. Depreciation instead is a non-cash expense that reduces taxable income. Depreciation therefore provides an incentive for firms to invest in new plants and equipment. However, firms are only allowed to depreciate based on the original plant and equipment costs. Therefore, current and future inflation during the asset's depreciation cannot be taken into account for these purposes. The rationale underlying depreciation concepts is that physical assets lose value over time as a result of such factors as deterioration, wear, and obsolescence.

Depletion is analogous to depreciation, but for natural resources. Thus, owners of a stone quarry, an oil well, or standing timber, can take depletion allowances based on the percentage of the resource used up in a given time period. However, common land used for agricultural or business purposes, such as the site for a building or lost parking, is not depreciable. Common land is an asset, but is not considered to wear out like plant or equipment assets.

Depreciation Techniques

Given that governments allow firms to depreciate their investments, there are numerous methods to accomplish the depreciation process. These standard methods simplify accounting for the depreciation expenses. Some of the more common depreciation techniques are: Straight-Line (SL) Method, Double-Declining Balance (DDB) Method, Sum-of-Years Digits (SOYD) Method, Modified Accelerated Cost Recovery System (MACRS), and Units of Production (UOP) Method.

Straight-Line Depreciation

Straight-Line (SL) Methods take an equal amount of depreciation every year. The SL Method takes the original cost, less the salvage value, divided by the number of years of life by the formula:

$$D = (C-S)/N$$

Where:

D = depreciation charge
C = asset original cost
S = salvage value
N = asset depreciable life (years)

Therefore, an asset with a $5,000 original cost, five-year life, and $1,000 residual salvage value, would have SL depreciation of $800 per year: ($5,000 - $1,000) / 5 years = $800.

Double-Declining Balance Depreciation
Double-Declining Balance Depreciation applies a constant depreciation rate to the assets' declining value. The DDB formula is:

$D = (2/N)\ (BV_{r-1})$

Where:

D = depreciation charge
C = asset original cost
BV_{r-1} = book value of asset at end of rth year
BV_0 = book value at beginning of first year = C
N = asset depreciable life (years)

In the SL example above, the similar DDB amounts would be as shown in Table 7.1, for an asset with a $5,000 original cost, five-year life, and $1,000 residual salvage value.

Table 7.1—DDB Amounts for an Asset with a $5,000 Original Cost, 5-Year Life, and $1,000 Residual Salvage Value

Year	DDB Calculated Amount	DDB Allowable Depreciation	Book Value at Beginning of Year	Book Value at Year End
1	(2/5)($5,000) = $2,000	$2,000	$5,000	$5,000 - $2,000 = $3,000
2	(2/5)($3,000) = $1,200	$1,200	$3,000	$3,000 - $1,200 = $1,800
3	(2/5)($1,800) = $720	$720	$1,800	$1,800 - $720 = $1,080
4	(2/5)($1,080) = $432	* $80	$1,080	$1,080 - $80 = $1,000
5	(2/5)($1,000) = $400	* $0	$1,000	$1,000
	Total = $4,752	$4,000		
***Note:** An asset cannot be depreciated below its salvage value. Thus, depreciation totals under the DDB method for Years 4 and 5 total $80. In the comparison between SL and DDB methods, DDB allows faster write-offs of the asset value. A general economic cost principle, given the time value of money, is that early money is of greater importance.				

Sum-Of-Years Digits Depreciation
Sum-of-Years Digits (SOYD) method allows depreciation to be taken at a faster rate than SL. This SOYD method takes depreciation in any one year, as the product of a fractional value times the total original depreciable value. The fractional value for any given year has as its numerator, the years of asset life remaining, while the denominator is the sum of digits, including one through the last year of the asset's life.

The SOYD depreciation formula is:

$D_r = (C - S)*[(2(N-r+1))/(N(N + 1)]$

Where:

D_r = depreciation charge for the rth year
C = asset original cost
S = salvage value
N = asset depreciable life (years)
r = r^{th} year

From the SL and DDB examples above, the similar SOYD amounts would be as shown in Table 7.2, for an asset with a $5,000 original cost, five-year life, and $1,000 residual salvage value:

Calculating the SOYD numerator: SOYD = [(N/2)*(N + 1)] = [(5/2)*(5 + 1)] = 15

Table 7.2—SOYD Amounts for an Asset with a $5,000 Original Cost, 5-Year Life, and $1,000 Residual Salvage Value

Year	SOYD Formula	SOYD Calculated Amount
1	(5/15)($5,000 - $1,000) =	$1,333.00
2	(4/15)($5,000 - $1,000) =	$1,067.00
3	(3/15)($5,000 - $1,000) =	$800.00
4	(2/15)($5,000 - $1,000) =	$533.00
5	(1/15)($5,000 - $1,000) =	$267.00
	Total =	**$4,000.00**

In the comparison between SL and SOYD methods, SOYD is similar to DDB in that it allows faster write-offs of the asset value in the early years. Again, as noted above, a general economic cost principle, given the time value of money, is that early money is of greater importance.

Modified Accelerated Cost Recovery System Depreciation
The Modified Accelerated Cost Recovery System (MACRS) method is unique to the U.S. Tax Code. Depreciation under this MACRS method is based on original asset cost, asset class, asset recovery period, and asset in-service date. Asset classes are differentiated based on three-year, five-year, seven-year, ten-year, and other property lives depending on asset type. In the U.S., depreciation rates are set by percentages allowed under the U.S. Tax Code. In other countries, depreciation may or may not be allowed under the applicable taxation regulations.

Units of Production Depreciation
The Units of Production (UOP) method is used when depreciation is more accurately based on usage instead of time. The UOP Method is particularly useful when an asset encounters variable demand. A piece of construction equipment may be used 1,200 hours in one year, 1,600 hours the next, and 900 hours the third year; the UOP Method recognizes that the equipment wears out based on usage and is therefore a more accurate barometer than years.

The overall goal for any organization in choosing a depreciation method is to choose the method that will depreciate an asset as quickly as possible. Depreciation is a non-cash expense that thereby reduces an organization's total taxable income in any given year. Based on the principles in this section, higher depreciable amounts in early years result in less taxable amounts due from the firm to a given entity. In addition, given the general inexorable trend of higher inflation, when the asset has been fully depreciated, there will be less funding available to buy the successor plant or equipment unit.

ECONOMIC ANALYSIS TECHNIQUES
There are a variety of economic analysis techniques available to enable accurate choices between competing alternatives. The general principle is that there are competing alternatives and the goal is to choose the alternative with the highest return. Competing alternatives may be advanced by those in an organization favoring one alternative course of action over another. Sound economic analysis that accurately assesses benefits and costs, along with the timing of these cash flows can avoid the organizational politics associated with the unjust selection of a particular course of action. In order to analyze returns from alternative choices, there are a number of techniques. These include: net present value, capitalized cost, annual cash flow analysis, rate of return analysis, benefit-cost ratio analysis, and payback period. These analysis techniques, except the payback period method, deal with a concept commonly referred to as the time value of money.

Time Value of Money
The Time Value of Money is a key area in economic cost analysis. Different alternatives will have differing amounts of cash income and cash expenses over their lifetime. In order to compare these different alternatives on the same basis, these cash amounts of income and expenditure must be set to equivalent terms. There is a common unit of measure set in a currency, whether it is dollars, Euros, pesos, yen, or some other measure; there will also be an interest rate set which provides the common basis for calculations. (When evaluating expected income/cost flows in the future, this interest rate can also be referred to as a “discount rate,” since money in the future is “worth” less than money in the present.) The interest rate may sometimes be the rate at which a firm is charged in the capital markets for borrowing funds to finance capital projects, such as through sales of corporate bonds. Certain firms may set an interest rate higher than their cost of borrowed capital and this may be referred to as a "hurdle" rate. Projects that meet this hurdle rate will be funded, whereas projects that do not meet these target rates will not be funded by the firm. However, usually investment capital is a scarce resource and only projects will be selected with the most beneficial returns. In evaluating an interest rate, it is comprised of two components: lender's long run expectations of the inflation rate and the risk of lending the capital. Therefore, if long-term inflation rates are projected to be three percent and the risk of the lender regarding a particular borrower is low at four percent, the derived interest rate would be summed at seven percent. For riskier ventures and riskier borrowers, interest rates will correspondingly trend upward.

In order to perform this time-value-of-money analysis, certain information will be available with other information requiring calculation. The problem facing the cost engineer must be capable of being expressed in quantitative terms. Therefore, a problem of whether it is best to paint a building blue or green cannot be judged on a quantitative basis. But if one type of building paint has a life of ten years with its own cost structure and another paint type has a life of seven years with a separate cost structure, this is a problem suitable for quantitative analysis. Similarly, consider a set of railroad tracks with the need for a road to cross over them. The most expensive alternative is to go to the expense of complete grade separation, such as to place a viaduct or a bridge whereby the road crosses underneath the railroad trackage or at elevation over the trackage respectively. The least expensive alternative is to have the road cross the railroad tracks at grade, with simple signage denoting the presence of the railroad tracks to vehicular traffic. In between these two extremes are other alternatives, such as flashing-light signage or crossing gates. The goal of any railroad track protection is to save lives and eliminate accidents, since fatalities and severe vehicle-damages result from train-vehicle contact. Economic studies can actually place a value on the probable cost of accidents eliminated based on roadway traffic volumes. Therefore, there is a cost placed on the potential saving of human lives. The economic calculation for potential vehicular damage can be taken from statistical sources.

Common language terms and their symbols for time value of money problems are [5]:

P = present value
F = future value
A = annual amount or annuity
G = uniform gradient amount
n = number of compounding periods or asset life
i = interest rate
S = salvage value

Table 7.3—Standard Formulas for Economic Analysis

Formula Name	Operation	Symbol	Formula
Single Payment			
Compound Amount	P to F	(F/P, i%, n)	$F= P(1 + i)^n$
Present Value	F to P	(P/F, i%, n)	$P = F(1 + i)^{-n}$
Uniform Series			
Sinking Fund	F to A	(A/F, i%, n)	$A = F[i /((1 + i)^n – 1)]$
Capital Recovery	P to A	(A/P, i%, n)	$A = P[(i(1 + i)^n)/((1 + i)^n – 1)]$
Compound Amount	A to F	(F/A, i%, n)	$F = A[((1 + i)^n – 1)/ i]$
Equal Series Present Value	A to P	(P/A, i%, n)	$P = A[((1 + i)^n – 1)/(i*(1 + i)^n)]$
Gradient Series			
Arithmetic Uniform Gradient Present Value	G to P	(P/G, i%, n)	$P = G[((1 + i)^n – i*n – 1)/(i^2(1 + i)^n)]$

Net Present Value Method
In the comparison of alternatives, a common basis is needed for this comparison. Alternatives typically will have different costs and benefits over the analysis period. The Net Present Value (NPV) method provides the platform to resolve alternatives into equivalent present consequences.

Example NPV Problem:
A firm is evaluating the potential purchase of one of two pieces of equipment. Unit A has a purchase price of $10,000 with a four-year life and zero salvage value. Annual maintenance costs are $500 per year. Unit B has a purchase price of $20,000 with a 12-year life and $5,000 salvage value. In year one, maintenance costs are zero. In year two, maintenance costs are $100 and increase by $100 per year thereafter. The firm's cost of capital is eight percent.

Example NPV Problem Solution:
This problem illustrates two common issues faced in comparison of alternatives. Units A and B have unequal lives of four years and 12 years respectively. Therefore, a common multiple of the respective unit lives must be selected which in this case will be 12 years. Economic analysis problems view re-purchase of the same unit, at four year intervals, will be at original cost unless there is concrete information to the contrary. The second issue this problem illustrates is that of a gradient where maintenance costs for Unit B are increasing on a year-to-year basis as opposed to steady costs.

NPV Unit A:

NPV_A = $10,000 + $10,000(P/F,8%,4) + $10,000(P/F,8%,8) + $500(P/A,8%,12)
= $10,000 + $10,000(0.7350) + $10,000(0.5403) + $500(7.536)
= $26,521

NPV Unit B:

NPV_B = $20,000 + $100(P/G,8%,12) - $5000(P/F,8%,12)-$100
= $20,000 + $100(34.634) - $5000(0.3971)-$100
= $21,378

Since, in this example problem, we are analyzing costs and not benefits, Unit B with the lower NPV cost structure is preferable ($21,378 < $26,521). If the calculation involved benefits greater than costs, then Unit A with the higher benefit NPV would be preferable.

Capitalized Cost Method
In some cases, problems present themselves which have an infinite analysis period. The need for a structure, such as a road or a bridge for example, is perpetual. With these types of situations, the capitalized cost method is chosen. Capitalized Cost (CC) represents the present sum of money that needs to be set aside now, at some interest rate, to yield the funds required to provide the service indefinitely. The capitalized cost formula is A = P*i.

Example Capitalized Cost Problem:
A bridge is built for $5,000,000 and will have maintenance costs of $100,000 per year. At six percent interest, what is the capitalized cost of perpetual service?

Example Capitalized Cost Problem Solution:
The $5,000,000 is a present cost with the $100,000 per year maintenance costs as ongoing. Therefore:

Capitalized Cost = $5,000,000 + ($100,000) / 0.06
= $5,000,000 + $1,666,667 = $6,666,667

Equivalent Uniform Annual Cost (EUAC) or Benefit
For some types of analysis, it may be preferable to resolve the comparison to annual cash flow analysis. The comparison may be made on the basis of Equivalent Uniform Annual Cost (EUAC), Equivalent Uniform Annual Benefit (EUAB) or on the EUAB-EUAC difference.

Example EUAC Problem:
Assume that the two units, Unit A and Unit B are compared on the basis of EUAC. Unit A has an initial cost of $20,000 and $3,000 salvage value, while Unit B has an initial cost of $15,000 and $2,000 salvage value. Unit A has a life of ten-years, whereas Unit B has a five-year life. Cost of capital is ten percent.

Example EUAC Solution:
The relevant EUAC formula is (P-S)(A/P,i,n) + SI

EUAC Unit A:
$EUAC_A$ = ($20,000-$3,000)(A/P,10%,10) + $3,000(0.10)
= ($17,000)(0.1627) + $300
= $2765.90 + $300 = $3065.90
EUAC Unit B:
$EUAC_B$ = ($15,000-$2,000)(A/P,10%,5) + $2,000(.10)
= ($13,000)(0.2638) + $200
= $3429.40 + $200 = $3629.40

On the basis of the EUAC comparison between Unit A and Unit B, Unit A has the lower EUAC by $563.50 ($3629.40 - $3065.90) and would be the choice.

Rate of Return Analysis
Many organizations in making investment choices often set hurdle rates. The hurdle rate is the benchmark rate of return that a capital investment decision must achieve to be acceptable. A Rate of Return (ROR) is computed from the projected cash flows of the project. ROR values provide a ready basis for the comparison of alternatives. In the case where capital investment funds are limited, projects with the highest ROR values can be selected for the organization.

Example ROR Problem:
Assume that in the comparison between Unit A and Unit B (each with a one-year life) that the cost of $20,000 for Unit A, versus a $10,000 cost for Unit B, also results in an incremental benefit of $15,000 for Unit A as compared to Unit B. If the organization has a hurdle rate of 20 percent for capital projects, which alternative is a better choice?

Example ROR Solution:
The NPV of the cost is set equal to the NPV of the benefit. The NPV Cost of Unit A versus Unit B is $10,000 ($20,000 - $10,000). The NPV of the Unit A – Unit B Benefit is $15,000.

ROR	NPV Cost	= NPV Benefit
	$10,000	= $15,000(P/F,i,1)
	$10,000/$15,000	= 0.6667

By inspection, if $10,000 cost difference increases benefits by $15,000 in one year, the ROR must be 50 percent. Consulting either compound interest factor tables or calculation by the P/F formula for one-year at a 50 percent ROR confirms this fact. Since the organizations' hurdle rate is 20 percent, the additional investment in Unit A meets the criteria.

Benefit-Cost Ratio Analysis Method
Benefit Cost (B/C) Ratio Analysis involves the simple comparison between benefits and costs of a proposed action. Benefits are placed in the numerator and costs are placed in the denominator. If the ratio of benefits to costs is greater than one, the project is viable. Comparisons can be made between projects to select those projects with the highest B/C ratio.

Example B/C Problem:
Project A with the NPV Benefits of $1,500,000 and NPV Costs of $1,200,000 is being compared to Project B with NPV Benefits of $2,000,000 and NPV Costs of $1,700,000. Which is the preferred project on a B/C analysis basis?

Example B/C Solution:
Project A:

B/C_A = $1,500,000 NPV Benefits/$1,200,000 NPW Costs = 1.25 B/C Ratio

Project B:

B/C_B = $2,000,000 NPV Benefits/$1,700,000 NPW Costs = 1.17 B/C Ratio

On the basis of the above B/C analysis, both Projects A and B have a positive B/C ratio, but Project A would be selected as its 1.25 ratio is greater than the Project B 1.17 ratio. The key question in this analysis is whether the benefits and costs have been evaluated on an accurate basis. If the benefits for Project A are really $1,300,000, then Project A is less attractive than Project B. B/C analysis is commonly used in the public sector in evaluating major capital projects such as dams, waterways, bridges, and other similar public works endeavors. In the public sector, politics can therefore unfairly select one project versus another more attractive alternative.

Payback Period Method
Payback Period is the period of time necessary for the benefits of the project to pay back the associated costs for the project. This is a very simple method and can prove to be inaccurate. A project with a payback period of three years may be selected over a similar project with a five-year payback period. Differences in the timing of cash flows are not considered, nor are the benefits and costs beyond the payback period. Payback period analysis is approximate and may not yield the same result as with other more precise methods, such as NPV or EUAC criteria.

As an example, an investment of $400,000 with benefits of $100,000 per year would have a payback period of 4 years ($400K/$100K = 4 years). However, a $4,000,000 investment with benefits of $800,000 per year and a payback period of 5 years ($4,000K/$800K = 5 years) might be the better choice on other criteria. The $400K investment may be scrap after the end of the fourth year, whereas the $4,000K investment, once the five-year payback period is complete, could yield comparable returns of $800K per year through ten years. Simple payback analysis in this case would lead to the fastest payback period rather than the best overall investment choice.

CONCLUSION
The area of economic costs presents problems ranging from simple to complex. In this era of intense global competition, economic cost decisions must be made on the basis of gathering as much relevant information as possible and then applying the most appropriate analysis technique. The accuracy of future cost and value streams is essential in properly applying these concepts. Numerous organizations have badly underestimated cost streams and overestimated value streams, resulting in significant losses. One large energy company in an industrialized country built a billion-dollar power plant in a lesser-developed country. However, the power produced from the plant was too expensive for the local market to afford, resulting in the abandonment of the investment. Enhanced knowledge of these economic cost analysis tools will lead to improved-decision quality for an organization.

RECOMMENDED READING

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Chapter 8—Activity Based Cost Management[3]

Gary Cokins

Abstract
The general ledger uses a chart of accounts, whereas Activity Based Cost Management (ABC/M) uses a chart of activities. Expenses occur at the point of acquisition with third parties. This is when money (or obligation) exits the company. From the expenses, all costs are *calculated* representations of how those expenses flow through work activities and into outputs of work. ABC/M ties each cost element to an activity and makes cost structure transparent. Low and non-value contributing activities can be reduced or eliminated to improve cost efficiency.

Keywords
ABC/M, Chart of Accounts, Cost, Direct Expenses, and Indirect Expenses.

INTRODUCTION
Why do some managers shake their heads in disbelief when thinking about the company's cost accounting system? I once heard an operations manager complain, "You know what we think of our cost accounting system? It is a bunch of fictitious lies—but, we all agree to them." This is certainly an exaggeration of the truth, but there is some merit to its point. It is sad to observe some users of the managerial accounting data resign themselves to a lack of hope. Unfortunately, some accountants are comfortable when the numbers all exactly reconcile in total but could care less if the parts making up the total are correct. For this type of accountant, the total is all that matters, and any arbitrary cost allocation can tie out to the total. The progressive accountants, who are increasing in numbers, recognize that indirect expenses should be traced and translated into product costs, based on cause-and-effect relationships. Progressive accountants realize that past practices of allocating costs using convenient base factors, such as direct labor or machine input hours, misrepresents economic reality. Progressive accountants use measurable cost drivers related to the unique and proportional support-related expenses to model the consumption relationships.

Imagine that you and three friends go to a restaurant. You order a small salad and they each order an expensive prime rib and a glass of wine. When the waiter brings the bill your three friends say, "Let's split the check evenly." How would you feel? It is not fair, not equitable.

This is how many products and service lines also feel when the accountants take a large amount of indirect and support overhead expenses and allocate them as costs without logic. There is minimal, or no causal link that reflects a true relative consumption of the expenses by the individual products, service lines, or end-users. Activity Based Cost Management (ABC/M) gets it right; it more fairly splits the waiter's check into four checks, based on what was individually ordered.

The sad truth is that when employees and managers are provided with reports that have accounting data in them, that information is used regardless of its validity, or skepticism of its integrity. They are using the data to gain insights, draw conclusions, and make decisions.

How can traditional accounting, which has been around for so many years, suddenly be considered so bad? This chapter on Activity Based Cost Management (ABC/M) systems explains the reasons leading to the interest and acceptance of ABC/M principles, as well as how to construct an ABC/M cost assignment model.

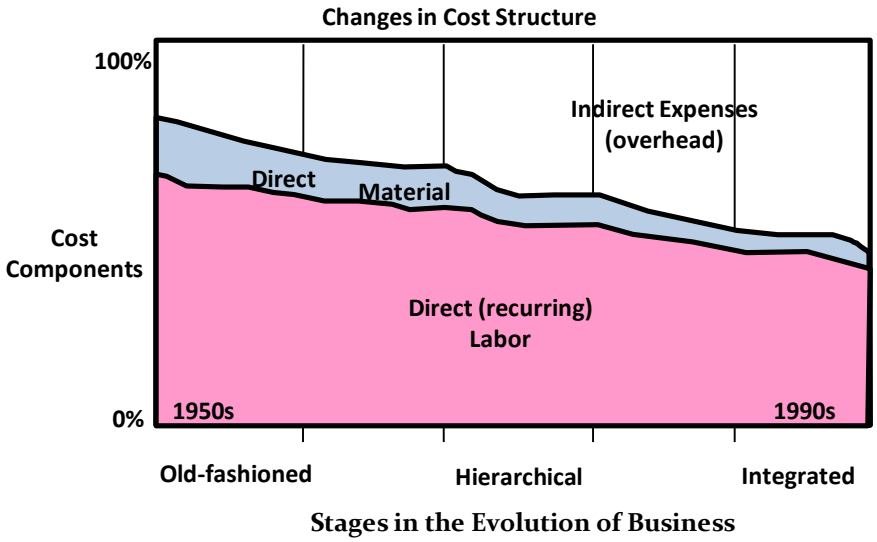
Learning Objectives
After completing this chapter, the reader should be able to:

- Understand why managers and employees are misled by arbitrary cost allocations with broad averages that violate the costing cause-and-effect principle.
- Understand how Activity Based Cost Management (ABC/M) transforms spending expenses on resources (e.g., salaries) into calculated costs of processes, and their work activities that belong to them, and then into products, service-lines, channels and customers.
- Identify how cost drivers cause costs to occur.
- Understand how attributes are tags or scores that are attached to activities to suggest actions.
- Understand how ABC/M is used in addition to strategic purposes, such as profit margin analysis, but also for cost management, productivity, and asset use.

[3] This chapter includes modified excerpts from **Activity-Based Cost Management: An Executive's Guide**; Gary Cokins; John Wiley & Sons; 2001; ISBN 0-471-44328-X)

INDIRECT (OVERHEAD) EXPENSES ARE DISPLACING DIRECT EXPENSES

The direct laborers in organizations are the employees who perform the frontline, repeatable work that is closest to the products and customers. However, numerous other employees behind the frontline also do recurring work on a daily or weekly basis. These employees' work is highly repeatable at some level, for example, a teller in a bank. Figure 8.1 is a chart that includes this type of expense, plus the other two major expense components of any organization's cost structure, its purchased materials and its indirect and shared expenses – commonly referred to as overhead[4].



Source: *Activity Based Cost Management: An Executive Guide; Gary Cokins; Permission to re-print granted.*

Figure 8.1—Indirect Expenses are Displacing Direct Expenses

Most organizations are experienced at monitoring and measuring the work of some of the laborers who do recurring work by using cost rates and standard costs. In the bottom layer of Figure 8.1 is the cost information that also reveals performance-related costs other than the period's spending, such as labor variance reporting. It is in this area of the chart, for example, that manufacturers use labor routings and process sheets to measure efficiency. These costs are well known by the name Standard Costs. Service organizations also measure this type of output-related information. For example, many banks know the standard cost for each deposit, each wire transfer, and so forth.

Problems occur in the indirect expense area appearing at the top portion of Figure 8.1. The figure reveals that over the last few decades, the indirect support expenses have been displacing the recurring costs. The organization already has substantial visibility of its recurring costs, but it does not have any insights into its indirect expenses, or what is causing the level of spending for these expenses. ABC/M can help provide for insights and learning.

In a bank, for example, managers and employee teams do not get the same robustness of financial information about the vice-presidents working on the second floor and higher up in the building as they do about tellers. The only financial information available to analyze the expenses of the vice presidents and other indirect support expenses is the annual financial budget data. These levels of expenses are annually negotiated. The focus is on spending levels, not on the various cost rates. The expense spending is monitored after the budget is published. Spending is only monitored for each department or function for each period to see if the managers' spending performance is under or over budget.

ABC/M extends to the indirect expenses the understanding and visibility of spending that is already applied to the recurring laborers. ABC/M can then become an organization-wide method of understanding work activity costs, as well as the standard costs of outputs.

Impact of Diversity in Products, Service Lines, Channels, and Customers

When asked why they believe indirect and shared expenses are displacing direct costs, most answer that it is because of technology, equipment, automation, or computers. In other words, organizations are automating previously manual jobs. However, this is only a secondary factor in the shift in organizational expense components.

[4] Organizations often refer to this support-related work as overhead. Overhead is also referred to as indirect costs. The two terms can be used interchangeably; however, the term "overhead" can be misleading and often has a negative connotation. In many cases overhead is crucial and is a very positive thing to have.

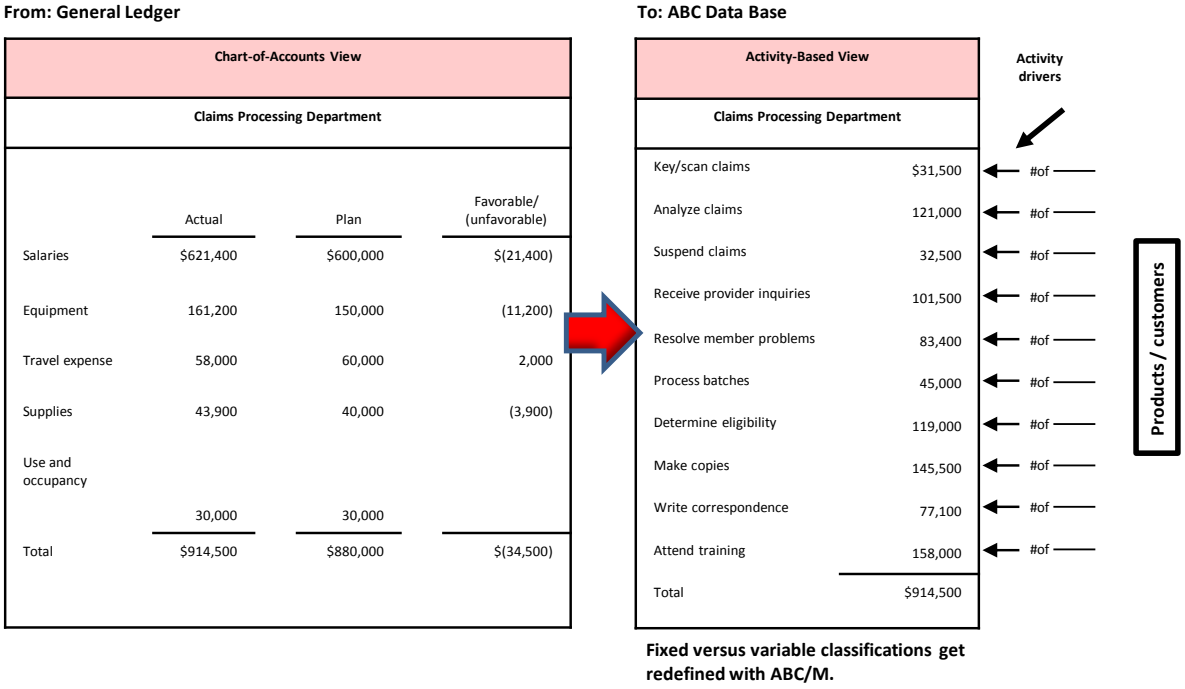
The primary cause for the shift is the gradual proliferation in products and service lines. Over the last few decades, organizations have been increasingly offering a greater variety of products (e.g., more colors, sizes, ranges, etc.) and services, as well as using more types of distribution and sales channels. In addition, organizations have been servicing more and different types of customers. Introducing greater variation and diversity (i.e., heterogeneity) into an organization creates complexity, and increasing complexity results in more indirect expenses to manage it. The fact that the indirect component of expenses is displacing the recurring labor expense does not automatically mean that an organization is becoming inefficient or bureaucratic. It simply means that the company is offering more variety to different types of customers.

In short, the shift to indirect expenses displacing direct labor reveals the cost of complexity. ABC/M does not fix or simplify complexity; the complexity is a result of other things. But what ABC/M does is point out where the complexity is and where it comes from.

ACTIVITIES ARE EXPRESSED WITH ACTION VERBS AND TRACE EXPENSES TO OUTPUTS

Figure 8.2 begins to reveal the explanation as to why traditional cost allocations of expenses are flawed and therefore misleading. The left side shows the classic monthly responsibility-center statement report that managers receive. Note that the example used is the back office of an insurance company. This is to demonstrate that, despite misconceptions, indirect white-collar workers produce outputs no differently than do factory workers.

If you ask managers, who routinely receive this report, questions such as, "How much of these expenses can you control or influence? How much insight do you get into the content of work of your employees?" They will likely answer both questions with, "Not much!" This is because the salary and fringe benefit costs usually make up the most sizable portion of controllable costs, and all that the manager sees are those expenses reported as lump-sum amounts.



Source: *Activity Based Cost Management: An Executive Guide; Gary Cokins; Permission to re-print granted.*

Figure 8.2—Each Activity Has its Own Driver

When the chart-of-account expenses are translated into the work activities that consume the financial general ledger's expenses, a manager's insights from viewing the activity costs begin to increase. The right side of Figure 8.2 is the ABC/M view that is used for analysis, and as the starting point for calculating the costs for both processes and diverse outputs. In effect, the ABC/M view resolves the deficiencies of traditional financial accounting by focusing on work activities. ABC/M is work-centric, whereas the general ledger is transaction-centric.

Expenses must be distinguished from costs. They are not the same thing. All costs are *calculated* costs. It is important to recognize that assumptions are always involved in the conversion and translation of expenses into costs. The assumptions stipulate the basis for the calculation. Expenses occur at the point of acquisition with third parties, including employee wages. This is when money (or its obligation) exits the company. At that special moment, value does not fluctuate; it is permanently recorded as part of a legal

exchange. From the expenses, all costs are *calculated* representations of how those expenses flow through work activities and into outputs of work.

A key difference between ABC/M and the general ledger and traditional techniques of cost allocation (i.e., absorption costing) is that ABC/M describes activities using an action-verb-adjective-noun grammar convention, such as inspect defective products, open new customer accounts, or process customer claims. This gives ABC/M its flexibility. Such wording is powerful because managers and employee teams can better relate to these phrases, and the wording implies that the work activities can be favorably affected, changed, improved, or eliminated. The general ledger uses a chart of accounts, whereas ABC/M uses a chart of activities. In translating general ledger data to activities and processes, ABC/M preserves the total reported revenues and costs, but allows the revenues, budgeted funding, and costs to be viewed differently.

Notice how inadequate the data in the chart of accounts view are for reporting business process costs that run cross-functionally, penetrating the vertical and artificial boundaries of the organization chart. The general ledger is organized around separate departments or cost centers. This presents a reporting problem. As a result of the general ledger's structure of cost center mapping to the hierarchical organization chart, its information drives vertical behavior, not the more desirable process behavior popularly displayed as value streams by the lean management community. The general ledger is a wonderful instrument for what it was designed to do, categorize and accumulate spending transactions into their accounts. But, the data in that format is structurally deficient for decision support other than the most primitive form of control, budget variances.

In effect, using traditional cost systems, managers are denied visibility of the costs that belong to the end-to-end value stream business processes. This is particularly apparent in the stocking, distribution, marketing, and selling costs that the traditional accounting "expenses to the month's period." With traditional cost allocations, these Sales, General, and Administrative expenses (SG&A) are not proportionately traced to the costs of the unique products, containers, services, channels, or customers that cause those costs to occur.

In summary, the general ledger view describes what was spent, whereas the activity-based view describes, what it was spent for, why, and how much. The ledger records the expenses, and the activity view calculates the costs of work activities, processes, and all outputs, such as products. Intermediate output costs, such as the unit cost to process a transaction, are also calculated in the activity view. When employees have reliable and relevant information, managers can manage less and lead more.

DRIVERS TRIGGER THE WORKLOAD

From Figure 8.2 much more information can be gleaned from the right-side view. Look at the second activity, analyze claims for $121,000, and ask: What would make that cost significantly increase or decrease? The answer is the number of claims analyzed, which is the work's activity driver. Figure 8.2 also shows that each activity, on a stand-alone basis, has its own activity driver. At this stage, the costing is no longer recognizing the organizational chart and artificial boundaries. The focus is now on the work that the organization performs, and what affects the level of that workload.

There is additional information. Let's assume there were 1,000 claims analyzed during that period for the department, shown in Figure 8.2. The unit cost per each analyzed claim is $100 per claim. If a specific group of senior citizens, over the age of 60, were responsible for half those claims, we would know more about a specific customer or beneficiary of that work. The senior citizens would have caused $60,500 of that work (i.e., 500 claims times $121 per claim). If married couples with small children required another fraction, married couples with grown children a different fraction, and so forth, ABC/M would trace all of the $121,000. If each of the other work activities were similarly traced using the unique activity driver for each activity, ABC/M would pile up the entire $914,500 into each group of beneficiary. This reassignment of the resource expenses would be much more accurate than any broad-brush cost allocation applied in traditional costing procedures and their broad averages.

In the past, calculating costs using volume-based allocations (e.g., direct labor or equipment input minutes; number of units produced) may have been acceptable and may not have introduced excessive error; but most organization's resource expense structure began to change in the 1970s. With greater indirect expenses relying on a basis for cost allocations, that were tied to unrelated volumes of usage, the traditional costing method had become invalid, relative to how the rich variation of products and services consumed costs. Therefore, the unfavorable impact of the costing errors was becoming much more intense than in the past. ABC/M resolves the problem of poor indirect expense allocations, but it also provides additional information for analysis to suggest what positive actions, strategic or operational, can be taken based on the new data.

To sum up Figure 8.2, when managers receive the left-side responsibility center report, they are either happy or sad, *but rarely any smarter*.

The Cost Assignment Network is one of the major reasons that ABC/M calculates more accurate costs of outputs. The assignment of the resource expenses also demonstrates that all costs actually originate with the customer or beneficiary of the work. This is at the opposite extreme of where people who perform cost allocations think about costs. Cost allocations are structured as a one source-to-many destinations redistribution of cost; but, the destinations are actually the origin for the costs. The destinations, usually

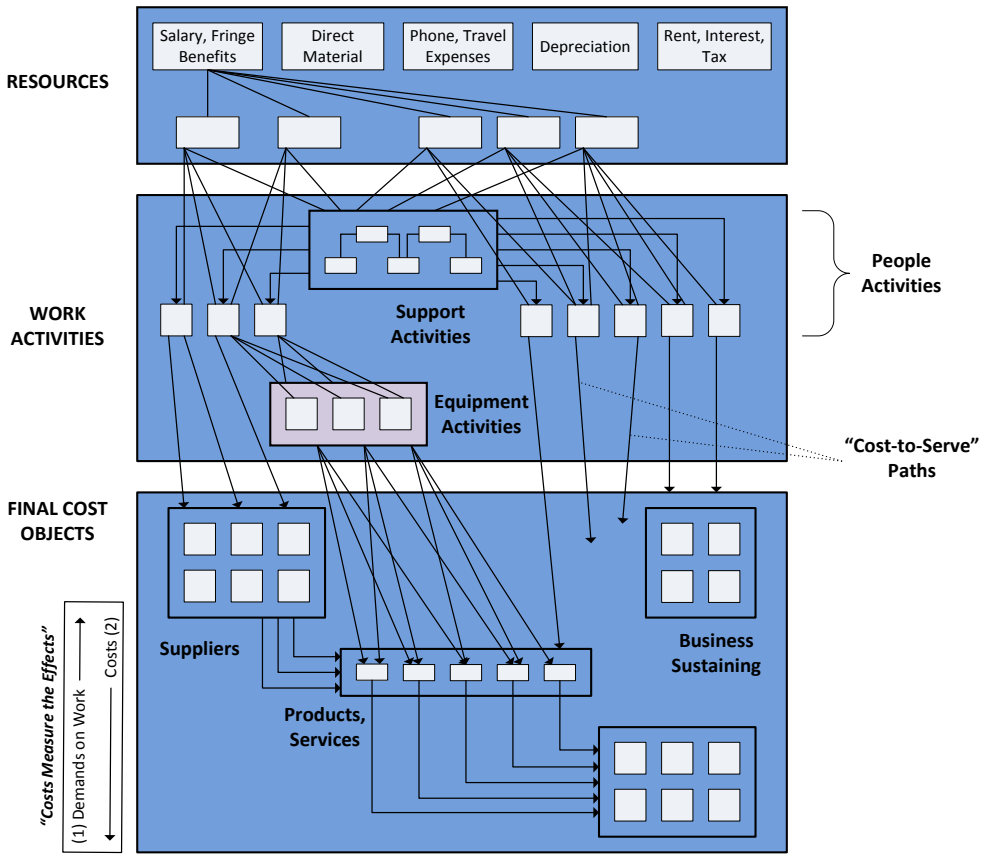
outputs or people, place demands on work, and the costs then measure the effect, by reflecting backward through the ABC/M cost assignment network. ABC/M complies with costing's cause-and-effect principle.

ABC/M IS A COST RE-ASSIGNMENT NETWORK

In complex support-intensive organizations, there can be a substantial chain of indirect activities prior to the work activities that eventually trace into the *final cost objects*. These chains result in activity-to-activity assignments, and they rely on *intermediate* activity drivers in the same way that final cost objects rely on activity drivers to re-assign costs into them based on their diversity and variation.

The *direct costing* of indirect expenses is no longer, as it was in the past, an insurmountable problem given the existence of integrated ABC/M software. ABC allows *intermediate* direct costing to a local process, or to an internal customer, or required component that is causing the demand for work. That is ABC cost flow networks no longer have to "hit the wall" from limited spreadsheet software that is restricted by columns-to-rows math. ABC/M software is arterial in design. Eventually, via this expense assignment and tracing network, ABC re-assigns 100 percent of the costs into the final products, service lines, channels, customers, and business sustaining costs. In short, ABC connects customers to the unique resources they consume—and in proportion to their consumption.

Let's review the ABC cost assignment network in Figure 8.3 that consists of the three modules connected by cost assignment paths. This network calculates the cost of cost objects (e.g., outputs, product lines, service lines, or customers). It is basically a snap-shot view of the business conducted during a specific time period.



Source: *Activity Based Cost Management: An Executive Guide; Gary Cokins; Permission to re-print granted.*

Figure 8.3—ABC/M Cost Re-assignment Network

Resources, at the top of the cost assignment network, are the capacity to perform work, because they represent all the available means that work activities can draw on. Resources can be thought of as the organization's checkbook; this is where all the period's expenditure transactions are accumulated into buckets of spending. Examples of resources are salaries, operating supplies, or electrical power. These are the period's cash outlays and amortized cash outlays, such as for depreciation, from a prior period. It is during this step that the applicable resource drivers are developed as the mechanism to convey resource expenses to the work

activities. In sum, resources are traced to work activities. It is during this step that the applicable resource drivers are developed as the mechanism to convey resource expenses into the activity costs.

A popular basis for tracing or assigning resource expenses is the time (e.g., number of minutes) that people or equipment expend when performing activities. Note that the terms *tracing* or *assigning* are preferable to the term *allocation.* This is because many people associate *allocation* with a redistribution of costs that have little to no correlation between source and destinations; hence to some organizations, indirect expense allocations are believed to be arbitrary and are viewed cynically.

The activity module is where work is performed and where resources are converted into some type of output. The activity cost assignment step contains the structure to assign activity costs to cost objects (or to other activities), using activity drivers as the mechanism to accomplish this assignment.

Cost objects, at the bottom of the cost assignment network, represent the broad variety of outputs and services where costs accumulate. The customers are *the final-final cost objects*; their existence ultimately creates the need for an expense and cost structure in the first place. Cost objects are the persons or things that benefit from incurring work activities. Examples of cost objects are products, service lines, distribution channels, customers, and outputs of internal processes. Cost objects can be thought of as what or for whom work is done.

Once established, the cost assignment network is useful in determining how the diversity and variation of things, such as different products or various types of customers, can be detected and translated into how they uniquely consume activity costs.

USING THE ATTRIBUTES OF ACTIVITY-BASED COSTING

One role for calculating costs is to help suppliers and service-providers identify which of their organization's work activities are:

- Not required at all and can be eliminated (e.g., a duplication of effort).
- Ineffectively accomplished and can be reduced or redesigned (e.g., as a result of outdated policies or procedures).
- Required to sustain the organization (i.e., the work is not directly caused by making a product or delivering services through channels to customers), and therefore it may not be possible to reduce or eliminate the work activity (e.g., provide plant security, compliance with government regulations, etc.).
- Discretionary and can potentially be eliminated (e.g., the annual employees' picnic).

Activity Based Cost Management (ABC/M) systems provide for distinguishing these work activities, either by including them in a cost assignment structure (i.e., sustaining cost objects) or, by tagging their costs as an overlay (i.e., attributes). Attributes are popular with the lean management and Six Sigma quality management community who identify waste and inefficiencies.

Organizations have very little insight about how their individual costs—whether in products, customers, or business processes—vary, aside from the amount of the cost. Traditional cost accounting methods do not provide any way for individual costs to be tagged or highlighted with a separate dimension of cost, other than the amount that was spent. An example of a range of a tag that can be scored for activities is: very important versus required versus postponable. These are popular ways of measuring how much value-added costs exist and where they are located. What this introduces is visibility to the colors of money.

In short, traditional accounting simply provides racked-and-stacked numbers; aside from the cost amount or emphasis in the appearance of the numbers, one cannot differentiate one cost from another. This is true whether one is examining resource expenditures or their calculated costs of activities, processes, and final cost objects (i.e., workflow outputs, products, or customers). Attributes solve this money-dimension-only limitation of traditional costing. One can think of attributes as offering many other dimensions to segment costs that are different from absorption costing's single dimension, which only reflects variation and diversity consumption of cost objects, like outputs, products, service lines, and customers. Attributes can be used as a grading method to evaluate the individual activities that contribute to a process output's goods or services. ABC/M attributes allow managers to differentiate activities from one another, even if they are equal in amount.

Advanced, mature users are masters at employing ABC/M attributes. A popular attribute involves scoring activities along the high-versus-low-value-adding scale. The idea is to eliminate low-value-adding activities and optimize higher-value-adding activities, thus enabling employees to focus on the worth of their organization's work. Employees can see how work really serves customers and which activities may be considered wasteful. Focus and visibility are enhanced because people can more easily see where costs are big or small, and also which costs can be managed in the future. Scoring costs with attributes invokes action beyond just gazing at and analyzing costs.

In the early days of ABC/M, the scoring choices for value-adding were limited to either Value-Added (VA) or Non-Value-Added (NVA). This either-or choice created problems. First, it was considered a personal insult to employees to tell them that part or all of what they do is non-value-adding. Employees are not real happy to hear that. But, even more restrictive is the ambiguity of scoring value that can lead to unsolvable debates. For example, take the activity expedite order to prevent a late shipment to an important customer. Is this VA or NVA work? A solid argument can support either case. It is better to simply discard the VA versus NVA dichotomy with a different set of words that scale along a continuum and better describe levels of importance (e.g., critical, necessary, regulatory, or postponable.)

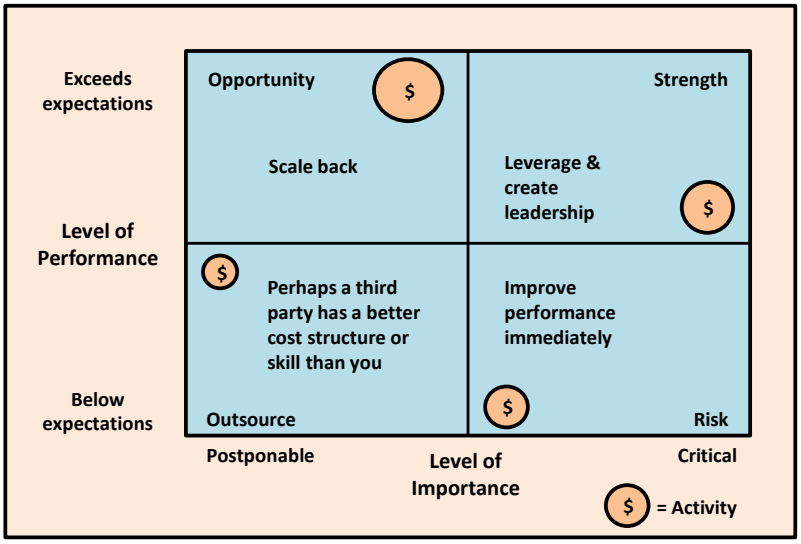
Regardless of what type of attribute scale you use to score or grade value, the objective is to determine the relation of work, or its output, to meeting customer and shareholder requirements. The goal is to optimize those activities that add value and minimize or eliminate those that do not. Following are some tips, but by no means hard rules, for classifying value attributes. High-value-adding activities are those:

- Required to meet customer requirements.
- That modifies or enhances purchased material of a product.
- That, if more of them are accomplished, the customer might pay more for the product or service.
- That are critical steps that cannot be eliminated in a business process.
- That are performed to resolve or eliminate quality problems.
- That are performed as a result of a request or expectation of a satisfied customer.
- That, in general, if time permitted, you would do more.

Low-value-adding activities are those that:

- Can be eliminated without affecting the form, fit, or function of the product.
- Begin with the prefix re- (such as rework or returned goods).
- Result in waste and add no value to the product or service.
- Are performed as a result of inefficiencies or errors in the process stream.
- Are duplicated in another department or add unnecessary steps to the business process.
- Are performed to monitor quality problems.
- Are performed as a result of a request of an unhappy or dissatisfied customer.
- Produce an unnecessary or unwanted output.
- If given the option, you would prefer to do less.

Another popular attribute scores how well each activity is performed, such as exceeds customer expectation, meets, or is below. Multiple activities can be simultaneously tagged with these grades from two or more different attributes. As an option, activities can be summarized into the processes the activities belong to. Using two different attributes along the process view, organizations can see, for example, that they are spending a lot of money doing things they are good at, but that they have judged to be unimportant. In this example, it is obvious the organization should scale back and spend less on that kind of work. Figure 8.4 illustrates the four quadrants that result from combining the two attributes for performance (vertical axis) and importance (horizontal axis). The activity costs for such unimportant activities would be in the upper-left quadrant.



Source: *Activity Based Cost Management: An Executive Guide; Gary Cokins; Permission to re-print granted.*

Figure 8.4—ABC/M's Attributes Can Suggest Action

Although most attributes are subjectively scored or graded by managers and employees, when the attributes' targeted activities or cost objects are grouped together, any subjectivity begins to become directionally reliable (assuming there was no bias in the scoring of every single attribute). As a result, the attributed costs introduce emotionally compelling business issues, like the example above. Attributes make ABC/M data come alive to some people. When the attributed ABC/M data are exported into Online Analytical

Processing (OLAP) Software, Business Intelligence (BI), and Executive Information System (EIS) Tools, they can have a very stimulating impact on users.

LOCAL VERSUS ENTERPRISE-WIDE ABC/M

A common misconception is that the scope of an ABC/M system must be enterprise-wide. That is, the expenses included in the system must account for *all* the employees in the organization and 100 percent of a time period's expenditures. (Or alternatively, the expenses must include *all* the people in a substantial portion of the organization, such as a factory or service-delivery arm.) People with this misconception have usually been exposed only to ABC/M models or systems that are used for calculating the total costs of a product or service line used to determine their total and individual profitability.

ABC/M can also be applied to subsets of the organization for process improvement, rather than revenue enhancement and profit margin increases. An example of a subset is an order-processing center or equipment maintenance function. These ABC/M models and systems are designed to reveal the cost structure to the participants in the main department and related areas. In ABC/M's cost assignment view, the cost structure is seen from the orientation of how the diversity and variation of the function's outputs cause various work to happen, and how much. The costs of the work activities that belong to the processes are also revealed in the ABC/M model, that relate in time and sequence. However, it is ABC/M's powerful revealing of the costs of various types of outputs that serves as a great stimulant to spark discussion and discovery. For example, if an order-processing center learns that the cost per each adjusted order is roughly eight times more costly than for each error-free or adjustment-free entered order that would get attention. This result happens even if the order entry process has been meticulously diagrammed, flowcharted, and documented.

Commercial ABC/M software now enables consolidating some, and usually all, of the local, children ABC/M models into the enterprise-wide, parent ABC/M model. The local ABC/M model data are used for tactical purposes, often to improve productivity. In contrast, the consolidated enterprise-wide ABC/M model is often used for strategic purposes because it helps focus on where to look for problems and opportunities. Also, enterprise-wide models are popular for calculating profit margin data at all levels, including channel-related and customer- and service-recipient-related profit contribution layers.

Table 8.1 illustrates how the unit costs of the output of work can be made visible for a government's highway maintenance department. The benchmarking of relative data can be more powerful than process flow charts in stimulating discussion about what to change.

Table 8.1—An Example of Unitized Costs

Type of Roadbed Costs						
# of Lanes	**Road Surface**	**Location**	**Total Cost**	**# of Miles**	**Work Activity**	**Unit cost/mile**
Four	Asphalt	Interstate	$270,137,078.40	125,342		$2,155.20
					Cut grass	$120.00
					Electronic signs	$334.25
					Fill potholes	$150.00
					Plow roads	$975.00
					Paint stripes	$450.00
					Replace signs	$124.85
Two	Bituminous	Rural	$29,783,384.10	43,578		$683.45
					Cut grass	$220.00
					Electronic signs	$0.00
					Fill potholes	$65.00
					Plow roads	$250.00
					Paint stripes	$112.20
					Replace signs	$36.25
Four	Asphalt	County	$95,567,207.84	65,672		$1455.22
					Cut grass	$180.00
					Electronic signs	$203.00
					Fill potholes	$110.00
					Plow roads	$658.40
					Paint stripes	$227.82
					Replace signs	$76.00

Source: *Activity Based Cost Management: An Executive Guide; Gary Cokins; Permission to re-print granted.*

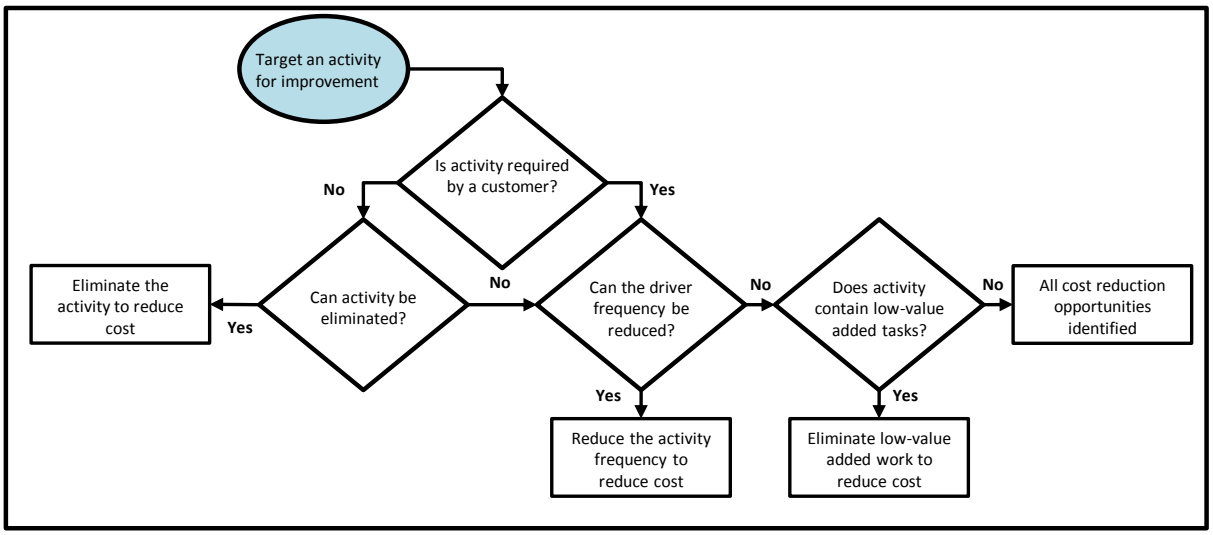
This approach places intra-ABC/M models within an enterprise ABC/M model. A large parent ABC/M model is simply subdivided into its component children ABC/M models. Commercial ABC/M software accommodates consolidations of children into parent ABC/M models. The costs and information are unaffected regardless of which ABC/M models you work with.

APPLICATIONS OF LOCAL ABC/M

Examples of locally applied ABC/M data, such as that for the purchasing process, are limitless. Whenever you have people and equipment doing work where the outputs have diversity, a local ABC/M model can be constructed. The objective of local ABC/M models is not to calculate the profit margins of products, service lines, and customers; it is to compute the diverse costs of outputs to better understand how they create the organization's cost structure.

An interesting application is when a marketing, recruiting, or promotion department has employees who are trying to generate new or continuing inbound orders. They may be trying multiple channels, such as newspapers, radio, television, tradeshows, websites, billboards, and so forth. The costs for advertising placements are different and so might be the results in terms of success (including any additional differences in the type of sale, recruit, or sale). This is an ideal case for an ABC/M calculation to determine the costs versus benefits of all the channel combinations to rank order which are more or less the best return on spending.

In addition to analyzing the impact of diverse cost objects, there is also the traditional activity analysis and cost driver analysis. Figure 8.6 reveals the link between an activity driver and work activity. In a simple fashion, it describes how each work activity can be judged, based on the need of the product or customer, its efficiency, and its value content.



Source: *Activity Based Cost Management: An Executive Guide; Gary Cokins; Permission to re-print granted.*

Figure 8.5—Activity Analysis

Some managers believe that the only way to truly cut costs is to remove the work activity altogether. Their reasoning is that to try to cut back on costs is rarely effective. They believe there is little point in trying to do cheaply what should not be done at all. That is, a job not worth doing is not worth doing well.

Regardless of how one attacks achieving improvements, the main message here is that work is central to ABC/M. What do we do? How much do we do it? Who do we do it for? How important is it? Are we good at doing it?

Some refer to the application of local models as Activity Based Management (ABM), an earlier generation term for ABC/M, because the uses of the ABC/M data are more operational than strategic. Instead, view the local ABC/M models using an analogy of a musical symphony orchestra conductor in rehearsal—first working the violins, then the trumpets, followed by all the string instruments, then all the brass instruments—and finally the entire orchestra in a live concert. The combined orchestra represents a consolidated parent ABC/M model, with local models rolled up into a parent model, then performing as a repeatable and reliable system.

When ABC/M is applied at all organizational levels—local departments, processes, enterprise-wide, or across the supply chain—it provokes intelligent actions and supports better decisions.

If ABC/M is the Answer, What is the Question?

In addition to the need to address the distortion of true costs that are flawed and misreported by traditional standard cost systems, the rise in ABC/M has resulted from external factors. The level of competition that most firms face has increased dramatically. In the

past, most organizations were reasonably profitable. They could make mistakes, and adequate profitability would mask the impact of wrong or poor decisions. But competition has intensified, especially with impact of globalization. A company can no longer carry unprofitable products and service lines, and unprofitable customers, hoping the profitable ones will more than offset and make up the difference. They can no longer survive with misleading cost allocations and without having visibility of their costs across their end-to-end business processes.

Today, the margin for decision errors is slimmer. Businesses cannot make as many mistakes as they could in the past and remain competitive or effective. Price quotations, capital investment decisions, product mix, technology choices, outsourcing, and make-versus-buy decisions today all require a sharper pencil. More competitors are better understanding the cause-and-effect connections that drive costs, and they are fine-tuning their processes, managing their Costs of Quality (COQ), and adjusting their prices accordingly. The resulting price squeeze from more intense competition is making life for businesses much more difficult. Budget tightening is similarly affecting government and not-for-profit organizations. Knowing what your real costs are for outputs, product costs, and the costs-to-serve channels and customers are becoming a key to survival. With activity-based costing visibility, organizations can identify where to remove waste, low-value-adding costs, and unused capacity, as well as understanding what drives their costs. They can also see the degree of alignment of their cost structure with their organization's mission and strategy.

Today, an organization's road is no longer straight and lengthy. It is windy, with bends and hills that do not give much visibility or certainty to plan for the future. Organizations need to be agile and continuously transform their cost structure and work activities. This is difficult to do when an organization does not understand its own cost structure and its economics.

ABC/M IN ADVANCED, MATURE USERS

The advanced and mature ABC/M users of ABC/M, such as the Coca-Cola Company, are interested in two goals: to institutionalize ABC/M enterprise-wide into a permanent, repeatable, and reliable production reporting system and to establish the ABC/M output data to serve as an enabler to their ongoing improvement programs, such as lean management and Total Quality Management (TQM) that include change management, cycle-time compression, core competency, business process re-engineering (BPR), product rationalization, target costing, and channel/customer profitability.

More recently, new issues for the advanced and mature ABC/M users are emerging; they include the following:

- Integrating the ABC/M output data with their decision-support systems, such as their cost estimating, predictive planning, budgeting, driver-based rolling financial forecasts, Customer Relationship Management (CRM), and balanced scorecard and dashboards performance measurement systems.
- Learning the skills and rules for re-sizing, re-shaping, re-leveling, and otherwise readjusting their ABC/M system's structure in response to solving new business problems with the ABC/M data.
- Collecting and automatically importing data into the ABC/M system.
- Automatically exporting the calculated data out of their ABC/M system.

For experienced ABC/M users, ABC/M eventually becomes part of their core information technologies.

More specifically, the output data of an ABC/M system is frequently the input to another system, such as a customer order quotation system. ABC/M data also complement other productivity or logistics management tools—such as simulation software, process modelers, business process flow charters, Executive Information Systems (EIS), Business Intelligence (BI), and Online Analytical Programs (OLAP). In the future, we will see a convergence of these tools that to many organizations are separate and disparate software applications, but now can be integrated to become part of the manager's and analyst's tool suite.

CONCLUSION

Expenses are different from costs. Expenses occur at the point of acquisition with third parties, including employee wages. This is when money (or its obligation) exits the company. From the expenses, all costs are *calculated* representations of how those expenses flow through work activities and into outputs of work. ABC/M focuses on work activities. ABC/M is work-centric, whereas the general ledger is transaction-centric. The general ledger describes what was spent, whereas ABC/M describes what is spent for. Using ABC/M cost assignment network, all final cost objectives (such as work packages, facilities, services, or products) are clearly linked to the resources expended through the related work activities. ABC/M makes the overall cost structure transparent and improve cost effectiveness by reducing or eliminating low or non-value added activities.

RECOMMENDED READING

1. AACE International (2010). Chapter 8, *Skills & Knowledge of Cost Engineering* (5th ed.). Morgantown, WV: AACE International.
2. Cokins, G. (2001). *Activity-Based Cost Management: An Executive's Guide*. New York, NY: Wiley & Sons.

SECTION 2 – COST ESTIMATING

Chapter 9: Cost Estimating
Chapter 10: Process Product Manufacturing
Chapter 11: Discrete Part Manufacturing

Chapter 9—Cost Estimating

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Abstract
Cost estimating is of integral importance to the quality of the cost and scheduling program on any project. Of paramount importance to the success of a project is the integrity of the cost estimate; and this is ensured by the use of the appropriate cost estimating methodology. The cost estimate becomes the basis for setting up the cost budget, resources and the ensuing progress and schedule monitoring processes during project execution. To achieve this, a cost estimating basis/framework is necessary.

The cost estimating framework that is developed in this chapter includes:

- A system for the classification of cost estimating.
- The methodologies used in the preparation of cost estimates.
- Establishing estimating accuracy as it relates to the scope definition.
- The application of risk analysis to contingency.

This chapter presents functional examples for the different cost estimating methodologies inclusive of:

- Conceptual Estimating
- Ratio or Factored Estimating
- Parametric Estimating
- Detailed or Deterministic Estimating

This cost estimating chapter also establishes the relevance of the Work Breakdown Structure (WBS) and Resource Breakdown Structure (RBS) to the structure of the estimate and the ensuing budget and schedule. Finally, good estimating practices are stressed such as; the inclusion of the Basis of the Estimate (BOE) document as a pre-requisite in presenting the final estimate.

In conclusion, this chapter provides a solid background in estimating, and establishes that estimates are a vital component to project success to establish project budgets and to provide accurate information to support scheduling, cost monitoring, and progress measurement of a project during execution.

Keywords
Basis of Estimate (BOE), Budget, Classification of Cost Estimates, Conceptual, Contingency, Cost Monitoring, Estimate Accuracy, Estimating Methodologies, Parametric and Deterministic Estimating, Ratio or Factored, Resource Breakdown Structure (RBS), Scheduling, and Work Breakdown Structure (WBS).

INTRODUCTION
Cost estimating is the predictive process used to quantify, cost, and price the resources required by the scope of an investment option, activity, or project. The output of the estimating process, the cost estimate, may be used for many purposes such as:

- Determining the economic feasibility of a project.
- Evaluating between project alternatives.
- Establishing the project budget.
- Providing a basis for project cost and schedule control.

Cost estimating may be used to quantify, cost and price any investment activity, such as building an office building or process power plant, developing a software program, or producing a stage play. The basic estimating steps are the same: Understand the scope of the activity to quantify the resources required, apply costs to the resources, apply pricing adjustments, and organize the output in a structured way that supports decision-making. Additional steps involve the assessment of risk associated with the estimate, and review and validation of the estimate. For the purposes of this chapter, the primary focus will be on estimating as applied to support the creation of capital assets (a building, industrial facility, bridge, highway, etc.); however, the estimating processes described can be applied to any investment activity.

As potential projects are considered, there are many decision points at which to decide whether a specific project should be continued to be developed. Each subsequent decision-making point, during the project life cycle, typically requires cost estimates of increasing accuracy. Estimating is thus an iterative process that is applied in each phase of the project life cycle as the project scope is defined, modified, and refined.

The cost estimate is obviously of paramount importance to the success of a project. The capital cost of a proposed project is one of the key determinants in evaluating the financial viability and business case of the project. From an owner's perspective, if the cost

estimate is not accurate, the financial return from the capital investment may not be realized; and compounding this problem is the fact that other deserving projects may not have been funded. It is obvious that estimating is critical for the economic and optimal use of an owner's limited capital budget.

From a contractor's perspective, accurate estimating is just as important. In a lump-sum bidding situation, the profit margin of the contractor is dependent on the accuracy of his or her estimate. If the project is exceptionally large, the loss from an inaccurate estimate on a lump-sum bid can potentially put a contractor out of business. For cost-plus projects, the contractor will face less direct economic risk from an inaccurate estimate, but the damage to the contractor's reputation can be severe.

The cost estimate, however, serves other purposes besides establishing the budget for a project. It also serves as a tool or resource used for both scheduling and cost control of projects. The estimate not only establishes a project budget, but plays an equally important role in providing the information required to monitor the budget during project execution. It is the relationship between estimating, scheduling, and cost control, which is typically identified by the term "cost engineering" that serves as a driver for successful and cost-effective projects. Thus, an effective estimate must not only establish a realistic budget, but must also provide accurate information to allow for scheduling, cost monitoring, and progress measurement of a project during execution.

Learning Objectives

Cost estimating is one of the cornerstones of cost engineering and total cost management. The objective of this chapter is to introduce the reader to the various classifications of cost estimates, and the estimating methodologies and procedures used to prepare cost estimates. After completing this chapter, the reader should be able to:

- Understand the classification of cost estimates.
- Understand some of the common methodologies used in preparing cost estimates.
- Relate estimate accuracy to the level of scope information and methodologies used in preparing cost estimates.
- Understand how to apply risk analysis to determine contingency in an estimate.
- Understand how to present and review estimates.
- Apply the knowledge gained to specific project estimating situations.

Estimate Classifications

Estimate classifications are commonly used to indicate the overall maturity and quality for the various types of estimates that may be prepared; and most organizations will use some form of classification system to identify and categorize the various types of project estimates that they may prepare during the lifecycle of a project. Unfortunately, there is often a lack of consistency and understanding of the terminology used to classify estimates, both across industries as well as within single companies or organizations.

AACE International developed Recommended Practice 17R-97, *Recommended Practice for Cost Estimate Classification*, to provide generic guidelines for the general principles of estimate classification that may be applied across a wide variety of industries [1]. This document has been developed to:

- Provide a common understanding of the concepts involved in classifying project cost estimates.
- Fully define and correlate the major characteristics used in classifying cost estimates, so that different organizations may clearly determine how their particular practices compare to the AACE guidelines.
- Use degree of project definition as the primary characteristic in categorizing estimate classes.
- Reflect generally accepted practices in the cost engineering profession.

AACE RP 17R-97 maps the phases and stages of project estimating with a maturity and quality matrix; providing a common reference point to describe and differentiate various types of cost estimates [1]. The matrix defines the specific input information (i.e., design and project deliverables) that is required to produce the desired estimating quality at each phase of the estimating process. The matrix defines the requirements for scope definition and indicates estimating methodologies appropriate for each class of estimate.

AACE identifies five classes of estimates, which it simply designates as Class 1, 2, 3, 4, and 5. A Class 5 estimate is associated with the lowest level of project definition (or project maturity), and a Class 1 estimate is associated with the highest level of project definition. For each class of estimate, five characteristics are used to distinguish one class of estimate from another. The five characteristics used in the AACE Recommended Practice are:

- degree of project definition
- end usage of the estimate
- estimating methodology
- estimating accuracy
- effort required to produce the estimate

The level of project definition is the primary (or driving) characteristic used to identify an estimate class. The other characteristics are "secondary," with their value typically determined by the level of project definition. Figure 9.1 shows the generic *AACE Cost Estimate Classification Matrix*.

	Primary Characteristic	***Secondary Characteristic***			
ESTIMATE CLASS	**MATURITY LEVEL OF PROJECT DEFINITION DELIVERABLES** Expressed as % of complete definition	**END USAGE** Typical purpose of estimate	**METHODOLOGY** Typical estimating method	**EXPECTED ACCURACY RANGE** Typical +/- range relative to index of 1 (i.e. Class 1 estimate) [a]	**PREPARATION EFFORT** Typical degree of effort relative to least cost index of 1 [b]
Class 5	0% to 2%	Screening or feasibility	Stochastic (factors and/or models) or judgment	4 to 20	1
Class 4	1% to 15%	Concept study or feasibility	Primarily stochastic	3 to 12	2 to 4
Class 3	10% to 40%	Budget authorization or control	Mixed but primarily stochastic	2 to 6	3 to 10
Class 2	30% to 75%	Control or bid/tender	Primarily deterministic	1 to 3	5 to 20
Class 1	65% to 100%	Check estimate or bid/tender	Deterministic	1	10 to 100

Notes: [a] If the range index value of "1" represents +10/-5%, then an index value of 10 represents +100/-50%.
[b] If the cost index value of "1" represents 0.005% of project costs, then an index value of 100 represents 0.5%.

Figure 9.1—Generic Cost Estimate Classification Matrix [1]

Typically, project owners (the owner organization responsible for the investment of funds in the project) are concerned with Class 5 through Class 3 estimates. Most capital projects base their final funding decisions on estimates associated with Class 3 level of project definition, which should provide a sufficient level of estimate accuracy for the commitment of final funding for the project. Often Class 5 and Class 4 estimates may be prepared by the project owner organization itself; but may also be prepared by their engineering contractors or other 3rd parties. In general, most Class 3 estimates are not prepared by the owner organization because of the resources required for preparation of these more detailed estimates; and these will typically be prepared by their engineering or other contractors. Despite which entity prepares the estimates up to and including the final funding estimate, the owner organization needs to accept final responsibility for these estimates through a thorough review and validation process (including the identification and assessment of cost risks associated with the estimate). In the end, the owner is responsible for project funding.

Class 2 and Class 1 estimates for capital projects are typically prepared to support construction bids or project execution change control (post-authorization changes); and for capital projects. These are most often prepared by construction contractors.

In addition to the generic estimate classification system, a more specific version [AACE RP 18R-97] has been created for the process industries (see Figure 9.2) [2]. The term "process industries" is intended to include firms involved with the manufacturing and production of chemicals, petrochemicals, pulp/paper and hydrocarbon processing. The commonality among this industry (for the purpose of estimate classification) is their reliance on Process Flow Diagrams (PFD's) and Piping and Instrument Diagrams (P&ID's) as primary scope defining documents. These documents are key deliverables in determining the level of project definition, and thus the extent and maturity of estimate input information, and subsequently the estimate class for an estimate for a process industry project.

	Primary Characteristic	*Secondary Characteristic*		
ESTIMATE CLASS	**MATURITY LEVEL OF PROJECT DEFINITION DELIVERABLES** Expressed as % of complete definition	**END USAGE** Typical purpose of estimate	**METHODOLOGY** Typical estimating method	**EXPECTED ACCURACY RANGE** Typical variation in low and high ranges [a]
Class 5	0% to 2%	Concept screening	Capacity factored, parametric models, judgment, or analogy	L: -20% to -50% H: +30% to +100%
Class 4	1% to 15%	Study or feasibility	Equipment factored or parametric models	L: -15% to -30% H: +20% to +50%
Class 3	10% to 40%	Budget authorization or control	Semi-detailed unit costs with assembly level line items	L: -10% to -20% H: +10% to +30%
Class 2	30% to 75%	Control or bid/tender	Detailed unit cost with forced detailed take-off	L: -5% to -15% H: +5% to +20%
Class 1	65% to 100%	Check estimate or bid/tender	Detailed unit cost with detailed take-off	L: -3% to -10% H: +3% to +15%

Notes: [a] The state of process technology, availability of applicable reference cost data, and many other risks affect the range markedly. The +/- value represents typical percentage variation of actual costs from the cost estimate after application of contingency (typically at a 50% level of confidence) for given scope.

Figure 9.2—Cost Estimate Classification Matrix for the Process Industries [2]

This Estimate Classification System for the Process Industries is meant to supplement the generic standard. Additional estimate classification recommended practices have been established for other industries as well (e.g., general construction, software development, etc.).

Included with the supplemental guideline for the process industries [AACE RP 18R-97] is a chart which maps the maturity of estimate input information (project definition deliverables) against the classes of estimates (see Figure 9.3) [2].

ESTIMATE CLASSIFICATION					
General Project Data:	**CLASS 5**	**CLASS 4**	**CLASS 3**	**CLASS 2**	**CLASS 1**
Project Scope Description	General	Preliminary	Defined	Defined	Defined
Plant Production/Facility Capacity	Assumed	Preliminary	Defined	Defined	Defined
Plant Location	General	Approximate	Specific	Specific	Specific
Soils and Hydrology	None	Preliminary	Defined	Defined	Defined
Integrated Project Plan	None	Preliminary	Defined	Defined	Defined
Project Master Schedule	None	Preliminary	Defined	Defined	Defined
Escalation Strategy	None	Preliminary	Defined	Defined	Defined
Work Breakdown Structure	None	Preliminary	Defined	Defined	Defined
Project Code of Accounts	None	Preliminary	Defined	Defined	Defined
Contracting Strategy	Assumed	Assumed	Preliminary	Defined	Defined
Engineering Deliverables:					
Block Flow Diagrams	S/P	P/C	C	C	C
Plot Plans		S	P/C	C	C
Process Flow Diagrams (PFDs)		S/P	P/C	C	C
Utility Flow Diagrams (UFDs)		S/P	P/C	C	C
Piping & Instrument Diagrams (P&IDs)		S	P/C	C	C
Heat & Material Balances		S	P/C	C	C
Process Equipment List		S/P	P/C	C	C
Utility Equipment List		S/P	P/C	C	C
Electric One-Line Drawings		S/P	P/C	C	C
Specifications & Datasheets		S	P/C	C	C
General Equipment Arrangement Drawings		S	P/C	C	C
Spare Parts Listings			S/P	P	C
Mechanical Discipline Drawings			S	P	P/C
Electric Discipline Drawings			S	P	P/C
Instrumentation/Control System Discipline Drawings			S	P	P/C
Civil/Structural/Site Discipline Drawings			S	P	P/C

Figure 9.3—Estimate Input Checklist and Maturity Matrix for the Process Industries

This is a checklist of basic deliverables found to be in common practice in the process industries. The maturity level is an approximation of the degree of completion of the deliverable. The degree of deliverable is indicated by the following letters:

- None (blank): Development of the deliverable has not yet begun.
- Started (S): Work on the deliverable has begun. Development is typically limited to sketches, rough outlines, or similar levels of early completion.
- Preliminary (P): Work on the deliverable is advanced. Interim cross-functional reviews have usually been conducted. Development may be near completion except for final reviews and approvals.
- Complete (C): The deliverable has been reviewed and approved as appropriate.

Note: It is only the level of project definition that defines the class of the estimate. Although typically the other characteristics associated with the estimate classification matrices are correlated with the level of project definition, this is not always the case and the other characteristics do not define the class of estimate. As an example, it is possible to prepare a Class 5 (conceptual) estimate on a very repetitive project for which the scope may be well understood (although not defined by technical deliverables), and for which very good reference cost data exists that may have a higher level of estimate accuracy than a Class 3 estimate for a project involving new technology in a new geographical location for which adequate reference cost information is not available.

Estimating Methodologies

In general, estimating methodologies commonly fall into two broad categories: conceptual and deterministic. As can be seen from the Cost Estimate Classification Matrices (see Figures 9.1 and 9.2), as the level of project definition increases, the estimating methodology tends to progress from conceptual (stochastic or factored) methods to deterministic methods.

With conceptual estimating methods, the independent variables used in the estimating algorithm are generally something other than a direct measure of the units of the item being measured. They usually involve simple or complex modeling (or factoring), based on inferred or statistical relationships between costs and other, typically design related, parameters. Often, the cost estimating relationships used in conceptual estimating methods are at least somewhat subject to conjecture.

For deterministic estimating methods, the independent variables used in the estimating algorithm are more or less a direct measure of the item being estimated, such as straightforward counts or measures of items multiplied by known unit costs. Deterministic estimating methods require a high degree of precision in the determination of quantities, pricing, and the completeness of scope definition. Of course, any particular estimate may involve a combination of conceptual and deterministic methods.

There is another key difference between conceptual and deterministic estimating methods. Conceptual estimating methods require significant effort in data-gathering and methods development before estimate preparation ever begins. There is a significant effort in historical cost analysis to develop accurate factors and estimating algorithms to support conceptual estimating. Preparing the conceptual estimate itself takes relatively little time, sometimes less than an hour.

In contrast, a deterministic (or detailed) estimate requires a large effort during the actual preparation of the estimate. The evaluation and quantification of the project scope can take a substantial amount of time, sometimes weeks or even months for extremely large projects. Research and application of accurate detailed pricing information, and application of specific estimating adjustments to the quantified scope can also take considerable time.

The estimating method used for any particular estimate will depend on many factors: the end use of the estimate, the amount of time and money that is available to prepare the estimate, the estimating tools and data available, and of course the level of project definition and design information on hand.

Conceptual Estimating Methodologies

Conceptual estimating methods are typically used for Class 5 and Class 4 (and sometimes for Class 3) estimates. They are often referred to as "order-of-magnitude" (OOM) estimates in reference to their typically wide range of estimate accuracy (as previously defined in the estimate classification matrices). They provide a relatively quick method of determining the approximate probable cost of a project without the benefit of detailed scope definition. As indicated in the estimate classification matrices, these estimates may be used for:

- Establishing an early screening estimate for a proposed project or program.
- Evaluating the general feasibility of a project.
- Screening project alternatives (such as different locations, technologies, capacities, etc.).
- Evaluating the cost impacts of design alternatives.
- Establishing a preliminary budget for control purposes during the design phase of a project.

Conceptual estimates are generally based on little project definition (i.e., engineering deliverables), thus typically subjecting them to a wide range of estimate accuracy. Their accuracy can depend on several factors—including the level of project definition, the quality of the past historical cost data used in development of the factors and algorithms, as well as the judgment and experience of the estimator. These limitations should, of course, be recognized in using conceptual estimating methods. Nonetheless, there are many cases where conceptual estimates can be very reliable, especially in estimating repetitive projects. Generally, the emphasis with conceptual estimating is not on detailed accuracy, but on obtaining a reasonable cost estimate of sufficient accuracy to insure that the results are meaningful for management to make the decision at hand. Early, conceptual estimates (Class 5 or Class 4 estimates) are typically used to determine whether the project may meet project financial thresholds, and thus support investment in additional design and engineering development to support an estimate that can be used to establish final project funding (typically a Class 3 estimate).

There are a wide variety of conceptual or OOM estimating methodologies. Several of the more commonly used methods are: end-product units, physical dimensions, capacity factor, various ratio or factor methods, and parametric modeling. Most conceptual estimating methods rely on relationships of one form or another.

End-Product Units Method

This conceptual estimating method is used when the estimator has enough historical data available from similar projects to relate the end-product units (capacity units) of a project to its construction costs. This allows an estimate to be prepared relatively quickly, knowing only the end-product unit capacity of the proposed project. Examples of the relationship between construction costs and end-product units are:

- The construction cost of an electric generating plant and the plant's capacity in kilowatts.
- The construction cost of a hotel and the number of guest rooms.
- The construction cost of a hospital and the number of patient beds.
- The construction cost of a parking garage and the number of available parking spaces.

To illustrate, consider a client that is contemplating building a 1,500 luxury hotel in a resort area. The client needs an approximate cost estimate for the proposed hotel as part of the feasibility study. Assume that a similar luxury hotel has been recently completed at a nearby resort and the following information is available.

The hotel just completed included 1000 guest rooms, as well as a lobby, restaurants, meeting rooms, parking garage, swimming pool, and nightclub. The total construction cost for the 1,000 room hotel was $67,500,000. The resulting cost per room is thus calculated as $67,500,000/1,000 = $67,500 per room.

Therefore, we can use this information to determine the cost of the 1500 room hotel, of comparable design and a nearby location, as $101,250,000 ($67,500/Room X 1500 Rooms). While this cost estimate may serve to meet the needs of the feasibility study, it has ignored several factors that may impact costs.

For example, it has ignored any economies of scale that may be generated from constructing a larger hotel; and has assumed that the cost of the common facilities (lobby, restaurants, pool, etc.) vary directly with the increase in the number of rooms. It is important to identify and assess the cost impact of these differences, and make the appropriate adjustments to the initial cost estimate. Similarly, if the location or timing of the proposed hotel had differed significantly from the known cost data point, then cost indices can be used to adjust for these differences.

Physical Dimensions Method

Somewhat similar to the end-products units method is the physical dimensions estimating methodology. The method uses the physical dimensions (length, area, volume, etc.) of the item being estimated as the driving factor. For example, a building estimate may be based on square feet/meters or cubic volume of the building; whereas pipelines, roadways, or railroads may be based on a linear basis.

As with the end-product units method, this estimating methodology also depends on historical information from comparable facilities. Consider the need to estimate the cost of a 3,600 SM warehouse. A recently completed warehouse of 2,900 SM in a nearby location was recently completed for $623,500, thus costing $215/SM. The completed warehouse used a 4.25M wall height, thus containing 12,325 CM, and resulting in a cost of $50.59/CM on a volume basis ($623,500/12,325 CM).

In determining the cost for the new warehouse, we can estimate the new 3,600 SM warehouse using the SM basis at $774,000 ($215/SM X 3,600SM). However, the new warehouse will differ from the one just completed by having 5.5M high walls; so we may decide that estimating on a volume basis may provide a better indication of costs. The volume of the new warehouse will be 19,800 CM (3,600 SM X 5.5M), and the new estimate will be $1,002,000 (rounded to the nearest $1000).

Again, we have ignored the cost impact of economies of scale in developing the estimate, and any other differences in quality between the two warehouses. If additional information is available to identify and assess these differences, then further adjustments should be made to the cost estimate. If location or timing differences had existed, we would also account for those cost impacts by using cost indices or other adjustments.

Capacity Factor Method

A capacity factored estimate is one in which the cost of a new facility is derived from the cost of a similar facility of a known (but usually different) capacity. It relies on the (typical) non-linear relationship between capacity and cost. In other words, the ratio of costs between two similar facilities of different capacities equals the ratio of the capacities multiplied by an exponent:

$$\$B/\$A = (Cap_B/Cap_A)^e$$

Where: $A and $B are the costs of the two similar facilities; and Cap_A and Cap_B are the capacities of the two facilities. This is shown in Figure 9.4.

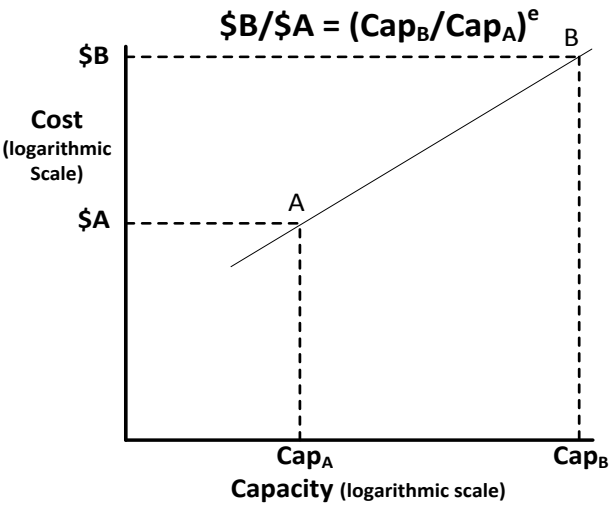


Figure 9.4—Capacity Factor Relationship

If we rewrite this equation to use as an estimating algorithm, it becomes:

$\$B = (\$A)(Cap_B/Cap_A)^e$

Where: $B is the cost of the facility being estimated, $A is the known cost of a similar facility, Cap_B is the capacity of the facility being estimated, Cap_A is the capacity of the similar facility, and "e" is the exponent or proration factor.

The exponent "e" typically lies between 0.5 and 0.85 depending on the type of facility, and must be analyzed carefully for its applicability to each estimating situation. The exponent "e" used in the capacity factor equation is actually the slope of the curve that has been drawn to reflect the change in the cost of a facility, as it is made larger or smaller (see Figure 9.4). These curves are typically drawn from the data points of the known costs of completed facilities. The slope will usually appear as a straight line when drawn on log-log paper. With an exponent value less than 1, scales of economy are achieved such that as facility capacity increases by a percentage (say, 20 percent), the costs to build the larger facility increase by less than 20 percent.

The methodology of using capacity factors is sometimes referred to as the "scale of operations" method, or the "six tenth's factor" method, because of the common reliance on an exponent value of 0.6 if no other information is available. With an exponent of .6, doubling the capacity of a facility increases costs by approximately 50 percent, and tripling the capacity of a facility increases costs by approximately 100 percent.

It is also important to realize that although the data, when plotted on a log-log graph, will usually appear as a straight line over a small range of capacity values; it is probably not constant over the entire range of possible capacities or facility sizes. In reality, as facility capacities increase, the exponent tends to increase as illustrated in Figure 9.5. As an example, between the capacities A and B (in Figure 9.5), the capacity factor exponent may have a value of 0.6; however between the capacities B and C, the exponent has a value of 0.65. Between the capacities C and D, the value of the exponent may have risen to 0.72. Eventually, as the facility capacity increases to the limits of existing technology, the exponent tends toward a value of 1. At this point, it becomes more economical to build two facilities of a smaller size than one large facility. In other words, cost becomes a linear function of capacity, and scales of economy are no longer obtained.

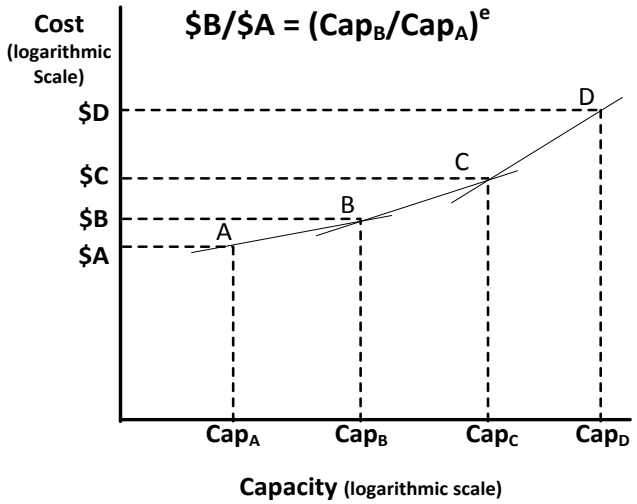


Figure 9.5—Capacity Factor Exponents are not Constant Across all Capacity Ranges

Capacity factored estimating can be quite accurate. If the capacity factor used in the estimating algorithm is relatively close to the actual value, and if the facility being estimated is relatively close in size to the similar facility of known cost, then the potential error from capacity factoring is quite small, and is certainly well within the level of accuracy that would be expected from such a conceptual estimating method. For example, if the new facility is triple the size of an existing facility, and the actual capacity factor is 0.80, instead of an assumed 0.70, you will have underestimated the cost of the new facility by only 10 percent, calculated as $(3^{.8} – 3^{.7})/3^{.8}$. Similarly, for the same threefold scale-up in facility size, but the actual capacity factor should be .60 instead of an assumed 0.70, you will have overestimated the facility cost by only 12 percent, calculated as $(3^{.7} – 3^{.6})/3^{.6}$.

Thus, if the facility size being estimated is reasonably close to the size of the known facility, and a realistic capacity factor exponent is used, error from the capacity factoring algorithm is small. However, this error can be compounded by other assumptions we must make in an actual estimating situation. Typically, we must also adjust for differences in physical scope, location and time between the estimated facility and the known facility. Each of these adjustments can also add to the level of error in the overall estimate.

Let's examine a typical situation where we need to estimate the costs of a 100,000 BBL/Day Hydrogen Peroxide unit, to be built in Philadelphia and completed in 2004. We have recently completed a 150,000 BBL/Day plant in Malaysia, with a final cost of $50 million in 2002. Our recent history shows a capacity factor of 0.75 is appropriate. The simple approach is to just use our capacity factor algorithm:

$\$B = (\$A)(Cap_B/Cap_A)^e$
$\$B = \$50M \text{ X } (100/150)^{.75} = \$36.9M$

This would be fine for as far as it goes, but as we have noted in our earlier discussions of estimating methodologies, we have thus far ignored differences in quality (or scope), location, and time.

For this example, let's adjust for the differences in scope, location, and time. The plant in Malaysia included piling, tankage, and owner costs that will not need to be included in the proposed plant for Philadelphia. Construction in Philadelphia is expected to cost 1.25 times the construction costs in Malaysia (location adjustment). Escalation will be included as a 1.06 multiplier from 2002 to 2004 (an obviously simple approach). There are costs for additional pollution requirements in Philadelphia that were not included in the cost of the Malaysian plant. Taking these into account, the estimate now appears like this:

150,000 BBL/Day Plant in Malaysia	$50M
Deduct Piling. Tankage, Owner Costs	***- $10M***
Adjusted Cost for Scope	**= $40M**
Malaysia to Philadelphia Adjustment (X 1.25)	= $50M
Escalate to 2002 (X 1.06)	= $53M
Factor = $53M X $(100/150)^{.75}$	= $39M
Add Pollution Requirements (+$5M)	= $44M

The key steps in preparing a capacity factor estimate are:

- Deduct costs from the known base case that are not applicable in the new plant being estimated.
- Apply location and escalation adjustments to normalize costs. This now determines what the adjusted scope for the base case will cost in the new location and time frame.
- Apply the capacity factor algorithm to adjust for plant size.
- Add any additional costs which are required for the new plant but which were not included in the known plant.

The capacity factor estimating method provides a relatively quick and sufficiently accurate means to prepare early estimates during the concept screening stage of a project. The method requires historical cost and capacity data for similar plants and processes. Although published data on capacity factors exists, the best data would be from your own organization and requires a level of commitment to collect, normalize, and maintain this information. When using this method, the new and existing known plant should be near duplicates, and reasonably close in size. You must account for differences in scope, location, and time. Each of the adjustments that you make adds additional uncertainty and potential error to the estimate. Despite this, capacity factor estimates can be quite accurate, and are often used to support decision-making at the pre-design stage of a project.

Ratio or Factor Methods

Ratio or factored estimating methods are used in situations where the total cost of an item or facility can be reliably estimated from the cost of a primary component. For example, this method is commonly used in estimating the cost of process and chemical plants where the cost of the specialized process equipment makes up a significant portion of the total project cost. This is often referred to as "Equipment Factor" estimating.

Equipment factored estimates are used to develop costs for process and utility units for which the behavior of the costs of the direct labor and bulk materials used to construct the facilities is correlated with the costs (or the design parameters) of the major equipment. Typically, this estimating methodology relies on the principle that a ratio or factor exists between the cost of an equipment item and costs for the associated non-equipment items (foundations, piping, electrical, etc.) needed to complete the installation.

An equipment factored estimate can typically be generated when project definition (engineering complete) is approximately one to 15 percent complete (Class 4). A sufficiently accurate equipment list should be available at this point in the project. This estimate is often a feasibility estimate used to determine whether there is a sufficient business case to pursue the project. If so, then this estimate may be used to justify the funding required to complete the engineering and design required to produce a funding or budget estimate (Class 3).

Depending on the particular factoring techniques and data used, the factors may estimate Total Installed Costs (TIC) or Direct Field Cost (DFC) for the facility. Usually, the factors generate costs only for the Inside Battery Limits (ISBL) facilities, and require the Outside Battery Limit Facilities (OSBL) costs to be estimated separately; however, sometimes appropriate factors are used to estimate the costs of the complete facilities. Therefore, it is extremely important to understand the basis of the particular factors being used in an equipment factored estimate.

In 1947, Hans Lang first published an article in *Chemical Engineering* introducing the concept of using the total cost of equipment to factor the total estimated cost of a plant [3]:

Total Plant $ = Total Equipment $ X Equipment Factor

Lang proposed three separate factors based on the type of process plant (see Table 9.1). Lang's factors were meant to cover all the costs associated with the total installed cost of a plant including the Battery Limits Process Units (ISBL Costs) and all Offsite Units (OSBL Costs).

Table 9.1—Lang Factors

Type of Plant	Factor
Solid Process Plant	3.10
Solid-Fluid Process Plant	3.63
Fluid Process Plant	4.74

An example of a Lang Factor estimate for a fluid process plant:

Total Estimated Equipment Cost = $1.5M
Total Plant Cost = $1.5M X 4.74
Total Plant Cost = $7.11M

Lang's approach was rather simple, using a factor that varied only by the type of process. Since that first publication, many different methods of equipment factoring have been proposed, and some methods have become very sophisticated. The term "Lang Factor," however, is often used generically to refer to all the different types of equipment factors.

In 1958, W.E. Hand elaborated on Lang's work by proposing different factors for each type of equipment (columns, vessels, heat exchangers, etc.) rather than process type [4]. Hand's factors estimated direct field cost, excluding instrumentation. Hand's published equipment factors ranged from 2.0 to 3.5 (which might correlate to approximately 2.4 to 4.3 including instrumentation). Hand's factors excluded Indirect Field Costs (IFC), Home Office Costs (HOC), and the costs for Offsite or Outside Battery Limit (OSBL) facilities. These costs would need to be estimated separately.

An example of an estimate prepared for a fluid processing plant using Hand's equipment factoring techniques appears in Figure 9.6. In this example, the total cost of all equipment items for each type of equipment was multiplied by a factor for that specific type of equipment to derive the DFC for that equipment type. For instance, the total cost of all vertical vessels ($540K) was multiplied by an equipment factor of 3.2 to obtain an installed DFC of $1,728K. The DFC costs for all equipment types totals $7,753K. Direct Field Labor (DFL) was estimated at 25 percent of DFC or $1,938K. The Indirect Field Costs (IFC) were then estimated at 115 percent of the DFL costs, totaling $2,229K. The sum of the DFC and IFC costs comprise the Total Field Costs (TFC) of $9,982K. Home Office Costs (HOC) are factored as 30 percent of DFC, which totals $2,326K. For this estimate, the Project Commissioning Costs were factored as three percent of DFC, and Contingency was factored as 15 percent. The Total Installed Cost (TIC) for this estimate totals $14,422K.

Acct No.	Item Description	Adj. Factor	Costs				Eqmt Mult	% Total
			Labor $	Eqmt $	Eqmt Factor	Total		
51	Columns			650,000	2.1	1,365,000		
52	Vertical vessels			540,000	3.2	1,728,000		
53	Horizontal Vessels			110,000	2.4	264,000		
54	Shell & Tube Heat Exchangers			630,000	2.5	1,575,000		
55	Plate Heat Exchangers			110,000	2.0	220,000		
56	Pumps, Motor Driven			765,000	3.4	2,601,000		
				2,805,000				
	DIRECT FIELD COSTS	**25% of DFC**	**1,938,000**			**7,753,000**	**2.8**	**53.8%**
10	Temporary Construction Facilities							
11	Construction Services/Supplies/Consumables							
12	Field Staff/Subsistence/Expense							
13	Payroll Burdens/Benefits/Insurance							
14	Construction Equipment/Tools							
15	International Expense							
	INDIRECT FIELD COSTS	**115% of DFL**				**2,229,000**		**15.5%**
TOTAL FIELD COSTS						**9,982,000**	**3.6**	**69.2%**
20	Project Management							
21	Project Controls/Estimating							
22	Project Procurement							
23	Project Cosntruction Management							
24	Engineering/Design							
25	Home Office Expenses							
	HOME OFFICE COSTS	**30%of DFC**				**2,326,000**		**16.1%**
TOTAL FIELD and HOME OFFICE COSTS						**12,308,000**	**4.4**	**85.3%**
30	Owner's Costs							
31	Project Commissioning Costs	**3%**	**Of DFC**			233,000		
32	Escalation							
33	Other Non-Assignable Costs							
34	Contingency	**15%**	**Of Above**			1,881,000		
35	Fee							
	OTHER PROJECT COSTS					**2,114,000**		**14.7%**
TOTAL PROJECT COSTS						**$14,422,000**	**5.1**	**100.0%**

Figure 9.6—Equipment Factored Estimate Example

Note the various equipment factors displayed in this example. Total equipment cost to DFC is a factor of 2.8 (a typical range would be 2.4 to 3.5). Total equipment cost to TFC is a factor of 3.6 (with a typical range being 3.0 to 4.2). Total equipment cost to total

project cost, including contingency, is a factor of 5.1 (a typical range would be 4.2 to 5.5). This correlates closely with Lang's original overall equipment factor of 4.74 for fluid plants.

Arthur Miller proposed another enhancement to the concept of equipment factors in 1965 [5]. Miller recognized the impact of three specific variables that affect the equipment material cost to a greater degree than they affect the cost of the associated bulk materials and installation. These three factors are: the size of the major equipment, the materials of construction (metallurgy) of the equipment, and the operating pressure. Miller noted that as the size of a piece of major equipment gets larger, the amount of corresponding bulk materials (foundation, support steel, piping, instruments, etc.) required for installation does not increase at the same rate. Thus, as the equipment increases in size, the value of the equipment factor decreases.

A similar tendency exists for metallurgy and operating pressure. If the equipment is made from more expensive materials (stainless steel, titanium, monel, etc.), the equipment factor will become smaller. If the operating pressure increases, the equipment factor gets smaller. Again, as the equipment becomes more costly because of expensive materials of construction or higher operating pressures, the costs for the associated bulk materials required for installation increase at a lower proportion or rate, and the resulting equipment factor becomes smaller.

Miller suggested that these three variables could be summarized into a single attribute known as the "average unit cost" of equipment. The "average unit cost" of equipment is:

Total Cost of Process Equipment/Number of Equipment Items

If the "average unit cost" of equipment increases, then the equipment factor is scaled smaller. The correlation between increasing average unit cost of equipment and decreasing equipment factors was statistically validated in subsequent studies by Rodl [6] and Nushimura [7].

Thus far, the equipment factors we have discussed have been used to generate all-in DFC or TIC costs. Another method of using equipment factors is to generate separate costs for each of the disciplines associated with the installation of equipment. Using this methodology, each type of equipment is associated with several discipline specific equipment factors. For example, one discipline equipment factor will generate costs for concrete, another factor will generate costs for support structural steel, another generates the cost for piping, etc. An advantage to this approach is that it provides the estimator with the capability to adjust the costs for the individual disciplines based on specific knowledge of the project conditions, and improves the accuracy of the equipment factoring method. It also allows the costs for each specific discipline to be summed, and compared to other similar projects. Miller, and later Guthrie, described this methodology [8].

An example of using discipline specific equipment factors is shown in Figure 9.7. The example shows discipline equipment factors for a 316SS Shell & Tube Heat Exchanger with a size range of 350 – 700 SM. In this example, the equipment cost of $10,000 is multiplied by each of the indicated factors to generate the DFC costs for that discipline. For example, the equipment installation labor is factored as $10,000 X 0.05 = $500; piping material and labor is factored as $10,000 X 1.18 = $11,800; etc. The total DFC costs for installation of this heat exchanger totals $28,600 (including the equipment purchase cost of $10,000). This equates to an overall DFC equipment factor of 2.86. These costs do not include IFC, HOC, or OSBL costs.

Exchanger, Shell & Tube, 316 Stainless Steel, 350-700 SM										
	Eqmt. Cost	Eqmt. Install Labor	Concrete	Structural Steel	Piping	Electrical	Instruments	Painting	Insulation	Total DFC Costs
Factor		0.05	0.11	0.11	1.18	0.05	0.24	0.01	0.11	2.86
Cost	$10,000	$500	$1,100	$1,100	$11,800	$500	$2,400	$100	$1,100	$28,600

Figure 9.7—Discipline Equipment Factor Example

Development of the actual equipment factors to be used in preparing process plant estimates is a tedious and time-consuming affair. Although some published data exists on equipment factors (see the articles included in the references), much of this data is old, and some of the assumptions in normalizing the data for time, location, and scope are incomplete or unavailable. A clear explanation of what is or what is not covered by the factors is sometimes missing. Lacking anything better, the published data provides a starting point for your database of equipment factors; however, the best information will be data that comes from your own organization's project history and cost databases, and which matches your engineering and construction techniques.

Overall equipment factors from total equipment cost to DFC/TIC (true "Lang" factors) are the easiest to generate. Historical data from completed projects should be normalized to a common time and location/labor productivity baseline. The total equipment costs, direct field costs, and total installed costs should be very easy to derive from the historical data. The results can then be analyzed, plotted, and tested to establish overall equipment "Lang" factors.

Developing individual equipment factors that vary based on the type of equipment, or separate factors for each discipline is much more complicated. Generally, it is difficult to derive the required data from the actual cost histories for completed projects. Project accounting and cost coding typically does not collect actual cost data in the necessary format. Instead, these types of equipment factors are typically developed by generating detailed estimates for a matrix of equipment types, size ranges, metallurgies, operating pressures, etc. The estimates are then carefully analyzed to develop individual equipment and adjustment factors, so that equipment size, metallurgy, and operating pressure can be accounted for.

The factors developed in this manner can then be tested and calibrated against actual project histories. The proposed equipment factors (and adjustment factors, if necessary) are applied to the actual equipment costs for completed projects, and the results from the factoring exercise are compared to the actual project costs to determine if a reasonable degree of accuracy has been obtained. If the factoring results vary widely from the actual costs, or are consistently low or consistently high, then an analysis to determine the reasons will need to be performed, and development of the factors will continue until sufficient accuracy can be obtained.

When preparing an equipment-factored estimate, the first step, of course, is to estimate the cost for each piece of process equipment. The list of equipment needs to be examined carefully for completeness and compared against the Process Flow Diagrams (PFD's) and/or the Piping and Instrument Diagrams (P&ID's). When an equipment factored estimate is prepared, the equipment list is often still in a preliminary stage. Although the major equipment is identified, it may be necessary to assume a percentage for auxiliary equipment that has not yet been defined. Equipment sizing should also be verified. At this preliminary stage of engineering, a common problem is that equipment is often sized at 100 percent of normal, operating duty. However, typically by the time the purchase orders have been issued, some percentage of oversizing has been added to the design specifications. The percentage of oversizing that occurs varies by the type of equipment, and an individual organization's procedures and guidelines. It is prudent to check with the process engineers and determine if an allowance for oversizing the equipment, as listed on the preliminary equipment list, should be added before pricing the equipment.

The purchase cost of the equipment may be obtained by several methods: purchase orders or cost information from recent equipment purchases, published equipment cost data, preliminary vendor quotations, or firm vendor quotations. Since the material cost of equipment can represent 20 to 40 percent of the total project costs for process plants, it is extremely important to always estimate the equipment costs as accurately as possible. When using equipment factoring methods to develop the project estimate, this becomes even more important. If historical purchase information is used, you must ensure that the costs are escalated appropriately, and adjusted for location and/or market conditions as required.

Once the equipment cost is established, the appropriate equipment factors need to be established and applied. Ensure that adjustments for equipment size, metallurgy, and operating conditions are included if necessary. Also, any specific project or process conditions need to be evaluated to determine if additional scope-based adjustments to the factored costs are required. For example, the particular plot layout of the project being estimated may require much closer equipment placement than is typical. Therefore, you may want to make some adjustment to account for the shorter piping, conduit, and wiring runs than the factors would normally account for. Locating a project in an active seismic zone may require adjustments to foundations, support steel, etc.

Once the equipment factored costs have been developed, you must account for the remainder of the project costs that are not covered by the equipment factors. Depending on the particular type of equipment factors used, this may require developing the costs for indirect field costs, home office (project administration and engineering/design) costs, outside battery limit costs, etc.

Equipment factored estimates are typically prepared during the feasibility stage of a project. They can be quite precise if the equipment factors are appropriate, the correct adjustments have been applied, and the list of process equipment is complete and accurate. They have an advantage over capacity factored estimates in that they are based upon the specific process design for the project. It is extremely important to understand the basis behind the equipment factors being used, and to account for all costs that are not covered by the factors themselves.

Ratio or factored methods may often be used in other situations, such as estimating the cost for Outside Battery Limit Facilities (OSBL) from the cost of Inside Battery Limit Facilities (ISBL); or estimating the costs of indirect construction cost from the direct construction costs. Derivation of the appropriate multiplying factors from accurate historical cost information is critical to the resulting accuracy from this estimating methodology.

Parametric Method

A Parametric Cost Model is an extremely useful tool for preparing early conceptual estimates when there is little technical data or engineering deliverables to provide a basis for using more detailed estimating methods. A parametric model is a mathematical representation of cost relationships that provide a logical and predictable correlation between the physical or functional characteristics of a plant (or process system) and its resultant cost [9]. A parametric estimate is comprised of cost estimating relationships and other parametric estimating functions that provide logical and repeatable relationships between independent variables, such as design parameters or physical characteristics and the dependent variable, cost.

Capacity factor and equipment factors estimates are simple examples of parametric estimates; however, sophisticated parametric models typically involve several independent variables or cost drivers. Yet, similar to those estimating methods, parametric estimating is reliant on the collection and analysis of previous project cost data in order to develop the Cost Estimating Relationships (CER's).

The development of a parametric estimating model can appear to be a daunting task; however, the use of modern computer technology (including popular spreadsheet programs) can make the process tolerable, and much easier than it would have been years ago. The process of developing a parametric model should generally involve the following steps [10, 11]:

- Cost Model Scope Determination
- Data Collection
- Data Normalization
- Data Analysis
- Data Application
- Testing
- Documentation

The first step in developing a parametric model is to establish its scope. This includes defining the end use of the model, the physical characteristics of the model, the cost basis of the model, and the critical components and cost drivers. The end use of the model is typically to prepare conceptual estimates for a process plant or system. The type of process to be covered by the model, the type of costs to be estimated by the model (TIC, TFC, etc.), the intended accuracy range of the model, etc. should all be determined as part of the end use definition. The model should be based on actual costs from complete projects, and reflect your organization's engineering practices and technology. The model should generate current year costs or have the ability to escalate to current year costs. The model should be based on key design parameters that can be defined with reasonable accuracy early in the project scope development, and provide the capability for the estimator to easily adjust the derived costs for specific complexity or other factors affecting a particular project.

Data collection and development for a parametric estimating model requires a significant effort. The quality of the resulting model can be no better than the quality of the data it is based upon. Both cost and scope information must be identified and collected. The level at which the cost data is collected will affect the level at which the model can generate costs, and may affect the derivation of the CER's. It is best to collect cost data at a fairly low level of detail [12]. The cost data can always be summarized later if an aggregate level of cost information provides a better model. It is obviously important to include the year for the cost data, in order to normalize costs later. The scope information should include all proposed design parameters or key cost drivers for the model, as well as any other information that may affect costs. The type of data to be collected is usually decided upon in cooperation with the engineering and project control communities. It is usually best to create a formal data collection form that can be consistently used, and revised if necessary.

After the data has been collected, the next step in the process of developing a parametric model is to normalize the data before the data analysis stage. Normalizing the data refers to making adjustments to the data to account for the differences between the actual basis of the data for each project, and a desired standard basis of data to be used for the parametric model. Typically, data normalization implies making adjustments for escalation, location, site conditions, system specifications, and cost scope.

Data analysis is the next step in the development of a parametric model. There are many diverse methods and techniques that can be employed in data analysis, and are too complex to delve into in this article. Typically, data analysis consists or performing regression analysis of costs versus selected design parameters to determine the key drivers for the model. Most spreadsheet applications now provide regression analysis and simulation functions that are reasonably simple to use. The more advanced statistical and regression programs have goal seeking capabilities, which can also make the process easier.

Generally, a series of regression analysis cases (linear and non-linear) will be run against the data to determine the best algorithms that will eventually comprise the parametric model. The algorithms will usually take one of the following forms:

Linear Relationship: $\$ = a + bV_1 + cV_2 + \ldots$

Non-Linear Relationship: $\$ = a + bV_1^x + cV_2^y + \ldots$

Where: V_1 and V_2 are input variables; a, b, and c are constants derived from regression; and x and y are exponents derived from regression.

The various relationships (cost versus design parameters) are first examined for "best-fit" by looking for the highest "R-Squared" value. R^2 has the technical sounding name of "coefficient of determination," and is commonly used as a measure of the goodness of fit for a regression equation. In simple terms, it is one measure of how well the equation explains the variability of the data. The

resulting algorithms from the regression analysis are then applied to the input data sets to determine, on a project-by-project basis, how well the regression algorithm predicts the actual cost.

Regression analysis can be a time consuming process (especially with the simple regression tools of a spreadsheet program), as iterative experiments are made to discover the best-fit algorithms. As an algorithm is discovered that appears to provide good results, it must be tested to ensure that it properly explains the data. Advanced statistical tools can quicken the process, but can be more difficult to use. Sometimes, you will find that erratic or outlying data points will need to be removed from the input data in order to avoid distortions in the results. It's also very important to realize that many costs relationships are non-linear, and therefore one or more of the input variables will be raised to a power (as in the equation above). You will need to experiment both with the variables you are testing against, and the exponential powers used for the variables. Regression analysis tends to be a continuing trial-and-error process until the proper results are obtained that appears to explain the data. Several individual algorithms may be generated and then later combined into a complete parametric model.

The data application stage of the development process involves establishing the user interface and presentation form for the parametric cost model. Using the mathematical and statistical algorithms developed in the data analysis stage, the various inputs to the cost model are identified; and an interface is developed to provide the estimator with an easy and straightforward way in which to enter this information. Electronic spreadsheets provide an excellent mechanism to accept estimator input, calculate costs based upon algorithms, and display the resulting output.

One of the most important steps in developing a cost model is to test its accuracy and validity. As mentioned previously, one of the key indicators of how well a regression equation explains the data is the R^2 value, providing a measure of how well the algorithm predicts the calculated costs. However, a high R^2 value by itself does not imply that the relationships between the data inputs and the resulting cost are statistically significant.

Once you have performed the regression analysis, and obtained an algorithm with a reasonably high R^2 value, you still need to examine the algorithm to ensure that it makes common sense. In other words, perform a cursory examination of the model to look for the obvious relationships that you expect to see. If the relationships from the model appear to be reasonable, then you can run additional tests for statistical significance (t-test and f-test), and to verify that the model is providing results within an acceptable range of error.

One of the quick checks to run is to test the regression results directly against the input data to see the percent error for each of the inputs. This lets you quickly determine the range of error, and interpreting the results can help you to determine problems and refine the algorithms. After all of the individual algorithms have been developed and assembled into a complete parametric cost model, it is important to test the model as a whole against new data (data not used in the development of the model). You should consult statistical texts for more information about testing regression results and cost models.

Lastly, the resulting cost model and parametric estimating application must be documented thoroughly. A user manual should be prepared showing the steps involved in preparing an estimate using the cost model, and describing clearly the required inputs to the cost model. The data used to create the model should be documented, including a discussion on how the data was adjusted or normalized for use in the data analysis stage. It is usually desirable to make available the actual regression data sets and the resulting regression equations and test results. All assumptions and allowances designed into the cost model, to include exclusions, should be documented. The range of applicable input values and the limitations of the model's algorithms should also be explained.

As an example of developing a parametric estimating model, we will examine the costs and design parameters of Induced Draft Cooling Towers. These units are typically used in industrial facilities to provide a recycle cooling water loop. The units are generally prefabricated, and installed on a subcontract or turnkey basis by the vendor. Key design parameters that appear to affect the costs of cooling towers are the cooling range, approach, and flow rate. The cooling range is the difference in temperature between the hot water entering the cooling tower and the cold water leaving the tower. The approach is the difference in the cold water leaving the tower and the design wet bulb temperature of the ambient air; and the flow rate measures the desired cooling capacity of the tower.

Table 9.2 provides the actual costs and design parameters of six recently completed cooling towers. The costs have been normalized (adjusted for location and time) to a Northeast US, Year 2000 timeframe.

Table 9.2—Cost and Design Information for Recent Cooling Tower Projects

Induced Draft Cooling Tower Costs and Design Parameters			
Cooling Range (°F)	**Approach (°F)**	**Flow Rate (1000 GPM)**	**Actual Cost**
30	15	50	$1,040,200
30	15	40	$787,100
40	15	50	$1,129,550
40	20	50	$868,200
25	10	30	$926,400
35	8	35	$1,332,400

This data provides the input to the data analysis steps of running a series of regression analyses to determine a sufficiently accurate algorithm for estimating costs. After much trial and error, the following cost estimating algorithm was developed:

Cost = $86,600 + $84500(Cooling Range in °F)$^{.65}$ - $68600(Approach in °F) + $76700(Flow Rate in 1000GPM)$^{.7}$

From this equation, we can see that the cooling range and flow rates affect costs in a non-linear fashion (i.e., they are raised to an exponential power), while the approach affects costs in a linear manner. In addition, the approach is negatively correlated with costs. Increasing the approach will result in a less costly cooling tower (as it increases the efficiency of the heat transfer taking place). These appear to be reasonable assumptions. In addition, the regression analysis resulted in an R^2 value of 0.96, which indicates the equation is a "good-fit" for explaining the variability in the data; and the F-Test shows statistical significance between the input data and the resulting costs.

In Table 9.3, the design parameters are displayed as used in the model (raised to a power where needed) and shown against the actual costs and the predicted costs from the estimating algorithm. In addition, the amount of the error (the difference between the actual and predicted costs), and the error as a percent of actual costs are shown. The percentage of error varies from –4.4 percent to 7.1 percent for the data used to develop the model.

Table 9.3—Predicted Costs for Cooling Tower Parametric Estimating Example

Induced Draft Cooling Tower Predicted Costs from Parametric Estimating Algorithm						
Cooling Range (°F)$^{.65}$	Approach (°F)	Flow Rate (1000 GPM)$^{.7}$	Actual Cost	Predicted Cost	Error	%Error
9.12	15	15.46	$1,040,200	$1,014,000	-$26,000	-2.5%
9.12	15	13.23	$787,100	$843,000	$55,900	7.1%
11.00	15	15.46	$1,129,550	$1,173,000	$43,450	3.8%
11.00	20	15.46	$868,200	$830,000	-$38,200	-4.4%
8.10	10	10.81	$926,400	$914,000	-$12,400	-1.3%
10.08	8	12.05	$1,332,400	$1,314,000	-$18,400	-1.4%

Using the estimating algorithm developed from regression analysis, we can develop tables of costs versus design parameters (see Table 9.4), and plot this information on graphs (see Figure 9.8).

Table 9.4—Data for Cost Graph Based on Parametric Estimating Example

Induced Draft Cooling Tower Costs Based on Parametric Model			
Cooling Range (°F)	**Approach (°F)**	**Flow Rate (1000 GPM)**	**Predicted Cost**
30	15	25	$559,000
30	15	30	$658,000
30	15	35	$752,000
30	15	40	$843,000
30	15	45	$930,000
30	15	50	$1,014,000
30	15	55	$1,096,000
30	15	60	$1,176,000
30	15	65	$1,254,000
30	15	70	$1,329,000
30	15	75	$1,404,000
40	15	25	$717,000
40	12	30	$816,000
40	15	35	$911,000
40	15	40	$1,001,000
40	15	45	$1,089,000
40	15	50	$1,173,000
40	15	55	$1,255,000
40	15	60	$1,334,000
40	15	65	$1,412,000
40	15	70	$1,488,000
40	15	75	$1,562,000

This information can then be rapidly used to prepare estimates for future cooling towers. It would also be very easy to develop a simple spreadsheet model that will accept the design parameters as input variables, and calculate the costs based on the parametric estimating algorithm.

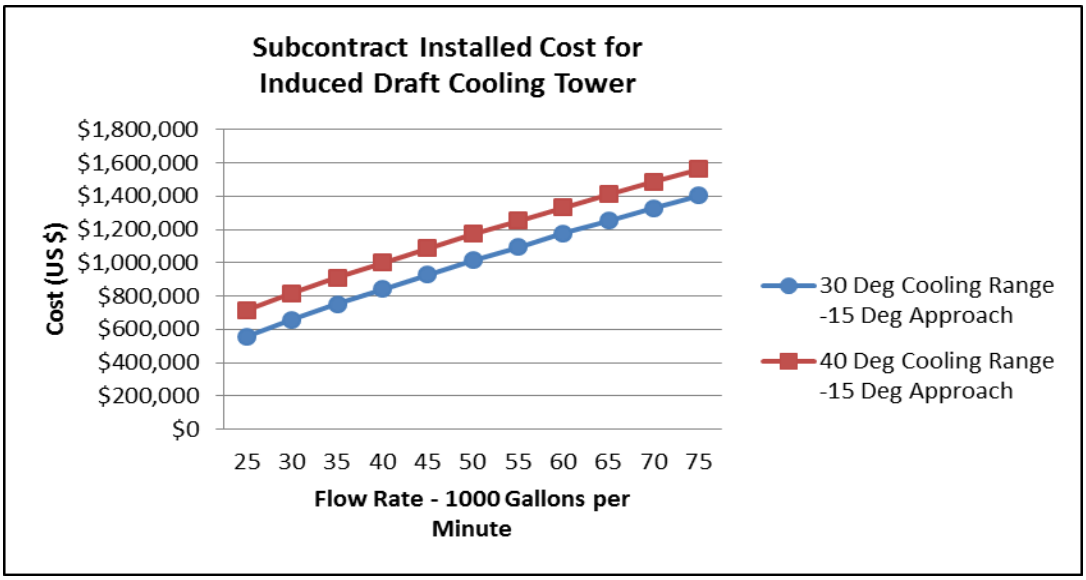


Figure 9.8—Graph of Cooling Tower Costs Based on Parametric Model

Parametric cost models can be a valuable resource in preparing early conceptual estimates. They are often used during both the concept screening and feasibility stages of a project. Parametric models can be surprisingly accurate for predicting the costs of even complex process systems. Parametric estimating models can be developed using basic skills in estimating, mathematics, statistics, and spreadsheet software. It is important to understand that the quality of results can be no better than the quality of the input data, and great care should be taken during the data collection stage to gather appropriate and accurate project scope and cost data.

Deterministic (Detailed) Estimating Methodologies

A detailed estimate is one in which each component comprising a project scope definition is quantitatively surveyed and priced using the most realistic unit prices available. Detailed estimates are typically prepared to support final budget authorization, contractor bid tenders, cost control during project execution, and change orders (Class 3 through Class 1 estimates). Detailed estimates use a deterministic estimating methodology and require a substantial amount of time and costs to prepare. It is not unusual for detailed estimates on very large projects to take several weeks, if not months, to prepare and can require thousands of engineering hours to prepare the required technical deliverables.

Detailed estimates are most often prepared by engineering or construction contractors, or other third parties, rather than the owner organization ultimately responsible for project funding. This is typically because of the large number of resources, time, and breadth and depth of detailed cost information required in the preparation of detailed estimates. In these cases, contract documents or some other form of estimate requirement documents may exist that specify how the estimate should be prepared; and these should be carefully reviewed.

The following is a description for detailed estimating activities associated with a process or industrial project, but could easily be adopted for other types of construction-related projects such as commercial construction. At a minimum, the required engineering and design data required to prepare a detailed estimate include process and utility flow drawings, piping and instrument diagrams, equipment data sheets, motor lists, electrical one-line diagrams, piping isometrics (for alloy and large diameter piping), equipment and piping layout drawings, plot plans, and engineering specifications. Pricing data should include vendor quotations, current pricing information from recent purchase orders, current labor rates, subcontract quotations, project schedule information (to determine escalation requirements), and the construction plan (to determine labor productivity and other adjustments).

In a completely detailed estimate, all costs are detailed including the direct field costs, indirect field costs, home office costs, and all other miscellaneous costs for both the ISBL and OSBL facilities. One variation of the detailed estimate is a semi-detailed estimate in which the costs for the ISBL process facilities are factored, and the costs for the OSBL facilities are detailed. Another variation is the forced-detailed estimate in which detailed estimating methods are used with incomplete design information. Typically, in a forced-detailed estimate, detailed takeoff quantities are generated from preliminary drawings and design information.

The following steps comprise the activities undertaken during preparation of a detailed estimate:

1. Prepare project estimate basis and schedule
2. Prepare Direct Field Cost (DFC) estimate
3. Prepare Indirect Field Cost (IFC) estimate
4. Prepare Home Office Cost (HOC) estimate
5. Prepare sales tax/duty estimates
6. Prepare escalation estimates
7. Prepare project fee estimate (for contractors)
8. Prepare cost risk analysis/contingency determination
9. Preview/validate estimate

The first step in preparing a detailed estimate is to begin establishing the project estimate plan and schedule. This is essentially the pre-planning phase for the estimate. As mentioned, a detailed estimate for a large industrial facility may take weeks to prepare, and involve several estimators and extensive support from engineering. The estimate plan documents the activities and course of action that will be used to prepare the estimate. The first activity is to review the organization's estimating guidelines and procedures with the estimating team. The project Work Breakdown Structure (WBS) should be reviewed with the project controls team, and agreement should be reached on the estimate format, structure, and deliverables. The detailed estimate is typically used to support cost control during execution of the project, and should be structured to accomplish that purpose. The listing of engineering and technical deliverables to be used to prepare the estimate should be reviewed, and the procedures for receiving and tracking the drawings and other design information established. The estimating team should identify the estimating resources, techniques, and data that will be used during estimate preparation. Any estimate exclusions that are known at this time should be reviewed and documented. Often, visits by the estimators to the project site are useful to understand specific site conditions.

The estimate schedule should be prepared, documenting when the various engineering deliverables are to be supplied, when each of the major sections of the estimate should be completed, and when estimate reviews will be scheduled. The estimate basis and schedule should be reviewed with the project team at an estimate kickoff meeting prior to estimate preparation. The estimate kickoff meeting provides an opportunity for the entire project team to understand the roles and responsibilities of the various participants, and to review the plans for the estimate preparation activities and estimate schedule. On very large projects, it is often beneficial to establish a few key contacts that will act as the liaisons between estimating and engineering. Any questions developed by the estimators during estimate preparation are funneled through a liaison that will then work with the responsible engineering representative to develop the answers.

Preparing the direct field cost estimate is the most intensive activity of preparing the detailed estimate. The project scope should be reviewed and understood, and all technical deliverables assembled. On large projects, the engineering drawings and technical information may be submitted to estimating over time. As each drawing or other information is received from engineering, it should be logged and kept track of. Performing the estimate takeoff (described in more detail below) should take place in accordance with the estimating department (and any special project) guidelines. This involves quantifying all the various material and labor components of the estimate. Care should be taken to ensure that all quantities are accounted for, but not double-counted. Material pricing is applied to the material quantities, using the best pricing information available. The labor worker hours are assigned to the labor activities, adjusted for labor productivity, and wage rates applied. Any estimate allowances are established. Any owner supplied materials, or other owner costs, are accounted for. The DFC estimate is then summarized and formatted. Finally, the DFC estimate should be reviewed for completeness and accuracy.

After the DFC estimate has been prepared, the Indirect Field Cost (IFC) estimate is started. The DFC estimate should be reviewed, and the total labor workhours identified. The labor workhours are typically a basis for factoring many of the of IFC costs. The indirect estimate factors should be determined, and applied. Indirect labor wage rates and staff labor rates are established and applied, and any indirect estimate allowances are accounted for. The IFC estimate is then summarized, formatted, and reviewed for completeness and accuracy. The construction manager should be specifically involved in the initial review of the IFC estimate.

The Home Office Cost (HOC) estimate is then prepared. For a detailed estimate, the various project administration and engineering disciplines should provide detailed workhour estimates for their project activities. The appropriate wage rates are then applied to the workhour estimates. Home office overhead factors are determined and applied to develop the home office overhead costs and expenses. The HOC estimate is then summarized, formatted, and reviewed.

Other miscellaneous activities and costs are then estimated. If sales tax is applicable to all (or portions) of the facility, they will need to be estimated using the appropriate local sales tax rates. If materials are to be imported, duties may be charged and will need to be estimated. Escalation costs should be estimated based on the project schedule. Depending on the project delivery method and contracting strategy, appropriate project fee estimates will need to be calculated and included. Finally, a cost risk analysis study should be performed and appropriate contingency is included in the estimate.

As with an equipment factored estimate, particular attention should be paid to pricing the process equipment for a detailed estimate as it contributes such a large share of the costs (20 to 40 percent of the total installed cost of the facility). Estimating the costs for process machinery and equipment requires many sources of input. The minimum information requirements for pricing equipment include the process flow drawings, the equipment lists, and the equipment process data sheets (usually prepared by the process engineering group). Often, the equipment process data sheets are provided to the mechanical/vessels engineering group to prepare narrative specifications and Request for Quotation (RFQ) packages.

Whenever possible, these engineering groups (perhaps in association with the procurement group) should be responsible for providing the equipment material purchase costs to the estimator for inclusion in the project estimate. Although estimating is typically responsible for pricing the material costs for bulk materials, the process and mechanical engineers are best able to accurately determine equipment material pricing, and are generally in close contact with potential equipment vendors. Slight differences in equipment specifications can sometimes result in large differences in pricing, which an estimator may be unaware of. Formal vendor quotes for equipment pricing are preferred; however, sometimes time constraints in preparing the estimate do not permit solicitation of formal vendor quotes. In this case, equipment pricing may depend on informal quotes from vendors (i.e., phone discussions), in-house pricing data, recent purchase orders, capacity factored estimates from similar equipment, or from parametric pricing models.

The estimator should be responsible for checking the equipment list against the flow diagrams (or P&ID's) to ensure that all equipment items are identified and priced. The estimator must also be responsible for verifying that the costs for all equipment internals and accessories (trays, baffles, ladders, etc.) are included with the cost of the appropriate equipment. As opposed to most bulk commodity accounts (where the materials are generally available locally), freight costs for equipment can be significant and should usually be identified explicitly. Also, any vendor assistance and support costs should be identified and included with the material costs of the equipment. Major spare parts for process equipment will also need to be accounted for and included in the estimate.

Equipment installation costs are usually prepared by the estimator, with assistance from construction where required. Construction assistance is usually needed for heavy lifts, or where special installation methods may be used. The placement of large process equipment in an existing facility may also require special consideration. Installation workhours for equipment installation are usually based on weight and equipment dimensions, which are obtained from the equipment process data sheets). Using the equipment weights (or dimensions), the installation workhours are typically determined from curves based on historical data. Other forms of in-house or published data may also be used. When referencing the labor workhour data for equipment, the estimator must be careful to include all labor associated with the pieces of equipment (vessel internals, etc.). Depending on the information available, the labor hours to set and erect a heavy vessel may not include the hours to erect, takedown, and dismantle a guy derrick, gin poles, or other special lifting equipment. Special consideration may also be required to ensure costs for calibration, soil settlement

procedures, special internal coatings, hydro-testing, and other testing costs are included in the estimate. Some equipment may be erected by subcontractors or the vendor, and included in the material purchase costs. Care must be taken to identify these situations.

As with the rest of the estimate, the responsibility of the estimator is to make sure that all costs are accounted for. For equipment in particular, this requires attention to detail, working closes with engineering and construction, and asking the right questions. With most process plants being so equipment-centric, the costs for purchasing and installing equipment comprise a significant portion of the total installed cost of the facility.

Detailed estimates are the most accurate of the estimating methods, but also require the most time and effort to prepare. Although detailed estimates are desirable for final budget authorization, the level of engineering progress needed and the time required for estimate preparation will sometimes preclude them from being used for this purpose. In today's economy, budgeting and investment decisions are often needed sooner than a detailed estimate would allow. Semi-detailed and forced-detailed estimates will often be employed for final budget authorizations, and a complete detailed estimate may be prepared later to support project control.

Take-Off

As mentioned above, estimating take-off is the process of quantifying the material and labor quantities associated with the project. The term take-off is also used to refer to the quantities themselves (often known as a Bill of Quantities). Take-off involves a detailed examination of the engineering drawings and deliverables to count the number of each item appearing on the drawings. The quantities of like items are then summarized according to the control structure (WBS/RBS) of the project. Once the take-off is complete, and total quantities for each like item summarized, the items can be costed (or priced), and the results added together resulting in the estimated direct field costs for the project.

Generally, the process of "take-off" for the estimate is much more efficient when standard estimating guidelines are established and followed. This provides advantage enough when a single estimator is preparing a specific estimate, but is even more important when multiple estimators are working on the same project. Guidelines for preparing an efficient take-off include:

- Use pre-printed forms for the orderly sequence of item descriptions, dimensions, quantities, pricing information, etc.
- Abbreviate (consistently) whenever possible.
- Be consistent when listing dimensions (i.e., length X width X height).
- Use printed dimensions from drawings when available.
- When possible, add up the printed dimensions for a given item.
- Measure all dimensions carefully.
- Use each set of dimensions to calculate multiple quantities where possible.
- Take advantage of design symmetry or repetition.
- List all gross dimensions that can be used again to rough check other quantities for approximate verifications.
- Convert imperial dimensions (feet/inch) to decimal equivalents.
- Do not round until the final summary of quantities.
- Multiply the large numbers first to reduce rounding errors.
- Do not convert the units until the final quantities are obtained.
- Items should be measured/converted to the same units consistently throughout the take-off.
- Mark the drawings as quantities are taken off. Use different colors to identify various types of components or items, as well as to identify items on hold, etc.
- Verify the drawings taken-off versus the approved drawing list to be used with the estimate. Check off drawings on the drawing list as take-off is completed.
- Keep similar items together, different items separate.
- Organize the take-off to match the control structure and format of the estimate.
- Identify drawing numbers, section numbers, etc. on the take-off forms to aid in future checking for completeness, and for incorporating late changes later on.
- Be alert for notes shown on drawings, changes in scale used on different drawings, drawings that are reduced from original size, discrepancies between drawings and specifications, changes in elevation that may not be obvious, etc.
- Be careful to quantify all labor operations that may not have a material component.

By keeping a uniform and consistent take-off process, the chance or error or omission is greatly reduced, and productivity is increased. Multiple estimators will find it easier to work on the same project; and if a personnel change takes place, it is much easier for a new estimator to pick up.

After the take-off is completed, the quantities can be extended, consolidated and priced. If a procurement department or other resources will be used to investigate certain pricing (major equipment, large bulk material purchases, subcontracts, etc.), a listing should be compiled and sent to the appropriate person.

With today's computerized estimating software, the process of take-off is often performed directly into the estimating software, rather than compiled manually onto forms. The software will often prompt for key dimensions and/or parameters for the specific item being quantified, and perform many of the required calculations automatically. In some cases, electronic digitizers can be used which automate the time-consuming task of measuring quantities from drawings, and can help to reduce errors. Using a digitizer, an estimator can measure the area of a concrete slab, the length of a piping run, or count a quantity of valves by tracing a boundary, touching end points, or selecting items from a paper drawing. In combination with the estimating software, the digitizer performs the required calculations required to accurately quantify the various items. The estimating software can also summarize quantities, and apply pricing.

In many construction projects, the material quantities used for estimating may be generated from the Computer-Aided Design (CAD) models developed by engineering. In such cases, the estimator must fully understand what has and has not been modeled at the time the quantities are generated by engineering to support estimate preparation. As an example, a process plant CAD model may only include the piping equal to or larger than 3" in diameter. In such a case, it is important for the estimator to account for all items (such as all piping items less than 3" in diameter) by performing a manual take-off, factoring from known relationships with other items, or by some other means. The estimator should spot check and validate that all quantities required by the scope of the project have been identified.

Take-off is performed for all of the activities and disciplines involved in the project scope. For example, for construction it may involve: site preparation and earthwork; foundations and concrete; structural steel; buildings; mechanical equipment; piping; electrical; instrumentation; and coatings. All other activities involved also need to be quantified including: project management; engineering and design; start-up and commissioning; etc. When required, activities performed by subcontractors or other third parties are identified separately from those activities being performed by the direct involvement of the party preparing the estimate.

Costing Vs. Pricing

Costing is the process of applying unit costs to the individual quantities of items associated with the estimate. For a detailed estimate, this is usually in the form of labor hours, wage rates, material costs, and perhaps subcontract costs. These costs may come from a variety of sources such as an estimating database (either in-house or commercial), vendor quotes, the procurement department, estimating experience, etc.

Pricing, on the other hand, is adjusting the costs that have been applied for specific project conditions, and commercial terms. Pricing includes adjustments to cost to allow for overhead and profit, to improve cash flow, or otherwise serve the business interests of the party preparing the estimate. Thus, the level and type of pricing adjustments depends on the particular party preparing the estimate.

For example, to a concrete contractor preparing a bid for a defined scope of foundation work, his or her costs will include the direct material and labor costs associated with pricing and installing the foundations. However, the price reflected by his bid will include not only his or her costs, but also an allowance for his or her overhead and profit; so the price reflected in his or her bid is higher than his or her cost.

Pricing also includes adjustments to costs for specific project conditions. Depending on the specific cost information used in preparing the estimate, material costs may need to be adjusted for location, materials of construction, or to account for differences between the item being installed and the item you may have an available cost for. Labor hours may require productivity adjustments for a variety of conditions such as weather, amount of overtime, interferences from production, material logistics, congestion, the experience of the labor crews, the level of contamination control, etc. Labor rates may also need to be adjusted for location, crew mix, open shop versus union issues, and specific benefit and burden requirements.

Material Costing/Pricing

The costs for material components of the project estimate may be obtained from internal estimating databases, quotations from suppliers, or from available vendor pricing catalogs. As referenced above, often accurate pricing information for the exact component may not be readily available for the required item, and the cost will need to be estimated and adjusted from the price for a closely related item. Also, pricing adjustments may be required to account for the overall quantity of items being procured (bulk purchase discounts). If pricing is obtained from internal estimating databases, constant calibration of the estimating database is required to account for changing market conditions.

Labor Costing/Pricing

The costs for labor activities depends on both the unit hours for an activity (e.g., hours to install a 10" welded valve) and the corresponding wage rates for the personnel involved with the labor activity. In construction for example, most labor activities are performed by a mixture of personnel, each of which may be compensated for at a different wage rate.

For example, a piping installation crew may involve a foreman, five pipefitter/welders, and two laborers. Thus, the unit hours for installing a 10-inch welded valve are typically accounted for as "crew" hours. If the installation of the welded valve requires eight

hours, it implies that on average five hours will be required by the pipefitters/welders to set and weld the valve; two hours will be required by the laborers to obtain the valve from the warehouse and prepare it for installation; and one hour is required by the foreman for overall supervision, planning, and record keeping. The unit hours (labor hour production rates) are typically documented as crew hours per unit (such as eight hours per 10-inch welded valve). Unit hours are typically obtained from documents known as Labor Productivity Norms (Unit Labor Charts), or may be documented within an estimating database. Labor Norms (whether as stand-alone documents or contained in estimating databases) may be developed and maintained in-house by the estimating department, or may be obtained from commercially available sources.

As defined in AACE RP 73R-13, "A labor productivity norm is a value that is established as the agreed upon reference or benchmark for the number of labor hours (work effort) required to complete a defined work activity, given the specific definitions associated with that value [13]. Each labor productivity norm is an agreed upon typical or average number of labor hours required by a worker or group of workers (i.e., crew) associated with the work activity performed under the stated qualifications. For simplicity, we define a labor productivity norm as a number of hours for an activity per unit of measurement (sometimes called unit hours)."

"Labor productivity norms should be an all-inclusive value for one item of work. Labor norms should include a preamble for the relevant disciplines where the scope of each individual labor norm is defined. A preamble in the context of a labor norm is a narrative to precisely describe the content (inclusion and exclusions), the labor normal conditions, the unit of measurement and method of measurement of an activity and its applicable labor norm. It should be clear from the definitions in the preamble when circumstances or working conditions will require adjustment factors to be applied to the labor productivity norm."

"The labor productivity norm establishes the baseline labor hours for a work activity (typically treated as the 1.0 baseline, to which further adjustments may be applied based on specific project conditions)." Applying labor norms (unit hours) within an estimate requires the application of labor productivity adjustment factors to account for the differences between the basis on which the labor norms were developed (and documented) and the specific project conditions, such as:

- Weather
- Work-week schedule and overtime
- Overall scale of project activity
- Interference from production operations (in an existing facility)
- Hazardous conditions
- Material logistics
- Workspace congestion
- Experience of the labor crews

Labor rates can also be a tricky issue. Labor rates may reflect varying combinations of the following:

- Underlying base wages (the wage rate earned by the individual worker)
- Benefits and burdens (fringe benefits, taxes, and insurances paid by the worker's employer on behalf of the worker)
- Crew make-up
- Work-week schedule and overtime shift differentials
- Overhead, profit, and risk premiums
- Labor indirects of many kinds

When the labor hours assigned to an activity in the estimate reflect unit crew hours, the corresponding labor rate should be a "crew labor rate." As described above, crew may involve several types of workers with each type earning a different base wage rate (and possibly different benefits and burdens). Complicating the matter further is that there may be multiple skill levels within each worker type. For example, in the piping crew described above, four of the five pipefitter/welders may be journeyman, while one is an apprentice.

A composite crew rate for the piping crew can be calculated as shown in the following example:

Table 9.5—Composite Wage Rate Calculation

	Percent	Base Wage Rate	Benefits				Subtotal Rate with Benefits	Statutory Burdens		Total Wage Rate with B&B
			Vacation 10%	Health & Welfare	Pension	Other		Employer Insurance 3.60%	Worker's Comp 1.95%	
Piping Foreman	12.50%	$29.94	3.00	1.33	3.70	1.02	$39.00	1.19	0.64	$40.83
Pipefitter/Welder JM	50.00%	$26.75	2.68	1.33	3.70	1.02	$35.48	1.06	0.57	$37.11
Pipefitter/Welder App	12.50%	$19.80	1.98	1.33	3.70	1.02	$27.83	0.78	0.42	$29.03
Laborer	25.00%	$24.20	2.42	1.25	3.30	0.76	$31.93	0.96	0.52	$33.41
Piping Foreman	12.50%	$3.74	0.38	0.17	0.46	0.13	$4.88	0.15	0.08	$5.11
Pipefitter/Welder JM	50.00%	$13.38	1.34	0.67	1.85	0.51	$17.75	0.53	0.29	$18.57
Pipefitter/Welder App	12.50%	$2.48	0.25	0.17	0.46	0.13	$3.49	0.10	0.05	$3.64
Laborer	25.00%	$6.05	0.61	0.31	0.83	0.19	$7.99	0.24	0.13	$8.36
Composite Piping Rate		**$25.65**	**$2.58**	**$1.32**	**$3.60**	**$0.96**	**$34.11**	**$1.02**	**$0.55**	**$35.68**

The top half of Table 9.5 identifies the types and skill levels of the workers in the crew, and the percent of each. It also lists the base wage rate, and the allocations for benefits and burdens for each. The last column in Table 9.5 identifies the total wage rate, including benefits and burdens, for each type and skill level for the crew workers.

The bottom half of Table 9.5 shows the weighted calculations (i.e., the cost from the top half of the table multiplied by the percentage of each type and skill level of crew member. Thus, the overall composite wage rate for the piping crew is identified as $35.68 per hour. The foreman contributes $5.11 toward the composite rate while the journeyman pipefitter/welder contributes $18.57 toward the composite rate, etc.

This is a simplistic example. Actual crew make-up will vary, as will individual base wage rates, and the types and allocations for benefits and burdens. Wage rates, benefits, and burdens will vary based on union versus open-shop construction. They will also vary by region and other considerations. The estimator will need to research current rates for wages, benefits, and burdens applicable to the particular project. This data may be maintained internally, or may be obtained from applicable union, trade, government, or other organizations. Occasionally, wage rates to be used may be dictated by contractual or governmental agreements.

Table 9.5 is also based on a typical 40-hour per week (straight-time) work week schedule. Overtime or other work shift schedules would also require adjustment. Also, the Table 9.5 excludes contractor overhead, profit, and risk premiums and any labor indirect factors. Owner organizations sometimes want to see “all-in” wage rates used in their estimates; including contractor overhead, profit, and risk premiums, and perhaps some level (or potentially all construction indirect costs). Additional calculations would be required to generate “all-in” costs.

Unit Rate Construction

Some projects are estimated on a unit-rate basis, in which an all-in cost including material and labor (hours and wage rate) costs are applied to each unit, instead of separating material and labor costs separately. For example, in a unit-rate estimate, and all-in rate of $435 per cubic meter of foundation concrete may be used, which includes all material and labor costs. Supporting cost information and estimating databases would have to be maintained appropriately to provide this information for use in a unit-rate estimate. Subcontract construction costs are often based on unit-rate costs.

ESTIMATE ALLOWANCES

Allowances are often included in an estimate to account for the predictable but undefinable costs associated with project scope. Allowances are most often used when preparing deterministic or detailed estimates. Even for this class of estimate, the level of project definition may not enable certain costs to be estimated definitively. There are also times when it is simply not cost effective to quantify and cost every small item included with the project. To account for these situations, an allowance for the costs associated with these items may be included in the estimate.

Allowances are often included in the estimate as a percentage of some detailed cost component. Some typical examples of allowances that may be included in a detailed construction estimate are:

- design allowance for engineered equipment
- material take-off allowance
- overbuy allowance
- unrecoverable shipping damage allowance
- allowance for undefined major items

A design allowance for engineered equipment is often required to account for continuing design development that occurs even after placement of a purchase order for the equipment. At the time of a detailed estimate, vendor quotes are usually available to account of the purchase cost of the equipment. However, for specialty engineered equipment, it is often likely that the quoted cost is not the final cost incurred by the project. We do not necessarily understand when or how the costs will increase, but we can often predict that they will be based on past project experiences. After initial placement of the order for specialty equipment, continuing design activities may tighten tolerances, increase the quality of finish required, change metallurgies, etc. The predicted additional cost will frequently be included in the estimate as an design allowance for engineered equipment (or design development allowance); and be applied as a percentage of the total cost of engineered equipment, or the total cost of specific engineered equipment types when the percentage allowance will vary by equipment type. Typical percentages are from two percent to five percent of engineered equipment cost.

Material take-off allowances are usually intended to cover the cost of undefinable materials at the time of estimate preparation. The completeness of bulk material take-off can vary widely, depending on the status of engineering deliverables at the time estimate preparation begins. For example, all of the small-bore piping may not be included on the design drawings, or perhaps not all of the embeds and related small accessories are identified in the concrete design. A material take-off allowance may be included to cover for the lack of complete project definition. It may also account for those small items it is simply not economical to take-off or detail in the estimate. Generally, material take-off allowances are included as both a material and labor cost. They are intended to cover materials which are an actual part of the project and will thus need to be installed. Material take-off allowances are typically applied as a percentage of direct commodity costs by discipline (or trade). The percentages will vary by discipline, and from project to project depending on the estimating methods used and the level of engineering deliverables to support the estimate. Percentages may run from two percent to 15 percent of discipline costs.

Overbuy allowances provide for inventory losses as a result of such things as damage at the jobsite, cutting loss or waste, misuse of materials, theft, etc. Every project experiences these types of losses depending on jobsite location and other project conditions. Some organizations may split these into several separate allowances (breakage, theft, etc.). Overbuy allowances usually apply to material costs only, may vary from two percent to ten percent of discipline material costs.

Damage to equipment and materials during shipment can be expected on virtually every project. Usually, the cost of damage is covered by insurance if detected upon arrival at the jobsite and dealt with expeditiously. An allowance for unrecoverable shipping allowance is intended to cover such losses that are not covered by insurance. This allowance will vary based on project conditions, project material delivery and handling procedures, and the types of material and equipment being shipped.

Occasionally, an order-of-magnitude cost for a major segment of scope must be stated before definition of that work has begun. A particular area of scope may simply not have progressed in design as far as the rest of the project, but a cost for that scope must be included in the estimate. In this case, the cost is included as an allowance, and may simply be a best “guestimate” to be included in the estimate until a later time when better definition can be obtained. This is sometimes referred to as an allowance for an undefined major item.

Other miscellaneous allowances may sometimes be included in an estimate for any situation where a statistical correlation is more reliable that a detailed quantification, or where it is not economical to perform a detailed take-off. Percentage allowances are often included for such items. Material and/or labor costs routinely covered by such items include:

- Percentage of hand excavation/backfill (vs. machine excavation/backfill)
- Formwork accessories
- Structural steel connection materials
- Bolts, gaskets, etc.
- Piping hangers, guides, etc.
- Miscellaneous welding operations
- Hydro-testing, other testing operations

Specific application of estimating allowances will depend on many things. For conceptual estimates, such as a capacity factored estimate, allowances may not be required as the estimating methodology itself covers all scope and costs included in the project.

Allowances are usually more applicable to semi-detailed and detailed estimates, with the cost value of allowances (or percentage costs) becoming less as project definition increases. The specific allowances and values will usually depend on specific organization estimating procedures and experience.

ESTIMATE ACCURACY

An estimate is a prediction of the expected final cost of a proposed project (for a given scope of work). By its nature, an estimate is associated with uncertainty, and therefore is also associated with a probability of over-running or under-running the predicted cost. Given the probabilistic nature of an estimate, it should not be regarded as a single point number or cost. Instead, an estimate actually reflects a range of potential cost outcomes, with each value within this range associated with a probability of occurrence.

Typically, the preparation of an estimate results in a single value. If we prepare a conceptual estimate using capacity factored techniques, we calculate a single point value as the estimated cost. When preparing detailed estimates, as the sum of many individual estimating algorithms, we also calculate the estimate total as a single point value. What we need to understand is the uncertainty associated with that single point value, and the true probabilistic nature of an estimate.

Most of the end uses of an estimate require a single point value within the range of probable values to be selected. For example, when used to develop a project funding amount or budget, we must select a single value to represent the estimate. When taking into account the uncertainty associated with an estimate, we will often add an amount (contingency) to the initially developed point value to represent the final estimate cost. When doing so, we must take into account such things as the accuracy range of the estimate, confidence levels, risk issues, and other factors in selecting the best single point value to represent the final value of the estimate.

Estimate accuracy is an indication of the degree to which the final cost outcome of a project may vary from the single point value used as the estimated cost. It should generally be regarded as a probabilistic assessment of how far a project's final cost may vary from the single point value that is selected to represent the estimate. Accuracy is traditionally represented as a +/- percentage range around the point estimate; with a stated confidence level that the actual cost outcome will fall within this range. This common +/- percent measure associated with an estimate is merely a useful simplification, given the reality that each individual estimate will be associated with a different probability distribution explaining its unique level of uncertainty.

Estimate accuracy tends to improve (i.e., the range of probable values narrows) as the level of project definition used to prepare the estimate improves. Generally, the level of project definition is closely correlated with engineering progress; thus, as the level of engineering progresses, estimate accuracy improves. This is shown in Figure 9.9.

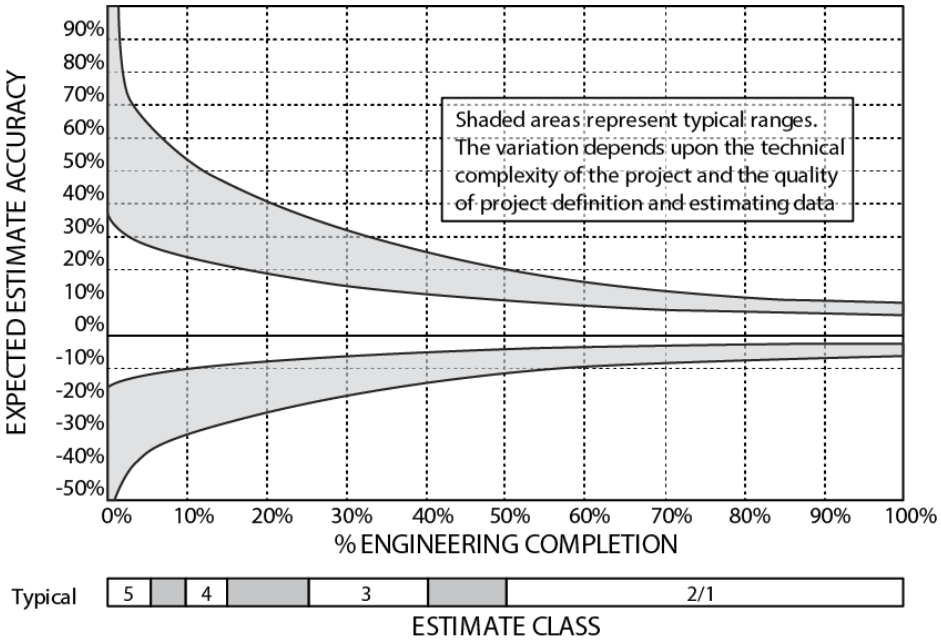


Figure 9.9—Relationship between Estimate Accuracy and Engineering Progress

This chart is intended only as an illustration of the general relationship between estimate accuracy and the level of engineering complete. As shown in Figure 9.9, and described in the AACE Recommended Practices on Estimate Classification, there is no absolute standard range on any estimate or class of estimate. For the process industries, typical estimate ranges are illustrated as:

- Typical Class 5 Estimate:
  - High range of from: +30 to +100 percent
  - Low range of from: –20 to –50 percent
- Typical Class 4 Estimate:
  - High range of from: +20 to +50 percent
  - Low range of from: –15 to –30 percent
- Typical Class 3 Estimate:
  - High range of from: +10 to +30 percent
  - Low range of from: –10 to –20 percent

Although the percent of engineering complete (or level of project definition) is an important determinant of estimate accuracy, there are many other factors which also affect it. Some of these other factors include the state of new technology in the project, the quality of reference cost information used in preparing the estimate, the experience and skill of the estimator, the estimating techniques employed, the level of effort budgeted to prepare the estimate, and the desired end use of the estimate. Other important factors affecting estimate reliability are the project team's capability to control the project, and the capability to adjust the estimate for changes in scope as the project progresses.

Consideration of all of these factors is the reason that the high and low ranges of typical estimate accuracy are themselves variable. It is simply not possible to define a precise range of estimate accuracy, based solely on the percentage of engineering complete or class of estimate. Any specific estimate may not exhibit the patterns shown above. It is possible to have a Class 5 estimate with a very narrow estimate range, particularly for repeat projects with good historical costs upon which to base the estimate. Conversely, it is possible to have a Class 3 or Class 2 estimate with a very wide accuracy range, particularly for first-of-a-kind projects or those employing new technologies.

The +/- percent accuracy range of the estimate should always be determined from an assessment of the design deliverables and estimating information used in preparation of the estimate. Cost risk analysis studies should always be used for individual projects to determine their accuracy range based on this type of information. The resulting output of the cost risk analysis model then establishes a final estimate cost based on the level of confidence (or risk) acceptable to management in order not to overrun the project budget.

It is very important to recognize that the estimate ranges above are simply indicative across a large portfolio of projects, and only if all of the estimating deliverables have been developed to the required level of maturity for the class of estimate; and only if appropriate contingency has been included in the project estimate. Every individual project should identify its potential estimate accuracy from a comprehensive and appropriate cost risk analysis study.

When discussing estimate accuracy, it is also important to realize that for early conceptual estimates, variations in the design basis will have the greatest impact on costs. Estimating tools and methods, while important, are not usually the main problem during the early stages of a project when estimate accuracy is poorest. In the early phases of a project, effort should be directed toward establishing a better design basis than concentrating on using more detailed estimating methods.

Contingency and Risk Analysis

Contingency is, in many respects, the most misunderstood element contained in an estimate. This is, in large part, because of how the different members of a project team view contingency from their own frame of reference. A project manager may want the project budget to include as much funding as possible in order not to overrun the budget, and may want as a large contingency value included in the estimate as he or she can get away with. An engineering manager may want contingency funds to cover any overruns in engineering, while the construction manager hopes that engineering does not use any of the contingency funding, so that he or she has the entire amount to use in funding construction overruns. Corporate management may think of all requests for contingency as "padding" the estimate, and may consider any use of contingency funds as only being required because a project is poorly managed.

To the estimator, contingency is an amount used in the estimate to deal with the uncertainties inherent in the estimating process. The estimator regards contingency as the funds added to the originally derived point estimate to achieve a given probability of not overrunning the estimate (given relative stability of the project scope, and the assumptions upon which the estimate is based). Contingency is required because estimating is not an exact science. One definition of an estimate is that it is the expected value of a complex equation of probabilistic elements, each subject to random variation within defined ranges. Since the value assigned to each individual component of an estimate is subject to variability, the estimate total itself is also subject to variation.

Figure 9.10 illustrates the potential variability of a single component of an estimate. In this example, the variability is shown as a normal probability distribution around the estimated value of $100. Since this is a normal probability distribution, the probability of underrun (shown as the area under the curve to the left of the vertical dotted line) equals 50 percent, the same as the probability of

overrun (the area under the curve to the right of the dotted line). The estimate line item has an estimated cost of $100; however, the accuracy range of the cost varies from $50 to $150, or an accuracy range of +/- 50 percent.

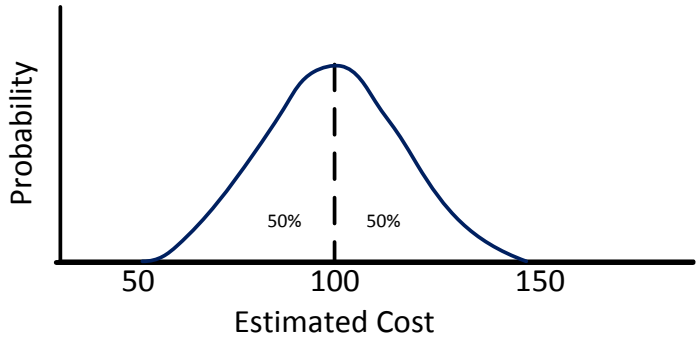


Figure 9.10—Variation of an Estimate Line Item with Normal Probability Distribution

Unfortunately, most items of cost in an estimate do not exhibit a normal probability distribution in respect to its potential variability. Most of the time, variability is more closely associated with a skewed distribution. Figure 9.11 shows the variability of an estimate line item for which the accuracy range of the cost is skewed to the high side.

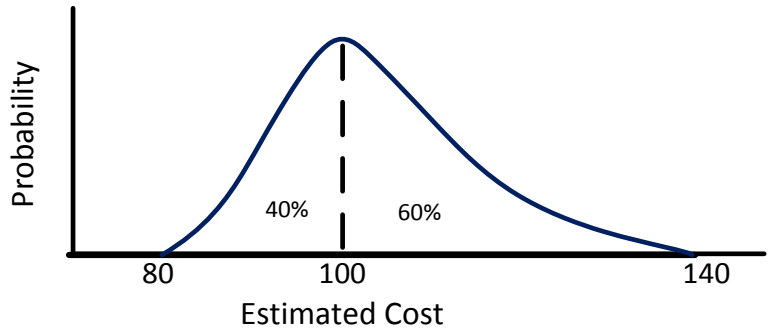


Figure 9.11—Variation of an Estimate Line Item with a Skewed Probability Distribution

In this example, the item has been estimated at $100; however, the accuracy range of the cost varies from $80 to $140, or -20 to +40 percent. With an estimated value of $100, this example shows that there is only a 40 percent probability of underrun, while there is a 60 percent probability of overrun. In order to equalize the probability of underrun and overrun, an amount would need to be added to the original point value of $100. This amount would be considered contingency. Contingency would not change the overall accuracy range of $80 to $140; however, it would increase the probability of underrun while decreasing the probability (risk) of overrun.

Most items of cost in an estimate will demonstrate some measure of skewness, usually to the high side where the probability of overrun is higher than the probability of underrun. There are usually items where the skewness will be to the low side as well. The variability of the total estimate is then a function of the variability associated with each individual line item. Since the probability distribution of most line items is skewed to the high side, the overall probability distribution for the estimate as a whole is also typically skewed to the high side. Contingency is thus usually a positive amount of funds added to cover the variability surrounding the point value of the estimate, and to reduce the chances of overrunning the point estimate to an acceptable level.

Items typically covered by contingency include:

- Errors and omissions in the estimating process.
- Variability associated with the quantification effort.
- At the time of estimate preparation, design may not be complete enough to determine final quantities.
- Some items required to be estimated may defy precise quantification.
- Some items to be quantified are generally computed by factored or other conceptual methods.
- Labor productivity variability.
- Labor availability, skills, and productivity may vary from that originally assumed.
- There is no such thing as an “average” tradesman that installs every incremental quantity of an item at the “average” rate typically used in preparing the estimate.

- Weather may vary from that assumed affecting labor productivity.
- Wage rate variability.
- Wages may vary from that assumed in the estimate because of inflationary reasons, changes in assumed crew mix, labor availability, and market conditions.
- Material and equipment costs.
- Material and equipment costs may vary from that in the estimate due to inflationary reasons and market conditions.
- Certain materials of construction may be substituted from that assumed in the estimate.
- Changes in actual quantities may change discount schedules from that assumed in the estimate.

Contingency specifically excludes:

- Significant changes in scope.
- Major unexpected work stoppages (strikes, etc.).
- Disasters (hurricanes, tornadoes, etc.).
- Excessive, unexpected inflation.
- Excessive, unexpected currency fluctuations.

Risk analysis is a process that can be used to provide an understanding of the probability of overrunning (or underrunning) a specified estimate value. It provides a realistic view of completing a project for the specified estimate value by taking a scientific approach to understanding the uncertainties and probabilities associated with an estimate, and to aid in determining the amount of contingency funding to be added to an estimate. Its purpose is to improve the accuracy of project evaluations (not to improve the accuracy of an estimate).

Risk analysis generally uses a modeling concept to determine a composite probability distribution around the range of possible project cost totals. It provides a way in which to associate a level of risk with a selected project funding value. If the original point value of an estimate is assumed to be approximately the midpoint of the possible actual cost outcomes of project cost, that means that there is a 50 percent probability that the final outcome will exceed the estimated cost (without contingency). In reality, there is usually a greater probability that costs will increase rather than decrease. This means that the distribution of project cost outcomes is skewed, and there is a higher than 50 percent probability that final actual costs will exceed the point estimate (and this is historically the case). Two types of risk analysis are commonly used:

- Strategic risk analysis models that evaluate the level of project definition and project technical complexity in determining the overall risk to project cost. And,
- Detailed risk analysis models that evaluate the accuracy range for individual or groups of estimate components in determining the overall risk to project cost.

Both forms of risk analysis models usually generate overall probability distributions for the expected final cost outcomes for the project, and tables equating confidence levels with specific final cost values. The resulting probability distributions of final cost outcomes can be used to determine an amount to be included in the estimate as contingency. Basically, management typically makes this determination based on the level of risk they are willing to accept. The difference between the selected funding value and the original point estimate is the amount of contingency.

Table 9.6 shows an example of a cumulative probability distribution table produced by a typical risk analysis model. In this example, the original point estimate (before contingency) is $23.3 million. As can be seen from this table, the point estimate of $23.3 million results in only a 20 percent probability of not exceeding (or underrunning) this value.

Table 9.6—Sample Cumulative Probability Distribution Table

Project Estimate			
Cumulative Probability of Underrun	**Indicated Funding Amount (Million $)**	**Estimated Contingency (Million $)**	**Percentage**
10%	$22.3		
20%	$23.3		
30%	$24.2		
40%	$24.8		
50%	$25.4	$2.1	9.0%
60%	$26.0		
70%	$26.6	$3.3	14.2%
80%	$27.4		
90%	$28.6		

If we wanted to achieve a 50 percent probability of underrun (and thus a 50 percent probability or overrun) we would need to fund the project at $25.4 million. This would mean adding a contingency amount of $2.1 million in the estimate, equivalent to nine percent of the original point value of the estimate. If we wanted to provide a 70 percent probability of not exceeding our project funding, we would need to fund $26.6 million, which would add a contingency amount of $3.3 million to the estimate (equivalent to 14.2 percent of the point estimate).

This can also be shown in a typical graphical output from a risk analysis model for the same estimate as shown in Figure 9.12.

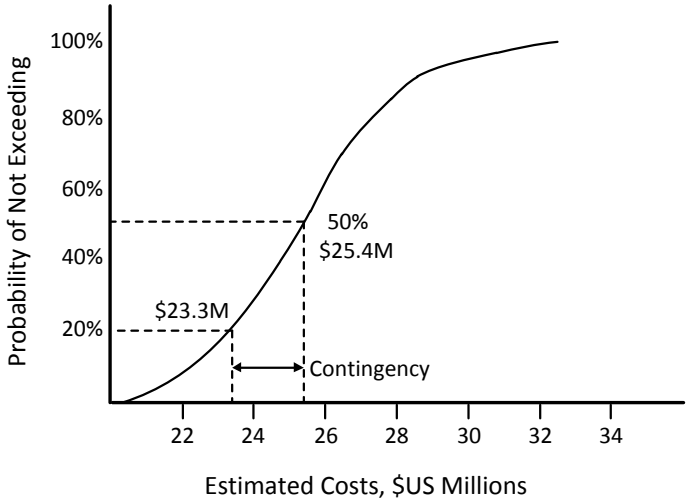


Figure 9.12—Graphical Cumulative Probability Distribution

As can be seen from this graph, increasing the amount of contingency increases the probability of not exceeding the project funding amount (the point estimate plus contingency). ***Contingency does not increase the overall accuracy of the estimate—it does not change the overall accuracy range of approximately of $18.5M to $32.5M***. However, contingency does reduce the level of risk associated with the estimate and improve project evaluations when properly used.

Appropriately applied, risk analysis provides an effective means of determining an amount for estimate contingency, and of providing management with information about the variability of project estimates. In addition, the process of preparing a risk analysis model typically identifies specific project areas associated with both risk and opportunity. Those areas identified with high risk can then become focus areas in order to reduce and mitigate any risk issues, and the areas of low risk can become focus areas in order to capitalize on the opportunities they may provide.

Structuring the Estimate

The control structure for a project is the breakdown of the total work into manageable units or packages for the purposes of estimating and control of cost and schedule. The structure will vary with the size and complexity of the project, as well as the reporting requirements. The proper structuring of a project for control purposes contributes greatly to the effective implementation of project control procedures and the success of the project itself.

To maintain some kind of order in the estimate (and later in project execution), it is necessary to segregate costs into various categories:

- Material vs. Labor vs. Subcontracts
- Direct Costs vs. Indirect Costs vs. Home Office Costs
- Concrete vs. Structural Steel vs. Piping vs. Other Construction Disciplines

The control structure should be established as early as possible in the project life cycle, because it sets the pattern for accumulation of project costs, and should be used to form the basis for the structuring the estimate. The process of producing the project's control structure, often known as Work Breakdown Planning, is often an ongoing process requiring updates as the scope of the project is refined during the project lifecycle. The segregation of costs can be referred to as establishing the project "coding" structure, and more specifically as the "code of accounts". Codes are the umbilical cords between cost accounting and cost engineering (estimating and cost control).

Large projects will often use Work Breakdown Structures (WBS) and Resource Breakdown Structures (RBS) as components of the overall coding structure. Smaller projects will often use a simpler code of accounts based simply on the disciplines or construction trades used on the project.

The WBS and RBS are basic project management tools that define the project along activity levels that can be clearly identified, managed, and controlled. The WBS is the division of a project, for the purposes of management and control, into sub-projects according to its functional components. The WBS typically reflects the manner in which the work will be performed, and should reflect the way in which cost data will be summarized and reported.

A WBS should be customized to be specific to a particular project, and is usually organized around the geographical and functional divisions of a project. It forms the high-level structure for an estimate. Figure 9.13 illustrates how a typical WBS might be organized.

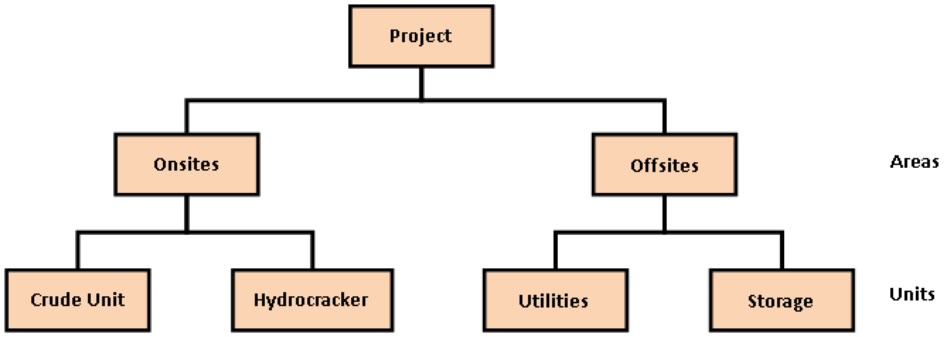


Figure 9.13—Sample WBS

The RBS is a breakdown of all labor and material resources required in the execution of the project. The RBS identifies functional lines of authority, and extends to the level at which work is actually assigned and controlled. The RBS typically remains consistent from project to project (at least for the same project types). Figure 9.14 illustrates a sample project RBS.

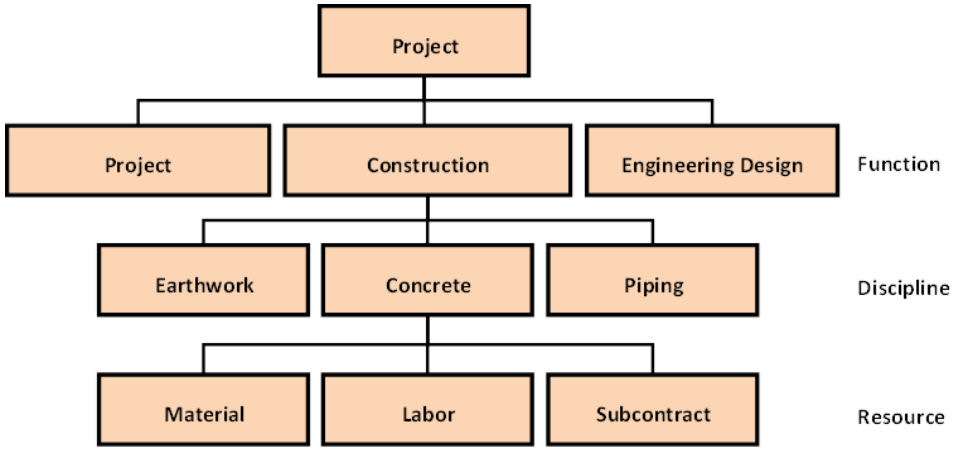


Figure 9.14—Sample RBS

The matrix of the WBS and RBS forms the full Project Control Structure or Project Breakdown System (PBS). The intersection points of the WBS and RBS structures is called a “cost center,” and corresponds to a defined unit of work and the resources involved in executing that work. Each cost center equates to a specific “cost code.” Figure 9.15 displays a sample Project Breakdown Structure.

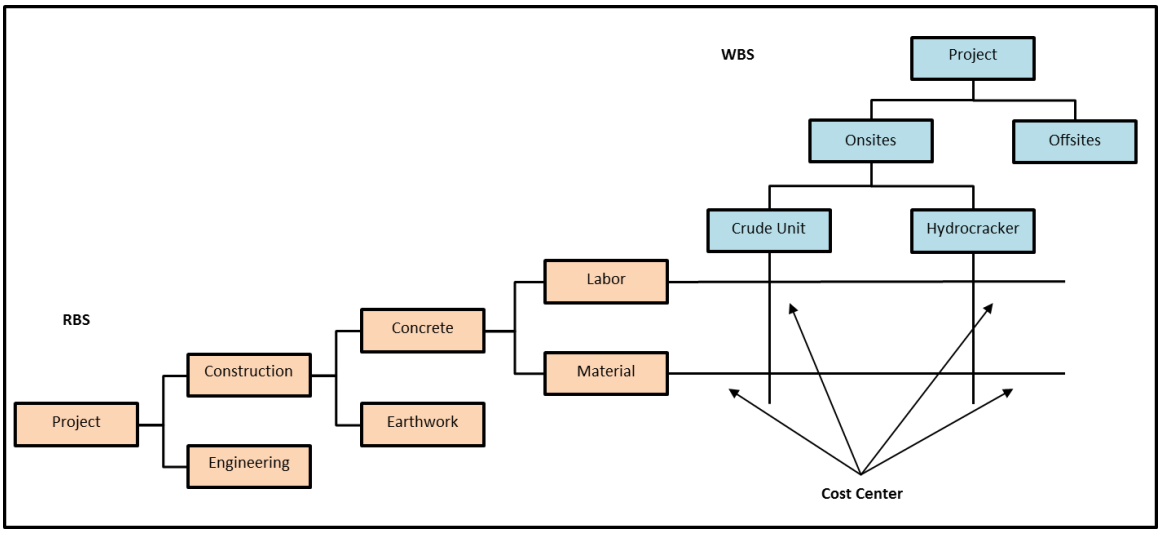


Figure 9.15—Sample Project Breakdown Structure

Corresponding with the PBS is a numbering system used to identify each cost center. The collection of codes used to designate the intersection of WBS and RBS identifiers forms the project's Code of Accounts. Figure 9.16 shows a sample coding structure.

Area	Unit	Function	Discipline	Detail	Resource
01-Onsites	01-Crude unit	A-Project Administration	1-Earthwork	001-Formwork	1-Labor
	02-Hydrocracker	B-Engineering/Design	2-Concrete	002-Rebar	2-Material
	03-Vacuum unit	C-Construction	3-Structural steel	003-Pour	3-Subcontract
			4-Piping	004-Embeds	
			5-Equipment	005-Finish	
02-Offsites	01-Utilities				
	02-Storage				
	03-Pipeway				

Figure 9.16—Sample Project Coding Structure

For a specific unit of work, the labor to pour concrete in the hydrocracker unit, the cost code would be: 01-02-C-2-003-1 (Onsite-Hydrocracker-Construction-Concrete-Pour-Labor).

The Code of Accounts formally refers to the full coding structure (including Project Identifier, WBS, and RBS elements), but the term is often used in regards to the RBS elements only. The coding structure must reflect the manner in which the project will be executed, and the way in which costs can reasonably be expected to be collected. The coding structure should also reflect the way in which your particular organization executes projects. The estimate, which predicts project execution, should be organized and structured to match the project code of accounts.

The coding structure adopted by an organization should be documented in detail. Typically, a code book is published and made available to all project personnel. The code book should contain a code by code listing which documents a description of not only what should be included under a specific code, but also what is excluded (for those items that could be easily misunderstood).

Estimate/Cost/Schedule Integration

The integration of the project cost estimate with the project schedule and cost control system is crucial for effective project management and control. Accomplishing this goal can be difficult at best; yet the estimate, schedule and cost system must share information with each other for each to be as accurate as possible. The schedule will provide dates that are essential to calculating escalation, cash flow, and commitment forecasts. The estimate provides labor hours and craft breakdowns essential to determining schedule activity durations and resource loading. The estimate also provides cost and quantities to the cost control system. The cost reporting system's record of labor and material expenditures needs to be correlated with schedule progress and remaining durations for schedule activities correlated to the forecasts-to-complete in the cost system.

The relationship between the cost estimate and schedule is not always straightforward. The natural breakdowns (or hierarchy) of cost and schedule structures are different. The cost system is organized to estimate, monitor and control dollars. The schedule system is organized to plan, monitor and control time. The control and monitoring of both variables are not necessarily compatible;

and most often it is not the same people performing both tasks. The goal, then, is to align estimate cost data and schedule data at a level to support integration.

One approach is to breakdown the estimate to the level of schedule activities. This can result in a tremendous amount of detail in the cost estimate, and compromise efficient cost and schedule control. Some of the problems resulting from the one-to-one approach are:

- Collecting costs by detailed schedule activities is generally not feasible.
- Schedule activities are subject to much more change within the project than traditional cost codes.
- Tracking bulk material costs by activity is cumbersome and requires high administrative costs. And,
- Costs are often not incurred at the same time as construction activities.

The goal must be to determine an appropriate level of detail to correlate cost and schedule. It is important not to let either the estimate or the schedule drive the other down to an inappropriate level of detail. It is also important not to integrate at too high of a summary level.

Integrating at a sufficient level of detail involves keeping the estimate and schedule structures the identical to a certain level of work breakdown structure. Below this level, additional cost accounts and schedule activities are defined separately as required by each. The desire is to interface at a level where meaningful relationships exist.

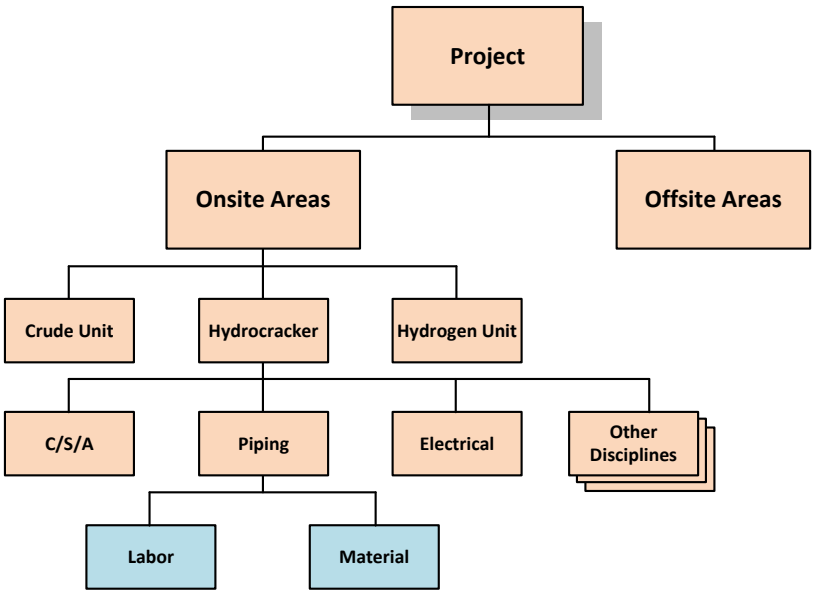


Figure 9.17—Cost/Estimate WBS Structure

Figure 9.17 illustrates a typical cost or estimate structure for a process plant, while Figure 9.18 illustrates a sample schedule structure for the same project. At some point, the cost and schedule WBS structures will diverge to meet each structure's particular control needs.

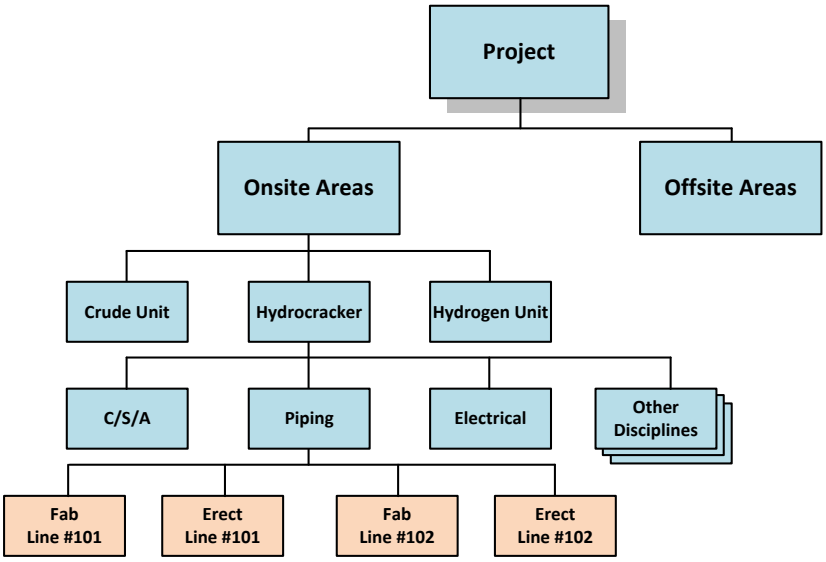


Figure 9.18—Schedule WBS Structure

The basic methodology for integrating the cost estimate and schedule therefore is to let the estimator and scheduler communicate on the high level WBS, and determine the levels at which cost items and schedule activities can be correlated. Then each further defines the lower level of detail required for particular needs. Meaningful information can then be transmitted at the appropriate level of detail between the two.

It should be acknowledged that one-to-one relationships between estimate cost items and schedule activities is not possible. Early and continuous communication between the estimator and scheduler can determine the best level at which to maintain compatibility and exchange information. The estimator can promote integration by assigning as many identification fields to the estimate line items as possible (i.e., building location, room number, system number, piping line number, foundation number, etc.). This will greatly assist in transferring estimate cost data and resource needs to the schedule. Computerized estimating and scheduling systems are making great strides in providing two-way communication between systems.

The cost estimate can be very sensitive to, and is usually prepared in correlation with a specific schedule. If the schedule is undefined or subject to change, the estimate is compromised and should reflect the appropriate cost risk. Changes to the project plan that affect either schedule duration or completion dates may significantly affect project cost. The basis estimating algorithm is:

Total $ = (Qty. X Unit Material $) + (Qty. X Unit Labor Hours X Wage Rate)

The Unit Material $, Unit Labor Hours, and Wage Rate can all be dependent on the assumed schedule and plan. Unit material costs are schedule dependent for impacts of inflation and seasonal variations. Unit labor hours are schedule dependent for seasonal labor availability, climate, and schedule impacts resulting from execution plan changes (affecting productivity). Wage rates are also sensitive for impacts of inflation, seasonal variation, and execution plan changes (affecting overtime and/or shift premiums).
Many costs in a project are very dependent on the duration. Project management and related costs are often estimated (and incurred) on a “level of effort” basis. If the project duration is extended, cost for these activities is directly affected. Construction indirect costs, such as construction management, field office, construction equipment rental, security office, site maintenance, etc., are also affected in a similar way.

Some costs are dependent on when they occur in the calendar year. Labor productivity can be adversely affected by weather (both snow and rain in the winter, or hot weather in summer). Construction indirects, such as weather protection or other construction support costs, can also be similarly affected.

Project costs can also be affected by schedule impact of execution plan changes. Changes to the execution plan to shorten the project duration may cause out-of-sequence construction, overtime, shift premiums, congestion, inefficient labor usage, etc. adversely impacting costs. A delay in equipment delivery may extend the project schedule increasing duration dependent costs. On the other hand, this may also result in increased efficiencies if labor resources can be allocated more efficiently, perhaps resulting in less overtime and shift premiums.

It is important to evaluate the effects of schedule and duration when preparing the estimate. Besides the obvious of accounting for the escalation costs to incorporate into the estimate, schedule impacts may directly affect labor productivity, as well as labor and

material pricing. It is also essential to plan early for estimate/schedule integration so that estimate results can be shared with the schedule as required to assist in resource loading, and to aid in earned value analysis and progress reporting.

Estimate Review

An estimate is of critical importance to a project's success, it makes sense that the estimate should undergo a rigorous review process. The estimate should be evaluated not only for its quality or accuracy, but also to ensure that it contains all the required information and is presented in a way that is understandable to all project team members and client personnel. A structured (if not formal) estimate review process should be a standard practice for all estimating departments.

The following sequence of steps will discuss a formal review process for an internally prepared appropriation grade estimate (an estimate submitted for capital budget authorization). The level of detail and diligence used during the estimate review cycle will vary both with the strategic importance, total value, and purpose of the particular estimate. These steps can be easily adapted on a fit-for-use basis. In this discussion, we are focused on reviewing and validating an estimate – we are not discussing bidding strategies which can involve many other factors and decisions.

Estimate Review Cycles

The principle purpose of an estimate review process is to present information about both the estimate and the project in a way that allows the reviewer to evaluate that the estimate is of sufficient quality to meet its intended purpose. The estimate review process is usually comprised of a series of estimate reviews, beginning with internal estimating department reviews, engineering reviews, project team reviews, and continuing with reviews by various levels of management, depending on the importance of the project.

Estimating Team/Estimating Department Review

The first review of the estimate should, of course, be conducted by the estimating team that prepared the project estimate. This is essentially a screening review to ensure that the math is correct (extensions of pricing are correct, summaries add up properly, etc.), that the estimate is documented correctly (comprehensive basis of estimate document is prepared), and that it adheres to estimating department guidelines. Typically, this review is conducted by the lead estimator with the members of his estimating team. On very large projects or those of significant importance, this review may be conducted by the estimating department manager or supervisor.

Check the Math

When starting the estimate, review the mathematical computations to ensure they are correct. With today's computerized estimating systems, this is much less of a concern than 20 years ago when estimates were primarily prepared by hand using simple calculators; however, math errors can still occur. This can be a major concern when using an electronic spreadsheet, such as Excel, for preparing the estimate (as opposed to a commercial computerized estimating system). Surprisingly, it is very easy to make a formula error in a spreadsheet, such as inserting a row or column which does not get included in a subtotal. All spreadsheet formulas, subtotals and totals should be examined carefully for correctness. From a client's point of view, nothing will help to lose credibility in the entire estimate faster than a finding a math error that went undetected.

Basis of Estimate

The comprehensive Basis of Estimate (BOE) document should be reviewed carefully to ensure that it is both correct and complete. The BOE is an extremely important document. The dollar amount indicated on an estimate is meaningless without knowing the parameters, or what is included and not included in the estimate. The BOE serves to clearly define the design basis, planning basis, cost basis, and risk basis of the estimate.

- **Design Basis**: The overall scope of the project should be summarized, with additional detail provided for each area/unit/work package of the project. Specific inclusions, and even more importantly, specific exclusions of items or facilities should be documented. All assumptions regarding project scope should be documented. If available, equipment lists should be attached or referenced; and a listing of all drawings, sketches, and specifications used in the preparation of the estimate should be documented, including drawing revision date and number.
- **Planning Basis**: This portion of the BOE should document information from the integrated project plan which affects the estimate. It should include specific information about any contracting strategies for engineering, design, procurement, fabrication, and construction. It should include information about resourcing and project execution plans such as the length of the workweek, use of overtime, number of shifts, etc. It should include information about the project schedule and key milestone dates affecting the estimate.
- **Cost Basis**: The source of all pricing used in the estimate should be documented in this section of the BOE. This would include the source of all bulk material pricing, the pricing of major equipment (referencing quotes or purchase orders if used), and all labor rates including office, engineering, fabrication, and construction. The source of all labor workhours should be documented, along with any assumptions regarding labor productivities. All allowances included in the estimate should also be clearly identified. It is also important to document the time basis of the estimate (i.e., what point in time is assumed), and the basis for cost escalation included in the estimate.
- **Risk Basis**: Since, by definition, every estimate is a prediction of probable costs, it is clear that every estimate involves uncertainty and risk. Contingency is typically included in an estimate to cover the costs associated with this uncertainty.

This section of the BOE should document how the contingency was determined, and identify key areas of risk and opportunity in the cost estimate.

It is important to ensure that the basis of estimate is clear and easily understood, and to verify that all information and factors documented in the BOE have been consistently applied throughout the estimate (i.e., wage rates, labor productivities, material pricing, subcontract pricing, etc.). Again, the estimate can lose credibility if different pricing or labor rates have been used for the same item within the estimate detail.

Estimating Department Guidelines
A careful review should be done to verify that the cost estimate follows standard estimating guidelines for the department. This would include a review to verify that standard estimating procedures were followed regarding estimate format, cost coding, presentation and documentation. This would include items such as:

- Verify that the proper estimating methods, techniques and procedures were used that match the stage of project completeness. In other words, different estimating techniques will be used depending on the type and completeness of the engineering documents and deliverables available to create the estimate.
- Confirm that the estimate summary and details are organized and presented in the proper format (i.e., following the project work breakdown structure and code of accounts); and that the format is consistent with the intended purpose of the estimate (i.e., an estimate serving as a basis for cost control contains sufficient detail).
- Ensure that all estimate backup information is organized properly. Can all values on the summary page of the estimate be traced to the estimate detail pages, and can all information on the estimate detail pages be traced to the estimate backup or source documents? And,
- Verify that all allowances and factors are appropriate for the type of estimate being prepared, and are consistent with comparable projects and estimates.

This level of estimate review helps to ensure that all estimates prepared by the department are using established guidelines, and are presented in a consistent manner from project to project.

Engineering/Design Review
The next level of estimate review should be conducted with the engineering team, and should evaluate the estimate in terms of accurately representing the project scope. The core members of the engineering team are key participants in this review, along with the lead estimator and estimating team.

Completeness of Engineering Deliverables
One of the first items to review is the listing of all drawings, sketches, specifications, and other engineering deliverables used in preparing the estimate to ensure that it is complete (see design basis above). The lead engineers need to cross-reference this listing against their own engineering drawing and deliverables lists to make sure that all relevant information was passed on to the estimating team. The revision numbers of drawings should be checked to ensure that they match the intended revision for the estimate. If late changes to the engineering drawings have occurred, and are intended to be incorporated into the estimate, this needs to be checked to ensure that all late changes have been included.

Equipment List
For those projects involving major equipment, the equipment list and equipment pricing should be double-checked by the engineering team for completeness and accuracy. Equipment is often one of the key drivers of cost and scope, and needs to be checked carefully for completeness and accuracy.

Design Basis of Estimate
The engineers should review the basis of estimate and summary of project scope carefully to verify and correlate their understanding of the project scope with that expressed in the estimate. All exclusions expressed in the BOE should be agreed to; and all allowances and assumptions verified. If an estimator has had any questions about interpretation of the drawings or engineering deliverables, now is the time to discuss the estimator's interpretation with the engineers, and to make sure that the project scope is accurately reflected in the estimate. All drawings used for the estimate should be available during this review. Sometimes, it can help to have the estimator explain how each drawing was used in the preparation of the estimate (i.e., was a hard takeoff performed from isometric drawings; was a quantity developed from a P&ID and plot plan, etc.).

Engineering/Design Costs
The engineering team should also review the assumptions and costs associated with the engineering and design portion of the estimate. The engineering team needs to feel comfortable that the amount of money included in the estimate for engineering, design, and support is adequate for the level of effort expected to be expended on the project.

Risk Basis of Estimate
Lastly, the engineering team should review the risk basis of the estimate, and be in position to agree with the analysis of cost risk associated with the estimate. The level of risk associated with scope definition, and with engineering/design costs should be of particular interest to the engineering team, and concurrence should be sought by the team.
As mentioned, the goal of this portion of the estimate review is to make sure that the scope of the project as understood by engineering is reflected in the estimate. At the end of the engineering review, the estimate should have the full support of the engineering team during subsequent reviews.

Project Manager/Project Team Review
Once the estimate has been reviewed closely by the estimating and engineering teams, it is ready for review by the project manager and the rest of the project team. The objective now is to gain the entire project team's support of the estimate, and especially that of the project manager. This is also the first point where the estimate should be able to pass overall validation tests, in addition to a quality review.

Estimate Documentation
The first part of this review should be the examination of the estimate documentation by the project team and project manager. This includes the basis of estimate, as well as the estimate summary and estimate detail pages. The purpose is to ensure that the estimate is presented in an understandable manner. If standard estimating guidelines have been followed (as discussed above), then all estimates should be presented in a consistent, understandable style. It is very important that the project manager fully understand how the estimate is prepared because he/she often becomes the person responsible for presenting (and defending) the estimate to upper management, and later to the eventual customer. The entire project team should also understand the entire estimate package, format and contents.

Cost Review
Engineering should have already reviewed the engineering, design, and associated support costs. Now is the time for the other key members of the project team (project manager, project controls, procurement, construction manager, commissioning manager, etc.) to examine their respective costs which are included in the estimate, and to obtain agreement that they are correct. Although primarily the responsibility of the estimating team, the scope related costs should also be reviewed by the rest of the project team to gain consensus. In particular, the following areas should be discussed:

- Verify that the latest project schedule agrees with the estimate (particularly as it relates to escalation).
- Examine the project administration, and other home office related costs (engineering/design costs should have already been reviewed) for reasonableness.
- Conduct a final constructability review to ensure that the methods of installation and construction assumed in the estimate are reasonable and cost effective.
- Review the construction indirect costs (field staff, temporary facilities, temporary services, construction equipment and services, construction tools and consumables, etc.) to make sure they are reasonable. And,
- Ensure that all required start-up and commissioning materials are included (if necessary). This is often an area of costs which is overlooked.

For international projects, there may be many more items of cost that should be carefully reviewed. These may include such items as international labor adjustments for productivities and wage rates, adjustments for workweek variations, material cost adjustments for both local and globally sourced materials, international freight costs, international duties and taxes, labor camp costs, premiums for expatriate costs, etc.

Estimate Validation
In most organizations, the project manager is ultimately held responsible for the execution of the project. Therefore, the project manager has a vested interest in performing "sanity checks" or otherwise validating the estimate as reasonable. Most experienced project managers will have various "rules-of-thumb" that they will want to use to verify against the estimate. Regardless, the estimate should include an estimate review "metrics" report which summarizes and compares several key benchmark ratios and factors versus historical (and sometimes estimated) values from similar projects.

The goal is to ensure that key metrics from the estimate are in line with the same metrics from similar projects. If there is a large discrepancy, it must be explainable by the particular circumstances of the estimated project versus the similar completed projects. Such comparison metrics may include values such as: percent of administration (home office) costs, percent of engineering/design costs, equipment to total field cost ratios, equipment to totals project cost ratios, cost per piece of equipment, worker hours per piece of equipment, cost to plant capacity ratios ($/BBL, $/SM), etc. Sometimes the metrics will be generated down to the discipline level where you may look at ratios such as cost per diameter inch of piping, cost per cubic meter of concrete, cost per ton of steel, etc.

In addition to examining key benchmark metrics and ratios, another form of estimate validation may involve preparing a quick check estimate using order of magnitude estimating methods. Again, any large discrepancies between the estimates should be explainable by the peculiarities of the project.

Estimate validation is a very important activity during the project review cycle, and the proper tools need to be in place to allow this to occur. Benchmarking key estimate ratios and metrics depends upon having a project history database in place to collect, analyze and present the required information. Similarly, the capability to provide quick check estimates depends on having the correct strategic and conceptual estimating information and tools ready for use.

Risk Basis of Estimate
The project manager and project team should again review the risk basis of the estimate, and agree with the analysis of cost risk associated with the project. The project manager, in particular, should agree with the risk assessment and contingency amounts, and be able to defend it in subsequent review to upper or corporate management.

Reconciliation to Past Estimates
Lastly, the project manager will usually be interested in reconciliation of the current estimate to the preceding estimate (or estimates). This is an important, but often overlooked, aspect to the overall estimate review process. The current estimate can gain credibility by comparing it with earlier estimates, and clearly explaining the differences and reasons for the differences. The reconciliation can usually be presented at a high level, without excessive detail, but the backup should be available in case it is required during the review.

Management Reviews
The last series of reviews is usually conducted by various levels of corporate management. The number of upper management reviews, and the level of management they are presented to, typically vary with the strategic importance and/or total estimated cost of the particular project. These reviews are typically conducted at a very high level of analysis, and usually do not involve the details of the estimate. Upper management reviews often focus on substantiating the overall adequacy of the estimate in regards to its intended use. In other words, can management be assured that the level of detail available for the estimate, the estimating methods employed, and the skills of the estimating and project teams support their decision making process on whether to proceed.

As with the project manager review, estimate validation is a key element of the upper management reviews. It is important to be able to explain and demonstrate that metrics for the current estimate are in line with data from other similar projects, (i.e., that the estimate is reasonable). It is also important to show where the metrics may be substantially different from other projects, and provide explanations for the differences.

Management will also be interested in the cost risk assessment. It is important to clearly and concisely explain how the contingency amount was developed, and what the levels of risk are. It is then up to management to accept the level of risk indicated, or change the amount of contingency and accept more or less risk for the project. When reviewing the risk analysis, it is always important to discuss the areas of high risk, and what is being done to mitigate those risks.

Up until the management reviews, the estimate review will have typically concentrated on the project as defined by the project scope documents. If the project was built according to the defined project scope alternative, what will it cost? Usually, the recommended alternative for project scope has long since been determined and agreed to by the project team; and the engineering deliverables created for preparing the estimate have been focused on a single design alternative. However, many times management will start asking questions concerning other alternative scopes or designs. One of the certainties is that management will always think the project cost is too high, and will now be probing to determine if there are lesser cost options. Therefore, it is important to have available for the management reviews any earlier design/cost alternatives, and the decision tree leading to the selected design.

The effectiveness of an estimate review relies on the information which is presented, and the manner in which it is presented. The above discussion has concentrated on how to structure a sequence of estimate reviews for internally prepared estimates to ensure that estimates are well documented, consistent, reliable, and appropriate for their intended use. After this review cycle, the level of estimate accuracy should be apparent, reflective of the scope information available for preparing the estimate, and capable of supporting the required decision making process for the project. Next, we will discuss techniques for reviewing estimates prepared by others.

REVIEWING ESTIMATES PREPARED BY OTHERS
The foregoing discussion has focused on structuring an estimate review process for the estimates that we internally prepare to ensure that the estimate is of a high quality and supports the decision making process of our management. Often, we may also find ourselves in a position to review (and/or approve) estimates prepared by others; and that may or may not have gone through a rigorous internal review cycle as described above. When reviewing estimates by others, we always want to keep in mind the basic fundamentals previously described. Complicating the matter, however, is the problem that many times the amount time allowed for a complete estimate review is very short. Thus, the review of an estimate prepared by others is usually accomplished by a critical

assessment of the estimate and its documentation, and a series of questions to assist in evaluating the level of diligence used in preparing the estimate. The following discussion centers on guidelines that we can use to efficiently review estimates prepared by others.

Basis of Estimate
The first thing to assess is the basis of estimate. Is it well organized and complete? Does it provide the required information regarding the design basis, planning basis, cost basis, and risk basis of the estimate? Does the design basis clearly document the scope of the project, and have all engineering deliverables used in developing the estimate been identified? Have all scope assumptions been acknowledged? Is the planning basis (schedule, resource plan, construction plan, etc.) reasonable? Is the basis of cost (material prices, labor rates, labor productivities) reasonable, in line with expectations, and consistently applied throughout the estimate? Has the risk basis been clearly defined, and is it reasonable for the level of information available to prepare the estimate?

Estimating Personnel Used
Next, you will want to know who prepared the estimate, and what their level of estimating experience is. Do they have established estimating procedures and guidelines? Was the estimate checked and reviewed before publication?

Estimating Methodology and Procedures
What estimating methods, techniques and procedures were used in preparing the estimate? Are they appropriate for the level of information available and project type? Were different estimating methods used for different parts of the estimate? Is the level of detail in the estimate sufficient for the purpose of the estimate? Were parts of the project difficult to estimate, and why? Was sufficient time available to prepare the estimate? What adjustments were made to the estimate for location, complexity, etc., and are they reasonable? Was the estimate prepared using a code of account structure?

Estimate Documentation
Is the estimate documented clearly? Are the estimate summary and detail pages well organized, and presented at an appropriate level of detail? Is every cost appearing on the estimate summary traceable to the estimate detail and other estimate backup?

Estimate Validation
Hopefully, the estimate for review will include a metrics report showing key estimating metrics and benchmark ratios for the estimate and similar past projects. You should review this report, and question any significant differences. You should also have your own set of metrics and statistics from your own project history to compare against.

At this point, you may also develop your own quick check estimate (using conceptual estimating techniques) for comparison purposes. This is always a good technique to see if the estimate being reviewed is reasonable. If there is a significant difference, then question the estimator and listen to their explanations and opinions for the deltas. Significant differences between the check estimate and the estimate being reviewed may indicate the need for taking a more thorough examination of the estimate detail.

Estimate Detail
If the preceding inquiry (or should we say interrogation) has gone well, and you are confident that the estimate appears to have been prepared in a professional manner, you are ready to delve into some of the estimate details to verify estimate quality. The goal is to check that selected areas of the estimate can withstand further scrutiny. The key here is to not get too deep into the details, and lose sight of the forest for the trees.

An important point to remember here is the 80/20 rule. This principle generalizes that 80 percent of the cost will come from 20 percent of the estimate line items. For any particular estimate, the significant cost drivers may vary. Sometimes, the main cost driver may be a particular process unit of the project; other times it may be the type of process equipment or machinery throughout the project, and still other times it may be the overall bulk material quantities or labor worker hours. You should examine the estimate summary and detail pages closely to ascertain which aspects of the estimate you may want to examine in closer detail. Basically, you should examine in detail those items of the estimate which will have the most significant cost impact if estimated incorrectly.

One review technique which is often employed is to thoroughly examine and review the estimating steps which were used for a particular part of the estimate. Select an area of the estimate, and ask how the quantities were derived. Don't just take their word for it however. Ask the estimators to show you the drawings from which the quantities were generated. Perform a quick takeoff to see if the quantities can be verified. Ask what the basis was for the unit material price and labor workhours. Have these been consistent throughout the estimate? What adjustments were made and why? If the answers to your questions are evasive, it may call into question the credibility of the entire estimate, and a more thorough review of the complete estimate may be necessary. If your questions are answered confidently, and the answers can be verified against the engineering deliverables and scope information, then you may decide to check the rest of the estimate details in a more cursory fashion.

Typically in this situation, once you have shown the wherewithal to compel the estimator to back up any claims or explanations, then he or she discovers he/she can't just "pull the wool over your eyes." From that point forward, you will usually find that you are getting honest answers to your questions.

The goal of an estimate is to predict the probable cost of a project. The goal of an estimate review is to determine that a high quality and sufficiently accurate estimate has been prepared. The review should ensure that the proper estimating methods, procedures, techniques, data, and guidelines have been employed in the preparation of the estimate. The use of a structured estimate review cycle and estimating review techniques will help to ensure that quality estimates are consistently prepared which effectively support the decision making process by management.

Presenting the Estimate

The method in which you present an estimate to your customer (internal company management or external client) is extremely important. An estimate should never be presented as just a list of numbers, or estimating calculations. A number (or even a range of numbers) is meaningless without the supporting information that describes what the number represents, and sometimes even more importantly what it doesn't represent. In general, a complete estimate report will include the following:

- Basis of Estimate (BOE)
- Estimate Summaries
- Estimate Detail
- Estimate Benchmarking Report
- Estimate Reconciliation Report
- Estimate Backup

We have previously talked about the basis of estimate in the prior discussion on estimate reviews. This is a critically important document in describing the scope that is represented by the estimated cost; and in conveying all the assumptions that have been embedded into the estimate. A well-written basis of estimate document can go a long ways towards providing confidence in the estimate itself.

Typically, various estimate summaries may be prepared according to the Project Work Breakdown Structure. For example, one estimate summary may be prepared by project area, and then broken down by process system; while another summary may be prepared by process system and then broken into project areas within each process system. The various parties interested in the estimate will all have different ways in which they want to see the estimate summarized, depending on the classification and end use of the estimate being prepared. It is very important that every value appearing on an estimate summary be easily tracked back to the estimate detail.

The estimate detail typically shows all of the individual Cost Estimating Relationships (CER's) used in preparing the estimate. For a conceptual estimate, it may be a page or less of calculations; however, for a large detailed estimate it may include hundreds of pages of individual line items. This report is also prepared according to the Project Work Breakdown Structure, and may be provided in a variety of different sort options.

An Estimate Benchmarking Report will often be included. It should show benchmark information and metrics with other similar projects. For example, for a building estimate, this report may show the cost per square meter of building area ($/SM) compared to recent similar projects. The key benchmark metrics and ratios presented may include:

- Project administration costs as percent of total project cost.
- Engineering costs as percent of total project cost.
- Ratio of equipment cost to total project cost.
- Construction labor as percent of total field cost.
- Total field costs as percent of total project cost.
- Project cost per unit of capacity.
- Average composite crew rate by trade.

An estimate reconciliation report should also be prepared which reconciles the current estimate with any previous estimates prepared for the same project. This report should identify the cost differences dues to changes in scope, changes in pricing, changes in risk, etc.

Lastly, all estimate backup should be compiled and available. This information may not need to be presented to the estimate customer, but should be available if questions arise. This will include all notes, documentation, drawings, engineering deliverables, vendor quotes, etc. that were used in preparation of the estimate.

ESTIMATING RESOURCES

Reliable estimate preparation depends on information. Besides the engineering and design information needed to quantify the scope of the project, other information is also required such as:

- conceptual estimating factors
- material cost and pricing information
- labor workhour charts and information
- labor productivity information
- labor wage rates, composite crew mixes, etc.
- other estimating factors and information

Successful estimators will rely on a myriad of resources to obtain this information. Estimating guideline and procedure manuals will be used to promote standard estimating methods and procedures. In-house cost history manuals will provide historical cost data for completed projects. Special cost studies may have been developed to serve as resources for particular estimating applications, such as special scaffolding studies, concrete placement studies, labor productivity studies, etc. Engineering and design manuals and specifications will be used to identify the specific materials of construction, and all related labor operations required to complete the scope of work.

Every completed project should be documented by a final job report covering everything about a project from design considerations to construction execution strategy to cost summaries. Selected data should from the final job reports should be collected and stored in a computerized database and made available to all estimators. Estimators continually rely on past project information and cost data in the preparation of new estimates.

Collections of labor charts will typically provide standard labor workhour units by task. These are generally normalized for location and time, and serve as a base for estimate preparation, and then adjusted for specific project requirements. They will often be supplemented by commercial estimating database publications. In-house and commercial material cost databases and publications will also be needed. Current wage rate information should be maintained, including union agreements, for all locations the estimator may be involved with. A library of vendor catalogs should also be maintained. Many of these are now available on the internet. These may provide technical information, pricing information, and other data required by an estimator.

There are hundreds of sources published every year that contain useful information for an estimator. This includes AACE publications (Recommended Practices, Professional Practice Guides, Cost Engineering journal, etc.), as well as publications from other professional organizations and commercial sources.

Estimating software is another important resource. Estimating software can enhance the accuracy and consistency of estimates, while reducing the time required for preparing estimates. The software may be commercial estimating software or be developed in-house. When using the cost databases supplied with commercial estimating software, it is always important to calibrate the data to your specific needs and estimating situations. Estimating software should also be regarded as simply a tool to facilitate the preparation of estimates by estimators. Estimating software doesn't convert a non-estimator to be an estimator.

All of the resources described above serve to help the most important resource to successful estimating –well trained and experienced estimators. Estimating is a profession requiring an ongoing commitment to training and development.

CONCLUSION

As potential projects are considered as investment opportunities, management will require various estimates to support key decision points. At each of these points, the level of engineering and technical information available to prepare the estimate will change. Accordingly, the techniques and methods to prepare the estimates will also vary. The basic estimating techniques are well established, and this chapter has been intended to review the estimating process and relevant estimating methodologies for the various types of estimates.

The determination of using a conceptual approach versus a detailed approach will depend on many factors: the end use of the estimate, the amount of time and money available to prepare the estimate, the estimating tools available, and the previous historical information available. A conceptual estimating approach (capacity factored, equipment factored, parametric) requires a significant effort in data-gathering and methods development before estimate preparation ever begins. In contrast, a detailed estimating approach requires a large effort during the actual preparation of the estimate.

With either approach, the challenge for the estimator is to evaluate the unique combination of required material and labor resources in order to prepare a cost estimate for a project to be completed in the future. The use of structured estimating techniques and tools, high quality engineering deliverables, good historical data and pricing information, combined with estimating skill and experience, will assure that the best possible estimate is prepared. The desired end result is to prepare estimates that are well documented, consistent, reliable, appropriate, accurate, and that support the decision making process for the project.

Estimating is obviously a vital component to project success. Estimates are used not only to establish project budgets, but also to provide accurate information to support scheduling, cost monitoring, and progress measurement of a project during execution. Estimating is thus but one component to total cost management – the integration of cost engineering and cost management principles used in managing the total life cycle cost investment in strategic assets.

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2. AACE Recommended Practice 20R-98, *Project Code of Accounts*. Morgantown, WV: AACE International
3. AACE Recommended Practice 20R-98, *Project Code of Accounts – As Applied in Engineering, Procurement, and Construction for the Process Industries*. Morgantown, WV: AACE International
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Chapter 10—Process Product Manufacturing[5]

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Abstract
In process/product manufacturing, the cost professional needs to consider the type of operation involved and its associated cost impacts. Relevant costs include labor, raw materials, byproduct (credits or debits), operating supplies, utilities, supervision, maintenance, royalties, rentals, insurance, overhead, and taxes associated with the operation. Another cost which is non-cash cost, but reduces income taxes, is government-allowed depreciation charges related to the capital investment. Government-allowed depreciation methods vary by country; therefore, tax laws will require analysis depending upon the location of the investment. Operational levels, whether at full capacity or less than that, will have an impact on unit costs.

Keywords
Accelerated Costs, Contingencies, Credits and Debits, Depreciations, Operating Cost Estimates, and Process Manufacturing.

INTRODUCTION

Definitions

For terminology in this Chapter, please refer to the AACE Recommended Practice 10S-90, *Cost Engineering Terminology*, available online at www.aacei.org.

To perform an operating or manufacturing cost estimate properly, and to determine the potential profitability of a process, all costs must be considered in certain specific categories. The distinction between the various categories is quite important as they are treated differently for purposes of calculating taxes and profitability.

- **Direct costs** are those which tend to be proportional, or partially proportional, to throughput or production. These costs include materials (raw materials, utilities, operating supplies, etc.) and labor (direct operating labor, operating supervision, maintenance supervision, payroll burden on labor charges, etc.). Direct costs include both variable costs and semi-variable costs. *Variable costs* are those, such as raw materials and by-product credits, which tend to be a direct function of production or plant output. In other words, these costs are maximized when a plant operates at full capacity and are not incurred when the plant is not operating.
- **Semi-variable costs** are direct costs which are only partially dependent upon production or plant output. These include such items as direct labor, supervision, maintenance, general expense, and plant overhead. While these costs increase with production rate, they do not increase in direct proportion to production rate. When the plant is not operating, a portion of the semi-variable costs continues to be incurred. At zero production or put-put, semi-variable costs generally total about 20 to 40 percent of the total semi-variable cost at full production.
- **Indirect costs**, or **fixed costs**, are those costs which tend to be independent of production. These costs are incurred whether or not production rates change. They include plant overhead or burden (plant administration, laboratory, shops and repair facilities, property taxes, insurance, etc.). Indirect costs include depreciation.
- **Contingencies** constitute an allowance which must be made in an operating cost estimate for unexpected or unpredictable costs or for error or variation likely to occur in the estimate. A contingency allowance is just as important in the operating cost estimate as it is in the capital cost estimate.
- Another operating and manufacturing cost category is **distribution costs**. These are the costs associated with shipping the products to market. They include containers and packages, freight, operation of terminals and warehouses, etc.
- **General and administrative expenses**, or G & A expenses, are costs which are incurred above the factory or production level and are associated with management. This category includes marketing and sales costs, salaries and expenses of officers and staff, accounting, central engineering, research and development, etc.

It is important in performing an operating cost estimate not to confuse general and administrative costs with *general work expense.* The latter item is a synonym for the fixed-cost factory overhead items, e.g., property taxes and insurance.

Learning Objectives

After completing this chapter, the reader should be able to:

- Understand how to determine the operating and manufacturing costs of a continuous process on a conceptual basis.
- Distinguish between direct and indirect costs in manufacturing as compared to construction.
- Relate operating costs at full production to reduced costs at less than full plant capacity.

[5] Excerpted by permission from *Basic Cost Engineering*, 3rd edition, by K.K. Humphreys and P. Wellman, Marcel Dekker, Inc., New York 1996

- Understand depreciation rules and their relationship to operating and manufacturing costs.

Types of Operating Cost Estimates and Estimating Forms

As is true for a capital cost estimate, the purpose of an operating cost estimate is the controlling factor in determining the type of estimate to be performed. Preliminary or order-of-magnitude estimates are often used to screen projects and to eliminate uneconomical alternatives. More detailed estimates are then applied when the screening process has reduced the choice to a relatively few alternatives. As more information becomes available, detailed estimates become feasible.

In performing the operating cost estimate, particularly on a preliminary basis, good judgment is necessary to avoid excessive attention to minor items which, even if severely over or under-estimated, will not have a significant effect on the overall estimate. Pareto's Law of Optimality also known as the "80-20 Rule" should govern here. Pareto's Law of Optimality states that 80 percent of the costs will typically be derived from 20 percent of the items. The estimator focuses primary attention on those high-cost areas for increased emphasis.

In performing operating cost estimates, it is also necessary to calculate costs at reduced production rates, as well as at design capacity. Operating costs are decidedly nonlinear with respect to production rate.

This fact and the fact that virtually no plant or process operates all the time at full design production rate makes it imperative that reduced production rates be considered. This subject is discussed in considerable detail later in this chapter.
Finally, when estimating the effect of changes or additions to an existing process, the cost analysis should be performed on an incremental basis to evaluate the effect of change, as well as on an overall basis to determine if the entire project is worthy of being continued even without change. Frequently a process change will not be economical, but the total project will be attractive. In other cases, the incremental costs of change will appear to be quite profitable; however, this profit will not be enough to offset losses entailed in the existing portion of the plant. Thus, both types of analyses must be made.

Operating cost estimates can be performed on a daily, unit-of-production, or annual basis. Of these, the annual basis is preferred for the following reasons:

- It "damps out" seasonal variations.
- It considers equipment operating time.
- It is readily adapted to less-than-full capacity operation.
- It readily includes the effect of periodic large costs (scheduled maintenance, vacation shutdowns, catalyst changes, etc.)
- It is directly usable in profitability analysis.
- It is readily convertible to the other bases, daily cost and unit-of-production, yielding mean annual figures rather than a potentially high or low figure for an arbitrarily selected time of year.

A basic flow sheet of the process is vital to preparation of an estimate. This flow sheet should detail to the maximum extent possible the quantity, composition, temperature, and pressure of the input and output streams to each process unit.

In addition, to properly prepare an operating or manufacturing cost estimate, a standardized estimating form should be used to assure that the estimate is performed in a consistent manner and to avoid omitting major items. Figure 10.1 is an example of a suitable form for this purpose.

						Date:
						By:
Location:			Product (s):			
Capital Investment			Process:			
Total:			Nelson Index:		CE Index:	
Less working capital:			M&S Index:		Annual:	
Operating						
Less salvage value:			ENR Index:		Days:	
Depreciation investment:			Annual production:			
	Raw Materials	**Annual quantity**	**Unit cost**	**$/year**		**$/**
(1)						
(2)						
(3)						
(4)						
(5)						
	Gross raw material cost (sum of lines 1 to 4):					
Miscellaneous credits and debits						
(6)						
(7)						
(8)						
(9)	Total debit (credit) (sum of lines 6 to 8):					
(10)	Net raw material cost (lines 5 + line 9):					
Direct expense		Unit	Quantity	Unit cost	$/year	$/
(11)	Steam	M lb.				
(12)	Water ( )	M gal.				
(13)	Water ( )	M gal.				
(14)	Electricity	kW-hr.				
(15)	Fuel ( )					
(16)	Fuel ( )					
(17)	Labor					
(18)	Supervision					
(19)	Maintenance					
(20)	Factory supplies					
(21)	Indirect overhead					
(22)	Payroll overhead					
(23)	Laboratory					
(24)	Contingencies					
(25)	Total direct conversion cost (sum of lines 11 to 24):					
Indirect expense						
(26)	Depreciation					
(27)	Real estate taxes and insurance					
(28)	Depletion allowances					
(29)	Amortization					
(30)	Total indirect conversion cost (sum of lines 26 to 29):					
(31)	Total conversion cost (line 25 + line 30):					
(32)	Total operating cost (line 31 + line 10):					
(33)	Packing and shipping expense:					
(34)	TOTAL COST FOR PLANT (line 32 + line 33):					

Figure 10.1—Typical Production Cost Estimating Form

The estimating form acts as a checklist and as a device for cost recording and control. It must include the date of the estimate, the capital investment information which was previously determined, an appropriate cost index value reflecting the date of the capital cost estimate, the plant location, plant design capacity, annual anticipated plant operating days and/or annual production rate, and the plant or product identification. In any event, if a form is not used, the cost engineer should be equipped with a checklist and should be familiar with the technical aspects of the process. An estimate should never be made without specific technical knowledge of the process. Last, wherever possible, cost data used in the estimate should be obtained from company records of similar or identical projects (with adjustment for inflation, plant site differences, and geography). For preliminary estimates, company records are probably the most accurate available source of cost data.

If not available from company records, cost data may also be obtained from literature sources. However, such data is not always reliable. Published information must always be used with care. It is often inadequately explained and frequently improperly dated. Date of publication is meaningless, because the data may be months or years old and may require adjustment to current cost levels. Too often it seems that in the rush to complete an estimate, people will grasp any number they can find without fully understanding how it was derived or what it represents. Differing data sources may include or exclude certain cost elements.

Cost of Operations at Less Than Full Capacity

The preceding discussion emphasized the necessity of performing operating and manufacturing cost estimates both at full plant capacity and at conditions other than full capacity. Frequently, the inexperienced estimator will perform an estimate assuming operations only at full design capacity. This approach is totally erroneous as it does not consider unscheduled downtime, market fluctuations in product demand, time required to develop markets for a new product, and other factors.

Figure 10.2 is an illustration of cost effects of operation at less than full capacity. This figure takes into account the fixed, variable, and semi-variable costs discussed earlier.

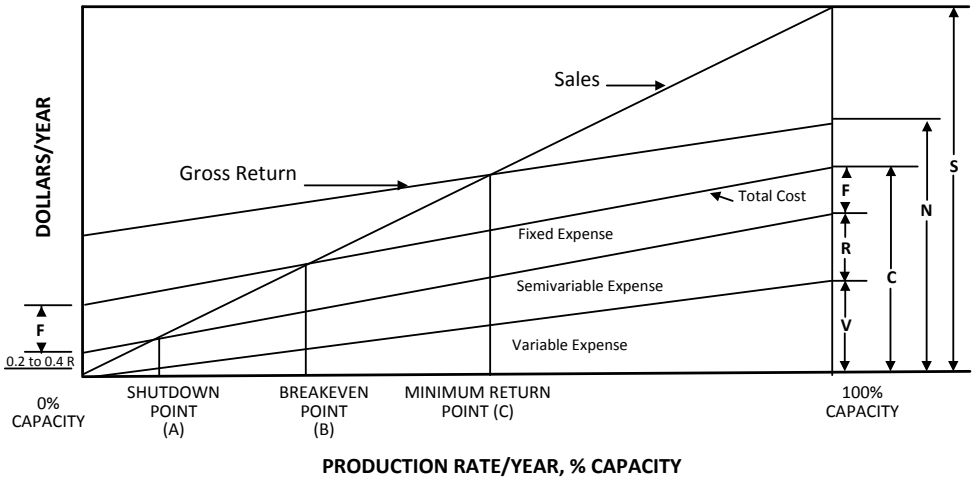


Figure 10.2—Cost Effects of Operations at Less Than Full Plant Capacity

Semi-variable costs, those which are partially proportional to production level, may include, among others, the following:

- Direct Labor
- Supervision
- General Expense
- Plant Overhead

Other costs which may be semi-variable, depending upon individual circumstances, are royalties and packaging. Packaging may be either variable or semi-variable depending upon the particular situation.

Royalties may be variable, semi-variable, fixed, or even a capital expense. Thus, they must be carefully examined to be certain that they are included in the proper cost category. A royalty fee, which is paid in a lump sum, should be capitalized. Royalties which are paid in equal annual increments are treated as fixed costs. Those paid as a fee per unit of production or sales are variable costs, and those which are paid in a sliding scale, (i.e., at a rate per unit of production which declines as production increases), are semi-variable. In certain cases, royalty agreements may contain elements of more than one cost category—for example, an annual fee (fixed) plus a charge per unit of production (variable).

Fixed cost items, in addition to royalties if applicable, include the following:

- Depreciation (a non-cash cost that reduces income taxes)
- Property Taxes
- Insurance

Variable costs generally include the following:

- Raw Materials
- Utilities
- Royalties (if applicable)
- Packaging (if applicable)
- Marketing
- Catalysts and Chemicals

Figure 10.2 graphically demonstrates the implications of operating at less than full capacity. In this figure, at 100 percent of capacity, the following apply:

- The fixed expense = **F**
- The variable expense = **V**
- The semi-variable expense = **R**
- Total operating cost = **T**
- Sales income = **S**
- The income required to achieve the minimum acceptable return on investment before taxes for the capital investment = **N**

As can be seen from the figure, the variable expense declines to zero at zero percent of capacity, fixed expense is constant, and semi-variable expense declines at zero percent of capacity from 20 to 40 percent of its value at full capacity.

This simple plot is used to determine the following:

- The minimum production rate at which the desired return on investment will be achieved, (C).
- The breakeven point, that point at which income will exactly equal total operating cost, (B).
- The shutdown point, that point at which it is advisable to shut down the plant rather than operating at lower production rates, (A).

The plot readily identifies the range of production rates at which the following apply:

- The return on investment will equal or exceed the desired minimum (all production rates ≥ C).
- The return on investment will be less than the desired value, but will be greater than zero (production rates < C but > B).
- The process will result in a loss, but losses will be minimized by continuing to operate the plant rather than shutting it down (production rates ≤ B but > A).
- Losses will become so large that it is less expensive to close the plant and pay fixed expenses out of pocket, rather than to continue operations (production rates ≤ A).

The breakeven and shutdown points can also be determined mathematically as follows:

$$B\ (breakeven\ point) = \frac{(\mathrm{F} + \mathrm{nR})}{\mathrm{S} + \mathrm{nR}(1 + \mathrm{n})\mathrm{R}}$$

Equation 10.1

$$A\ (shutdown\ point) = \frac{\mathrm{nR}}{\mathrm{SRVR}(1\mathrm{Rn})\mathrm{R}}$$

Equation 10.2

Where:

n = decimal fraction of semi-variable costs incurred at 0 production (usually about 0.3)

Similarly, the total cost line can be expressed as:

$$C_p = [V + (1 - n)R]p + F + nR$$

Equation 10.3

Where:

C_p = total cost at production rate p
p = actual annual production rate as a fraction of plant capacity

Since total annual sales are proportional to production (assuming no stock-piling of production), and therefore have no value at zero output, the Equation for the sales line is:

$$S_p = (S\ x\ p)$$

Equation 10.4

Where: S_p = sales income at production rate p.

The point when the sales and total cost lines cross is the breakeven point for the plant and is equal to the level of output at which sales is equal to total cost.

RAW MATERIALS COSTS

Depending upon the particular process, raw materials costs can constitute a major portion of operating costs. For this reason, a complete list of all raw materials must be developed using the process flow sheet as a guide. In developing the raw materials list, the following information must be obtained for each raw material:

- Units of purchase (tons, pounds, etc.)
- Unit cost
- Available sources of the material
- Quantity required per unit of time and/or unit of production
- Quality of raw materials (concentration, acceptable impurity levels, etc.)

In estimating the quantity of each raw material, appropriate allowance must be made for losses in handling and storage, process waste, and process yield.

Price data for purchased raw materials are generally available to a high level of accuracy from many sources. Purchased price information can generally be obtained from the suppliers. Alternatively, supplier catalogs and price lists can be used, as can published data. The *Wall Street Journal*, European Chemical News, and similar trade and business publications are all good sources of spot data.

In estimating the cost of any raw material, it must be remembered that in general raw material costs vary with quality (concentration, surface finish of metals, impurities, etc.) and generally decrease in unit cost as quantity increases. There is little sense, for example, in relying on a cost figure for reagent-grade hydrochloric acid in five pound bottles, when the process can use much-lower-cost commercial-grade hydrochloric acid (muriatic) in tank car or tank truck lots.

Another major factor to be considered is availability of the raw material. Does sufficient productive capacity exist such that the market can supply the demands of the proposed process? A sudden large new demand for a raw material can, and often does, cause substantial price increases, particularly if the material is available only in small quantities, or as a by-product of another process.

In pricing raw materials, it must also be remembered that the prices are generally negotiated and the discounts obtained can result in prices substantially less than quoted or published data. Where available, company experience in negotiating supply orders for the same or similar materials should be used to estimate the amount of any such probable discounts.

A common pitfall in operating and manufacturing cost estimates is to neglect the cost of raw materials manufactured, or obtained in-house, or from another company division, because they are not purchased. Such raw materials do represent a cost to the company and have value in the estimate; therefore, they should be included as a cost at their market value or company book value. The market value to be used is the going market price, corrected for any direct sales costs which are not incurred as a result of internal use. In addition, internal company freight, handling, and transfer costs must be added.

If the captive raw material is an intermediate product which has no established market price, the cost should be based upon the value of the nearest downstream product or material for which an established market price exists. The cost is equal to the value of the downstream product, less direct sales costs not incurred, plus internal company transfer costs, less manufacturing costs for operations avoided by using the intermediate product instead of processing it further.

Other raw materials cost items which are easily overlooked are fuels which are used as raw materials (e.g., natural gas in methane conversion processes) and periodic makeup of losses to catalysts and other processing materials. In the case of catalysts and similar materials, the initial fill of such items is usually treated as a capital expense, if its useful life exceeds one year. Otherwise, it is considered as a start-up expense.

Fuels which are used as a raw material should be treated in the same manner as any other raw material. Those which are used for utility purposes should be treated as a utility cost.

Finally, in estimating raw materials costs it must be recognized that prices are usually quoted Freight on Board or Free on Board (FOB) to the supplier's plant or basing point, not at the point of use. Thus, freight to the point of use and local handling costs must be added to the quoted prices.

Current freight tariffs for each commodity should be checked in preparing the estimate, since they are complex and often illogical and cannot be generalized.

By-Product Credits and Debits

All process by-products, including wastes and pollutants, must be considered in the operating cost estimate. Thus, every output stream shown on the process flow sheet must have a cost assigned to it. Obviously, these costs may be credits (in the case of salable or usable by-products) or debits (in the case of wastes or unsalable by-products).

Items which are not immediately obvious as by-products are nuisance expenses. No longer can a plant discharge pollution at will into the air or water; nor can high noise levels, thermal discharges, odors, etc., remain unabated. These are all by-products, albeit undesirable ones, and the cost of treating them must be included in the estimate—either in the overall process costs, or as a by-product debit. Similarly, the capital cost estimate must include the costs of associated pollution and nuisance abatement equipment.

Nuisance costs are also not confined to things which leave or are discharged from the plant. In many cases, nuisances which are entirely confined to the plant premises, (e.g., high noise levels) must be eliminated because of safety and employer liability considerations.

Sellable by-products have values which can be established in the same manner as determining the cost of raw materials. The by-product credit then may be estimated from the market prices or anticipated selling prices of the by-products; less any costs of processing, packaging, selling and transporting them to market. In many cases, it is necessary to do a complete capital and operating cost estimate on the by-product processing facilities in order to determine the latter costs.

If the by-product is not sold but instead is converted to another salable product, it is valued at the market value of the subsequent product less the associated conversion and transfer costs. Wastes similarly carry the negative value associated with their treatment and disposal. Again, it may be necessary to perform another complete capital and operating cost estimate to determine these costs.

Credits should be taken for by-products with great care. Many processes have been economic failures because assumed by-product credits were not realized. In many cases, the production of a by-product can glut its market, particularly if the current market is a small one. Similarly, introduction of the by-product into competition with other materials can depress prices as a result of competitive pressures.

A notable example of this type of situation is the sulfur market. In the mid-1960s, sulfur was in short supply and carried an inflated price. At that time, many estimates were being made on pollution-control devices and systems which would produce sulfur as a by-product and were justified on the basis of the high by-product credits for sulfur. However, as a result of improved methods of sulfur production and the additional sulfur being produced as a by-product, in a very short period of time the market for sulfur and sulfur products, such as sulfuric acid, changed to a state of oversupply and depressed prices. As a result, many supposedly profitable processes actually became uneconomical.

Utility Costs

The estimation of utility costs, particularly in light of rapidly increasing energy costs, is another critical area of operating cost estimation.

In estimating utility costs, it is necessary first to determine the requirements for each utility, including a reasonable allowance for nonproduction items—such as plant lighting, sanitary water, etc. An allowance should also be made for miscellaneous usage and contingencies. Table 10.1 is a typical utility summary of the nature that is required.

Table 10.1—Typical Utility Summary

	Power (kWh/hr)	Water required (gpm)	Water recovery and makeup (gpm)
Mine	1,830	1,000	
Crushing plant	1,920		
Concentrating plant	18,550	38,650	36,720
Pelletizing plant	4,050	1,330	
Subtotal:	**26,350**	**40,980**	**36,720**
Utilities:			5,580
Makeup water	400		
Plant lighting	300		
Sanitary water	150	500	
General facilities	150	300	
Misc./contingencies	150	520	
Total	**27,500**	**42,300**	**42,300**

In addition to the utility summary, consumption patterns should be examined to determine if consumption will be at a uniform rate during each day and from day to day. If consumption rates fluctuate, utility pricing may be based not only on total consumption but also in the peak demand rate; and, in some cases, the time of day which the peak occurs. A further consideration is whether or not the utility must be available on an uninterruptible basis. If the process has an alternate utility source or can tolerate periodic cutbacks in utility supplies, rates may be available which are somewhat lower in cost than uninterruptible rates. The estimate should also consider the fact that in certain areas, uninterruptible rates are not available and electric power, natural gas, etc., can be curtailed by the utility companies at any time if residential demands exceed available supplies. In these situations, both the capital and operating cost estimates must consider standby alternative utility sources; such as emergency power generators, combustion units which can operate on alternate fuels, etc.

In general, utility costs decrease as demand increases (although there is a growing tendency toward national and state policies to increase unit cost with quantity artificially, in order to encourage reductions in consumption).

Electric power charges are usually based upon a demand factor—the maximum power draw during a 15- to 30-min period in any given month. A load factor is computed as the ratio of average usage to the demand factor and rates are established to give preference to high load factors (i.e., steady consumption). The rate schedules also generally include an escalation factor which is tied to increases in fuel costs.

Electric rates in the past were remarkably stable for many, many years, and it was common estimating practice to arbitrarily assume a cost of the order for a cent or two per kW-hr for preliminary estimating purposes. This is no longer true, and no generalization can be made about electric rates. The estimator must obtain current rates from the utility companies serving the proposed plant and can no longer safely assume any "rule-of-thumb" figure for power costs. If the power is not purchased and instead is obtained from a captive company-owned generating system, utility costs must be based upon a study of the system itself.

Natural gas prices depend on quantity required. Steam costs are dependent upon many factors including pressure, cost of fuel, temperature, credit for heating value of condensate, etc. If company data is not available on steam cost, it must be estimated taking into account fuel cost, boiler water treatment, operating labor, depreciation on investment, maintenance and other related costs of steam production. Black has suggested that steam costs can be approximated as two to three times the cost of fuel [1].

Water costs are highly variable depending upon the water quality needed and the quantity required. Purification costs, if contamination occurs before disposal, must also be included, as must cooling costs if the process results in heating of process water. In most jurisdictions, water may not be discharged into streams, or the natural water table, unless it is equal or better in quality and temperature as when it was withdrawn from the stream.

Fuel costs vary with the type of fuel, the Btu value of the fuel, and the source of supply. Careful consideration should be given in the estimate, not only to fuel cost but to the type of firing equipment required, and to required fuel storage facilities. Often, these factors can rule out what would otherwise be the least expensive available fuel.

Another factor to be considered is that, as mentioned earlier in this chapter, certain fuels, although lower in cost than alternative fuels, may not be available in sufficient quantities, if at all. In some cases, a fuel may be abundant in warm weather, but be in short supply during the heating season, necessitating use of alternate fuel supplies, or planned production cutbacks, or stoppages, during periods of severe winter weather.

In estimating utility requirements, equipment, efficiency losses, and contingencies must be considered. Utility consumption generally is not proportional to production, as a result of economies of scale and reduced energy losses per unit of volume or

production on larger process units. Black has suggested that utility consumption varies to the 0.9 power of capacity, rather than in direct proportion to capacity [1].

Last, while not normally thought of as a utility, cost of motor fuels and greases for all mobile equipment must be estimated. Fuel costs are based upon annual operating hours for each piece of equipment, times fuel consumption per hour, times the prevailing cost for the fuel to be used. Greases and lubricant costs are directly related to fuel costs, and for most types of equipment amount to approximately 16 percent of the fuel cost.

Labor Costs

Labor costs, particularly in a labor-intensive process, may be the dominant cost factor in an operating or manufacturing cost estimate. To properly estimate these costs, a staffing table must be established in as detailed a manner as possible. This table should indicate the following:

- The particular skill or craft required in each operation.
- Labor rates for the various types of operations.
- Supervision required for each process step.
- Overhead personnel required.

It is not always possible to determine the extent of supervision and overhead personnel which are required. In such cases, alternate methods of estimating these costs may be used. These are discussed later in this book. However, if sufficient data is available, these factors should be included in the staffing table for maximum estimate accuracy.

Table 10.2 is a typical staffing table for a complete estimate. This table illustrates the detail which is required in a complete staffing table. Note that the table includes general and administrative personnel, production workers, maintenance workers, and direct supervision.

Table 10.2—Typical Complete Staffing Table

	Shifts per day	Days per year	# of labor personnel	Wages for labor ($)			# of salaried personnel	Salaries ($)		Total wages or salaries
				Hour	Day	Year		Per person	Total	
GENERAL										
Administrative & services										
General supt.	1						1	80,000	80,000	80,000
Chief engineer	1						1	60,000	60,000	60,000
Personnel officer	1						1	36,000	36,000	36,000
Personnel clerk	2						2	24,000	48,000	48,000
Payroll clerk	4						4	40,000	160,000	160,000
Accountant	3						3	36,000	108,000	108,000
Guard	6	365	8.4	11.04	529.92	193,600				193,600
Stock clerk	6	365	8.4	11.04						193,600
Safety engineer	1						1	48,000	48,000	48,000
Chief chemist	1						1	52,000	52,000	52,000
Drafter	3						4.2	28,000	117,600	117,600
Sample collector/tester	12						16.8	36,000	604,800	604,800
Stenographer	3						3	24,000	72,000	72,000
SUBTOTAL									**1,386,400**	**1,773,600**
MINE										
Administrative & services										
Mine supt.	1						1	72,000	72,000	72,000
Mining engineer	2						2	56,000	112,000	112,000
Geologist	1						1	48,000	48,000	48,000
General foreman	1						1	52,000	52,000	52,000
Rod carrier	2	260	2.0	12.04	192.64	50,000				50,000
Clerk	2						2	28,000	56,000	56,000
Warehouser	3	260	3.0	13.04	312.96	81,200				81,200
Janitor	2	260	2.0	11.04	176.64	46,000				46,000
Party chief	1						1	48,000	48,000	48,000
SUBTOTAL						**177,200**			**388,000**	**565,200**
PRODUCTION										
Shovel operator	9	260	9.0	18.36	1321.92	343,600				343,600
Shovel oiler	6	260	6.0	12.04	577.92	150,400				150,400
Truck driver, haulage	24	260	24.0	14.04	2695.68	700,800				700,800
Powderer	1	260	1.0	13.72	109.76	28,400				28,400
Powderer helper	2	260	2.0	12.04	192.64	50,000				50,000
Truck driver, service	6	260	6.0	13.04	625.92	162,800				162,800
Operator, crane	3	260	3.0	13.04	312.96	81,200				81,200
Operator, drill	9	260	9.0	17.07	1226.88	318,800				318,800
Operator helper, drill	9	260	9.0	12.04	866.88	225,200				225,200
Operator, tractor	9	260	9.0	13.40	964.80	250,800				250,800
Laborer	18	260	18.0	11.72	1687.68	438,800				438,800
Shift foreman	3						3	52,000	156,000	
Dumper	3	260	3.0	11.40	273.60	71,200				71,200
Operator, front end loader	1	260	1.0	13.72	109.76	28,400				28,400
Operator, primary crusher	3	260	3.0	13.56	325.44	84,400				84,400
General laborer, primary crusher	4	260	4.0	11.04	353.28	91,600				91,600
SUBTOTAL						**3,026,400**			**156,000**	**3,182,400**

Once the staffing table is developed (at a minimum including all direct production labor), labor costs can readily be estimated from company records of wages and salaries by position, union wage scales, salary surveys of various crafts and professions, or other published sources. Labor rates are prone to rapid inflation, often at rates sharply different from general inflation rates, care must be taken to obtain current figures and to properly project future wage rates.

Generally, data on wage rates includes shift differentials and overtime premiums. If not, these factors must be added to the extent applicable. Further, when estimating around-the-clock, 168-hr/wk operations, allowance must be made for the fact that a week includes 4.2 standard 40-hr weeks. Even with four work crews on "swing shift," one crew must work eight hr/week of overtime to keep the plant in steady operation. Depending upon local custom, laws, and union contracts, this overtime is generally payable at one-and-a-half to three times the normal hourly rate.

An alternate method of calculating labor requirements, if sufficient data are not available to establish a staffing table, is to consider a correlation of labor in workhours per ton of product per processing step. This relationship, which was developed by Wessell, relates labor requirements to plant capacity by the following equation [6]:

$$\frac{Operating\ workhours}{tons\ of\ product} = t\left[\frac{number\ of\ processing\ steps}{(capacity, tons/day)^{0.76}}\right]$$

Equation 10.5

Where:

t = 23 for batch operations with a maximum of labor
t = 17 for operations with average labor requirements
t = 10 for well-instrumented continuous process operations

As pointed out previously, the relationship between labor requirements and production rate is not usually a direct one. The Wessell Equation recognizes that labor productivity generally improves as plant throughput increases. It can also be used to extrapolate known workhour requirements from one plant to another of different capacity.

Another shortcut method of estimating labor requirements, when requirements at one capacity are known, is to project labor requirements for other capacities to the 0.2 to 0.25 power of the capacity ratio.

A significant factor to be considered in estimating labor costs is overtime. As mentioned earlier, around-the-clock, 24-hr/day, seven-day/week operations have an inherent overtime penalty of eight-hr/week. With this exception, and occasional overtime to cover for absent workers, overtime is usually not a major consideration in manufacturing and production operations.

However, in estimates involving construction projects or those which anticipate regular scheduled overtime, these costs can be substantial and must be carefully evaluated.

Scheduled overtime over an extended period can result in substantial decreases in worker productivity, resulting in a major cost penalty for productivity losses, in addition to the higher direct costs of premium pay at one-and-a-half, two, or even three times normal hourly rates.

Scheduled overtime involves a planned, continuing schedule for extended working hours for individual workers, or even entire crews. It is not occasional overtime caused on an irregular basis by absenteeism, equipment malfunctions, etc.

Unfortunately, scheduled overtime rarely saves money or accelerates production. Two articles, *Effect of Scheduled Overtime on Construction Projects* [2] and *Overtime in Construction* [5] which appeared in the AACE Bulletin in 1973, amply illustrated this point. These articles, prepared by representatives of the Construction Users Anti-Inflation Roundtable (now Business Roundtable), clearly demonstrated that scheduled overtime rarely, if ever, is beneficial, and that overtime should be avoided in favor of additional employees working normal shifts or partial shifts.

Quoting from the article, *Overtime in Construction*, by W. McGlaun [5]:

> "Studies indicate that when a job is placed on overtime, there is a sharp drop in productivity during the first week, with a substantial recovery which holds for about two weeks and is followed by a fairly steady decline. At the end of seven to nine weeks, the productivity on an overtime basis is no greater than the productivity would be on a 40-hour week. In this period, there is an increase in work accomplished of about 12 percent. After seven to nine weeks of operation, productivity continues to decline and the work accomplished is less than would have been accomplished on a 40 hour per week schedule. After 18 to 20 weeks, there is no gain in total work accomplished...."

Supervision and Maintenance Costs

As discussed above, supervision costs should be established, if at all possible, through a staffing table and tabulation of associated costs. Unfortunately, for most preliminary estimates, particularly those for proposed new processes, this is not possible.

In such cases, costs of supervision can be roughly estimated by taking a fixed percentage of direct labor costs, based upon company experience. In the absence of prior data on similar operations, a factor of 15 to 20 percent is generally satisfactory.

The validity of the latter factor can readily be seen when considering the fact that one front line supervisor can effectively manage no more than eight to ten workers, (i.e., supervision workhours of 0.100 to 0.125 per direct labor workhour). With front line supervision (i.e., foremen or forewomen) at labor rates approximately 50 to 60 percent above general labor rates, a 15 to 20 percent supervision factor is evident.

Maintenance labor costs, like supervision costs, should be delineated in the labor staffing table if at all possible. However, other than company records of existing similar plants, reliable data on maintenance costs are generally not available, and the staffing table approach is usually not feasible. For this reason, maintenance costs are often estimated as a fixed percentage of depreciable capital investment per year. For complex plants and severe corrosive conditions, this factor can be ten to 12 percent or higher. For simple plants with relatively mild, noncorrosive conditions, three to five percent should be adequate.

Maintenance costs are a semi-variable category, which is generally distributed about 50 percent to labor and 50 percent of materials. For a preliminary estimate, the various factors making up plant maintenance can be back calculated from the total maintenance number, using the following approximate percentages:

- Direct maintenance labor, 35 to 40 percent;
- Direct maintenance labor, supervision, seven to eight percent;
- Maintenance materials, 35 to 40 percent; and,
- Contract maintenance, 18 to 20 percent.

Then, as the project evolves toward a final staffing plan, the factors can be improved and are finally replaced with numbers generated from the staffing table. The back calculation allows one to estimate the number of people required early in the project.

When operating at less than 100 percent of capacity, maintenance costs generally increase per unit of production. Such operations can be estimated as follows:

Percent of Capacity	Maintenance cost as a percentage of cost at 100 percent capacity
100	100
75	85
50	75
0	30

Maintenance generally increases with age of equipment, although most estimates use an average figure to simplify the estimate. This apparent error is offset in the overall estimate by use of average or "straight-line" depreciation, whereas accelerated depreciation is in fact generally used for tax purposes. Figure 10.3 shows the validity of using an average maintenance figure over the life of a project.

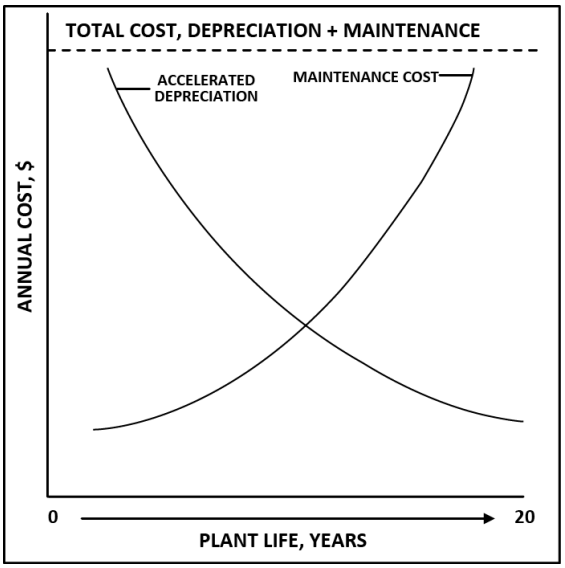


Figure 10.3—Comparison of Maintenance and Depreciation Costs over the Life of a Plant

Finally, for major projects, it may be necessary to include costs for additional maintenance supervisors. However, for small plant additions, additional supervision is generally not required.

OPERATING SUPPLIES AND OVERHEAD COSTS

Generally, operating (or factory) supplies are a relatively minor cost of operations. Nevertheless, these must be included in the operating or manufacturing cost estimate. Such costs include miscellaneous items such as lubricating oil, instrument charts, wiping cloths, etc. Lacking more detailed information, they may be estimated as a percentage of payroll. This percentage can range from a few percent to 20 percent or more, depending upon plant complexity and whether or not routine maintenance items and supplies are included or accounted for separately. For example, six percent of payroll is probably an adequate allowance for operating supplies in a coal preparation plant, while 20 percent is probably more reasonable for an oil refinery, a more complex operation requiring a cleaner environment. The best source of such costs, however, is always company records of similar past projects.

Overhead or burden costs are operating and manufacturing costs which, while not directly proportional or related to production, are associated with payroll or general and administrative expense. Such costs, depending upon what they represent, are either semi-variable or indirect costs.

The major semi-variable overhead costs are so-called payroll overheads. These are costs which are associated with employee "fringe benefits." They include workers' compensation, pensions, group insurance, paid vacations and holidays, Social Security, unemployment taxes and benefits, profit-sharing programs, and a host of others. The extent of these costs varies from industry to industry and company records are the best measure of their magnitude. However, in the absence of company data, payroll overheads may be roughly estimated at 25 to 40 percent of direct labor, plus supervision, plus maintenance labor costs for the U.S. For other countries, this factor must be adjusted to suit local conditions. In heavily socialized nations, payroll overheads can exceed 100 percent of the basic labor costs.

In addition, payroll overhead must be applied to indirect overhead (clerical, administrative, etc., personnel) if not previously included in cost estimates for this item.

It must be noted that, if company data is used, care must be exerted to avoid including items in payroll overhead which are, by accounting definition of the company, included in general expense.

Further, it should be recognized that in some industries, notably the U.S. and Canadian coal mining industry, a major portion of fringe benefits is based upon royalties levied by the unions on production; rather than being a function of labor costs. Such royalties are variable production costs and must be treated as such.

The expense of operating company testing and research laboratories is another overhead expense which must be included in the estimate. Generally, such costs are indirect costs, although in the case of product laboratories they may be direct semi-variable costs.

Laboratory overhead is best estimated based upon company experience. Lacking suitable data, these costs can be estimated as follows:

- From workhours required, plus associated overhead;
- From literature sources; and,
- As a percentage of direct labor costs.

If the last is used, laboratory overhead costs may range from three to 20 percent or more for complex processes. A suitable figure for average situations might be five to ten percent.

ROYALTIES AND RENTALS

As was discussed earlier in this chapter, royalties may be variable, semi-variable, fixed, or capital costs (or a combination of these) depending upon the conditions of the royalty agreement. The same is true of rental costs.

Single-sum royalty, rental, or license payments are properly considered as capital investment items, whereas payments in proportion to production or fixed payments per annum are treated as direct operating costs.

Royalty expenses, in the absence of data to the contrary, are treated as a direct expense and may be estimated at one to five percent of the product sales price. As a result of the complexity of agreements for royalty payments and variations in tax laws and accounting methods, extreme care should be exercised to be sure that such costs are properly included in the appropriate expense category.

Contingencies
As is true with a capital cost estimate, any operating or manufacturing cost estimate should include a contingency allowance. This accounts for those costs which cannot readily be determined or defined, or which are too small to estimate individually but may be significant in the aggregate. The contingency allowance applies both to direct and indirect costs and ranges from one to five percent (and more in some cases) depending upon the uncertainty in the data used to prepare the estimate and the risk associated with the venture.

Hackney has suggested the following guidelines for contingency allowances in operating and manufacturing cost estimates [3]:

- Installations similar to those currently used by the company, for which standard costs are available: one percent.
- Installations common to the industry, for which reliable data are available: two percent.
- Novel installations that have been completely developed and tested: three percent. And,
- Novel installations that are in the development stage: five percent.

General Works Expense
General works expense or factory overhead represents the indirect cost of operating a plant or factory and is dependent upon both investment and labor. Black suggested that factory overhead be estimated by the sum of investment, times an investment factor and labor, times a labor factor [1]. In this case, labor is defined as total annual cost of labor; including direct operating labor, repair and maintenance, and supervision; and labor for loading, packaging, and shipping.

Black's suggested labor and investment factors for various industries are as follows:

Industry	Investment factor (Percentage per year)	Labor factor (Percentage per year)
Heavy chemical plants, large-capacity	1.5	45
Power plants	1.8	75
Electrochemical plants	2.5	45
Cement plants	3.0	50
Heavy chemical plants, small capacity	4.0	45

Alternately, for preliminary estimates, indirect overhead may be approximated at 40 to 60 percent of labor costs, or 15 to 30 percent of direct costs. Humphreys has suggested 55 percent of operating labor, supervision, and maintenance labor for the mineral industries [4]. Again, these factors may be somewhat higher outside the US, depending upon local customs and laws.
It is important to note that indirect or factory overhead (general works expense) does not include so-called general expense (i.e., marketing or sales cost) and administrative expense.

Depreciation
Depreciation, while not a true cash-cost, is considered a business expense for income tax purposes. The purpose of depreciation is to allow a credit against operating costs, and hence taxes, for the unrecoverable capital expense of an investment.

The basis for computation of depreciation is the total initial capital expense for tangible assets, including interest during construction and start-up expense. The depreciable portion of capital expense is equal to the total initial investment, less working capital and salvage value as shown in Figure 10.1.

In theory, working capital can be totally recovered at any time after the plant or process is shut down. Similarly, salvage value, the scrap or sales value of the process equipment at the end of its useful life, can, in theory, be recovered at any time after plant shutdown. Thus, the sunk and permanently lost capital is the total initial investment, less working capital and salvage value (including land value). Through depreciation, this investment may be recovered as an operating expense over the useful life of the project.

Unfortunately, true useful life of a project generally does not correlate with the permissible depreciation period dictated by tax laws. The U.S. Internal Revenue Service (IRS) establishes criteria for useful life of various investments, which must be observed in cost and tax calculations whether or not these criteria actually reflect the true projected life of the plant or process. Table 10.3 lists typical permissible depreciation periods (the class life) for various plants and investments, as approved by the Internal Revenue Service. In countries other than the U.S., local taxing authorities should be consulted to determine the permissible life which can be used in depreciation calculations for any particular type of investment.

Table 10.3—Depreciation Class Lives and Recovery Periods

	Recovery periods (in years)		
	Class life	GDS (MACRS)	ADS
SPECIFIC DEPRECIABLE ASSETS USED IN ALL BUSINESS ACTIVITIES, EXCEPT AS NOTED:			
Office furniture, fixtures, and equipment	10	7	10
Information system	6	5	5
Data handling equipment, except computers	6	5	6
Airplanes (airframes and engines), except those used in commercial or contract carrying of passengers or freight, and all helicopters (airframes and engines)	6	5	6
Automobiles, taxis	3	5	5
Buses	9	5	9
Light general purpose trucks	4	5	5
Heavy general purpose trucks	6	5	6
Railroad cars and locomotives, except those owned by railroad transportation companies	15	7	15
Tractor units for use over-the-road	4	3	4
Trailers and trailer-mounted containers	6	5	6
Vessels, barges, tugs, and similar water transportation equipment, except those used in marine construction	18	10	18
Land improvements	20	15	22
Industrial steam and electric generation and/or distribution systems	22	15	22
DEPRECIABLE ASSETS USED IN THE FOLLOWING ACTIVITIES:			
Agriculture	10	7	10
Cotton ginning assets	12	7	12
Cattle, breeding, or dairy	7	5	7
Any breeding or work horse than is 12 years old or less at the time it is placed in service	10	7	10
Any breeding or work horse than is more than 12 years old at the time it is placed in service	10	3	10
Any race horse than is more than 2 years old or less at the time it is placed in service	None	3	12
Any horse than is more than 12 years old at the time it is placed in service and that is not a race horse , breeding horse, nor a work horse	None	3	12
Any horse not described above	None	7	12
Hogs, breeding	3	3	3
Sheep and goats, breeding	5	5	5
Farm buildings except single-purpose agricultural or horticultural structures	25	20	25
Single-purpose agricultural or horticultural structures GDS = 7 years before 1989	15	10	15
Mining	10	7	10
Offshore drilling	7.5	5	7.5
Drilling of oil and gas wells	6	5	6
Exploration for and production of petroleum and natural gas deposits	14	7	14
Petroleum refining	16	10	16
Construction	6	5	6
Manufacture of grain and grain mill products	17	10	17
Manufacture of sugar and sugar products	18	10	18
Manufacture of vegetable oils and vegetable oil products	18	10	18
Manufacture of other food and kindred products	12	7	12

Table 10.3—Depreciation Class Lives and Recovery Periods (Continued)

	Recovery periods (in years)		
	Class life	GDS (MACRS)	ADS
DEPRECIABLE ASSETS USED IN THE FOLLOWING ACTIVITIES:			
Manufacture of food and beverages—special handling devices	4	3	4
Manufacture of tobacco and tobacco products	15	7	15
Manufacture of knitted goods	7.5	5	7.5
Manufacture of yarn, thread, and woven fabric	11	7	11
Manufacture of carpets and dyeing, finishing, and packaging of textile products and manufacture of medical and dental supplies	9	5	9
Manufacture of textured yarns	8	5	8
Manufacture of nonwoven fabrics	10	7	10
Manufacture of apparel and other finished products	9	5	9
Cutting of timber	6	5	6
Sawing of dimensional stock from logs, permanent or well established	10	7	10
Sawing of dimensional stock from logs, temporary	6	5	6
Manufacture of wood products and furniture	10	7	10
Manufacture of pulp and paper	13	7	13
Manufacture of converted paper, paperboard, and pulp products	10	7	10
Printing, publishing, and allied industries	11	7	11
Manufacture of chemicals and allied products	9.5	5	9.5
Manufacture of rubber products	14	7	14
Manufacture of rubber products—special tools and devices	4	3	4
Manufacture of finished plastic products	11	7	11
Manufacture of finished plastic products—special tools	3.5	5	3.5
Manufacture of leather and leather products	11	7	11
Manufacture of glass products	14	7	14
Manufacture of glass products—special tools	2.5	3	2.5
Manufacture of cement	20	15	20
Manufacture of other stone and clay products	15	7	15
Manufacture of primary nonferrous metals	14	7	14
Manufacture of primary nonferrous metals—special tools	6.5	5	6.5
Manufacture of foundry products	14	7	14
Manufacture of primary steel mill products	15	7	15
Manufacture of fabricated metal products	12	7	12
Manufacture of fabricated metal products—special tools	3	3	3
Manufacture of electrical and non-electrical machinery and other mechanical products	10	7	10
Manufacture of electronic components, products, and systems	6	5	6
Any semiconductor manufacturing equipment	5	5	5
Manufacture of motor vehicles	12	7	12
Manufacture of motor vehicles—special tools	3	3	3
Manufacture of aerospace products	10	7	10
Ship and boat building machinery and equipment	12	7	12
Ship and boat building dry docks and land improvements	16	10	16
Ship and boat building—special tools	6.5	5	6.5
Manufacture of locomotives	11.5	7	11.5
Manufacture of railroad cars	12	7	12
Manufacture of athletic, jewelry, and other goods	12	7	12

Table 10.3—Depreciation Class Lives and Recovery Periods (Continued)

	Recovery periods (in years)		
	Class life	GDS (MACRS)	ADS
RAILROAD TRANSPORTATION			
Railroad machinery and equipment	14	7	14
Railroad structures and similar improvements	30	20	30
Railroad wharves and docks	20	15	20
Railroad track	10	7	10
Railroad hydraulic electric generating equipment	50	20	50
Railroad nuclear electric generating equipment	20	15	20
Railroad steam electric generating equipment	28	20	28
Railroad steam, compressed air, and other power plant equipment	28	20	28
Motor transport-passengers	8	5	8
Motor transport-freight	8	5	8
Water transportation	20	15	20
Air transport	12	7	12
Air transport (restricted)	6	5	6
Pipeline transportation	22	15	22
TELEPHONE COMMUNICATIONS:			
Telephone Central office buildings	45	50	45
Telephone Central office equipment	18	10	18
Computer-based telephone central office switching equipment	9.5	5	9.5
Telephone station equipment	10	7	10
Telephone distribution plant	24	15	24
Radio and television broadcasts	6	5	6
TELEGRAPH, OCEAN CABLE AND SATELLITE COMMUNICATIONS (TOCSC):			
TOCSC—Electric power generating and distribution systems	19	10	19
TOCSC—High frequency radio and microwave systems	13	7	13
TOCSC—Cable and long-line systems	26.5	20	26.5
TOCSC—Central office control equipment	16.5	10	16.5
TOCSC—Computerized switching, channeling, and associated control equipment	10.5	7	10.5
TOCSC—Satellite ground segment property	10	7	10
TOCSC—Satellite space segment property	8	5	8
TOCSC—Equipment installed on customer's premises	10	7	10
TOCSC—Support and service equipment	13.5	7	13.5
CABLE TELEVISION (CATV):			
CATV—Headend	11	7	11
CATV—Subscriber connection and distribution systems	10	7	10
CATV—Program origination	9	5	9
CATV—Service and test	8.5	5	8.5
CATV—Microwave systems	9.5	5	9.5

Table 10.3—Depreciation Class Lives and Recovery Periods (Continued)

	Recovery periods (in years)		
	Class life	GDS (MACRS)	ADS
ELECTRIC, GAS, WATER, AND STEAM UTILITY SERVICES:			
Electric utility hydraulic production plant	50	20	50
Electric utility nuclear production plant	20	15	20
Electric utility nuclear fuel assemblies	5	5	5
Electric utility steam production plant	28	20	28
Electric utility transmission and distribution plant	30	20	30
Electric utility combustion turbine production plant	20	15	20
Gas utility distribution facilities	35	20	35
Gas utility manufactured gas production plants	30	20	30
Gas utility substitute natural gas (SNG) production plant (naphtha or lighter hydrocarbon feedstocks)	14	7	14
Substitute natural gas-coal gasification	18	10	18
Natural gas production plant	14	7	14
Gas utility trunk pipelines and related storage facilities	22	15	22
Liquefied natural gas plant	22	15	22
Water utilities	50	20 (1)	50
Central steam utility production and distribution	28	20	28
Waste reduction and resource recovery plants	10	7	10
Municipal wastewater treatment plant	24	14	24
Municipal sewer	50	20 (2)	50
Distribution trades and services	9	5	9
Distribution trades and services—billboard, service station buildings and petroleum marketing land improvements	20	15	20
Recreation	10	7	10
Theme and amusement parks	12.5	7	12.5

Notes: (1) 25-year straight line may apply if placed in service after June 12, 1996. See IRS Publication 946.
(2) In those cases where guidelines are not listed for any given industry or type of equipment, or where the listed guidelines are clearly inappropriate, the depreciable life of each such property shall be determined according to the particular facts and circumstances.
Source: Depreciation, U.S. Department of the Treasury, Internal Revenue Service, Publication No. 534, 1994.

Taxing authorities usually permit the use of any generally accepted method of depreciation calculation provided that it is applied in a consistent manner to all investments applicable to the plant or process being considered. Different depreciation techniques may not be applied to various portions of the total investment. Further, effective in 1981 in the U.S., a specialized system known as the Accelerated Cost Recovery System (ACRS) was mandated by law. Subsequently, in 1986, the U.S. tax laws were revised again, and the ACRS system was replaced by a system called the Modified Accelerated Cost Recovery System (MACRS). Both systems are described later in this chapter.

Most industrial firms use accelerated depreciation in their actual operating cost and tax calculations. The time value of money dictates that for income tax purposes, depreciating an asset as quickly as possible, consonant with applicable tax laws in effect, results in an enhanced financial position for the firm. Depreciation to the firm is not considered an out-of-pocket cash cost, but is instead, a book cost. This book cost helps to minimize income taxes. Such techniques permit a major portion of the investment to be deducted from costs in the early years of the life of the plant, thus deferring taxes to the latest possible date. For the purpose of making preliminary operating and manufacturing cost estimates, straight-line depreciation is normally used, even though the company may in fact use accelerated depreciation in its books.

The reason for this apparent anomaly is that, as explained in the previous discussion, maintenance costs, which are known to increase with time, are generally assumed to be constant with time. As shown in Figure 10.3, accelerated depreciation increases with time. Assuming both to be constant, i.e., assuming straight-line depreciation, generally results in offsetting errors, as the actual sum of the two factors, for most plants, tends to be constant, or essentially constant, over the life of any given plant.

To calculate straight-line depreciation, annual depreciation is simply made equal to the depreciable portion of the initial capital investment divided by the depreciable life of the project.

In the case of projects with components having different depreciable lives, each component may be depreciated separately or the weighted average life may be used on the total depreciable investment.

Mathematically, annual straight-line depreciation is equal to:

$$D_{SL} = \frac{C}{Y}$$

Equation 10.6

Where:

D_{SL} = annual straight-line depreciation
C = depreciable portion of capital investment
Y = IRS-approved life, in years

There are many other acceptable depreciation techniques, of which the double-declining balance and sum-of-years-digits methods, both forms of accelerated depreciation, are the most commonly used.

In the double-declining balance method, an annual depreciation deduction is permitted on the undepreciated portion of the investment at a rate equal to twice the straight-line rate.

For example, if an investment has an approved life of five years for depreciation purposes, the straight-line deduction is 20 percent of the original depreciable investment per year. Thus, the double-declining balance deduction is twice this rate, or 40 percent.

For example, with an investment of $1 million, a salvage value of zero, and a five year life, annual double-declining balance depreciation allowances are as follows:

Year	Investment	Depreciation at 40%	Undepreciated Balance
0	$1,000,000	-	$1,000,000
1	$1,000,000	$400,000	$600,000
2	$600,000	$240,000	$360,000
3	$360,000	$144,000	$216,000
4	$216,000	$86,400	$129,600
5	$129,600	$129,600[a]	0
Total		**$1,000,000**	

[a]In the final year of the life, the total remaining undepreciated balance may be deducted.

Mathematically, the double-declining balance method is:

$$D = \left\{\frac{2(F - CD)}{n}\right\}$$

Equation 10.7

Where:

D = depreciation in any given year
F = initial asset value
CD = cumulative depreciation charged in prior years
N = asset life, in years

Note that in the double-declining balance method, the basis for depreciation ordinarily is the total depreciable investment *without* deducting for salvage values other than land value. In this method, the investment is depreciated down to the salvage value, and the final undepreciated balance may not be less than salvage value.

Another common method of computing accelerated depreciation is the "sum-of-years-digits" method. This technique is based upon the depreciable portion of the investment, i.e., excluding land *and* salvage value.

To use this technique, the approved years in the life of the plant are summed. Deductions are based on the remaining years of plant life divided by the sum of years. For example, with a five year life, the sum of years digits equals 5 + 4 + 3 + 2 + 1 = 15. In the first year, 5/15 of the depreciable investment is deducted; 4/15 in the second year, 3/15 in the third year; and so forth.

Using the same example as given above for the double-declining balance method, sum-of-years-digit depreciation deductions would be as follows:

Year	Investment	Depreciation ($)	Undepreciated Balance
0	$1,000,000	-	$1,000,000
1	$1,000,000	5/15 = 333,333	666,667
2	$1,000,000	4/15 = 266,667	400,000
3	$1,000,000	3/15 = 200,000	200,000
4	$1,000,000	2/15 = 133,333	66,667
5	$1,000,000	1/15 = 66,667	0
Total		**$1,000,000**	

The sum-of-years-digits method is expressed mathematically as:

$$D_y = C\left[\frac{2\,(n - Y + 1)}{n\,(n+1)}\right]$$

Equation 10.8

Where:

D_y = depreciation in year Y
C = depreciable portion of investment
n = asset life, in years

There are numerous other acceptable depreciation methods including some which are combinations of the above methods. However, the three methods described above are the most commonly used techniques.

As mentioned earlier, if constant maintenance costs are assumed, no matter which depreciation technique is actually used by the company, the straight-line technique should be generally used for are all preliminary estimates.

Accelerated Cost Recovery System

In 1981, a major revision of tax laws in the U.S. replaced the pre-existing depreciation systems described above with a system known as the Accelerated Cost Recovery System (ACRS). ACRS is mandatory for all capital assets acquired after 1980 and before 1987, when another system of depreciation, the Modified Accelerated Cost Recovery System (MACRS) became mandatory. However, the tax laws specify that any acquisition must continue to be depreciated on its original basis. Thus, since many capital assets have depreciable lives of up to 60 years, the old depreciation systems, MACRS, and ACRS will coexist and be used by cost professionals for many years into the future.

Under ACRS, capital assets are not subject to depreciation in the customary sense. It is not necessary to estimate salvage values or useful lives for equipment. Instead, the law establishes various property classes and provides for deductions calculated as specific percentages of the cost of the asset. Property classes are:

- **Three-year property:** "Property that has a mid-point class life of four years or less, or, is used for research and experimentation, or, is a race horse more than two years old when placed in service, or any other horse that is more than 12 years old when placed in service." This obscure and somewhat confusing definition includes automobiles, light trucks, and short-lived personal property.
- **Five-year property:** property not otherwise defined and which is not real property. This class includes most types of equipment and machinery.
- **Ten-year property:** public utility property having a midpoint class life of more than 18, but not more than 25 years. Also included are manufactured homes, railroad tank cars, certain coal use property, theme and amusement park property, and other property, as defined in the act.
- **Fifteen-year property:** long-lived public utility property.
- **Ten-year and fifteen-year real property classes**. The fifteen-year class consists of real property with a midpoint class life in excess of 12.5 years. All other real property falls into the 10-year class.

Detailed descriptions of each property class may be obtained upon request from the US Internal Revenue Service.

Under the 1981 law, ACRS deductions were phased in on a gradual basis over a period of years. Allowable deductions for personal property are listed in Table 10.4. Fifteen-year class real property deductions vary from Table 10.4, and are based in part on the month in which the property was placed in service. Internal Revenue Service regulations should be consulted to determine applicable deduction rates.

Table 10.4—ACRS Deductions (Percentage) for Personal Property and 10 Year Real Property

	Year property was put in service		
	1981-1984	**1985**	**1986**
3-year property			
1st year	25	29	33
2nd year	38	47	45
3rd year	37	24	22
5-year property			
1st year	15	18	20
2nd year	22	33	32
3rd year	21	25	24
4th year	21	16	16
5th year	21	8	8
10-year property			
1st year	8	9	10
2nd year	14	19	18
3rd year	12	16	16
4th year	10	14	14
5th year	10	12	12
6th year	10	10	10
7th year	9	8	8
8th year	9	6	6
9th year	9	4	4
10th year	9	2	2
15-year personal property			
1st year	5	6	7
2nd year	10	12	12
3rd year	9	12	12
4th year	8	11	11
5th year	7	10	10
6th year	7	9	9
7th year	6	8	8
8th year	6	7	7
9th year	6	6	6
10th year	6	5	5
11th year	6	4	4
12th year	6	4	3
13th year	6	3	3
14th year	6	2	2
15th year	6	1	1

It should also be noted that the ACRS requirements place certain limitations on deductions for disposition of property prior to the end of its class life, and also provide for optional use of alternate percentages based on the straight-line method of depreciation.

The logic of the standard ACRS percentages, as outlined in Table 10.4, becomes apparent when it is realized that the 1981 to 1984 rates are approximately equal to those for 150 percent declining-balance depreciation with a switch to straight-line depreciation in later years. The 1985 rates similarly approximate 175 percent declining-balance depreciation, and the rates for 1986 are essentially those for the double-declining-balance method, both with a switch to the sum-of-years-digits method at the optimum point in time. Thus, ACRS is merely a combination of previously used and widely accepted methods of computing accelerated depreciation.

MODIFIED ACCELERATED COST RECOVERY SYSTEM

When the ACRS depreciation system was adopted in the U.S., it was phased in over a period of five years ending in 1986, and was originally anticipated to remain in effect after that time. The 1986 tax reform act further revised the depreciation regulations and, effective in 1987, implemented a new system called the Modified Accelerated Cost Recovery System (MACRS).

MACRS expanded the ACRS property classes from five to eight, revised the depreciation periods for most items (see Table 10.3), and redefined the classes. The definitions including minor tax law changes, since 1987, are as follows:

- **Three-year property**, which was redefined as including, “tractor units for use over the road, any race horse over two years old when placed in service, and any other horse over 12 years old when placed in service." Also included in this class is

qualified rent-to-own property. Three-year property generally includes those items with an IRS-approved class life of four years or less. The major change in this class was the elimination of automobiles, light trucks, and other short-lived personal property which was moved to the five year class.

- **Five-year property**, which was totally redefined as including "trucks, computers, and peripheral equipment, office machinery, and any automobile." Most items with an IRS-approved class life of four-plus years and less than 10 years are included under this category. This class includes taxis, busses, property used in research and experimentation, breeding cattle, dairy cattle, and furnishings (furniture, rugs, etc.) used in residential rental real estate.

- **Seven-year property**, which is defined in part as including, "office furniture and fixtures, any property that does not have a class life, and that has not been designated by law as being in any other class; and, if placed in service before 1989, any single purpose agricultural or horticultural structure." This category covers any items with an IRS-approved class life of 10 years or more and less than 16 years. It effectively includes almost all industrial machinery and equipment. It also includes agricultural machinery and equipment.

- **Ten-year property**, which includes "vessels, barges, tugs and similar water transportation equipment, and, if placed in service after 1988, any single-purpose agricultural or horticultural structure, and any tree or vine bearing fruit or nuts. This category includes those items with an IRS-approved class life of 16 years or more and less than 20 years.

- **Fifteen-year property**, which includes items with an IRS-approved class life of 20 years or more and less than 25 years, plus wastewater treatment plants and equipment used for two-way exchange of voice and data communications. This class includes improvements made directly to land or added to it (such as shrubbery, fences, roads, and bridges). It also includes any retail motor fuels outlet (such as a convenience store).

- **Twenty-year property**, which includes items with an IRS-approved class life of 25 years or more, excluding real property and including sewer systems. Farm buildings (other than single-purpose agricultural or horticultural structures) fall into this category.

- **Twenty-seven and one-half year property**, which includes residential rental property.

- **Nonresidential real property**, which includes real property other than residential rental property. For this category, the recovery period for depreciation is 31.5 years for property placed in service before May 13, 1993, and 39 years for property placed in service after May 12, 1993.

The depreciation allowances under the MACRS system, like ACRS, do not consider salvage value. For the three, five, seven, and ten-year categories, the MACRS depreciation schedule (see Table 10.5) is equivalent to 200 percent declining-balance switching to straight-line at the optimum point to maximize the deduction. In the first year, a half-year convention applies. The assumption is made that the item being depreciated was placed in service for only six months, no matter what the actual date of service was. If more than 40 percent of the cost basis was placed in service during the last three months of the year, a mid-quarter convention applies (i.e., it is assumed that the equipment was in service for only 1.5 months of the first year).

Table 10.5—MACRS Deduction Rates

If the recovery Year is:	3-year	5-year	7-year	10-year	15-year	20-year
1	33.33	20.00	14.29	10.00	5.00	3.750
2	44.45	32.00	24.49	18.00	9.50	7.219
3	14.81	19.20	17.49	14.40	8.55	6.677
4	7.41	11.52	12.49	11.52	7.70	6.177
5		11.52	8.93	9.22	6.93	5.713
6	5.76	8.92	7.37	6.23	5.285	
7			8.93	6.55	5.90	4.888
8			4.46	6.55	5.90	4.522
9				6.55	5.91	4.462
10				6.56	5.90	4.461
11				3.28	5.91	4.462
12					5.90	4.461
13					5.91	4.462
14					5.90	4.461
15					5.91	4.462
16					2.95	4.461
17						4.462
18						4.461
19						4.462
20						4.461
21						2.231

For the 15- and 20-year categories, the depreciation schedule, as shown in Table 10.4, is 150 percent declining-balance with a switch to straight-line at the optimum point. The half-year and half-quarter conventions also apply to these categories.

For the real property categories, straight-line depreciation must be used with a mid-month convention in the first year.

If all of the foregoing seems confusing and convoluted, it is. The logic of tax laws and their complexity is rarely clear, and the MACRS system is a significant example of how the political process can complicate what should otherwise be an easily understood subject.

To further complicate the calculation of depreciation, the MACRS system allows for an alternate method. As shown in Table 10.3, the Alternate Depreciation System (ADS) permits straight-line depreciation over specified periods which are equal to, or longer than, the regular MACRS recovery periods (General Depreciation System (GDS)). Generally, the taxpayer must specifically choose to use the alternate method; otherwise the regular MACRS system applies. In a very few cases, the alternate system must be used. The exceptions are small in number and generally are not of concern to the cost engineer.

The alternate system, however, is always a bad choice economically. The regular system permits far more rapid deductions and resultant tax savings. The alternate system is easier to understand and is simple to calculate, but is a poor choice—it costs money. Nevertheless, the government permits poor economic decisions. The wise cost engineer should avoid them.

Amortization, Depletion, Insurance, and Real Estate Taxes

Amortization is a term which is applied to writing off or recovering any portion of the initial capital expense, which is intangible in nature, and as such has no definable useful life. A lump-sum royalty payment is an example of such an investment. Such intangible assets are written off as an operating cost over the life of the plant or process, using exactly the same calculation techniques as used for the depreciation of tangible assets. Alternately, intangible assets may be written off as a function of production over the life of a project. In some cases, amortization is, for simplicity, included in the depreciation charge (albeit erroneously). However, for practical purposes the distinction between depreciation and amortization is usually of no consequence.

Depletion allowances, while not considered to be an operating cost, must be included in estimates involving extraction of a natural resource, e.g., in coal mining. These allowances are deductions from gross income prior to calculation of taxes on income. Thus, they are, in effect, tax credits granted by law to compensate for eventual exhaustion of an irreplaceable natural resource such as coal or oil. They are computed as a fixed percentage of the market value of the resource in its first usable and salable form, even though the resource may be further processed and eventually sold in another form at a different cost. Depletion rates are established by law and are periodically changed. It is necessary when performing an estimate first to determine the currently applicable rate.

Insurance and real estate (or property) taxes must also be included in the estimate if not previously considered in determining general works expense. In most areas, these costs total about one-and-a-half to three percent of investment per year. Two percent is about average for locations in the U.S.

Distribution Costs

The costs of packing and shipping products to market (i.e., distribution costs) are highly variable and are dependent upon product characteristics. In many cases, especially with consumer products, these costs often exceed the cost of producing the product itself.

Distribution costs may include the following:

- Cost of containers, including their repair, testing, cleaning, etc., if reusable, and their depreciation or rental, if nonexpendable;
- Transportation costs; and,
- Applicable labor and overheads for packing and shipping.

If the product is sold FOB the plant, the cost of transportation is borne by the customer and need not be considered in the estimate. If, however, it is sold on a delivered basis, transportation costs must be included.

The mode of transportation and the shipping distance drastically affect transportation costs. In general, pipelines, barges, and tankers are the least expensive forms of transportation. Rail shipment is somewhat higher in cost, and truck shipment is the most expensive. Freight tariffs are regulated by the states (intrastate shipments) and the federal government (interstate shipments) and vary from product to product. In general, for any given product, freight costs per ton-mile are lowest for long hauls (250-300 miles or more) and high volumes. Short-distance haulage and small shipments incur considerably higher freight rates.

CONCLUSION

Manufacturing and operating cost estimation is a key element toward determining project feasibility. Facilities encounter variable demand, based on product success and seasonality factors along with calculation of the breakeven point. Therefore, cost estimates in this area have to be based on full-capacity operations along with points at less than this output level. Since a facility will often be in operation over a number of years, inflation factors may be part of this analysis. Typical cost elements for the estimate include labor, material, supplies, utility, and overhead costs. Added to this should be elements such as royalties, taxes, insurance, and byproduct costs such as waste/waste disposal. All of these above are cash costs. To this analysis, non-cash costs for depreciation must be included in the estimate. Depreciation allowances vary by country according to their tax laws and typically the analysis will attempt to depreciate plant assets on an accelerated basis.

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Chapter 11—Discrete Part Manufacturing

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Abstract
The following chapter reviews and discusses discrete manufacturing, or the production of the separate and individual pieces. It is focused on the production of separate, individual products and uses various manufacturing philosophies and terms for estimating in discrete manufacturing. The ladder of cost illustrates the interrelationships between the various cost terms and a cost estimating form has been presented for discrete manufacturing.

Keywords
Computer Aided Process Planning (CAPP), Concurrent Engineering, Cost Point, Group Technology, Just In Time, Lean Manufacturing, Materials Requirement Planning (MRP), Required Return, and Supply Chain Management.

INTRODUCTION

Learning Objectives

After completing this chapter, the reader should be able to:

- Understand the fundamentals of basic operations in discrete part manufacturing.
- Comprehend the discrete part manufacturing philosophies.
- Identify the basic cost terms and relationships in discrete manufacturing.
- Understand cost-estimating-item components in discrete manufacturing.
- Recognize differences between time-based and quantity-based approaches to break-even analysis.
- Identify the four break-even points in discrete part manufacturing.
- Calculate the four break-even points in discrete part manufacturing.

Discrete Part Manufacturing

Discrete part manufacturing is concerned with the production of separate, individual products, whereas continuous manufacturing is concerned with large units to be further processed, such as a roll of sheet steel, or units in fluid form with no distinct shape. Discrete parts typically are solid products which have the dimensions and sizes for final use, whereas solid continuous products will be further processed into specific shapes. Discrete production often produces low quantities of production; the average lot size is less than 75 units. Discrete manufacturing has special tools for the various products, so set-up and tooling changes are much more frequent in discrete manufacturing than in continuous manufacturing. Rapid manufacturing is a discrete manufacturing process where most parts are produced as single units and are often prototypes or replacement parts.

Another variation in discrete manufacturing is whether the manufacturing is that of an individual component part, of a product, or with the assembly or joining of parts to form the completed product. Assembly is often the final operation in the production of a manufactured product, before it goes to the customer. For example, assembly lines are the final phase in the manufacture of automobiles and this involves the assembly of many components. However, one of those components, the engine, is a sub-assembly of various components, one of which is the engine block. The engine block is considered a discrete manufactured part, whereas the automobile is considered as a discrete manufacturing assembly. The focus of this section is on the discrete manufactured part, although the approach is similar for discrete manufactured assemblies.

Operations in Discrete Part Manufacturing

There are a wide variety of products produced in discrete manufacturing, and thus an extreme variety of operations performed to obtain the desired shape and properties required of the product. The operations performed vary considerably depending upon the material being used for the specific component. Seven major groups of component operations and a few of the manufacturing operations of each group are presented in Table 11.1. Note that some manufacturing operations are repeated in different major component groups and the machining operations, which are listed in the metal component manufacturing group, would also be heavily used in the plastic component manufacturing, and also used in some other groups. Several rapid manufacturing and rapid prototyping processes can be used for a variety of materials.

Table 11.1—Component Operation Classification for Discrete Manufacturing [2, 7, 10]

1. Metal Component Manufacturing	
Casting	Sand casting, investment casting, die casting, permanent mold casting, squeeze casting, full mold casting, slush casting, etc.
Forming and Shaping	Rolling, open-die forging, impression die forging, impact extrusion, drawing, shearing, bending, powder processing, etc.
Machining	Turning, drilling, milling, sawing, planning, shaping, grinding, water-jet machining, electrical discharge machining, etc.
2. Plastic Component Manufacturing	Injection molding, extrusion, blow molding, pultrusion, thermoforming, compression molding, filament winding, etc. and many of the machining operations
3. Ceramic and Glass Component Manufacturing	
Ceramic	Slip casting, extrusion, pressing, injection molding, hot pressing, drying, firing, drawing, etc.
Glass	Rolling, float forming, pressing, blowing, etc.
4. Component Surface Modification	Case hardening, hot dipping, porcelain enameling, electroplating, thermal spraying, vapor deposition, anodizing, painting, cleaning, peening, etc.
5. Assembly and Joining	Arc welding, resistance welding, laser welding, brazing, soldering, adhesive bonding, wave soldering, etc.
6. Micro-Electronic Component Manufacturing	Crystal growth, wafer slicing and polishing, photolithography, doping, sputtering, chemical vapor deposition, etching, adhesive bonding, wave soldering, etc.
7. Rapid Manufacturing and Rapid Prototyping	Selective Laser Sintering, Melted and Extrusion Modeling, Direct Metal Laser Sintering, Electron Beam Melting, Filament Fused Fabrication, Laminated Object Manufacturing, Powder Bed Printing, Digital Light Processing

Discrete Part Manufacturing Philosophies

There are several manufacturing philosophies/techniques which have been introduced during the last 50 years to assist in the reduction of manufacturing costs. Some of these philosophies/techniques are: Computer Aided Process Planning (CAPP), Concurrent Engineering, Group Technology, Just-in-Time, Lean Manufacturing, Materials Requirement Planning, Supply-Chain Management, and Total Quality Management. Each of these techniques will be briefly presented to indicate the goals of that philosophy/technique.

Computer-Aided Process Planning (CAPP)—The goal of Computer-Aided Process Planning is to be able to automatically generate the process plan to produce the component from the component drawing and specifications. This would include the sequence of the operations, as well as the particular operation parameters, and it optimizes the processing time, operation costs, and product quality. The two approaches to CAPP are: the variant approach which searches a database for similar parts and modifies the closest similar component plan for the new component; and the generative approach which designs the process plan starting from "scratch," that is generating a completely original approach.

Concurrent Engineering—Concurrent engineering, as defined by the Institute for Defense Analysis is a systematic approach to the integrated, concurrent design of products and their related processes, including manufacturing and support [9]. This approach is intended to cause the developer (designers), from the outset, to consider all elements of the product life cycle from conception through disposal, including quality, cost, schedule, and user requirements.

Group Technology—Group technology is a manufacturing philosophy that identifies and exploits the underlying sameness of component parts and manufacturing process [3]. There are two primary approaches which are: Classifying parts into families which have similar design features, and classifying parts into families which have similar processing operations. This permits the standardization of parts in the design process and in the second case, production of parts as families by permitting cell formation and reducing the set-up times, via fewer set-up changes.

Just-in-Time—Just-in-time is the manufacturing philosophy which requires that the supplies (raw materials) are delivered when required and inventory costs are theoretically driven to zero, as there is no inventory. This is not only for external suppliers, but for internal use as parts go from one operation to the next, so the work-in-process inventory is a minimum. This is closely related to the "Kanban" system or "pull" system in which parts are not produced until ordered.

Lean Manufacturing—Lean Manufacturing is a manufacturing philosophy to shorten lead times, reduce costs, and reduce waste. This philosophy is implemented by reducing waste through scrap reduction, improved yields and developing new products from waste stream materials, improving employee performance, skills, and satisfaction via training, recognition, and employee

involvement and empowerment, and investing capital to improve processes, process rates, and capabilities. Lean Manufacturing is not "mean" manufacturing, and it is not a short term process but is a continuous improvement process.

Material Requirements Planning (MRP)—Materials Requirements Planning is process of keeping complete records of inventories of materials, supplies, parts in various stages of production, scheduling of production, purchasing of parts, and orders and delivery dates for customers and from suppliers. This information is used to create the master production schedule for machine loading, purchasing schedules, and delivery dates for suppliers and customers. The MRP system often results in various inventories because of scheduling of a variety of products with limited resources (machines).

Supply Chain Management—The production of complex products requires the integration of many different components from a variety of suppliers. Supply Chain Management involves the assurance that the parts will arrive from the suppliers when required, to avoid large inventories or production stoppages from a lack of parts. Supply chain management also requires the involvement of suppliers in the design process to eliminate unnecessary operations and inefficient designs of components, or even unnecessary components. Although the focus is on the movement of materials, it also involves the transfer of information on the status of delivery and financial flow of credit, terms and conditions, and payment schedules, as the materials move through the various stages of the supply chain. The goals are to reduce inventory, reduce the time-to-market, reduce costs, and improve quality.

Total Quality Management—Total Quality Management is a leadership philosophy, organizational structure, and working environment that fosters and nourishes a personal accountability, responsibility for quality, and a quest for continuous improvement in products, services, and processes [8].

Total Cost Management (TCM)—Total Cost Management is the effective application of professional and technical expertise to plan and control resources, costs, profitability and risk. TCM is a systematic approach to managing cost throughout the life cycle of any enterprise, program, facility, project, product or service [1].

Manufacturing Cost Terms and Definitions

Definitions of some of the basic manufacturing terms are presented with respect to discrete manufacturing. Some of these terms are used elsewhere, but will be repeated and may differ slightly from the other definitions.

Administrative Expense—This includes administrative costs that are not at the factory level, but at the corporate level. This includes executive salaries, legal and accounting expenses, basic research and development, and all other expenses that are not factory or marketing, selling, and distribution expenses.

Contingency—An amount added to an estimate to allow for changes that experience shows will likely be required. May be derived either through statistical analysis of past project costs, or by applying experience from similar projects. Usually excludes changes in scope, or unforeseeable major events such as strikes, earthquakes, etc. (**Note**: contingencies have generally been applied to process industries rather than discrete production products, but this is primarily because values have not been determined. Examples of contingencies for discrete manufacturing would be for changes in design or process, such as modification of a foundry pattern to solve a gating problem.

Cost—The amount measured in money, cash expended, or liability incurred, in consideration of goods and/or services received.

Cost Estimating—The determination of quantity and the predicting or forecasting, within a defined scope, of the costs required to construct and equip a facility, to manufacture goods or products, or to furnish a service. Costs are determined using past experiences and calculating and forecasting the future cost of resources, methods, and management within a scheduled time frame. Included in the costs are assessments and evaluation of risks and uncertainties. Cost estimation provides the basis for project management, business planning, budget preparation, and cost and schedule control.

Costing (Cost Accounting)—Costing, also called cost accounting, is the methodology for classifying, recording, and allocating the appropriate expenditure incurred for the determination of the cost of construction, production, or service performed. Costing is achieved by keeping a continuous record of all the costs involved in the constructed facility, goods manufactured, or service provided. It is the historical reporting of disbursements, costs, or expenditures for a project, product, or service.

Direct Costs—Costs that directly contribute to the final product and can be directly allocated to the manufacture of a specific product. Direct costs can frequently be divided into the following elements—material costs, labor costs, engineering costs, and burden costs (other expenses). In many instances, the only direct costs available are the material and labor costs, with the engineering and burden costs incorporated into the overhead costs. The sum of the direct costs is also referred to as the prime cost.

- **Direct Material Costs**—Costs of those materials used directly for the manufacture of the product and which become part of the finished product.
- **Direct Labor Costs**—Costs of labor performed on the product to convert it to its final shape.
- **Direct Engineering Costs**—Costs of engineering to design, prototype, test, and evaluate the product.

- **Direct Burden Expenses**—All expenditures, other than direct labor, direct material, and direct engineering, that can be directly allocated and charged to the product. This can include tooling such as jigs and fixtures, patterns, etc., that can be charged to the specific product.

Factory Expense—Costs that cannot be allocated directly to the manufacture of a particular product. Indirect costs include the wages of supervisory and inspection staff, indirect materials such as grease, lubricants, coolants, and repair materials, and maintenance costs.

Indirect Costs—The indirect costs also can be divided into the four elements used for the direct costs—material costs, labor costs, engineering costs, and indirect burden costs.

- **Indirect Material Costs**—Oils and lubricants for machining or forming, sand in casting, nails and screws for packaging, etc.
- **Indirect Labor Costs**—Labor costs that are not directly tied to a specific product, but whose labor is used indirectly in the manufacture of the product. This may include maintenance staff, material handling staff, etc.
- **Indirect Engineering Costs**—The cost of computers and software packages that are used for the design of various parts, not just one part.
- **Indirect Burden Expenses**—All burden expenses, except direct burden expenses. This could include janitorial staff expenses, factory clerical staff, etc. Rent, utilities, depreciation, and insurance are often indirect factory expense items.

Marketing, Selling, and Distribution Expense—All expenses related to marketing, sales, and distribution of the product from the factory. This includes salaries, commissions, travel expenses, advertising expenses, and delivery of the product from the factory to the customer.

Mark-up Rate (Profit)—The profit and taxes that must be paid when a profit is made. There are three different types of profit generally recognized, and the definitions of each are:

- **Gross profit***—Earnings from an ongoing business after direct costs of goods sold have been deducted from sales revenue for a given period.
- **Operating profit***—Earnings of income after all expenses (selling, administrative, depreciation) have been deducted from the gross profit.
- **Net profit***—Earnings or income after subtracting miscellaneous income and expenses (patent royalties, interest, capital gains) and federal income tax from operating profits.

*Terms are from AACE Recommended Practice 10S-90, *Cost Engineering Terminology* [1].

BASIC COST RELATIONSHIPS

The basic relationships between the various manufacturing cost terms are illustrated in Figures 11.1 and 11.2. Figure 11.1 shows the relationships between the various cost terms in a flow chart form [4], and Figure 11.2 shows the terms in a stepwise fashion, referred to as the "ladder of costs." The sum of direct material, direct labor, direct engineering, and direct expense is the prime cost. However, many companies do not keep adequate records of direct engineering and direct expenses, and in many instances the prime cost is the sum of the direct material and direct labor costs. Usually the design engineering and direct expense are often included as overhead costs and not allocated to the specific product. This is partly because only ten percent of new products make it into the production stage, and most of the design work cannot be assigned to the specific products. The relationships between the cost terms can also be illustrated by equations such as:

*Prime Cost = direct material cost + direct labor cost + direct engineering cost + direct expense

Equation 11.1

**Manufacturing Cost = prime cost + factory expense

Equation 11.2

Production Cost = manufacturing cost + administrative expense

Equation 11.3

Total Cost = production cost + marketing, selling, and distribution expense

Equation 11.4

Selling Price = total cost + mark-up (profit and taxes)

Equation 11.5

**Prime cost is also called direct cost*
***Manufacturing cost is also called factory cost*

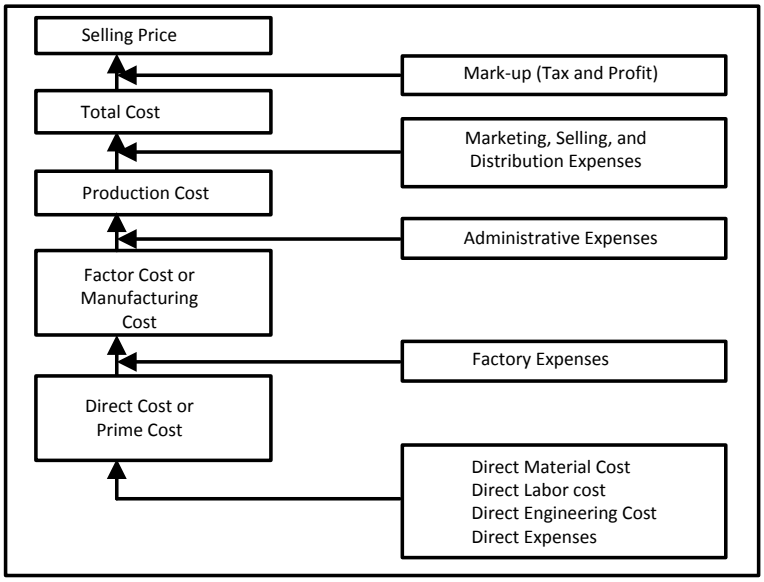


Figure 11.1—Cost Diagram of Basic Cost Relationships for Discrete Part Manufacturing

The profit should be related to the value to the customer. Some items will generate more profit than others; those parts which one is skilled at producing should yield more profit than those items which are new or for those items which you do not have the ideal equipment to produce. Equal profits indicate your prices will be too high for those items which you have difficulty making and too low for those items which are your specialty area.

Profit is usually meant to imply net profit; that is the profit after all expenses have been incurred and after taxes have been paid. The other profit terms, that is gross profit and operating profit, do not include all the expenses and thus are larger than the net profit term.

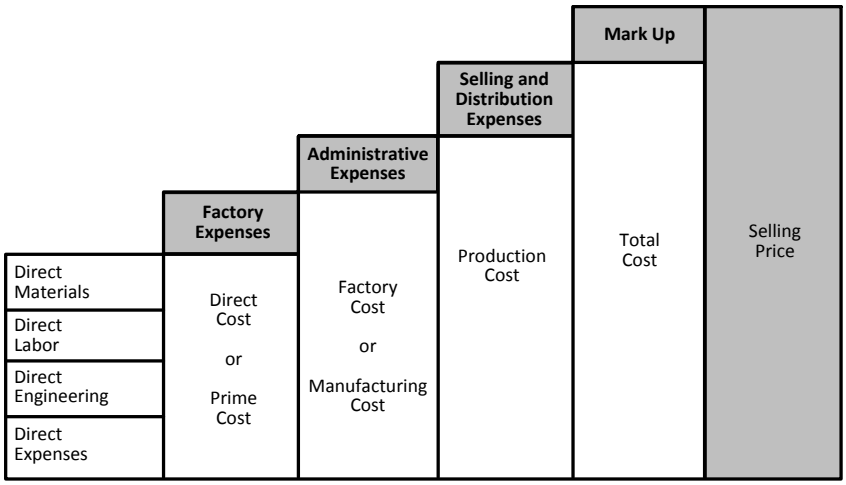


Figure 11.2—Ladder of Costs for Discrete Part Manufacturing

Cost Estimating for Discrete Part Manufacturing

Direct and Indirect Costs

Direct costs are those costs that can be directly related to a specific part and these most commonly are direct materials and direct labor. There can be other direct costs, such as direct engineering or direct burden expenses, but these are often not separated (even though they should be) and are thus included in the overhead components. The direct materials include all the direct material consumed in the product unless they are small and cost more to track. The direct labor and direct material costs are also referred to

as the “out-of-pocket” costs; they are costs which are being directly paid to others and do not cover any of the direct or indirect overhead costs. The direct labor represents the fully burdened labor costs, including benefits as well as wages.

For example, in the copying of a report on a copy machine, the costs would be the paper cost, the toner cost, and machine rate costs, the operator cost, and the staple cost. The paper and toner would be direct material costs, the operator cost would be a direct labor cost, but the cost of an individual staple is so small compared to the other costs, it typically would be included as part of the indirect burden costs. The machine rate cost includes the operating cost plus capital costs, so it would be an indirect cost, but it is applied directly to the product. If one did not make the copy, the direct costs of the operator and the paper would be saved and these would be the “out-of-pocket” costs.

There are different degrees of the indirect costs. In the copying machine example, there is a certain amount of energy consumed to operate the machine each time a copy is made. However, when the machine is on idle, it also consumes some energy and this cost is built into the burden based upon its expected usage. The capital costs of purchasing and installing the machine are another level of burden, based upon the expected life of the machine and the expected copies produced per year. Although these are direct costs, they are considered indirect costs as the machine is used for a wide variety of reports and not only one report.

Other indirect costs are those which cannot be directly tied to the product, such as supervision, administrative salaries, maintenance, janitorial, material handling, legal and etc. These costs have different degrees of indirectness. For example, the immediate supervisor may have five people working for him whereas the plant superintendent may have 500. More of the immediate supervisors cost must be allocated in the overhead burden than the plant superintendents overhead for particular component part, but the plant superintendents overhead will be applied over many more component parts. In large companies, indirect costs also include items such as basic and applied research and development which must be done to develop future products. However, the costs of such activities must be recovered on the current products being produced and these R&D costs for future products are indirect burden costs for the current products.

Cost Estimating Guide Form

There are so many different types of operations and items that can be included on a cost estimating form, that only a general guide can be given. A form particular to the particular products produced must be designed to obtain good cost estimates. The general form is illustrated in Table 11.2. The mark-up amount is calculated as:

Mark-up (amount) = Total Cost x [% MU / [1 - % MU]]

Equation 11.6

Where:

Total Cost	= total costs
%MU	= decimal percent of mark-up for profits and taxes

For example, if the total costs are $10,000 and the mark up is 20 percent, the mark-up amount would be:

Mark-up (amount) = $10,000 x [0.20 / [1.00 – 0.20]] = $2,500

The selling price would be $10,000 plus $2,500 mark-up or $12,500.

The selling price can be determined directly from the total costs by:

Selling Price = Total Cost / [1 - % MU]

Equation 11.7

Selling Price = $10,000 / [1 - 0.20] = $10,000/ 0.80 = $ 12,500

Table 11.2—General Discrete Costs Estimating Form with Illustrative Example Rates

1. Manufacturing/Factory Cost							
a. Prime cost							
Direct Labor	Amount (hours/unit)	Rate ($/hr)	Quantity	Cost			
Operation 1							
Operation 2							
Total Cost							
Direct Materials	Amount (weight/unit)	Rate ($/weight)	Quantity	Cost			
Material 1							
Material 2							
Total Cost							
Direct Engineering/Expense	Amount (units)	Rate ($/unit)	Quantity	Cost			
Item 1							
Item 2							
Item 3 (subcontracted)							
Total Cost							
Total Prime Cost							
b. Indirect Costs							
Indirect Materials		Rate (% of direct materials)					
Indirect Labor		Rate (% of direct labor)					
Indirect Engineering/Expense		Rate (% of prime cost)					
Contingency Costs							
Process Contingency		Rate (% of prime cost)					
Product Contingency		Rate (% of prime cost + contingency cost)					
Direct Supervision		Rate (%of direct labor)					
Total Indirect Cost							
Manufacturing/Factory Cost							
2. Production Plant Cost							
Plant Administrative							
Administration, Property Taxes, and Insurance		Rate (% of manufacturing costs)					
Total Plant Administrative Cost							
3. Selling Expenses							
Marketing Costs		Rate (% of manufacturing costs)					
Selling Commisions and Salaries		Rate (% of production costs)					
Shipping Expenses		Rate (% of prime cost)					
Warehousing		Rate (% of prime cost)					
Total Marketing, Selling, and Distribution Costs							
Total Costs							
4. Mark-Up							
Profit and Taxes		Rate (% of total cost)					
Selling Price							

To illustrate the use of the form, an example is presented in Table 11.3. The values represented are for illustrative purposes, and not specific values or rates to be used. These values must be evaluated for each individual company. An order has been received for 100 units of a component which will require 240 lbs. of Material-1 at $0.75 per pound and 5 lbs. of Material-2 which is $4.00 per pound for each unit. The product requires two operations, Operation-1 which takes 0.20 hrs./unit at $20/hr. and Operation-2 which takes 0.40 hrs./unit at $25/hr. The special tooling for the job is $3,600. The mark-up for profits and taxes is 20 percent of the selling price. The example costs and selling price are presented in Table 11.3, with the unit selling price of $596.38 and a unit cost of $477.10. With a tax rate of 40 percent, one must charge $1.67 for every $1.00 of after tax profit desired and this is one reason why businesses dislike the taxes so much. The contingency costs are primarily for new products and processes, and would be low for standard products. The contingency costs are for expected tooling changes and process changes that would be required from expected design errors or incorrect process parameters.

Table 11.3—General Discrete Costs Estimating Form with Illustrative Example Rates

1. Manufacturing/Factory Cost						**% of Total Cost**	
a. Prime cost						Component	Cumulative
Direct Labor	Amount (hours/unit)	Rate ($/hr)	Quantity	Cost			
Operation 1	0.20 hr/unit	$20/hr	100	$400			
Operation 2	0.40 hr/unit	$25/hr	100	$1,000			
Total Cost					$1,400	3	
Direct Materials	Amount (weight/unit)	Rate ($/weight)	Quantity	Cost			
Material 1	240 lbs/unit	$0.75/lb	100	$18,000			
Material 2	5 lbs/unit	$4.00/lb	100	$2,000			
Total Cost					$20,000	41.9	
Direct Engineering/Expense	Amount (units)	Rate ($/unit)	Quantity	Cost			
Item 1	1 tooling set	$3,600	1	$3,600			
Item 2							
Item 3 (subcontracted)							
Total Cost					$3,600	7.5	
Total Prime Cost					$25,000		52.4
b. Indirect Costs							
Indirect Materials		20% of direct materials			$4,000		
Indirect Labor		100% of direct labor			$1,400		
Indirect Engineering/Expense		10% of prime cost			$2,500		
Contingency Costs							
Process Contingency		5% of prime cost			$1,250		
Product Contingency		10% of prime cost + contingency cost			$3,750		
Direct Supervision		20%of direct labor			$280		
Total Indirect Cost					$13,180	27.6	
Manufacturing/Factory Cost					$38,180		80
2. Production Plant Cost							
Plant Administrative							
Administration, Property Taxes, and Insurance		5% of manufacturing costs			$1,909		
Total Plant Administrative Cost					$1,909	4	
Production Costs					$40,059		84
3. Selling Expenses							
Marketing Costs		3% of manufacturing costs			$1,145		
Selling Commissions and Salaries		10% of production costs			$4,006		
Shipping Expenses		4% of prime cost			$1,000		
Warehousing		6% of prime cost			$1,500		
Total Marketing, Selling, and Distribution Costs					$7,651	16	
Total Costs					$47,710		100
4. Mark-Up							
Profit and Taxes		20% of selling price			$11,928		
Total					$11,928		
Selling Price					$59,638		
Unit Selling Price = $59,638/100 = $596.38 or $600/unit							

The form is illustrative to indicate the various items that must be tracked to determine the total and unit costs. A form must be designed to the particular operations and materials of the particular company, and as shown in Table 11.1, there are a wide variety of possible operations and it is not practical to design a form to include all types of operations and materials. The costs in the form follow the ladder of costs and the percentages of the total cost can be included in the form as shown in Table 11.3.

BREAKEVEN ANALYSIS

Introduction

There are two critical issues in breakeven analysis [5, 6] that must be considered and they are: (A) the cost base and (B) the various breakeven points. The two different cost bases are the time base and the quantity base. The quantity based breakeven analysis determines the production quantity at the specific breakeven point and this has worked for marketing, sales, and top-management for forecasting yearly sales and other long range planning activities. However, it provides little assistance at the plant management level where the production quantity is not a variable, but is a quantity specified by the customer. Time based breakeven analysis focuses on the time to produce the order, which is something under the control of the plant supervision. Time based breakeven analysis determines the production time for the specific breakeven point and this is what can be controlled at the plant level. The same breakeven points can be used in either system, but the costs must be considered carefully as the different bases—time and quantity—result in different conclusions with respect to the variability of the costs.

Costs are generally classified as fixed costs, variable costs, and semi-variable costs; but whether as cost is fixed, variable or semi-variable depends upon the cost base used. One of the difficulties in promoting the time based system is that what has been treated as a variable cost in the quantity based system is often fixed in the time based system and vice-versa. The second issue is the

breakeven points is that increased quantities are desired in the quantity based system, whereas decreased times are desired in the time based system.

Cost Bases

Costs are generally classified into three major groups: fixed, variable, and semi-variable. How the costs are assigned to a group depends upon the cost base that is whether the cost base is the quantity base or the time base. Since production quantity has been the standard base, this base will be considered first. Costs that do not vary with respect to production quantity are considered as fixed costs and some of the commonly designated fixed costs are: property taxes, administrative salaries, research and development expenses, and insurance. Variable costs are those that vary linearly with production quantity and the most common variable cost items in the production quantity base are direct material costs and direct labor costs. The costs that do not fit into either of the fixed or variable costs are classified as semi-variable costs. One example of semi-variable cost is maintenance cost.

When a time based system is used many of the cost components in the fixed and variable categories change when compared to the quantity based system. Costs that do not vary with respect to time are considered as fixed costs, and these would be items such as the direct material costs, as the production quantity is fixed, the material costs would be fixed. On the other hand, costs such as property taxes, administrative salaries, research and development expenses, and insurance costs would be considered as variable, as they must be recovered over time. Direct labor may be a fixed or variable quantity depending upon the policies used. If the direct labor does not vary per unit of production, then with a fixed quantity the direct labor cost would be a fixed cost on the time basis. If the labor force were fixed, such as when management does not lay off employees, then it would be similar to a fixed salary and the cost would be variable with respect to the time to do the work.

Breakeven Points

The four Breakeven Points that are considered in the profitability evaluation of products or operations are the shutdown point, the cost point, the required return point, and the required return after taxes point. These points can be evaluated on either the time based or quantity based system. The points can be defined as:

- **Shutdown Point (SD)**—The Shutdown Point is the quantity or time where the manufacturing costs equals the revenues. In the production quantity system, it is the production quantity at which the revenues equal the manufacturing costs and in the production time system, it is the production time at which the revenues equal the manufacturing costs. The manufacturing costs include the material costs, tooling costs, labor costs, and plant/shop overhead costs.

- **Cost Point(C)**—The Cost Point is the quantity or time where the total costs equals the revenues. In the production quantity system, it is the production quantity at which the revenues equal the total costs and in the production time system, it is the production time at which the revenues equal the total costs. The total costs include the manufacturing costs, plus all other costs; such as: the administrative costs, selling and marketing, research and development expenses, and etc.

- **Required Return Point (RR)**—The Required Return Point is the quantity or time where the revenues equals the total costs plus the required return. In the production quantity system, it is the production quantity at which the revenues equal the total costs, plus the required return.

- **Required Return After Taxes Point (RRAT)**—The Required Return After Taxes point is the quantity or time where the revenues equal the total costs, plus the required return and the taxes on the required return. In the production quantity system, it is the production quantity at which the revenues equal the total costs, plus the required return, plus the taxes on the required return. In the production time system, the required return after taxes point is the time at which the revenues equals the total costs, plus the required return, plus the taxes on the required return.

The Breakeven Points increase in quantity as one proceeds from the shutdown point to the required return after taxes point, in the production quantity based system, which implies higher production quantities are desired. However, the Breakeven Points decrease in time as one proceeds from the Shutdown Point to the required return after taxes point in the time based system. The decrease in time indicates the importance of decreasing production to increase profitability and is similar to the "Just-In-Time" concept that focuses on time.

Illustrative Breakeven Example Problem

A metal casting example will be used to illustrate both the production quantity based approach and the time based approach to determining the four breakeven points. The same data will be used to illustrate that both methods can be used, but the time based system gives results that are more meaningful.

A new job is being considered in the foundry. The order is for 40,000 castings and the tentative price is $3.00/casting. The pattern will be designed for four castings per mold and the pattern cost has been quoted at $10,000. The molding line is the rate controlling step in the production process in this particular foundry and the production rate is 125 molds/hr. The estimated time for the production of the 40,000 castings would be determined by:

(40,000 castings) / (4 castings/mold x 125 molds/hr.) = 80 hr.

The costs and overheads are included in Table 11.4 and the corporate tax rate is estimated at 40 percent.

Production Quantity Based Calculations

The calculations for the four breakeven points will be made using "X" as the variable representing the number of units of production.

Shutdown Point

Revenues = Production Costs
3X = Material Costs + Labor Costs + Tooling Costs + Plant Overhead Costs
3X = 1.50X + 0.33X + 10,000 + 8,800
3X = 1.83X + 18,800
1.17X = 18,800
X = 16,068 units

Cost Point

Revenues = Total Costs
Revenues = Production Costs + Overhead Costs
3X = 1.83X + 18,800 + 12,000
3X = 1.83X + 30,800
1.17X = 30,800
X = 26,324 units

Required Return Point

Revenues = Total Costs + Required Return
3X = 1.83X + 30,800 + 9,600
3X = 1.83X + 40,400
1.17X = 40,400
X = 34,530 units

Required Return After Taxes

Revenues = Total Costs + Required Return + Taxes for Required Return
3X = 1.83X + 30,800 + 9,600 + 9,600 x (TR/ (1-TR))
3X = 1.83X + 40,400 + 6,400
1.17X = 46,800
X = 40,000 units

The results are from the production quantity model can be summarized as:

A. If the production quantity is less than 16,068 units, do not accept the order as the manufacturing costs will not be recovered.
B. If the production quantity is between 16,068 and 26,324 units, the manufacturing costs will be recovered, but not all of the overhead costs.
C. If production quantity is between 26,424 and 34,530 units, all costs will be recovered, but not all of the required return will be recovered.
D. If the production quantity is between 34,530 and 40,000 units, all of the costs and the required return will be recovered, but not all of the taxes for the required return will be recovered. (Thus, the required return will not be recovered after taxes as the government will take its share for taxes).
E. If the production quantity is more than 40,000 units, the required return will exceed the desired required return on an after tax basis.

Table 11.4—Cost Data for Time Based and Quantity Based Breakeven Example Problem

Cost Item or Revenue	$/unit		$/hr	$	Decimal
Revenue		3.00*		120000	
Manufacturing Costs					
Direct Costs					
Material costs					
Hot metal cost	1.00				
Core costs	0.35				
Filter cost	0.10				
Sand preparation cost	0.05				
	1.50	1.50*		60000	
Pattern cost				10,000**	
Labor cost					
Melting	0.10				
Molding and coremaking	0.08				
Finishing	0.15				
	0.33	0.33*	(165)	13,200*	
Indirect Costs					
Plant overhead rate			(110)	8,800*	
Overhead Costs					
General administrative, sales, and					
marketing overhead rate			(150)	12,000*	
Required Return and Taxes					
Required Return			(120)	9,600*	
Tax rate (40%)					0.40**
Taxes for required return			(80)	6,400*	
*values used for quantity based model					
**values used for both models					
() values used for time based model					
Conversion factors for data:					
Production rate: 500 units/hr or 0.002 hours/units; thus, 0.33 $/unit x 500 units/hr = 165 $/hr					
Production time: 40,000 units/500 units/hr = 80 hours; thus, 110 $/hr x 80 hrs = $8,800					

The results can be graphically illustrated using total costs versus production quantity, as illustrated in Figure 11.3. The various breakeven points are shown increasing in quantity from the shutdown point to the required return after taxes. Figure 11.4 is a plot of unit cost versus production quantity and illustrates the various breakeven points. The Breakeven Points on the two graphs are the same values, but the two figures are quite different. The symbols SD, C, RR, and RRAT represent the Breakeven Points at the Shutdown, Cost, Required Return, and the Required Return After Taxes.

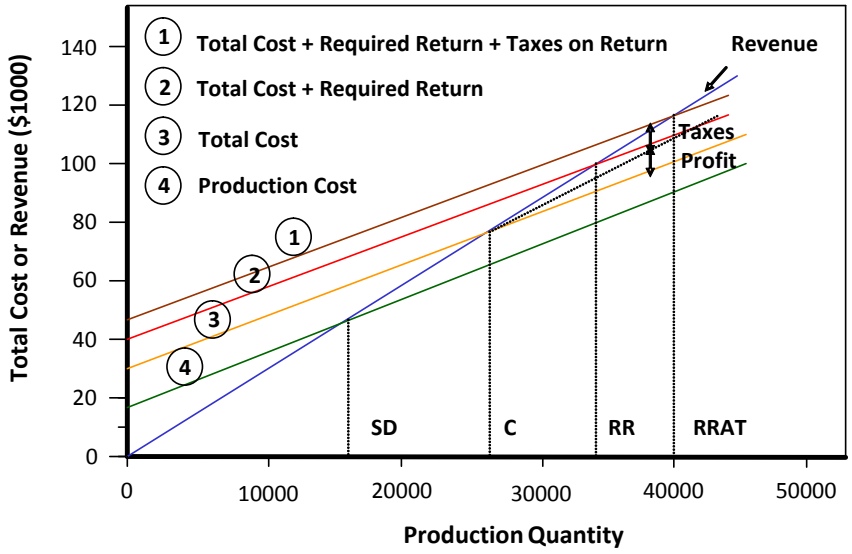


Figure 11.3—Total Cost and Revenues versus Production Quantity

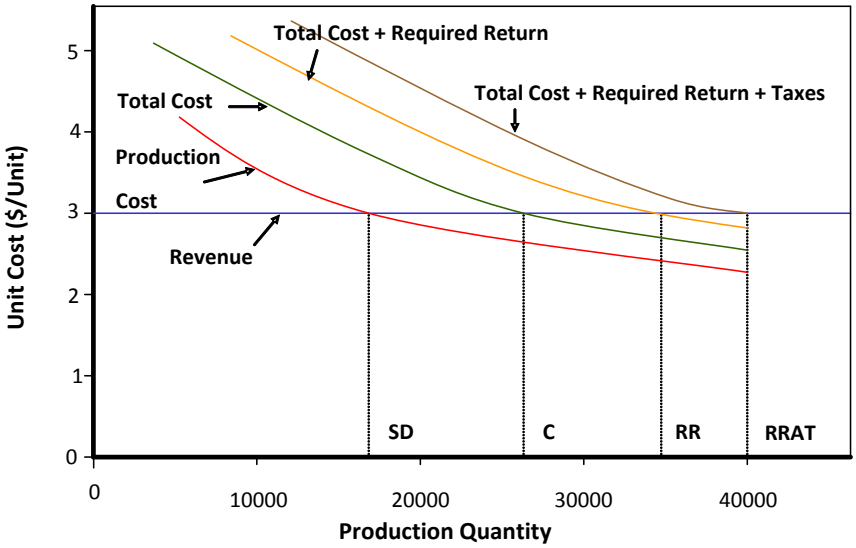


Figure 11.4—Unit Cost versus Production Quantity

Time Based Calculations

The calculations for the four Breakeven Points will be made using "Y" as the variable representing the hours of production.

Shutdown Point

Revenues	= Production Costs
Revenues	= Material Costs + Labor Costs + Tooling Costs + Plant Overhead Costs
120,000	= 60,000 + 165Y + 10,000 + 110Y
120,000	= 70,000 + 275Y
275Y	= 50,000
Y	= 181.8 hours

Cost Point

Revenues	= Total Costs
Revenues	= Production Costs + Overhead Costs
120,000	= 70,000 + 275Y + 150Y
120,000	= 70,000 + 425Y
425Y	= 50,000
Y	= 117.6 hours

Required Return Point

Revenues	= Total Costs + Required Return
120,000	= 70,000 + 425Y + 120Y
120,000	= 70,000 + 545Y
545Y	= 50,000
Y	= 91.7 hours

Required Return after Taxes

Revenues	= Total Costs + Required Return + Taxes for Required Return
120,000	= 70,000 + 425Y + 120Y + 120Y x (TR/ (1-TR))
120,000	= 70,000 + 425Y + 120Y + 120Y x (0.4/ (1-0.4))
120,000	= 70,000 + 425Y + 120Y + 80Y
625Y	= 50,000
Y	= 80.0 hours

The results are from the production time based model can be summarized as:

A. If the production time is more than 181.8 hours, do not accept the order as the manufacturing costs will not be recovered.
B. If the production time is between 181.8 and 117.6 hours, the manufacturing costs will be recovered, but not all of the overhead costs.
C. If production time is between 117.6 and 91.7 hours all of the costs will be recovered, but not all of the required return will be recovered.
D. If the production time is between 91.7 and 80.0 hours, the costs and the required return will be recovered, but not all of the taxes for the required return will be recovered. (Thus, the required return will not be recovered after taxes as the government will take its share for taxes).
E. If the production time is less than 80.0 hours, the required return will exceed the desired required return level on an after tax basis.

The results can be graphically illustrated using total costs versus production time as illustrated in Figure 11.5. The various breakeven points are shown decreasing in time from the shutdown point to the required return after taxes. Figure 11.6 is the profitability plot, which shows the profitability as a function of the production time and shows the various breakeven points. The profitability plot illustrates the importance of reducing production time for increasing profitability. The plot can illustrate either constant amount or constant rates of required return on the Breakeven Points.

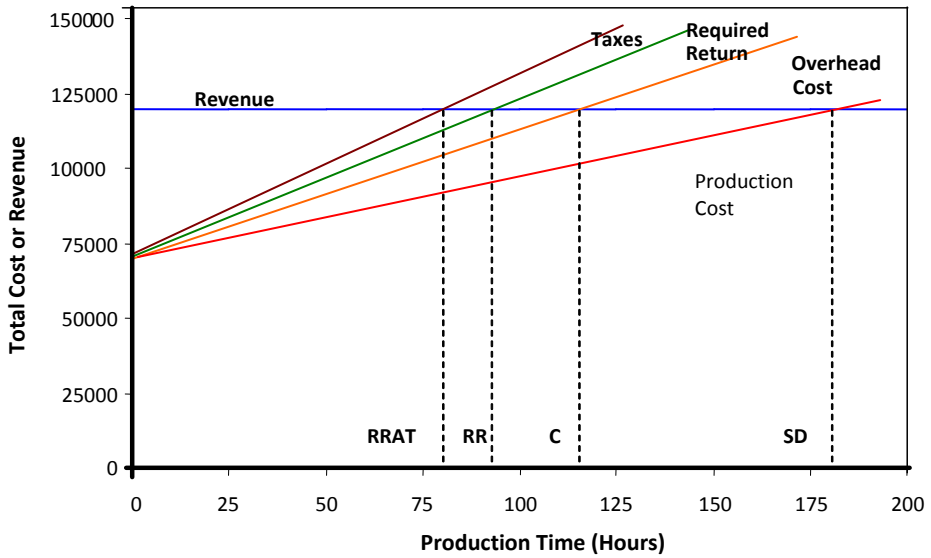


Figure 11.5—Total Cost and Revenues versus Production Time

The advantage of the time based Breakeven Analysis is that it can answer questions such as what is the effect of a four hour delay resulting from a machine breakdown. The effect is not obvious from the quantity breakeven analysis, but the time based Breakeven Analysis indicates that 84 hours is between the required return and required return after taxes breakeven times; that is all costs are recovered and the required return will be exceeded before taxes, but not after taxes. This can be evaluated by determining the profit from:

Profit = Revenues – Costs
Profit = $120,000 – ($70,000 + $425/hr. x time (hr.))
Profit = $50,000 - $425/hr. x 84hr.
Profit = $14,300
Profit after taxes = (1-TR) x 14,300 = 0.6 x 14,300 = 8,580

Since the required return after taxes was 9,600, and the required return before taxes was $16,000, the $14,300 amount is between the two expected values. The loss on the time base system could also be evaluated at $ 425 (165 + 110 + 150 = 425) per hour, and for four hours down the loss would be $1,700.

The loss in the quantity based system can be obtained by using some of the conversion factors; the loss of four hours is equivalent to the production loss of 2,000 units. This loss would be the labor lost, plus the plant overhead and the overhead costs for four hour; thus, the loss would be:

2,000 units x 0.33$/unit + 4hr. x (110 + 150) = $ 660 + $ 1,040 = $ 1,700

The Time Based Approach is much easier to use to determine this impact and is more straightforward.

Time Based Breakeven Analysis would also be useful in evaluating the cost of bottleneck delays and provide data for the economic justification of new equipment to improve productivity. The high cost of delays indicates that one of the factors to consider in the evaluation management's performance and this can be done using a time based system. The evaluation of bottlenecks and delays is critical in the Theory of Constraints and Supply Chain Management, as the focus is upon time in these situations.

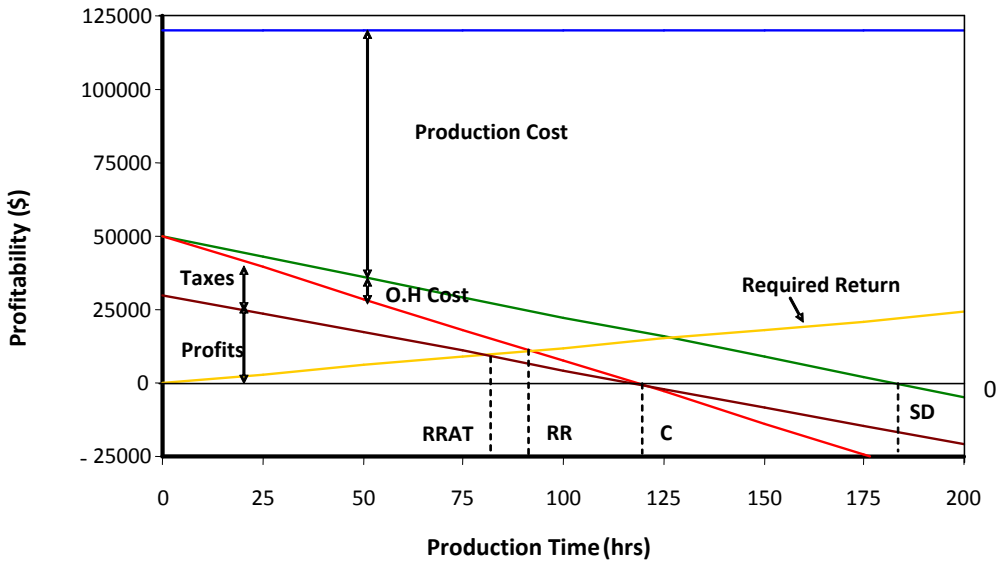


Figure 11.6-Profitability Plot for Time Based System

CONCLUSION

Discrete part manufacturing involves the production of separate, individual products, usually in small batches of 75 units or less. A wide variety of operations are used for the manufacturing of discrete parts, as well as a wide variety of materials. Various manufacturing philosophies and terms for estimating in discrete manufacturing have been presented. The ladder of costs illustrates the interrelationships between the various cost terms and a cost estimating form has been presented for discrete manufacturing.

Time Based Breakeven Analysis has been presented along with the traditional Quantity Based Breakeven Analysis to illustrate the differences and similarities in the two approaches in analyzing the same set of data. The classification of costs as to whether they are fixed, variable or semi-variable depends upon which cost base is used, and that is the time base or the quantity base. The four basic

Breakeven Points of shutdown, cost, required return, and required return after taxes, are illustrated in both the time based and production quantity based systems.

The Quantity Based approach is appropriate for marketing and sales forecasting as they need to estimate the revenues obtained and predict the sales quantities. However, at the plant level, where scheduling of the daily operations are concerned, the focus is upon time and Time Based Breakeven Analysis is more appropriate. Time Based Breakeven Analysis is also more appropriate to the newer approaches to production management, such as the Theory of Constraints, Just-In-Time, Supply Chain Management and Lot-Size-Of-One in that the focus is upon time rather than quantity. The profitability plots indicated the importance of production time reductions upon improving profits, which complements the Just-In-Time philosophy for manufacturing.

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SECTION 3 – PLANNING AND SCHEDULING

Chapter 12: Project Planning
Chapter 13: Scheduling

Chapter 12—Project Planning[6]

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Abstract
The purpose of this chapter is to discuss the importance of project planning for Engineering Procurement and Construction (EPC) type capital projects. The chapter is based partly on AACE Recommended Practice 39R-06, *Project Planning—As Applied in Engineering and Construction for Capital Projects*, the former S&K5, *Chapter 20, Project Planning*, which was based on, *"Effective Project Management Through Applied Cost and Schedule Control,"* by James A. Bent and portions of S&K 5, *Chapter 12, Planning*, by Jennifer Bates. This chapter will discuss the major features of project planning.

Keywords
Constraints, Constructability, Contract Documents, Cost Baseline, Cost Breakdown Structure (CBS), Goals, Objectives, Organization Breakdown Structure (OBS), Phase Relationships, Risk Planning, Schedule Baseline, Scope of Work, Sequencing, Stakeholders, Value Engineering, Variables, and Work Breakdown Structure (WBS).

INTRODUCTION
It is vital that the management of projects include identifying risks, maximizing cost savings, minimizing schedule delays, and improving economic return. This can only be achieved with a quality program of project planning and project management based on the planning outcomes and deliverables.

Project planning, as outlined in the *Total Cost Management (TCM) Framework*, is a process comprised of a series of integrated steps which begin with conceptual planning, continues with project scope development and execution planning, schedule planning and schedule development, resource and procurement planning, cost estimating and budgeting, risk management, and value management. After approval, project planning ultimately concludes with the implementation of project controls through the development of a detailed Critical Path Method (CPM) schedule and related deliverables. Other key components of the planning process for construction that need to be understood are logistics planning and planning documentation.

This chapter is based on a number of chapters from the *Total Cost Management (TCM) Framework* pertaining to planning. The TCM chapters are as follows:

- **Chapter 3**—Strategic Asset Planning
  - 3.1—Requirements Elicitation and Analysis
  - 3.2—Asset Planning
- **Chapter 4**—Project Implementation
  - 4.1—Project Implementation
- **Chapter 7**—Project Control Planning
  - 7.1—Project Scope and Execution Strategy Development
  - 7.2—Schedule Planning and Development
  - 7.3—Cost Estimating and Budgeting
  - 7.4—Resource Planning
  - 7.5—Value Analysis and Engineering
  - 7.6—Risk Management
- **Chapter 8**—Project Control Plan Implementation
  - 8.1— Project Control Plan Implementation

For the most current process maps, please refer to the *TCM Framework*, 2nd Edition.

Project Planning, as defined by AACE is, "the determination of a project's objectives with identification of the activities to be performed, methods and resources to be used for accomplishing the tasks, assignment of responsibility and accountability, and establishment of an integrated plan to achieve completion as required" [1, 5].

[6] This chapter is a merger of the former S&K 5*, Chapter 20, Project Planning*, which was based on, *Effective Project Management Through Applied Cost and Schedule Control*, by James A. Bent, and portions of S&K 5, *Chapter 12*, *Planning*, by Jennifer Bates. The chapter has been modified and revised by Peter W. Griesmyer for S&K 6.

Learning Objectives
After completing this chapter, the reader should be able to:

- Understand the role of the project manager and the project planner in project planning.
- Understand planning strategies.
- Understand the importance of planning.
- Understand planning deliverables.

THE IMPORTANCE OF PLANNING
Planning is extremely important to setting the stage for successful outcomes on projects. In its planning, an organization makes implicit assumptions about its future so it can take action today. It would be convenient if cost/benefit studies were available to prove the value of planning. Theoretically, the planning payoff could be measured by relating the value of what was achieved through planning to the cost of the planning. Unfortunately, the opportunity to quantify outcomes attained with or without the use of a planning process is very rare.

Accordingly, we must rely on other indicators. Given any project or opportunity, a company develops a plan to maximize opportunity. In doing so, it uses the best information available. When the plan is implemented, activities are carefully monitored and controlled, using the plan as a reference baseline. Complete records are maintained through the execution phase. Finally, the experience gained is fed back to the company to increase its knowledge base for the next planning action. This cycle represents the learning curve in action—each repetition makes planning for and achieving the next opportunity much easier, and the opportunity for success much greater. Without a firm commitment to the planning cycle, a company is continually "reinventing the wheel," frequently wasting time, money, and jeopardizing its place in the competitive marketplace. On EPC projects, this is often realized by having to perform "rework", with attendant lost productivity and increased costs; not to mention, critical path delay.

Establishing a Planning Culture
Planning is not done by upper management alone; it exists in a hierarchical structure made up of policies, strategic plans, and operational plans. Different organizational levels produce plans that are quite different in type and scope. Nevertheless, everyone involved in an undertaking must plan, whether their charge is to develop a long-range plan for company growth, or to develop a personnel procurement plan for a specific project. There are numerous reasons why a company that encourages a proactive, structured approach to planning will reap significant benefits over a company whose planning approach is reactive or random:

- Preparing a clear scope definition minimizes the potential for overlooking an aspect critical to success.
- Undertaking planning as a team effort permits various viewpoints and ideas to be expressed.
- Documenting the resultant plan provides a means and structure for communication between the participants.
- Providing a baseline plan prepares for managing the execution phase.
- Developing a baseline plan also provides the basis for measuring progress and controlling change. And,
- Conducting incremental phase and post-completion reviews greatly reduces the potential for planning errors on activities in subsequent phases.

Effective planning becomes routine when planning is an integral part of the company's culture. This begins with commitment by top management, continues with communication of that commitment to mid-level managers, and becomes rooted when every employee relates unequivocally with the company's goals. As with any operation, if those who are to manage a plan do not participate in its preparation, their level of commitment to success may be less than total. Therefore, using a team approach to planning—builds participant confidence in the organization, stimulates communication among the parties, and promotes their feelings of ownership in the outcome.

It also demonstrates that top management has a direction, that decision-making is under control, and that the total organization is working to achieve the same objectives. An additional, but no less important, result of the team approach is the training that lower-tier managers receive as they participate with upper management in the planning process. They are thus better able to assume higher levels of responsibility as opportunities develop, bringing with them a planning philosophy that is fully ingrained [2].

Planning Tools
When planning tools are mentioned, the tendency is to think in terms of hardware, software, and procedures. The most fundamental and useful planning tool available is the experience the planning team participants have gained during previous undertakings. While not readily quantifiable, this experience provides a sound basis for using the other, more tangible planning tools. These include the following.

- **Commercial handbooks and software programs**, a variety of which are available—These should be used with an understanding of their basis and limitations, rather than applying them across the board.
- **Standard, company-wide, policies and operating procedures** that have been officially issued—Planners can then feel free to use them without having to continually seek management guidance and approval.

- **Model plans (sometimes referred to as "go-bys" or "templates")** that can be adapted as necessary to specific undertakings—Organizations that tend to undertake repetitive work should develop a model project and plan for each type of work.
- **Checklists** that will support planning and help prevent overlooking key items that may have cost or schedule implications.
- **Historical databases** cataloging company experience on past projects in a standard format for use in new endeavors. Along with database cost and schedule performance information, valuable insight can be gained from interviewing company staff involved with those similar past projects.
- **Codes of Accounts** structured to catalog work, cost accounts, resources, and other information uniformly across all projects. These are essential if planning is to take advantage of previous planning experiences and available software. Codes of Accounts should be standardized to the extent practical to ensure consistency of data cataloging and use. For work breakdown structures and cost breakdown structures, the codes should be hierarchical, to permit capturing information at various levels of detail. [2]

The following are the major constituents of project planning:

- Establishing project objectives and defining how to achieve the objectives.
- Identifying project stakeholders their roles, responsibilities, and their effect on the scheduling planning process.
- Identifying contract requirements, including project delivery methods under the terms of the contract. The contract delivery method is one factor that will determine the extent of the planning effort by the project team.
- Identifying the constraints and variables that will be considered by the project team during the planning process.
- Determine the scope of work, client requirements, schedule hierarchy, division of responsibility, project plan review, approval requirements, and distribution.
- Identifying major work activities (phases) and deliverables (goals) and the preferred sequence in which they are to be accomplished.
- Establishing an integrated, time phased plan for the major work activities to achieve project completion as required.
- Identifying a process for further scope of work element cost/schedule definition.
- Identifying project management coordination necessary to track and report on cost/schedule progress.
- Developing non-schedule related planning methodologies, such as logistics planning. These planning methodologies are not discussed in detail in this chapter; however, typical non-schedule planning initiatives/methods required on most engineering and capital construction projects are noted further in this chapter.
- Preparing and achieving approval of the "Project Execution Plan", which documents all aspects of the resultant plan developed by the project management team. This living document becomes the framework of many other documents and is used in software tools such as the CPM Schedule; which is in turn used to monitor and control the work.

Planning and Scheduling are distinct but related processes for capital engineering and construction projects. Schedule planning (planning) and schedule development (scheduling) usually require a different set of skills and knowledge. Planning is strategic, where scheduling is tactical.

Planning consists of planning the work, the resources, and the estimated cost over time to complete the scope of work, as defined during the early phases of the project. Scheduling includes the identification of many elements associated with the scope of work which are developed into work packages, sequenced into phases, and their discrete performance activities. Prior to the project execution phase, planning uses an iterative planning process to identify the means, methods, and resources necessary to produce the project plan. Scheduling continues to evolve a detailed sequence of work during the life of the project and incorporates the experience and knowledge gained from previous project successes and failures.

The purpose of planning is to establish an acceptable course of action ("plan") to perform the defined scope of work of a project; in an efficient, coordinated manner, based on a review of project requirements and responsibilities. Key elements in the initial planning effort include the identification of stakeholders, contract requirements, and the project delivery method.

Planning precedes the scheduling effort. The project schedule is detailed during the schedule development phase of the project. Using the plan developed and approved during the planning phase, the project management team develops a time-phased management framework to help guide the project toward completion of the entire scope of work. The transition from project planning to schedule development occurs seamlessly after the plan has been finalized and reviewed by the appropriate stakeholders to become the basis for the schedule. Planning is a continuous process that never stops until the project is completed.

ESTABLISHING OBJECTIVES AND DEFINING HOW TO ACHIEVE THE OBJECTIVES

General

The process of developing objectives can assist in building team commitment and understanding of the project's scope, goals, and objectives. Objectives balance scope, quality, cost, and schedule; and are used as a guide to make decisions. These major objectives then guide development of more detailed goals, procedures, technical criteria, cost targets, and individual milestones. Ideally, a

common set of objectives should guide the owner, engineer, and constructor. These objectives provide the work direction to all parties, and as such, would have to be compatible and acceptable. The key to successful acceptance, by all stakeholders, is a set of well-defined goals and objectives which should be identified early in the planning process.

Stakeholder Satisfaction
Criteria and a measurement program of key cost and schedule performance indicators (KPI's), acceptable to the client, should be developed to a quantitative measure for periodic performance reports. Stakeholder communication should be the single most important objective for these reports. Performance should be measured against this objective; and in the case of variance over a threshold limit a plan for corrective action should be developed immediately.

Scope Objective
The initial scope objective should be developed by the owner and/or client. As the planning process evolves, the project team will expand the scope definition. A well-written, but brief, scope definition should be developed so that the definition directly supports the project scope, and the technical scope definition also relates to the contract payment schedule or the approved project budget appropriation, as discussed.

Cost and Schedule Baselines
The required quality and formats of the estimating and scheduling programs should be identified in conjunction with the associated databases and computer systems. All internal and external constraints, interfaces, risks, and influences should be carefully evaluated for both cost and schedule baselines. A CPM scheduling system is recommended. Levels of schedule detail, codes of accounts, and work breakdown structures for the cost/estimating program, all require careful consideration. Overall and intermediate milestone objectives should be developed. An ongoing risk analysis program should identify the ranges of risk for both cost and schedule. The responsibility for the management of contingency (cost and schedule) should be clear and precise. Maintenance and allocation of project contingency should be the project manager's responsibility, with management reserve treated as the prerogative of the next highest level of management. **Note**: AACE has a number of Recommended Practices (RPs) on project contingency. For the latest RP on contingency, please visit the AACE website at: www.accei.org. Two RPs that are relevant are:

- AACE Recommended Practice 40R-08, *Contingency Estimating—General Principles.*
- AACE Recommended Practice 70R-12, *Principles of Schedule and Cost Management as Applied in Engineering, Procurement and Construction.*

Quality
Clear and unambiguous quality control criteria should be developed and be fully acceptable to all project parties. The criteria need to be measurable, so that quality control reports can be issued on a regular basis, confirming the conformance with quality control criteria. Quality of project operations as well as quality of design and construction needs to be covered.

Communicating Project Objectives to Project Team
If this is not done and constantly reiterated, acceptance leading to commitment will be lacking. The establishment of clear priorities, with each objective having its relative priority, will allow multiple groups to work in harmony with each other. Thereafter, a constant effort (as part of team building) must be made to keep the project objectives visible. Such communication is best facilitated through the development of a "Project Execution Plan" and team alignment sessions throughout the life of the project.

Effective Project Team Building
Assembling a group of individuals, especially on large projects with large companies, does not make a team. Personnel can come from different locations (worldwide) and different cultures. Time is needed for individuals to recognize and control their individualities, as appropriate, and learn to work together. Individuals usually accept project assignments with little or no knowledge of the individuals with whom they will be working with. Therefore, it is essential that management and project leaders develop a team building program and maintain it throughout the duration of the project. Developing project commitment and cost consciousness in the team, helping team members achieve personal satisfaction, etc., are the deliverables.

Effective Community Relations
There is an ever-increasing opposition from local communities to process projects. This is because of the hazardous nature of many of these projects. An effective and positive public relations effort in conjunction with direct financial investment in local matters is necessary and essential.

IDENTIFY THE STAKEHOLDERS

General
There are many different stakeholders on a project and each has an interest (stake) in the endeavor with different goals and objectives. Stakeholders may have some influence on the development of the plan, either directly or indirectly. The stakeholder's interests may have a positive or negative influence on the planning process. Regardless, stakeholder consensus (buy-in) during the planning process is important to obtain.

Effective Interface with Stakeholders, Operations, and Maintenance for Scope Approval
Achieving pertinent design input from all project parties is a formidable task. This work is usually the direct responsibility of the project engineering manager, and strongly supported by the project manager. The design basis must be fully communicated and receive approval from all design decision-makers. After approval, the design basis must be shared openly with all participating parties. When the design basis is sensitive or proprietary, security procedures must be established. In addition to design, the project execution plan and financial program must be part of the approval process.

Stakeholder Consideration
Stakeholders may be internal or external to the project. Stakeholders, whether major or minor, have a significant influence on the success or failure of the project. Each stakeholder's concerns and limitations must be identified, evaluated, and reconciled during the planning process.

The project team must communicate the project plan, goals, and objectives to all stakeholders to ensure the stakeholders understand the plan, goals and objectives, as there will a number of different views. The project team, including internal sponsors, is responsible to mediate any differences between stakeholders—as soon as they are identified, in order to ensure eventual success. Failure to address all stakeholder concerns could result in delays, cost overruns, or even cancellation of the project.

Stakeholders include, but are not limited to, the following:

- Owners
- Designers, architects, engineers
- Contractors, subcontractors
- Supplier, vendors
- Government or public agencies
- Local utilities
- Community and special interest groups

IDENTIFY THE CONTRACT REQUIREMENTS

General
A primary focus of the project team during the planning phase is to understand the various aspects of the contract, and the attendant risks, obligations, and responsibilities it imposes on the contracting parties. In order to achieve a successful outcome, each project team member should fully comprehend the contract terms, conditions, project delivery method, technical requirements, and their respective responsibilities toward performing the work.

Contract Documents and Specifications
The governing document for any project is the project contract. The requirements of the contract define the scope of work, the performance standards, the compensation and perhaps the processes and procedures to be used. The contract requirements may define the minimum planning and schedule requirements.

Requirements in the contract documents cannot be disregarded. If any contract specifics do not align with the proposed plan, the variances must be promptly identified and resolved with the stakeholders to exclude the cause of the variance, or modify the compensation.

There are a number of contract types and delivery methods that can be used on projects.

Contract types are as follows:

- Stipulated Sum (value) contracts
- Fixed Price
- Unit Price
- Cost Reimbursable contracts
- Cost-Plus (with fixed, incentive, or award fee)
- Time and Material (T&M)

Project Delivery Methods are as follows:

- Design-Bid-Build
- At risk Construction Management
- Multi-Prime Contracting
- Design-Build

- Engineering-Procurement-Construction (EPC)
- Design-Build-Operate-Maintain
- Integrated Project Delivery (IPD)
- Alliance
- Variations of the above

IDENTIFY CONTRAINTS, RISKS, AND VARIABLES

General
Each project will experience constraints, risks, and variables that go beyond the availability of resources and will need to be mitigated. These include, but are not limited to, site conditions and restrictions, financial and economic factors, culture and religious factors, and local laws and regulations. They must be promptly identified and the impact evaluated, documented, and resolved or mitigated in accordance with the contract requirements. As the planning effort continues to evolve, the constraints, risks, and variables may change, be resolved, magnify or disappear. As a result, the current plan must be re-evaluated as the project conditions change.

Project constraints, risks, and variables are events, elements, or features that will have an impact on project execution and performance. Identification and resolution of these items during the planning phase is essential for success of the project. Significant project constraints, risks and variables include:

- Physical location and space
- Geological limitations
- Transportation limitations/restrictions
- Inefficient sequencing
- Productivity restrictions
- Resource availability
- Project labor agreements and their restrictions
- Funding methods, timing and phases
- Contracting and project delivery method limitations
- Testing and commissioning requirements
- Closeout and handover requirements

Historical Data
To assist in the mitigation of constraints, risks, and variables, the project team will often use historical and physical project data—such as conceptual plans, pertinent project correspondence, project variations and constraints, and impacts from similar projects, as a guide.

Examples of relevant historical data on a typical capital process plant include:

- Type of project
- Throughput/Capacity
- Technology
- Global market conditions
- Labor resources
- Project team organization
- Lessons learned
- Actual cost and schedule data

Examples of relevant physical data include:

- Geographic and other location factors
- Geological
- Terrain
- Utilities
- Transportation
- Environmental
- Cultural
- Climate Factors

Note: The historical and physical data identified above can be used in any other type of projects, with some additions and/or deletion of identified items.

Resource Planning
Resource Planning is a critical aspect of the planning process, and if not considered, may adversely impact the project. During the early stages of the planning process, the project team must identify the key resources and their availability, along with considerations for each, as they support the overall project goals and objectives. Resources include, but are not limited to, the following:

- People
- Equipment
- Materials
- Technology
- Capital funding

DETERMINE THE SCOPE OF WORK

General
Scope definition is a matter of project discipline and design control necessary to identify and prevent scope changes that are all too common on fast track projects. A Construction Industry Institute (CII) publication 43-2, "*Quantitative Effects of Project Change*," published in May 1995, reports, ". . . Poor scope definition and loss of control of the project scope rank as the most frequent contributing factors to cost overruns." A report developed by the Construction Owners Association of Alberta (COAA), entitled the *COAA Major Project Benchmarking Summary*, [6] emphasizes the importance of defining the scope of the project early, as well as the top five factors affecting cost, schedule or productivity. The report noted that performing a constructability review can reduce schedule growth by at least 50 percent.

Scope is Well Defined before Start of Detailed Engineering
The purpose of the feasibility study is to define the scope. However, the quality and extent of the early design and project work are a matter of management decision which can vary widely. A poor design package at the start of detailed engineering will result in significant delay, change, excessive rework, and a substantial cost increase. The major deliverable of the feasibility study is the basic design package—Statement of Requirements (SOR), which should be well-written documents that properly define the technical requirements and have sufficient depth to provide clear direction for all major design issues. The SOR should clearly communicate the scope of work to the designers and set appropriate boundaries on the project design to support detailed decision-making.

The Construction Industry Institute (CII) developed the Project Definition Rating Index (PDRI) [3, 4] program to identify inadequately defined scope early in the planning process. During the planning process, PDRI indexes the project scope definition against similar projects to determine the degree of design definition completeness, or maturity. The PDRI also supports the initial risk management definitions, as incomplete design is indicative of future risk events. It is used for the following:

- A scope checklist during early project development.
- A "gate" check before moving into the next project design phase.
- An indication of the level of additional front end efforts that is required.
- An indication of the front-end performance and/or benchmarking.
- A means of identifying scope which may later be subject to overrun.
- A means of capturing maturity indexes from successful projects, to be used as references in the PDRI and as lessons learned.

The scope document should cover:

- **Project Description**
  Project justification, project objectives, economic justification, and if pertinent, facility description.
- **Design Basis and Specifications**
  Process definition:
  - Description of process
  - Process flow diagrams
  - Tabular heat and material balance
  - Process conditions, special conditions
  - Construction of materials
  - Startup and shutdown requirements

  Mechanical definition:
  - P&ID drawings—preliminary sizing and piping tie-ins
  - Preliminary plot plan
  - Preliminary general arrangement
  - Preliminary equipment list

Electrical definition:
- Electrical one-line diagrams
- Electrical three-line diagrams
- Electrical panel diagrams
- Electrical schematic diagrams
- Preliminary equipment list

Instrument definition:
- Primary control points and purpose
- Instrument set points, low level alarms, etc.

Safety System:
- Hazards analysis (Hazops)
- Safety devices and their design criteria
- Interlock logic description and diagram

Project Location—Elements:
- Engineering and construction productivity factors (versus database)
- Logistics reviews, delivery to and at-site
- Infrastructure requirements, at-site
- Weather concerns and impacts

Project Conditions:
- Offshore installations-suppliers
- Prefabrication and modules (if needed)
- Operational restraints-conditions
- Site and access problems

Estimate—Definition:
- Work quantities and takeoffs
- Engineering, labor, staff hours
- Contingency and budget limitations
- Risk analysis and identification
- Work Breakdown Structure (WBS)

Schedule—Definition:
- Difficulty of proposed completion
- Phase definitions
- Project sequencing and phase relationships
- All constraints, restraints and critical relationships; and,
- Appropriate levels of detail.

Decisions on Scope Changes are made in a Timely Manner

This can only be achieved if there is a cohesive, dynamic trending program. Effective communication and team coordination are direct contributors. A timely change approval process is also a factor, as unapproved changes will result in cost and schedule impacts.

Dynamic Design Change Control—Formal Program

Project trending and reporting systems, such as the design change order log, are essential. The weekly trend meeting and regular progress meetings provide an early identification of change. An effective design control program is centered on an engineering milestone(s), called the Design Gate Control Points (DGCP). If the feasibility study is extensive, the design control point could be operational at the end of the study, and thereafter, all changes would be formally documented. If the feasibility work were minimal, then the design control point would be established at the early part of the project engineering phase. The design control point is reached when the project's original scope is properly defined, agreed to, and approved by the relevant stakeholders. As this approval is reached, the project manager will inform all appropriate parties that the design control point has now been implemented. On large projects there can be multiple design control points. The design control point is not a design freeze, as subsequent viable design changes may be evident and should be approved and implemented.

IDENTIFY THE MAJOR DELIVERABLES (GOALS) AND ACTIVITIES (PHASES)

General

Each project must establish goals that are achievable in order to be successful. Early in the project, the owner and/or client will establish a set of initial conceptual goals. The project team will be responsible to maintain, update and identify the client goals, review and recommend incremental and supplementary goals for all stakeholders to act upon.

A project goal is a concept that is identified and expressed as an achievable end product. Goals can be either contractual or non-contractual depending on the nature of the project. Contractual goals should only represent definable or measurable goals that can be achieved.

A phase is a significant period or group of work within the project. A phase may include several stages of planning and work. Phase definition involves the initial identification and outlining of major relationships and sequence planning, as defined by the scope of work. As the scope of work becomes more refined, the planning effort includes the identification or breakdown of the scope of work into more finite tasks that will cover the entire project.

As the phases become more defined they will lead to the initial development of the cost estimate, control level schedules, and the identification of risk elements that will need to be mitigated in order to achieve success in those defined phases.

Phases are based on and derived from the scope of work, sequence of work, phase relationships and the Work Breakdown Structure (WBS). Phases might be defined or modeled by a fragnet and may have distinct start and finish milestones.

ESTABLISH AN INTEGRATED TIME PHASE PLAN

General
A project execution plan is the most significant output of the planning process. Through a series of reviews and revisions the project team's planning effort and actions develop into the project baseline schedule. This baseline plan becomes the framework and benchmark for all project measurement evaluations.

The scope of work that was identified early during the planning process typically determines the basic means and methods necessary to develop the project plan as the project moves forward.

A combination of an organization's policies and procedures, along with those of the project, may dictate the development, review and approval of the baseline plan.

Sequencing and Phase Relationships
Phase relationships are the foundation for developing the framework used for monitoring, analyzing, controlling and reporting project execution. Phases will eventually become broad groupings within the schedule and ultimately result in a nesting of specific work packages and activities.

Sequencing is the part of the planning process that begins to bring together the interrelated phase of the project, using simple logic relationships. Phases may overlap each other within the project timeframe.

Identification and sequencing of phase relationships comprise the initial development of the schedule, as the project plan is transformed from a concept. Both sequencing and phase relationship identification are outputs of the planning process, and are products from the project team's definition of various phases of the project.

Breakdown Structures
As the various phases of project plan are developed, a mechanism needs to be developed to communicate the project's schedule, cost, and organization structures that will facilitate reporting against the baseline schedule and cost estimate or budget.

Work Breakdown Structure (WBS)
The Work Breakdown Structure (WBS) is a hierarchical division of the work elements to be performed on a project. The function of the WBS is to divide the scope of work into manageable parts that correspond to key deliverables, phases and/or milestones with the intent to avoid the omission of key elements and assist in the communication of cost, schedule, and resource performance data to stakeholders and team members. Phases may group WBS elements as a means of producing summary statistics, but are probably not true deliverables as they do no work. Milestones may indicate that one or more WBS elements are completed, but they are not WBS elements, as they do no work.

The level of WBS deliverable definition balances the requirement to clearly define the scope of work to the level that discrete elements allow effective management of the work, while not driving detail down to a level where managing that detail is inefficient and obtrusive. If the level is summarized too high, performance will not be determinable. If the level is detailed too low, performance will be impacted by excessive reporting. Summary level reporting simplifies analysis. Detailed level reporting allows for accuracy. WBS deliverable elements decompose the scope down to work packages, the lowest WBS level. Each work package is performed by one or more activities. Typical rules of thumb for defining activity scope and complexity are:

- Any activity should require at least a day in duration. Short duration, high-intensity projects may have activities in hours. These are typically "outage" projects. Long-term conceptual programs have durations in years.
- Ideally activity duration should not exceed the routine reporting period, for example, one month.
- As a work package is only performed by a single entity, an activity cannot mix multiple contractors or suppliers.
- Activities are the continuous performance of a deliverable, and are not interrupted by other activities. When this condition logically occurs, break the activity into multiple activities.

Organization Breakdown Structure (OBS)
The Organization Breakdown Structure (OBS) is a hierarchical division of the producers of the work to be performed by a particular organization. The function of the OBS is to identify the entity responsible for performing each deliverable of the scope of work as identified in the WBS. The lowest level of an OBS is the organizational entity performing the work package. As with the WBS, the level of breakdown is a balance between convenience and effectiveness.

Cost Breakdown Structure (CBS)
A Cost Breakdown Structure (CBS) is a hierarchical definition of the key elements of a project. At the highest level these are: labor, materials, direct, and indirect costs. For a complex project there will be many sub-levels of these major elements.

The Cost Breakdown Structure relates to the WBS and the OBS in the following manner:

- A work package is the lowest level WBS and OBS where performance is measured from the component activities, and resources/cost can be budgeted or estimated in the work packages.
- The Cost Account (CA) code value assigned to the work package at the intersection of WBS and OBS components where actual costs can be forecasted and collected.
- The work package is the intersection of what is being performed (WBS), who is doing it (OBS), and how much it is costing (CBS).
- The WBS, OBS, and CBS may be rolled up for summary reports at any appropriate intermediate level in each structure.

The CBS is a key element of control for a project and a planning requirement which allows all levels of management to track the cost for the project to meet their particular control and reporting needs. The CBS allows management to evaluate the effectiveness of the estimate versus the work in place, remaining work, and overall costs.

Every project requires a CBS in order to define the full range of elements of cost associated with all phase of project implementation. The development of the CBS allows tracking and forecasting of estimated costs versus actual costs. A CBS also makes it possible to capture the discrete cost to perform additional, unplanned work on a cost for labor and material (T&M) basis, when required.

Before project implementation commences, the various WBS/OBS/CA assignments for the work must be clearly communicated with the project team, so that cost and performance of each element of the work may be captured and reported. Roll-up summaries for management review can also be defined for higher level review. Accurate accountability for direct and indirect cost elements associated with the project is essential.

Cost Estimate Development
During the planning phase of the project, goals, objectives, phases, sequences, and the scope of work, are identified and developed into the project plan. As the goals, objectives, phases, sequences, and scope of work develop, the budget process begins that will incorporate the stakeholders expectations.

As the planning process evolves, so does the project estimate based on identified constraints and considerations. The project estimate is often revised so it is in alignment with the project goals, objectives, and timeline. As resource constraints and consideration are identified and their impacts will influence the planning, schedules, and budgeting process.

As the scope is decomposed, the WBS, OBS, and CBS for each work package is identified and incorporated into this integrated structure to form the basis for the preliminary cost estimate and the initial schedule model.

Contract documents may dictate when and how the cost estimate is developed. As a result, the contract may define the timing of certain cost estimates for a particular phase, or the entire project.

The cost estimate, plus contingencies, along with other risk factors, should lead to the final cost estimate. Budget development and its evolution are coordinated with the development of the project schedule. This budgetary process development provides the underlying basis of cost and schedule baselines used for project monitoring, controlling, and reporting. The budget should represent the value of all elements of the scope of work and the schedule should represent the performance of all the elements of the scope of work. They are complementary and integral, and both are driven by the scope of work.

During the planning process, alternative resource options are developed and evaluated based on timelines and the projected budget considerations. These “time versus cost” considerations include value engineering and constructability assessments, and may be affected by approved change orders to the project scope of work.

The schedule elements and associated budget are developed concurrently. At this point, major elements of the work scope are known in sufficient detail to graphically depict on a timeline—allowing an analysis of cash flow, overall resource use and further optimization of the plan. If necessary, revisions are developed and evaluated to reflect meeting the overall project goals.

Baseline Plan
The Baseline Plan is the final output of the initial planning process for a project. The baseline provides the framework for all progress and measurement evaluation, as the plan is implemented, and is approved by management prior to commencement of the work. The baseline provides a basis from which project management decisions are then made to exercise control of ongoing work in order to achieve successful outcomes.

Unless significant modifications to the scope of work occur, the baseline plan is a 'fixed' or unchanging document. Significant changes to the scope of work can cause the previous baseline to be an inadequate representation of the work, and of little value for work progress reporting. When that occurs, a planning effort to re-baseline is recommended. Budget, schedule, and percent complete status are all typically affected. This revised baseline should be approved by the level of management which authorized and approved the original baseline.

The schedule evolves during the planning process, until this final stage when the planning process goals, objectives, and overall scope of work are defined in sufficient detail.

Sequencing and phasing of major tasks allow planning to evolve into conceptual schedule model development. Key milestones can be identified and established. The project schedule model begins with a planning timeline and conceptual budget. These high level presentations allow stakeholders to evaluate the project planning status and likelihood of meeting desired planning goals.

The proposed planning timeline and cost estimate/budget for completion are evaluated to determine feasibility of meeting stakeholder requirements. If necessary revisions are developed and evaluated to reflect meeting overall project goals.

Risk Management Planning
Risk Management Planning includes the development and implementation of risk response plans to mitigate cost/schedule/technical impacts caused by risk and/or change. Changes may have a positive or negative impact on the current plan.

Procedures are developed to identify and mitigate the risk impacts. Early in the planning process, the areas of potential risk and the appropriate means and methods to mitigate those risks are identified. Planning for change should be an integral part of the planning process as change is a risk.

As the plan develops, the project team should be aware of the potential risks and implement a risk management plan to mitigate risk or retire those risks that have not occurred.

Risk may be reduced or increased as the plan is refined, as new variables are identified, as the plan is revised, or as there is a change in scope. Risk must be continually monitored as the project scope and planning evolve. In some cases, the risk may be too great for the project to proceed, thus resulting in re-scoping or even cancellation of the project.

Schedule risk should be evaluated once the entire scope of work has been incorporated, but before the project plan has been finalized, as mitigating schedule risk may have significant impact on the initial cost estimate and schedule development.

Value Engineering
Value Engineering (VE) should begin during design and focuses on maximizing value, and can be expressed as the sum of design item functionality and quality, per unit cost.

Value engineering during project design planning is intended to optimize product and production system design, by considering the effective performance of manufacturing and related activities. Alternate materials, or technologies and standardization (e.g., use common parts for different products) are some of the considerations. In relation to the planning and estimating processes, value engineering considers stakeholder requirements, prioritizes required functions, and evaluates their cost and schedule impact. VE involves thinking "out-of-the-box" to simplify and improve the flow and configuration of the work. As VE implies, this optimization usually occurs before or during the design phase, when changes may be implemented with minimal cost and savings may be maximized.

CONSTRUCTABILITY PLANNING

General
Constructability and construction pre-planning are often used interchangeably. Constructability is largely concerned with the technology, methods of installation, and the associated costs. Pre-planning deals with the scheduling of resources, organization, site access, and infrastructure. Both are intended to reduce time and costs by considering alternative design and/or installation methods. Typical examples would be steel or precast concrete structure frame, site fabrication or prefabrication and/or modularization. Constructability is not the same as value engineering, but the terms overlap as each tends to optimize time and cost performance. Constructability usually occurs after design is finalized; however, constructability can involve changes to the design

details to optimize physical performance of the work. Frequently, trade contractors and material suppliers will suggest minor design changes which result in significant savings in time and cost.

Early Economic "Path of Construction" Program
This is an evaluation of the physical sequence of construction work to produce the lowest cost. Many factors are involved, such as:

- Physical site conditions, weather
- Restraints of drawings, material delivery, schedule critical path sequences
- Economics of crew sizes and supporting resources
- Existing plant operations, safety regulations, regulatory requirements, etc.

Effective early planning to review design and/or material alternatives can occur at minimal cost, but with significant time and cost impact. The data that is developed is then used to optimize the project schedule and cost estimate.

Formal Constructability Programs are an Integral Part of Project Execution
Constructability assessments should occur periodically, beginning during the initial planning phase and continuing through the implementation phase of the project to project completion. Constructability assessments can reduce both time and cost impact to the project. To optimize the cost and schedule planning, while mitigating risk constructability, assessments are often continued throughout the life of a project. These assessments have the significant positive impact when implemented during the earliest planning stages of the project.

Front-End Planning Actively Incorporates Construction Input
It is essential that capable and experienced construction personnel are assigned to the project at this early stage and that their constructability and preplanning evaluations are a proper part of project development.

Construction Driven Scheduling as Key to CPM Program
This is also known as "backwards" scheduling, meaning that the project CPM schedule is structured around the construction schedule, assuming that the construction schedule has been developed on a best economic basis. Engineering and material deliveries then can be matched to the economic construction program, at no cost penalty. Research has clearly shown that the cost benefits of this approach are considerable. Process and industrial projects should always be planned and sequenced considering which WBS elements are necessary to complete each plant system. The overall project will be completed sooner and at less cost than other approaches if driven by system.

PROJECT MANAGEMENT COORDINATION

General
Management expects the project team to turn actual performance data into useful and relevant information. Contemporary computer programs can be configured to allow the gathering and collation of data to become a relatively simple task, with proper management support, organization and funding. Correctly establishing the requirements for input data, results in obtaining the required output and information. With the establishment of effective processes and communication channels, the information may then be directed to the correct stakeholder.

Execution Plan/Formal Written Program
The execution plan should be a dynamic document, which is revised and updated as conditions/scope change, and based on proper/timely inputs from all parties. Commitment to the plan must then be achieved by all project stakeholders, and especially management to ensure the data will be actionable at various management levels.

Typical major categories of a good execution plan include:

- What is the scope of work? (see information in earlier paragraph)
    - How is the work to be executed? Statement of project objectives and priorities.
    - Proposed division of work:
        - Work to be done in-house, by company; or,
        - Work contracted out.
    - Development of work packages.
- What will be the cost of the project and how does it compare to the initial budget?
    - When is the work to be carried out? Schedule and probability of schedule adherence.
    - Budget limitations versus acceleration. And,
    - Critical path and float analysis.
- Who is performing the work?
- What is the expected quality of the project and how will it influence the cost?
- Contract strategy

    - Required scopes and degree of definition
    - Forms of contract
    - Risk allocation versus cost of liability
- Permitting and licensing
    - Federal
    - State
    - Local
- Detailed Engineering
    - Third-party licensers
    - Environmental and regulatory; permits
- Procurement program
    - Types of contracts
    - Competitive bidding
    - Domestic and international
    - Single source; negotiated
    - Plant compatibility and spares requirements
- Construction
    - Preplanning program-critical highlights
    - Prefabrication, modules
    - Pre-commissioning and testing program
- Commissioning and startup
    - Quality assurance—control and inspection
    - Project organization (see earlier chapter)
    - Project coordination procedure
- Resource analysis
    - Engineering, construction availability,
    - Skills and trade union climate
- Marketing Interface
    - Limitations or constraints
- Cash flow limitations
    - Fiscal year requirements
- Access problems
    - Weather windows, traffic limitations
- Shutdown—retrofit program

All Scope Schedule and Cost Matters are Routed Through the Project Manager
This is to ensure project best practices and dynamic trending. This presupposes that the project manager has not just the responsibility, but also the appropriate decision-making authority for all scope, schedule, and cost matters. An organization may assign this responsibility and authority to a different position, but the concept is the same, regardless of the title.

A Team and Cost Culture is Developed on the Project
The project manager is directly responsible for creating an environment that will enable project control to be properly exercised. The project manager must be a cost leader, encourage project cost consciousness, seek counsel from all appropriate sources, accept sound advice, and lead cost/schedule staff to the extent of their capability and expand their performance. Team building and team leadership are key elements of successful project management.
On smaller projects, where the project manager is also the project control engineer for example, the project manager must possess project control skills and motivate the supporting/service groups to provide the quality information that is needed for effective analysis and decision-making.

Effective project control requires the timely evaluation of potential cost and schedule risks and the prompt presentation of recommended solutions to project management. Thus, the cost/schedule specialist must be a skilled technician, able to effectively communicate with management at any level. As in all staff functions, the ability to “sell” a situation or concept may be as important as the ability to perform the service.

On larger projects, multi-discipline project teams are mobilized to form “melting pots.” The difficulty of establishing effective and appropriate communication among all levels should not be underestimated. In such cases, the project manager must promptly establish a positive working environment where the separate functions of design, procurement, construction, and project control are bonded into a unified, performance-conscious team.

Internal Project Charter Program
This is a method of motivational team building reducing the entire execution scope to just a few key objectives. This one or two page document lists the major project objectives, the parties and perhaps their responsibility/accountabilities for the project objectives.

All parties sign the charter; thus, demonstrating commitment to the project plan. This document is also known as a partnering agreement, with an enlarged copy typically displayed in the stakeholder's offices.

Open Communication Lines at all Times Between all Project Parties
The project manager and the project team should iteratively review and confirm the project plan, and stakeholder goals and objectives. The stakeholders should fully understand the plan, goals, and objectives of the project as various stakeholders may not always be in alignment with the other project stakeholders, the project manager should identify and actively address any such variances.

Project Coordination Procedure (PCP)
The Project Coordination Procedure clearly defines communication channels within the project team and stakeholders. This procedure would typically include:

- Limits of authority
- Responsibilities of parties
- Execution processes
    - Project systems, processes, and procedures
- Company culture
    - The attitudes, values, behavior, and the environment of the company and project team
- Correspondence procedures including e-mail
- Information
    - Data elements, including project objectives, used to define the scope of the project
- Barriers/constraints
    - Objectives to creating and maintaining project alignment
- Filing and reporting codes
- Document and action schedule (for all drawings, documents, reports)
- Public relations procedures
- Security and safety procedures
- Tools
    - Software programs, checklists, and historical data which are typically used to develop and manage projects
- Project close-out report

Periodic Forecasts and Planning
A forecast is a prediction, or an estimate, of future conditions and events based on factual information and observable trends, professional judgment, and documented assumptions available at the time of the evaluation. A Periodic Forecast update is typically produced on a regular basis for a given future period. Periodic Forecasts are usually limited in scope. Forecasts fall into categories of periodic or special.

During the planning process, all stakeholders should be kept informed on a periodic basis of the status of the project. A special purpose forecast is created by one of the stakeholders to provide a proposed solution to a specific problem and to keep all stakeholders informed. A special purpose forecast might be used by managers to assess the risk associated with a contingent event.

Trend and Progress Meetings are Held on a Weekly Basis
These meetings are a vital communications tool, and the trend meeting must be attended by the project manager and the scheduler along with other members of the project team as appropriate. Of the many meetings conducted during the execution of a project, the weekly trend meeting is probably the most important. This is not a decision-making meeting, but is where information is gathered and shared by key technical/services specialists. The project manager leads the meeting, and the project cost engineer serves as secretary. All current and potential influences, changes, extras, and trends are reviewed and discussed. The key meeting objective is the common sharing, gathering, communicating, and coordinating of all project influences that are developing at the time. Each party to the contract should hold its own weekly trend meeting, followed up by a joint meeting of the main parties, (i.e., the owner, engineer, and general contractor).

The Project Management Information System (MIS) has Effective Levels of Detail
The development of reporting levels of detail must ensure that the information available to each management and supervisory function is accurate and timely. In order for the reporting process to be effective, summary level reports are adequate for higher management levels, while detail reports are necessary for front line managers to deal with the actual work.

NON-SCHEDULE PLANNING METHODOLOGIES

As part of construction planning there is a need for non-scheduling related logistics planning methodologies. Typically, these methodologies are not specially itemized in the schedule, but are critical functions of the project that need to be planned out for project success. These methodologies include, but are not limited to:

- Design interface and execution plans
- Site access plan
- Temporary facilities plan
- Crane and heavy lift plans
- Placement of cranes
- Engineered and long lead material/procurement plans
- Owner provided material/procurement planning
- Packaging, handling, storage, and transportation logistics plans
- Emergency reaction plans
- Other such specific purpose plans

It is important for planning of these logistic methodologies to ensure smooth construction operations. Logistics planning is comprised of planning, application, coordination, and supervision necessary to manage labor, material and equipment to/from and around the project site. The result of this logistics planning will be efficient and productive construction operations to support the overall project schedule and potentially reduce cost.

Why Plans Fail

Because of various factors and circumstances, projects may not succeed. Some of the oversights which might lead to failure include:

- Failure of the project team to understand and fully comprehend the scope of the project and/or the attendant risks.
- Failure or poor execution by the project team in developing and implementing a comprehensive plan.
- Failure by the project team to revise the plan when major changes have been identified and incorporated into the scope of work.
- Failure of the major stakeholders (owner, underwriter, and bonding company) to financially support the project as programmed, thus, delaying or stopping the work.
- Failure of the stakeholders to communicate their different interpretations of the project plan and/or goals. Alternatively, failure of the project team to reconcile known interpretation of the differences. And,
- Failure of the project team to develop essential non-planning methodologies during project development and execution.

CONCLUSION

Project planning is the activity with the most impact on the ultimate success or failure of the project. When this work is properly executed, the financial return is clear and substantial.

The objective of planning by a project management team can be summarized as:

- Identify the stakeholders, contract requirements, and project delivery method.
- Identify the scope of work and the constraints/variables that may impact the project.
- Establish an acceptable course of action ("plan") to perform the project (scope of work) in an organized and coordinated manner through the review of project deliverables, requirements, roles and responsibilities. And,
- Respond to events during project execution and develop alternate plans to keep the project on schedule.

Project planning begins as soon as the project is identified and continues as the project progresses through the various phases of the project life cycle from project conception to project completion and closeout. Rather than a definitive sequential process, planning is best thought of as an evolving planning cyclical process. Scheduling is the process of creating discrete activities, durations and relationships between activities that represent the plan and can be communicated effectively.

If one starts a project with a sound scope definition and a well aligned organization with good team communication, then project success can be a reality.

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Chapter 13—Scheduling[7]

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Abstract
The following chapter is an introduction into the skills and knowledge of scheduling. Scheduling includes the concept of updating, logic, and date processing, in accordance with the project execution plan that is a deliverable from the planning process (See Chapter 12, *Project Planning.*) Scheduling provides a way of contributing input during project execution concerning means, methods, techniques, sequences, or other conditions affecting outcomes. It uses the baseline established in planning and aligns the update of the schedule as a methodology of updating and analysis to allow for periodic progress and forecast reporting. Additionally, it establishes a basis for change management and project execution control to achieve successful outcomes.

Keywords
Activity, Activity Description, Activity Identification, Arrow Diagramming Method (ADM), Backward Pass, Bar Chart, Baseline, Calendar Days, Constraints, Critical Path, Critical Path Method (CPM), Early Finish (EF), Early Start (ES), Forward Pass, Free Float, Gantt Chart, Late Finish (LF), Late Start (LS), Masking, Milestone, Network, Original Duration (OD), Overlapping Scheduling Technique, PERT Diagram, Planning, Precedence Diagramming Method (PDM), Relationships, Remaining Duration, Resources, Resource Calendars, Resource Leveling, Schedule Calendar, Schedule Update, Scheduling, Scheduling Levels, Status, Target Schedule, Time Scaled Network, Total Float, Work Breakdown Structure (WBS), and Work Days (WD).

INTRODUCTION

Why Scheduling is Important

Scheduling is the process that converts the project's work execution plan into a detailed road map, which if followed and adjusted to account for change and/or delay as it may occur, will assure timely project completion. Scheduling is one of the key tools used for monitoring and controlling projects to ensure the objectives of cost, quality, and time are met. Schedules provide the baseline against which progress is measured and reported. Schedules are used to assess time impact of changes to the work scope as well as forecast completion dates for key milestones; and if necessary, re-plan and schedule the work to meet changing conditions and risks.

Effective project management involves coordinating activities such as planning, organizing, implementing, and controlling time and cost. Time control is usually achieved by preparing and using schedules to make the most efficient use of available time. Scheduling is an important and integral part of the overall planning effort, since the scheduling process forces people to quantify their efforts in discrete terms and to place activities in proper relationships with each other. The planning (as discussed in Chapter 12) and scheduling functions are usually performed iteratively in order to accomplish all required tasks within the specified time frames.

Benefits from Scheduling

Scheduling provides a basis for management of the work, improves communications, and facilitates coordination. Using a schedule improves the effective use of resources. The project schedule gives the user a baseline to monitor and control the work.

Scheduling provides a way of contributing input during project execution concerning means, methods, techniques, sequences, or other conditions affecting the plan's outcome. Scheduling provides a means for obtaining feedback since the development and use is a team effort incorporating the ideas and objectives of those responsible for the work. Schedules are good motivational tools providing intended work plans to those having to perform the work and reporting progress against them.

Schedules provide a baseline for measurement and a means for collecting and recording progress. Budgets, costs, and resources can be integrated into project schedule activities providing a basis for measuring cost as well as time performance.

Schedules are used as a forecasting tool to better plan, manage, and execute work; in particular, the interfaces and dependencies between parties.

Schedules may be used as a basis for payment applications supporting work completed. Critical path schedules are used and relied upon by courts for amending contract completion dates. When projects are faced with significant cost and time overruns, schedules are used as analytical tools to support assessments of labor efficiencies resulting from compression or extension of time, congested work areas, and disruption to planned work.

The cost engineer must have an understanding of the importance of developing and using schedules effectively. Knowing only the cost and cost estimating functions limits the ability of the cost engineer to function as a true project controls professional.

[7] Portions of this chapter are from previous editions authored by Dr. Brisbane H. Brown, Jr.; Dr. James M. Neil, PE CCP; and Jennifer Bates, CCP.

Learning Objectives

The purpose of this chapter is to provide a basic understanding of scheduling for the seasoned, as well as new, project staff and other department personnel that rely on timely project completion. After completing this chapter, the reader should be able to:

- Identify scheduling terms.
- Understand scheduling methods and techniques including each one's benefits and risks.
- Understand the most commonly used method and technique that will meet your project objectives.
- Understand Work Breakdown Structures (WBS) and the dependencies between work tasks to enhance team efficiencies.
- Apply overlapping schedule techniques and calculations that reflect real world management applications.
- Understand how to manage changes to the schedule.

To be clear, the subject of schedule development and use is extremely complex and cannot be completely addressed in a single chapter. Subjects such as schedule change management, resource loading and leveling, along with forensic delay analysis using CPM schedules, line of balance or linear scheduling techniques, and other higher uses of CPM schedule models and tools, are not addressed herein. Additionally, there are a significant number of computer CPM/project management software programs and ancillary tools, which are not addressed; however, the basics of scheduling should carry across the project management spectrum regardless of the tool(s) utilized.

SCHEDULE DEVELOPMENT

Tools for Developing Schedules

Computer software for developing, progressing, and updating schedules is affordable and readily available. Although computers are the tools and software provides the means for developing schedules, the individual user must understand what the computer is performing.

While numerous scheduling methodologies exist for developing project schedules, two of the most common are bar charts and critical path. A third method, Project Evaluation Review Technique (PERT), is mostly used in some software tools when dealing with schedule risk analysis and calculating the most likely duration for networks based on the variability of activity durations.

Bar Chart (Gantt Chart) Method

The Bar Chart, also called a Gantt Chart, is primarily meant to control the time elements of a program or project. However, since there are no direct relationship ties depicted between the activities, it is not possible to directly assess the impact of one activity on another or on the time of completion of the project.

When preparing a bar chart, the work effort must be divided into components, which are then scheduled against time. Preparing a bar chart involves the following steps:

- Analyze the program or project and specify the basic approach to be used in its execution.
- Segment the program or project into a reasonable number of activities that can be scheduled.
- Estimate the time required to perform each activity.
- Place the activities in time order, considering both sequential and parallel performance.
- Adjust the diagram until the specified completion date, if one exists, is satisfied.

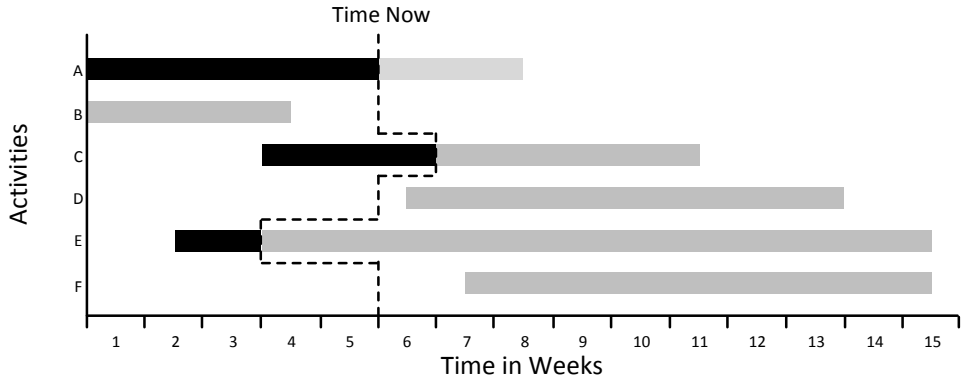


Figure 13.1—Typical Bar Chart

The primary advantage of using a bar chart is that it is simple to read. The plan, schedule, and progress of the program or project can be depicted graphically on a single chart. Figure 13.1 shows the six-activity plan, 15-week schedule, and current status. The current

status shows that Activity B has not started and is behind schedule (by five weeks), Activity C is slightly ahead of schedule (by one week), Activity E is slightly behind schedule (by two weeks), and all other activities are on schedule. However, it cannot be determined if Activity B or Activity E will have an impact on another activity or on the project completion. This graphical representation of work versus time is easy to read and provides a simple, understandable way to schedule small undertakings.

Bar charts have not been used successfully for large-scope, one-time-through projects primarily because of the following reasons:

- The inherent simplicity precludes including sufficient detail to enable timely detection of schedule slippages on activities of relatively long duration.
- The dependent relationships between activities cannot adequately be shown; thus, it is difficult to determine how progress delays in individual activities affect project completion.
- Developing bar charts is essentially a manual, graphical procedure, which makes them difficult to establish and maintain for large projects; they also tend to become quickly outdated, thus diminishing their usefulness.

Many large and technically demanding undertakings, such as developing weapons systems or constructing power plants, require schedules showing thousands of activities that take place in widely dispersed locations. Manually developed bar charts cannot adequately display this data, and are thus unsuitable for anything other than a summary display of information.

It should be noted that the printed/visual reporting of CPM based schedule models is through a report that includes a Bar or Gantt Chart depiction. Accordingly, in that instance, while the logic ties are often not shown, there are logical dependencies between activities that drive time of completion of intermediate and final completion milestones. The reviewer of a Bar/Gantt Chart report, must understand the underlying logic being used; whether through CPM calculations or implied without actual logic ties between activities.

If a network diagram (as discussed in the following sections) is prepared and the work scheduled, the display of relationships between the activities can be turned off or masked. This masking produces a bar chart, at any level of the project schedule, which can be used as a communication tool for the most complex and largest of projects.

Critical Path Method (CPM)
The disadvantages of manually developed bar charts, coupled with other disadvantages that became evident during the mid-1950s, set the stage for development of network-based project management methodology. One of the methods that emerged to overcome these weaknesses was critical path scheduling. The Critical Path Method (CPM) is a scheduling technique using arrow, precedence, or PERT diagramming methods to determine the length of a project through activity and phasing modeling, which can then be used to identify the activities and constraints on the critical path.

CPM enables a scheduler to do the following:

- Determine the shortest time in which a program or project can be completed.
- Identify those activities that are critical and that cannot be slipped or delayed.
- Show the potential slippage or delay (known as float) available for activities that are not critical.

The CPM was designed for, and is useful on, projects where the duration of each activity can be estimated with reasonable certainty; it predicts how long an endeavor will take to complete. It also identifies the activities that control the overall length of the project. CPM is widely used in the design and construction of process and industrial plants, buildings large and small, infrastructure projects, prototype development, and for controlling plant outages and shutdowns.

CPM computer software, typically known as project management software, in addition to modeling activities and their phasing, allows for the assignment of resources to activities. Assigning resources to the activities and allowing the resources to accomplish their assigned work based on their availability provides another variable in the overall project duration. Since the software has this capability, CPM scheduling is also used by industries with fixed pools of resources such as maintenance programs and information technology projects.

Project Evaluation Review Technique (PERT)
Project Evaluation Review Technique (PERT) is a probabilistic technique, originally developed and used by United States governmental agencies, for calculating the "most likely" durations for network activities. While no longer used strictly in that capacity, the underlying structure and techniques form the basis of modern schedule risk analysis.

During development of the Navy's Polaris Missile Program in the late 1950s, the team had no historical basis to draw upon when estimating the length of time it would take to accomplish certain tasks. For each of the activities, the developers of PERT estimated a best or shortest time, worst or longest time, and the most probable time to accomplish the tasks defined. Concurrent with the PERT network development, the team also developed computer software to run a probability analysis to arrive at a "most likely" duration for each activity and the overall project.

PERT is considered an indeterminate process for activity and project durations, while CPM is considered a deterministic process. The network of activities developed for PERT are similar to the Arrow Diagramming Method (ADM) and Precedence Diagramming Method (PDM) networks. Because of the similarity and resemblance of a CPM network to PERT, the term PERT has been used as a synonym for CPM.

Discussion of Critical Path Method

The most commonly used scheduling method and the technique that meets the objectives of most capital projects, is a CPM schedule model incorporating overlapping logic.

CPM is a scheduling technique using arrow or precedence diagrams (networks) to determine the length of a project and to identify the activities and constraints on the critical path. The critical path is defined as the longest chain or chains of activities, in terms of time or duration, through a network.

The two basic methods of critical path scheduling are the following:

- Arrow Diagramming Method (ADM) (also called Activity-on-Arrow, or the "i" - "j" Method).
- Precedence Diagramming Method (PDM) (also called Activity-on-Node).

Arrow Diagramming Method (ADM)

In Arrow Diagramming, nodes in the network identify the beginning and end of each activity. The activity beginning node is commonly referred to as the "i" node and the ending node as the "j" node. Each activity has a unique two number identification. This is referred to as the activity "i - j" number.

The Arrow Diagramming Method (ADM) is a method of constructing a logical network of activities using arrows to represent the activities and connecting those head to tail. This diagramming method shows the sequence, predecessor, and successor relationships of the activities. In ADM networks, project activities are shown on the arrows, with a node or event at each end. The tail of the arrow represents the beginning of the activity and the head of the arrow represents completion of the activity. The activity number, or identifier, consists of tail and head numbers, which are commonly referred to as the “i” and “j” nodes. This node numbering system is used to number activities.

Each node is uniquely numbered and is represented by a circle. Nodes have no duration. A node represents a particular point in time during the course of a project. It is appropriate to have a convention of identifying each node, which is used to organize or track activity data. Figure 13.2 depicts a typical arrow diagram.

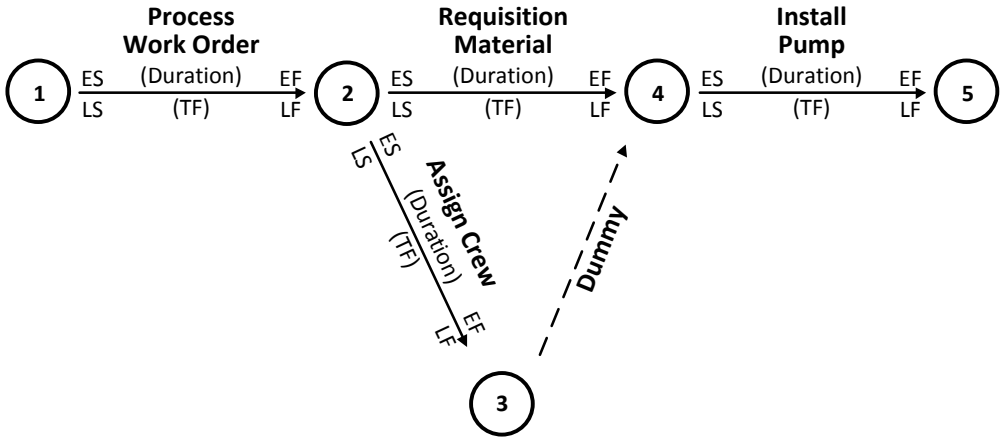


Figure 13.2—ADM Typical Activity Convention and Dummy Activity

- Activity description (requisition material)—above the activity line.
- Activity duration—centered above the activity line.
- Activity Number or ID—a unique “i—j” node number—within node. The “i” node is two (2) and the “j” node is four (4).
- TF (activity total float)—centered below the activity line.
- ES (activity early start)—upper left side of the activity in question. (Note that this represents the largest value ES of all the activities preceding the node.)
- EF (activity early finish)—upper right side of the activity line.
- LS (activity late start)—left side below the activity line.
- LF (activity late finish)—right side below the activity line.

The sequencing of activities in an ADM network must adhere to the following rules:

- All activities that immediately precede other activities must be complete before the latter activity can be commenced. No activity can start before its predecessors are complete. If this occurs, the activity must be subdivided.
- Neither the length of an arrow nor its direction in the network has any meaning—arrows imply logical relationships only.
- Each activity must have a different activity number.
- Duplicate activity numbers are not permitted.

A unique feature of ADM networks is the use of “dummy” activities. These are activities that have no time duration; they are used only to show relationships between activities that have more than one predecessor and/or to give each activity a unique "i - j" designator. Figure 13.2 also shows an ADM network using a dummy activity.

Precedence Diagramming Method (PDM)

In the Precedence Diagramming Method (PDM), the logic network is constructed using nodes to represent the activities and connecting them by lines that show logic relationships. The nodes (activities) can be circles or boxes. Activities that precede other activities are known as predecessor activities. Activities that follow other activities are known as successor activities. Figure 13.3 shows a typical PDM network diagram. In PDM, the activities are graphically represented by boxes that are assigned the properties of the activity they represent.

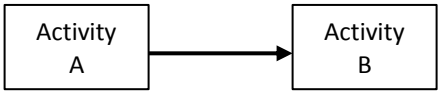


Figure 13.3—PDM Network

The lines in the network represent the interrelationships between activities. These relationships are referred to as links, or constraints. Arrows are not needed; however, arrows are more descriptive than lines. The nodes are sketched large enough to include certain information about the activity. This practice follows a convention so that others may easily understand the network while also facilitating manual calculations through the CPM model.

It is common to show the following information for an activity for each PDM node:

ACT ID	OD	TF
ACTIVITY DESCRIPTION		
ES	EF	
LS	LF	

Figure 13.4—Typical Precedence Diagram Activity

- **Activity Description**—A unique activity name and word description used to generally define the work to be accomplished; which easily identifies the activity to any recipient of the schedule.
- **Activity ID**—A unique activity number for identification and computer usage.
- **Activity Duration**—Length of time from start to finish of an activity, estimated or actual in working or calendar units. Figure 13.4 refers to the Original Duration (OD).
- **Activity Schedule Dates**—Typically, both early and late dates are shown.
- **ES (Early Start)**—The earliest date or time an activity may start is calculated by the schedule during the forward pass.
- **EF (Early Finish)**—The earliest point, date, or time an activity may finish is calculated by the schedule during the forward pass.
- **LS (Late Start)**—The latest date or time an activity may start so the project may be completed on time as calculated during the backward pass.
- **LF (Late Finish)**—The latest date or time an activity may finish so the project may be completed on time as calculated during the backward pass.
- **Activity Float Values**—Typically, Total Float (TF) is shown as in Figure 13.4. Total Float the maximum number of work periods by which an activity can be delayed without delaying project completion or violating a target (milestone) finish date. Total Float is equal to the late finish minus the early finish or the late start minus the early start of the activity.

Optional data that may be depicted within the PDN node descriptor box include resource requirements, codes, and percent complete.

The development of the network involves the identification of the project activities and the relationship between these activities. It is recommended that an activity list be initially developed which identifies the intended relationships. The activity list serves as the tool to be used in the development of the network. Table 13.1 is a typical activity list identifying project activities and their relationships between activities. It is common to modify or expand upon this list as the network is developed and later updated as the project moves toward completion.

Table 13.1—Precedence Network Activity List

Activity ID	Description	Predecessor
100	Activity A_{START}	-
200	Activity B	100
300	Activity A_{FINISH}	100
400	Activity C	300
500	Activity D	200, 400

Development of a precedence network diagram represents a graphical depiction of the work plan. The network, shown in Figure 13.5, is based on Finish to Start (FS) relationships.

The successor cannot start until the predecessor is finished. For example, there are five activities, A_{START}, A_{FINISH}, B, C, and D. Activity A_{START} is the first activity and can start anytime, Activity A_{START} must be finished before Activity A_{FINISH} and Activity B can start. Activity C cannot start until Activity A_{FINISH} is complete. Activity D cannot start until Activity B and Activity C are finished. Figure 13.5 represents the network of this work plan.

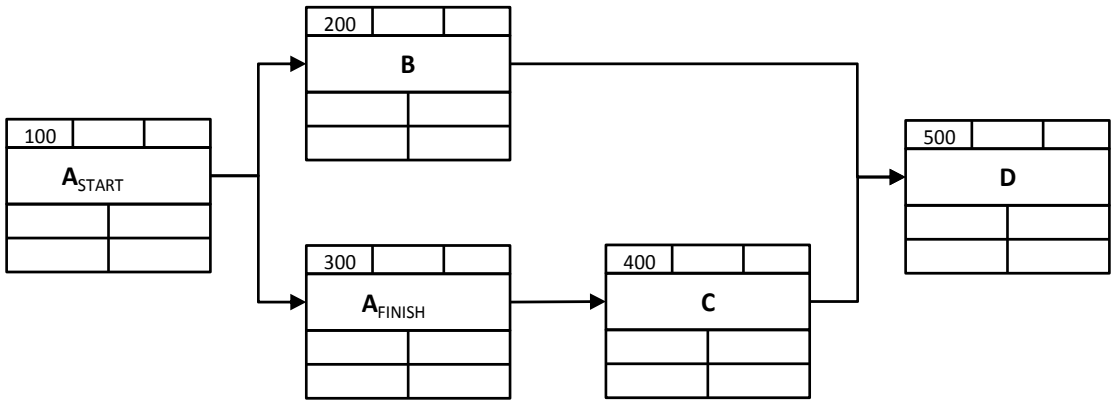


Figure 13.5—Precedence Network Diagram

The benefit of using the PDM for networking and scheduling is the ease of applying overlapping techniques to the activity relationships.

Overlapping Networks Techniques

The use of Overlapping Network Techniques is common in PDM applications. Overlapping network techniques allow activities to be grouped together, which reduces the number of activities in a network and can reduce the overall time of performance.

The overlapping scheduling technique allows for the development of a schedule, which more closely represents how a planner visualizes actual work execution conditions and constraints. For example, rather than wait for an activity to complete before starting the succeeding activity, it can be said that a successor activity can start a number of days after the start of its predecessor, or that it can finish a number of days after the finish of its predecessor.

Overlapping consists of two parts: a Relationship and a Lag Value or Constraint. Four basic types of overlapping relationships exist in most modern project management software. While specialized software may incorporate additional relationship types or variations on the basic types, the primary relationships are:

- Finish-to-Start + lag (FS+ N) Where “N” is lag
- Finish-to-Finish + lag (FF + N)
- Start-to-Start + lag (SS +N)
- Start-to-Finish + lag (SF + N)

In each case, a number of days (work periods) "N" is indicated to define the overlapped time frame, or the lead-time, between the activities in question. Lags can be either positive or negative, but are assumed to be zero if not specified. The use of activity relationship lags allows the CPM schedule model to minimize the number of activities that must be initially identified and created. It also allows for cleaner reporting graphics. Furthermore, updating for progress and changes is often easier to implement and maintain.

A **finish-to-start + lag (FS + N)** links the finish of the preceding activity with the start of the succeeding activity and indicates that the successor activity cannot begin until the preceding activity is complete. A lag "N" can be placed on the relationship to indicate that the succeeding activity cannot begin until a given time after the preceding activity has finished. A finish-to-start relationship with a lag value of zero is considered the default if no other value is specified. For example, in Figure 13.6, Activity 20 cannot start until "N" work periods after Activity 10 is complete.

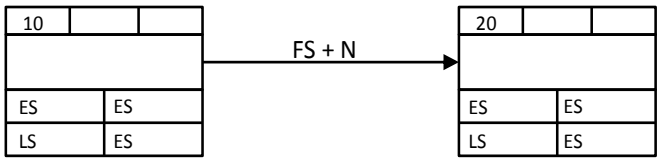


Figure 13.6—Finish to Start + Lag Relationship

Typical applications include cure time between the placement of concrete and the stripping of formwork, queuing time between a request for action and when the action takes place, and time for the approval process after a report has been submitted and the action is taken following approval. The alternate approach to this problem would be to include an activity in the network called concrete cure, review request, or review and approval, and assign the added activity a duration of "N" days.

A **start-to-start + lag (SS + N)** relationship links the start of the preceding activity with the start of the following or succeeding activity. It indicates that the successor activity cannot begin until the preceding activity has been started and the specified work periods (lag) or overlap time after the start of the preceding activity has elapsed. For example, in Figure 13.7, Activity 20 cannot start until "N" work periods after the start of Activity 10.

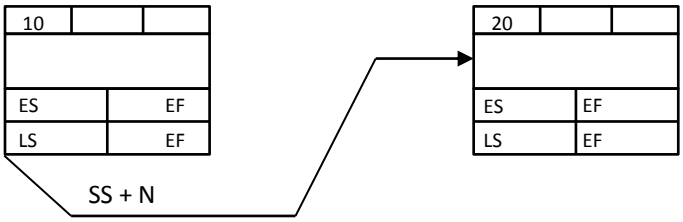


Figure 13.7—Start-to-Start + Lag Relationship

Typical applications include the relationship between the "pulling wire and cable" and "wire terminations," or starting the "report preparation" before all the "research information" has been completed. These relationships assume that if the work can begin before the preceding activities are complete, a SS + N relationship between them can be used. The schedule computations will indicate that the start of the succeeding activities can begin "N" days after the preceding activity has started. The alternate approach to this problem would be to include additional activities in the network to show the start and finish of both the predecessor and successor activities.

A **finish-to-finish + lag (FF + N)** relationship links the finish of the preceding activity with the finish of the following or succeeding activity and indicates that the latter activity cannot be completed until the preceding activity has been completed and the specified work periods (lag) or overlap time has elapsed. For example, in Figure 13.8, Activity 20 cannot finish until "N" work periods after Activity 10 is finished.

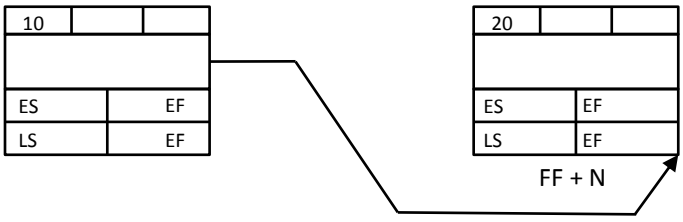


Figure 13.8—Finish-to-Finish + Lag Relationship

Typical finish-to-finish applications include the relationship between the finish of "wire terminations" and the finish of "test equipment" or between finish of "research information" and finishing the "report preparation." These relationships assume that if the successors work can finish "N" work periods after the finish of the preceding activities, an FF + N relationship between them can be used. The schedule computations will indicate that the finish of the succeeding activities requires "N" days after the finish of the preceding activity. The alternate approach to this problem would be to include additional activities in the network to show the start and finish of both the predecessor and successor activities.

A **start-to-finish + lag (SF + N)** relationship links the start of the preceding activity with the finish of the following or succeeding activity and indicates that the successor activity cannot finish until the preceding activity has started and the specified work periods (lag) or overlap time has elapsed. For example, in Figure 13.9, Activity 20 cannot finish until "N" work periods after Activity 10 is started. This relationship is seldom used because of inherent problems associated with the start of the successor activity and the finish of the predecessor activity. Generally, computer software will not allow this relationship.

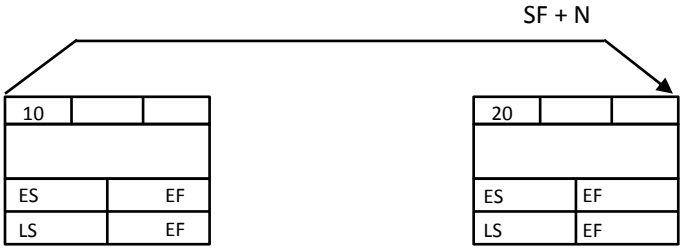


Figure 13.9—Start-to-Finish + Lag Relationship

The development of an overlapping network involves the same efforts as discussed previously for precedence networks. The difference is that during the development of the activity list, overlapping relationships and lead time periods, where applicable, are shown.

The activity list that illustrates overlapping is shown in Table 13.2. It is similar to the network previously used. The differences are that certain activities have been combined and overlapped and overlapping relationships are included. Note that where no overlapping relationship exists, a FS relationship with a lag of zero is assumed.

Table 13.2—Overlapping Network Activity List

Activity ID	Description	Predecessor
100	Activity A	-
200	Activity B	100 (SS + 10)
300	Activity C	100 (FS)
400	Activity D	200, 300 (FF + 3)

Figure 13.10 demonstrates that by using overlapping network techniques we can combine the two activities "A_{START}" and "A_{FINISH}" into one Activity "A" and show a relationship that Activity "B" can start 10 days after the start of Activity "A". Other changes to the network include that Activity "D" can finish three days after the finish of Activity "C".

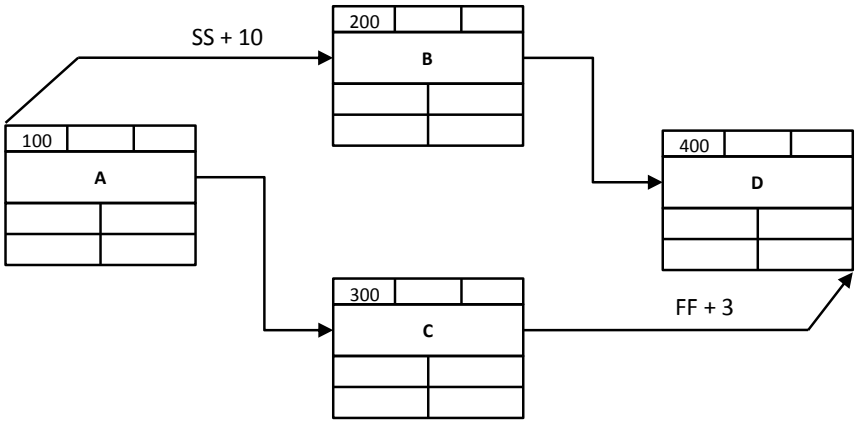


Figure 13.10—Overlapping Network

Overlapping network techniques in turn reduces the number of schedule computations required.

WORK BREAKDOWN STRUCTURE

The most effective tool to use in ensuring that all work scope is planned and modeled in the CPM schedule, is the Work Breakdown Structure (WBS). The WBS is a valuable management tool for planning, organizing, implementing, and controlling projects. The WBS is a tree structure of successively further breakdowns of work scope into component parts for planning, assigning responsibility, managing, controlling, and reporting project progress. The top of the tree represents the whole. Subsequent levels represent divisions of the whole on a level by level basis until the smallest element desired is defined.

Defining Work Breakdown Structure

The best approach to developing a WBS, as shown in Figure 13.11, is to first choose the desired hierarchy (process, organization, or product) and represent the entire project by a specific "project" block. The next step is to branch out beneath the "project" block into several levels and components, which are equivalent to the "project" block when combined.

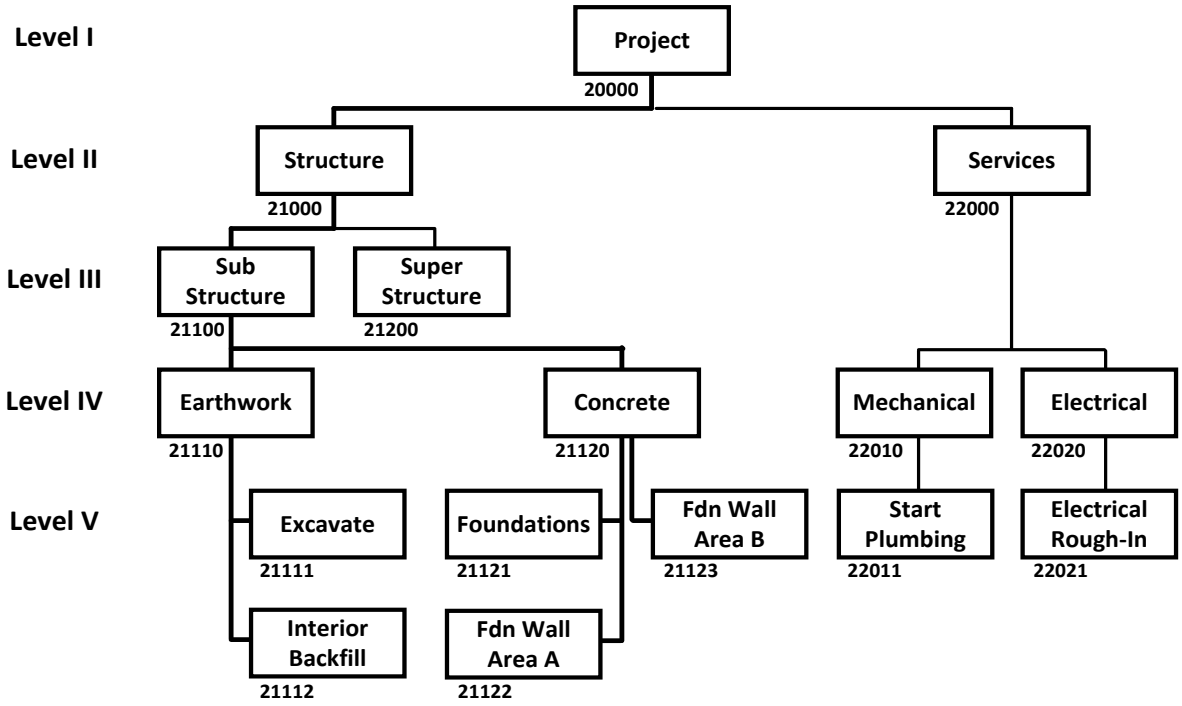


Figure 13.11—Process WBS

For example, structure and services define the second level of the process hierarchy for the project. All project process information can be further defined by these two classifications. For prototype development, other systems classifications may include research and manufacturing.

WBS techniques are valuable tools because they allow project details to be summarized into different groupings for analysis and control purposes.

It is not necessary that all WBS have the same number of levels. For example, the sample breakdown has five levels and a product based WBS may have four levels.

Coding Techniques

Most computer software provides a function for WBS coding that is based on the previously discussed hierarchical structure.

The fundamental element of any WBS is the detailed work activity. What enables the WBS technique to function is proper coding of detailed work activities. Activity definition and coding is best accomplished by numbering from the highest level of each WBS to the lowest level.

The project process-based WBS (see Figure 13.11) is used for illustration purposes to demonstrate the method of developing a WBS and its coding structure in detail from top to bottom. As stated previously, the Level II breakdown (structure and services) is completely representative of the Level I total project process activity. For illustration purposes, the structure component of the

project process is selected for further hierarchy definition and the line through structure is darkened to show the breakdown process.

Figure 13.11 shows that the Level II structure component is further defined by Level III. The Level III components are substructure and superstructure.

This technique of WBS development is very important. Without using it, the development process usually gets unnecessarily bogged down with efforts to achieve perfection before proceeding to subsequent levels.

The Level III sub-structure component is further defined by Level IV consisting of specific processes related to the sub-structure. One of those, concrete, is selected for further definition of the structure process.

In Level IV, concrete is further defined by three Level V components: Foundations, Foundation Walls Area A, and Foundation Walls Area B.

Once the levels and their components have been defined, it is best to begin numbering them. Count the number of levels and establish a code scheme with an equal number of digits. For example, there are five levels to the process WBS, and, consequently, a five-digit number is used. The number 20,000 is the highest level for the process WBS. Working downward and across in the WBS, the Level II components are numbered 21,000 and 22,000. Summarizing on WBS 2, where the first digit is equal to 2, provides the total project process information.

The lowest level of any WBS of the project constitutes the detailed work activities. Higher levels constitute summaries of the detailed activities.

Activity Coding

Table 13.3 illustrates a listing of detailed schedule activities. Included in Table 13.3 is the process WBS coding. The code for each WBS can be input into one of many activity code fields when using computer-based project management systems. It is necessary that every activity have a unique alphanumeric identifier. It is generally better to use the process-based WBS since it is usually the most detailed WBS.

Table 13.3—Detailed Activity List with WBS Coding

Activity Number	Activity Description	Process WBS
10	Excavate	21111
20	Foundations	21121
40	Foundation Walls Area A	21122
50	Foundation Walls Area B	21123
70	Backfill Interior of Walls	21112
80	Start Plumbing Slab Rough-In	22011
110	Electrical Rough-In	22021

The WBS for any particular project is usually prepared by project control personnel. However, project control personnel do not perform the actual work. Consequently, it is imperative that project control personnel and the project personnel who actually do the work reach an agreement concerning the WBS.

In addition to the previous construction example above, Figure 13.12 is provided to display a non-construction WBS example.

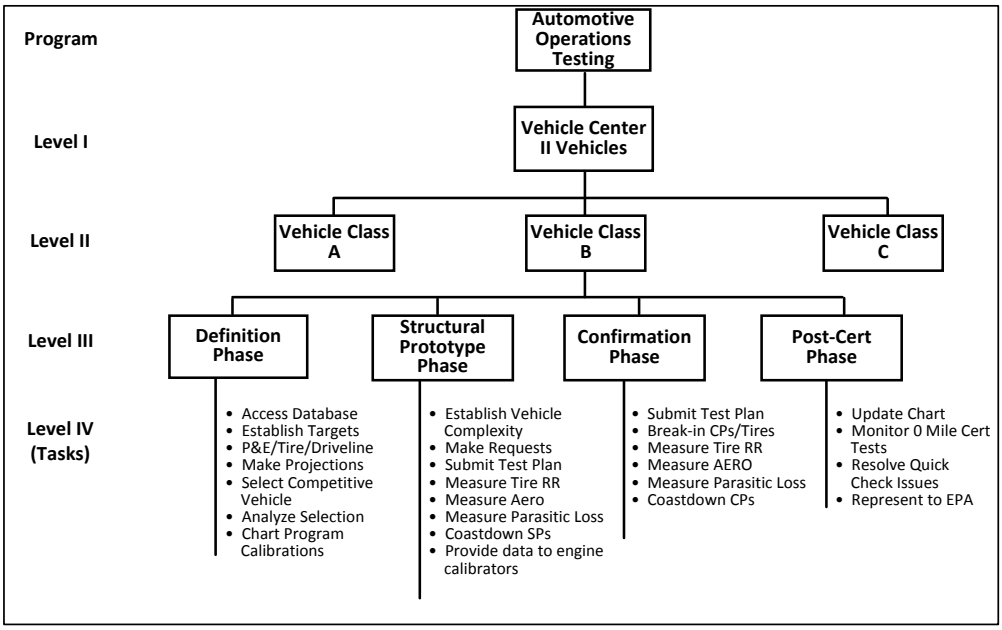


Figure 13.12—Non-Construction WBS

SCHEDULING TECHNIQUES

Making Time Calculations

Once a network has been created and the duration of each activity has been established, both the total time required to reach project completion and the individual start and finish times for each activity can be calculated. The four time values as associated with each activity are Early Start (ES), Early Finish (EF), Late Start (LS), and Late Finish (LF).

The computations required to calculate the above times involve simple addition and subtraction. Manual computation is easy and logical, but it can become tedious and time consuming when done for large networks.

Forward and Backward Pass

The forward pass through the network determines each activity's ES and EF and the project's duration or the earliest date a project can finish. The backward pass through the network determines each activity's LS and LF as well as providing a basis for calculating Float values for each activity. The calculations assume that activities begin on the morning of the scheduled start date and end in the evening of the scheduled finish date and that an event or milestone occurs on the evening of the day its last predecessor finished.

Before starting the network calculation, the precedence network list of activities is revisited to include the activity duration, as shown in Table 13.4. The activity duration is the length of time from start to finish of an activity, estimated or actual, generally quantified in working day or calendar day time units. Activity duration estimates are developed from historical experience or estimated time to perform the work.

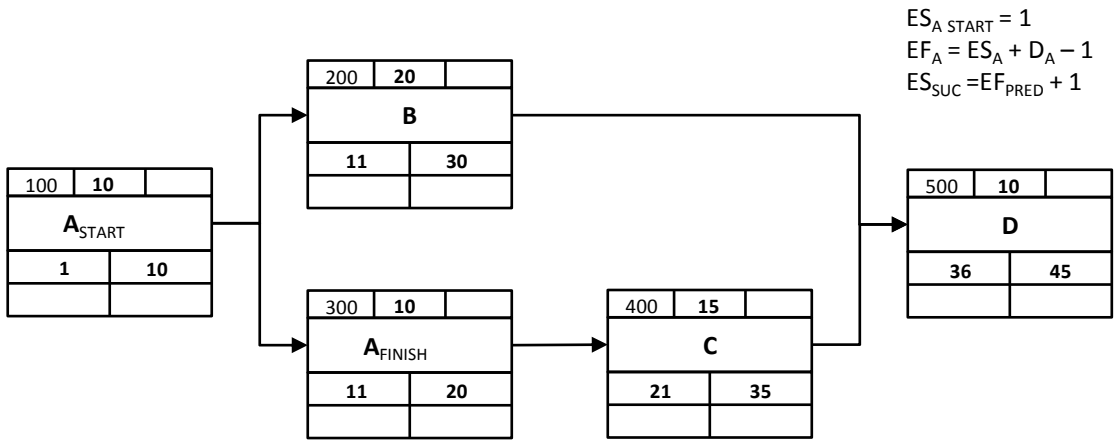
Table 13.4—Precedence Network Activity List With Durations

Activity ID	Description	Predecessor	Duration
100	Activity A_{START}	-	10
200	Activity B	100	20
300	Activity A_{FINISH}	100	10
400	Activity C	300	15
500	Activity D	200, 400	10

The durations are assigned to the activities as shown in Table 13.4 and Figure 13.13. In the forward pass, the earliest start and finish times for each activity are calculated, observing the following rules:

- Day 1 is the earliest start date for Activity 100.
- The ES of Activity 100 (A_{START}) is equal to one.
- The EF of the activity is equal to the ES of that activity plus the duration minus one.

- The ES of any succeeding activity is the EF of the predecessor activity plus one.
- The ES of an activity is equal to the largest of the EF times of the activities merging to the activity in question plus one.



In Finish-to-Start relationships, the early start of an activity is equal to the largest of the early finish times of the activities merging to the activity in question, plus one.

Figure 13.13—Forward-Pass Precedence Network FS Relationships

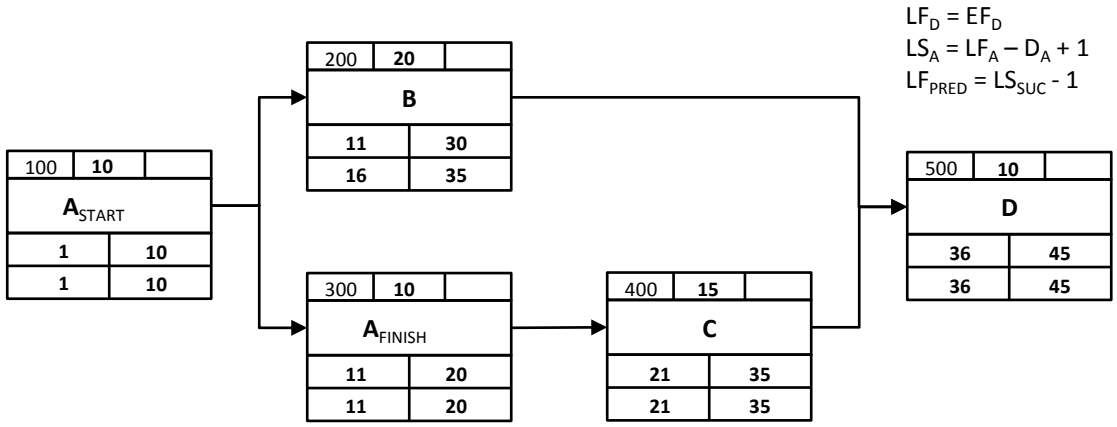
The early finish date for Activity A_{START} is day 10 (ESA_{START} + D -1). Day 11 is the early start date for Activities B and A_{FINISH}.

The early start for Activity D is day 36 since Activity D's early start is controlled by the largest early finish date of all predecessors (B and C) which is Activity C. The total project duration is forty-five days.

If the total project duration exceeds the available time, planners must reevaluate their work and take whatever action is needed to meet time objectives: perhaps optional activities can be dropped and others can be shortened by applying more resources or other schedule compression techniques.

In the backward pass, the latest allowable start and finish times for each of the activities are calculated, observing the following rules:

- The LF of the terminal activity in the network is either assigned as being equal to its EF or assigned the value established by the contract documents.
- The LS of an activity is its LF, minus its duration, plus one.
- The LF for all other activities is equal to the numerically smallest LS of succeeding activities, minus one day. Where two or more activities burst from or leave an activity, the numerically smallest LS of the successor activities, minus one day, is the LF of the activity in question.



In Finish-to-Start relationships, the late finish of an activity is equal to the smallest of the late start times of the activities bursting from the activity in question, minus one.

Figure 13.14—Backward-Pass Precedence Network FS Relationships

Figure 13.14 depicts the results of the backward pass through the network. Note that Activity A_{START}'s late start is controlled by Activity A_{FINISH}'s late finish.

Work Days and Calendar Days

The calculated dates are shown in consecutive calendar days. This calculation is used for all examples. However, it must be understood that many projects are worked on a five-day work week with weekends and certain recognized holidays not normally planned as working days. These nonworking days must be accounted for in the total project duration. Most contracts that stipulate contract time, do so in total calendar days, including weekend and holidays; regardless of the work day calendar used in the CPM schedule model.

Computerized scheduling applications allow for setting calendars and automatically accounting for weekends and holidays. Also, the dates provided by computerized schedules may, by choice, be in calendar days or dates. Multiple calendars may be necessary to model the work schedules of the engineering staff (working a five day work week); whereas procurement may be modeled on a seven day work schedule (calendar basis) and work at the site is on a two week fortnight basis (a rotation of ten work days followed by four non-work days).

Forward and Backward Passes for Overlapping Relationships

The forward and backward passes for the overlapping relationships are shown in Figures 13.15 and 13.16, respectively. The rules discussed earlier are shown with each figure.

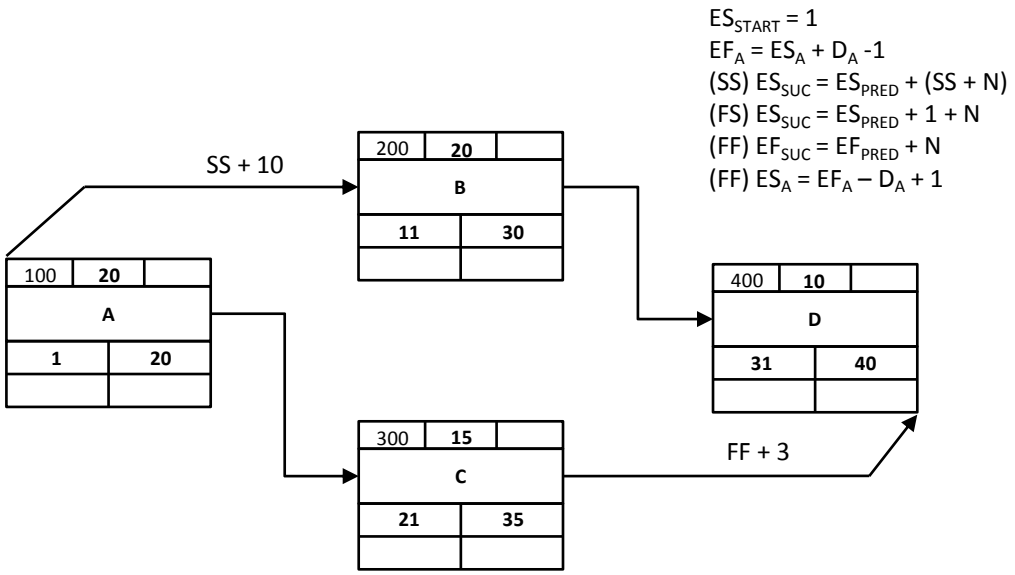


Figure 13.15—Forward-Pass Overlapping Technique Relationships

In Figure 13.15, the total project duration is 40 days, five days shorter than Figure 13.13.

From Figure 13.13, Activity A_{Start} and A_{Finish} were combined into Activity A. To account for the overlapped time, the relationship between Activity A and Activity B was changed to SS+10. The FS relationship between Activity A and Activity C remained unchanged. These changes result in the same EF for Activities B and C in Figure 13.15, as in Figure 13.13.

Changing the relationship between Activity C and Activity D from FS to FF + 3 allows Activity B to control the finish of Activity D, which results in a savings of five days.

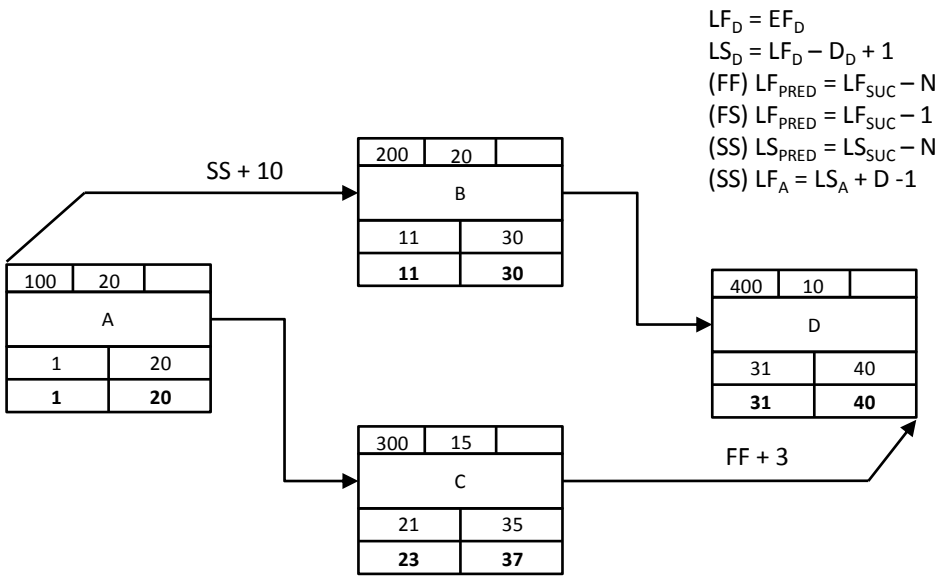


Figure 13.16—Backward-Pass Overlapping Technique Relationships

Note: Activity A's late start and late finish dates are controlled by the late start of Activity B.

Float

Free float is defined as the amount of time that the completion of an activity can be delayed without delaying any other following or succeeding activity.

Free float is equal to the difference between an activity's EF and the ES of the following or succeeding activity minus one. In the event that two or more activities succeed or follow an activity, the succeeding activity with the smallest ES is used to determine the activity free float amount. The Free Float of an activity is equal to the smallest value between the activity in question and all succeeding activities minus one.

Free Float belongs solely to the activity. As a result of the nature of the computations, Free Float is reserved for only the last activity in a linear chain (only one set of relationships driving each activity in the linked activity chain) of activities. Custom or preference may dictate whether Free Float is shown on the network or included in computerized reports.

Total Float (TF) is defined as the amount of time that the completion of an activity can be delayed without delaying the completion of the project's terminal activity. Total Float is equal to the difference between the activity's LF and EF, or the difference between the activity's LS and ES.

Total Float is shared by the activities in a chain. For this reason, it is a better indicator of the float time an activity possesses. It is cautioned that TF is shared. If a chain of activities possess 15 days of TF and there are four activities in the chain, each activity does not possess 15 days of TF independent of the other activities in the chain. If the first activity in the chain uses all 15 days, the TF along the chain is reduced to zero. Any further delays to any activities in this chain will result in a delay to the project's completion.

Critical Path

Except for activity date constraints and other CPM schedule model imposed elements such as leveled resources, Total Float defines the critical path of the project. The critical path is defined as the longest chain or chains of activities, in terms of time or duration, through a network. It is the chain or chains of activities through a network with the smallest total float value. If the network is continuous, there will be at least one continuous chain through the network. A discontinuous network is one where an imposed activity time constraint has disrupted the chain of activity date calculations.

Figures 13.17 and 13.18 depict the networks with the TF values shown. These figures also represent the networks with the critical path identified. It is shown as the heavier lined path on the following diagrams.

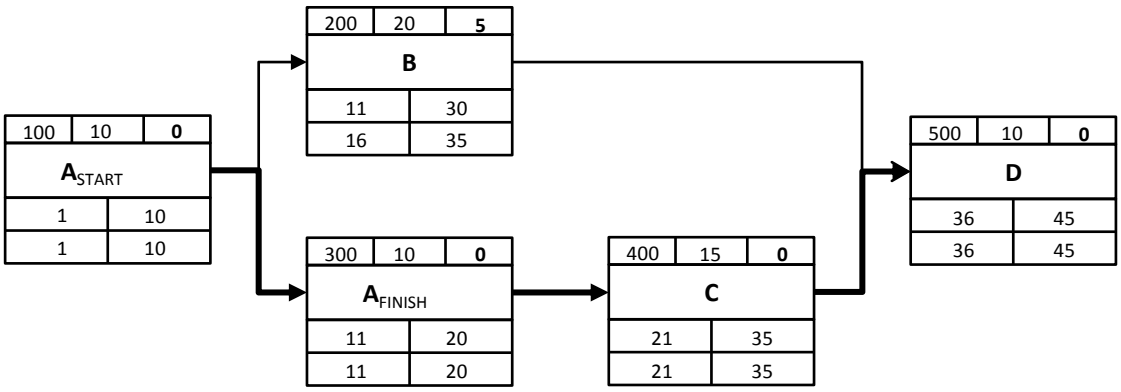


Figure 13.17—Precedence Network FS Relationships Total Float and Critical Path Shown

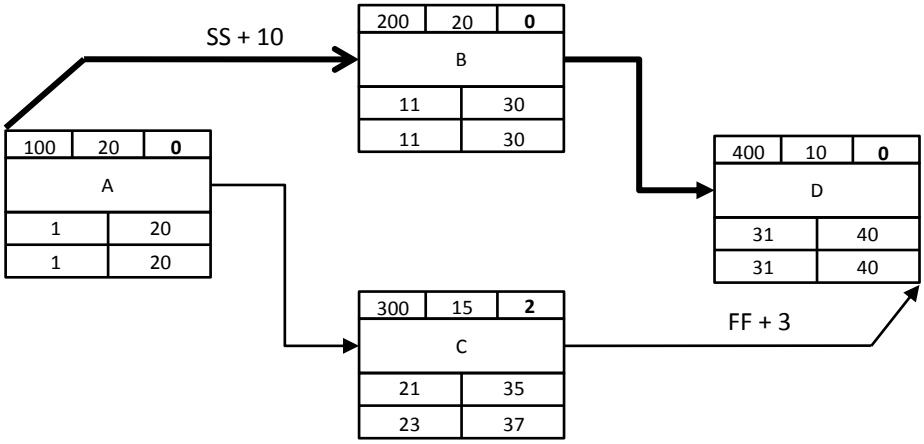


Figure 13.18—Overlapping Technique Relationships Total Float and Critical Path Shown

The late finish for Activity A is controlled by the late start of Activity A. The late start of Activity A is critical and controlled by the critical path coming from Activity B. Activity A's late start is $LS_A = LS_B – SS$: ($LS_A = 11 - 10$. $LS_A = 1$.) $LF_A = LS_A + D - 1$.

Constraints

The start and finish of certain activities, at times, must be constrained in order to represent what will actually occur. For example, based on network logic, the installation of a pump is scheduled to begin on September 17, but the actual pump delivery from the vendor is not scheduled until October 23. Therefore, the installation of the pump is constrained by the delivery of the pump.

There are seven major types of constraints:

- Start-on
- Start-no-earlier than
- Start-no-later than
- Finish-on
- Finish-no-earlier-than
- Finish-no-later-than
- As late as possible

Each of the above constraints affects the schedule differently. No-earlier-than (NET) constraints affect only the forward pass calculation in the network. No-later-than (NLT) affects only the backward pass calculation. “On” is a combination of NET and NLT and affects both the forward and backward pass.

A constraint may or may not be upheld, depending on network logic. For example, a Start NET April 7 constraint will control the network calculations if the early start time is before April 7. If the network logic produces an early start time on or after April 7, the network logic will be observed and the constraint will be ignored.

Similarly, if the constraint is “Start NLT April 7” and network logic dictates a late start on or before April 7, the network logic again will control and again the constraint will be ignored. If the network logic produces a late start date after April 7, the constraint will be observed, and the activity will be scheduled to have a late start date of April 7.

Scheduling Levels and Reporting

Scheduling levels are schedules used by various management echelons to manage the project. Senior management may require a summary level referred to as a milestone schedule. Project management and key department interface may only require a summary level of the project activities while hands-on managers require detailed project schedules and short-interval schedules for day-to-day management. Each level is an integral subdivision of the previous level and presents more detailed activities and relationships.

Level 1—Milestone Level Schedule—Level 1 Milestone Schedules comprise key events or major milestones selected as a result of coordination between the client’s and the contractor’s management. These events are generally critical accomplishments planned at time intervals throughout the project and used as a basis to monitor overall project performance. The format may be a list, summary network, or bar-chart and may contain minimal detail at a highly summarized level.

Significant events may include begin program definition, preliminary design complete, purchase major equipment, mobilization, foundations complete, delivery of major equipment components, and installation complete. Company management is usually apprised of the project's implementation progress with milestone level schedules.

Level 2—Project Summary Level Schedule—Level 2 Project Summary schedules are composed of summary project activities depicting critical work and other management selected activities generally indicating the activities' ES and EF dates. Key restraints and relationships between activities are identified and defined. This level of planning is represented by Level 2 schedules and provides an integral plan of the project activities for project management. Milestone schedule dates are compared to those derived from the project summary schedule. Upon review (making adjustments, if necessary) and acceptance, the dates from the project summary are used for the milestone schedule.

When using a network-based schedule, the detailed activities can be rolled up to a summary level and milestone level schedule. As the detailed schedule is developed, it must be summarized to replace the independently developed project summary and milestone schedules.

Typical summary level activities include engineering and design, procurement, major equipment fabrication and delivery, major structures, installation, start-up, and commissioning.

Figure 13.19 shows both a Level 1 and Level 2 schedule.

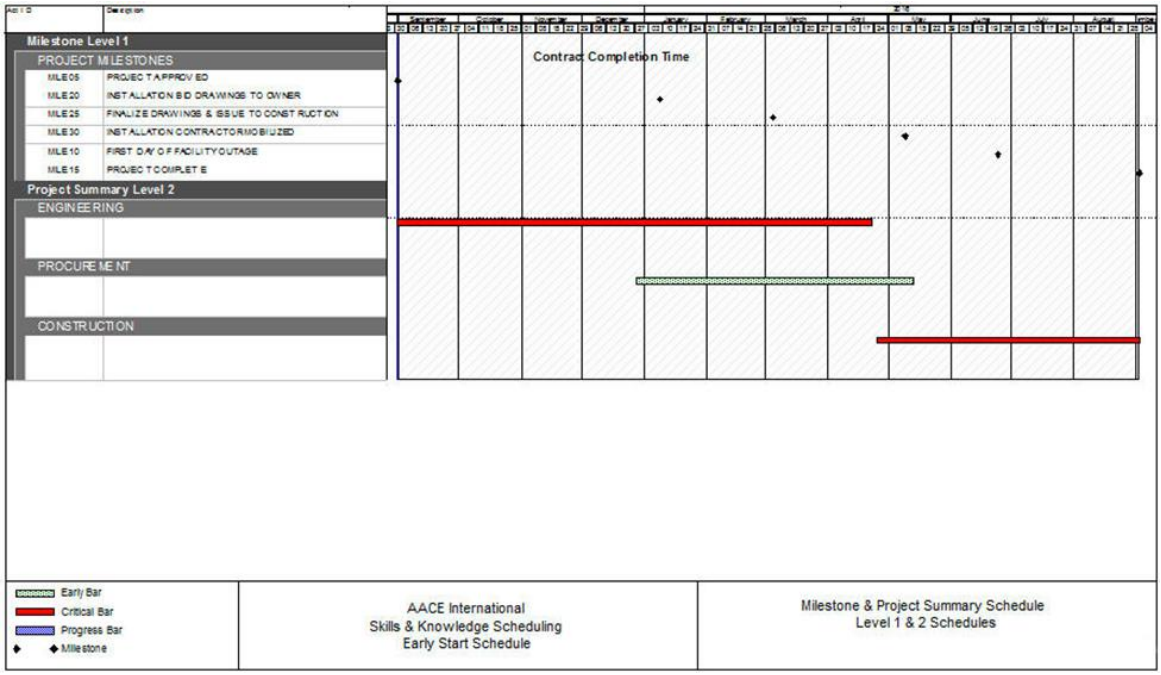


Figure 13.19—Level 1 and Level 2 Schedules

Level 3—Project Detailed Schedule—Level 3 schedules (see Figure 13.20) display the lowest level of detail necessary to control the project through job completion. The intent of this schedule is to finalize remaining requirements for the total project. Detailed scheduling identifies and defines activities that are more detailed than the project summary level. For example, an activity in the project summary level, such as structural steel engineering and design, would be represented by more meaningful detailed activities of shorter durations such as: define and collect loads, perform analysis, prepare drawings and specifications, and issue documents for procurement. This level of planning also provides better networking capabilities. This Level 3 schedule supports the planning effort for determining and assigning resources.

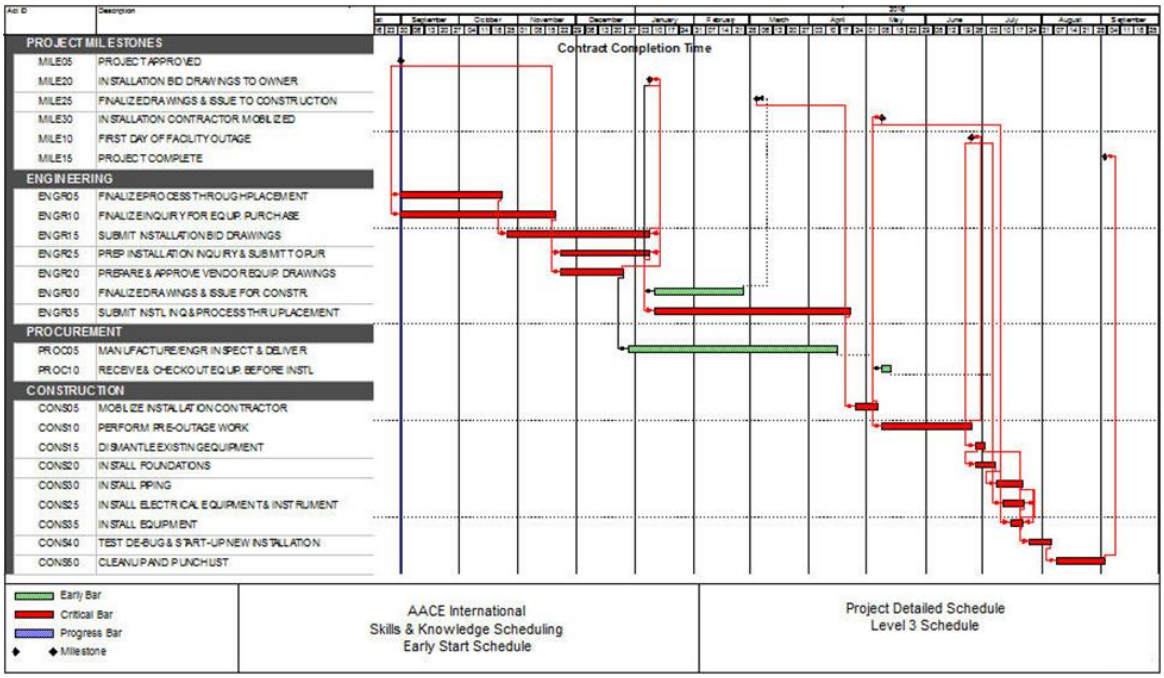


Figure 13.20—Level 3 Schedule

Level 4—Short-Interval Schedule—A Level 4 schedule is a two-to-six week look-ahead (duration is dependent upon complexity and overall duration of the project) schedule that shows resource(s) assigned and detailed work activities and is used for planning and progress reporting purposes, review and assignment of current week work plans, and advance planning for the near term future week work (see Figure 13.21). This level is sometimes referred to as a short-cycle schedule since the process for its use is a weekly cycle of collecting progress, working the current week, and planning future work assignments.

Short-interval Level 4 schedules are derived from the detailed Level 3 schedule network. These schedules are usually bar charts as they are best used for communicating information and are developed by masking the relationships. It is one of the best tools for conveying the planning requirements to those performing the work.

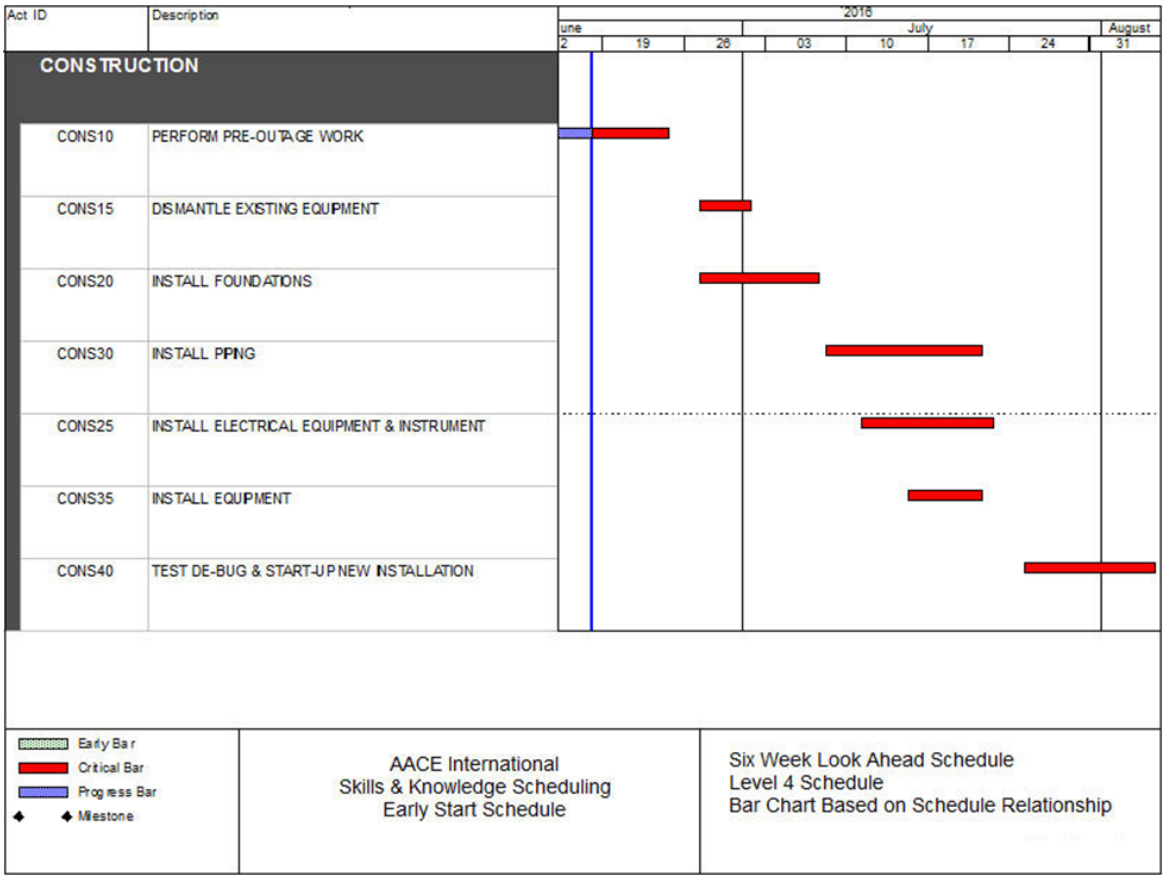


Figure 13.21—Level 4 Schedule

Schedule Reporting

The following discussion of reports is provided to familiarize planners and schedulers with selected types of computerized report products. These are basic reports. There are numerous other reports of interest to planners, supervisors, and project managers. Schedules comprised of hundreds or thousands of activities can be made manageable and meaningful. Schedules can be selected for each party involved in the project to minimize the number of pages for review that may seem to be overwhelming.

Early Start Dates Report—A listing of activities sorted by early start dates. This listing provides the scheduler and management with the activities that are scheduled to start by ascending dates. Short interval planning uses these lists to prepare for current period and future look-ahead periods. The same report for the overlapping activity schedule identifies the lead time relationships.

Total Float Report—The activities are sorted by total float in ascending value beginning with values of TF = 0. The report first lists all activities that are on the critical path (TF = 0), and then lists all other activities grouped by total float values.

Precedence Report—This is a listing by activity early start dates. However, the significance is the identification of all predecessor and successor activities for each activity. This report is used by planners for debugging schedules and comparing relationships on the network diagrams to those in the schedule reports.

Schedule Plots

- **Logic Diagrams**—These have been used in the sample problems.
- **Time-scaled Logic Diagram**—This type of plot shows activity relationships and displays the activities in their scheduled place in time, see Figure 13.20.
- **Early Start Date Schedule (Bar Chart)**—Bar charts without logic relationships shown. These types of charts are used more frequently by supervision and management to track work. An example of this is the six week look ahead in Figure 13.21, also see Figure 13.22.

Managing Changes in the Schedule

A schedule is simply a time-phased plan for accomplishing all of the specific activities that have been defined for a project. The schedule is derived and developed from estimated activity durations and the logical relationships or working sequence between activities. Unfortunately, in the real world, the actual work does not always progress in accordance with the original plan and a schedule disruption and slippage is usually the outcome. When this occurs, control measures must be employed to mitigate overall cost and delay impact where possible.

Actual activity durations may be greater (and sometimes even less) than the original estimated durations. In addition, the working sequence of activities will not always follow the sequence from the schedule. The correction of these changes through schedule updating can forecast any schedule slippage or delay and, hence, project management personnel can then react to bring the project back on schedule.

The updated schedule, when compared to the original schedule, becomes an indispensable management tool that can be used to assess the overall project impact of any change. For example, the owner or other party following the issuance of the original schedule may impose changes to the original work scope. This may include any engineering design changes, emergent or additional work scope activities, and any change orders that are issued during the performance of the project.

Nearly all projects require special material or equipment to complete. The procurement of these items often accumulates and at times, becomes the driving activity of the critical path. The updating process assesses the impact of the procurement items against the schedule and allows management to expedite the critical items and/or resolve them through activity logic changes (mitigation).

In addition, owners and government agencies may be slow in issuing permits or approvals. Labor strikes, legal disputes, resource availability, accidents, and weather conditions are all real dilemmas that many times can only be solved by rescheduling the work. While re-planning is necessary to mitigate the encountered issue(s), it can never regain lost work; however, re-planning can minimize the overall negative impact to final completion.

Consequently, the original schedule should be updated regularly to reflect current information. Updating is a control process, which implies that adjustments in the network may be necessary and forewarns of potential revisions of major consequences.

Reasons for Updating

Four major reasons a schedule should be updated regularly are to:

- Reflect current project status
- Keep the schedule as an effective management tool
- Document performance
- Provide documentation to plan for changes and support delay analysis

The contractor and owner need to be aware of the current status throughout the project duration. Contractors are concerned with status of their submittals, delivery of equipment and materials, resource availability, coordination and performance of subcontractors, and timely payment for progress completed. Owners are concerned with the status of work including quality and whether the contractor's progress is adequate to meet "turnovers" and completion dates. Both parties need to be aware of changes or delays as they occur and how they affect the project completion date. This will minimize any "surprises" and allow the parties to take necessary actions to get the project back on schedule.

An updated schedule is a tool that provides the project's status at a given time and is used to assess the performance of the owner, designer, and contractor to meet schedule commitments. It provides a record of the accomplishments as to timeliness and completeness. Changes to the work scope or methods of performance need to be included in the schedule updates. This provides management the opportunity to assess impact and plan remedial measures if necessary. In developing a project history, the causes for delays can be identified and measured from the updated schedules to support delay analysis and negotiations.

Updating Intervals for Managing Changes

Generally, project management requires formal updates to be performed at least monthly and it is not unusual to require weekly status and updates in a less formal manner. Most projects facilitate the need to update somewhere between the two extremes. These intervals generally coincide with routine business reporting periods such as monthly progress reports, loan payment schedules, and other fixed reporting requirements. The project status and progress should also be reported periodically with these update reports to provide complete decision making information. Although periodic updates may support business and financial needs, the project manager needs more frequent and routine updating to perform effective management of changes; such as weekly updates with detailed reporting and forecasting for the next three to six weeks to allow for immediate planning and control.

Updating Procedures for Managing Changes

Updating and revising a schedule may require several iterations before management and supervision decides on an acceptable plan for implementation.

When status and progress are input into the schedule, an analysis is performed to determine the impact on remaining activities and project completion. Adverse trends may have to be mitigated by adding resources or reviewing logic relationships. Reports, tabular and graphic, should be reviewed with supervision and management and adjustments, if necessary, should be made.

When changes become known, planning sessions must be initiated to determine their impact. For example, how will the change affect current progress? What new activities are needed to define and document the change and its impact on downstream activities? What are their durations and relationships to existing activities? What additional or new resource requirements must be considered?

This process may result in several alternate plans that may require the preparation of “what if” schedules. The schedules result in graphical depictions of the alternatives. These schedules assist management in making decisions in selecting optimum solutions.

The following steps are generally performed during the update process:

- Gather all current information in accordance with routine priorities.
- Identify and plan for any changes to the work which affect activity duration, logic, work scope, and any other significant information.
- Input these changes into the project schedule (any additional work or delay should be coded appropriately to reflect "unforeseen" or "delay" work activity).
- Recalculate the project schedule.
- Perform analysis and prepare reports for management review.
- Evaluate and adjust the updated schedule according to management’s and supervision’s review and direction.
- Issue updated schedule to all interested parties.

These procedures are generally performed throughout the duration of the project. Standard updating may be performed monthly. Revisions to schedule baseline models may be required as they become necessary due to significant change or delay, or routinely required on a bi-monthly or quarterly basis. It is important that the updated schedule (reforecast or revised baseline schedules) be issued to all parties in a timely fashion in order to plan and expedite the work effectively and minimize any future delay.

As an example of managing changes and updating a schedule, the schedule, shown in Figure 13.22, has been generated for a slab on grade construction project. The planned start for the project is May 30, and it has total project duration of 68 days, finishing on August 7.

There are two bars shown for every activity. The scheduling software allows a target bar to be inserted under the schedule bar for each activity.

The target bar does not move or change as a result of status or progress. Therefore, the target schedule is a static representation of the approved project schedule (the “baseline”).

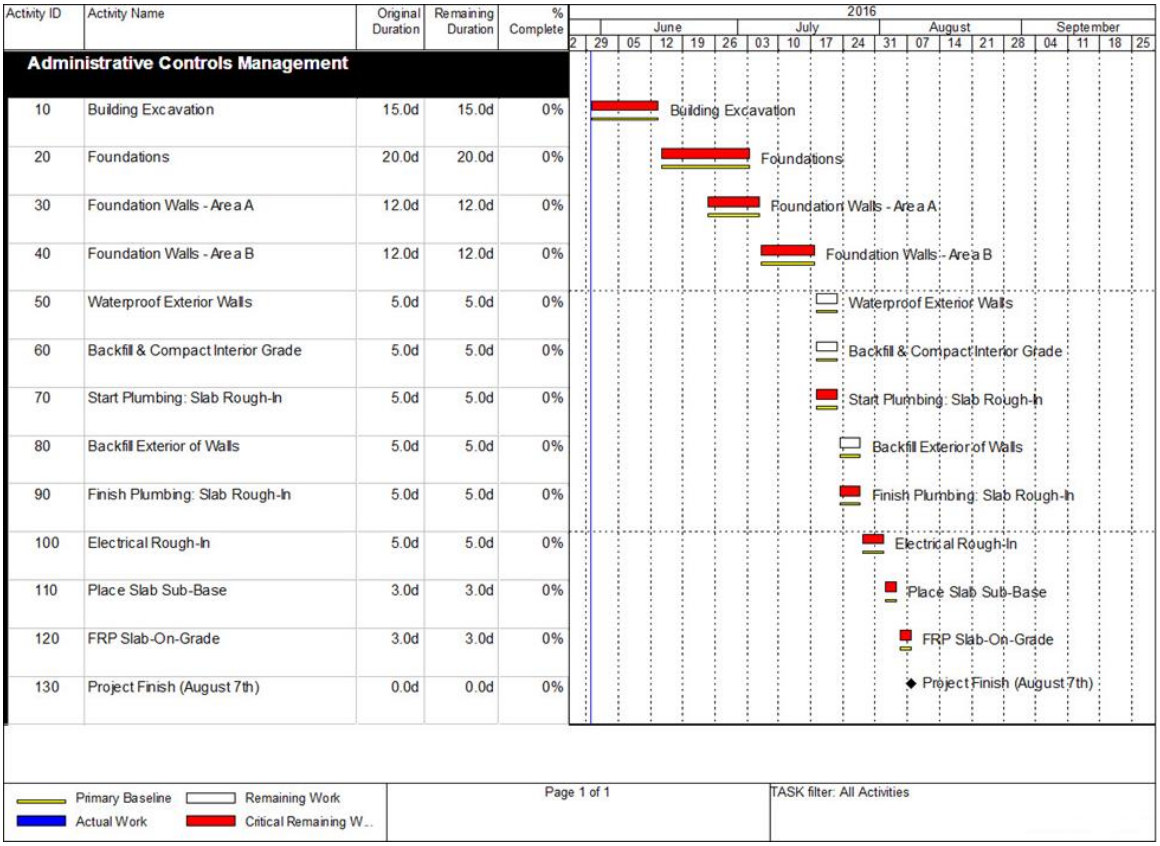


Figure 13.22—Slab on Grade Project Schedule

On July 1, a schedule update was conducted to show the current status of the job, see Figure 13.23. The following information was given and entered, and resulted in the following schedule.

- Building excavation started on June 4 and was completed on June 18 (late start).
- Foundations started on June 19 and are 25 percent complete (in-progress).
- A correction was made to increase the duration of the waterproofing to ten days (duration change).

The update shows the project now finishing on August 18, with a total duration of 79 days, 11 days later than the target.

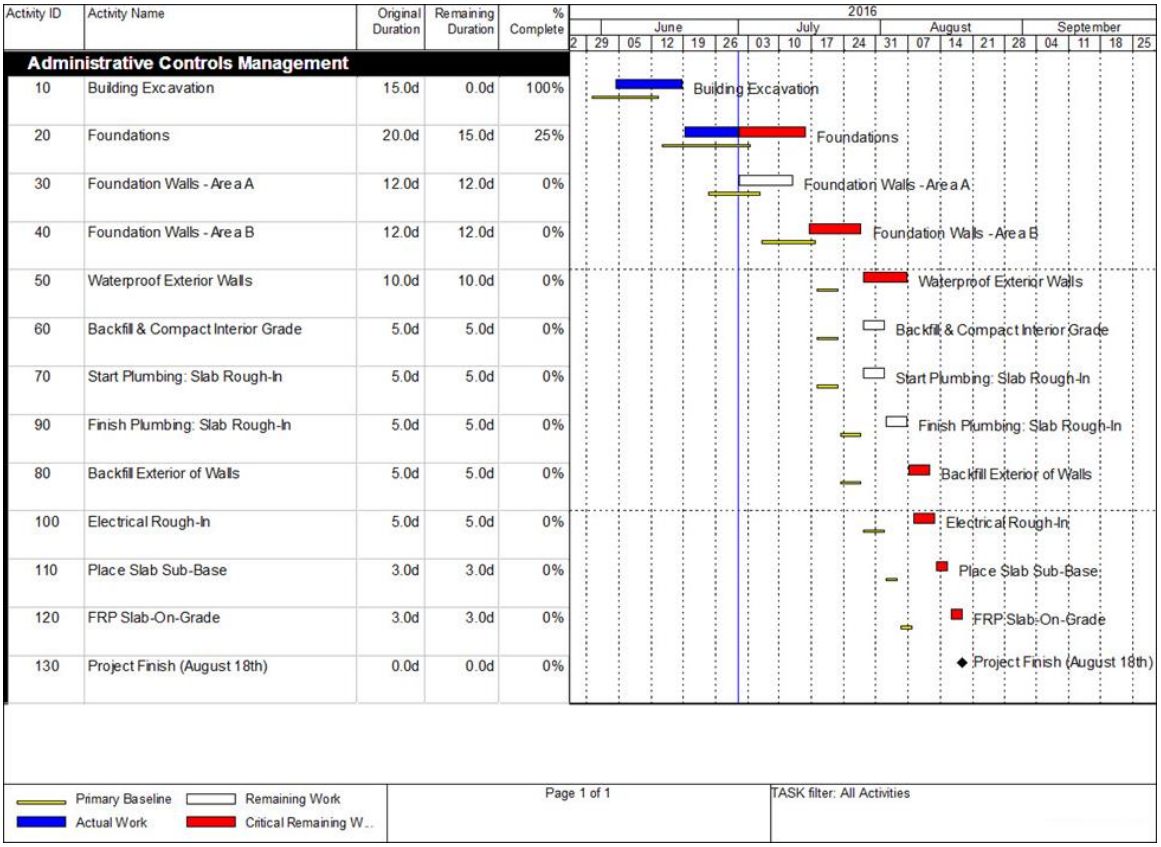


Figure 13.23—Slab on Grade Schedule Update 1

After reviewing the update, it was decided to make changes to durations and logic in an effort to regain some time in the schedule. This was done after management consulted the various subcontractors and discussed alternatives. The following changes were made, which resulted in Figure 13.24.

- Waterproof tasks will be changed to finish five days after foundation walls area B (revised relationship).
- Backfill interior wall and backfill exterior walls will be shortened to three days each (revised durations).
- Electrical rough-in will be changed to start the same day as finish plumbing: slab rough-in (revised relationship).

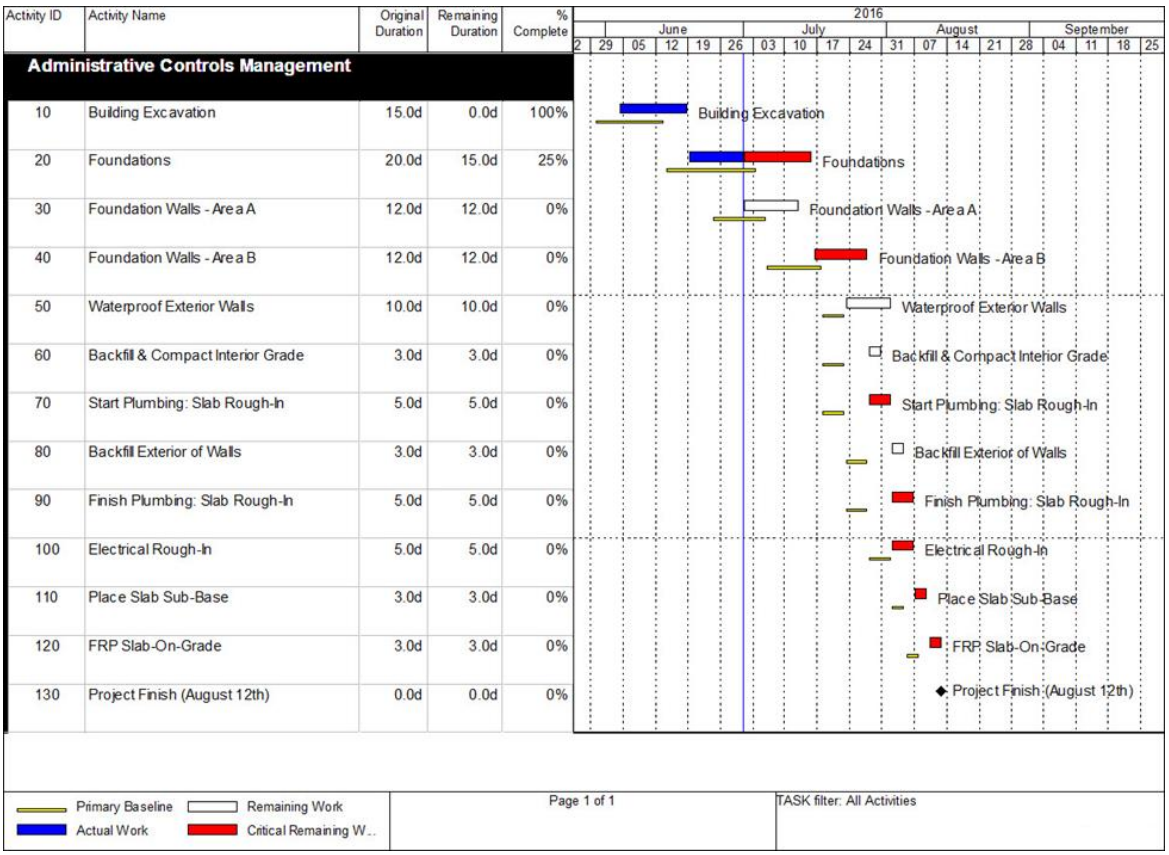


Figure 13.24—Slab on Grade Schedule Update 2

The schedule now shows total project duration of 73 days, with a completion date of August 12. When approved in accordance with project and contract procedures, the new/updated schedule may be designated as the revised baseline going forward.

CONCLUSION

The project schedule represents a communication tool that presents the project plan (complete work scope), the order in which it will be worked, and the length of time it will take to complete the activities and the project. The schedule represents the earliest dates the activities and project can occur and the latest dates for activities and project completion that must occur. As a work execution management updating, progress reporting and control tool, the CPM schedule is an indispensable tool for aligning the project team on the effort necessary for successful and timely project completion.

RECOMMENDED READINGS

1. East, W. (2015). *Critical Path Method (CPM) Tutor for Construction Planning and Scheduling*. New York, NY: McGraw Hill.
2. Hinze, J. W. (2012). *Construction Planning and Scheduling*, (4th ed.). Upper Saddle River, NJ: Prentice Hall.
3. O'Brien, J. J. & Plotnick, F. L. (2005). *CPM in Construction Management*, (7th ed.). New York, NY: McGraw Hill.
4. AACE Recommended Practice 27R-03, *Schedule Classification System*. Morgantown, WV: AACE International.

SECTION 4 – PROGRESS AND COST CONTROL

Chapter 14 – Earned Value Overview
Chapter 15 – Performance and Productivity Management

Chapter 14—Earned Value Overview

Sean T. Regan, CCP, CEP, FAACE

Abstract
This chapter reviews the current ANSI 748 Standards, as well as the commercial, international and standard U.S. government implementation and understanding of the Earned Value Management Systems.

AACE International Technical Basis:

- AACE International Recommended Practice 11R-88, *Required Skills and Knowledge of Cost Engineering*. Topic: 2, Process and Functional Skills and Knowledge; (2.1.3.) Project Control Process; (2.1.3.2) Earned Value Management Process.
- AACE International *Earned Value Professional Certification Study Guide*.

Keywords
Actual Cost Work Performed (ACWP), Actual Cost (AC), Budgeted Cost Work Performed (BCWP), Budgeted Cost Work Scheduled (BCWS), Control Account (CA), Control Account Manager (CAM), Control Account Plan (CAP), Cost Performance Index (CPI), Cost Performance Report (CPR), Cost Variance (CV), Critical Path Method (CPM), Earned Value (EV), Estimate at Completion (EAC), Estimate to Complete (ETC), Integrated Baseline Review (IBR), Level of Effort (LOE), Organizational Breakdown System (OBS), Planning Package (PP), Planned Value (PV), Project Manager (PM), Responsibility Assignment Matrix (RAM), Schedule Performance Index (SPI), Schedule Variance (SV), Statement of Work (SOW), Work Authorization (WA), Work Authorization Directive (WAD), Work Breakdown System (WBS), and Work Package (WP).

INTRODUCTION
This section of project control process is based upon earned value. The intent is to provide a summary analysis of earned value with a direct tie to the AACE International Earned Value Professional Certification Study Guide and the AACE International Total Cost Management (TCM) Framework, An Integrated Approach to Portfolio, Program, and Project Management.

Earned Value (EV) is a process for monitoring a project's performance against a baseline. Earned Value Management Systems (EVMS) are project progress control systems that integrate work scope, schedule, and resources to enable objective comparison of the earned value to the actual cost and the planned schedule of the project. This chapter provides a background for earned value on fixed and variable budgets and the use of EV for government and commercial projects.[8]

Earned Value can be based upon a fixed or variable budget. The US government applies a strict interpretation of earned value based on the *ANSI EIA 748* standards. US government applications are based on a fixed budget; however, changes are allowed, including re-baselining, as long as strict procedures are followed. Industry applications of earned value can vary from those in government. As an example, one of the changes to the ANSI/EIA 748, Earned Value Management Systems Revision - C2013 was, "Additional clarity on items of debate between the government and industry" [1].

EV terminology is not necessarily unique. There are references to schedule, cost, and performance, where the meaning is the same; but, variations exist between the old and new terms. Organizations often use similar names for the same thing. For instance, one organization may use the term Budgeted Cost of Work Scheduled (BCWS) while another uses the term Planned Value (PV). In the end, the EVMS uses the terminology based on schedule performance and cost of the work. Terminology has little effect on the future of EV. It is not the formulae, the software, or how much data can be loaded into the database. Neither is it the current trend of project controls professionals in becoming data entry, database managers, or IT specialists. Simply put, EV is the development of data which provides an early warning system. This allows the project to remain on track toward a timely completion, within budget.

This chapter uses terminology that is consistent with the ANSI 748, AACE Recommended Practice 10S-90, *Cost Engineering Terminology*.

Learning Objectives
After completing this chapter, the reader should be able to:

- Identify the six methods used for measuring work progress.
- Understand the concept of earned value and how to use it in fixed budgets to analyze cost and schedule performance.
- Understand how to evaluate worker productivity.

[8] NOTE: Please refer to S&K 6, Chapter 19 (Project Labor Cost Control) for basic earned value calculation.

ESTABLISHING EARNED VALUE

The work process identified in the TCM Framework represents the AACE International earned value process (Figure 14.1). This process is continued on into the certification process. Key to this framework is the establishment of basic systems, documentation, tools, and organization structures with appropriately assigned roles and responsibilities.

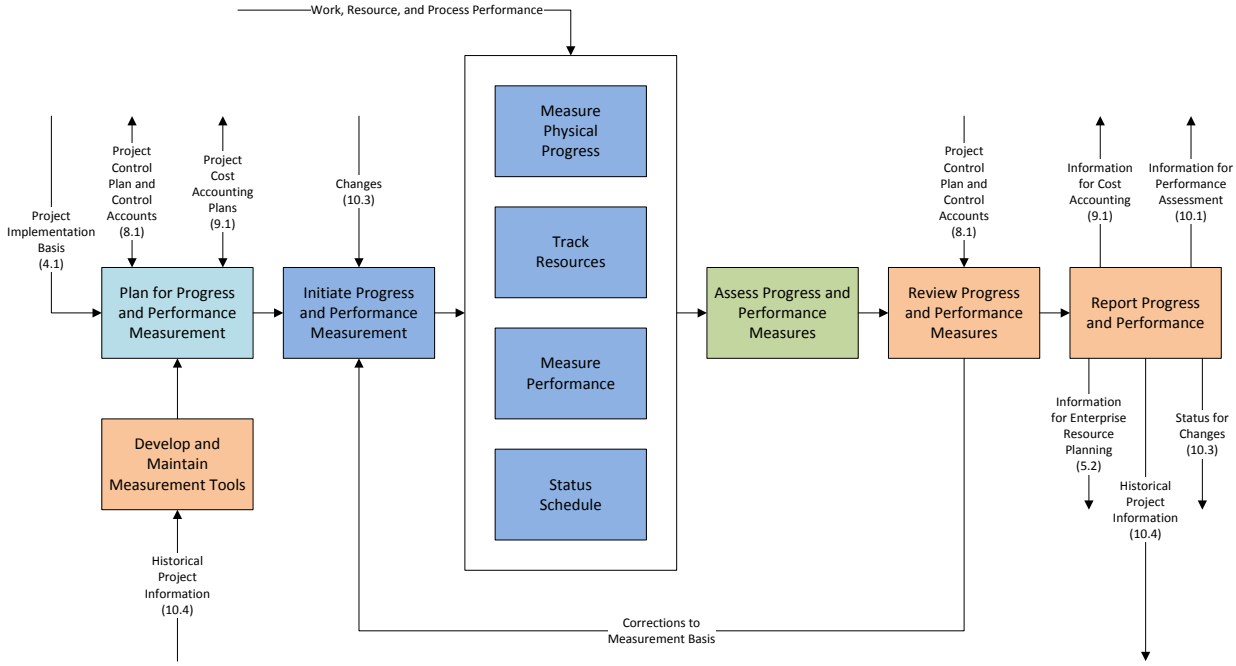


Figure 14.1—TCM Process Map for Project Performance Measurement

The following elements are required for establishing an Earned Value System [2]:

- Organization of Program
- Work Breakdown Structure (WBS)
- Organizational Breakdown Structure (OBS)
- Responsibility Assignment Matrix (RAM)
- Work Authorization (WA)
- Planning, Scheduling, and Budgeting
- Performance Measurement Baseline (PMB)
- Control Accounts (CA)
- Risk Management
- Work Package (WP)
- Planning Package (PP)
- Management Reserve (MR) and Contingency
- Earned Value Definitions
- Variance Reporting
- Change Management
- Performance Review

Project or Program Organization

The project or program organization may be managed by the Project Management Office (PMO). The project itself is organized according to the Work Breakdown Structure (WBS), while the assignment of people is described in the Organization Breakdown Structure (OBS) and the Responsibility Assignment Matrix (RAM), described below. Elements of work may be broken into Work Packages (WP) and Planning Packages (PP). In addition, there are further breakdowns described as: Control Accounts (CA), Control Account Manager (CAM), and Work Authorization (WA).

Work Breakdown Structure (WBS)

The Work Breakdown Structure is the fundamental definition of the program. The WBS provides the framework by which all project effort is organized. This is so it can be scheduled, budgeted, authorized, measured, and reported. The WBS is a product oriented,

top-down hierarchy, that is used as a framework for organizing and ordering the activities required to achieve the end goal of a project. It is a systematic approach to divide and subdivide a project's work into manageable components, such that it helps in the planning and control of technical aspects, cost and schedule of the project [3].

The WBS Dictionary provides a cross reference between WBS elements and the corresponding project Statement of Work (SOW) (mission statement, or other customer documents that define the scope of work). It includes the SOW, the work content of each WBS element, and a basis of estimate. It may also include the responsible organization, contract number, and other data.

The WBS is an organized, hierarchical decomposition of project scope that serves as the backbone for the planning and management of the project. The WBS should be designed with the intent that each WBS level may be used to report information of a similar level of detail consistently across the WBS. The WBS should be numbered so that each element is associated with a unique identifier.

Organizational Breakdown Structure (OBS)
The Organizational Breakdown Structure (OBS), or the project organization structure, as it is called by some companies, reflects the company's organization as integrated with a particular PMO to support a specific project. The OBS is usually illustrated by a graphic or tabular listing of the responsible and/or performing organizations, structured in a hierarchical manner. The OBS provides flexibility in reporting the internal organization of a company. It shows the hierarchical relationship of the organization, including subcontractors, for managing a scope within a WBS. The following are types of organizational structures:

- The **Functional Organizational Structure** is very rigid. Each functional group has a manager over the group's function, such as engineering services or information technology, and experienced individuals reside in the functional group performing tasks assigned to the group. Individuals may be assigned to support different projects; however, they remain part of the functional organization.
- The **Matrix Organization** is one in which an individual in a functional organization is assigned or matrixed to another organization, or group, to provide functional support to that project or organization. The matrixed individual receives their assignments from the project manager for the project they are supporting. The administrative needs of the matrixed individual remain the responsibility of the functional manager, while the employee is on assignment to another group or organization.
- The **Project Organization** is one in which a project is authorized to proceed, given a budget to allocate expenditures, and has a definitive start and finish date. The project organization has an appointed manager or task leader and develops an organizational structure from which specific resources are drawn in order to execute the project. The individuals assigned to the project support the project execution until they are released by the manager to return to their respective functional organization for re-assignment. For further information, refer to AACE Recommended Practice 83R-13, regarding the Organizational Breakdown Structure [4].

Responsibility Assignment Matrix (RAM)
The Responsibility Assignment Matrix (RAM) is a resource matrix of the organizational breakdown structure with the WBS, which identifies the resources associated with the control account levels established for the project, and cross references these levels to the appropriate WBS elements. At lower levels of the OBS, the intersection with the WBS defines control accounts. The control account is important, because it is the point at which technical performance, schedules, and costs are addressed. The RAM includes the functional organization and the named individual responsible for the Control Account, the Control Account Manager (CAM).

The RAM can be expressed as a graphic or tabular listing of the department, the individual, and the WBS element. For further information, refer to AACE Recommended Practice 83R-13, regarding Responsibility Assignment Matrix.

Work Authorization (WA)
Work Authorization is a formal process, which must be documented by a Work Authorization Directive (WAD). Typically, a WAD contains a scope of work, start and finish dates, WBS, responsible person, amount of budget authorized, and signatures of at least the CAM and the Project Manager. The signed WAD must be issued prior to the start of any work scope being authorized. Each level of authorization is agreed to by the parties involved, so that there is no question as to what is required to authorize work. The work authorization approval normally involves two types: outside or customer/client authorization and internal authorization within the organization performing the work. The customer/client authorization is applicable to items such as the basic contract, contract change notices, and supplemental agreements. The internal authorization varies depending on individual company policies and procedures starting at the highest level of management down to the CAM (sometimes referred to as a Cost Account Manager). The sublevel managers have an established level of authorization based on company policy and can delegate their authority to subordinate managers as they deem appropriate.

PLANNING, SCHEDULING, AND BUDGETING
Initial planning of a project requires a good definition of the scope of the project and the steps and resources required to accomplish the work. Part of the initial planning includes a list of deliverables and milestones for the project. The list indicates which deliverables and milestones are mandatory, as well as the relative weight or importance of the deliverables.

The term deliverable is often used interchangeably with milestones; but they are distinct. A deliverable is a contract or management requirement that identifies a physical product. A milestone is a scheduled event that marks the accomplishment of a specified effort or objective. In scheduling terms, a milestone generally is an event with zero duration that marks the start, interim step, or the end of one or more activities. A deliverable may be a milestone, but a milestone is not necessarily a deliverable.

Both deliverables and milestones are used to measure progress. Normally, a deliverable is composed of several activities and milestones. The activities are weighted to their relative importance and progress is measured against each activity. The weighted average of the activities comprising the deliverable is then used to determine the deliverable's overall progress. Key milestones can help manage the critical path of the project or program.

From the development of the milestone schedule, the project or program will develop a critical path based schedule, based on the WBS, with input from the project team. After several iterations and approvals, the schedule should be resource-loaded, identifying the labor, materials, equipment, subcontractors, and other required resources. The project team will develop a budget, based on the required resources, or will be given a budget and asked to complete the scope within the assigned budget.

The work on a critical path schedule is usually scheduled based on WBS, so that costs, work, and schedule can be easily integrated. Once all the activities and relationships are developed, the early and late start and finish dates are calculated for all the schedule activities, using a forward and backward pass analysis through the network paths. Calculated early and late dates may or may not be the same on any network path, since total float may be positive, negative, or zero. The schedule flexibility is determined by the difference between the early and late dates total float. Critical paths have zero or negative float and activities on the critical path are called "critical activities." Adjustments to activity durations and logic relationships may be necessary to produce network paths with zero or positive float. The schedule should initially be run without constraints, so that float is zero, and a true completion date is identified. If there are mandatory constraints imposed by management, regulatory bodies, or budget cycles, they should be accurately reflected; however, the use of constraints should be rare, as this prevents calculating a true critical path. If a completion date has been imposed and a negative float results, the schedule should be evaluated to either change the logic to run more activities in parallel, or management should be notified and the scope should be reduced.

The integration of the technical scope, planning, scheduling, budgeting, and work authorization results in the Performance Measurement Baseline (PMB). The PMB is the sum of the time phased budgets for all the control accounts, and summary level planning packages, plus the applicable indirect budgets and any undistributed budget. Management Reserve (MR) is added to the PMB to reach the Contract Budget Base. The Contract Budget Base, plus any profit or fees, equals the contract price.

The PMB becomes the plan by which to measure, monitor, and control overall cost performance on the project. The baseline is developed by totaling estimated costs by period; and is usually displayed in the form of an S-curve against which contract performance is measured. If a formal estimate is developed, the baseline is developed by integrating the estimate with the approved baseline schedule.

Key Characteristics of a PMB

- Baseline is reasonable in terms of resources and schedules to achieve the plan.
- Each budget is traceable to the WBS, OBS, scheduling system, and cost collection system.
- Budgets and schedules are established for all authorized work, at the control account level, and for undefined work at a summary level.
- Baseline reflects the work scope, time phased and consistent with the integrated schedule.
- The budget value for the work scope, in control accounts and higher level summary planning accounts, equals the sum of its work packages and planning packages budgets.
- The budgets are stated in either dollars, hours, or other measurable units.
- Both owner and contractor approve the budget baseline.
- Baseline cannot change except through a formal change control process.

Performance Measurement Baseline (PMB)
Simply defined, the PMB reflects the approved baseline of the project, time phased and loaded to cost accounts into which the program is managed. The baseline is set based on the original contract values and can only be adjusted through an approved change, via the program change management program.

Control Account (CA)
A Control Account (CA) is a management control point, at which budgets (resource plans) and actual costs are accumulated and compared to earned value for management control purposes. A CA is a natural management point for planning and control, since it represents the work assigned to one responsible organization element and one project or program WBS.

The purpose of defining CA's is to decompose the project scope into work elements that can be planned accurately and managed effectively. CA's should not span multiple WBS elements.

The CA's are defined by integration of the OBS and the WBS, and represent a defined work scope given to a single organization unit. Multiple CA's may exist within a single WBS, depending on the number of organizations authorized to do work within the WBS element scope.

CA's are where cost, schedule, and work scope requirements are integrated, planned, and managed. Integrating schedule activity duration and cost estimates results in a time-phased budget and resource plan. The application of earned value methods to CA's allows measurement of work scope cost and schedule performance that are summarized and reported to management as project performance.

Risk Management
Risk Management is the review of the schedule, budget, process, and management of the program, to determine the exposure to risk on the program. The risk impacts to both cost and schedule need to be quantified and managed, and must be included in the PMB. Contingency and MR should be identified to ensure adequate budget and time for the project. EVMS requires a proactive analysis of the risk data to determine the viability of the program.

Work Package (WP)
A Work Package (WP) is task or set of tasks performed within a control account. A WP has both schedule and resources.

A WP is the point at which work is planned, progress is measured, and earned value is computed. It is also the point at which schedule variance is calculated. WP is a generic term used to describe working level jobs, tasks, or processes that are natural to enterprise operations.

Work Packages are an element of control within control accounts. The number, size, and duration of Work Packages, in a CA, are defined by management need, company policy, as well as size and complexity of the project. Not all CA's will require multiple WPs for effective planning and control.

Planning Package (PP)
A Planning Package (PP) is a logical aggregation of work, usually future efforts, that can be identified and budgeted, but which is not yet planned in detail at the work package or task level.

Planning Packages are used as budget placeholders within a control account for future work that has not been defined enough to plan as a work package. The planning package budget is time-phased consistent with schedule due dates. The Planning Package is refined, as deliverables and products are defined in more detail, as the time to start work draws nearer. There is no standard advance planning look-ahead period for conversion of planning packages into work packages. Each enterprise will select the conversion methods that best meet the needs of each project.

Summary Planning Packages are used as high-level budget placeholders for work scope that is time-phased for interim resource management and budgeting; but has not been assigned to organizational units and control accounts. The summary planning package must be assigned to organization units and control accounts before the work is planned or actually begins.

Management Reserve (MR) and Contingency
Management Reserve (MR) is an amount added to the budget to allow for discretionary management purposes, outside the scope of the project. It may include amounts that are within the defined scope; however, management does not want to fund these as contingency.

MR is created from the contract cost baseline to provide management flexibility to allocate budget to cover unknown contingency at the total project or program level, rate changes, and other project unknowns. Prudent use of the MR prohibits use to offset accumulated over/under runs. MR may be held at the total project level or distributed and controlled at lower levels. It is reconcilable to the total contract level. MR is not used to absorb contract changes and must not be viewed by the customer as a source of funding for added work scope. MR does not have scope and is not assigned to specific risks.

MR is not part of the PMB and is excluded from earned value calculations. MR is normally used on large, complex federal projects.

Contingency is an allowance or reserve added to the cost estimate and schedule duration for anticipated, but not certain events. These "known-unknown" events are identified during a risk assessment, or by experienced estimators and project managers that have delivered similar work scopes to that being estimated. Enterprise management may add a contingency to increase their confidence and the probability that the project will be delivered on or under budget and no later than the planned project completion date.

Contingencies are part of cost and schedule baseline plans to deliver the authorized work scope. When the reserves for the known-unknown events are removed from the baseline plans, the PMB is created. The PMB is the schedule of expenditures for the estimated resources to deliver the project scope against, which project performance is measured using earned value techniques.

Using earned value techniques to measure performance against the baselines that include contingency will distort variances, because the earned value will be derived from the baseline plans that include the reserves.

Earned Value Definitions
Definitions of Earned Value are based on the development of the program Budget, Actual Cost, and Performance. The value is then divided into measurement classifications which determine the performance index and variances of the program.

Planned Value – Budgeted Cost Work Scheduled (BCWS)/Planned Value (PV): Represents the time phased value of the PMB. This is phased and placed into the CA and represents the total for the period and to date. The PV is calculated by multiplying the budget by the planned physical progress (percent complete).

Table 14.1 shows an example budget of $1 million over six months:

Table 14.1—BCWS/PV Spread

BCWS (PV)	Jan	Feb	Mar	Apr	May	Jun
Planned Monthly Progress	10%	20%	25%	20%	15%	10%
This Period	$100,000	$200,000	$250,000	$200,000	$150,000	$100,00
To Date	$100,000	$300,000	$550,000	$750,000	$900,000	$1,000,000

Earned Value – Budgeted Cost Work Performed (BCWP)/Earned Value (EV): Represents the time phased value of the performed scopes of work and the assigned definitions of work completed. The EV is calculated by multiplying the budget by the actual physical progress (percent complete).

EV performance can be measured at various levels and is based on the WA work in the CA. Credit cannot be earned for work not assigned and released with a WA. Any work assigned in a planning package must be released with a WA prior to being included in EV calculations.

Level of Effort (LOE) refers to the support effort, such as supervision, that does not readily lend itself to measurement of discrete accomplishment. It is generally characterized by a uniform rate of activity over a specific period of time for EV calculations on LOE activities; the earned value equals the actual costs. LOE should be limited to the absolute minimum on projects, since it obscures the actual progress. LOE activities should be in separate CA's and not be mixed with discrete, measurable activities.

Milestone Value: Percentage of earned is achieved, based upon completion of a specific milestone on a percentage assigned by the CAM.

Physical Progress: Value is divided based upon quantity of scope of work. This can be tons of steel, number of drawings, cubic yards/meters of concrete, number of piles, etc.

Incremental Phasing: The CAM assigns a value for completion of a certain scope, such as the steps on equipment shown in Table 14.2.

Table 14.2—CAM Incremental Phase Table

Description	Percentage Earned	Total Earned
Set Equipment	30%	30%
Electrical	20%	50%
Instrumentation	20%	70%
Grouting In	5%	75%
Bump and Run	20%	95%
Full Commissioned	5%	100%

Actual Value – Actual Cost Work Performed (ACWP)/Actual Cost (AC): This is based on the actual cost incurred on the project. This includes the period cost and accruals. ACWP/AC is not the same as the accounting value. The accounting value does not take into consideration accrued costs. Traditionally, actual cost is recognized on procurement when the item is received from the vendor. The only other exception is when progress payments are made, in which the payment can be recognized when they're made.

Variance Analysis
A variance analysis is the evaluation of the differences between the actual and expected project performance. Projects typically set the trigger points for requiring Variance Analysis Reports, such as >10 percent variance or >$50,000, for example. Earned value analysis segregates cost and schedule variances by comparing the budgeted cost of work performed (earned value) to both the actual cost of work performed (actual costs incurred) to determine the cost variance, and to the budgeted cost of work scheduled (planned expenditures or time-phased budget) to determine the schedule variance.

Reporting variance analyses is intended to provide early and improved visibility of project performance problems, issues, and effectiveness. The variance analysis report will describe the cause, impact and corrective action. The cause of a variance describes the problems or reasons for the performance variance from plan. The impact of a variance will evaluate how the current work and future plans may be affected by the performance that caused the variance. The corrective action describes what will be done to realign actual and planned performance, and who is accountable to execute the corrective actions.
Variance analysis and control begins at the lowest practical level of the WBS. For most projects, variance analysis is conducted at the control account level. Variance thresholds are established to focus the analysis on significant variances and to expedite the completion of the variance analysis reports. Variance is not bad per se, but the causes should be understood and actions taken to correct or mitigate the variances where possible.

CHANGE MANAGEMENT
Change management is the formal process for the review and approval of new/revised work scope versus original contract base scope. Only approved and signed contract change orders will be incorporated into the original baseline contract monetary levels on a regular basis. The revised contract baseline will contain the revised monetary base, revised/amended work scope, revised/amended schedule and current management plan. This results in the basis of the progress measurement for a given reporting period on CPI and SPI calculations that reflect the most current information. The Change Management Program will address other project activities/changes that are ongoing which reflect the balance of the program. These changes will reflect scope, process and administration changes, which must be accurately reflected in the change management program. Some programs require the approval of a Change Control Board, while others allow the Project Manager to approve changes up to a certain limit. Changes should not be allowed to "fix the variances" in cost or schedule EV metrics.

Performance Reporting
At least on a monthly basis, generate the following information at the control account and higher levels as necessary for management control, using actual cost data (plus accruals) from, or reconcilable with, the accounting system:

- Comparison of the amount of planned budget and the amount of budget earned for work accomplished, both for the month and cumulative. This comparison provides the schedule variance.
- Comparison of the amount of the budget earned vs. the actual (applied where appropriate) direct costs for the same work for both monthly and cumulative. This comparison provides the cost variance.

Also, accomplish the following:

- Identify, at least monthly, the significant differences between both planned and actual schedule performance and planned and actual cost performance, and provide the reasons for the variances in the detail needed by program management.
- Identify budgeted and applied (or actual) indirect costs at the level and frequency needed by management for effective control, along with the reasons for any significant variances.
- Summarize the data elements and associated variances through the program organization and/or WBS to support management needs and any customer reporting specified in the contract.
- Implement managerial actions taken as the result of earned value information.
- Develop revised estimates of cost at completion, based on performance to date, commitment values for material, and estimates of future conditions. Compare this information with the performance measurement baseline to identify variances at completion important to company management and any applicable customer reporting requirements, including statements of funding requirements.

Integrated Baseline Reviews (IBR)
An IBR is a formal review in the Department of Defense, conducted by the customer and prime contractor, and in some cases major subcontractors. IBRs are typically required following contract award to verify the technical content of the Performance Measurement Baseline (PMB) and the accuracy of the related resource (budgets), schedules, and technical ground rules and assumptions.

An IBR will also be performed when work on a production option of a development contract begins or, at the discretion of the program manager, when a major modification to an existing contract significantly changes the existing PMB or significant shift in the content and/or time-phasing of the PMB occurs. The intent is for the IBR to be a continuous part of the process of program management by both the client and the contractor.

General Guidance: IBRs should be conducted within six months of contract award and are continued based on changes to the performance measurement baseline. They are determined by a set value in the program/Client.

The IBR Process consists of the following jointly executed project management activities:

- PMB Assessment
- IBR Preparation
- IBR Execution
- Management Processes

CONCLUSION

This chapter provided an overview of the EVMS based on ANSI-748. EVMS is an integrated process of application of the Budget, Actual Cost and Performance Measurement. In many EPCM contracts, the evaluation methodology is still based on the performance of hours. There is an ongoing debate as to the benefit of EVMS, since hours are the metric of the engineering and construction program. There is also continued discussion that Engineering is not measured by hours. Analysis of these programs has shown that the metric of assigned currency, based on EV versus the traditional EPCM hour, has significantly reduced and allowed for programs to adjust.

Often, the general comments found in performance reviews of various projects mention that engineering does not use performance metrics. When asked to name the steps they execute, four to six steps, including dates, percentage weighting, and hours in which they can apply these metrics are normally given. In these cases, the only viable metric is the use of an EV system, which can relate to the value of the program on a consistent basis.

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Chapter 15—Performance & Productivity Management[9]

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Abstract
Success index, net profit, and total costs set the stage for a comprehensive discussion of the performance and productivity management necessary for organizational profitability outlined in this chapter.

Improving organizational performance is suggested by an examination of inefficiencies in areas, including those that the employee has control over. Suggested actions for the elimination or reduction of these areas are given, and an awareness of the areas that one does not have control over is highlighted.

A comprehensive treatment of inefficiencies and suggestions for improvement, specifically in the construction site, but also some of which is applicable in all environments, have been identified and include:

- Labor/contract disputes
- Meeting
- Personality issues: owner-engineer-contractor
- Late material/installed equipment deliveries

Productivity is discussed and is related to the variabilities that may occur on projects as it relates to:

- Sociological (area) factors
- Location factors
- Human factors
- Field organization and management factors
- Accounting and estimates formats

Productivity and performance analysis is also discussed in terms of the following:

- Determining percent complete
- Productivity measurement of individual work tasks
- Use of productivity data

In essence, this chapter fills a gap in the cost engineering knowledge base by the inclusion of skills and knowledge that is often missed in the technical scope, but is important; knowledge base that includes business, sociological, and organizational skills that are necessary for the success of the cost engineering professional.

Keywords
Human Resources, Inefficiencies, Location Factors, Net Profit, Organizational Skills, Performance Analysis, Productivity Management, Productivity Measurement, Sociological (Area) Factors, and Success Index.

INTRODUCTION
Companies in the business world are constantly concerned with improving their bottom line—increasing the rate of return on investment, increasing the ratio of profit to revenues, or simply increasing total profit. Using programs with buzzword titles; such as productivity improvement, total quality management, re-engineering, time-based competition, horizontal management, down-sizing, and right-sizing, they reorganize, trim staffs, invest in training, automate, computerize, and otherwise do whatever is considered necessary to optimize or maximize the company's performance and beat the competition. But, whatever the name of the program, the goal is the same; spend less to make more money or spend less to provide the same or better service. For production type activities, this translates into reducing worker and equipment hours per unit of output—i.e., improving productivity. For support and professional activities, it means improving efficiency. For all activities, it includes reducing waste of time, materials, and equipment. Altogether, it means improving the outcome of the total organization.

[9] Dr. James M. Neil authored the original chapter in S&K 5. Rohit (Roy) Singh made revisions and minor updates to the manuscript for S&K 6.

Learning Objectives
After completing this chapter, readers should be able to:

- Analyze worker productivity and performance.
- Identify ways to increase productivity, improve performance, and minimize waste in the workplace.

SUCCESS INDEX
Numerical evaluation of total organizational performance is possible using the Success Index (SI). It could be called the performance index, but doing so might cause it to be confused with the Productivity Index (PI), to be described and used later. Equation 15.1 is the formula for the success index for a profit-oriented business. Equation 15.2 is for a service organization, such as a government.

Success Index = (net profit) / (total costs)

Equation 15.1

Success Index = (value of services rendered) / (costs of providing services)

Equation 15.2

It should be noted that the success index is really an expression of organizational productivity, because it relates a form of output (profit or value) to a form of input (cost).

To continue the discussion, the denominators of Equations 1 and 2 can be re-expressed as shown in Equations 3 and 4:

Success Index = (Net profit) / (essential costs + cost of waste)

Equation 15.3

Success Index = (Value of services rendered) / (essential costs + cost of waste)

Equation 15.4

The denominators in both equations now divide total costs into two broad categories—essential costs and cost of waste. Essential costs are those personnel, material, equipment, tax, and other costs that would be incurred if the organization were efficiently organized and running perfectly. As for waste, these are the major categories:

- Inefficiencies inherent in the design and operation of the work place
- Individual inefficiencies
- Non-contributing (wasted) time by individuals
- Waste of materials, supplies, and services (misuse, overuse, loss)
- Waste of equipment (abuse, misuse, loss)
- Functions that no longer add value to the output of the organization

In the past, management tended to focus on productivity improvement as the key to reducing costs and/or improving the bottom line, and that subject was and still is given significant attention in technical literature. This is to be expected, since production activity can be readily measured, it can be expressed in hard numbers, its trends are easily noted, and it lends itself to detailed analysis and improvement studies. The problem is that there are many people and much equipment within a company performing functions whose effectiveness and contributions are not properly measured on the basis of output per unit of input. Personnel in this category include most support and professional staff—secretaries, design engineers, managers, etc. Equipment types include word processors, tower cranes, and administrative vehicles.

True, there are outputs associated with many of these individuals and pieces of equipment, but productivity is not the basis for their selection. For example, a receptionist or a security guard must be present to handle whatever comes up; their performances would not be evaluated on the basis of quantity output. Similarly, a tower crane at a building construction site is selected on the basis of lifting capacity at various boom radii—one does not think in terms of tons per hour. There have been efforts to apply productivity measurement concepts to individuals in this category who do have products (e.g., secretaries and design engineers), but with little or no success. In fact, doing so may create stress and cause quality to be compromised, as individual goals shift from quality to quantity production of the item designated for measurement (e.g., correspondence processed or drawings produced).

To expand on the above, within an organization's population are people who produce things and people who perform things. Most individuals do both to some degree. Performance may be associated with units of output, but the real performance standard is something other than quantity (e.g., quality of engineering drawings, ability to write, or responsiveness in an emergency). Performance is evaluated subjectively (e.g., above average or seven on a scale of ten).

One would like to assume that every organization seeks to do everything possible to promote performance and productivity. Unfortunately, the real-life situation tends to be as depicted in Figure 15.1. It shows that an individual has a basic capability resulting from many factors. What that individual can produce becomes restricted by organizational constraints.

What should an organization do? To borrow an expression from a U.S. Armed Forces recruiting commercial, the organization should do whatever is necessary to make each individual, "the best that he/she can be." That is done by eliminating or minimizing conditions within an organization that limit performance and productivity and by creating conditions that promote them. Remaining sections of this chapter provide guidance for doing this. The first section will focus on the challenge of improving performance of a total organization. That will be followed by a discussion relating specifically to those personnel in the workforce involved in production activity. Finally, the role of incentives in performance and productivity management will be reviewed.

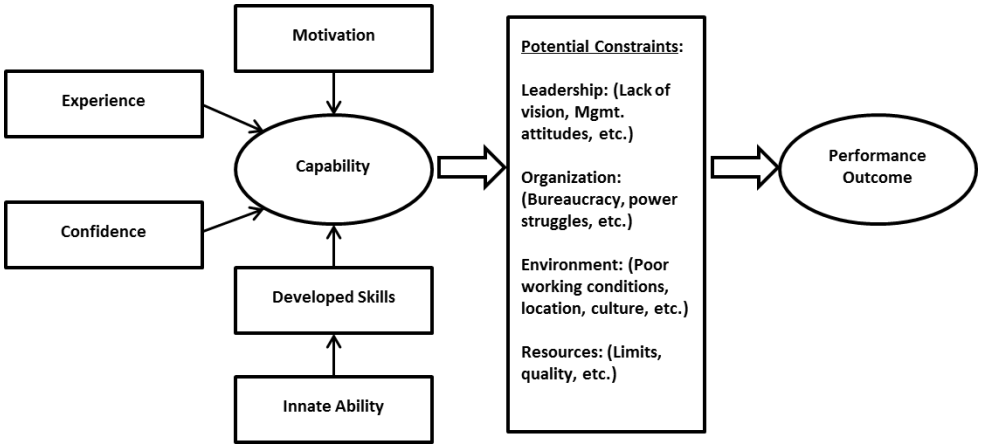


Figure 15.1—Performance Expectancy Model

THE OVERALL PERFORMANCE ISSUE

The Challenge

An organization's success index always will be less than what is potentially available had perfection prevailed, because human beings are involved and Murphy's Law (If anything can go wrong, it will) has yet to be repealed. The problem is illustrated in Figure 15.2, which illustrates how performance potential is lost through inefficiency and waste. The goal must be to eliminate or minimize the factors contributing to that degradation.



Figure 15.2—The Performance Problem

Losses Through Inefficiency

Inefficiencies are both organizational and individual. Inconvenient positioning of office reproduction equipment, shortages of equipment or materials, lack of procedures, excessive management layering, and poor lighting are typical organizational inefficiencies. Failure to plan, refusing to use labor-saving equipment (such as a word processor), and sloppy filing are typical

individual inefficiencies. All of these translate into time loss and higher costs. The problem with inefficiencies is that the losses tend to be hidden—an observer watching an individual doing what appears to be contributing work may not realize that the work is being done very inefficiently.

Waste Through Interruptions

Everyone acknowledges that interruptions are disruptive, but interruptions are seldom treated as a subject area with significant potential for improving productivity and performance. Take the typical office situation, shown in Figure 15.3, where an individual is trying to write a report: a series of interruptions in the form of telephone calls and visitors reduces the individual's average productivity significantly.

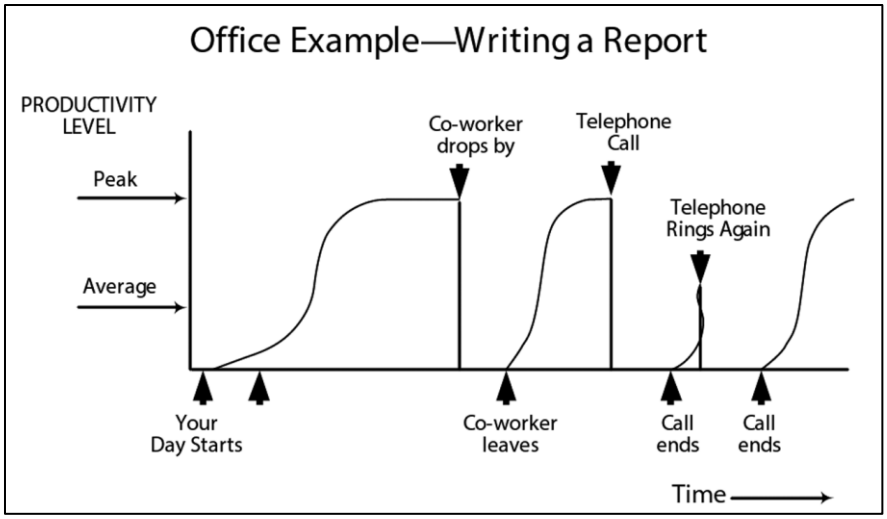


Figure 15.3—Impact of Disturbance on Performance

If something could be done to reduce these interruptions (e.g., an electronic mailbox, visitor screening, providing better office privacy), the individuals potential output would be improved. The lesson to be learned is simple: review work practices in an organization to determine where avoidable interruptions occur, and then take corrective action.

Other Time-Wasters

Interruptions are but one form of time waste; there are many more. First, is a list of events or situations that are accepted parts of life in most organizations, but each cause interruptions, and some result in wasted time:

- Official meetings and appointments
- Telephone calls
- Personal breaks and lunch breaks
- Official visitors
- A need to interrupt current activity to make a copy of something, send a fax, or coordinate with another worker
- Fire drills, hazard alarms, or other emergencies
- Adverse weather
- Power outages
- Equipment breakdowns
- Holds for quality checks or coordination
- Absentees whose work must be absorbed by others
- Turnover of key personnel—new ones must be brought up to speed
- Higher headquarters or outside agency inspections, audits, and reviews
- Secretaries /clerks delivering mail and messages
- Noise and conversations from adjacent work areas
- Unusual activity outside office windows
- Running out of something—paper, staples, etc.
- Misplacing something
- Forgetting something

Certain actions or policies may minimize the disruption and time loss effect of some of the above items, but the potential is not significant. The next list contains more events and situations that create time loss in an office setting. In this case, all of them have significant potential for elimination or reduction through better planning and management:

- Unnecessary and unstructured meetings;
- People late for meetings;
- Social visits or greetings from passing fellow employees;
- Sales calls without appointments;
- Waiting for engineering and vendor information;
- Errors or omissions on engineering drawings;
- Lack of communication—somebody didn't "get the word;"
- Too many people or organizations involved in getting an answer, approval, or decision;
- Excessive time taken to make decisions, or approve/coordinate something;
- Too few support personnel available (e.g., clerical), so professional staff must perform own support; and,
- Inadequate support equipment (e.g., copy machines) causes waiting.

In the case of construction field sites, these are controllable time-wasters:

- Ill-defined scope forces constant reworking of schedule;
- Contractual disputes;
- Labor disputes and adverse union activity;
- Arbitrary work rules;
- Personality problems among key personnel on owner, engineer, and contractor staffs;
- Late materials or installed equipment deliveries;
- Materials and equipment for installation do not meet specifications or have fabrication errors;
- Materials and equipment allowed to deteriorate in storage, so as to not be usable;
- Materials and equipment listed on warehouse inventory cannot be found;
- Failure to pick up all needed materials the first time;
- Excessive distances between work areas and tool rooms, warehouses, and laydown areas;
- Wrong or defective tools issued;
- Waiting for support equipment (e.g., crane);
- Waiting for an approval to do something;
- Lack of information or waiting for instructions;
- Issuing instructions after work has started;
- Waiting for other crews to get out of way;
- Individuals don't understand their roles or responsibilities—must always ask questions;
- Limited availability of a critical skill that must be shared among crews (e.g., competent person required by OSHA for certain operations);
- Late starts/early quits;
- Absentees—work must be reorganized;
- Discipline problems;
- Permits (such as hot work permits) not available;
- Daily renewal of permits;
- Conflicts with operating plant personnel on revamp work;
- Operating personnel, having not been consulted during development of the project, make changes on the fly;
- Changes are issued—both formal and constructive;
- Unexpected conditions require work reorganization;
- Waiting for access or removal of lockouts;
- Over-inspections;
- Outdated policies or procedures that must be interpreted to fit current needs;
- Work is started before being fully planned and without all resources needed;
- Safety incidents;
- Construction mistakes.

Many actions can be taken to eliminate or minimize the time-wasters listed above. For many, the nature of the problem makes the solution obvious. However, to provide several ideas with respect to one major time waster, consider the problem of meetings—too many, too big, and too unstructured. Following are some ideas that have worked for others to correct the situation:

- Prepare and implement a written policy/procedure for conduct of meetings.
- Train meeting sponsors on the policy.
- Prepare and work from an agenda for all meetings.
- Establish a limit of time and start promptly.
- Prepare minutes of meetings to include all decisions made, items remaining open, and actions assigned to individuals (with target dates for completion).
- As an occasional attention-getter, require meeting sponsors to prepare a timesheet for each meeting that lists individuals

attending, time spent, and their hourly billing rates (wages + fringes). The sponsor must extend and total the cost figures and submit the summary to his/her supervisor. This makes meeting sponsors think twice about scheduling questionable meetings, encourages them to better plan the meeting, and forces them to think in terms of benefits and costs.

- For any individual late to a meeting, fine them $5, and put it in the coffee or flower fund.
- Arrange the tables and chairs with respect to the entrance so that a latecomer cannot "sneak in." He/she must walk by the chairman and everyone else so that he/she will be totally embarrassed.
- Schedule meetings at beginning of day, just before lunch, just after lunch, or just before quitting time. Scheduling them in the middle of the work day creates a major interruption.

Waste Through Rework

Rework is a special form of waste. One tends to apply the term only to redoing work because the work is flawed or changed. But, one will find countless other forms of rework going on within organizations every day when you use the more general definition of rework: the repeating of an activity (and consequent expenditure of resources) with no value added to the final output. Because activity during rework usually looks the same as when work is done the first time, it is easily overlooked as an area of waste with tremendous potential for reducing costs. Following are common examples of rework in an organization.

- **Marketing rework**—Constantly looking for new work because the organization cannot attract significant repeat business.
- **Management/Supervision Layering**—maintaining excessive levels of supervision—a higher level essentially repeats the work of the lower level.
- **Materials Management**—Double (or more) handling of materials before use.
- **Reorganizations**—Reconfiguring an organization with no significant change in missions or workload.
- **Physical Relocations**—Moving personnel and equipment to accommodate a new organizational structure or otherwise.
- **Lack of Electronic Data Links**—Receiving data in hard copy and reentering it into another computer system, instead of electronically linking computer systems.
- **Computer Illiteracy**—A manager or other professional staff member who is computer averse still does everything long-hand and turns it over to a clerk for entry into a computer.
- **Excessive Administrative Review**—Requiring excessive numbers of approvals on documents such as purchase orders or travel claims.
- **Failure to Provide Management Guidance**—A manager failing to provide guidance when tasks are assigned and then rejecting the output as not being what he/she was looking for.
- **Excessive Quality Control**—Maintaining separate contractor and owner quality control operations on a project site.
- **Post-Production Engineering Review**—Performing a review of engineering deliverables after the deliverables have been fully drafted by the engineering staff. After being marked up, drawings must be redone.
- **Reinventing the Wheel**—Failing to conduct post-project reviews to develop experience data and lessons learned that can be used in future planning.
- **Scope Revision During Detailed Engineering**—Failing to completely define scope during conceptual engineering. Detailed designs must be reworked with each scope change. May create construction rework.
- **Claims**—Expending significant resources in the pursuit of claims, particularly the research and reconstruction of records to find out what really happened.
- **Estimating Formats**—Developing an estimate against one format and then reconfiguring it for project control.
- **Continual Hiring and Training of New Personnel**—Experiencing high turnover because the organization is unable to retain trained personnel.
- **Misuse of Fax**—Using a fax to transmit a copy of something that also is being transmitted in hard copy.
- **Not Invented Here**—Refusing to acknowledge good ideas that have been demonstrated by others, and, as a matter of hard-headedness, doing it another way.
- **Using Second Shift to Continue Work of First Shift**—Passing work from one crew to another at a shift change results in lost time as the new shift determines the status of work in place. They also may redo some work.
- **Out-of-Date or Incorrect Specifications**—Designing against out-of-date or incorrect specifications results in design rework and can create field rework or delays.
- **Resolution of Time-Card Discrepancies**—Resolving time-card discrepancies because of wrong coding, wrong totaling, etc.
- **Untimely Input on Design**—Introducing additional design requirements after design development is under way.

The Solution

As one reads through the lists of time and cost wasters above, potential corrective actions are almost obvious. The first step in waste elimination or minimization is to acknowledge that these conditions exist. Through surveys or group discussions, lists of negative conditions can be identified. Usually the list will be too long to attack in total at one time, so the list should be narrowed down to those with the greatest potential for improvement. Specific solutions can be generated through group problem-solving sessions using the various problem-solving tools associated with Total Quality Management (TQM)—flow charts, cause and effect diagrams, force-field analysis, and various statistical analyses. As problems from the original list are solved, return to the list and

determine if others should be added and select new targets for improvement. The result of these efforts will be continuous improvement, the ultimate goal of any TQM program. Of course, a proactive approach to waste control is always better than a reactive one. The following specific guidelines are appropriate:

- **Plan! Plan! Plan!**—This is universal guidance for any operation.
- **Establish written policies and procedures**—These become the standard references for how things are to be done.
- **Involve users** (e.g., operators) and constructors in design decisions.
- **Control changes**—Changes degrade performance because they delay and demoralize.
- **Give priority emphasis to safety and quality**—Many claim that performance is directly related to quality and safety.
- **Control disturbances and interruptions**—Examples have already been given. This should be an area of major emphasis.
- **Take advantage of modern technology**—Most productivity gains in the industrial world result from use of better technology.
- **Employ partnering and team building**—The team approach is always better.
- **Communicate**—An essential element within a true team.
- **Involve employees in planning**—This establishes their commitment.
- **Use employee group problem-solving techniques**.
- **Make your work place a good place to work**—This promotes employee loyalty and stability, and limits distractions and inefficiency.
- **Recognize employee achievements**—Let employees know you appreciate their contributions; this will stimulate continued achievement (see later discussion of incentives).
- **Promote first-level quality control**—This is the best way to minimize rework.
- **Train managers, supervisors, and workers**—This promotes professionalism and consistency within the organization, while also showing you care. One major industrial firm claims that they get $30 in benefits from every dollar spent on training.
- **Be selective in hiring**—Quality control of personnel cannot be overemphasized.

THE PRODUCTIVITY ISSUE

The Challenge

For any business involved in producing goods or providing services, the productivity of its production personnel and equipment directly influences that business' competitiveness and profitability. It follows that these businesses continually seek ways to improve their productivity. Usually, production is dependent upon some combination of machines and personnel, so both must be examined when seeking productivity improvements. In some situations, a company's production potential is totally constrained by the machines being used—they can produce only so many items per unit of time. If so, the solution is to either add more machines or find higher output machines. If human beings are a factor in the rate of production, improving their productivity is more complex.

The construction industry has a somewhat unique challenge when it comes to productivity. It is a fact that a large percentage of construction work is awarded on a fixed-price, target-price or target-work hour basis. In this arena, competing contractors must base their bids or proposals on productivity assumptions for all crafts involved. Then, once the contract is awarded, the contractor has the challenge of meeting or beating the productivity assumptions in order to make a profit or at least not lose money. With labor costs often being 40 percent or more of the total installed cost, and with profit margins in construction often being less than five percent, it is easy to see how errors in productivity estimation and management can ruin a contractor. Remaining discussion in this section will be examples on recognizing the particular challenges of the construction industry.

A major point to be made and emphasized is that productivity on the same type of work varies significantly from location to location within a country and from country to country. That variation is caused by many factors, which may be grouped as follows.

Variability—Sociological (Area) Factors

Some variation can be attributed to differences in the sociological makeup of the local population, local work ethic, level of mechanization, the education and training levels of workers, the climate, the organized labor situation, and urban vs. rural factors. Recognizing this, most major construction contractors and some owners maintain proprietary data on area productivity differences to be accounted for in their estimating of construction costs. Typically, they will select one area as the base area and give it an index of 1.00. Other areas are given indices that relate their general productivity to the base area, with indices less than 1.00 being less productive and those with indices greater than 1.00 being more productive.

For example, these are extracts from an index register used at one time by one-owner company:

Houston	1.00
Baton Rouge	0.85
Corpus Christi	1.10
Chicago	0.80
Denver	0.95

Internationally, the variation is even greater. An article, *International Labor Productivity*, in the January 1993 issue of AACE International's **Cost Engineering** journal, by J.K. Yates and Swagata Guhathakurta, provides relative productivity data for many countries. Its indices use a format that is the inverse of the above and it provides ranges for each country. Examples:

Washington, D.C. (base area)	1.00
Belgium	1.25 – 1.52*
Jamaica	1.49 – 3.05
China	2.60 – 4.50

**Interpretation*—Comparable work in Belgium will require 25-52 percent more work hours than in Washington, D.C.

Variability—Location Factors
As location varies, so do these factors:

- Weather patterns
- Altitude
- Access
- Availability of skills
- Availability of logistical support
- Trafficability of site
- Attitude of nearby communities
- Transportation network
- Local economy

Variability—Project and Contract Characteristics
No two projects or contracts are exactly alike. These differences definitely influence productivity potential.

- Project size; single craft size
- Schedule constraints
- Adequacy of scope definition
- Constructability of design
- Exposure to hazards
- Environmental requirements
- Height or depth of work
- Form of contract
- Budget constraints
- Quality of engineering
- Degree of congestion or confinement
- Relationship to existing facilities
- Relationship to other construction

Variability—Human Factors
The ultimate determinants of project performance are the human beings doing the managing and building. Overall performance is a function of these human factors:

- Management competence
- Supervisor competence
- Individual worker skills
- Work rules
- Personal pride
- Stability of employment
- Overtime
- Experience/point on learning curve
- Worker attitudes
- Crew stability/key personnel turnover
- Owner/contractor relationships
- Value system
- Personalities

Impact of Disturbance on Performance
Finally, these are those factors which are most completely in the hands of management to control:

- Site layout for construction
- Support equipment availability
- Project controls system
- Quality management program
- Technology/methodology used
- Subcontractor performance
- Degree of communication
- Crew balance
- Materials availability and quality
- Tool availability and quality
- Safety program
- Adequacy of support facilities
- Degree of planning
- Vendor performance
- Control of interruptions

Variability—Accounting and Estimates
Acknowledging that there are many variables that influence overall productivity on a project, contractors bidding on fixed-price or target-price work must somehow determine how these variables will interact to affect worker productivity on that project. Ideally, a contractor will maintain historical data files containing actual productivity data from past projects. For this data to be useful on future projects, several criteria apply:

- A standard chart of accounts for crew tasks must be used for all projects so that data from one project realistically can be compared to data from another.
- The breakdown of crew tasks for purposes of estimating must be the same as that used for reporting, so that estimated and actual performance can be truly compared. And,
- In addition to the numerical data collected on each project, the conditions under which work was performed (e.g., weather, congestion, materials shortages) should be described, since those conditions affect the outcome.

When preparing bids for a new project, estimators will research the historical files to find productivity data on similar work performed under similar conditions. Unfortunately, such efforts will be only partially successful, so judgment decisions must be made to adapt data on hand to the new project. Fortunately, there are some tools available to facilitate this process.

- **Range Estimating**—*Range estimating* is a generic term applied to several commercial and company-developed computer programs that use a Monte Carlo statistical modeling technique to deal with events where the outcome of each event can occur over a range represented by a frequency curve. It is particularly useful for evaluating the combined effect of multiple independent variables on measures of performance such as productivity. The point to be made is that range estimating can be used to quantify the risk associated with productivity variability on a number of different work tasks.

- **Checklists and Worksheets**—Some individuals and companies have developed structured approaches in the form of checklists or worksheets to help them in coming up with productivity estimates. As an example, Figure 15.5 is a description and sample of a Productivity Index Evaluation Worksheet developed by the author.

Promoting Productivity
To promote productivity on a project, managers must first be aware of the many factors that can affect it. These have been listed in previous paragraphs. During the pre-mobilization stage and using these lists as checklists, managers can identify those factors with potential to adversely affect productivity. From this list, they can identify those factors that cannot be controlled, those that can be partially controlled, and those that can be completely controlled. It is then a matter of prioritizing the controllable factors and developing positive programs to eliminate or minimize the effects of these factors.

As implied in the previous paragraph, a proactive approach to promoting productivity will yield the greatest return. If, during the course of a project, productivity is not what managers believe it should be, reactive action is required, but it will follow the same steps.

Additionally, since productivity is but a subset of performance, the guidance contained in the solution sub-section of the overall performance issue section, above, is fully applicable to productivity management programs.

PRODUCTIVITY ANALYSIS

Determining Percent Complete

The primary purpose of this section is to explain methods for measuring and analyzing productivity. However, use of these methods requires an understanding of the methods for measuring percent complete of work activities, so these will be described first. There are six methods:

Units Completed—This method is suitable when the total scope of an activity consists of a number of equal or nearly equal parts, and status is logically determined by counting parts completed and comparing that to the total number of parts in the total activity. Ideally, each unit is of relatively short duration. In engineering, a possible application is in the writing of a number of specifications of a given type where all specifications are considered to have essentially equal weight. In construction it is useful in activities such as earthwork, concrete work, and wire pulling.

Incremental Milestone—This method is appropriate for activities of significant duration that are composed of easily recognized, sequential sub activities. Percentage completion values are established based on the effort estimated to be required at each milestone point relative to the total for the activity. This method is ideal for control of engineering drawings and can be used in procurement. A typical example for drawing control is:

Start drafting	0 percent
Drawn, not checked	20 percent
Complete for office check	35 percent
To owner for approval	70 percent
First issue	90 percent
Final issue	100 percent

Vessel installation and assembly is a classic example in construction. For example:

Received and inspected	15 percent
Setting complete	35 percent
Alignment complete	50 percent
Internals installed	75 percent
Testing complete	90 percent
Accepted by owner	100 percent

Start/Finish Percentages—This method is applicable to activities that lack readily definable intermediate milestones, and/or the effort/time required is very difficult to estimate. For these tasks, controllers credit 20 to 50 percent when the activity is started and 100 percent when finished. The reason that a percentage is assigned for starting is to compensate for the period between start and finish when no credit is being given. In engineering, this method is appropriate for work such as planning, designing, manual writing, model building, and studies. It also can be used for specification writing. In construction, it is appropriate in any situation where scheduling is detailed with multiple, short-term tasks.

Ratio—This method is applicable to tasks such as project management, constructability studies, project controls, and comparable activity that involve a long period of time, have no particular end product, and are estimated and budgeted on a bulk allocation basis rather than on some measure of production. It also can be used on some tasks for which the start/finish method is appropriate. Percent complete at any point in time is found by dividing hours (or dollars) spent to date by the current estimate of hours (or dollars) at completion. This method is useful on any project where the status of non-production accounts (such as overhead) must be individually summarized with production accounts to determine the overall percent complete.

Supervisor Opinion—This is a subjective evaluation of percent complete and should be used only where more discrete methods cannot be used. There is a natural tendency to over-estimate the level of completion of an activity in its early stages.

Weighted or Equivalent Units—This method is applicable where the task is a major effort involving a long period of time and composed of two or more overlapping subtasks, each with a different unit of measurement (e.g., each, yd3). To set this up all subtasks are listed along with their respective units of measure and quantities. The subtasks are then weighted, using relative work-hours as weighting standards—the total of all weights equals 1.00 or 100 percent. The progress of each subtask is reported using one of the five measurement techniques described previously. When this percentage is multiplied by that subtask's weighting factor, its contribution to overall task completion is calculated. Those for all subtasks are added to give the overall percent completion of the major activity. A classic example is concrete placement, which is frequently estimated and reported in terms of cubic yards in place; it can be broken up into the subtasks of base preparation, forming, re-steel installation, concrete placement, curing, form stripping, and patching. Another example is steel erection, which is traditionally estimated and controlled in terms of tons of steel. The process is illustrated in Table 15.1.

Table 15.1 – Steel Erection as Traditionally Estimated and Controlled in Terms of Tons of Steel

Wt.	Subtask	UM	Quantity Total	Equiv. Steel TN*	Quantity to date	Earned tons**
0.02	Run foundation bolts	Each	200	10.4	200	10.4
0.02	Shim	Percent	100	10.4	100	10.4
0.05	Shakeout	Percent	100	26.0	100	26.0
0.06	Columns	Each	84	31.2	74	27.5
0.10	Beams	Each	859	52.0	0	0.0
0.11	Cross braces	Each	837	57.2	0	0.0
0.20	Girts/sag rods	Bay	38	104.0	0	0.0
0.09	Plumb and align	Percent	100	46.8	5	2.3
0.30	Connections	Each	2977	156.0	74	3.9
0.05	Punch list	Percent	100	26.0	0	0.0
1.00	Steel	Ton		520.0		80.5

***Equiv. Steel TN = (Wt.)(520 Ton)**
****Earned Tons = (Percent complete)(Equiv. Steel Tons)**

Percent complete = (earned tons) ÷ (total tons) = (80.5 tons) ÷ (520.0 tons) = 15.5 percent

Notice in this example how tons of steel is the account's unit of measure, and all subtasks are converted to equivalent tons. It also may be noted that percent complete could have been calculated by the following formula:

Percent complete = Σ [(weight) x (percent complete each subtask)]

Productivity Measurement of Individual Work Tasks—Owners and contractors are always interested in comparing actual field productivity to that estimated and budgeted. When dealing with a single work activity, the calculation of productivity is very simple:

Productivity = (number of units completed) ÷ (work-hours consumed)

What is more difficult, the calculation of productivity at a summary level or for an entire project?

Productivity Analysis at a Summary Level
While a comparison of earned to actual work hours is used by some practitioners to provide an evaluation of productivity at a summary level, that approach is valid only if actual quantities of work are exactly equal to those budgeted. This is not always true, particularly on fixed-price, lump-sum contracts, so another tool is needed to evaluate productivity. That tool is credit work hours. Credit Work Hours (CWH) are derived quantities and are found using this formula for work items completed:

CWH = (budgeted unit rate*) x (units completed to date)

**Budgeted Unit Rate = budgeted hours per unit of work*

For individual work packages in progress (not yet complete), this formula is appropriate:

CWH = (percent complete) x (budgeted unit rate)

The Productivity Index (PI) for a single work package is found by this formula:

Productivity Index = (CWH to date) ÷ (actual WH to date)

The productivity index (PI) for a combination of work packages or for a total project uses this formula:

Productivity Index = (CWH ÷ a c t u a l work hours).

The format of these equations is such that an index of less than 1.0 is unfavorable, while one equal to or greater than 1.0 is favorable.

Use of Productivity Data
It is a waste of time to collect data that is not used for the benefit of the project or the company. Recalling that project estimates include productivity assumptions for the various work tasks, a very important use of the field data is to compare estimated with actual productivity. It is unlikely that estimated and actual productivity associated with a single work task will ever be exactly equal, but

significant variations should be cause for concern—the difference may be attributable to a poor estimate and/or it may be attributable to field performance. In any event, significant variations should be investigated and the results shared with the estimators, since their databases may need updating.

Significant variations in the productivity index at the project level may or may not be of concern, depending on the phase of the project. Figure 15.4 shows example plots of a productivity index on both a period and cumulative basis. A productivity index of 1.0 is the datum line. Also shown is the expected cumulative plot. As drawn, that curve reflects the typical course of the cumulative productivity index over the life of gradually to a peak of 1.15-1.20 about the 50 percent complete point and then decreases, ideally becoming 1.0 at the 100 percent complete point. On this example, the actual cumulative productivity index has been running consistently below the expected cumulative curve, meaning that, in spite of the fact that at some points the period PI was above 1.0, the cumulative productivity index probably will be less than 1.0 when the project or activity is complete.

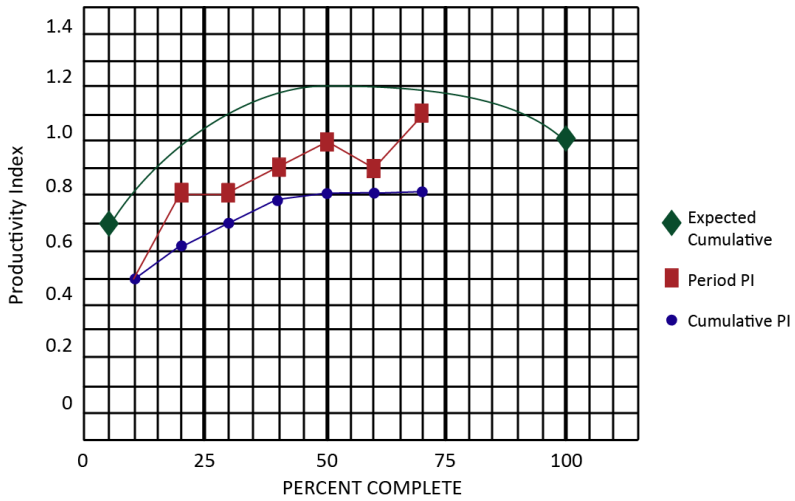


Figure 15.4—Productivity Index

INCENTIVES

Why Incentives?

Incentive programs must be included in any discussion of performance and p roductivity. Such programs have the potential to:

- Increase performance and productivity
- Reduce waste
- Reduce absenteeism
- Improve employee morale
- Promote teamwork
- Identify more cost-effective work procedures
- Improve equipment design
- Improve quality
- Share business risks with employees

In doing this the profitability of the organization is certainly improved; but, increasing profitability is not the only potential benefit. Users have found incentive programs to be excellent tools for opening lines of communication between managers and employees and for committing employees to the goals of the organization.

The Stimuli

If incentives are intended to stimulate employees to support management goals, it is important that management understands the stimuli that can be mobilized. These may be grouped into two categories from the perspective of the employee.

- Possibility of winning:
  - Excitement of winning something
  - Personal satisfaction in achieving a goal
  - Euphoria of being singled out for recognition
  - Financial gain
  - Career enhancement
  - Pride of association with a winning team

  - A chance to do something different
- Fear of losing:
  - Potential embarrassment
  - Potential loss of status, job, potential for promotion, etc.

Certainly, the best incentive programs are based on the concept of win-win; (i.e., both the employer and employee are potential winners). Those that capitalize on the employee's fear of losing are more fragile and can be counterproductive.

Rewards Within the Winning Scenario

Each incentive program in the win category has some reward associated with the achievement of some objective. In designing incentive programs and incentive awards it is important to realize that rewards have two values—intrinsic and extrinsic. The intrinsic value is essentially the exchange or cash value of the reward. The extrinsic value is that value above and beyond the cash value that accrues to the recipient because of what the award means to him or her—some might call this esteem value.

It is essential that every reward have some value—it may have either or both intrinsic and extrinsic value, but it is not necessary that it have both. A medal for heroism and the Eagle badge in Scouting have little intrinsic value, but tremendous extrinsic value to the awardees. If an employer gives a Rolex watch to every employee completing 25 years of service as an incentive to reduce turnover, that watch has considerable intrinsic value, but minimal extrinsic value because the quality of an employee's performance during those 25 years is not a factor. Achievement of professional registration has high extrinsic value, and also can have significant intrinsic value, if it means a raise in pay or chance for promotion.

Rewards, whose value is almost totally extrinsic, are certainly the most cost effective. The fact that such rewards can also be effective stimuli, puts incentive programs within reach of every employer. Specific examples of rewards in both categories will be incorporated within the following discussion of specific programs.

Example Incentive Programs

The following summaries of programs or activities that have been or are being used successfully illustrate the range of incentive program options that may be considered.

- **Open-Book Management**—This incentive program is really a revolutionary way of doing business and might be considered an advanced form of total quality management. As the name suggests, the company's books, strategies, good news and bad news are fully shared with employees, the theory being that employees will make better decisions and perform better if they know exactly how the company operates and what contributes to profits and losses. The incentive involved is a sharing of annual profits among employees, typically 25 percent.

- **The Green Stamp Program**—Under this program, employees earn credits (or green stamps) for achievement of various objectives. Typical objectives are zero defects, no accidents, no late starts/early quits, no absenteeism during a given period, achievement of a production or productivity goal, approval of a suggestion, etc. The number of credits awarded is commensurate with the achievement. Credits are allowed to accumulate in the employee's account for conversion to gift certificates at his/her convenience. Each credit is usually worth $1. This program has several advantages: the employee can pick the reward, the accumulation feature stimulates continuing achievement, it brings in the influence of an employee's family (they cheer the employee on), since awards can be significant and of the type the whole family chooses, and it is open to all employees.

- **Suggestion Program**—These programs have been around a long time. Employees make suggestions that are reviewed by selected committees for possible adoption. Adopted suggestions usually result in a cash award that is based on anticipated savings. If a suggestion is not adopted, or the benefits are other than cash savings, the reward is usually a letter of appreciation, but may include some token merchandise item. Suggestion programs have enjoyed mixed success. A high rate of suggestion rejection or excessively complex and time-consuming submission and processing procedures can quickly dim employee enthusiasm.

- **Sharing Savings**—On fixed-price or target-price contracts, an incentive program can be established, whereby field personnel will share in any savings realized. These are usually distributed based on salaries or wages paid during the life of the contract. An interesting form of this has been used by an open shop contractor to keep fixed-price projects within budget. First, all budgeted direct costs within the control of construction crews were allocated to their individual work packages. Then, as crews completed the work, actual costs were accumulated. These costs included labor costs, materials costs, equipment and tool costs, costs of accidents, and any other costs attributable to the crew's assigned work. At well-defined milestone points in the project, a tally was made of budgeted and actual costs in the covered period. Any savings were distributed totally to the workers, the distribution being proportional to worker hours or earnings during the period involved. There were no penalties for overruns—these are assumed to be a result of bad estimates and budgets. This program promotes teamwork and crew balancing, safety, conservation of materials and improved productivity, all while preserving company profit.

- **Target Bonuses**—Often an owner will establish a target completion date or a target cost for a project, knowing those targets can be met only with exceptional effort. To stimulate this effort, they will set aside a sum of money to be divided among the field personnel if the target is met.

- **Honoraria**—These are given to individuals for specific achievements relating to professional development, such as professional registration/certification, writing and publishing a professional paper, or representing the company in a professional forum.

- **Service Awards**—These are usually a combination of a certificate and a merchandise prize to recognize years of service with a company. The value of the merchandise increases with length of service. Often, a special luncheon or dinner is held to distribute these awards.

- **Merit Raises**—Either a portion or all salary increases in a given year are tied to performance evaluations. It is very difficult to create an impartial system for these since evaluations are very dependent upon subjective judgments of individual managers and these can be influenced by politics and prejudices. A selection board approach can minimize this problem.

- **Cross-Training**—An employer who provides cross-training for workers provides a measure of job security for those workers and this is motivating for the employees.

- **Special Training**—If a limited number of individuals are selected each year for some special training, competition for selection becomes a strong motivator for excellent performance.

Many successful incentive programs capitalize on the extrinsic value of the rewards involved; and, in doing so, achieve results at low cost. Examples:

- **The Simple "Atta Boy!"**—A simple pat on the back or word of appreciation, particularly when given in front of everyone in a work unit, can do wonders to motivate many people.

- **Management by Walking Around**—It is good management practice to maintain visibility with employees through frequent visits to work areas during which they chat with employees. By showing sincere interest in the individuals and their work, a manager effectively motivates employees.

- **Letter or Certificates of Appreciation and Achievement**—A document that commends an individual for an accomplishment has high, long-term value since it is written proof of special capability and may be the document needed in some future job search.

- **Certificates of Completion**—These recognize completion of some training program. They have significant value only if the participants in the program had to pass some meaningful test to graduate.

- **Decals**—These are usually used in conjunction with other awards. For example, someone completing a first aid or CPR course would receive both a certificate of completion and a decal to put on their hard hat.

- **Token Awards**—Awards in this category include inexpensive items such as t-shirts, coffee cups, baseball caps, calculators, and pen knives. These are appropriate for individual or crew minor achievements such as short-term safety, quality, or attendance records. Slightly higher-cost items, such as a wind breaker jacket, engraved desk sets, and clocks are suitable for more significant achievements, such as long-term safety or quality records.

- **"Exclusive Clubs" on the Job**—Individuals take pride in being part of a group whose membership criteria is exclusive. A group of earth movers had a "Million Yard Club" on their project. Production, safety, and quality goals can be set to qualify for membership in comparable clubs. Achievement of membership in the club is recognized through certificates, decals on the hard hat, t-shirts, bumper stickers, etc.

- **Employee or Crew of Month**—This program is very common in the service industry. A committee selects the recipients based on recommendations from managers, customer comment forms, or other criteria. The reward is usually a picture of the individual or crew displayed in a prominent location, plus a certificate. It can include a cash award or special luncheon/dinner. This program must be carefully managed so it does not degenerate into a popularity contest or "whose turn is it this month?" form of selection.

- **Problem-Solving Teams**—These are similar to quality circles except they are ad hoc and are given a specific problem to solve by management. Their work can result in cash or credit awards; however, a letter of appreciation or commendation

may be adequate. These teams are motivators since they are another form of participative management.

- **Team Builders**—There are a number of relatively inexpensive actions that can be taken to stimulate group morale and team spirit (and thus productivity and quality) on a project or in other workplaces.

    - Creating a project logo and using this logo on signage, hard hat decals, bumper stickers, stationery, etc. It is recommended that a project-wide contest be held to design the logo.
    - Publish a newsletter.
    - Have a contest to name the newsletter.
    - Use the newsletter or bulletin boards for publication of "Hats Off" type notices to recognize accomplishments of individuals.
    - Occasionally put out coffee for workers as they check in for work or cool beverages as they leave work on a hot day.
    - Have the project photographer take pictures of individuals and crews on the job. Display these pictures on a bulletin board near the check-in area. Perhaps make copies available to pictured individuals.
    - An alternate to the above is to provide video coverage of the project with the product being a weekly tape of about 15 to 30 minutes in length. On this tape, review project status, show crews at work, etc. Show the tape during lunch in protected break areas.
    - Use a special message board in a prominent location on which the project status is displayed, special accomplishments are announced and human interest stories told about project participants.
    - Sponsor charity work by the workers—food and toy drives, painting or repairing homes for the needy, building playgrounds, etc.
    - When a major project milestone is reached, allow an extra hour for lunch and have a catered lunch for the workers. Use this opportunity to give out safety and other awards.
    - Sponsor "family day" at the project, plant, or nearby park with a picnic lunch, tours of the project/plant, and games.
    - Put first names of workers on their hard hats.
    - Sponsor bowling, softball, and other teams in local leagues.
    - Issue press releases on project and employee accomplishments.
    - Recognize birthdays or other events with a congratulatory letter.
    - Do whatever you can to provide job security for employees—cross-training, information on upcoming jobs, out placement service, etc.
    - If the project receives some cash award for safety or other achievement, divide the award up into $50 packages and give them away in a raffle. All workers who contributed to the achievement are included in the drawing.
    - Anything to make the site "a good place to work"—a strong safety program, decent worker facilities, good layout, dust control, etc.

Incentive Program Guidelines

In analyzing the many individual and team incentive programs that have enjoyed success, a number of guidelines evolve.

- Learn from the experiences of others.
- Program must balance both employer and employee goals.
- Get workforce into the planning of program if possible—if union personnel are involved, the union must be involved.
- Keep each program element as simple as possible.
- Criteria for awards must be specific and understandable.
- Performance criteria must be achievable.
- Successful achievement of goals must be within control of target individual or group.
- Programs based on subjective rather than objective criteria are more difficult to manage impartially.
- The program will be most effective if the awards resulting from an accomplishment directly accrue to the individual or team making the accomplishment.
- Mobilize as many of the stimuli as possible in establishing the reward structure.
- Avoid any potential for discrimination in determining award recipients.
- Make certain your program is well publicized.
- Publicize achievements by individuals and teams.
- Ensure that the program is continuously well managed.
- Incorporate potential for many winners.
- Provide opportunities for the entire workforce.
- Don't turn off non-winners—"maybe next time."
- What works in one environment won't necessarily work in another. An example can be cited where a contractor used preferential parking as a reward and the program was very successful. Another contractor tried it and the rewarded workers found their tires slashed.
- Be aware of the tax implications of awards. Merchandise awards of nominal value (example: turkeys, coffee cups, etc.) are not taxable. Cash awards or awards equivalent to cash (example: gift certificates) or costly merchandise awards

(example: TV, pickup truck) are taxable.

- Proceed with caution when launching an incentive program. Start small and work up to more ambitious programs that build on the success of early programs. A failed incentive program can have totally negative effects.

Incentive programs have established a place for themselves in the business world. A variety of programs already have been successful. Companies can learn from these programs and design adaptations of them to fit their particular environments.

The question is inevitably asked, "What is the benefit: cost ratio for incentive programs?" Unfortunately, the author is not aware of any research on the subject and such data would be difficult to compile. However, several individuals with experience with incentive programs have expressed the opinion that the payoff is in the range of 4:1 to 10:1. Whatever the benefit to cost ratio, the results so far have shown that well-managed incentive programs can positively influence team- work, safety, quality, and overall performance.

PRODUCTIVITY INDEX EVALUATION WORKSHEET

Purpose

The Productivity Index Evaluation Worksheet depicted in Figure 15.5, is intended to facilitate a comparison of the productivity potential of a proposed project with respect to a completed project. For this purpose, a productivity index of 1.0 is average, a productivity index less than 1.0 is less than average (unfavorable), and a productivity index greater than 1.0 is better than average (favorable).

PRODUCTIVITY INDEX EVALUATION WORKSHEET						
Productivity Element	**Weight**	**75-99 (Low)**	**100 (Average)**	**101-125 (High)**	**Score**	**Product**
1. General Area Economy	**10**	**Prosperous**	**Normal**	**Depressed**		
construction volume in area	4	high	average	low		
unemployment situation	4	low	average	high		
local business situation	2	stimulated	Normal	dead		
2. Project Character	**25**	**Complex**	**Average**	**Favorable**		
schedule	6	compressed	normal	ample slack		
complexity of work	6	complex	average	simple		
contract form	5	reimbursable	fixed-price	incentive		
project type	5	revamp	new work	repeat work		
size	3	mega	average	small		
3. Craft Workers and Foremen	**25**	**Poor**	**Average**	**Good**		
quality and availability	8	poor	average	excellent		
distance to project	5	more than 60 min	30-60 min	less than 30 min		
substance abuse program	5	none	policy only	full program		
use of overtime and multiple shifts	4	much	some	exception		
rate of force build-up	3	fast	comfortable	(not used)		
4. Project Operating Conditions	**20**	**Poor**	**Average**	**Good**		
congestion and hazards	6	considerable	average	little		
management quality	6	inexperienced	average	highly qualified		
materials and tools availability	3	shortages	average	adequate		
required workmanship	3	exceptional	normal	(not used)		
site access	2	restricted	normal	open		
5. Weather	**10**	**Poor**	**Average**	**Good**		
amount of protected work	2	limited	normal	significant		
precipitation days	2	frequent	normal	occasional		
cold and wind days	2	often	average	rare		
days of extreme heat	2	many	average	rare		
days of extreme humidity	2	many	average	rare		
6. Construction Equipment	**5**	**Poor**	**Average**	**Good**		
condition	3	poor	average	excellent		
maintenance/repair availability	2	remote	nearby	onsite		
7. Delays and Interruptions	**35**	**Numerous**	**Some**	**Minimum**		
rate of changes expected	10	high	normal	low		
materials deliveries	6	uncertain	normal	timely		
operating plant/other interferences	6	frequent	a few	none possible		
site work permits	6	frequent	occasional	not applicable		
labor unrest potential	5	could happen	none expected	(not used)		
public protest potential	2	could happen	none expected	(not used)		
TOTALS	**130**					
PRODUCTIVITY INDEX = (Σ PRODUCT) ÷ (13,000) =						

Figure 15.5— Productivity Index Evaluation Worksheet

Evaluating productivity variation among projects is not an exact science. This worksheet serves only to force planners to seriously consider many conditions that can affect productivity and to evaluate their individual effects, as well as their cumulative effect. The productivity elements and the weighting factors used are not fixed—users should adjust them to reflect experience over time.

Use of Worksheet

- For a reference (completed) project, complete an evaluation of each of the seven categories of productivity elements. This is best done by several individuals familiar with the project so that the results represents group consensus. Note that each category is made up of two or more subcategories so that evaluations can be made at the subcategory level to yield the category score. For example, note that the first category, general area economy, has three sub-elements. Assume that the group makes the following analysis of a completed project:

    - Construction volume in the area at the time of the project was somewhat low compared to previous years when several major plants were built. Now, most construction activity involves homes and small commercial projects. This subcategory is given an index of 110.

- The unemployment rate in the area was about average for the state, but better than the national average. There were jobs available, but most were of the minimum wage category. This subcategory is given an index of 100.
- The local business situation was basically healthy, neither robust nor depressed. This subcategory is given an index of 100.
- The resultant score for Category 1, general area economy, is calculated from the above three index totals.

110 x 4 = 440
100 x 4 = 400
100 x 2 = 200
1,040 ÷ 10 = 104

- Continue the evaluation of the remaining categories to develop the score for the completed project.
- Make a similar evaluation for the proposed project. Then compare the scores to determine a multiplying factor to use in estimating productivities on the new project using productivities for similar work on the completed project as a reference.
- Productivity multiplying factor for Proposed Project = (PI proposed project) ÷ (PI completed project)
- The above does not consider regional differences in general workforce productivity resulting from sociological and other differences among worker populations. If planners believe such differences exist, they must further modify the multiplier obtained in paragraph three by multiplying it by a factor found by dividing the area productivity index of the proposed project by the area productivity index of the reference project.
- Use the resultant productivity multiplier in conjunction with relative wage rates to determine relative labor costs for the same volume of work.
- This worksheet also can be used to normalize data from past projects for entry into the historical database. Since the raw data from each project is distorted because of numerous project-unique conditions, normalizing it has the effect of bringing the data down to a baseline not affected by those conditions.

CONCLUSION

The ultimate performance index for an organization is the one that relates its net profit or value of services to the costs of achieving that profit or providing those services. An organization seeking to maximize that index must examine the operations of its total workforce, not just those of its production units. It must target waste in all forms—not only materials or equipment waste, but the waste associated with inefficiencies, interruptions, rework, and an assortment of other time-wasters, all of which effectively constrain their employees' ability to produce, perform, and achieve. And, most of all, that organization must provide a workplace with the facilities, procedures, atmosphere, and attitudes that stimulate performance.

SECTION 5 – PROJECT MANAGEMENT

Chapter 16 – Project Management Fundamentals
Chapter 17 – Project Organization Structure
Chapter 18 – Project Communications
Chapter 19 – Project Labor Cost Control
Chapter 20 – Leadership and Management of Project People
Chapter 21 – Quality Management
Chapter 22 – Value Engineering
Chapter 23 – Contracting for Capital Projects
Chapter 24 – Strategic Asset Management
Chapter 25 – Change Management Practical Guide
Chapter 26 – Overview of Construction Claims and Disputes

Chapter 16—Project Management Fundamentals[10]

James A. Bent, CCP

Abstract
The following chapter is an updated status of the Project Management Fundamentals required for Cost Management. The incorporation and updating of project management fundamentals is based on the updating of technology. Project management has been a continually evolving skill of theory that assists in identifying risks, maximizing cost savings, minimizing time delays, and improving economic return. The objective of the chapter is to introduce a proactive system of fundamentals that uses the time methodology and culture to develop a balanced program for execution, analysis, and management.

Keywords
Building Information Modeling (BIM), Cost Management (CM), Critical Path Methodology (CPM), Project Manager (PM), Project Performance Measurement (PPM), Time Management (TM), and Value Engineering (VE).

INTRODUCTION

Over the past 40 years there have been major changes in project execution as a number of techniques have become increasingly used across a broad spectrum of projects. The most significant developments can be narrowed to less than fifteen. These developments are:

- Building Information Modeling (BIM)
- Design Quality Assurance—Value Engineering
- Project Performance Measurement
- Functional Matrix Organization
- Critical Path Method (CPM) Scheduling
- Fast Track Scheduling Program
- Fast Track Trapezoidal Technique
- Independent Construction Management
- Integrated Project Delivery (IPD)
- Partnering; Contracting Arrangement
- Strong Matrix
- Total Quality Management (TQM)
- Participative and "Open Book" Management

Learning Objectives
After completing this chapter, readers should be able to:

- Understand the role of the project manager in project planning.
- Understand the major factors effecting successful project execution.
- Understand the overall company projects life cycle.
- Understand the major phases of projects.

THE CHANGING ROLE OF MANAGING PROJECTS

Project Execution, Old versus New

Figure 16.1 highlights the major changes in project management from the mid-1950s to the present. The traditional program of sequential completion with the individual phases of engineering, procurement, and construction, where construction management rarely got involved until four weeks prior to starting the project, has been replaced by the very challenging, but very efficient, fast track program.

Fast tracking, Critical Path Method (CPM) scheduling, and the trapezoidal technique program are the greatest advances in methodology that have occurred over the past 50 years or so, and followed the explosion of work that took place after World War II. Today's computer systems greatly assist in the collection and collation of data for these three techniques. This data, in turn, is developed into specific cost and schedule project baselines and other essential information.

The development of the Fast Track—Trapezoidal Technique (F.T.—T.T.) for plant projects, which was first published in 1982, was of particular importance. This technique is used in conjunction with construction complexity and labor density. Construction complexity

[10] The original chapter was written by James A. Bent and published in S&K 5 as Chapter 18, "Project Management Fundamentals." Madhu P. Pillai, Sean T. Regan and Neil D. Opfer made revisions and minor updates to the manuscript for S&K 6.

and labor density are essential in developing, or verifying, the quality of a conceptual estimate and planning schedule for process plant projects. When the appropriate factors are properly developed, the resulting cost and schedule numbers can have a probability of 90 to 95 percent of being on target for the project.

A Proactive Role

The best of today's construction management take a proactive role through the newly developed programs of construction preplanning and constructability; which results in a strong construction involvement at the early stage of the project to ensure that engineering design and early planning fully recognize the requirements of an economic construction program. An example of construction preplanning is backwards scheduling, where the overall project schedule is structured around the construction schedule, with design drawing issues and material deliveries being matched to construction needs. If this is done early in the design stage, there will be no cost impact on design engineering or purchasing, and the construction cost savings can be considerable, even with the added cost of early construction involvement.

Business management is now considered of greater value than the old standard of aggressively pushing the work, with cost and contractual considerations of lesser consequence.

Note: The timing of new major programs and elements, as shown in Figure 16.1, represents the approximate date when the individual categories were widely used (proven), not when they were first developed.

Project Performance Measurement—Company

The development of a program to measure project performance and personnel skills is essential for any quality program. By the same token, a full benchmarking evaluation is also essential. The real value or added value of a benchmarking program is to provide a state-of-the-art skills base, which is then used to constantly measure the total project performance of the group over time. Benchmarking targets can come from other project performance criteria within the organization, as well as from other organizations where comparison data are relevant. With a good technical program, the correct balance/mix of skilled personnel, and effective training, there should be a steadily improving project performance that directly results in lower cost/higher quality of a company's capital projects. The measurement program needs to be simple, but effective, using existing information and being directed by the project's quality assurance group. The overall performance goal needs to reach a rating of approximately 80 percent.

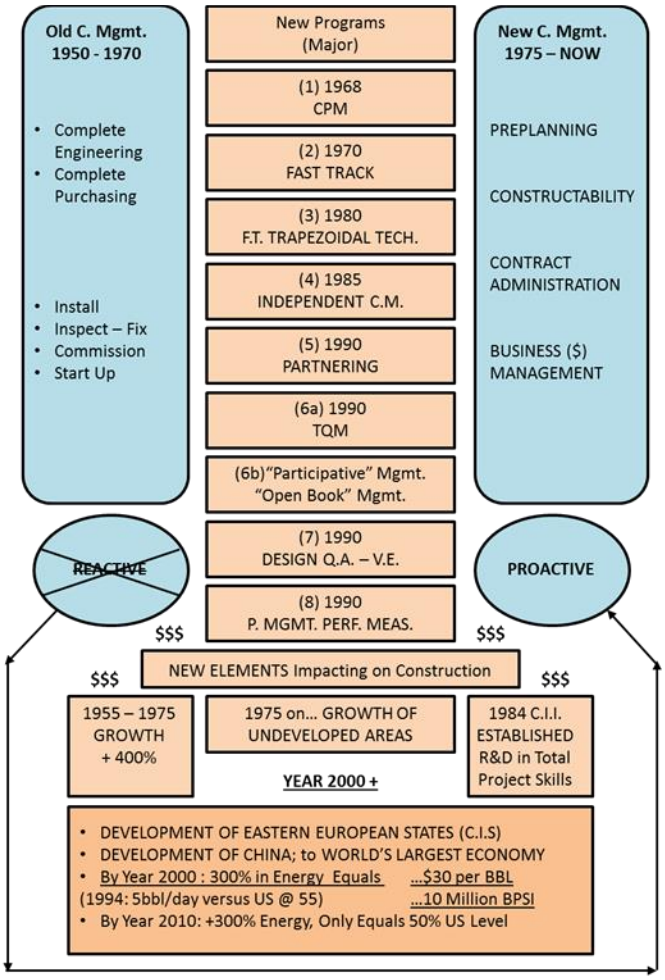


Figure 16.1—The Changing Role of Project Management

Key Essentials
With the difficult and challenging business environment today, it is vital that the management of projects focuses on:

- Identifying risks
- Maximizing cost savings
- Minimizing time delays
- Improving economic return

These results can only be achieved through:

- Effective management of people
- Tough but fair project objectives
- Efficient business techniques
- Outstanding leadership skills

The following project management chapters cover these subjects in considerable detail. The roles, functions, and interfaces of company management, project management, engineering management, construction management, and support service groups are explored, in sufficient detail. Overall relationships and personnel relationships, essential for successful project execution, are outlined. Throughout, the emphasis is on practical approaches and the relationships of personnel in the project team.

The emphasis is on the need for and substance of early project planning and the related scheduling of time and development of costs. It is essential that all project leaders be effective professional project managers—they need to be organizers, planners, motivators, communicators, and business persons. Today, project management has moved very close to business management.

Current studies by the Construction Industry Institute (CII) at the University of Texas, into state-of-the-art project management methodologies, have shown that the top category for successful project execution is front-end planning and project organization. The CII studies have concluded with the following simple but true premise: "If we get it right at the front end, we have a chance of success, though not guaranteed. If we do not get it right at the front end, then we have little chance of success."

The following material, mostly flowcharts, illustrates the major factors, functions, and project phases of the project life cycle. Full understanding of all these elements can ensure effective communication channels, tight schedules, low cost, and a clear path for efficient decision making. But it is in these very elements where many problems develop and become impossible to solve or reduce as the project progresses. The result is more money is spent than necessary, more time is consumed than needed, and there is a constant recycling of options and alternatives that should have been eliminated at an earlier stage. Project ignorance, lack of skill, poor cooperation, and refusal or reluctance to participate properly is commonplace. Biases, prejudices, and personal interests are also part of the equation. All of these elements are compounded by a lack of up-to-date project management training throughout the industry.

DEFINITION OF A PROJECT

A project can be defined loosely as an item of work that requires planning, organizing, dedication of resources and expenditure of funds in order to produce a concept, a product, or a plant. It is a temporary endeavor with a definite starting point and ending point undertaken to create a unique product or service. This chapter focuses on plant projects, all of which require design engineering, the purchase of material, and the installation of that material in adherence to the previously completed design plans and specifications.

PROJECT MANAGEMENT FUNCTION

Almost all companies have personnel who are trained, skilled, and dedicated to the execution of the firms' projects. The individuals who lead these efforts are called project engineers and/or project managers. Supporting these project managers are such personnel as design engineers, procurement personnel, contracts officers, estimators, cost engineers, planners, construction managers, and a variety of technical specialists. In many cases, the type, size, and complexity of projects vary greatly, and therefore, the skills and experience of project engineers, project managers, and support personnel can, similarly, vary in capability.

One of the keys to the project management function is the recognition of the type of organization that the project manager may have to work with. Two generally accepted organizations are the Strong Matrix, where the project manager is the primary source of control with a direct team reporting to the organization and the Functional Matrix in which the project manager works with the functional disciplines such as engineering, procurement, construction, and business management with the resources more aligned to the function then internal to the team. Figure 16.2 shows the major factors essential to the successful execution of projects.

Cost Management

Many projects have project cost as the top objective, and this requires the project to be completed at, or less than, the budgeted cost. Significant business skills are essential to meet this objective. There is an industry-wide company policy that the approved project budget can be exceeded by ten percent, without there being a supplemental funds request.

Time Management

To meet the cost objective, it is necessary to manage time efficiently. This means the pre-determined schedule, upon which the cost was based, must be met and met economically. Some projects may have schedule as the top objective. In such cases, acceleration programs are planned and it is probable that there will be corresponding cost increases to the economic-based project.

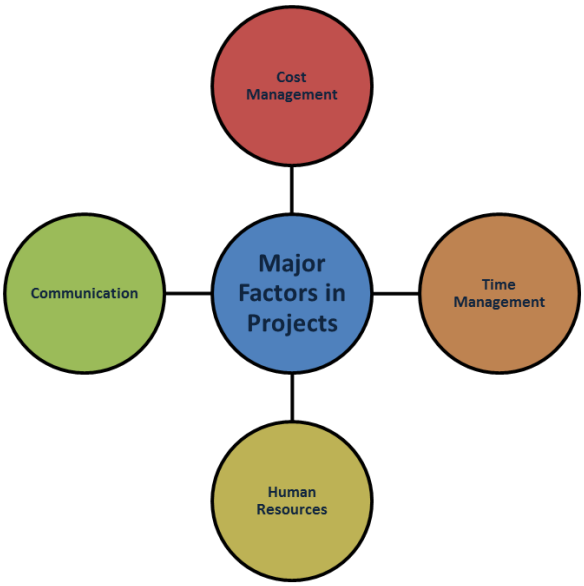


Figure 16.2—Major Factors That Are Essential for the Successful Execution of Projects

Human Resources

Of all the resources required for plant projects, human resources are the most difficult to manage. Interpersonal skills and the effective motivation of people, at all levels, are essential for successful project execution. Lack of human resources, plus a corresponding lack of the correct mix of people skills, is becoming an increasing issue with projects. One of the most abused human resource concepts is the "lean and mean" program. Some managers believe that a reduced group of people, through advanced skills, can execute as effectively as a larger group; thereby, reducing costs by using a lower number of people. There is some merit in this concept; but in many cases, it is a "device" used by poor management to cut costs. If there is a significant lack of people, there is almost certain to be a corresponding inefficiency in project execution, coupled with an increase in costs.

Communication

A formal and informal structure of effective communication is absolutely essential for successful project execution. In addition to weak people skills, many company organizations and cultures have poor administrative practices that also form barriers to project success. These barriers are common to all companies and are generally referred to as Matrix Interface Conflicts (MICs). The conflicts or barriers are caused by departmental jealousies, rivalries, and failures by management to create a culture where project consciousness and esprit de corps are common to all personnel. The total quality management programs sweeping the industry are an attempt to solve these problems.

OVERALL COMPANY PROJECTS LIFE CYCLE

Figure 16.3 shows the general steps common to all plant projects. Experience in this process, recognition of each company program's individualities, and the skills of bridging the matrix interface conflicts, are necessary for project success. Getting the front-end planning right is the key to success.

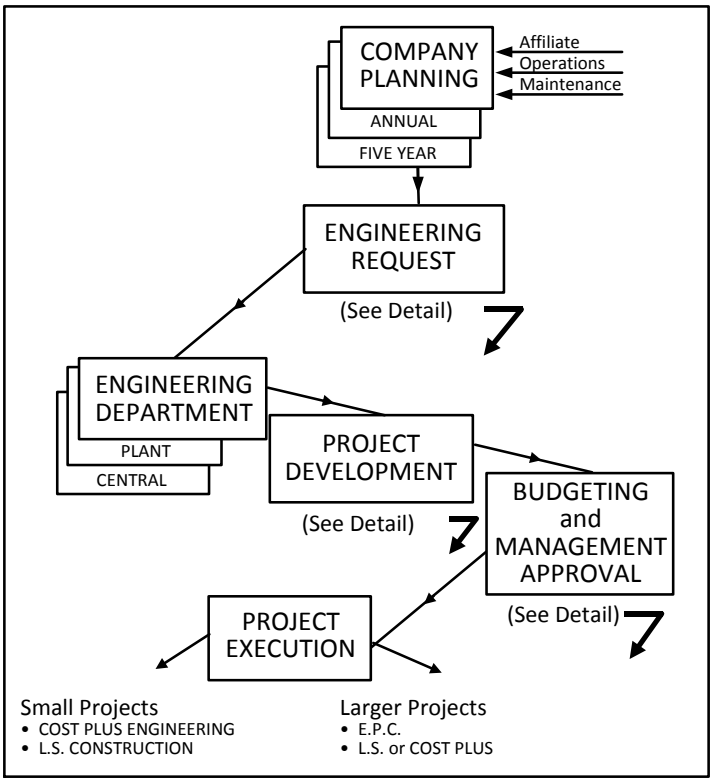


Figure 16.3—Overall Company "Projects" Life Cycle Functions Flowchart

Engineering Request

Figure 16.4 illustrates the major factors that generate the capital project work. Time and quality assessments of plant requirements are difficult to achieve but are essential for company profitability. Such assessments result in formal engineering requests for the project work.

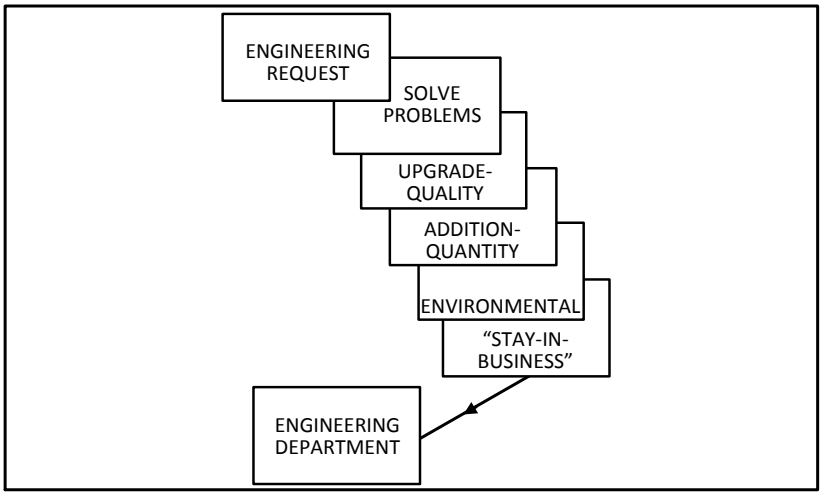


Figure 16.4—Engineering Request—Functions Flowchart

Project Development

Figure 16.5 shows the major components for developing the scope of each project. Each of these components is then further defined and prioritized. It is vital that the priority be clearly established.

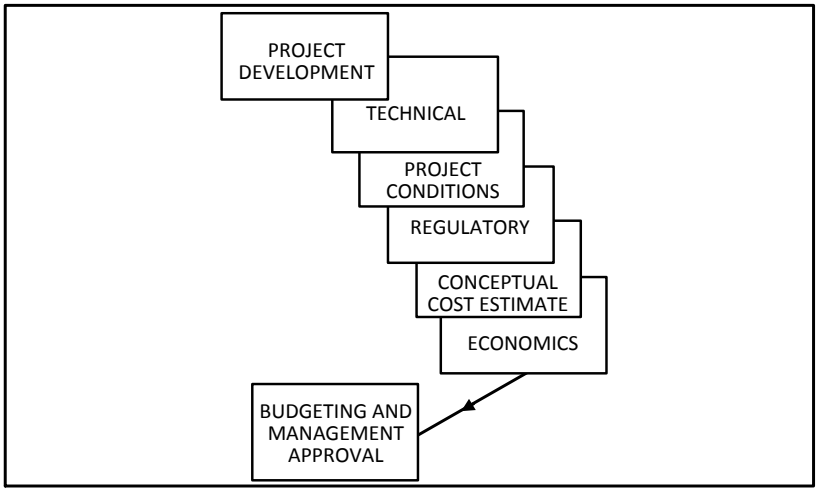


Figure 16.5—Project Development; Functions Flowchart

Budgeting and Management

Development of the scope in terms of risk, cost, time, and resources is followed by approval, partial approval, or rejection of each proposed project. Figure 16.6 shows the budgeting and management process, closely followed by the development of the project strategy and project organization. The correct assessment of the human resources, especially the key people, is essential at this early stage.

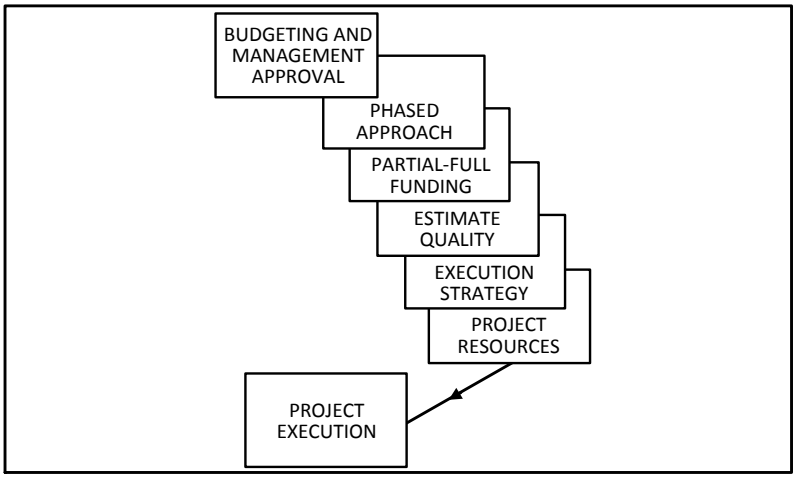


Figure 16.6—Budgeting and Management; Functions Flowchart

Typical Project Phases and Lifecycle

The time and interface relationship of major project phases is shown in Figure 16.7. Assuming a fast track program, most of these phases will overlap, and the degree of overlapping will depend on the work content of each phase and the efficiency of decision-making present in the project.

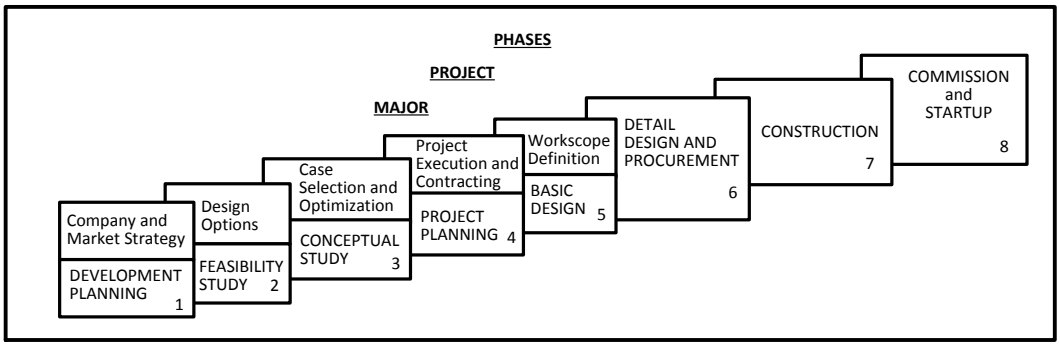


Figure 16.7—Major Phases Flowchart

PROJECT TEAM CULTURE

Finally, there is the question of company personnel working as a team. Without question, this matter has become the vital issue to profitability; especially as companies downsize and reduce the core. Greater personnel efficiency and increased operational quality are essential requirements in today's difficult business environment. The “Bean Counter” Syndrome is a wide spread practice, where effective cost control is absent or greatly diminished. The bean counter syndrome is a dangerous and unacceptable practice. Cost control efforts on a project should be proactive. The term “bean counter” is utilized for those that take only a reactive stance to cost control and believe that cost control is merely historical recordkeeping of costs. Instead cost control on proactive terms seeks to forward plan for lower costs on a systems basis. This “bean counter” practice has two major contributing factors. First, the project manager does not want an aggressive, creative, analytical function for the cost engineer and, therefore, relegates the work to a retroactive, and recordkeeping function. Second, the cost engineer can be directly responsible for this practice; as the individual may not possess the essential analytical skills, or may not believe in an aggressive trending approach and/or does not possess the essential people and communication skills.

These individuals are in fact, content with just a cost recordkeeping role. There is a much wider acceptance today of the need for dynamic, proactive cost engineering-trending and the function has become a pivotal project role, as effective cost trending is essential for project success.

“Projects are designed and built by people, not companies. People do it singly, or in multiple groups; and if there are skilled people and good relationships, there is a chance of success. If the people and relationships are poor, there is little chance of success.”

CONCLUSION

In the end, the objective of sound project management fundamentals is based on a system, concepts and utilization of personnel to maximize the project management process. It is the utilization of an organization via a functional or matrix concept to develop a program of comprehensive skills and systems to maximize the potential of the project. As noted in the chapter, a key for the program is that you can have systems, procedures and processes, but the project is built by a team; and as such, the characteristics of the program must be in line with both tools and personnel.

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Chapter 17—Project Organization Structure[11]

James A. Bent, CCP

Abstract
This chapter is based on the international business market that project organizations are required to work with. A basic review of matrix organization and functional organization in relationship to the changes that have been seen through new management theories, such as Demingisim, Total Quality Management, and other similar approaches are presented in this chapter.

Keywords
Conflict Management, Continuous Improvement Program (CIP), Equal Partner Relationship, Matrix Organization, and Total Quality Management (TQM).

INTRODUCTION

In today's difficult global business environment, it is vital that the management of projects identify risks, maximize cost savings, minimize schedule delays, and improve economic return. This can only be achieved with a quality program of project planning and project organization.

The most comprehensive study done to date of the construction industry was carried out by the Construction Industry Institute (CII), and they concluded that the number one category of project management methodology is project planning and project organization (note that the Construction Industry Institute refers to it as strategic project organizing). Very simply: If one gets it right at the front end—one has a better chance of success, though not guaranteed. However, if one does not get it right, then one has no chance of success.

Learning Objectives

After completing this chapter, the reader should be able to:

- Describe a project organization structure.
- Understand how to establish objectives.
- Define matrix structures.
- Use communication and information effectively.
- Identify a variety of contracting arrangements.
- Identify pre-contract activities for contractor evaluation.

Careful attention to the following details of these constituents will provide a good start to any project.

BACKGROUND TO PAST ORGANIZATIONAL STRUCTURE

The Matrix Structure

Over the past 30 years, the most widely used organization structure has been, and still is, the matrix organization. Most projects are executed with the "matrix," where multiple projects are executed by many departments carrying out the work at the same time, resulting in the project manager having inadequate decision-making authority. Both academics and professional project managers agree that the "matrix organization" is the most complex form of organization structure.

Matrix structures were developed to more efficiently use common resources and work on many projects at the same time, with the same staff. This structure provided effective information exchange and allowed for efficient management coordination of the total project workload. Matrixes achieve this by having the working personnel be simultaneously accountable to both the project manager and the departmental manager. In the "matrix," both project managers and departmental managers have authority and responsibility over the work, albeit, there is an agreed division of responsibility. The departmental manager is generally responsible for the technical content and working resources, and the project manager decides on the cost and time baselines.

Unfortunately, the person who comes off worst in the "matrix" is the individual who is actually doing the work. He/she reports to two bosses: the project and functional departmental managers. This leads to divisions of responsibility, problems of loyalty, and differences over priorities, poor communications, and lack of single and direct "line authority." The management of personnel and departmental interfaces is a demanding task, and in the "matrix," is often referred to as "conflict management."

[11] The original chapter was written by James A. Bent and published in S&K 5 as Chapter 19, "Project Organization Structure." Madhu P. Pillai and Sean T. Regan made revisions and minor updates to the manuscript for S&K 6.

The fundamental of "matrix theory" in a project environment, requires the project execution plan to be clearly defined, so that all working groups would then accept, commit to, and work to the agreed execution plan. There would be unanimous support from all and all would be working to the same "plan."

With a strong project management culture and effective company leadership, it was anticipated that the "matrix" would be effective. Initially, it was, but as time passed, the "matrix" failed and the research carried out by Peters and Waterman (in their book, *In Search of Excellence* [1]), and many other management experts, has clearly demonstrated this failure. The answer, agreed by all, was a new approach, called quality management. This new approach was spearheaded by Dr. Edward Deming, working in Japan in the 1960s and 1970s.

Demingism and Total Quality Management (TQM)
This made its debut in the U.S. in 1981, at the Ford Motor Corporation. Yet, the transformation of the U.S. industry to Demingism has been slow, even though there is wide acceptance that his quality management approach is essential. TQM, a version of Demingism, has been implemented in the manufacturing industries since the late 1980s, and in the engineering/construction industry in the early 1990s. Dr. Deming has developed 14-key sets of criteria for developing a quality management program. This criterion is referred to as "Demingism" and is summarized as follows.

- **Client Satisfaction**—For the services provided or for the product sold.
- **Understanding and Reducing Variation**—Every management process, practice, procedure, policy must be evaluated for its effectiveness in allowing the company's individuals to work at maximum effectiveness.
- **"Top-Down" Management Leadership and Commitment**—Improvement cannot come merely from middle managers and workers "trying harder." There must be full understanding of, and total commitment to, the necessary systematic change and the planned improvements.
- **Change and Improvement Must Be Continuous**—It must be all-encompassing, involving every "process," individual and outside services and suppliers.
- **Ongoing Training and Education is Essential for all employees** and it must be of a high technical quality so that high standards of skills and practices can be implemented by all personnel.
- **A Culture of Personnel Pride and Job Satisfaction**—is required at all levels. This requires leadership, program champions, the development of trust and loyalty, personnel empowerment, and the elimination of inadequate performance measurement schemes that can create more losers than winners, resulting in lowering of morale. Such schemes, says Deming, do not account for the "variations" and weaknesses in the company process and can be inaccurate and unfair, and are perceived as such by the employees.

Performance measurement is essential, but it should be of the system or process, and individuals should be paid for experience and responsibilities. Not all personnel will be "star" performers, and the contribution of the company janitor can be equal to that of the chief engineer, when there is a commitment to excellence from both. However, most companies do not agree with Deming on this particular issue and use a "personnel performance" awards program that gives individuals recognition and awards for superior performance. Such a program is Fluor Corporation's "MVP" (most valuable player) program, which was developed through an extensive personnel survey where the staff stated that the company salary program did not properly reward superior performance and that individual awards would be a viable program. Thus, the practice of employee empowerment showed a major "need," and Fluor instituted the program in 1994. They currently report that the program is successful and is a key feature of their Continuous Improvement Program (CIP). Fluor Corporation is an international contractor, with headquarters located in Irving, TX.

General
It is an obvious, but not well-understood fact, that it is people in single, medium, or large size groups who design and build projects, not companies. If there are people available, if they have the required skills, if they have a positive working environment, then success is possible. If none of these conditions exist, or only partially exist, then success is very questionable. Therefore, there must be a consistent and long term interest in people needs, their development and their training. When there is little interest, or the interest is not genuine, the long term success of the company is unlikely. The entire Total Quality Management (TQM) program is built around the needs and development of people, and there is unanimous acceptance by industry that total quality management is the key to success. In essence, develop the people, and in turn, the people will develop the profits.

Is the Owner Company Qualified to be its Own Project Manager?
A very fundamental consideration in today's world of company reengineering is the question of an owner functioning as his or her own project manager. Too often, owners arrive at an affirmative answer through poor analysis. It is a matter of previous experience of the specific project (particularly size), having adequate in-house or consulting resources (skills and numbers), and a good project management program. This is, currently, a major consideration with many operating companies as they downsize their operations. Many companies confuse the issue as a result of technical/engineering considerations. Having competent engineering personnel, they take on the project management responsibility, but without adequate project experience or project resources. It needs to be emphasized that engineering design competence does not necessarily translate into project capability.

Note: This question was a major issue during the early development of the North Sea Oil and Gas industry (1972-1977). The answer at that time, was an emphatic "no," from all the large oil companies (as all had limited resources), except for the rare case where contractors declined to bid because of a lack of capacity. The problem of limited contractor resources and inadequate technical and engineering expertise, was resolved with a Project Services Contract (PSC), and ultimately, partnering and integrated project teams.

Does the Organization Structure Properly fit Contracting Arrangements?
Different skills and different numbers of personnel are directly related to contracting arrangements; i.e., lump sum, reimbursable, unit price, project services contract, agent, and independent contractor. From an owner's perspective, reimbursable contracts can require three times as many owner people as a lump sum contract, and would require personnel with extensive analytical skills. For lump sum, a good design package and strong project discipline (no/little design changes) are essential. Often, there is a mismatch of people resources, in relation to contract arrangements. Having both the wrong contract and the wrong organization/people is a recipe for disaster. Also, a poor management application of the lean and mean principle will result in a serious lack of resources, leading to poor project execution and cost over-runs/schedule slippage.

Is the Project Manager (the Person) Qualified?
The answer to whether a project manager is qualified, should address technical expertise, project experience, business capability, leadership ability, facilitating skills and people skills, including mentoring. Project, business, and people expertise should have greater consideration, especially for larger projects. On smaller projects and feasibility studies, technical skills would be more important.

Does the Owner Project Manager Report to the Client or Projects/Engineering?
In contractor groups, who the owner project manager reports to is not normally a problem. In owner organizations, it is often a problem because the owner project manager normally reports to both groups. A generally accepted practice is to report to the internal company client, so as to "follow" the financial responsibility, and report to projects/engineering for direction on technical methods.

Should the Project Task Force (PTF) Approach be Used?
Significant experience has now shown that the project task force is more efficient for larger projects. The close working relationships allow more efficient communication channels and a more efficient decision-making process. The challenge of welding together many individuals from many parts of the company is a substantial task. The organization structure should follow the current trend for larger projects, which has added the new function of a business manager.

Business Management Must Receive the Correct Emphasis
On economically important projects, the emphasis should be on business considerations. There has to be a correct "balance" of technical versus business (estimating, cost control, scheduling, purchasing, contract preparation, contact/construction administration), with emphasis on business considerations.

Efficiency and Effectiveness of the Project Team
There is often the conflict of quality versus quantity. On large projects, it is easy to make the mistake of over-substitution of numbers of people to satisfy lack of skill. It is a question of degree, as some substitution is common-place. In most cases, lack of good contracting personnel is a major problem. Personnel planning on a project should be undertaken in the early stages of implementation for it to be successful. This results in effective scheduling of all required personnel resources. Careful consideration should be given to the mobilization and demobilization timing of all key managers and supervisors.

Project Organization Structure—Owner Team (Reimbursable Contract)
On a reimbursable project, the owner should have a directing position in order to manage the contractor and to avoid inefficient project execution. The balancing force to contain risk is the quality and skill of the owner's project team; both in the pre-contract activities and the post-contract work execution program. An Equal Partner Relationship (EPR) is an essential requirement and should be built into the agreement with the appropriate contract clause. These pre-contract activities should be undertaken by the project team to ensure that the contractor provides competent personnel. The use of new or inexperienced personnel by the contractor should be weighed carefully by the owner when approving contractor personnel.

Pre-Contract Activities for Contractor Evaluation

- An effective proposal evaluation program should evaluate the quality of the contractor's program, with individual criteria for technical, project management, commercial/pricing, project control, contractual, and construction.
- If the project is of substantial size, interviews should be carried out with key personnel.
- Ensure that correct contracting arrangement/conditions are in the contractor's proposal; especially the equal partner relationship clause. Assess the required liability of an agent or independent contractor.
- Evaluate the contractor's proposal program/execution plan and key interfaces with local, corporate, and government entities.

Project Organization Charts
These should be dynamic, up-to-date documents, used to identify owner and contractor positions. During execution of the work, these will be the current and future personnel plan as generally agreed in pre-contract meetings. These charts need to be properly recognized and the organization clearly understood by all project team members. The use of formal job descriptions and duties is recommended.

Project Manager Authority
The project manager has full authority to make both design and cost decisions, with appropriate limits of authority and management reporting requirements. On reimbursable projects, the authority of the contractor project manager must be adequate to allow efficient day-to-day operations.

Project Control Function Reports Directly to Project or Business Manager
Many hold the concept that cost control should be an audit function of the project, and therefore report to higher/senior management. AACE International does not support that concept, since it can lead to an adversarial relationship and dilute the trust and cooperation that is absolutely essential in the cost effort in the project. There are always independent, periodic cost reviews by senior home office personnel that should be more than adequate for a management audit.

CONCLUSION
Project Organization structures have become dynamic as a result of the numerous international perspectives that have been applied. The successful companies embrace a quality of empowerment and reward, which allows them to maximize the organizational programs. Many of these new organizational changes come from the Demingism programs which developed the Total Quality Management (TQM) philosophy. Project organizations can be as strong as the processes that have been put in place, but in the end—success is measured by the quality of the leadership and the members of the project team.

REFERENCES
1. Peters, T. J. & Waterman, R. H. Jr. (1982). *In Search of Excellence – Lessons from America's Best-Run Companies,* London, England: HarperCollins Publishers.

Editor's Note: This chapter is based upon, or portions are excerpted by permission from, *Effective Project Management Through Applied Cost and Schedule Control*, by James A. Bent, Marcel Dekker, Inc., New York, 1996.

Chapter 18—Project Communications

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Abstract
Project personnel involved in total cost management spend considerable time working with project stakeholders and use extensive verbal and written communications. Effective communication with project stakeholders is a vital element to help ensure project success.

This chapter explains the project communications process with a focus on the problems in today's work environment. The importance of knowing your project stakeholders will be discussed, along with a proven method to analyze the importance and interest of the stakeholders. Elements of active listening will be covered, including good listening habits. How to read body language will be explained. Best practices for written communication will be covered along with the essential elements of a concise message. Finally, the keys to making virtual teams work will be discussed. Implementing the suggestions in this chapter will help you become a better communicator on your projects.

Keywords
Active Listening, Body Language, Communication, Communications Process, Emotional Intelligence, Paralinguistic, Personality Style, Stakeholders, Stakeholders Analysis Matrix, Team Charter, and Virtual Teams.

INTRODUCTION
Communication is the exchange of information using a system of symbols resulting in shared meaning between the sender and the receiver. We communicate on projects to exchange information, provide feedback, solve problems, make requests, reach decisions and influence the actions of other project stakeholders. Effective communication is a vital skill for anybody working with other people on projects.

Long ago business communication, if not done by face-to-face conversation, was accomplished by written letters and supplemented in the 1800s by telegraph. With the increased use of the telephone at the start of the 20th century the use of written letters dropped, along with the need for face-to-face meetings. More recently, the introduction of email in the early 1980's sparked a major change in project communications. Email allows quick dissemination of information to a large group of people, and the use of email reduced the need for telephone conversations and face-to-face meetings.

How we communicate is constantly changing due to cultural norms and technology advances. Figure 1 shows past and future trends in the use of communication technology for business purposes.

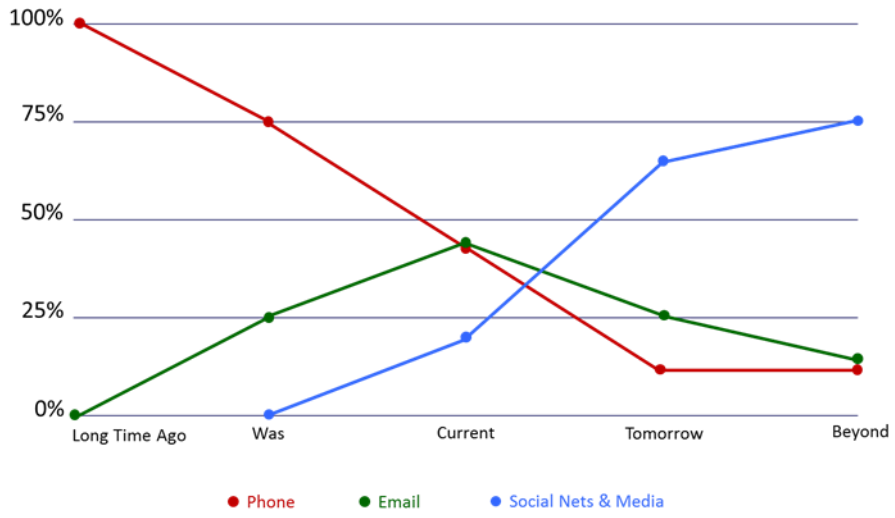


Figure 18.1: Use of Technology for Business Purposes: Past and Future Trends [1]

What we are witnessing today is the expanding use of social networks and media for business communications. Text messaging is becoming more prevalent between project team members, along with the use of other social networks such as LinkedIn and Facebook. The forecast is for the use of social networks and media to expand and the use of email and the telephone to drop.

Unfortunately, the advance in communication technology has not resulted in better communication. Plenty of evidence indicates that just the opposite is occurring. Effective communication is suffering in today's fast-paced work environment, and the bad connections are more than a mere inconvenience. Poor communication can impact morale, erode productivity and contribute to project failure.

Learning Objectives

Reading and studying this chapter will help total cost management personnel communicate more effectively with project stakeholders. After completing this chapter, the reader should be able to:

- Describe the key elements of the communication process.
- Identify the filters and barriers that affect communication.
- Interpret the Stakeholder Analysis Matrix.
- Summarize the purpose of the Team Charter.
- Differentiate between positive/negative nonverbal communication clues.
- Differentiate between internal/external barriers to active listening.
- Describe the elements of the acronym **LISTEN.**
- Understand best practices for written communication.
- Describe the top five communication barriers for virtual teams.

COMMUNICATIONS PROCESS

By definition, communication occurs when ideas are transmitted from one person to another. The communication process is the way in which the message is delivered and received. There are three key elements to the communication process:

1. **Sender**: this person transforms the ideas into words and has responsibility for making the information clear, unambiguous and complete.
2. **Message**: this is the content, context and medium to convey the information.
3. **Receiver**: this person restructures the ideas from the words and has responsibility for making sure the information has been received in its entirety and is understood correctly.

The communication process loop is shown in Figure 2. The first two steps in the communication loop starts when the sender encodes the message and then transmits the message. The symbols are the actual words used, but even more important than words alone are the use of nonverbal cues and paralanguage. A discussion on the importance of nonverbal cues and examples of positive and negative signals will be covered later in this chapter. Paralanguage is "the set of non-phonemic properties of speech, such as speaking tempo, vocal pitch, and intonation contours, that can be used to communicate attitudes or other shades of meaning" [2]. A person who states "I'm upset" in a normal voice is quite different from a person shouting "I'm upset". Non-verbal and paralanguage are both factors in face-to-face conversations, but you lose the non-verbal element with telephone and voice mail, and also lose paralanguage when using written communication.

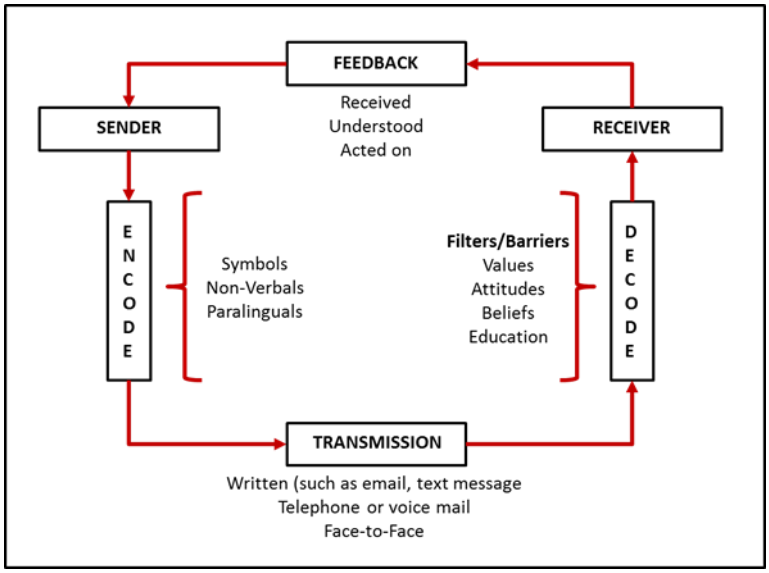


Figure 18.2: The Communication Loop

The next two steps in the communication loop are when the receiver decodes the message and provides feedback. In decoding the message, the receiver will have filters and even barriers that can impact their understanding of the message. Filters and barriers examples that can impact the decoding of the message include:

- **Values**: if the receiver's value is that family comes first on week-ends and office work will not be done, it can influence the decoding of the message 'right now' sent 4 PM on a Friday afternoon as meaning Monday morning while the project manager value is putting the project first with action expected immediately.
- **Attitude**: if the receiver doesn't accept the importance of the project and is not fully committed, he/she might not believe and act on the urgent request from the project manager for an estimate update.
- **Beliefs**: if the receiver doesn't trust the sender of the message, he/she may not believe the content of the message.
- **Education**: if the receiver has a limited vocabulary or lacks technical expertise related to the project, he/she might not fully understand the message due to the words used.

The point is communications can easily break down at any point in the communication loop. Problems can occur with the sender, the message and/or with the receiver. Problems for the sender include making assumptions on the audience knowledge related to the message, providing the right amount of information, lacking time to send a complete message, and lacking language skills. Problems with the message can include difficulties with the medium (such as bad reception on a cell phone call), delivery (lost mail, firewalls), two or more definitions for some words and false assumptions such as assuming the receiver of the message already has the needed supporting information to take action on the message. Finally, problems for the receiver include physical problems (hard of hearing), inattention (multi-tasking while on a conference call), language skills and misinterpretation of the message. There can also be listening barriers for the receiver, but we will talk more about those in the section of this chapter on active listening. In summary, most project personnel can easily cite multiple examples where the communications loop broke down, and it typically leads to poor project results.

KNOW YOUR STAKEHOLDERS

A key first step for effective communications is identifying the stakeholders you need to communicate with on the project. Stakeholders are those people who are directly or indirectly affected by the project or who can affect the project. Stakeholders can view the project as positive or negative, and can be either internal or external to the organization. Examples of stakeholders can include project team members, end users, customers, management, legal and/or regulatory agencies, contractors and suppliers, and neighbors living close to the project site.

Ideally the project team should work together to develop a list of project stakeholders. If a list exists, it should be distilled down to the people you will have contact with on the project. This is basically a subset of the project stakeholder list. If the project team has not developed a list of project stakeholders, take the initiative and develop your own personalized list of stakeholders.

Keep in mind not all stakeholders are created equal! For each stakeholder, you should assess their power and control over project deliverables, along with their interest in the project. An effective tool for doing this is the Stakeholder Analysis Matrix shown in Figure 3. The vertical axis is a measure of the power the stakeholder has over project deliverables. For example, the project sponsor has high power, but a supplier of a standard commodity such as drywall has low power. The horizontal axis is a measure of interest the stakeholder has regarding the project. Obviously a project sponsor would have high interest in the project. However, the supplier of drywall may have low interest in the project if the amount of drywall needed is minimal.

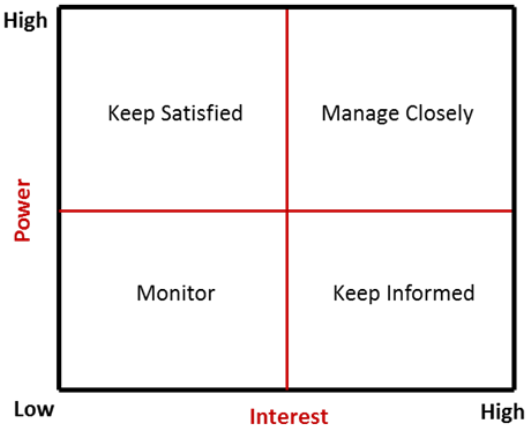


Figure 18.3: Stakeholder Analysis Matrix [3]

For each identified stakeholder, a determination should be made on their interest in the project, and their power over project deliverables. Stakeholders with high interest and high power should be closely managed, which may include frequent project

updates and regular face-to-face meetings. Stakeholders with high interest but low power need to be kept informed, while stakeholders with high power but low interest merely need to be kept satisfied.

An example of a stakeholder with high power but low interest is a town inspector doing a code inspection of the plumbing installed for a new office building. The inspector undoubtedly has many projects requiring inspection so his/her interest in this one particular project is probably low, but the inspector has high power and can stop the project if the plumbing fails inspection. Finally, stakeholders with low interest and low power really don't require any attention. At most you should monitor this group in case any pertinent issues develop.

In summary, communications won't happen if you don't know the stakeholders on your project, so identifying the stakeholders is a critical first step.

THE TEAM CHARTER

Establish the expectations for communications at the start of your project. The project manager should do this with the entire project team, but if not you should do this with your work group on the project. An effective tool for setting the expectations for communications is the Team Charter, which is basically an upfront agreement on operating guidelines for the team. The Team Charter covers more than communications, and can include roles and responsibilities for team members, administrative procedures such as reporting hours worked, team performance measures and the process for making decisions. The Team Charter should also contain 'ground rules' for team conduct such as openly dealing with issues and conflicts, meeting commitments, returning phone calls and emails, and being on time for meetings. Most important, the Team Charter should cover the five W's of communications:

- Who are the recipients for the various project communications (this is from your list of analyzed stakeholders)
- What information do the stakeholders need (such as status reports, schedule updates, cost reports)
- Where will the communications be done (face-to-face, email, project website posting)
- When will communications take place (frequency of creating, updating and disseminating information)
- Why will each communication be done (for information only, or for review and approval)

The five W's are simply common sense. Unfortunately on many projects communications suffers, and the primary reason is a lack of understanding on the communication requirements for the project. The way to avoid this is use of the Team Charter!

BODY LANGUAGE AND PERSONALITY TYPES

Effective communication depends on social skill, which is the ability to read situations and the feelings of people you are communicating with. This is one of the five components of emotional intelligence (EI); the other four are self-awareness, self-regulation, motivation and empathy [4]. Critical to developing your EI is having a good knowledge of personality styles. The better you understand personality styles (both your own and the people you work with), and the better you are at adapting your personality style, the more successful you will be as a cost management professional in your communications with project team members and project stakeholders. For more information on personality styles, refer to the paper presented at the 2013 AACE International Annual Meeting, *Pardon Me – Your Personality is Showing* [5].

Experts on nonverbal behavior say we literally leak our true or masked feelings through our body language and movements. Professor Albert Mehrabian, PhD., is well known for his pioneering studies regarding the effectiveness of spoken communications. His research resulted in the statistics shown in Figure 18.4 regarding the effectiveness of spoken communications of feelings and attitudes. Non-verbals, such as gestures and body language, were most important, closely followed by paralinguistic clues such as tone of voice. The actual words used were of least importance for communication of feelings and attitudes.

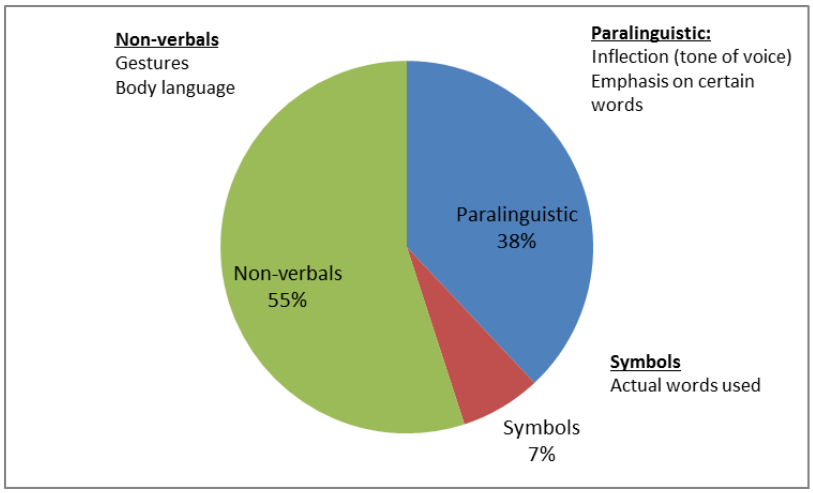


Figure 18.4: Communications of Feelings & Attitudes [6]

Some people mistakenly quote this breakdown as applying to all communications, but the research was done specifically for communications of feelings and attitudes. However, with all communications nonverbal and paralinguistic are important even if not as high a percentage when the conversation is about feelings and attitudes. Words alone are not always sufficient to understand the message a person is conveying. Words can mislead, but the body does not lie. The nonverbal and paralanguage components are important in communications.

With that in mind, this section of the chapter will focus on nonverbal communications and tips that can help you interpret the message a person is conveying. There are four key principles for nonverbal communications:

1. **Context is important**: people will be more relaxed and open at a bar or restaurant compared to being at a job interview.
2. **Observe behavior holistically**: gestures and individual movements should be viewed as a whole since multiple gestures will clarify what the person is really feeling.
3. **Watch for changes in body language**: also observe what causes the change; if a person's body language becomes negative when you ask them to do something, it could be an indicator he/she is uncomfortable with the request.
4. **Watch for congruence**: this occurs when the person's words, paralanguage (such as tone of voice) and body language all match; when incongruence occurs use the body language (not the words) for your default interpretation.

Nonverbal signals can be positive or negative. Positive signals suggest openness, acceptance and comfort. Listed below are some positive nonverbal clues to watch for in your communications:

- **Uncrossed arms and/or legs**: an indicator the person is becoming more comfortable and being more open with the conversation
- **Open hands**: Allstate Insurance uses this as a corporate symbol because it sends a powerful message of acceptance so watch for this in your conversations
- **Rubbing the chin**: an indicator of being engaged, the person is thinking and evaluating
- **Moving closer**: this could be a subtle forward lean towards you or moving their chair towards you; this is a strong indicator of interest
- **Eye contact**: good eye contact is a very positive signal
- **Matching**: this occurs when the person follows or matches your body language and indicates he/she is in sync with you

Negative signals indicate discomfort, distrust and indifference. Listed below are some negative nonverbal clues to watch for in your communications:

- **Crossed arms and/or legs**: an indicator the person is becoming more uncomfortable and less open with the conversation
- **Clenched or hidden hands**: this is an indicator of discomfort
- **Moving away**: this could be a slight lean back or turning away; this is an indicator of discomfort
- **Limited eye contact**: this can be an indicator of boredom or disagreement; but staring could be an indicator of distrust or even anger

- **Excessive body motions**: this can include fidgeting or tapping the fingers or feet and is an indicator of impatience, boredom or even nervousness

When watching for these positive and negative nonverbal clues remember how people behave based on their personality style. For example, an analytical personality style (introvert and thinker) maintains minimal eye contact in conversations, so a positive nonverbal clue would be more than the usual eye contact, but the amount of eye contact would still be much less compared to the other three personality styles. Also be careful of your nonverbal actions in a conversation. If you move closer to an analytical or amiable personality style (both are introverts), their natural reaction is to step back to maintain their distance and it may not indicate discomfort with the conversation, but a discomfort based on their personality style!

Also, a note of caution is needed because reading body language is complicated by cultural factors, social dynamics, personal habits, temporary state of mind and even environment factors such as room temperature [7]. For example, you are pitching your product to a client and she leans back, looks away briefly and crosses her arms. You may read this as unspoken resistance to your statement on pricing. However, in reality she likes the product and pricing but is simply uncomfortable due to the cold temperature in the room. The point is you need to look at the nonverbal clues in the context of other factors that may be present. It takes practice to correctly read nonverbal clues and just reading this chapter will not make you proficient in reading nonverbal clues in all situations!

In summary, nonverbal clues are an important component in communications, but it only exists with face-to-face conversations. Further, when using email or text messaging the paralinguistic component is also lost. Losing both components and relying on just words such as when using email or text messaging can make successful communications more difficult!

Active Listening

People typically talk at around 120 words per minute (WPM), and the fastest speaking debaters can speak 350 WPM or more [8]. However, our thinking capacity is 600 WPM, so in conversations it's easy to multi-task or let your thoughts wander instead of focusing on the message the sender is communicating to you.

There are many internal and external barriers to active listening. External barriers can include:

- **Noise:** Equipment, traffic, other people talking, radio, TV
- **Visual Distractions:** Scenic window views, passersby, cluttered desk or work area
- **Physical Setting:** Temperature too hot or cold, uncomfortable seating, odors, distance from speaker
- **Interruptions:** Unexpected visitors or phone calls
- **Fidgeting with Objects:** Pocket change, pens, paper clips, jewelry
- **Multi-Tasking:** Doing email or texting while conducting a phone call
- **Speaker's Personal Appearance:** Grooming, hairstyle, clothing
- **Speaker's Mannerisms:** Nervous habits, dramatic gesturing, overly expressive face

Internal barriers can include:

- **Anxiety:** Personal worries and concerns crowding in
- **Self-centered:** Prefer own thoughts or daydreaming to speaker's message
- **Closed-Minded / Judging:** Don't personally like speaker or disagree with his / her ideas
- **Laziness:** Speaker's subject matter difficult and takes effort to follow
- **Boredom:** Speaker's subject matter dry or uninteresting
- **Superiority:** Feel as though you have nothing to learn from the speaker
- **Preconceptions:** Listen only for what conforms with your current beliefs and expectations
- **Ego:** Prefer talking to listening

With all of these potential internal and external barriers, it's easy to see the difficulty in practicing active listening. Practice the following six key elements from the acronym **LISTEN** (explained below) and it will help ensure that you hear and understand the other person, plus let the other person know you are hearing what he/she said [9].

1. **L**ook interested and give the speaker your undivided attention. If the communication is face-to-face practice using the positive nonverbal clues discussed in the last section of this chapter. Examples include nodding occasionally, use of facial expressions such as smiling, and making sure your posture is open and inviting. Giving your undivided attention means putting aside distracting thoughts and avoiding the internal and external barriers discussed earlier in this section.
2. **I**nvolve yourself by responding to show you are listening. Encourage the speaker to continue with small verbal comments such as 'yes', or 'please continue' or even 'uh huh'.
3. **S**tay on target. This means listening to the message and not changing the topic because you find the message uncomfortable or uninteresting. The message is important to the sender or they would not be communicating it to you, so regardless how you feel about the message stay on target and complete the communication loop for the topic being discussed!

4. **T**est your understanding since your role as a listener is to understand what is being stated. Reflect on what was said by paraphrasing using words such as "what I'm hearing is" or "it sounds like you are saying".
5. **E**valuate the message and respond appropriately. Think before responding and be candid and honest in your response. Present your opinions respectfully and treat the other person in a way you think he/she would want based on their personality style.
6. **N**eutralize your feelings. Defer judgment and avoid interrupting since it will frustrate the speaker and will limit your understanding of the message. Allow the speaker to finish each point before asking questions, and don't interrupt with counter arguments. If you find yourself reacting emotionally to what the speaker said, ask for more information. For example, one response could be "I may not be interpreting what you said correctly and I find myself taking it personally; what I thought you said is this…is that what you meant?"

In summary, it takes concentration to be an active listener. Most people are not good active listeners, so follow the suggestions described here and break your existing bad habits. Remind yourself that your goal with any conversation is to hear and understand what the other person is saying. Active listening will make you a better communicator, help improve your work productivity and build better relationships.

WRITTEN COMMUNICATIONS

Unfortunately, effective written communications is becoming a lost art. In addition, grammar and spelling mistakes can be found in many emails, letters and other documents such as newspaper articles. Spelling and grammar checks are available with software so this should not be occurring; and mistakes can reflect badly on the sender of the message. However, many times mistakes are present in documents and we don't even see them. Refer to the brief paragraph in Figure 5 and despite the numerous misspellings you can probably easily read the words.

> Aoccdrnig to a rscheearch at Cmabrigde Uinervtisy, it deosn't mttaer in what oredr the ltteers in a word are, the olny iprmoetnt tihng is that the frist and lsat ltteer be at the rghit pclae. The rset can be a total mses and you can still raed it wouthit a porbelm. This is bcuseae the huamn mind deos not raed ervey lteter by istlef, but the word as a wlohe. Amzanig, huh?

Figure 18.5: Is Spelling Important? [10]

The point is we often ignore or subconsciously correct grammar and spelling mistakes found in documents. However, set your goal to being a good writer so you don't have to hope people won't notice your bad grammar and spelling mistakes! To become a good writer, practice will improve your writing skills and can be achieved by writing papers for publication and presentation. Technical writing classes can also be helpful.

Since email is currently the most commonly used medium for written communication, this chapter includes best practices for writing and managing emails. On the plus side, email is a quick and efficient means of communications, especially when communicating with multiple people. However, on the negative side email is not always the best or most effective communications medium, is prone to some of the worst features of written communications and is arguably the most dangerous communication tool invented to date.

Some best practices to follow when writing and managing email include:

1. Consider your audience before writing the email. If most of your audience only needs a summary, with a few people needing more details, consider sending two separate emails or have a brief summary at the beginning of the email and state 'more information below for those who want more details'.
2. Use the subject line to let the reader know what information the email contains. A common bad practice is forwarding and replying to emails multiple times where it gets to the point the email content has nothing to do with the subject line.
3. Keep your email short and concise but coherent. A good rule of thumb is keeping emails to less than one page long. If longer consider using an attachment with the additional information.
4. Don't type in all caps which can be considered 'shouting' and don't type in all lower case.
5. Proofread your document before sending and use grammar and spell check.
6. Don't attach files unnecessarily. A common mistake is sending an email with an attachment containing the meeting agenda, which turns out to be much less than one page. In this situation, the agenda should be in the email text body.
7. Remember that email is not necessarily confidential, so if the subject is confidential consider picking up the phone or walking down the hall to talk with the person. If you receive a confidential email never print a copy to a public use printer.
8. Don't send emails when angry. A good technique is to write the email and save it as a draft until the next day. Most often you'll end up deleting the email or changing it to be a more effective communication.

In summary, establish a goal to become an excellent writer. As mentioned earlier, practice will improve your skills, along with creative and technical writing classes. In addition, there are many free resources for improving your writing skills available on the internet [11].

VIRTUAL TEAMS

A virtual project team is a group of individuals geographically dispersed who work across space and organizational boundaries to complete projects. More than 66% of business professionals engage in virtual work, and that number is rapidly growing [12]. The danger for virtual teams is communication barriers that can impede the virtual team from being a high performance team. The top five communication barriers are [13]:

1. **Lack of a common language**: Many project team members across the globe speak English, but the proficiency level can greatly vary. It's important to ask team members and other project stakeholders their comfort level with English.
2. **Use of slang and terminology**: We all use slang; not realizing that what's 'known' locally may not be understood by people from different locales. The better approach when working with a diverse virtual team is to avoid the use of slang. Another issue can be the use of acronyms, so make sure the stakeholders you are communicating with are familiar with any acronyms used.
3. **Loss of nonverbal communication clues**: This was discussed earlier in the chapter and using a conference call for virtual team meetings eliminates the nonverbal aspects of communications.
4. **Lack of team interaction**: With a co-located team, the frequent team interaction can help create a positive team environment. It's hard for virtual teams to bond on a personal level with other team members, but it's critical to find ways to achieve this or the result will be ineffective communications.
5. **Use of information filters**: There is a fine balance between not enough communications and too much communications. Too many emails tend to get ignored, but not keeping team members informed results in misunderstandings and a lack of information can result in bad decisions.

Virtual teams can be successful provided effective communications is achieved. Some suggestions for achieving better communications with virtual teams include:

- **Connect with all team members**: Find reasons to communicate with each team member, not just the group as an entity. Ask for information, ask their opinion and just connect to say thanks for getting an important task done. When working with virtual team members a helpful tool is keeping a contact log which lists the date of communications and the topic. This provides a reminder that you haven't talked to a team member recently.
- **Prepare e-communications plan**: Every project should have a communications plan, but on virtual projects it is a critical element for success. The e-communications plan describes who needs to receive information, what information will be shared, when the information will be shared, and how the information will be delivered.
- **Use effective communication tools**: Video conferences are a very effective communications tool with virtual teams, but not all organizations have the necessary hardware to support this technology. In this case, a good technique is using screen sharing software so everybody sees the same information, and some software even has polling capabilities. This can be used to keep team members engaged by 'voting' on ideas and suggestions that come up during the meeting. Another effective communications tool is a shared project website, where project documents can be uploaded and shared. This ensures all team members have access to needed project information. In addition, project progress photos can be included on the site.

In summary, virtual teams are becoming the new norm for projects, and virtual teams can be very successful at delivering projects. The danger for virtual teams is that communication barriers can prevent achievement of being a high performance team. Project success is possible with a virtual team provided effective communications is established.

CONCLUSION

Project personnel involved in total cost management spend considerable time communicating and effective communication with project stakeholders is a vital element to help ensure project success. Unfortunately, communication often suffers in today's fast-paced work environment, and poor communications can impact morale, erode productivity and contribute to project failure. Effective communications starts by knowing your stakeholders and understanding the communications process, especially the problems that can occur in the process. Establish the expectations for communications at the start of your project by use of a Team Charter. When using face-to-face conversations watch for nonverbal clues, which are an important component in communications. Use active listening to become a better communicator and build better relationships. Establish a goal to become an excellent writer. Finally, make sure effective communications is established when working with a virtual team.

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Chapter 19—Project Labor Cost Control[12]

Dr. Joseph J. Orczyk, PE, CCP

Abstract
This chapter addresses the aspects of construction labor costs as a variable element of the construction program. Currently labor is considered one of the most direct impacts on the cost of a program, due to the ever changing variables of the programs and their impact on cost through natural areas such as inflation. These areas can also be impacted by variables such as low cost efficiency centers, regional price changes, and an aging skilled construction trade market in specialties.

Keywords
Actual Cost of Work Performed (ACWP), Budgeted Cost of Work Performed (BCWP), Budgeted Cost of Work Scheduled (BCWS), Earned Value Management (EVM), Labor Cost Control, Unit Rates, and Variance.

INTRODUCTION
Construction labor costs are the most variable element of the project construction budget. Therefore, labor cost control is paramount to profitability for all contractors. Owners also need to control labor costs for work performed in-house and for work performed by contractors on a reimbursable basis. In order to control costs, project management must first develop a realistic budget. Just as a yardstick that is not exactly 36 inches long is of little use for measuring distances, an inaccurate budget is useless for measuring labor cost performance. Secondly, in order to maintain an accurate budget, project management must continually compare the actual dollars and workhours to the budget dollars and workhours to identify deviations. Once deviations are identified, project management must take swift corrective action to minimize cost overruns. Creating realistic budgets and maintaining them requires choosing an excellent and efficient cost control system for controlling construction labor costs.

Good construction labor cost control methods use the feedback and corrective action elements of the control cycle. The two prevalent construction labor cost control reporting systems are the earned value method and the unit rates method. Each method has the following elements: measuring inputs, measuring outputs, and report processing. The discussions in this chapter are illustrated using data from the concrete accounts of a pre-engineered warehouse project. The labor cost estimate for the concrete accounts is shown in Table 19.1.

[12] Dr. Joseph J. Orzyck authored the original chapter in S&K 5. Sean T. Regan made revisions and minor updates to the manuscript for S&K 6.

Table 19.1—Labor Cost Estimate for Concrete Accounts

WAREHOUSE PROJECT BUILDING CONCRETE LABOR ESTIMATE & ACTUAL QUANTITY @ DAY 30							
COST CODE		DESCRIPTION	QTY.	UNIT	UNIT COST	LABOR COST	DAY 30 ACTUAL QTY
	Column Pads 5'-6" x 5'-6" x 1'-0"		64	Each			
03120		Forms 4 uses	1,408	SF	$1.50	$2,112	Prod Rpt
03110		Anchor bolts	160	EA	$2.00	$320	132
03210		Rebar 8- #5 bars each way	5,340	LB	$0.15	$801	4,380
03310		Concrete 4,000 PSI	72	CY	$5.00	$360	56
03340		Finish	1,936	SF	$0.15	$290	1,510
	Frost wall 1305' x 2'-0" x 2'-6"		1,305	LF			
03120		Forms Top 4"	870	SF	$1.50	$1,305	Prod Rpt
03310		Concrete 3,000 PSI	242	CY	$5.00	$1,210	242
03340		Finish	2,610	SF	$0.15	$392	2,610
	Slab on Grade 360' x 360' x 6"		129,600	SF			
03140		Forms	2,880	LS	$0.50	$1,440	Prod Rpt
03220		6 x 6 #6/6 Welded wire fabric	129,600	SF	$0.13	$16,848	18,150
03330		Concrete 4,000 PSI	2,400	CY	$5.00	$12,000	240
03360		Finish	129,600	SF	$0.25	$32,400	12,960
03150		Control joints	8,640	LF	$0.50	$4,320	520
03160		Expansion Joints & Column diamonds	1,800	LF	$1.00	$1,800	180
	Dock wall footings 135' x 4'-0" x 1'-0"		135	LF			
03120		Forms	270	SF	$1.50	$405	Prod Rpt
03210		Rebar 3-#5 Bars Cont. & #4 Bent @ 18" OC	600	LB	$0.15	$90	540
03310		Concrete 4,000 PSI	20	CY	$5.00	$100	15
03340		Finish	540	SF	$0.15	$81	405
	Dock walls 135' x 1'-6" x 5'-6"		135	LF			
03130		Forms	1,485	SF	$2.50	$3,713	Prod Rpt
03210		Rebar 4-#4 Bars Cont. & #4 @ 18" OC	659	LB	$0.15	$99	264
03320		Concrete 4,000 PSI	42	CY	$7.00	$294	0
03350		Finish Top	203	SF	$0.25	$51	0
03370		Patch & rub walls	1,485	SF	$0.15	$223	0
TOTAL ESTIMATED LABOR COST						**$80,653**	

Learning Objectives

After completing this chapter, the reader should be able to:

- Calculate installed quantities (progress) for construction activities.
- Define how actual labor workhours are collected using time cards.
- Analyze labor cost performance using earned value.
- Analyze labor cost performance using unit rates.
- Define the three components of labor costs—quantities installed, production rates and wage rates.
- Analyze labor cost performance using variance analysis.

LABOR COSTS AND PRODUCTIVITY

Every construction project incorporates a large amount of skilled and unskilled craft labor. The greatest variable in the final cost of a construction project is the labor cost. Labor cost is a function of worker hourly wage rate and worker productivity; although hourly rates are relatively easy to predict, productivity is the real variable (see, "*Factors Affecting Construction Craft Productivity*" later in this chapter). Therefore, a contractor must monitor both worker hours expended and productivity as major elements in the cost control program.

The element of quantity control is also included as a basis for progress reporting, as well as for estimate verification. The craft labor cost may be paid directly by the project owner, the prime contractor, or the subcontractor. If the craft labor is paid by the

subcontractor, the prime contractor experiences the craft labor cost as a subcontract cost. Depending upon whether the subcontract is cost plus or lump sum, the prime contractor may or may not be at risk for labor inefficiencies. A similar relationship exists between the prime contractor and the project owner. The labor cost control methods included in this chapter are applicable to those organizations at risk (financial or otherwise) as a result of labor inefficiencies.

Some factors affecting construction craft productivity include the following:

- Crew sizes and craft composition
- Craft density (area per worker)
- Interference with other crews
- Scheduling
- Material availability
- Equipment and tool availability
- Information availability
- Rework as a result of design, fabrication, and field errors
- Site layout
- Weather
- Constructability

Implementing the labor cost control methods included in this chapter can be costly. A cardinal rule of cost control is that the cost of the control techniques be less than the money saved by using the cost control techniques. Cost control should be approached as an application of Pareto's Law, which essentially states that 80 percent of the outcome of a project is determined by only 20 percent of the included elements. Thus, in establishing a cost control system, the idea is to isolate, and control in detail, those elements with the greatest potential impact on final cost, with only summary-level control on the remaining elements.

The methods presented in this chapter use two-dimensional tables or spreadsheets to present data. These spreadsheets lend themselves to computerization. In fact, many job cost control and project management software will perform the calculations described in this chapter.

Measuring Inputs and Outputs

In the example of a home heating system, only one measurement, temperature, is required to control the system. Controlling construction labor costs requires many measurements. The ultimate goal of labor cost control is to expend the fewest dollars to complete the project. In order to achieve this goal, project management must measure the efficiency or cost-effectiveness of each dollar spent. This requires hundreds of measurements of labor dollars and workhours expended and quantities produced. Labor cost efficiencies are measured as a ratio of inputs (workhours or dollars) and outputs (quantities produced).

A construction project is complex and must be broken into controllable parts. This is accomplished using a Work Breakdown Structure (WBS). The WBS is the classification of each project element along activity levels where the activity outputs can be measured and then compared to the resources expended for that activity. Each classification is assigned a cost code for identification. A construction project has hundreds of cost codes. A study of 30 building contractors in Atlanta showed that for a two-million-dollar project, the median number of line items was 400 [1]. This translates into approximately the same number of cost accounts. The WBS must be carefully constructed and documented so that all members of the project team consistently use the correct cost codes for inputs and outputs.

Labor input is measured by workhours expended, or by labor dollars spent. Workhours are measured directly using cost codes and time cards. Dollars are calculated by multiplying each workhour expended by the appropriate wage rate (i.e., dollars per workhour). Unlike construction inputs that have the common unit of measurement (i.e., dollars and workhours), the output cannot be measured with a common unit of measure. Consequently, a large number of measures are used for construction outputs. Examples of these measures include cubic yards of excavation, square feet of concrete formwork, tons of structural steel, lineal feet of pipe, and number of electrical terminations.

Cost control requires matching each unit of output to the input (resources) that was required to produce it. Each category of output requires a separate cost account. The input is separated into the appropriate cost account in order to match each unit of output to the resources (inputs) that produced the output. The breakdown of the inputs into the individual cost accounts is accomplished by observing and recording the number of workhours expended each day by cost account. If workhours are not accurately recorded in the correct cost account, the cost control system will be ineffective. The accuracy of cost coding workhours is improved by the following:

- Training all personnel in the use of company cost accounts to correctly code time cards.
- Checking time cards for correct cost codes before recording in the cost control system. And,
- Developing and maintaining a well-documented WBS.

A major consideration for measuring construction quantities is to determine if an item (such as cubic yards of concrete placed or lineal feet of wire pulled) is installed in one step or several steps. Quantities installed in one step are the easiest to measure. The item is either installed or it is not installed. The project management team can physically measure the output for the reporting period. It is more difficult to measure progress when quantities are installed in several steps.

One method to measure progress is to assign each sequential step its own cost account. However, this would burden the labor cost control system with too many cost accounts. A better way to handle this situation is to use the equivalent units method to report the partially completed units as equivalent units completed. In the equivalent units method, each step is assigned a weight based upon the percentage of the activity's budget dollars or workhours required to complete that step.

Table 19.2—Daily Production Report

WAREHOUSE PROJECT BUILDING CONCRETE FORMWORK: DAILY PRODUCTION REPORT																					
COST CODE	DESCRIPTION	UNIT	DAY 16	DAY 17	DAY 18	DAY 19	DAY 20	DAY 21	DAY 22	DAY 23	DAY 24	DAY 25	DAY 26	DAY 27	DAY 28	DAY 29	DAY 30	TOTAL @ DAY 30	RULES OF CREDIT	SUBTOTAL	TOTAL QUANTITY
03120	Column Pad Forms																				
	Erect	SF	154		154		176		154		154		176			154	22	1144	0.6	689	862
	Wreck	SF				154		154		176								484	0.3	145	
	Clean & Oil	SF				154		154										308	0.1	31	
03120	Frost Wall Forms																				
	Erect	SF	200			200			270			200						870	0.6	522	870
	Wreck	SF					200			200			270				200	870	0.3	261	
	Clean & Oil	SF					200			200			270				200	870	0.1	87	
03120	Dock Wall Footing Forms																				
	Erect	SF								108					108		28	244	0.6	146	190
	Wreck	SF											108					108	0.3	32	
	Clean & Oil	SF											108					108	0.1	11	
03130	Dock Wall Forms																				
	Erect	SF												297			297	594	0.6	356	396
	Wreck	SF																0	0.3	0	
	Clean & Oil	SF																0	0.1	0	
03140	Slab on Grade Edge Forms																				
	Erect	LF											370				397	767	0.6	460	608
	Wreck	LF													370			370	0.3	111	
	Clean & Oil	LF													370			370	0.1	37	

The breakdown of the effort that is required for each step is called the "Rules of Credit." Following is an example of piping rules of credit [3].

- Pipe placed in the permanent location—60 percent of the work.
- Pipe end connections are welded or bolted—20 percent of the work. And,
- Pipe trim installed and pipe is ready for hydro-test—20 percent of the work.

An example of a daily production report that measures work progress by cost account using the equivalent units (rules of credit) method can be found in Table 19.2. The detailed description of each field in the daily production report for Cost Account 03140 is as follows:

- The description column lists all subtasks for Account 03140. The field engineer or superintendent records the quantity completed each day for each of the following subtasks: erect forms, wreck forms, and clean and oil forms. The subtask quantities are totaled through the report cut-off date.

- Each subtask is weighted according to the estimated level of effort required for that subtask. These weights are the "rules of credit" and are listed for each subtask.

- The actual quantity for the subtask is multiplied by the subtask's weight, in order to obtain the subtotal of quantity completed for the account. For this example, 767 Linear Feet (LF) of edge form was erected by day 30. This amount is multiplied by 0.6, the weight for erecting edge forms, to obtain the equivalent completed units for the account of 460 LF.

- The subtotal for each subtask is totaled to obtain the equivalent actual quantity for the account. The total of 608 lf is the sum of 460 LF (erect form) plus 111 LF (wreck forms), plus 37 LF (clean and oil forms).

Earned Value Method

Once the actual inputs and outputs are measured, the project management team compares the actual inputs and outputs to the project budget inputs and outputs. This comparison occurs at both the cost code and project levels (or at any level in the WBS). The dollars and workhours cost cannot be compared directly to the budget dollars and workhours because the actual is only for completed work, whereas the budget dollars and workhours are for the entire project.

In the earned value labor cost control system, the budget dollars or workhours are multiplied by the percent of work completed to calculate the earned value. The percent complete for the cost account is the actual quantity divided by the forecasted total quantity (see Equation 19.1). The forecasted total quantity is the project management team's current assessment of the total quantity included in the cost account. The earned value is compared directly to the actual cost to evaluate project cost performance. Earned value is measured by either workhours or labor dollars. Earned value is also referred to as the Budgeted Cost of Work Performed (BCWP).

Percent complete (single account) = (actual quantity) / (forecasted total quantity)

Equation 19.1

The relationship between the earned value and the budget is expressed in Equation 2:

Earned value (BCWP) = (actual percent complete) x (budget for the account)

Equation 19.2

As can be seen from this equation, a portion of the budget dollars or workhours are earned as a task is completed up to the total budget in that account. One cannot earn more than has been budgeted. If an account has a budget of $3,200 and 100 workhours and the account is now 25 percent complete, then $800 and 25 workhours have been earned. Cost performance is measured by comparing the earned value to the actual cost. Earned value and actual cost can be measured in either dollars or workhours. Also, earned value and actual cost can be for a period or the total to-date. The actual cost comparison can be a ratio or a variance as illustrated in Equations 3 and 4.

Cost Variance (CV) = (earned value) - (actual cost)

Equation 19.3

Cost Performance Index (CPI) = (earned value) / (actual cost)

Equation 19.4

Note that a positive variance and an index of 1.0 or greater indicate a favorable performance.

Since progress in all accounts can be reduced to earned workhours and dollars as illustrated, then multiple accounts can be summarized and overall progress calculated as shown in Equation 5.

Percent Complete (multiple accounts) = (earned value all accounts) / (budget value all accounts)

Equation 19.5

The estimated total dollars or workhours at completion (EAC) is determined by predicting the overall cost performance index at the completion of the cost account. There are several sophisticated forecasting techniques that are explained in management science texts, but few constructors are comfortable with them and their reliability appear no better than using a few simple approaches. One fact is certain; no two methods, sophisticated or otherwise, produce the same answer. Three basic approaches are provided here [2].

Method 1: assumes that work from this point forward will progress at the budget (CPI = 1) whether or not this performance has prevailed to this point (See Equation 19.6.)

Estimate at Completion (EAC) = (actual cost to-date) + (budget - earned value)

Equation 19.6

Method 2: assumes that the performance to-date will continue (See Equation 7.)

Estimate at Completion (EAC) = (budget) / (CPI)

Equation 19.7

Method 3: Uses historical curves that show the normal variation in the CPI as the cost account progresses. The forecaster simply makes the best extrapolation possible using the typical shapes of such curves and whatever other information may be available to the forecaster to make the projection.

It is recommended that no single forecasting method be used. Rather, include a forecast by each of the above methods in order to provide a range of possibilities.

Both the quantities installed and the labor unit rates determine labor dollars or workhours. The labor unit rate is expressed in either dollars per unit or in workhours per unit. Multiplying the workhours per unit yields the dollars per unit. Overall cost performance depends on both quantity variances and unit rate variances.

As can be seen from the calculation of earned value, cost overruns occur when actual quantities installed total more than the budget quantities. This occurs under circumstances such as when the budget quantities is based upon an inadequate scope, when there are mistakes in the estimate takeoff, when scope creep occurs after the budget is prepared, when the field force installs more quantities than what was required, and when there is rework as a result of poor workmanship. Cost overruns could also occur when the actual unit rate (dollars per unit or workhours per unit) is greater than the budget unit rates.

The credit value is the budget unit rate multiplied by the actual quantities of work installed [3]. The credit value represents what the cost would have been if the actual quantities were installed at the budget unit rate. The credit value is computed using Equation 19.8a or 19.8b, and the unit of measure can be dollars or workhours. Comparing the credit value to the actual cost measures the performance of the unit rate alone without the compounding effect of changes in quantities. Use the unit cost (dollars per unit) for the budget unit rate to calculate credit dollars (C$). For an analysis by workhours, use the production rate (workhours per unit) to calculate Credit Workhours (CWH). The unit cost index is the ratio of the credit dollars (C$) to the actual dollars (See Equation 19.9.) The productivity index is the ratio of the Credit Workhours (CWH) to the actual workhours (See Equation 19.10.)

Credit dollars = (actual quantity) x (budget unit cost)

Equation 19.8a

Credit workhours = (actual quantity) x (budget production rate)

Equation 19.8b

Unit Cost Index (UCI) = (credit dollars) / (actual dollars)

Equation 19.9

Productivity Index (PI) = (credit workhours) / (actual workhours)

Equation 19.10

A significant feature of an earned value analysis is that the calculated earned value can be compared to the scheduled value to measure schedule performance. The scheduled value is the value in dollars or workhours of work scheduled. It is also known as the Budgeted Cost of Work Scheduled (BCWS). The BCWS is computed using either Equation 19.11 or 19.12.

Scheduled value (BCWS) = (scheduled percent complete) x (budget dollars or workhours)

Equation 19.11

Scheduled value (BCWS) = (quantity scheduled) x (budget unit cost or production rate)

Equation 19.12

Schedule performance is measured by comparing the earned value to the scheduled value. This comparison can be a variance, as in Equation 19.13, or a ratio, as in Equation 19.14.

Schedule Variance (SV) = (earned value) - (scheduled value)

Equation 19.13

Schedule Performance Index (SPI) = (earned value) / (scheduled value)

Equation 19.14

Note: A positive variance and an index of one or greater is a favorable performance.

Table 19.3—Schedule Report

WAREHOUSE PROJECT BUILDING CONCRETE BCWS @ DAY 30							
COST CODE		**DESCRIPTION**	**LABOR BUDGET**	**START DATE**	**DURATION**	**SCHED. DAY @30**	**BCWS @30**
	Column Pads 5'-6" x 5'-6" x 1'-0"						
03120		Forms 4 uses	$2,112	18	16	13	$1,716
03110		Anchor bolts	$320	20	11	11	$320
03210		Rebar 8- #5 bars each way	$801	20	11	11	$801
03310		Concrete 4,000 PSI	$360	22	11	9	$295
03340		Finish	$290	22	11	9	$238
	Frost wall 1305' x 2'-0" x 2'-6"						
03120		Forms Top 4"	$1,305	16	15	15	$1,305
03310		Concrete 3,000 PSI	$1,210	19	10	10	$1,210
03340		Finish	$392	19	10	10	$392
	Slab on Grade 360' x 360' x 6"						
03140		Forms	$1,440	26	20	5	$360
03220		6 x 6 #6/6 Welded wire fabric	$16,848	28	15	3	$3,370
03330		Concrete 4,000 PSI	$12,000	29	15	2	$1,600
03360		Finish	$32,400	29	15	2	$4,320
03150		Control joints	$4,320	31	15	0	$0
03160		Expansion Joints & Column diamonds	$1,800	28	15	3	$360
	Dock wall footings 135' x 4'-0" x 1'-0"						
03120		Forms	$405	21	13	10	$312
03210		Rebar 3-#5 Bars Cont. & #4 Bent @ 18" OC	$90	23	8	8	$90
03310		Concrete 4,000 PSI	$100	24	8	7	$88
03340		Finish	$81	24	8	7	$71
	Dock walls 135' x 1'-6" x 5'-6"						
03130		Forms	$3,713	26	15	5	$1,238
03210		Rebar 4-#4 Bars Cont. & #4 @ 18" OC	$99	28	10	3	$30
03320		Concrete 4,000 PSI	$294	30	10	1	$29
03350		Finish Top	$51	30	10	1	$5
03370		Patch & rub walls	$223	32	10	0	$0
TOTALS			**$80,653**				**$18,150**

In Table 19.3, the scheduled value (BCWS) is calculated at the end of day 30 of the project. Day 30 is the cut-off date for the example used in this chapter.

The detailed description of each field in the schedule report for Cost Account 03140 appears as follows:

- The budget labor dollars are listed for each account. This is $1,440 for Account 03140.
- The start date is expressed as the beginning of the day. The slab edge forms are scheduled to start at the beginning of day 26.
- The number of workdays scheduled by the cut-off date, day 30, is five for Account 03140. The five days are days 26 through 30 inclusive (26, 27, 28, 29, and 30).
- The BCWS is calculated by multiplying the budget labor dollars or workhours by the percentage of work that should have been accomplished by the cut-off date. For this example, the assumption is that an equal amount of work is scheduled for each day. Therefore, the scheduled percentage complete is equal to the scheduled days at day 30 divided by the total duration. The scheduled percent complete for account 03140 is five days divided by 20 days, or 25 percent. The BCWS is 25 percent multiplied by $1,440, or $360.

Table 19.4—Labor Cost Report Using Earned Value

WAREHOUSE PROJECT BUILDING CONCRETE LABOR COST REPORT USING EARNED VALUE														
COST CODE	DESCRIPTION	UNIT	ACTUAL QUANTITY	BUDGET QUANTITY	FORECAST QUANTITY	LABOR BUDGET	ACTUAL COST OF WORK PERFORM	BUDGETED COST OF WORK PERFORM	BUDGETED COST OF WORK SCHED	CREDIT DOLLARS	ESTIMATE AT COMPL.	COST PERFORM INDEX	SCHEDULE PERFORM INDEX	UNIT COST INDEX
03110	Anchor bolts	EA	132	160	160	$320	$285	$264	$320	$264	$346	0.926	0.825	0.926
03120	Form footings	SF	1,922	2,548	2,602	$3,822	$3,200	$2,823	$3,333	$2,883	$4,332	0.882	0.847	0.901
03130	Form walls	SF	356	1,485	1,782	$3,713	$582	$742	$1,238	$890	$3,416	1.087	0.599	1.304
03140	Form edge of slab	LF	608	2,880	2,880	$1,440	$288	$304	$360	$304	$1,364	1.056	0.844	1.056
03150	Control joints	LF	520	8,640	8,640	$4,320	$267	$260	$0	$260	$4,436	0.974		0.974
03160	Expansion joints & column dia.	LF	180	1,800	1,800	$1,800	$205	$180	$360	$180	$2,050	0.878	0.500	0.878
03210	Rebar #3 to #7	LB	5,184	6,599	6,851	$990	$805	$749	$921	$778	$1,064	0.930	0.813	0.965
03220	Welded wire fabric	SF	18,150	129,600	129,600	$16,848	$2,425	$2,360	$3,370	$2,360	$17,316	0.973	0.700	0.973
03310	Place footing concrete	CY	313	334	338	$1,670	$1,670	$1,546	$1,593	$1,565	$1,804	0.926	0.971	0.937
03320	Place wall concrete	CY	0	42	50	$294	$0	$0	$29	$0	$350			
03330	Blace slab on grade, concrete	CY	240	2,400	2,400	$12,000	$1,375	$1,200	$1,600	$1,200	$13,750	0.873	0.750	0.873
03340	Finish top footings	SF	4,525	5,086	5,194	$763	$638	$665	$701	$679	$732	1.042	0.948	1.064
03350	Finish top of walls	SF	0	203	244	$51	$0	$0	$5	$0	$61			
03360	Finish slab on grade	SF	12,960	129,600	129,600	$32,400	$3,522	$3,240	$4,320	$3,240	$35,220	0.920	0.750	0.920
03370	Patch & rub walls	SF	0	1,485	1,782	$223	$0	$0	$0	$0	$267			
TOTALS						$80,654	$15,262	$14,333	$18,150	$14,603	$86,508	0.933	0.790	0.950

In Table 19.4, labor cost and schedule performance is analyzed using the earned value method. The detailed description of each field in the labor cost report using earned value for Cost Account 03140 appears as follows:

- The actual quantity is transferred to this report from the actual quantities provided in Table 19.1, or from the quantities calculated on the daily production report, Table 19.2. Note that some cost accounts appear in more than one work package. The cost account total on this report is the summary of all work packages. For Cost Account 03140, the actual quantity is calculated at 608 LF. See the daily production report in Table 19.2.
- The budget quantity is transferred to this report from the labor estimate. For Cost Account 03140, the budget quantity is 2,880 LF.
- The Estimate At Completion (EAC) quantity is the budget quantity plus or minus any changes or corrections. There are no changes or corrections for Cost Account 03140.
- The labor budget in dollars is transferred from the labor estimate. For Cost Account 03140, the labor budget is $1,440.
- The Actual Cost of Work Performed (ACWP) is gathered from the daily time cards. For Cost Account 03140, the ACWP is $288.
- The Budgeted Cost of Work Performed (BCWP) is also known as the earned value. The BCWP is calculated by multiplying the budget labor dollars by the percentage of work actually completed. The actual percent complete is the actual quantity divided by the forecast quantity. For Account 03140, the actual percent complete is 608 lf divided by 2,880 LF, or 21.11%. The BCWP is $1,440 multiplied by 21.11%, or $304.
- The Credit Dollar (C$) is equal to the budgeted unit cost multiplied by the actual quantity. This is the amount that would have been spent to produce the actual quantity at the budget unit cost. The budget unit cost is the budget dollars divided by the budget quantity. For Cost Account 03140, the budget unit cost is $1,440 divided by 2,880 LF, or $0.50 per LF. The C$ is $0.50 per Lf multiplied by 608 LF, or $304. Note that the C$ is equal to the BCWP. This will always be the case when the forecast quantity is equal to the budget quantity.
- The Cost Performance Index (CPI) is a measure of the cost performance. It is calculated by dividing the BCWP by the ACWP. Indices > one indicate good cost performance, while indices of < one point toward poor cost performance. For Cost Account 03410, the CPI is equal to $304 divided by $288, or 1.056, which indicates good performance.
- In Table 19.4, the estimated total dollars or Workhours at Completion (EAC) is computed assuming that the cost performance to-date will continue until the cost account is completed. The EAC is calculated by dividing the budget labor dollars by the CPI. The CPI is equal to 1.00 for accounts that have no actual data. For Cost Account 03410, the EAC is equal to $1,440 divided by 1.056, or $1,364.
- The Schedule Performance Index (SPI) is a measure of schedule performance. The SPI is the ratio of the BCWP to the BCWS. Indices > one indicate good schedule performance, while indices < one point toward poor schedule performance. For Cost Account 03410, the SPI is equal to $304 divided by $360, or 0.844, which indicates poor schedule performances; i.e., the edge forms are behind schedule.
- The Unit Cost Index (UCI) is a measure of unit cost performance. The labor unit cost is a combination of both labor productivity and wage rates. Variances between the budget unit cost and actual unit cost are generally caused by differences in productivity. Therefore, the UCI is an indirect measure of labor productivity. The UCI is the ratio of the C$ to the ACWP. Indices > one indicate good unit cost performance, while indices < one point toward poor unit cost performance. For Cost Account 03410, the UCI is equal to $304 divided by $288, or 1.056. Note that the UCI is equal to the CPI for this account. Since there is no variance between the budget quantities and actual quantities, the overall cost performance is determined solely by the unit cost variance.

Unit Rates Method

The unit rates labor cost control system is another method for the project management team to compare the actual inputs and outputs to the project budget inputs and outputs. This method makes comparisons at the cost code level, total project level, or at any level in the WBS. In the unit rates labor cost control system, actual dollars or workhours to-date are used to calculate actual unit rates (dollars per unit or workhours per unit). The actual unit rates are then analyzed to forecast the unit rates at the completion of the account. The estimated total dollars or workhours at completion (EAC) is calculated by multiplying the estimate at completion unit rates by the forecasted quantities. The EAC is then compared to the budget dollars or workhours to determine cost performance. Note that at the account level, to-date unit rates can be compared directly to the budget unit rates to determine performance.

The estimated total dollars or workhours at completion (EAC) is analogous to finding the EAC in the earned value method. It is determined by predicting the overall unit rates at the completion of the cost account. The three basic approaches for determining EAC are restated here for the unit rates method.

Method 1: assumes that work from this point forward will progress at budget unit rates whether or not these rates have prevailed to this point (See Equation 19.15.)

Estimate at Completion (EAC) = (actual dollars or workhours to-date) + [(to go quantity) x (budget unit rate)]

Equation 19.15

Method 2: assumes that the unit rate prevailing to-date will continue to prevail (See Equation 19.16.)

Estimate at Completion (EAC) = (total quantity) x (actual unit rate)

Equation 19.16

Method 3: uses historical curves that show the normal variation in unit rates as the cost account progresses. The forecaster simply makes the best extrapolation possible using the typical shapes of such curves and whatever other information may be available to make the projection.

As with the earned value method, it is recommended that no single forecasting method be used. Rather, include a forecast by each of the above methods in order to provide a range of possibilities.

The labor cost report using forecasted unit rates, which analyzes labor cost performance using the forecasting unit cost method, can be found in Table 19.5. The detailed description of each field in the labor cost report using forecasted unit rates for Cost Account 03140 follow:

- The actual quantity is transferred to this report from the actual quantities provided in Table 19.1, or from the quantities calculated on the daily production report, Table 19.2. Note that some cost accounts appear in more than one work package. The cost account total on this report is the summary of all work packages. For Cost Account 03140, the actual quantity is calculated at 608 LF. See the daily production report in Table 19.2.

- The budget quantity is transferred to this report from the labor estimate. For Cost Account 03140, the budget quantity is 2,880 LF.

- The forecasted quantity is the budget quantity plus or minus changes or corrections. There are no changes or corrections for Cost Account 03140.

- The percent complete is the actual quantity divided by the forecast quantity. For Account 03140, the actual percent complete is 608 lf divided by 2,880 LF, or 21.1%.

- The actual cost of work performed is gathered from the daily time cards. For Cost Account 03140, the actual cost of work performed is $288.

- The budget labor dollars is transferred from the labor estimate. For Cost Account 03140, the labor budget is $1,440.

- The actual unit cost is the actual dollars divided by the actual quantity. For Cost Account 03140, the actual unit cost is $288 divided by 608 LF, or $0.4737 per LF. Note that the report is formatted to show only two decimal places.

- The budget unit cost is the budget dollars divided by the budget quantity. For Cost Account 03140, the budget unit cost is $1,440 divided by 2,880 LF, or $0.5000 per LF. By comparing the budget unit cost to the actual unit cost, the project management team can determine the cost performance of the account. The project management team has to take into account that cumulative to-date unit rates at the beginning of an account are typically higher than the unit rate at completion.

Table 19.5—Labor Cost Report Using Unit Rates

WAREHOUSE PROJECT BUILDING CONCRETE												
COST CODE	DESCRIPTION	UNIT	% COMPL.	ACTUAL QUANTITY	BUDGET QUANTITY	FORECAST QUANTITY	ACTUAL COST OF WORK PERFORM	LABOR BUDGET	ESTIMATE AT COMPL.	ACTUAL UNIT COST	BUDGET UNIT COST	FORECAST UNIT COST
03110	Anchor bolts	EA	82.5%	132	160	160	$285	$320	$346	$2.16	$2.00	$2.16
03120	Form footings	SF	73.9%	1,922	2,548	2,602	$3,200	$3,822	$4,332	$1.66	$1.50	$1.66
03130	Form walls	SF	20.0%	356	1,485	1,782	$682	$3,713	$3,416	$1.92	$2.50	$1.92
03140	Form edge of slab	LF	21.1%	608	2,880	2,880	$288	$1,440	$1,364	$0.47	$0.50	$0.47
03150	Control joints	LF	6.0%	520	8,640	8,640	$267	$4,320	$4,436	$0.51	$0.50	$0.51
03160	Expansion joints & column dia.	LF	10.0%	180	1,800	1,800	$205	$1,800	$2,050	$1.14	$1.00	$1.14
03210	Rebar #3 to #7	LB	75.7%	5,184	6,599	6,851	$805	$990	$1,064	$0.16	$0.15	$0.16
03220	Welded wire fabric	SF	14.0%	18,150	129,600	129,600	$2,425	$16,848	$17,316	$0.13	$0.13	$0.13
03310	Place footing concrete	CY	92.6%	313	334	338	$1,670	$1,670	$1,804	$5.34	$5.00	$5.34
03320	Place wall concrete	CY	0.0%	0	42	50	$0	$294	$350		$7.00	$7.00
03330	Place slab on grade, concrete	CY	10.0%	240	2,400	2,400	$1,375	$12,000	$13,750	$5.73	$5.00	$5.73
03340	Finish top footings	SF	87.1%	4,525	5,086	5,194	$638	$763	$732	$0.14	$0.15	$0.14
03350	Finish top of walls	SF	0.0%	0	203	244	$0	$51	$61		$0.25	$0.25
03360	Finish slab on grade	SF	10.0%	12,960	129,600	129,600	$3,522	$32,400	$35,220	$0.27	$0.25	$0.27
03370	Patch & rub walls	SF	0.0%	0	1,485	1,782	$0	$223	$267		$0.15	$0.15
TOTALS							$15,362	$80,654	$86,508			

In Table 19.5, the estimated total dollars or workhours at completion (EAC) is computed assuming that the actual unit cost to-date will continue until the cost account is completed (Method 2). Therefore, the forecasted unit cost is equal to the actual unit cost. For cost account 03410, the forecasted unit cost is $0.4737 per LF. Method 2 is just one of the methods used to forecast the final unit cost. The accuracy of the forecast generally improves as more information is used in the forecast.

The EAC is calculated by multiplying the forecast quantity by the forecast unit cost. For Cost Account 03410, the EAC is equal to 2,880 LF multiplied by $0.4737 per LF, or $1,364.

TWO-WAY VARIANCE ANALYSIS

A frequent question is what or who is responsible for the total difference between the budget and the EAC? Variance analysis is one method for answering this question. The project manager allocates the total difference between the budget and the EAC to the appropriate variance category, quantity, or rate. This is an important step in determining the appropriate corrective action for attaining budget performance for the account. Corrective actions to correct quantity variances are different than the corrective actions to correct rate (dollars per unit or workhours per unit) variances. Table 19.6 contains the necessary formulas to carry out the variance analysis. Table 19.7 illustrates an example of a variance analysis of a pipe installation account.

Table 19.6—Variance Analysis Formulas—Two-Way

Estimated Total at Completion (EAC)	$Q_F \times P_F$
Budget	$Q_B \times P_B$
Quantity Variance	$C_Q \times P_B$
Rate Variance (Production Rate or Unit Cost)	$Q_F \times C_P$

Legend

- Q_B Budget Quantity
- Q_F Forecast Quantity
- P_B Budget Production Rate or Unit Cost
- P_F Forecast Production Rate or Unit Cost
- C_Q Change in Quantity = $Q_B - Q_F$
- C_P Change in Production Rate or Unit Cost = $P_B - P_F$

Table 19.7—Variance Analysis Example

COST CODE	DESCRIPTION	ACTUAL QUANTITY LF	BUDGET QUANTITY LF	FORECAST QUANTITY LF	LABOR BUDGET WH	ACWP WH
15170-20	Chilled water, large bore welded	221	365	395	256	140

	Quantity		WH/LF		Workhours		
Budget	365	LF	0.701	WH/LF	256.0	WH	
EAC	395	LF	0.633	WH/LF	250.2	WH	
Change	-30	LF	0.068	WH/LF	5.8	WH	
Quantity Variance =	-30	LF	0.701	WH/LF	-21.0	WH	
Production Rate Variance =	395	LF	0.068	WH/LF	26.8	WH	
Check					5.8	WH	OK

CONCLUSION

Which method, earned value or unit rates, is best for controlling construction labor dollars or workhours? Each method analyzes the construction project and identifies the deviations from the budget for corrective action by the project management team. Using the assumptions stated in this chapter yields identical estimates at completion for each method. While both methods, when diligently applied, will control project costs and produce identical estimated total at completion, each one has a unique advantage. The earned values in the earned value method can be compared to the value of work scheduled as part of an integrated project control system. The advantage of the unit rates method is that the unit costs and production rates are used for estimating and are therefore familiar to most managers.

Labor cost control is best achieved by using a feedback loop. As construction activities proceed, both actual dollars or workhours and actual progress are measured. The actual performance is compared to the budgeted or planned performance. The project manager concentrates corrective efforts on those activities whose actual performance deviate from the budget. The effectiveness of the corrective action is monitored by the feedback loop. This chapter presented an integrated system for implementing a feedback loop control system for labor cost control.

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Chapter 20—Leadership & Management of Project People [13]

Dr. Ginger Levin

Abstract
The following chapter recognizes some of the key contributions to the field of leadership and management. It describes the challenges of working with multicultural teams and identifies the advantages they afford. Recognition of motivational theories can reveal the challenges associated with business ethics at the individual and organizational levels, as well as appreciation of critical mistakes to avoid.

Keywords
Culture, Empowerment, Ethics, Hierarchy of Needs, Leadership, Management, Motivation, Motivation-Hygiene Theory, Team, Theory X Management, and Theory Y Management

INTRODUCTION
It has been recognized for some time that management professionals are faced with an unprecedented rate of technological change and growing competitiveness in the marketplace. It is also recognized that technology and competition can be more easily managed than the human element of the enterprise. Some have noted that the management of change would present few problems if it were not for the people who create, and are affected by, the change [24]. This presents even greater challenges for today's leaders.

Leadership Styles
Today's leaders must work to promote a team culture and establish partnerships with customers and suppliers. This has become all the more important with many organizations having global teams which spread over various geographical locations across multiple time zones with heterogeneous multi-cultural teams in each of these locations. Communication and information sharing among all stakeholders is critical. Leaders are not always required to directly control team members, but may seek commitment from them to support goals and objectives by fostering open communication, increased productivity through group efforts, and participatory decision making. The leader does not necessarily have to be a technical expert; his or her expertise is in leading the team to successfully attain all established goals and objectives. A number of theories and studies by behavioral scientists have influenced the development of leadership styles; five key contributions are discussed in this chapter.

Learning Objectives
After completing this chapter, the reader should be able to:

- Identify the key contributors to the field.
- Understand the advantages and challenges to a multicultural team.
- Recognize the various theories of motivation.
- Identify how to avoid motivational mistakes.

CONCEPTS

Concept 1—Douglas McGregor, in *The Human Side of Enterprise,* was an early proponent of management as a profession [11]. McGregor stated that management demands a scientific base of research and application to make it a successful profession. He said that to develop the professional manager, the manager should first examine how he or she saw himself or herself in relation to the job of managing human resources. The starting point is a set of fundamental beliefs or assumptions of what people are like. McGregor developed two theoretical constructs of the nature of a man or woman in relation to his or her work, known as Theory X and Theory Y. Theory X assumptions include:

- The average person has an inherent dislike of work and will avoid it if possible.
- The average person must be coerced, controlled, directed, or threatened with punishment to put forth adequate effort toward achievement of organizational objectives.
- The average person prefers to be directed, wishes to avoid responsibility, has relatively little ambition, and wants security.
- Control should be externally imposed.

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Theory Y assumptions include:

- People are self-motivated and will exercise self-direction and self-control toward achieving objectives to which they are committed.
- Average people learn to not only accept, but also seek responsibility.
- People are capable of a high degree of imagination, ingenuity, and creativity in solving organizational problems.
- The average person's intellectual potential is only partially used.

Central to a discussion of McGregor's two theories is the matter of control. Under Theory X, control is externally imposed, while Theory Y emphasizes self-control or an internal control. Theory Y implies that within a climate of trust and respect, the employee is capable of putting forth willing effort and controlling work habits. Theory Y presented a flexible view and opened up a wide range of possibilities for new managerial policies and practices.

Concept 2—Frederick Herzberg studied the relationship between the role of work and working conditions [17]. During his studies, he found that certain job characteristics led to satisfaction while other factors led to dissatisfaction. These findings led to the development of the Motivation-Hygiene Theory. Herzberg found that real motivation resulted from the worker's attainment of motivating factors (satisfaction) such as achievement, recognition, responsibility, and advancement. Conversely, hygiene factors such as company policies, work conditions, salary, and status, led to dissatisfaction. Herzberg also noted that addressing the hygiene factors alone would not necessarily lead to job satisfaction. Herzberg's emphasis on job enrichment (motivating factors) stated that increasing the challenging content of the job would cause the employee to grow both in skill and in a feeling of accomplishment; however, this growth could not happen unless the hygiene factors (dissatisfaction) had been eliminated. Managers must be able to recognize and understand the disparate nature of both hygiene and motivating factors in order to increase job enrichment.

Concept 3—Chris Argyris advanced some of McGregor's theories and said that the organization may be the source and cause of human problems [2]. He believed that individual needs and organizational needs were not met effectively in most organizations, as he described the dichotomy between these two sets of needs. Part of the problem, Argyris noted, was a result of the bureaucratic nature of organizations and their hierarchical structures. He was an early proponent of the concept of ad hoc work groups, or project teams that crosscut organizational lines. Argyris believed that the organization must change to conform to human needs, and that the organization should offer meaningful challenges and opportunities for responsibilities. A climate of open communication and trust is needed in all interpersonal relationships. Argyris advocated the development of interpersonal competence and authenticity in relationships as the first step in dealing with any personal differences that may block information flow and understanding of objectives at the individual, unit, and organizational levels.

Concept 4—Rensis Likert, well known for the development of an attitude measurement approach known as the Likert-type scale, also developed the concept of the linking pin – a person who belongs to two groups in the organization [13]. The linking pin shows that the entire organization is viewed as a set of overlapping and interacting groups. Likert advocated open communication within groups, development of mutual trust, consensus decision-making, group goal setting, definition of roles, and shared responsibility. He said that real authority is not just official or formal authority, but is dependent on how much authority a manager's subordinates allow the manager to exert over them, regardless of formal authority position. As Likert stated, the amount of influence a manager exerts over subordinates is determined by how the manager allows himself or herself to be influenced by them. The degree of group commitment and involvement is based on the extent to which the manager considers the opinions of subordinates in reaching a decision whose outcome has impact upon the group. Likert further developed four basic styles of leadership, as follows, related to a wide range of organizational variables.

- **Exploitive-Authoritative** – The leader has a low concern for people and uses such methods as threats and other fear-based methods to achieve conformance. Communication is almost entirely downwards and the psychologically distant concerns of people are ignored.
- **Benevolent-Authoritative** – When the leader adds concern for people to an authoritative position, a "benevolent dictatorship" is formed. The leader now uses rewards to encourage appropriate performance and listens more to concerns lower down the organization, although what they hear is often rose-tinted, being limited to what their subordinates think that the boss wants to hear. Although there may be some delegation of decisions, almost all major decisions are still made centrally.
- **Consultative** – The upward flow of information here is still cautious and rose-tinted to some degree, although the leader is making genuine efforts to listen carefully to ideas. Nevertheless, major decisions are still largely centrally made.
- **Participative Group** – The leader makes maximum use of participative methods, engaging people lower down the organization in decision-making. People across the organization are psychologically closer together and work well together at all levels [14].

These four styles exist in everyday practice, but Likert directed his attention toward the Participative Group, which he believed was ideal for a human-concerned organization.

Concept 5—Drs. Robert Blake and Jane Mouton developed the Managerial Grid (See Figure 20.1) in 1962 [4]. They believed there was an unnecessary dichotomy in the minds of most managers between concern for people problems and concern for production problems. These concerns are complementary. They said that each manager has a discernible style of management based on the degree of concern for production and people. At one end is the manager who is only concerned with production; at the other end is the manager who coddles people at the cost of lost production. There are 81 possible positions on their Managerial Grid representing leadership styles; of these, though, there are five key styles. Ideally, on the Grid, a manager should be a nine and nine. This manager stresses team management. Concern for people and production are interdependent. The manager's job is one of a coach, an advisor, or a consultant.

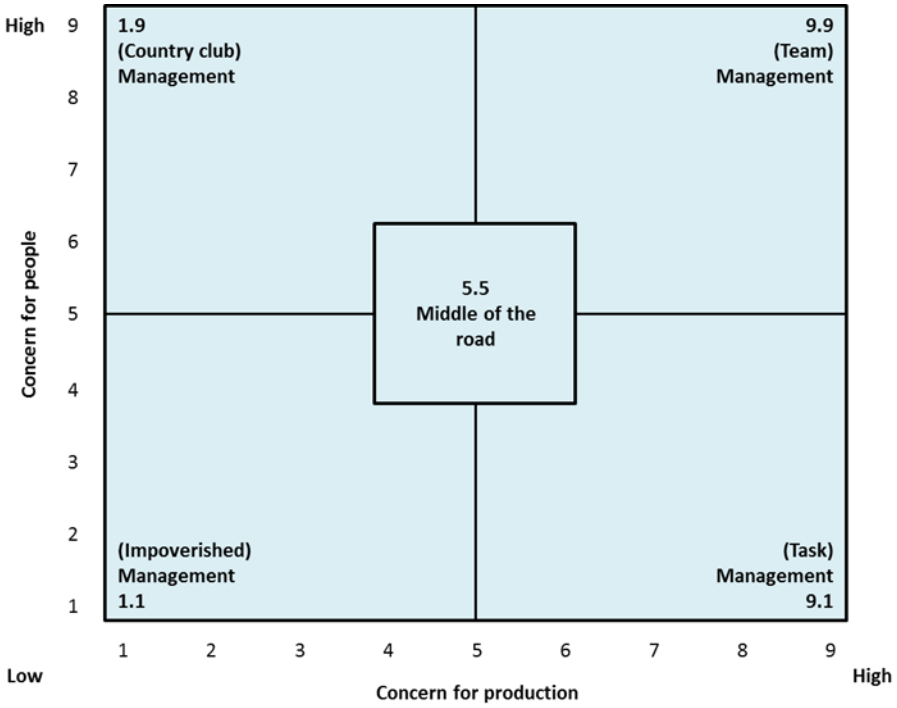


Figure 20.1—The Managerial Grid

Teams

It has long been recognized that teams perform better than individuals acting alone especially when performance involves multiple skills and areas of expertise. However, groups of people do not become a team merely because they are assigned to one. They must collectively commit to each other and become mutually accountable. As noted by J.R. Katzenbach and D.K. Smith, "the wisdom of teams comes with a focus on collective work products, personal growth, and performance results [12]. However, a meaningful "team'" is always a result of pursing a demanding performance challenge."

A team begins as a group of individuals with different motivations and expectations. Some people are pleased to be part of the team; others are not. Some people want to have significant responsibility on the team and want to lead it, while others want to follow. People also bring views as to how teams should operate and stereotypes that reflect one's views and attributes toward members of various groups. Not everyone will view the project with the same attitudes.

These conditions create an environment in which individuals determine the conditions on the team. Individual accountability must be merged with mutual accountability. Team members must be committed to a common approach as to how they might best work together. Performance challenges should energize teams, as a team's performance goals must always relate to its overall purpose [8]. Clear goals can help reduce the potential for future disruptive conflicts and minimize any past differences among the various people represented on the team. As G.M. Parker notes, the people who come together to be part of the team, will be effective to the extent that they agree on a common goal, set aside their individual priorities and agendas, develop a plan to reach that goal, and then commit to work together to attain it [20].

Cross-Cultural Concerns

Culture, as defined by the American Heritage Dictionary is, "The totality of socially transmitted behavior patterns, arts, beliefs, institutions, and all other products of human work and thought"[1]. It includes political, economic, demographic, educational, ethical, ethnic, religious, and other areas, including practices, beliefs, and attitudes, that affect the way people and organizations interact [22].

All organizations are becoming more culturally diverse with each passing business day. People working on teams in the field of cost management, for example, often consist of members representing many different nationalities, languages, and cultures. This cultural richness brings many advantages to a team in the form of different backgrounds, values, norms, and perspectives. However, the characteristics that add richness to a team also increase its complexity. With multicultural teams, the team leader must create vehicles to bridge the cultural gap and bring the team together. A manager working with a multicultural team needs to be aware of these cultural differences and take special care to avoid the potential risks associated with them.

Culture affects our work in many ways. Research has shown that birth culture has a greater effect on a worker's frame of mind than organizational culture. A worker, no matter how well he or she adopts the organization's culture; he or she is still motivated primarily by the cultural environment in which he or she was raised.

In such a multicultural world, the manager or team leader faces a variety of issues, such as managing the team member who may be bicultural and dealing with a culture that may be different from that of the organization. This includes a wide range of challenges from language barriers and time differences to religious diversity and differences in food preferences. For example, in some cultures that value harmony, indirectness, and shared identify, conflict is seen as a loss of face. Open discussion and resolution of conflict is viewed as negative, and a direct approach to conflict resolution, such as a confrontational style, is considered threatening. Here, conflict is best handled behind the scenes, using a smoothing or compromising method. Other cultures that value confrontation see conflict as a positive force as it allows ideas to be aired and insights to be shared in an open fashion. As another example, some cultures view risks as only the responsibility of the executives in the organization, while others view it as the team's responsibility.

For a manager working in settings that involve multicultural issues, it is important to recognize the effect of cultural factors. In the area of communications, the manager must be aware of verbal and nonverbal differences, and recognize that cultural differences can result in misunderstandings. The manager should remain conscious of the fact that cultural differences do exist and try to accommodate these differences if possible. Diversity itself, although a challenge, is also the source of many creative issues. A multicultural team does have greater potential for higher productivity and innovation. Cultural differences should not be ignored or minimized, and if a cultural difference does cause a problem, it should be addressed. Awareness of cultural differences among team members may even make the difference between success and failure.

Leading, Managing, Facilitating, and Mentoring

Similar to many leaders, cost engineers and other professionals are often promoted into leadership roles only for reasons related to technical competency, have not demonstrated leadership and management skills. This technical professional then must acquire functional knowledge of basic leadership and management skills, because ultimately the success and failure of all projects can be traced to the "people" component. D.K. Carr, K.J. Hard, and W.J. Trahant and J. Fitz-Enz offer resources related to generic managerial and leadership skills [6,7]. The importance of this elusive, people component remains constant in today's complex world in which sophisticated technology and software resources are available to manage the intricate processes of any project.

Working with a team of people, the manager faces many challenges. These include the following:

- Uncertain organizational resource support for the project.
- Extreme time pressures.
- First-time challenges to solve unique and complicated problems.
- A wide variety of personnel and other resource interdependencies.
- Challenges of obtaining resources from senior managers who may not totally support the project.

As a result, the successful manager must bring special skills and abilities to the organization. He or she must be able to:

- Apply both technical and managerial skills in addition to operating as a generalist.
- Motivate the team toward the goals and objectives of the project, while still attempting to meet each individual's professional goals.
- Create group cohesion without succumbing to "group think."
- Think and thrive under pressure, while integrating and resolving conflicting priorities and goals of other stakeholders.
- Drive the team toward excellence.
- Work with the emotional, intellectual, and physical challenges in the startup and close out phases of the project.
- Think in terms of three dimensions – timely delivery, cost compliance, and task performance.
- Create mechanisms within the team that encourage the discussion of conflict and balance the process through methods that motivate the team toward decisive action.

The role of the project manager is a multifaceted one, and during a project, the project manager then must be able to assume the following four different roles.

Leadership

Being a leader of a project is a more subtle, complicated role than simply being the person who is in charge of the project and is supposed to deliver it on time and within budget. True leadership involves the ability to conceptualize the vision and direction of the project, and be able to communicate and sell the vision to the team members and other stakeholders. In this context, vision is not an idealistic, amorphous concept of the project, but involves identifying the purpose of the project. This involves listening to the customer to determine the added value the project will bring and recognizing what the customer is not saying. Once the project manager has discussed the purpose of the project, the next step is to create a personal vision of its purpose. The key point is to create a personal representation of the true purpose of the project, noting subtle goals and the customer's true requirements. This then enables the project manager to be confident and motivated to begin the project and to determine how to best sell this process to the team and required stakeholders. The next step is to begin a dialogue with the team members on the subject of the project's purpose. The project manager must create an atmosphere in which all team members are encouraged to ask questions about the purpose of the project and to offer opinions and clarification. Additionally, the project manager must gain credibility and must demonstrate managerial actions and behaviors that are consistent with verbally espoused values. Congruence in actions and stated values is crucial because it creates a state of comfort and trust for team members that enable the person to take the leader at face value and become involved in the task without holding any reservations, doubts, or hesitation. Leadership also involves an active role in being the team's voice to the outside world. The leader needs to communicate actively with outside participants who affect the success of the project to address stakeholders in terms of supporting and buying into the project goals, obtaining needed project resources, providing updates and progress reports, and addressing conflict in a productive and forthright manner. Active communication between the project manager and various stakeholders maintains sponsor support, creates needed ongoing liaisons, and helps reduce the risk of unexpected obstacles hindering the project.

Management

The manager role is one that ensures the project is completed on time, within budget, and at acceptable levels of performance. It involves creating the administrative procedures and structure to monitor completion of the work. The manager role, viewed from the perspective of the people challenges, involves creating an administrative system with enough structure and discipline to complete the project without having the structure stretch into the realm of excessive bureaucracy. It is important to balance the need for structure and the need for autonomy and flexibility.

Facilitation

Facilitation is one of the more subtle, yet profound roles for the project manager. Facilitation is those behaviors and attitudes that help others get their work done and is often achieved through the art of influencing others. It involves communication abilities, conflict resolution skills, the ability to actively procure necessary supplies and resources for the team as a hole, and the ability to motivate both individual team members and the team as a unit. Team-focused motivation involves the creation of strategies that unite the team in common action and rewards that are realistic. The goal is to provide team members with choices, options, and a conductive setting and then trust that the team members will create the desired outcome. A project manager who is adept at helping team members address and resolve conflict in a productive manner promotes facilitation.

Mentor or Coach

The roles of mentoring and coaching are becoming increasingly important areas in the workplace. They can be defined as those processes in which one person (the mentor or coach) assists another person, either formally or informally, in various tasks related to the general purposes of professional growth and development. This assistance takes the form of guidance and encouragement and may or may not be directly tied to an actual project issue being faced by the individual but may be directed at assisting the individual in attaining a broader view of future career directions or advancement. Mentoring and coaching involves the following:

- Being a role model in which the project leader demonstrates desired skills, behavior, and attitudes whose adoption may benefit team members.
- Demonstrating a genuine, personal interest in the welfare and professional growth of team members.
- Offering suggestions, possibilities, resources, problem-solving approaches, and opportunities to think-out-loud with team members regarding current or future issues.
- Providing feedback that is supportive and also frank and accurate.
- Offering motivation directed toward assisting team members in identifying and achieving long-term professional goals.

The Four Key Roles

The most effective project manager thus is able to assume these four roles – leader, manager, facilitator, and mentor—throughout the project and has competency in each of the four areas. Additionally, the project manager needs the skill of timing to determine when to move from one role to another, since projects have different needs at different times.

MOTIVATORS AND DE-MOTIVATORS

What is Motivation?
Used in the context of this chapter, motivation is defined as, "That process, action, or intervention that serves as an incentive for a project team member to take the necessary action to complete a task within the appropriate confines and scope of performance, time, and cost"[8].

The impetus for taking action may come from either intrinsic or extrinsic sources of motivation. Intrinsic motivation is that which arises from a source within a team member, such as a desire to obtain new skills or the need to confront a stimulating personal challenge. Extrinsic sources of motivation involve a force outside of the individual, such as recognition from one's peers in a professional association, or at a conference; or a manager providing a sizeable pay increase for a job well done. Members of a high-performance team tend to be motivated by both sources. The manager or leader, working with his or her team, should strive to search for both intrinsic and extrinsic sources of motivation when working with each team member.

Motivation Challenges
Motivation is particularly difficult because of three strong forces and trends:

- The continuing ongoing reductions in force.
- The unspoken contract between the employee and employer, which has changed dramatically over the past 20 years.
- The increase in the number of team members that are from different backgrounds and viewpoints.

Organizations in both the public and private sectors continue to downsize. Even organizations that are experiencing growth in one sector of their operations may downsize in other sectors. Nearly all downsizing results in situations in which those who survive are required to, "do more with less." Frequently, the survivors have experienced feelings of anger and guilt that clearly decrease motivation [19]. It is not unusual to find pervasive cynicism and skepticism among the surviving employees, which creates an environment in which motivation can be difficult at best.

The contract between the employer and the employee in today's operating environment primarily focuses on the organization owning the job with the employee owning the career. This differs from that before the 1970s, in which employees often perceived an unspoken contract between themselves and their employer, grounded in the belief that quality job performance and loyalty would be in turn rewarded by job security. In today's environment, the manager must instead be creative in developing motivational approaches and processes that are part of the current reality.

The richness of team members from different backgrounds, as noted in the previous section, brings many positive contributions to the work environment. However, it can be a richness that also complicates the process of motivating team members since the "norms" for what is viewed as motivating can be so different across the different cultural groups and locations involved. Greater sensitivity is required of the manager who is responsible for motivating people from different cultural backgrounds. Knowledge must be acquired as to what is motivating for each individual while at the same time avoiding the pitfall of generalizing as to what will be motivating for specific cultural groups.

Other Motivational Considerations
It is important to recognize that people bring with them a certain amount of "baggage" that will affect their motivation. This baggage can be feelings, attitudes, or expectations that have a negative tone and are the result of previous negative personal or professional experiences of the individual involved. The baggage then becomes an impediment to the person's positive engagement with the work to be done. Sources of such baggage include:

- Previous or ongoing organizational problems;
- Industry changes;
- Health issues;
- Career stalling; and,
- Personal problems.

It is also important to be aware of some motivational mistakes such as:

- What motivates me will probably motivate others.
- People are primarily motivated by money.
- Everyone wants to receive a formal award.
- Team members are motivated by quotas.
- Each person needs a rally slogan.
- The best leader is a strong cheerleader.
- People who are professionals do not need motivating.

- People only need to be motivated if there is a problem.
- Everyone should be treated the same.
- Just find one thing that motivates each person and then stay with it.

Motivation of project team members is one of the most challenging tasks of managers. It must address individual issues, as well as organizational issues. Sources of motivation are both fluid and dynamic. As a manager, a good practice to follow is to ask each person what he or she finds to be particularly motivating. This practice, although simplistic, will provide the manager with a wealth of current and specific information that cannot be obtained through any other method.

Theories of Motivation
Traditionally, theories of motivation have characterized the subject from the perspective of evolution, biology, drives, needs, and social influence. Each of these perspectives is in agreement that individuals display a wide range of motives. An overview of these theories of motivation begins with the premise that motivation involves goal-directed behavior.

Biology Theory—The biologic perspective is considered an evolutionary approach. It asserts that actions or behaviors that contribute favorably to the preservation and expansion of the species will produce motivation. It is appropriate when confined to the more basic aspects of human behavior, such as hunger and thirst, reproduction, and the need for affiliation for the goal of basic survival.

Drives Theory—Drive theories state that certain behaviors are the result of individuals meeting the requirements of specific drives. Drives are considered complex combinations of internal stages of tension that cause the individual to take action to reduce the level of tension. The goal of reducing tension is to achieve an internal state of equilibrium or balance or homeostasis. Individuals in this model are believed to desire homeostatic states in their lives, and behavior is motivated because of attempts to maintain this balance. Similarly to evolutionary theories, these drive theories work best when applied to the most basic human behaviors. They are often insufficient when trying to explain the complex behavior and skills involved in management.

Incentive Theory—Incentive theories state that individual behavior is pulled in certain directions based on the external conditions in the specific setting. Much of the field of learning theory and instrumental learning work conducted by the noted psychologist, B.F. Skinner, is based on the incentive type of motivation. These approaches can work in settings when the manager and team member have the ability and the resources to identify a desired behavior that can be awarded by providing the identified incentive. The incentives must be valued by the group and may need to come directly from the group members. The incentives also need to be appropriate to the culture of the organization.

Needs Theory—Another approach is to consider motivation as based on a theory of needs. This approach is primarily based on work done by David McClelland, who developed the concept that people who value the need for achievement are often those people who are the leaders in the areas of creativity and economic growth [16]. This approach is based on premises that, as humans, challenging environments provide us with an opportunity to achieve excellence or to compete against others successfully will provide motivation. The need to achieve and compete within one's own professional discipline can self-motivate many individuals.

Fear of Failure Theory—Another approach is to describe the motivation to act and succeed based on a fear of failure. This approach can be a strong motivator in situations when the consequences for failure are especially distasteful or catastrophic. However, it should be employed only in unusual circumstances, such as if a project is headed for crisis, and immediate action is required. If it is employed too frequently, it may create a crisis management, disaster-avoiding environment, which will easily lead to employee burn out and dissatisfaction.

Hierarchy of Needs Theory—Abraham Maslow set forth a hierarchical theory of motivation that adopts the premise that the basic physical needs and more subtle social or psychological needs will motivate people [15]. Maslow states that people are motivated by the desire to satisfy these various needs according to a hierarchy, with the most basic needs placed at the bottom of a "needs pyramid." When one need is satisfied, the individual will then move upward to the next need. Maslow's hierarchy of needs can be described as follows:

- Level 1 – Physiological needs (food, thirst).
- Level 2 – Security and safety needs (stability, survival).
- Level 3 – Belonging needs (affiliation, love).
- Level 4 – Esteem needs (achievement and the acquisition of recognition).
- Level 5 – Cognitive needs (knowledge).
- Level 6 – Aesthetic needs (beauty, order).
- Level 7 – Self-actualization needs (the realization of one's personal potential).

Career Theory—E. Schein presents a different approach through a model that describes major stages in a personal career [23]. An understanding of an individual's current career stage by the leader can be used in developing tangible approaches to individual motivation. This model has ten career stages.

- **Stage One and Stage Two**—These two stages occur in a person's life before entering the world of work and involve early years of career exploration, followed by formalized career preparation such as higher education and specialized training.
- **Stage Three**—The first formal entry into the workplace where real world sills of the profession are acquired.
- **Stage Four**—Training in the concrete application of skills and professional socialization occur as the identity of being a professional is becoming established.
- **Stage Five**—The individual is observed as having gained full admission into the profession based on competency and performance.
- **Stage Six**—The individual gains a more permanent membership in the profession.
- **Stage Seven**—The natural mid-career assessment or crisis in which questions are asked as to the value of the career and what has been accomplished.
- **Stage Eight**—The challenge of maintaining momentum as the career starts to move into its final chapters.
- **Stage Nine**—The individual beginning to disengage from the profession and the world of work.
- **Stage Ten**—The retirement stage in which the individual must come to some form of closure of employment with a specific organization or membership in a certain profession.

Schein also stated that our personal values affect our enjoyment and pursuit of various tasks in the workplace, and that as a result, the more we understand our own values in specific areas, the better we are able to achieve work satisfaction. Therefore, our motivation will be greatest when we pursue tasks and functions consistent with our values. Schein identified eight of these values, which he describes as career anchors; the word anchor suggesting a person's self-image of what is important for them as they consider the aggregate of their skills, motives, and values. These eight values are:

- Technical-Functional
- General Managerial
- Autonomy and Independence
- Security and Stability
- Entrepreneurial Creativity
- Service and Dedication to a Cause
- Pure Challenge
- Lifestyle

Empowerment Theory—J.R. Meredith and S.J. Mantel suggest that a team environment be established in which the members experience a strong sense of empowerment through the use of participatory management methods [18]. Empowerment is defined as an approach that stresses individual initiative, solution creation, and accountability. The team is then motivated by the opportunity to be self-determinative in creating the structure and methods to achieve its goals.

Ethical Theories and Applications

For any professional, in any discipline, ethics is an emotionally and intellectually charged word. It prompts images of moral responsibility and obligation, scholars debating the intricacies of profound issues, and arguments between professionals and social commentators about right and wrong behavior. Other images are those such as a professional oversight board ruling on professional conduct or misconduct; discussions about financial or corporate malfeasance, and the like [9].

Business ethics is considered a management discipline because of the social responsibility movement that began in the 1960s. During the 1960s, social awareness movements emphasized the expectations of businesses to use their influence to address social problems. People asserted that since businesses were making profits, it was also their responsibility to work to improve society. Many replaced the word "stockholder" with "stakeholder," including employees, customers, suppliers, and the wider community. By the 21st century, 90 percent of business schools provided some type of training in business ethics. However, philosophers, academics, and social critics traditionally have handled the field of business ethics. Much of the literature that is available is not geared to the practical requirements for the behavior of leaders and managers. And, while there is no shortage of differing opinions about what businesses should do in various situations, there is little information on ways to actually implement ethical practices.

One definition is that ethics "is the science of judging specifically human ends and the relationship of means to those ends" or, "the art of controlling means so that they will serve specifically human ends." It therefore involves techniques of judging and decision making, as well as tools of social control and personal development. Accordingly, it is involved in, or should be involved in all human activities. In terms of business, ethics is concerned with the relationship of business goals and techniques to human ends. It studies the impacts of acts on the good of the individual, the organization, the business community, and society as a whole [10]. Another definition is that it is, "the guidelines or rules of conduct by which we aim to live" [5]. As A. Cadbury explains, while it is difficult enough to resolve dilemmas when one's personal rules of conduct conflict, the real difficulties arise when one must make decisions affecting the interests of others. Often it is necessary to balance the interests of employees against those of shareholders and the differing views that exist among the shareholders. What matters most is how one behaves when faced with decisions that involve combining ethical and commercial judgments.

Most organizations have ethics programs but many are unaware of them. These ethics programs typically are composed of values, policies, and activities, which affect the propriety of organization behavior. Ethics is a matter of values and associated behavior. Several principles have been set forth for highly ethical organizations:

- They easily interact with diverse internal and external stakeholder groups. The ground rules of these firms make the good of the stakeholder groups as part of the organizations' own good.
- They are obsessed with fairness. The ground rules emphasize that other peoples' interests count as much as their own.
- Responsibility is individual, not collective; individuals assume personal responsibility for the actions of the organization. The ground rules state that individuals are responsible to themselves.
- Activities are viewed in terms of purpose; this purpose is a way of operating that, members of the organization value highly. The purpose ties the organization to its environment [21].

There are few, if any, ethical truths or standards that can be memorized and applied in a concrete fashion in all settings applicable to cost and project management. Ethical behavior is difficult to qualify and operationalize. It can be viewed as a process that one goes through, a method to consider the conflicting and often contentious agendas of those involved. It is not an action taken after the consideration of memorized rules of conduct. It is, instead, the actions that are taken as a result of the individual having engaged in a process of considering the needs of the various stakeholders, thinking through the consequences of various actions, and arriving at an action that is grounded in a good faith approach to respect the rights of those involved [9].

The difficulty of establishing sound ethical norms for an organization cannot be underestimated, as the ethical climate of an organization is extremely fragile. The task requires unremitting effort, and ethical codes can be helpful, although not decisive [3]. In the end, society sets the ethical framework within which those who run companies must work out their own codes of conduct. Business must take into account its responsibilities to society in reaching its decisions, but society must accept its responsibilities for setting the standards against which decisions are made [5].

CONCLUSION

The challenges associated with management of project people are numerous and complex. But, as a leader or manager, one must establish direction, communicate this direction to others, and motivate and inspire people to achieve goals and objectives. This involves the necessity of developing a leadership style that is appropriate to the specific organizational situation, working more frequently with diverse groups of people representing many different cultures, using the most appropriate motivational approaches, and also taking professional responsibility for one's actions.

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Chapter 21—Quality Management[14]

Gary Cokins

Abstract
This chapter presents concepts in quality management. It extends to the concepts presented by Professor Cokins, as well as added AACE International materials and references. At its basic precepts is the AACE International definition: "Quality Management: concerns the optimization of the quality activities involved in producing a quality product, process, or service. As such, it includes appraisal, training, and prevention activities" (AACE International Recommended Practice: 10S-90, *Cost Engineering Terminology* [as amended: January 14, 2014]. Additional AACE International sources include:

- *Total Cost Management Framework*, Chapter 11, Subchapter 11.4, *Quality and Quality Management*.
- *Total Cost Management Framework*, Chapter 3, *Materials*.
- Various AACE International Recommended Practices [1].

The text of the article provides an historical background on the development of quality controls and management and modern and renewed emphasis on quality management. Quality management is one of a suite of management tools essential for producing a quality product meeting the expectation of all parties. Quality management accounting is emphasized wherein building the facts associated with the project is the key for success[15].

Keywords
Benchmark Index, Benefit Cost Analysis, Chart of Accounts, Cost, Cost of Lost Business Advantage, Cost of Quality, Cost of Quality Conformance, Cost of Quality Non-Conformance, Functional Worth, Indirect Costs, Overhead, Prevention, Quality, Quality Performance Tracking System, Total Quality Management (TQM), and Value Activity.

INTRODUCTION
This chapter on quality management introduces quality management concepts and current concepts (such as Six Sigma), as well as their implications for cost professionals. It provides both a background on how the quality controls and management developed in the past and the emphasis on quality management as of this document's publication date.

Learning Objectives
After completing this chapter, the reader should be able to:

- Understand a brief history of the quality management, continuous improvement, and benchmarking movements.
- Appreciate why there is renewed interest in quality management, now emerging as the convergence of lean management and Six Sigma.
- Understand why traditional managerial accounting has failed the quality management movement.
- Understand the Cost of Quality (COQ) categories: error-free, conformance related, and non-conformance related.
- Understand how Activity Based Cost Management (ABC/M) provides a foundation for repetitively and reliably computing COQ.
- Appreciate the goals and uses of COQ and benchmarking data.

CONCEPTS OF QUALITY MANAGMENT

Was the Total Quality Management Movement a Fad?
In the 1980s, the Total Quality Management (TQM) movement, which included vast collection of philosophies, concepts, methods, and tools, grew increasingly popular. It received substantial business media attention and was intellectually appealing as a way to remove waste and reduce costs. At an operational level, TQM was effective at identifying waste and accelerating problem solving for tactical issues. However, at a more strategic level, it was believed by many that TQM was not the magic pill for which senior executives always seem to be searching.

TQM usually did not double or triple an organization's profits. In many cases, implementation of TQM probably prevented greater financial losses from customer defections caused by quality problems, or from waste and inefficiencies. Unfortunately, the avoidance of reduced profits is not measured, or reported, by the financial accounting system. No one could easily assess TQM's

[14] This chapter includes modified excerpts from Activity-Based Cost Management: An Executive's Guide; Gary Cokins; John Wiley & Sons; 2001; ISBN 0-471-44328-X) [3].

[15] Professor Cokins' body of information presented in his article originally appeared in AACE International, *Skills and Knowledge*, 5th ed. The body of information was reviewed and updated by him for this *Skills and Knowledge*, 6th ed. John Evans made additional minor revisions to the manuscript for S&K 6.

benefits. As a result, in the 1990s, TQM was regarded by senior management of some organizations as another check-in-the-box improvement program that they needed to have in place, along with other programs. But TQM was not viewed as foundational.

What led to the initial interest in TQM? By the 1980s it had become evident to senior executives and the federal government that Japan was winning market share with better quality. What began as a competitive nuisance quickly became feared as a serious threat. Japan's economy had miraculously transitioned from a low- to high-quality reputation. In hindsight, we now realize it was not miraculous, but a result of plain commonsense business practices. What had occurred was that consumers began to recognize Japanese products as either superior or a bargain. Consumers realized they did not need to resign themselves to accepting shoddy workmanship.

North American executives countered this threat and began to realize that quality management initiatives improve productivity, while concurrently defending their market share positions—a win-win. Executives were learning that there is no trade-off of extra cost for greater market share.

In the 1980s, TQM got its opportunity to shine as a leading change initiative. Popular quality management consultants raised awareness and educated businesses. Joseph M. Juran, W. Edwards Deming, Phillip Crosby, and others became leading experts and guides for organizations struggling with how to turn them around. Quality management programs became prevalent and often institutionalized via accepted standards, such as the ISO 9000 Quality System Standard. In 1987, the US Congress passed a law establishing the Malcolm Baldrige National Quality Award. In 1988, the European Foundation for Quality Management (EFQM) was founded; and in 1992, it introduced the European Quality Award. It appeared as if industry was solving its "quality crisis."

In the early 1990s, skepticism about TQM began to take the bloom off the TQM rose. A disappointing pattern from past TQM projects had emerged. Results from TQM were below possibly inflated expectations. Regardless of the explanation, after initial improvements from TQM, executives began to question if there were enough results. Some managers questioned the payback from TQM. In short, there was an ominous disconnection between quality and the bottom line, as increasingly more companies adopted quality programs, yet few could validate much favorable impact on profitability.

About the same time, other change initiatives, such as just-in-time production management, Business Process Reengineering (BPR), and lean management began capturing management's attention. Total Quality Management (TQM) settled in as a necessary-but-not-sufficient backseat program.

Renewed Emphasis on Quality Management

During the Middle Ages, many guilds of craftsmen were established to guarantee the quality of workmanship and to define standards by the purchaser. During the Industrial Revolution, many of the technological advances, such as the development of the steam engine, were made possible through developments in metrology and the standardization and interchangeability of engineering components, such as screw threads.

The advent of mass production during the 20th century increased the demands for control of product quality. During the 1940s and 1950s, the techniques of quality control became an increasingly important element of business management as organizations sought to gain competitive advantages. The success of Japanese manufacturers during the 1960s and 1970s changed the emphasis from a quality control approach to a quality assurance approach that involved more of an organization's functions.

Organizations worldwide began recognizing that quality management need not operate in isolation from other change initiative programs. Executives admitted to themselves that there had been drawbacks that had harmed quality management's reputation, such as non-verifiable measures, claimed but unrealized cost savings, and small projects that were too local and tactical. However, these same executives realized that with corrections, what was earlier referred to as TQM could be repositioned with new branding.

In the face of increasing pressure, organizations have often launched massive, but usually uncoordinated, change initiatives that may or may not achieve their goals. Each effort in isolation may have shown results, but collectively the initiatives can fall well short of their potential. Despite the temptation for management to continue this search for that special improvement program, system, or change initiative to cure their ills, pragmatic executives realize that there is no single program. Multiple concurrent change initiatives are needed, and they require integration.

A variety of programs and management systems began to emerge. Balanced scorecards became accepted as a solution for aligning organizational execution with strategy. Information systems, such as Enterprise Performance Management (ERP) and Advanced Planning and Scheduling (APS) improved execution, compressed lead times, and reduced unused capacity. Customer relationship management (CRM) systems connected the sales force to customer needs, value, satisfaction, and loyalty. Activity Based Cost Management (ABC/M) systems improved the visibility and understanding for management to gain insights, understand and believe their profit margins, draw conclusions, and make better decisions.

The rate of change began to accelerate. The strong force of recognizing customer satisfaction and loyalty, as being essential to move organizations from hierarchical structures toward process-based thinking, is important. The business process reengineering message

was to worry about the outputs, not the functions: Do not get entangled in the politics of the hierarchical organization chart. Power was shifting from sellers to buyers, and organizations had to shift their orientation.

Quality management qualified as one of the essentials in the new suite of management tools and methodologies. Corporate role models emerged. Six Sigma programs with "black belt" quality training at General Electric and Motorola were heralded as keys to their successful performance. To validate an organization's claim to having achieved high quality, quality assessment mechanisms have been developed. The Malcolm Baldrige Award, established in 1987, has become coveted as a sign of excellence in the US. Europe honors its winners of the European Quality Award (EQA), and Japan has honored winners of the Deming Application Prize. Figure 21.1 illustrates the stages of maturity that quality programs have progressed through. The figure emphasizes that the benefits for any program stem from when customer and/or shareholder value are created.

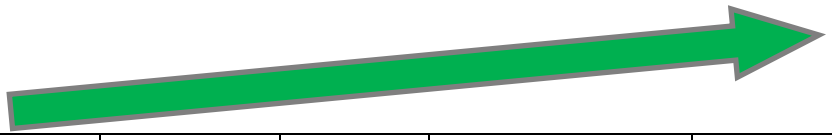
Work Focus	**Customer Value-adding**	**Non-value-adding Waste**		**Business Value-adding**	
Management Objective	**Cost Reduction**	**Waste Reduction**	**ROI**	**Business performance competitiveness**	**Response to unpredictable change**
RESULTS Investor's Value-add					
Management Approach	**Rationalize, Automate, Downsize**	**Quality Improvement**	**TQM**	**Business process reengineering**	**Process management; Six Sigma**

Figure 21.1—Evolution of Value Realization

There have been and will continue to be endless debates about which management techniques matter and are effective—and which don't matter. There is an increasingly prevailing consensus that strategy and mission are essential; after strategy and mission are defined by senior management, the core business processes take over to execute the strategy. The core business processes are now accepted as the mechanism to deliver the value (both customer and stockholder value) defined by the strategy. Time, flexibility, quality, service, and cost are all derivatives of the business process. They are inextricably braided together and should not be addressed in isolation from each other. Programs such as lean management, core competencies, organizational learning, and cycle time compression are considered to be important enablers.

At the same time that management tools and methodologies were being blended, new thinking about how to achieve competitive advantage began displacing old thinking. Quality management has a golden opportunity to be part of this new managerial thinking. As an example, at some companies the Michael Porter "competitive advantage" strategy model is being abandoned. Companies can no longer compete by concentrating on "low price versus high product or service-line differentiation." This is because companies that are successfully sustaining their competitiveness have achieved competencies in new product development. Those "first-to-market" suppliers are quickly met by competitors with rapid "me-too" capabilities—and with lean cost structures. There is no place left to stake out a competitive edge. The only option that competitors have is to adopt aggressive, confrontational management styles.

This is a major challenge for executives, and it involves increasing value. *Value* is an ambiguous term and can be highly subjective. A primary responsibility for executives is to create value for the customer while increasing economic wealth for employees and shareholders—all at the same time! The capability of producing value is a prerequisite to growth, and pressure is mounting to increase the rate of value creation.

A simple equation for value is: Value = Performance/Cost

Where performance loosely refers to the right type of results aligned with the organization's strategy.

With this math, value increases if the numerator goes up or the denominator goes down.

In some ways, executives feel boxed-in given that pricing is market-driven. They are realizing that profit margin management will require visibility and relentless management of costs. Quality management is an essential element for managing costs.

PRODUCTIVITY PARADOX

To complicate matters, some companies that have been "re-engineered" may have become leaner and smaller from downsizing, but not necessarily fitter. It may have helped them to survive, but they may still not have a distinct competitive or quality advantage. In many cases, you cannot simply remove bodies if you do not also reduce the work; otherwise, service levels erode and deteriorate. In

addition, the old methods and old systems usually remain in place. Management may have met some short-term objectives, but the surviving workforce is hopefully operating with the long term in mind.

As a result of these types of changes, so-called improvements in productivity do not always translate into a more profitable business. This has been referred to as the productivity paradox.

Some organizations have invested in improving processes that were not critical to their strategic success. Such processes may have been improved, sometimes dramatically, but they did not turn out to be sufficiently relevant to the organization's long-term performance and success. Process performance improvement, cost reduction, and the like are managerial terms, but they are not necessarily indicative of value added. Value is an economic concept. However, with an advanced managerial accounting system, increases or decreases to shareholder wealth can be traced to the changes in features, functions, and processes aimed at altering customer satisfaction.

In addition, simply being lean and agile will no longer be sufficient for success. Companies' success will depend on all the trading partners in their supply chains behaving similarly. Waste and redundancy created by inter-organizational mistrust must be removed via collaboration. Ideally, quality management can be a shared experience among trading partners and a basis for communications.

THE ROLE OF CONTINUOUS IMPROVEMENT PROGRAMS

Quality management encompasses many tools and techniques under the umbrella of what is popularly referred to as continuous improvement, including Statistical Process Control (SPC), quality control, quality improvement, quality assurance, and benchmarking. The emphasis in this section of this study guide is on measuring the financial accounting aspects, because Six Sigma's key differentiator is its emphasis on financial returns justification (in contrast to its predecessor quality management programs) and because AACE International is an organization that approaches problem solving and management from a cost viewpoint. There are substantial materials about continuous improvement tools and techniques at the website of the American Society for Quality at: www.asq.org.

WHY IS TRADITIONAL ACCOUNTING FAILING QUALITY MANAGEMENT?

One of the obstacles effecting quality management and other initiatives has been the shortcomings of the financial accounting field. Part of the problem is the traditional emphasis of accounting on external reporting.

A significant reason why traditional accounting fails quality managers is that the initial way in which the financial data are captured is not in a format that lends itself to decision making. It is always risky to invest in improving processes for which the true cost is not well established, because management lacks a valid cost base against which to compare the expected benefits of improving or reengineering the process. Gabe Pall, in *The Process-Centered Enterprise* states:

> "Historically, process management has always suffered from the lack of an obvious and reliable method of measurement that consistently indicates the level of resource consumption (expenses) by the business processes at any given time—an indicator which always interests executive management and is easily understood. The bottom line is that most businesses have no clue about the costs of either their processes or their processes various outputs [4]."

Another part of the problem involves attitudes. For some quality professionals, using quality to connect with the bottom line or with executive thinking may seem irrelevant or, worse yet, destructive. These quality professionals fear the danger of managers who myopically focus on short-term results.

In short, understanding the economic contribution toward increasing shareholder wealth from individual business processes is a significant concern for management. When the costs of processes and outputs can be adequately measured financially, two things can happen:

- The data can gain management's attention and confidence that they can depend on these managerial accounting data as reliable business indicators. And,
- Management can more reliably assess the different worth of processes and how they contribute to the overall performance of the business.

Finally, another part of the problem is accountants and deficiencies with their financial accounting system. The accountants' traditional general ledger is a wonderful instrument for what it is designed to do: post and categorize transactions into their specific ledger account balances. But, the cost data in this format (e.g., salaries, supplies, depreciation) are structurally deficient for decision support, including measuring Cost-of-Quality (COQ). The accounting community has been slow to understand and accept this problem.

The quality professional's focus should be on the quality of cost, as well as the cost-of-quality. That is, focusing on the quality of cost ensures that any money spent on the business produces its equivalent in value for the customer and the supplier's employees and

shareholders. The cost-of-quality measures how much cost is caused by poor quality. Both are important. This is a before-investment view in contrast to an after-the-fact view. The next section discusses the issues related to measuring the financial dimensions of quality.

BRING FACTS, NOT HUNCHES

To some people, it is obvious that better management of quality ultimately leads to goodness that in turn should lead to improved financial health of an organization. Perhaps some of these same people have difficulty imagining a bridge of linkages that can equate quality improvements with exactly measured costs or profits. However, for them this does not matter very much. These types of people operate under the belief that if you simply improve quality, good things, such as happier customers and higher profits, will automatically fall into place.

Other types of people prefer having fact-based data and reasonable estimates with which to evaluate decisions and prioritize spending. These types of people do believe in quality programs, but in complex organizations with scarce idle resources, they prefer to be more certain they know where it is best to spend the organization's discretionary money.

Some quality managers have become skeptical about measuring the COQ. They have seen increasing regulations and standards, such as the ISO 9000 series, where installing any form of COQ measurement was perceived as more of a compliance exercise to satisfy documentation requirements to become "registered," rather than a benefit to improve performance.

Some people perceive quality and cost as an investment choice, implying that there is a trade-off decision. This thinking assumes that achieving better quality somehow costs more and requires more effort. This is not necessarily true. If quality programs are properly installed, productivity can be improved while also raising customer satisfaction. These two combined eventually lead to increased sales, market penetration, and higher profits and returns.

Managers in the quality field have seen a number of quality programs and tools come along. Some have fallen short of their initial promise. The ISO 9000 series is the popular international standard. It addresses not only products and service lines, but also the processes and policies of an organization. The benefits of the ISO 9000 accreditation included relieving buyers of redundant supplier assessments, expansion of assessments to the suppliers' suppliers, protection against product liability litigation, and a firm foundation upon which organizations could potentially further develop their quality development. However, a disadvantage of ISO 9000 is that it represents only a minimum standard, perhaps insufficient to induce competitive-advantage behavior. Also, because it is written in general terms, with a manufacturing origin, it is open to interpretation with ambiguities for service sector organizations. Some complain that ISO 9000 serves as a documentation tool with little extension to apply as a managerial tool.

Now Six Sigma is vying to exhibit staying power as a quality management program. Will it succeed, or is there an inherent flaw? Six Sigma is viewed as a paradigm shift in the quality arena. Veterans of quality management believe that quality just for quality's sake—meaning conformance to standard—is not good. This sounds paradoxical. Quality is obviously needed to capture and retain customers, but quality must also be applied to the business itself. Six Sigma ensures that there is emphasis on the conversion and the paperwork-related transaction processes as well. But Six Sigma goes much farther and also suggests consideration of the business' financial health.

A popular definition of quality preferred by Six Sigma advocates is quality is a state in which *value* entitlement is realized for the customer and the supplier (i.e., employees and shareholders) in every aspect of the relationship. It is predictable that there will be debates about trade-offs among shareholders, customers, employees, taxpayers, and the environment. The methods of COQ measurement will be useful to convert debates into agreements [5].

This new perspective acknowledges that investing additional capital intended to reduce defect rates will not be sustained unless shareholders and lenders are assured of high-quality financial returns. In Six Sigma, financial data to support manager proposals for projects are absolutely required. So, just like customers who demand utility-value, owners, investors, and lenders have a rightful expectation of profit-value and wealth creation.

This broader notion of quality is well beyond the narrower TQM of the 1990s. For producers, it is no longer enough to just make and deliver quality goods and services. A quality business must exist as well. The intent of Six Sigma is to refocus on business economics as the driver of quality improvements.

WHAT IS QUALITY?

Before discussing the various costs of quality and how to measure them, one should have a definition of quality itself. To some quality might mean durability or richness in a product or a pleasurable experience. This is a "fitness for use" definition that relates to a customer's needs. In the 1980s, a predominant supplier-oriented view defined quality as being a high conformance to the buyer's requirements or specifications, usually measured at the time of final product test. One of the risks of limiting the definition of quality to a supplier "doing things right" is that it can miss the customers' *real* needs and preferences.

More recently, quality has been considered from a customer satisfaction and loyalty orientation to meet or exceed customer requirements and expectations. This shifts the view from the sell side to the buy side. There has been substantial research about customer preferences, both stated and subconscious, with elaborate survey questionnaires, diagnostics, and conjoint statistical analysis. For example, "food" may be a customer's stated need, but "nourishment or a pleasant taste" is the real primary needs. A customer's ultimate perception of quality involves many factors. In short, the universally accepted goals of quality management are lower costs, higher revenues, delighted customers, and empowered employees.

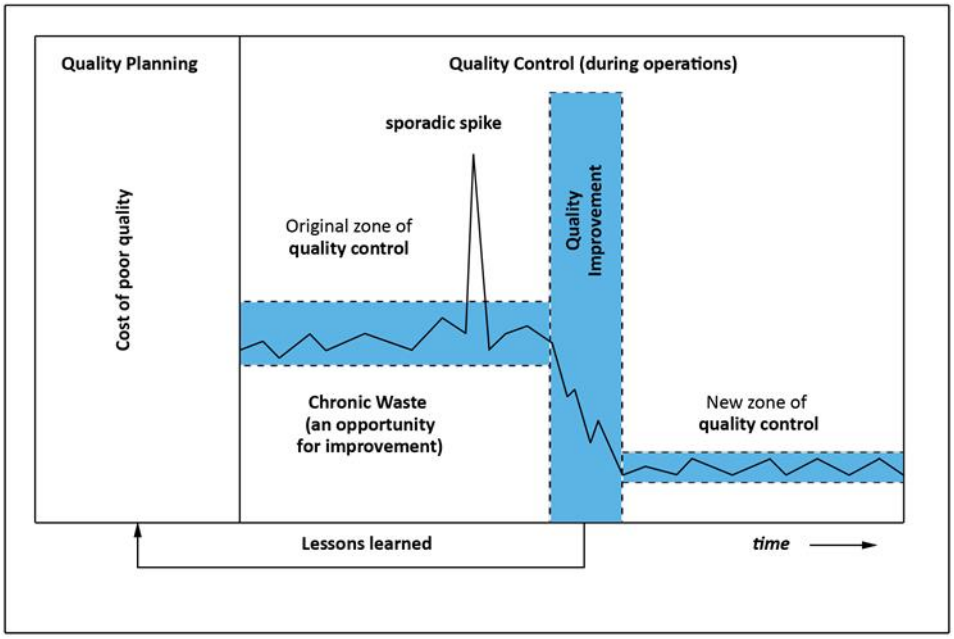
Impact of Poor Quality

Almost every organization now realizes that not having the highest quality is not even an option. High quality is simply an entry ticket for the opportunity to compete. Attaining high quality is now a must. Anything less than high quality will lead to an organization's terminal collapse. In short, high quality is now a prerequisite for an organization to continue to exist. The stakes are much higher.

The quality techniques that have been applied in the past, however, are still relevant. One of leaders in the quality movement, Joseph M. Juran, has described managing for quality by using three managerial steps, called "Juran's Trilogy":

- **Quality planning**: Translating customer needs into characteristics of products and service lines (e.g., quality function deployment analysis).
- **Quality control**: Measuring quality levels and comparing them against desired levels (i.e., removing *sporadic* deficiencies).
- **Quality improvement**: Implementing incremental improvements to attain better levels of control (i.e., removing *chronic* deficiencies) [6].

Figure 21.2 shows that each step leads to a result used in the next step.



Source: *Juran's Quality Handbook, McGraw-Hill, 1999. Permission to re-print granted.*

Figure 21.2—Juran's Trilogy

In Figure 21.2, sporadic problems are those that periodically occur and are dealt with shortly after they happen. In effect, the problem is quickly corrected until the process or off-spec output is returned to an acceptable level. Sporadic problems will likely continue to recur because the solution is usually more a bandage than a real cure.

Chronic problems, in contrast, have usually existed for an extended period of time, and may be accepted or tolerated by the organization as known but unresolvable. Examples include poor communications or inadequate tools for workers. Employees are often resigned to the existence of chronic problems. They are undesirable, but expected to persist, because they have been subconsciously designed and planned into the processes and procedures.

Organizations tend to concentrate on sporadic problems, because when they occur there is usually an adverse consequence, such as a customer complaint or a missed delivery date. But the fix may not necessarily be lasting. In contrast, the elimination of chronic

problems requires greater effort. The solution may be the result of forming a project team that produces an innovative solution. The problem analysis will likely be more intent on truly understanding the root cause. When root causes of chronic problems are removed, improvements in performance and costs can be substantial.

In Figure 21.2, sporadic problems can spike from an unplanned event, such as a power failure. Immediately following these events teams troubleshoot and "put out the fire," which restores the error level back to the status quo—the planned chronic level. The figure also reveals that after a quality improvement initiative addresses the process, the level of error is driven downward.

Another leader in the quality movement, W. Edwards Deming, advocated a similar and now well-accepted set of steps with his "Plan-Do-Check-Act" (PDCA) cycle, an iterative approach to achieving preventive and corrective solutions [8]. Some now have reduced PDCA to a more simple "Do-and-Reflect." Regardless of the quality techniques applied, financial measures will be increasingly relevant as organizations move from decisions based on instinct and intuition toward fact-based decisions.

CATEGORIZING QUALITY COSTS

To some people, quality costs are very visible and obvious. To others, quality costs are understated; and they believe that much of the quality-related costs are hidden and go unreported.

There are several levels of non-error-free quality costs, as illustrated in Figure 21.3. The following discussion of scope is restricted to the inner concentric circles, although there are additional quality costs.

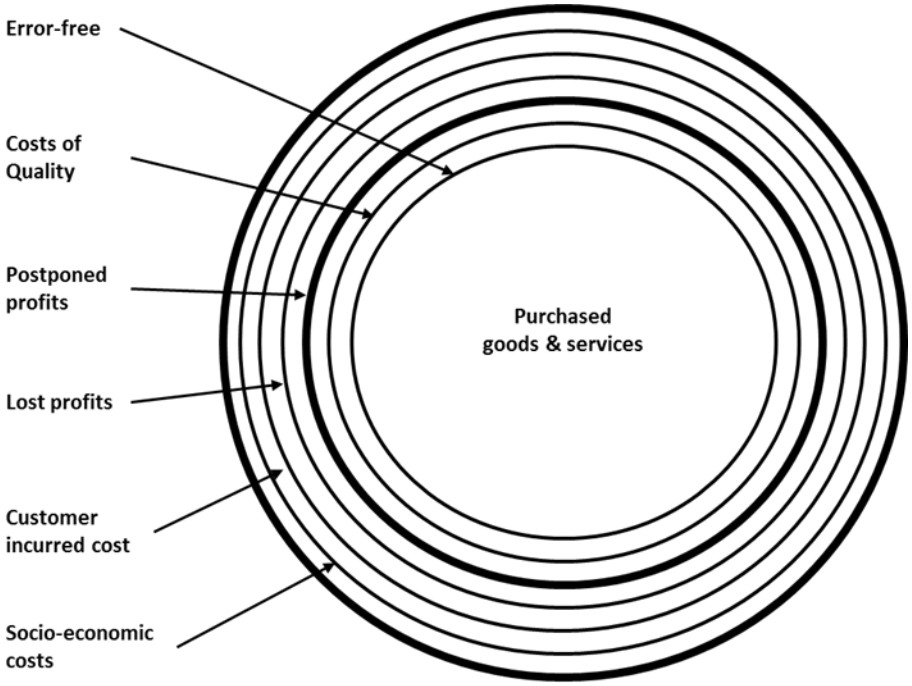


Figure 21.3—Levels and Scope of Quality Costs

Figure 21.3 begins to reveal that there are other hidden financial costs and lost income opportunities beyond those associated with traditional obvious quality costs. Examples of obvious quality-related costs are rework costs, excess scrap material costs, warranty costs, and field repair expenses. These typically result from errors. Error-related costs are easily measured directly from the financial system. Spending amounts are recorded in the accountant's general ledger system using the "chart-of-accounts." Sometimes the quality-related costs include the expenses of an entire department, such as an inspection department that arguably exists solely as being quality-related. However, as organizations flatten and de-layer and employees multitask more, it is rare that an entire department will focus exclusively on quality.

The hidden poor quality costs, represented in the figure's outer circle, are less obvious and are more difficult to measure. For example, a hidden cost would be those hours of a few employees' time sorting through paperwork resulting from a billing error. Although these employees do not work in a quality department that is dedicated to quality-related activities, such as inspection or rework, that portion of their workday was definitely quality-related. These costs are not reflected in the chart-of-accounts of the accounting system. That is why they are referred to hidden costs.

The lack of widespread tracking of the Cost of Quality (COQ) in practice is surprising because the tools, methods, and technologies exist to accomplish reporting COQ. A research study investigating the maturity of COQ revealed that the major reason for not tracking COQ was lack of management interest and support in tracking COQ and a belief that quality costing is paperwork that does not have enough value [9]. Other major reasons for not tracking COQ were a lack of knowledge of how to track costs and benefits of COQ, as well as a lack of adequate accounting and computer systems. Given the advances in today's data collection, data warehousing, data mining, and ABC/M system implementations, these reasons begin to appear as lame excuses—technology is no longer the impediment for reporting COQ that it once was.

Providing employee teams with visibility of both obvious and hidden quality-related costs can be valuable for performance improvement. Using the data, employees can gain insights into causes of problems. The hidden and traditional costs can be broadly categorized as:

- **Error-free costs**: Costs unrelated to planning of, controlling of, correcting of, or improving of quality. These are the did-it-right-the-first-time (nicknamed "dirt foot") costs.
- **COQ**: Costs that could disappear if all processes were error-free and if all products and services were defect-free. COQ can be subcategorized as:
    - **Costs of conformance**: the costs related to prevention and predictive appraisal to meet requirements.
    - **Costs of non-conformance**: the costs related to internal or external failure, including detective appraisal work, from not meeting requirements. The distinction between internal versus external is that internal failure costs are detected prior to the shipment or receipt of service by the customer. In contrast, external failure costs result usually from discovery by a customer.

An oversimplified definition of COQ is the costs associated with avoiding, finding, making, and repairing defects and errors (assuming that all defects and errors are detected). Cost-of-quality represents the difference between the actual costs and what the reduced cost would be if there were no substandard service levels, failures, or defects.

Simple examples of these categories for a magazine or book publisher might be as follows:

- **Error-free**— "First time through" work without a flaw.
- **Prevention**—Training courses for the proofreaders; preventive maintenance on the printing presses.
- **Appraisal**—Proofreading.
- **Internal** —Unplanned printing press downtime; correction of typographical errors.
- **External failure**—Rework resulting from a customer complaint.

There are other quality-related costs depicted in the outer levels that are somewhat more difficult to measure, but may be relevant in decision analysis. These additional concentric rings of costs are supply chain-related:

- **Postponed profits (current)**: Profits that could not be formally recognized during a specific financial accounting period, because the goods and services did not satisfy all of the customer's requirements. The impact is deferred cash inflow.
- **Lost profits (permanent)**: The sales and profit opportunity permanently lost when a customer elects to switch to a competitor, or substitute, or no longer purchase because of a bad experience.
- **Customer incurred costs**: All of a customer's COQ (plus postponed and lost profits from the customer's customers) caused by the supplier's nonconformance. Examples include the customer's own rework, its equipment repair, or its tarnished name as a result of reduced service levels.

Some people may argue that an additional level of socioeconomic costs exists where the public and community are affected, such as when an oil spill or pollution occurs. This is represented in Figure 21.3 as the most outside concentric ring of costs.

Figure 21.4 uses a pie chart to portray, in financial terms, how an organization's sales, profits, purchased materials, and COQ expenses might exist in an organization. In principle, as the COQ expenses are reduced, they can be converted into higher bottom line profits.

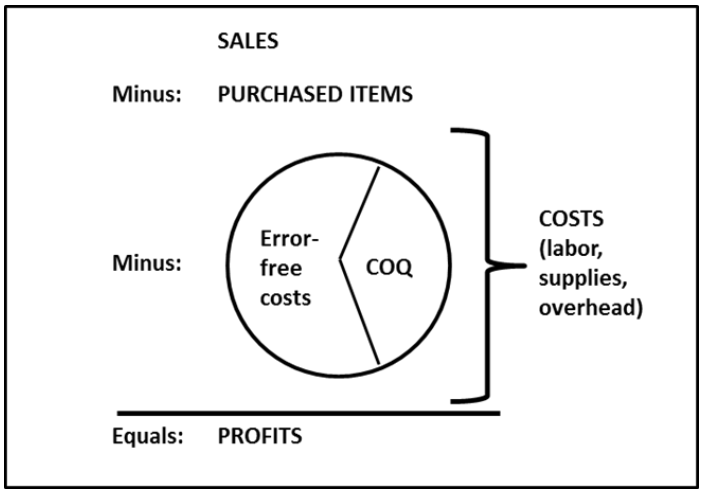


Figure 21.4—Sales, Costs, and Profits

Figure 21.5 illustrates how attributes can be tagged or scored into increasingly finer segments of the error-free and COQ subcategories. Attributes are tagged to individual activities, for which the activities will already have been costed using Activity Based Cost Management (ABC/M). Hence, the subcategory costs can be reported with an audit trail back to which resources they came from. Each of the sub attributes can be further subdivided with deeper "indented" classifications.

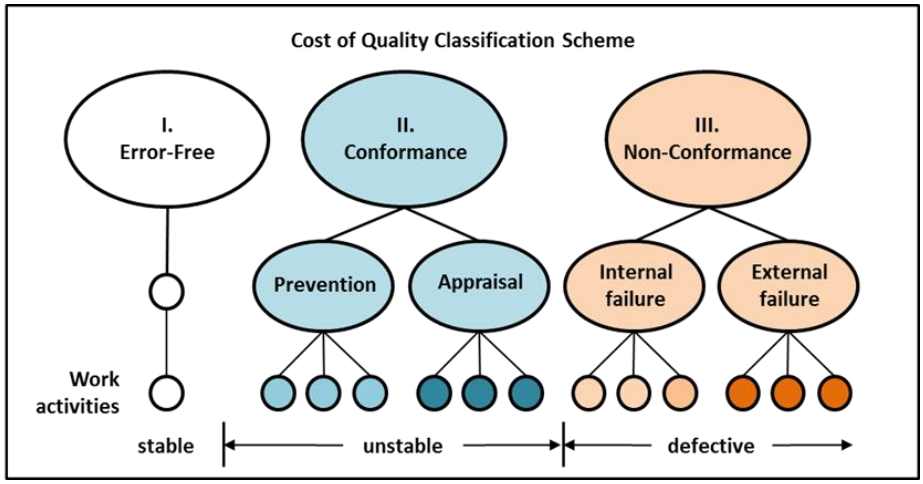


Figure 21.5—ABC/M's Attributes Can Score and Tag Costs

Because 100 percent of the resource costs can be assigned to activities, 100 percent of the activities can be tagged with one of the COQ attributes, since the activities have already been costed by ABC/M. (Read the section at the end of this chapter, "ABC/M is a Cost Re-Assignment Network" to learn more about ABC/M). The attribute groupings and summary roll-ups are automatically costed as well.

Life would be nice in an error-free world, and an organization's overall costs would be substantially lower relative to where they are today. But, all organizations will always make mistakes and errors. They will always experience some level of errors. However, the goal is to manage mistakes and their impact. Cost-quality serves to communicate fact-based data—in terms of money—to enable focusing and prioritizing to manage mistakes.

As previously mentioned, unless an entire department's existence is fully dedicated to one of the COQ subcategories, or an isolated chart-of-account expense account fully applies to a COQ category, most of the COQ spending is hidden. That is, the financial system cannot report those costs.

A danger exists if only a fraction of the quality-related costs are measured and their amount is represented as the total quality costs—this is a significant understating of the actual costs. Unfortunately, there are as many ways of hiding quality costs as there are

people with imagination. Organizations that hide their complete COQ from themselves continue to risk deceiving themselves with an illusion that they have effective management. ABC/M is an obvious approach to making visible the missing COQ amount of spending.

WHERE DO QUALITY COSTS RESIDE IN THE ACCOUNTANT'S FINANCIAL STATEMENT?

It may make it easier to think of the sum total of all of the cost categories (i.e., error-free or COQ) as equating to the total expenditures during a time period, less purchased material costs, that comprise a company's budget statement. For any non-profit organization, their budget-funded expenditures "equal" error-free costs "plus" COQ. If there are fees or revenues, then these are simply added to the budget-funded amount.

One hundred percent of the total expenditures can be accounted for and included. Figure 21.6 illustrates the distribution of quality costs using the sub-categories that were described in Figure 21.5. An example of a commercial company will be used to demonstrate how quality can be measured financially. The parallels to a government or non-profit organization are high.

In this example, a fictitious manufacturer with revenues of $200,000 enjoys a healthy five percent profit-to-sales rate and purchases roughly half of its expenditures, $90,000, from its suppliers. The remaining expenses of $100,000 were not directly purchased from external suppliers; 80 percent of these were error-free, but 20 percent involved quality-related work. In advanced COQ analysis, portions of the external purchases, such as for contract labor, can also be included for segmentation as error-free versus COQ. Externally purchased services are rarely perfect.

Sales $200,000
Expenses
Purchased direct material < 90,000>
Labor, supplies and overhead <100,000> this equates to the total expenses not purchased from suppliers (i.e., error-free + COQ)
Profit $ 10,000 (5 percent of sales)

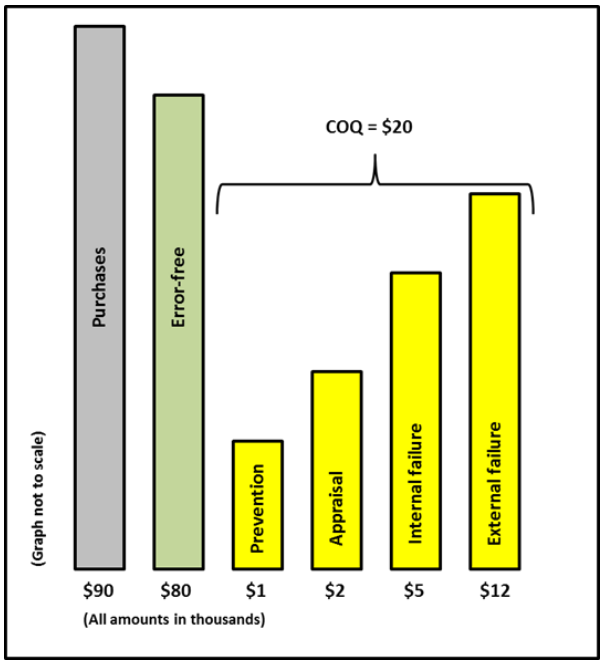


Figure 21.6—COQ Histogram

In Figure 21.6, the majority of the costs of $20,000 quality-related costs are classified as external failure ($12,000). That is, the product or service-line was unsuccessful after being received by the end-customer. In theory, if half of that type of COQ expense (i.e., $6,000) could be eliminated, and presuming the freed-up capacity can be re-deployed elsewhere, then the profits would rise by $6,000 from $10,000 to $16,000 – a 60 percent jump! The profit rate would increase from five percent to eight percent of sales.

Figure 21.7 displays how the COQ histogram from Figure 21.6 could ideally appear after both prevention and appraisal spending is initially increased, and then process efficiencies are applied to drive all of the COQ costs down.

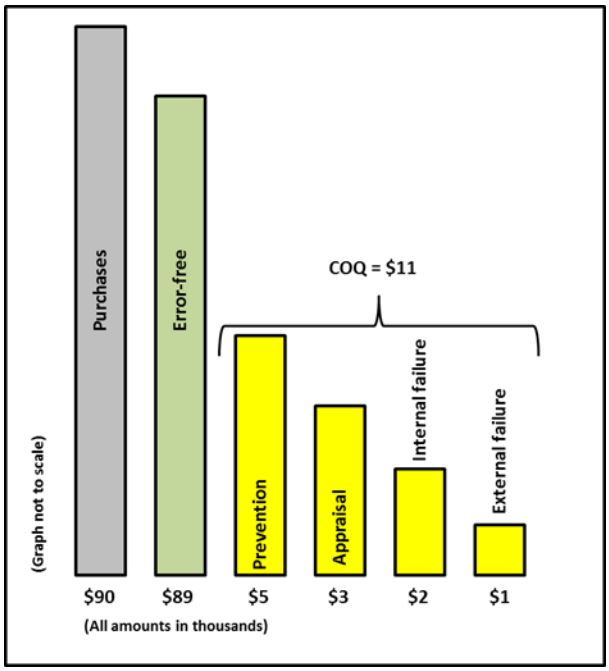


Figure 21.7—Getting Efficient with Conformance-related COQ

BENEFITS FROM INCLUDING TOTAL EXPENDITURES WHEN MEASURING QUALITY

Starting the measurement by assuming a 100 percent inclusion of the total expenditures, then subsequently segmenting those expenses between the error-free costs and the COQ:

- **Reduces Debate**: With traditional COQ measures, people can endlessly debate whether a borderline activity is a true COQ, such as scrap produced during product development that may arguably be expected. Including such a cost as COQ may inflate a measure that is of high interest. By excluding it, that expense melts away without any visibility into all the other total expenditures of the organization. It can be tempting for controversial costs to be excluded as a quality-related cost category. By starting with the 100 percent expenditure pool, *every* cost will fall into some category and always be visible. Each type of cost can always be reclassified later on, as people better understand how to use the data.
- **Increases Employee Focus**: By developing classifications into which all costs can be slotted, organizations will hopefully focus much less on their methods of measurement and focus much more on their organization's problems and how to overcome them.
- **Integrates With the Same Data Used in the Boardroom**: When traditional and obvious COQ information is used, only portions of the total expenditures are selected for inclusion. This invites debate about arbitrariness or ambiguity. However, when 100 percent of expenditures are included, the COQ plus error-free costs exactly reconcile with the same data used by executive management and the board of directors. There is no longer any suspicion that some COQ has been left out, or that the COQ data are not anchored in reality. By starting with 100 percent expenditures, the only debate can be about misclassification, not omission.

The capture of COQ can be further refined if it is worthwhile for the organization.

When making decisions, the universally popular costs-versus-benefits test can be applied with COQ data. If either subcategory of COQ is excessive, it draws down profits for commercial companies or draws down resources in government agencies that could have been better deployed on higher-value-added activities elsewhere.

GOALS AND USES FOR THE COST-OF-QUALITY INFORMATION

If an organization makes the effort to collect data, validate the information, and report it, it might as well *use* the information. In fact, to state the obvious, the amount of use of and utility in the information will be proportional to the length of life of the COQ measurement system. In short, the uses of a COQ measurement system can range from favorably influencing employee attitudes toward quality management by quantifying the financial impact of changes to assisting in prioritizing improvement opportunities.

The rationale for implementing COQ is based on the following logic:

- For any failure, there is a root cause.
- Causes for failure are preventable.
- Prevention is cheaper than fixing problems after they occur.

If you accept the logic that it is always less expensive to do the job right the first time, than to do it over; the rationale and goal for quality management and using COQ to provide a quality program with concrete and fact-based data should be apparent.

Implementation involves the following steps:

- Directly attack failure costs with the goal to drive them to zero.
- Invest in the appropriate prevention activities, not fads, to effect improvements.
- Reduce appraisal costs according to results that are achieved.
- Continuously evaluate and redirect prevention efforts to gain further improvement.

Figure 21.8 illustrates the direction in which quality-related costs can ideally be managed. Ideally all four COQ cost categories should be reduced, but one may initially need to prudently increase the cost of prevention to dramatically decrease the costs of and reduced penalties paid for nonconformance. This makes COQ more than just an accounting scheme; it becomes a financial investment justification tool.

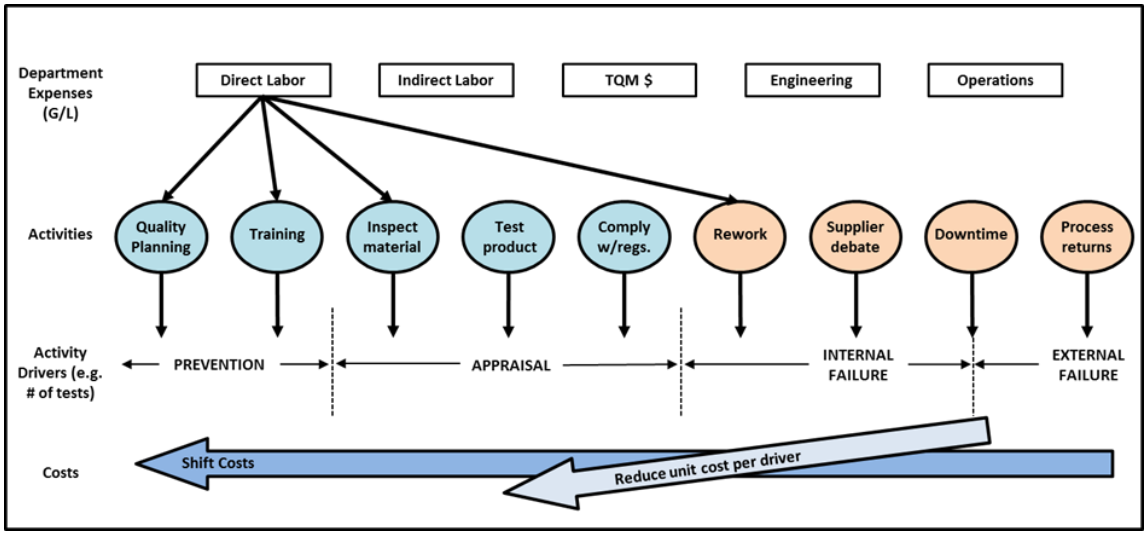


Figure 21.8—Driving Cost-of-Quality Downward

A general corrective operating principle is that as failures are revealed, for example via customer complaints, the root causes should be eliminated with corrective actions. A general rule-of-thumb is that the nearer the failure is to it being used by the customer, the more expensive it is to correct. The flip side is that it is less expensive—overall—to fix problems earlier in the business process. As failure costs are reduced, appraisal efforts can also be reduced in a rationale manner.

QUANTIFYING THE MAGNITUDE OF THE COSTS OF QUALITY

The formal COQ measurement system provides continuous results. In contrast to a one-time assessment, it requires involvement by employees who participate in the business processes. More important, these employees must be motivated to spend the energy and time, apart from their regular responsibilities, to submit and use the data.

Commercial ABC/M software products were designed for frequent repeated updating. For such a COQ system to be sustained longer-term, the system requires senior management's support and interest as well as genuinely perceived utility by users of the data to solve problems.

Regardless of the collection system selected, it is imperative to focus analytical and corrective time and energy on the area of failure costs. As Dr. Joseph Juran discussed in his highly popular article, "*Gold in the Mine*," there is still much "mining" that can be performed [7]. This mining should be considered a long-term investment, because failure costs, when starting a quality management program, usually constitute 65 to 70 percent of a corporation's quality costs. Appraisal costs are normally 20 to 25 percent and prevention costs are five percent.

CONTINUOUS IMPROVEMENT WITH BENCHMARKING AND UNIT COSTS

Tagging attributes onto costs is obviously a secondary purpose for measuring costs. The primary purpose for costing is to simply know what something costs. This data allows you to measure profit margins, to focus on where the larger costs are that may be impacted, or to estimate future costs to justify future spending decisions (e.g., ROI). In short, managerial accounting transforms resource expenses into calculated costs. That is, expenses are posting's to the general ledger bookkeeping system to recognize exchanges of money to vendors and employees. Expenses are purchases of resources. In contrast, costs are the uses of that spending. Costs are always calculated. Many organizations arbitrarily allocate costs based on broadly-averaged volume factors, but the proper rule is to trace and assign the expense based on a one-to-one cause-and-effect relationship.

When an organization has good costing, then it can use calculated cost consumption rates, such as the cost per processed invoice, as a basis for comparison. In short, the (*unit cost per each output of work)* is computed and then this data is usable for benchmarking—either internally or with other organizations.

The use of ABC/M data is becoming more popular as a metric for benchmarking. Often in benchmarking studies, there can be a bad case of apples-to-Oreo's. That is, there is lack of unrecognized comparability among the participating organizations. There is lack of consistency among what work activities or outputs are to be included or excluded in the study. An ABC/M methodology and system introduces rigor and is sufficiently codified and leveled for relevancy as to remove this nagging shortcoming of benchmarking. For additional information on Activity Based Cost Management, also refer to Chapter 8 "Activity Based Cost Management."

In practice, the vast majority of ABC/M is applied to subsets of the organization for process improvement, rather than revenue enhancement and profit margin increases. An example of a subset is an order-processing center or equipment maintenance function. These ABC/M models and systems are designed to reveal the cost structure to the participants in the main department and related areas. In ABC/M's cost assignment view, the cost structure is seen from the orientation of how the diversity and variation of the function's outputs cause various work to happen, and how much. The costs of the work activities that belong to the processes are also revealed in the ABC/M model as they relate in time and sequence. However, it is ABC/M's powerful revealing of the costs of various types of outputs that serves as a great stimulant to spark discussion and discovery. For example, if an order-processing center learns that the cost per each adjusted order is roughly eight times more costly than for each error-free or adjustment-free entered order that would get people's attention. This result happens even if the order entry process has been meticulously diagrammed, flowcharted, and documented.

DECOMPOSING COST-OF-QUALITY CATEGORIES

In effect, the technique to calculate a reasonably accurate COQ is to apply ABC/M and ABC/M's attribute capability. Figure 21.9 shows categories for work activities that are one additional level below the four major categories of COQ. This figure reveals how each of these subcategories can be tagged against the ABC/M costs. This provides far greater and reliable visibility of COQ without the great effort required by traditional cost accounting methods.

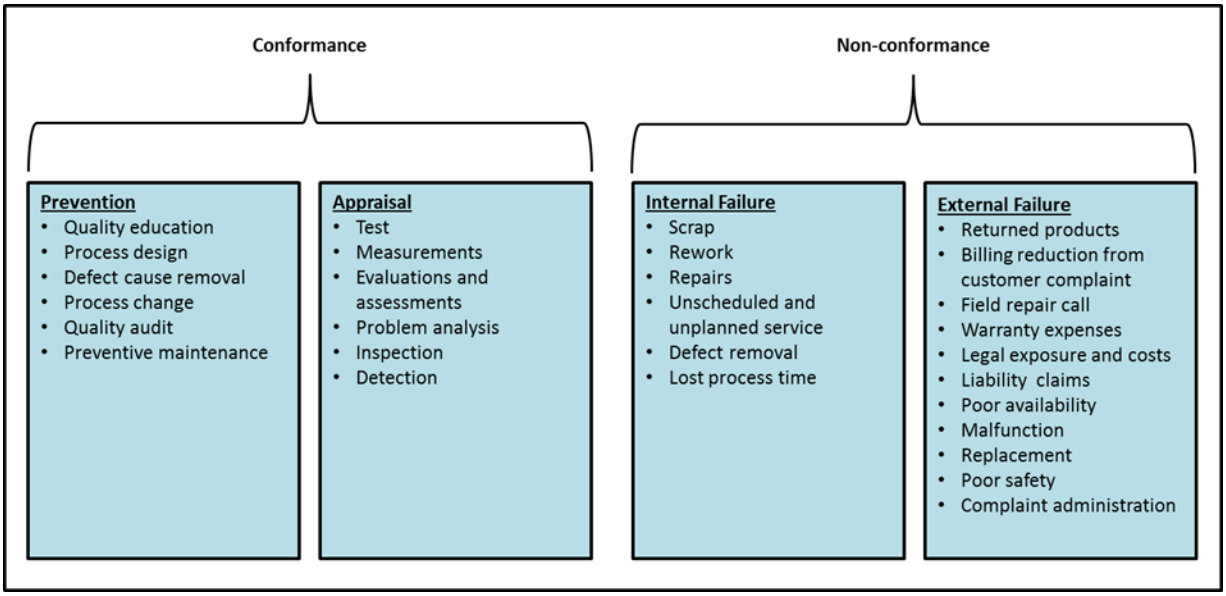


Figure 21.9—Typical Cost-of-Quality Components

The quality movement has been a loud advocate for measuring things, rather than relying on opinions. It would make sense that measuring the financial implications of quality will become an increasingly larger part of the quality management domain.

CONCLUSION

The reputation of quality management movement has experienced a few waves of ups and downs. It had become almost religious-like for some years and then ridiculed. Hopefully, the addition of valid costing data will give the quality movement more legitimacy. In a publication from one of the key sanctioning quality societies, The American Society for Quality (ASQ), there was a key definition. ANSI/ISO/ASQ Q9004-2000 suggests financial measurement as an appropriate way to assess, "the organization's performance in order to determine whether planned objectives have been achieved" [2]. Hopefully, there will be increased coordination among the quality, managerial accounting, and operations system.

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RECOMMENDED READING

1. AACE Recommended Practices:
   [**Note**: There are multiple Recommended Practices that apply both directly and indirectly to Quality Management concepts and practices].
2. AACE International (2011). Virtual Total Cost Management (VTCM). Morgantown, WV: AACE International.
3. AACE International (2015). *Total Cost Management Framework*, (2nd ed.). Morgantown, WV: AACE International.

Chapter 22—Value Engineering

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Abstract
Value engineering techniques came about from the problem with wartime shortages encountered during World War II. In order to meet the all-consuming demand for military hardware and supplies, industries with non-military production needs were placed on allocation, or simply could not procure needed items. Because of this dilemma, the techniques of value engineering were born of necessity. If a needed item could not be procured, then what could take its place? An example might be the lack of a pump to fill an industrial storage tank. If a pump was not available, would gravity flow or some other method solve the problem? After World War II, these techniques migrated into other areas of endeavor, including government, construction, and retail, along with spreading to other countries around the world. Now many owners, including government agencies, mandate the application of value engineering procedures as part of the procurement practices.

Keywords
Cost Value, Esteem Value, Exchange Value, Level of Influence, Pareto's Law, and Value Engineering.

INTRODUCTION
This chapter presents the concepts of value engineering. The term "value engineering" lacks a common definition. According to some people, value engineering refers to only design concepts, whereas the other similar terms, such as value analysis or value management, are in the business and management areas. However, in this chapter, value engineering is considered synonymous with value analysis or value management. This chapter offers a definition for value engineering, describes the benefits of implementing value engineering concepts by different organizations, discusses the different types of values as it relates to engineering, and provides an overview of the value engineering job plan.

Learning Objectives
After completing this chapter, the reader should be able to:

- Express a more detailed understanding of value management.
- Identify the detailed steps of the value management process.
- Conduct a value management workshop.
- Conduct a value management review process.
- Identify the practical steps for implementation of the value management process.
- Understand value management implementation and its impact on the budget and schedule.

VALUE ENGINEERING DEFINED
Some use the term value engineering as shorthand for a cheapening process. Instead, value engineering has the goal of lowest life cycle costs. The benefits of value engineering accrue to a variety of parties, including the producer and owner. Constructability is considered by some as synonymous with value engineering, and in limited respects it is, but actually it is more focused on only the field construction process in reducing overall construction costs. Value engineering is broader in scope than just constructability, wherein this could instead be viewed as a subset.

For these purposes, value engineering is best defined as an organized systematic approach that delineates the essential functions required by the user of the project, product or service, while establishing appropriate function costs based on a life-cycle analysis.

Value Engineering Benefits
If an organization engages in value engineering, what is the expected return on this investment? As a general rule, the more complex the project, product, or service, then there will be more opportunity for value engineering to prove successful. Evidence has shown that simple projects, such as warehouse construction, benefit much less from value engineering as opposed to a more complex building type. While almost anything, (whether product, project, or service), can benefit from value engineering, Pareto's Law of Optimality, which states that 80 percent of the costs come from 20 percent of the items, is an important caution given limited resources. Therefore, to achieve maximum benefit in value engineering, focus on those high-cost areas. Over the years, studies by private sector and public sector owners have found typical returns of 300 percent to 400 percent or more, as compared to the value engineering costs. That is to say that for every dollar spent on value engineering, a return of from three to four dollars can be expected from this activity. Therefore, value engineering can prove to be a powerful tool in an increasingly-competitive world market.

Why is There Poor Value?

The goal of improved value from a production, construction, or service process, would seem to be just common sense. It would be thought that improved value is a key goal that is at the forefront of all efforts. If this is the case, then why would organizations find that value engineering can return significant benefits? It has been found that poor value can result from one or a number of factors. Consider the following:

- **Poor Attitudes**—In some cases people and their organizations become locked into certain poor attitudes, and thus are not receptive to better ideas. People may reject an idea out of hand, that was successfully used in another field of endeavor, because "it was not invented here."
- **Poor Habits**—Everyone is a creature of habit. Some habits are good and some are poor. Automatically specifying a high level of concrete finish everywhere on a project, may be appropriate in many circumstances, but unnecessarily drives up costs when significant areas of that concrete work will be backfilled or covered up with other finishes anyway.
- **Poor Ideas**—In any organization, poor ideas may surface in a project, product, or service, because no one at the time has come up with a better concept. Creative and organized analysis applied to the issue should result in a better idea than the existing situation.
- **Poor Information**—Technology is advancing every day. An optimal solution yesterday may be obsolete or supplanted by a better method today. For example, in consumer electronics, VHS tapes have been supplanted by DVD formats that in turn will be supplanted tomorrow by new advances. The DVD format offers higher quality, more durability, faster production and enhanced navigation characteristics.
- **Time Constraints**—Time pressures abound in many organizations. As a consequence, the first solution thought of becomes implemented and work progresses onto other areas. Even though the solution may not be optimal, this then may act as a template in the future.
- **Temporary Circumstances**—As a result of a temporary situation, a problem solution is set in place. At the time it may be recognized as non-optimal, but it is done on a temporary basis. Those responsible move on, or are re-assigned, and later the temporary "solution" becomes incorporated as a standard procedure.
- **Mistaken Beliefs**—In some cases, decisions are made, not on the basis of facts, but because someone wrongly believes something that is not true. One designer had summarily dismissed plastic piping for an application, in lieu of a more expensive metal alternative, in the belief that the metal pipe was superior. In fact, for the particular application in a corrosive soils environment, the plastic pipe was actually superior to the metal alternative on a life-cycle basis.

VALUE TYPES

In considering value engineering techniques, practitioners must be cognizant of the four types of value. These four value types are cost value, exchange value, use value, and esteem value.

- **Cost Value**—This refers to the totality of labor, material, equipment, and overhead costs, involved in actually producing and bringing to market the project, product, or service.
- **Exchange Value**—This refers to the characteristics of the project, product, or service which allow it to be traded for something of value, such as currency.
- **Use Value**—This refers to the characteristics of the project, product, or service which enable it to accomplish its intended purpose.
- **Esteem Value**—This refers to the characteristics of the project, product, or service which make ownership attractive.

While some involved with value engineering would only focus on cost value, the other three characteristics have proven to be important metrics for success in an increasingly competitive marketplace. For illustrative purposes, some products such as autos, esteem value allows one vehicle to be sold for a significant premium above its actual cost. Moreover, manufacturers can damage the esteem value of an auto product, such as by poorly-chosen substitutes of plastic for chrome or plastic for leather, in this area. Several manufacturers have sought on a world-wide basis to differentiate their luxury auto products by creating entirely separate brands for this category: such as Honda (Acura), Toyota (Lexus), GM (Cadillac), and Ford (Lincoln). For example, BMW differentiates product classes by the three-series, five-series, and seven-series monikers. The seven-series has more esteem value than the five-series. Some equipment manufacturers, such as CAT and Komatsu, note that their construction equipment has a higher value at trade-in time, than their competitor's trade-in values, as seen in equipment auction results (exchange value comparison).

Level of Influence Curve

The Level of Influence Curve makes one of the strongest arguments for involving value engineering techniques at the start of a project, during the early design concept stage. This Curve was originally conceptualized by the U.S. General Services Administration, in the early 1970s time frame. While it is intended to apply to construction projects, the concept can be successfully applied to a variety of products and services with concept, production, and use/maintenance phases.

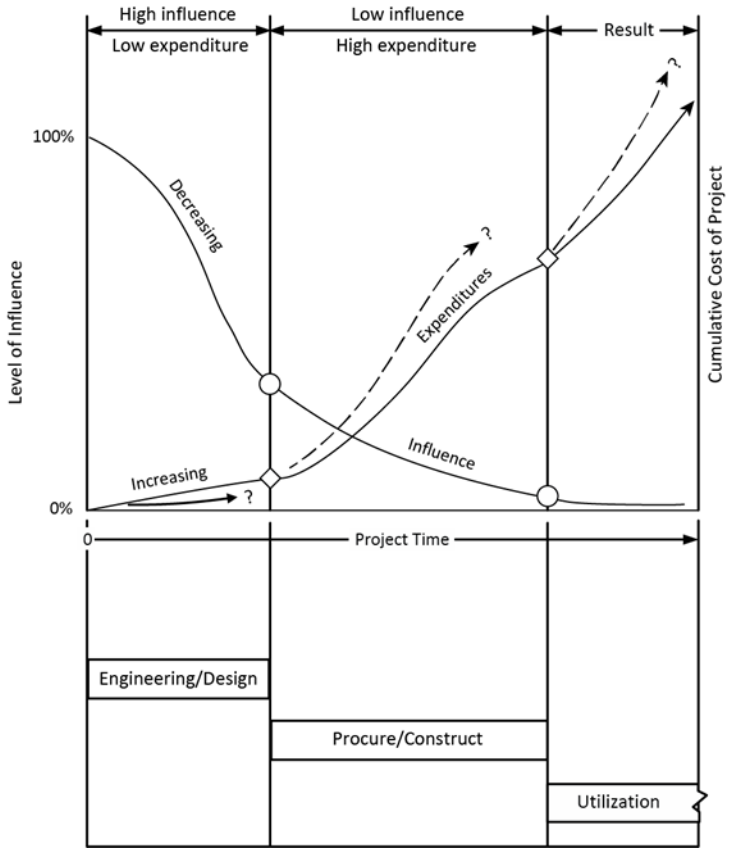


Figure 22.1—Level of Influence

For the purposes of this discussion, the focus will be on a typical building construction project. At the start of a project, there is 100 percent influence over the project cost. The decision is to build the building or not to build it. As decisions proceed in the design phase, certain cost elements of the project will be set for the building. Will the building be steel frame or reinforced concrete frame? How many floors will be in the building? What type of cooling and heating system will be in the building? While the design phase in itself is low cost, in terms of the expenditure of design funds, most of the high cost decisions are made in this phase. Hence, the categorization at the top of the graph, wherein the design phase is categorized as high influence but low expenditure. Perhaps 65 percent of the expenditure of funds is set at this point, but less than ten percent of the project cost is spent in this phase. The procure/construct phase is obviously heavily influenced by the design phase decisions. While 60 percent of the total project costs are spent in this procure/construct phase, the level of influence is approximately 35 percent over the expenditures. Finally, in the utilization phase, a significant amount of funds will be expended for building maintenance and utilities, but there is a comparatively low level of influence as to these costs. In utilization, if a high-energy cost HVAC system or high-energy cost lighting system is selected, this decision made in the design phase is not readily subject to change.

The key point of the Level of Influence Curve is that early application of value engineering techniques in the design phase makes the most sense. VE application during the construction process can be problematic. Numerous design decisions have already been reached and the construction process can see significant delays if value engineering changes require time-consuming design changes. In addition, designs have both an investment by the designers and a “pride of authorship.” Therefore, VE changes made after final design are expensive and resisted by designers for these reasons. The impact of one design change can result in The Law of Unintended Consequences where it triggers additional design changes and problems. Early-on design changes, at the schematic/conceptual phases, have a much less traumatic impact on the project.

Project Schedule

Figure 22.2 illustrates the relationship between project duration and project cost. What does this have to do with value engineering? Usually, value engineering is concerned with the cost relationships of physical items as to lowest life cycle cost.

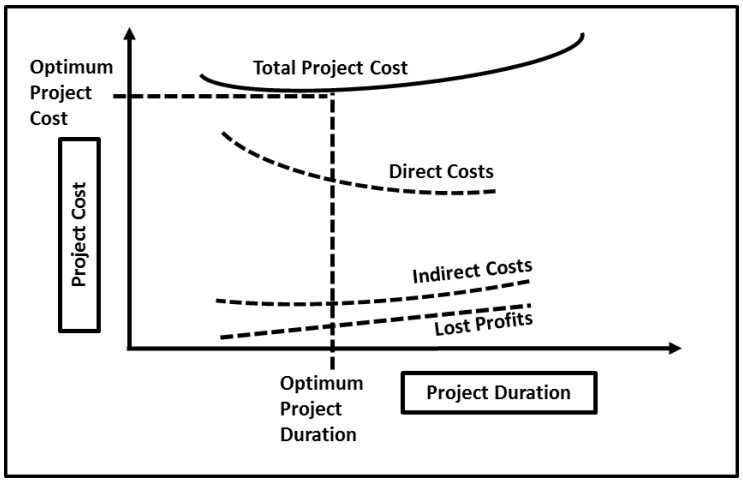


Figure 22.2—Value Engineering Project Cost Life Cycle

However, Figure 22.2 can be used to illustrate a schedule's impact on lowest life cycle cost. At the lowest point, on the total project cost curve (bottom of the "bathtub shape"), will be the optimum project duration. As the project's duration is extended, as a result of schedule delays or other reasons, increases are then seen in the total project's cost as a move up on the total cost curve. Similarly, if reduced project duration is required, costs will also be seen to increase as a move up on the total cost curve. Ultimately, there is no "free lunch," with the increase in these total project costs. One example was a 91-day construction union's strike that held up a high-rise office building project in Chicago, Illinois. At the time, the developer's financing interest costs were $110,000 a day. This strike resulted in an extra $10 million of costs having to be incorporated into the office project through higher rents. This extra rent tariff resulted in a decreased overall value for the office building project. During the strike, direct on-site costs were almost non-existent, except for security staff. However, the indirect costs of the financing, along with the lost profits of months of lost office rent, continued to increase for the developer. However, this subject strike was an unpredictable event and took place near the end of the project when substantial funds had been expended in construction. How could value engineering provide a solution in this instance? One solution might have been to agree with the unions on a "no-strike" clause for the duration of the project.

"FAST" Diagram/Functional Approach

The Functional Analysis Systems Technique (FAST) diagram was developed in the early 1960s by Charles Bytheway. The FAST diagram is designed to show the specific relationships of the various functions in relation to each other. In value engineering, the FAST Diagram acts to define, clarify, and simplify value engineering problems. The FAST diagram lies at the heart of value engineering which is that of a functional approach. In the functional approach, FAST effectively identifies unnecessary costs. In normal VE studies, sticky-backed large-size notes would be placed up on a room wall with design objectives, secondary functions, and other functions to provide communication benefits for the VE team. This is an iterative process, with numerous changes taking place, as the team continues to study the problem.

A function in the FAST diagram methodology would be defined by a verb-noun combination. To be more specific, there should be an active verb and a measurable noun. As an example, the function of a storage tank could be defined as "contain liquid." A secondary function is one that depends on another higher order function for its existence. The basic functions make the product work or accomplish the task. The supporting functions make the product sell. As an example, in one value engineering study, a refrigerator was broken down into one task with seven functions and 26 supporting functions. The FAST diagram provides a functional "family tree" listing of the functions of a product, project, or service—listed in order of importance. Other functions that happen at the same time, or are "caused by" another function, can be listed below the noted secondary function areas. The "Why" question would be as to why are we doing something? The "How" question would be as to how something is proposed to be accomplished? The "Why" and "How" analysis is the key to the power of this thinking/diagramming process. The failure of some lies in just asking the "How" questions. If one seeks to control the mice population in an area, the "How" perspective would suggest a more improved mousetrap. Consideration of the "Why" side of the analysis finds that the goal is to reduce/eliminate the mice population. This opens up possibilities, including the utilization of natural predators such as cats, poisons, or other means to interfere with their breeding cycle. Anyone of these solutions may be more effective than just the narrow focus on an improved mousetrap. The same could apply to the invention of the grass trimmer with a string line. The older method was to shear grass. The string-line grass trimmer whacks grass which provides a significant improvement in productivity and flexibility.

An example of a value engineering problem to be diagrammed, using a FAST diagram, could be a water meter. The higher order function would be "meter water." However, there are various required design objectives including: to support the meter, admit water, expel water, be convenient, attract user, and be dependable. Secondary functions under a heading, such as be convenient, would include: install meter, understand meter, service meter, and read meter. For a task, such as install meter, one has to connect

the meter to the inbound and outbound pipe sections. One would also want the meter to be relatively light weight to aid in ease of the installation process. By the use of the FAST Diagram, communication is enhanced to all those specialists participating in the value engineering process.

Pareto's Law of Optimality states that 20 percent of the elements in a product, project, or service account for 80 percent of the costs. Given this, the FAST Diagram should focus most attention of these 20 percent high-cost elements. As time permits, other functions can be included, but the high-cost areas are an obvious priority.

Value Engineering Study Team

Rather than having one person focus on a value problem, it has been found that value engineering efforts are more successful via a team approach. The team should be diversified, representing a variety of relevant specializations for the particular scope of study. As an example, the VE study of a light-rail transit project with trackage, bridges, stations and a maintenance facility should include representative specialists. These specialists could include experts from geotechnical, civil/structural, electrical, mechanical, architectural, facilities, construction, estimating, operational, and end-user areas. The blend of specialists should be broad enough to cover the key areas of the value engineering study subject.

THE VALUE ENGINEERING JOB PLAN

The systematic approach to value engineering dictates the implementation of a value engineering job plan. The heart of the value engineering job plan is the study phase, whereby specialists are convened together for a concentrated 40- to 50-hour study period. It has been found over the years, from trial-and-error experience, that this 40- to 50hour window produces the best results in an optimal time frame for the typical study. A simpler analysis, on one part or a more complex analysis, on a large project, by necessity, may shorten or widen this time frame.

Prior to this study phase is the pre-study portion of the job plan, whereby team members and a facilitator are selected for the study phase. Necessary data is also developed for the team member's use, including plans, shop drawings, specifications and costing information as part of the pre-study work. This pre-study also defines the scope limits for the VE effort. Since once the VE team starts the study phase, they will be limited to the selected time frame, every effort must be extended to have this necessary pre-study data available for the team. Ideally, this data will be provided prior to convening the team, so that they can gain familiarity prior to the study period. If this data is not available, the effectiveness of the team is compromised in its efforts. The actual study phase of the VE job plan includes six phases, which are:

- Information Phase
- Speculation Phase
- Analysis Phase
- Development Phase
- Presentation Phase
- Follow-up and Implementation Phase.

The fundamental aspects of these phases are listed in Table 22.1.

Table 22.1—Value Engineering Job Plan

Phase 1: Information	Phase 2: Speculation	Phase 3: Analysis	Phase 4: Development	Phase 5: Presentation	Phase 6: Follow-up/ Implementation
What is it? What does it do? What does it cost? What is the worth of the basic functions?	What else will perform the basic functions?	What does each feature cost? Will each perform the basic functions?	Will it work? Will it meet requirements? What are the problems? What are the costs? What are the savings?	Make the presentation with written and verbal information Present the problem Explain before and after Realistically list advantages and disadvantages Remove roadblocks Secure commitments Use sound human relations	Follow-up to facilitate the implementation Offer assistance to the decision maker Note potential roadblocks Suggest an action plan

Phase One—Information Phase

Phase One provides the foundational basis for the value engineering study. This phase investigates the purpose of the element or project or sub-project. From this point, costs and methods are then analyzed for the purposes of the study. While questions such as: (a) What is it? (b) What does it do? (c) What does it cost? and (d) What is the worth of the basic functions? seem to be easy questions; answering them can often take a considerable research effort. Answers to these questions can be gained from a variety of sources such as design, production, and manufacturing, purchasing, and marketing areas. There is no limit to the amount of information that can be gathered in Phase One. Sometimes even seemingly irrelevant data may ultimately prove to be helpful to the VE study effort. Costing information, while critical, has various levels of required accuracy. On a major construction project, a five-dollar cost item may be viewed as trivial. However, on a manufactured part that sells for 20 dollars with hundreds of thousands of pieces produced, a one-dollar cost item is crucial. For the construction situation, the five-dollar cost is akin to a rounding error in an accounting statement; whereas, in the manufacturing situation, the one-dollar cost represents 10 percent of the production price. To help sort these issues out, a FAST Diagram is developed and accompanied by relevant functional cost information. The FAST Diagram then serves to highlight cost visibility.

Phase Two—Speculation Phase

Phase Two is the creative section of the analysis work. This phase is also referred to as "brainstorming." The key question is what else could do the job? The strength of the VE study effort is the diversified team that aggregates a variety of specialists from different areas. All involved should be free to and encouraged to propose alternate ideas. The overarching principle in this phase is that there are no bad ideas. No criticism of anyone's suggestion is allowed in this phase. In fact, historically, what at the time may seem an outlandish idea has, through later refinement, proved to be part of the overall solution on many value engineering efforts. Team members must learn to be creative and free their minds in the search for potential solutions. Team members cannot be critical and creative at the same time. The mantra is to try everything. Create by function, eliminate functions, and simplify are all techniques used in this phase. What can substitute for something else, or can there be a combination of functions. All of the ideas generated are recorded during this speculation phase.

Phase Three—Analysis Phase

Phase Three looks at what something will cost and if it will work? Dollar signs are placed on every decision. A common technique is to group ideas by similarity, such as all ideas that deal with the mechanical areas of a construction project, would be grouped into piping, HVAC, fire protection, and other relevant categories. Ideas generated in the speculation phase are now looked at with the determination to eliminate nonsense. The goal is to determine if something is feasible, along with suitability for the intended purpose.

Phase Four—Development Phase

This is the phase where the development of final ideas takes place. The team selects the best ideas in each category along with the best alternates to those ideas. In order to create change, those in charge must be convinced that the new idea represents a better concept than the existing situation. In order to do this, decision makers must be confronted with convincing facts. Roadblocks must

also be anticipated by the VE study team members. These roadblocks include well-worn responses as, "it's not practical," or "we've tried that in the past and it didn't work," or "that just won't work," or "if this idea is so good, how come nobody is using it?" The team needs to overcome these common objections in order to be able to implement the value engineering changes. It may be that something in the past was tried and did not work, but it may be because of improper application, lack of real change effort, or inadequate commitment of necessary resources. Teams must be realistic in the development phase and avoid overstating benefits.

Phase Five—Presentation Phase
Material is prepared and presented to the decision maker(s) in this phase. The groundwork for success was laid in the work involved with the development phase. The presentation phase is where the results of the value engineering effort are "sold" to those responsible for accepting the change. The emphasis has to be on realistic benefits. Decision makers will demand proof of life cycle benefits, and exaggerated claims in this phase are spectacularly unsuccessful. Moreover, in some instances, exaggerated claims and a "blue smoke and mirrors" approach may sour those in charge on the whole concept of implementing value engineering. Ratios of return to investment must be grounded in reality.

The presentation phase is most effectively accomplished with a verbal presentation, accompanied by written documentation. The best spokesperson should make the presentation and the other members of the team should be present to answer technical questions. Benefits require strong backup and documentation. The spokesperson and team members need to avoid being argumentative, to yield a successful overall VE effort.

Phase Six—Follow-Up/Implementation Stage
Sometimes value engineering efforts have succeeded through the above five phases and then are never implemented by the organization. Top and middle management may be overloaded by existing work demands, or for some other reason, suggested changes have not been made to the existing project, product, or service.

Value Engineering Success Steps
Organizations have found that for value engineering efforts to be successful, value engineering must provide the following:

- An organized creative approach to cost reduction.
- Targets function versus technique.
- Targets areas of unneeded costs.
- Enhance the value of the product or service.
- The same level, or improved performance level, at reduced cost.
- Does not harm quality or reliability.

CONCLUSION
Value engineering techniques have been a proven technique for achieving optimum value and lowest life cycle costs for well over half a century. The techniques have been successful in a wide variety of both public and private sector efforts in varied fields from service, manufacturing, and construction industries across the globe. Often, investments in value engineering efforts have yielded returns of four to eight times each dollar spent. Value engineering efforts have proven most successful, when using an organized approach with FAST diagrams, in a team setting with a multi-disciplinary effort. In addition, value engineering efforts have proven most successful with new products, projects, or services, when initiated at the early design/concept stages.

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Chapter 23—Contracting for Capital Projects

James G. Zack, Jr., CFCC FAACE AACE Hon. Life

Abstract
All capital improvement projects are executed through written contracts which bind both the owner and the contractor to certain terms and conditions and performance requirements. Cost professionals working on projects must read, understand, and comply with these contract documents. Contract administration is an integral part of a cost professional's job. Therefore, cost professionals must learn the fundamentals of contracts and contract management in order to properly perform their jobs.

Keywords
Contract, Mistakes, and Risk.

INTRODUCTION
The purpose of this chapter is to discuss the elements of a contract, various contracting arrangements, changes to contracts, and disputes arising under contracts. Cost management is an integral part of good contract administration. Thus, practicing cost professionals must learn the fundamentals of contracts in order to fulfill their role properly.

Learning Objectives
After completing this chapter, the reader should be able to:

- Understand the basic requirements of a contract.
- Understand how contracts may become defective and, possibly, unenforceable.
- Understand the types of contracts typically employed in capital projects, their requirements and the potential advantages and disadvantages of each.
- Understand typical project delivery methods and how contracts are employed in each method.
- Understand various key clauses in contracts.

Definition of a Contract
A contract is simply an agreement between two or more entities that is enforceable within law. It is a business agreement (as opposed to a social transaction), whereby one party agrees to perform work or services for the other party for some consideration. Depending upon the nature of the business to be transacted, and the jurisdiction in which the work or services are to be performed, contracts may be either written or oral. That is, in some jurisdictions certain types of contracts must be in writing, otherwise they are not enforceable. For example, in California, all contracts for sales of property must be in writing by statute.

The difference between a contract and an agreement is the element of legal enforceability. Whenever two parties have a meeting of the minds on a subject, an agreement exists. It is only when the two parties agree and intend to be legally obligated to perform to the terms and conditions of an agreement, does a contract arise. Such agreements are most often reduced to writing and are executed by authorized representatives of the parties.

Requirements of a Contract
Regardless of the type of contract, or the nature of the contractual arrangements; to be enforceable, the basic elements of contract formation must be met. The basic elements of a contract are:

- **Offer—**To be enforceable, there must be a clear, unequivocal offer to perform the work or services by one party. The offer to perform must be definite, seriously intended and communicated to the other party.
- **Acceptance**—Once an offer has been clearly communicated to a party and that party, or someone authorized to act on their behalf, accepts the offer, a contract can be formed. As with the offer, the acceptance must be communicated to the party making the offer. Counteroffers do not constitute acceptance. If a party receives an offer and agrees to accept the offer with a condition not included in the original offer, this is a rejection of the original offer and a counteroffer. In such a case, the offer has not been accepted and no contract exists. Of course, the party receiving the counteroffer may accept the counteroffer and thus form a contract.
- **Legality of Purpose**—To be an enforceable contract, the work or service to be performed must involve legal activities. For example, a contract to design and construct a laboratory to manufacture illegal drugs is likely to be considered unenforceable, even if it is in writing and meets all other conditions.
- **Competent Parties**—In order to have an enforceable contract, all parties to the contract must be competent—that is, possess the legal and mental capacity to form a contract. Typically, contracts with minors, insane individuals, intoxicated persons, convicts (in some states in the U.S.) and enemy aliens are not legally binding. These are, however, fairly rare situations. What is more common is the issue of whether both parties executing a contract have the legal authority to do so. In private contracts, the person asking for an offer is presumed to have the legal authority to contract (apparent authority). A party dealing with such an individual has the legal right to assume authority and competency to contract on

the part of that individual. In public contracts, however, the risk is shifted by statute to the party making the offer. That is, there is no apparent authority doctrine applicable to public contracts. The party making the offer has an affirmative obligation to determine whether the public official asking for the offer has the authority to execute a contract and legally bind their public agency to the terms and conditions of the contract.

- **Consideration**—Courts will enforce contracts only when there is consideration. Consideration is another differentiating factor between mere agreements and contracts. Consideration, under the law, is whatever one party demands and receives in exchange for the work or services performed. Typically, consideration for most contracts is monetary in nature. That is, "We will design and construct the specified processing facility for $205,000,000." However, consideration may be anything the receiving party perceives as having sufficient value to warrant performing the work or services offered.

Mistakes That Make Contracts Defective

Despite all of the above, certain mistakes may occur during the contract formation stage, which will render a contract unenforceable. These mistakes include:

- **Mistakes as to the Nature of the Transaction**—A mistake as to the nature of the transaction will render a contract void if the mistake was brought about by fraud by one of the contracting parties (e.g., express misrepresentation or concealment of material facts). For example, if the parties agree to the design and construction of a facility at a specific location, and it turns out that the property is not zoned for such a facility, then a mistake as to the transaction has arisen. However, to raise this defense, the party asserting fraud must demonstrate that they, themselves, were not negligent during contract formation (e.g., never read the terms and conditions of the contract).
- **Mistakes as to the Identity of a Party**—Freedom to contract brings with it the right to refuse to contract with some parties. The law cannot compel any party to contract with another party they have chosen not to do business with. If one party is mistaken as to the identity of the party they are contracting with, then the contract is unenforceable. It is, however, incumbent upon contracting parties to perform some due diligence during the contract formation stage to determine who they are dealing with and deciding whether they will, in fact, execute a contract.
- **Mutual Mistakes as to the Identity of the Subject Matter**—Unlike the mistakes set forth above, (which were unilateral mistakes or mistakes on the part of only one party), mistakes as to the identity of the subject matter must be mutual—made by both parties. For example, a party may plan to design, construct, and operate two similar facilities in two different locations. During the bidding process, engineering drawings may be inadvertently switched, such that drawings for Facility A may be substituted for those of Facility B, a location at which Facility A cannot be constructed. Any contract arising under these circumstances is likely to be unenforceable.
- **Mutual Mistakes as to the Existence of the Subject Matter**—If two parties contract for the remodeling of an existing facility; but, unknown to either party the facility is destroyed by fire, the contract is unenforceable.

Mistakes That Do Not Make Contracts Defective

There are other mistakes, again arising during the contract formation stage, which do not make the contract unenforceable and include:

- **Mistakes as to Value, Quality, or Price**—A contract is not rendered unenforceable simply because one of the parties was mistaken about value, quality, or price of the contract. If a contract clearly requires full compliance with a particular building code, for example, and one of the parties is not knowledgeable of the requirements, or expense involved with conforming to such code, this type of mistake does not render the contract unenforceable.
- **Mistakes as to the Terms of the Contract**—Mistakes such as this typically result from a failure to read the terms and conditions of the contract, or a failure to understand the meaning of the provisions of the contract. For example, a party, after contract execution, learns that they are required to provide weekly project schedules (at a cost of $1,000,000 over the life of the contract). That party's failure to understand the requirement at the time of contract execution, or include the cost in their bid, will not excuse them from compliance nor render the contract unenforceable.

Other Factors Affecting the Enforceability of Contracts

Depending upon the nature and type of contract and the jurisdiction (location) where the work is to be performed, there may statutory or regulatory provisions applicable to contracts, the effect of which may determine the enforceability of a contract. For example, in many states throughout the U.S., to be a legally enforceable construction contract, the contractor must possess a valid contractor's license at the time the contract is executed and throughout the entire duration of the project. Failure to do so may render the contract unenforceable and compensation otherwise owed, unrecoverable. Likewise, to enforce design contracts, in some states, a design professional may be required to possess a valid professional registration in the state where the project is to take place. Other jurisdictions may have other statutory or regulatory requirements, which must be complied with fully in order to have an enforceable contract. It is, therefore, incumbent upon contracting parties to be fully aware of the laws affecting various types of contracts in the jurisdiction where the work is to be performed. To quote John Selden—"Ignorance of the law excuses no man; not that all men know the law, because 'tis an excuse every man will plead, and no man can tell how to confute him."

Parties to a Contract

As described above, there must be a minimum of two parties in a contractual arrangement. The first, for the purposes of this chapter, will be referred to as the owner. The owner is the party who wants a capital project completed (i.e., design and construction of a petrochemical processing facility, airport, bridge, aircraft, missile, power plant, etc.). The owner is typically the party issuing the Invitation to Bid (ITB) or Request for Proposal (RFP)—that is, asking other parties to make offers to perform the project. The second party will be referred to in this chapter as the contractor. The contractor is the party offering to perform the work or service the owner is seeking. The contractor may be an Architect or Engineer (A/E) who offers to perform design and design related services. The contractor may be a constructor who offers to physically build the facility, or a manufacturer who offers to fabricate the items requested, or a Design Build (D/B) entity who offers to design and construct the facility.

That is the simplest form of contracting. Contracting may, however, be considerably more complex and may involve many more parties. The owner may be a consortium of individual companies who have joined together as an operating entity, or the owner may be a subsidiary or special purpose entity established by a parent company for the purposes of accomplishing this project. Contractors may also act in consortiums, joint ventures, prime contract arrangements with multiple subcontractors, an Integrated Project Delivery (IPD) entity, etc.

Because of the potential complexity of parties to contracts, to avoid any possibility of mistake as to the identity of a contracting party, contracting parties have an obligation to perform some due diligence to determine everyone involved and decide whether they want to contract with the other parties. Failure to perform some level of due diligence may put one of the contracting parties at risk.

Why Have Written Contracts?

In some cases, a written contract is a legal requirement of the jurisdiction in which the project is being performed. Such a legal requirement is typically found in a State's Statute of Frauds, in the U.S., wherein specific types of contracts must be in writing in order to be enforceable (i.e., sales of property, contracts of more than one year's duration, etc.). In more general terms, however, a written contract is necessary to record the conditions of the contract, the commercial terms and pricing arrangements, the scope of work to be performed, and other necessary project execution provisions. That is, the contract sets forth the duties, obligations, and responsibilities of the parties involved, as well as the commercial terms and conditions under which the project is to be performed.

In the event of a disagreement during the performance of the work, the parties must be able to look to a written document to determine what is, and what is not, required of each party to the contract. If the disagreement grows into a legal dispute, then the trier of fact (an arbitrator, judge or jury) should have a written document framing the original agreement of the parties in order to render a decision on the dispute.

Contents of a Contract

The term contract is typically a defined term in most contracts. That is, in the definitions section of a contract the term is defined to include, generally, the following:

- Invitation to Bid or Request for Proposal
- Instructions to Bidders
- Addenda Issued During the Bid Period
- Bid or Proposal
- Contract
- Bonds
- General Conditions
- Special or Supplemental Conditions
- Scope of Work (plans, drawings, specifications, special provisions, etc.)
- Change Orders or Contract Modifications

In some contracts (generally engineering and construction contracts) other items, which may be included in the definition of the term contract include:

- Permits
- Operating Licenses
- Environmental Agreements
- Codes
- Geotechnical or Subsurface Conditions Reports
- Other technical requirements incorporated by reference

It is important for the contracting parties to define the term contract carefully at the outset so everyone involved in the project clearly understands what documents make up the contract. Poor definition of this term often leads to disputes.

Scope of Work

Perhaps the most critical part of a contract is the scope of work. This is most often referred to as the "technical requirements" of the contract. The work scope should define, in fairly precise terms, what work is to be accomplished by which party, when, and to what level of quality. Disagreement over what is in or out of scope is one of the most frequent causes of disputes. The number of disputes, typically, is in inverse proportion to the amount of time spent defining, negotiating, and recording the scope of work. That is, the more time and effort spent on defining and understanding the scope of work, the lower the number of disputes over work scope.

While contract form, contractual terms and conditions, and commercial arrangements are frequently "off the shelf items" and often not subject to a great deal of negotiation, the scope of work is almost always individually crafted and often subject to a great deal of negotiation. Even in public contracting (the hard dollar, low bid form of contracting) plans, specifications, and technical requirements of the contract are individually drafted. Thus, the more time and effort expended on defining scope clearly and properly, the less likely it is that a scope dispute will arise. Proper scope definition requires that the project owner clearly determines and articulates their needs prior to or during the drafting of the scope of work. Owners who do not make up their own minds on what it is they want are much more likely to face multiple changes during the construction or fabrication of the project. Such "scope creep" in turn often leads to cost overruns and later disputes.

Uniform Commercial Code (UCC) Contracts

Every state in the U.S. has adopted the Uniform Commercial Code (UCC) in one form or another. The UCC is intended to be a common set of legal principles governing the sale of goods. In capital projects, the UCC rarely applies to the prime contract between the owner and the contractor since the transaction in this contract deals with a capital project. However, procurement of materials or equipment, for example, either by the owner or the contractor on a capital project is likely to be subject to the terms of the UCC in the state where the capital project is to take place. While it is beyond the scope of this chapter to discuss the UCC in any detail at all (especially since the UCC differs to some extent in every state), the reader is cautioned that contracts for goods (materials or equipment) may be subject to this statute. In such situations, the reader would be well advised to obtain appropriate legal information and advice on the application of the UCC to the transaction the reader is involved with.

TYPES OF CONTRACTS

It would be simple if there was such a thing as a uniform standard contract regardless of what is being contracted between the parties. For a whole host of reasons, this is impracticable. The primary underlying reason for the genesis of multiple contracting forms is how the parties wish to deal with the issue of risk transfer. That is, a definition of which party agrees (not always willingly) to undertake responsibility for some aspect of the work scope such that they now own the risk of performance or non-performance. There are, generally, four basic types of contracts (with an almost unimaginable number of variations) used on capital projects. The basic types of contracts are the following:

1. **Fixed Price–Lump Sum Contracts**—Under this contractual arrangement, the scope of work is well defined, the contracting risks are clearly identified, the price is fixed, and the time of performance is typically also fixed. The contractor is generally free to select their own means and methods to perform the work provided that the final result conforms to the technical, quality, schedule, and cost requirements of the contract. This results from the fact that under a fixed price contract, the contractor assumes risk for the fixed price. That is, if the bidding documents are accurate but the fixed price too low, it is the contractor who suffers the penalty. To balance this out somewhat, contractors are most often allowed the freedom to select their own means and method of performing the work. Some variations of lump sum, fixed price contracts include:

    - **Fixed Price With Economic Adjustment**—Under this form of contract, some portions of the fixed price may be adjustable under certain conditions. For example, if the price of asphalt, copper tubing, computer chips, electronic boards, etc. increases or decreases more than 15 percent from the bid price, payment for the specified item(s) will be adjusted in accordance with the terms and conditions of the Economic Price Adjustment (EPA) Clause of the contract.
    - **Fixed Price With Incentives**—In this variation, while the base price of the contract itself is fixed, certain incentive clauses may be inserted into the contract related to time, cost savings, project performance, etc. For example, the contractor may be entitled to additional payment for every day the project is completed prior to the contract completion date. Or, a contractor may be entitled to keep 50 percent of cost saved on a completed project when the final cost is less than the original contract cost. The concept is to provide an economic incentive for the contractor to meet or exceed certain baseline objectives.

2. **Fixed Price - Unit Price Contracts**—This is another form of a fixed price contract. The distinction between this contract form and the lump sum form discussed above is that the price is fixed for each unit of work rather than the entire scope of work – thus, the name, unit price. The total value of the contract is a function of the number of units fabricated, delivered, installed, etc. Similar to the lump sum form of contract, a unit price contract may also be incentivized. Unit priced contracts generally also contain a Quantity Variation Clause to the effect that if any of the unit quantities vary by +/- 15 percent, say, from the estimated quantities, then the unit price is subject to adjustment on that portion of the work. The risk of changes to units provided is, therefore, shared between the owner and the contractor under such a clause. Since

the total quantity of units may not be known at the outset of the contract, the final cost of the contract is generally subject to adjustment once the final quantity is known—at the end of the contract.

3. **Cost Reimbursable Contracts**—As opposed to fixed price contracts, cost reimbursable contracts are not fixed. Rather, the essence of a cost reimbursable contract is the contractor is paid for their legitimate actual cost incurred in performing the work, plus a stipulated amount for profit. Such a contract is often referred to as a Cost Plus Fixed Fee (CPFF) contract or a Time and Material (T&M) contract. Costs are ordinarily classified as either direct or indirect costs under such contracts.

    - **Direct Costs**—These are costs incurred for the direct and sole benefit of the project and can be associated directly with work scope items or activities under the contract (such as labor, materials, equipment, subcontractors or sub-consultants, consumables, etc.).
    - **Indirect Costs (Overhead Costs)**—These are associated with the management of the work (field overhead or general conditions costs) which cannot be allocated to specific items of work or costs allocated by the contracting organization's headquarters (i.e., salaries of corporate executives), as a cost of doing business (home office overhead). Home office overhead costs are usually calculated as a percentage of direct costs.
    - **Mark Up Costs**—Mark up typically includes the fee (profit) and bond costs if the contract is secured by performance and/or payment bonds.

Cost reimbursable contracts may also be incentivized for meeting or exceeding selected project objectives—time or costs, for example. Incentivizing, in capital projects, typically takes the form of the owner offering to pay additional money to the contractor in the event of early delivery of the project, underruns on the budget, performance of the completed project above specified standards, etc. Such contracts are often referred to as Cost Plus Incentive Fee (CPIF) contracts.

4. **Target Contracts**—Target contracts are generally intended to provide an economic inducement to the contractor to entice completion of work at the lowest possible cost and least amount of time. Often, the contractor will perform the early part of the work (i.e., planning and design) on a cost reimbursable basis. At some point, the contractor will prepare and negotiate with the owner, a detailed estimate, with a not to exceed cost and time of performance. Once agreement on time and cost is reached, these become the contractor's targets. (Such contracts are also referred to as Guaranteed Maximum Price [GMP] contracts.) At the end of the work the contractor's actual costs are compared to target cost and underruns, if any, are shared between the owner and the contractor on the basis stipulated in the contract. Overruns, unless caused by owner intervention or other excusable events set forth in the contract, are generally assessed to the contractor. Similarly, a comparison of targeted versus actual schedule is made and early completion bonuses are often paid to the contractor for completing the project earlier than the target schedule.

REQUIREMENTS, ADVANTAGES AND DISADVANTAGES OF CONTRACT TYPES

The following list of requirements, advantages and disadvantages for each of the contract types has been extracted from Chapter 19, *Contract Packages—Contracting Arrangements*, prepared by James A. Bent, CCP, and included in the AACE International *Skills & Knowledge of Cost Engineering*, 4th edition.

Fixed Price—Lump Sum Contracts—In this type of contract, the contractor is generally free to employ whatever methods and resources it chooses in order to complete the work. The contractor carries total responsibility for proper performance of the work, although approval of design drawings and the placement of purchase orders and subcontracts can be monitored by the owner to insure compliance with specifications. The work to be performed must be carefully defined. Since the contractor will not perform unspecified work without requiring additional payment, a fully developed specification is vitally important. The work has to be done within a specified period of time, and the owner can monitor status/progress to ensure that completion meets the contractual requirements.

The lump sum/fixed price contract presents a low financial risk for to the owner, and the required investment level can be established at an early date. This type of contract allows a higher return to the contractor for superior performance. A good design definition is essential, even though this may be time consuming. Further, the bidding time can be twice as long as that for a reimbursable contract bid. For contractors, the cost of bidding and the high financial risk are factors in determining the lump sum approach.

Requirements

- Good definition (of work scope) and stable project conditions.
- Effective competition in a buyer's market.
- Several months for bidding and appraisal.
- Minimum scope changes.

Advantages

- Low financial risk to owner since maximum risk is on the contractor.
- Cost and project viability are known before a commitment is made.
- Minimal owner supervision – mostly quality assurance and schedule monitoring.
- Contractor will usually assign its best personnel to the work.
- Maximum financial motivation of contractor—maximum incentive for the contractor to achieve early completion at superior performance levels.
- Contractor has to solve his/her own problems and do so quickly.
- Contractor selection by competitive bidding is fairly easy, apart from the deliberate low price.

Disadvantages

- Variations (changes) are difficult and costly—the contractor, having quoted keenly when bidding, will try to make as much as possible on extras.
- An early start is not possible because of the time taken for bidding and for developing a good design basis.
- The contractor will tend to choose the cheapest and quickest solutions, making technical monitoring and strict quality control by the owner essential; schedule monitoring is also advisable.
- The contractor has a short term interest in completing the job, and may cause long term damage to local union relationships by doing such things as setting poor precedents/union agreements.
- Bidding is expensive for the contractor, so the bid invitation list will be short; technical appraisal of bids by the owner may require considerable effort.
- Contractors will usually include allowances for contingencies in the bid price and they might be high.
- Bidding time can be twice that required for other types of contracts.

Fixed Price – Unit Price Contracts—These require sufficient design definition or experience in order to estimate the unit/quantities for the work. Contractors then bid fixed prices for each unit of work. The time and cost risk is shared, with the owner responsible for total quantities, and the contractor assumes the risk of a fixed unit price. A quantity increase greater than 10 to 15 percent can lead to increases in unit prices.

Requirements

- An adequate breakdown and definition of the measured units of work.
- A good quantity surveying/reporting system.
- Adequate drawings and/or substantial experience for developing the bill of quantities.
- Financial/payment terms that are properly tied to the measured work and to partial completion of work.
- Owner supplied drawings and materials must arrive on time.
- Quantity sensitivity analysis of unit prices to evaluate total bid price for potential quantity variations.
- Ability to detect biased bidding and/or front end loading.
- Contractor experience with this contracting arrangement.

Advantages

- Good design definition is not essential—typical drawings can be used for the bidding process.
- Very suitable for competitive bidding and relatively easy contractor selection subject to sensitivity evaluation.
- Bidding is speedy and inexpensive, and an early start is possible.
- Flexibility—depending on the contract conditions, the scope and quantity of work can be varied.

Disadvantages

- Final cost is not known at the outset, since the bills of quantities have been estimated on incomplete engineering.
- Additional site staff is needed to measure, control, and report on the cost and the status of the work.
- Biased bidding and front end loading may not be detected.

Cost Reimbursable Contracts—These contracts require little design definition, but need to be drawn in a way that allows expenditures to be properly controlled. The major advantage of a reimbursable cost contract is time, since a contract can be established during the early stages of a project. This type of contract presents a disadvantage to an owner, however, since poor performance by the contractor can result in increased costs and because final costs are the owner's responsibility. Further, the final or total investment level is not known until the work is well advanced.

Reimbursable cost contracts can contain lump sum elements such as the contractor's overhead charges and profit, which is usually preferable to calculating these costs on a percentage basis. Reimbursements may be applied to such items as salaries, wages, insurance and pension contributions, office rentals, and communication costs. Alternatively, reimbursement can be applied to all-inclusive hourly or daily rates for time spent by engineers on the basis that all office support costs are built into these rates. This form of contract is generally known as a fixed fee or reimbursable cost contract, and can be used for both engineering and other office services as well as for construction work.

Such arrangements give the owner greater control over the contractor's engineering work, but reducing the lump sum content of the contractor's remuneration also reduces its financial incentive to complete the work economically and speedily. Further, it lessens the owner's ability to compare and evaluate competitive bids, since only a small percentage of the project cost is involved. Finally, it is possible that the "best" contractor may not quote the lowest price.

Requirements

- A competent and trustworthy contractor.
- Close quality supervision and direction by the owner.
- Detailed definition of work and payment terms covered by lump sums and by all-inclusive rates.

Advantages

- Flexibility in dealing with changes (which is very important when the job is not well defined), particularly if new technology development is proceeding concurrently with the design.
- An early start can be made.
- Useful where site problems, such as trade union actions (i.e., delays or disruptions) may be encountered.
- Owner can control all aspects of the work.

Disadvantages

- Final cost is unknown.
- Difficulties in evaluating proposals—strict comparison of the amount quoted may not result in selecting the "best" contractor or in achieving the lowest project cost.
- Contractor has little incentive for early completion or cost economy.
- Contractor may assign its "second division" personnel to the job, make excessive use of agency personnel, or use the job as a training vehicle for new personnel.
- Owner carries most of the risks and faces the difficult decisions.
- Biased bidding of fixed fees and reimbursable rates may not be detected.

Target Contracts—Target contracts are intended to provide strong financial incentive for the contractor to complete the work at minimum cost and within minimum time. In the usual arrangement, the contractor starts work on a reimbursable cost basis. When sufficient design is complete, the contractor produces a definitive estimate and project schedule for owner review, mutual negotiation, and agreement. After agreement is reached, these become targets. At the end of the job, the contractor's reimbursable costs are compared with the target and any saving or overrun is shared between the owner and the contractor on a prearranged basis. Similarly, the contractor qualifies for additional payment if it completes the work ahead of the agreed upon schedule. The main appeal this form of contract has to the contractor is that it does not involve competitive bidding for the target costs and schedule provisions.

Requirements

- A competent and trustworthy contractor.
- Quality technical and financial supervision by the owner.
- Competent estimating ability by the owner.

Advantages

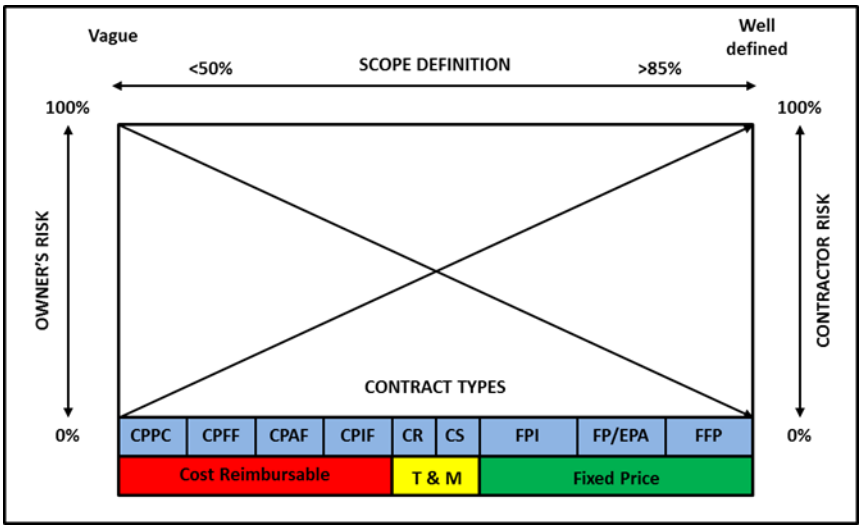
- Flexibility in controlling the work.
- Almost immediate start on the work, even without a complete scope definition.
- Encourages economic and speedy completion (up to a point).
- Contractor is rewarded for superior performance.

Disadvantages

- Final cost initially unknown.
- No opportunity to competitively bid the targets.
- Difficulty in agreeing on an effective target for superior performance.
- Variations/changes are difficult and costly once the target has been established—contractors tend to inflate the costs of all variations/changes to increase profit potential with easy targets.
- If the contractor fails to achieve the targets, it may attempt to prove that this was as a result of owner interference or to factors outside the contractor's control; hence, effective control and reporting are essential.

Summary of Risks by Contract Type

Figures 23.1 and 23.2 depict the allocation of risk between owners and contractors under the types of contracts discussed above.



Source: World Class Contracting, Gregory Garrett. Permission to reprint granted.

Figure 23.1—Contract Risk Allocation [1]

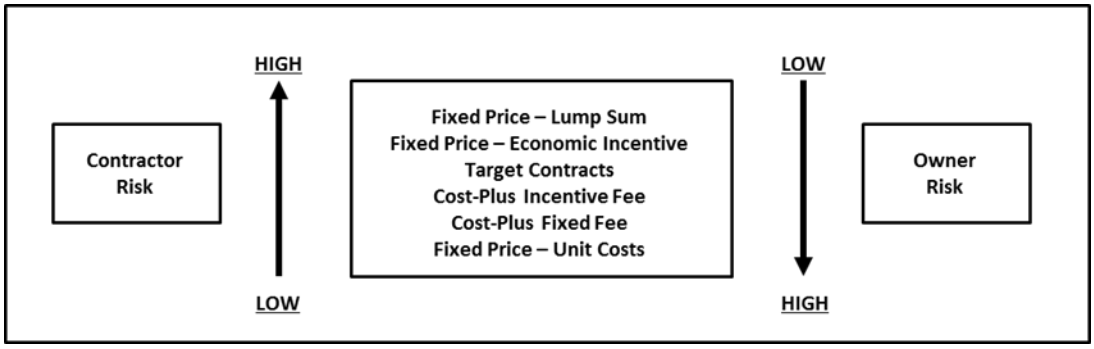


Figure 23.2—Risk Levels

PROJECT DELIVERY METHODS AND RESULTING CONTRACTING ARRANGEMENTS

Project Delivery Methods—Contracting arrangements between a project owner, the Architect/Engineer (A/E) and the contractor are substantially influenced by the project delivery system chosen; which in turn, is frequently driven by the time available in which to accomplish a project. That is, some project delivery systems are inherently faster than others and as owners perceive the time available to deliver a project shortening, they are more likely to choose some project delivery methods over others.

First, some comments concerning project delivery method. In most industries, there are three general project delivery methods applicable to almost all projects. The first and perhaps, most common, is known as the Design-Bid-Build Method. This methodology is shown in Figure 23.3.

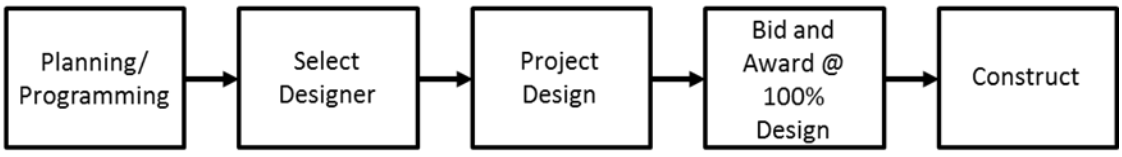


Figure 23.3— Design-Bid-Build Project Delivery Method

The fundamental concept involved in the Design-Bid-Build project delivery method is that the project is fully designed prior to the time the contractor is employed. Owners who employ a lump sum, fixed fee or a lump sum, unit price contract as the contracting vehicle, frequently opt for the design-bid-build project delivery method in order to get some assurance that the total project cost is a known quantity prior to the start of construction.

The other project delivery method used on many projects is generally referred to as the Design-Build Method. This project delivery method is also referred to by many other names including the Engineer-Procure-Construct (EPC), Fast Track, Flash Track, or Turnkey Methods. This delivery method is shown in Figure 23.4.

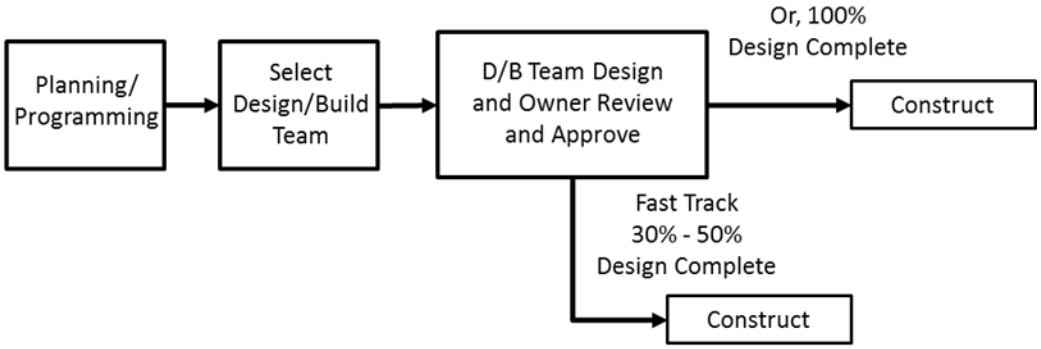


Figure 23.4—EPC Project Delivery Method

The essence of this project delivery method is to shorten the timeframe between the planning and the construction process, thus allowing for a faster project delivery. Since the same team that designs the project will construct the project then construction can start prior to the completion of the design as the owner is not at risk for coordination between the designer and the contractor. This method also eliminates the need for a separate bid/award phase for construction, again shortening the timeframe for project delivery. Of course, the total project cost is not known to the owner at the time the Design-Build Team is selected, and may not be known for some time after.

The third project delivery method is generally referred to as the Indefinite Quantity Contract Method. This project delivery method has also been referred to as Task Order Contracting, or Job Order Contracting. The method is shown in Figure 23.5.

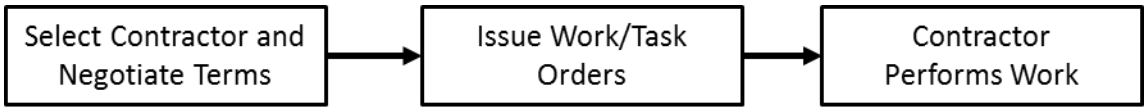


Figure 23.5—Task Order Delivery Method

This project delivery method is typically used on repetitive, non-capital projects such as routine repair and maintenance projects. Under this method, the owner and contractor establish set prices up front for labor, equipment, markups, etc. As a result, the scope of each work assignment does not need to be fully known prior to commencing work. Since work is performed on a set price basis, scope can be developed as work proceeds. Accordingly, project delivery time can be significantly shortened once the contractor is selected and terms negotiated, because there is no need to design and bid subsequent work.

While there are some variations of these three project delivery methods, the above three fairly well generically describe the most common project delivery methods.

Contracting Arrangements—As with project delivery methods, there are five basic contracting forms or arrangements used in most industries to develop and deliver projects. Not addressed in this chapter is whether the owner staffs the project with internal staff personnel or directly hires outside personnel (consulting or others) to perform such functions.

The first arrangement is referred to most often as Single Prime Contractor and is most often used with the Design-Bid-Build project delivery method. Under this arrangement, the owner first contracts with a design professional to perform the requisite planning and design work. Once design is complete, the owner then contracts with a single contractor to deliver the project. All procurement and subcontracts are with the single prime contract, thus insulating the owner, to a limited extent, from claims or disputes arising from vendors, suppliers and/or subcontractors. The arrangement is shown in Figure 23.6.

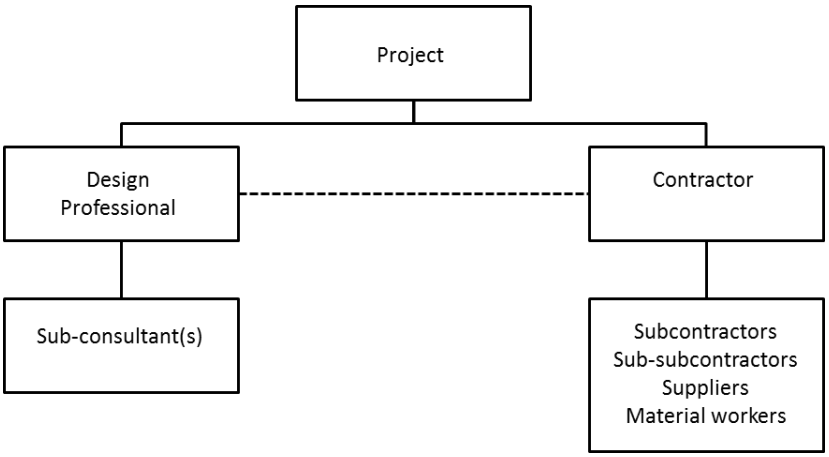


Figure 23.6—Single Prime Contractor Organization

In this contracting arrangement, the owner has a single relationship for project design and a separate, single contractual relationship for construction or project delivery. The owner's liability exposure for coordination is substantially reduced under this arrangement.

The second basic contracting arrangement is most often referred to as the Multiple Prime or Independent Prime Contractors arrangement. Under this contracting model, the owner still retains a design professional as the single point of responsibility for planning and design of the project. However, the owner then issues a series of construction contracts to several prime contractors. The multiple prime contracting arrangement may be organized by craft or trade (i.e., civil/structural, mechanical, electrical contracts) or may be organized by project phase (i.e., site grading, prep and utilities; building A, Building B, Parking Structure, etc.). In either event, the concept is that project delivery can be sped up by allowing construction to begin prior to completion of the full design. This contracting arrangement is shown in Figure 23.7.

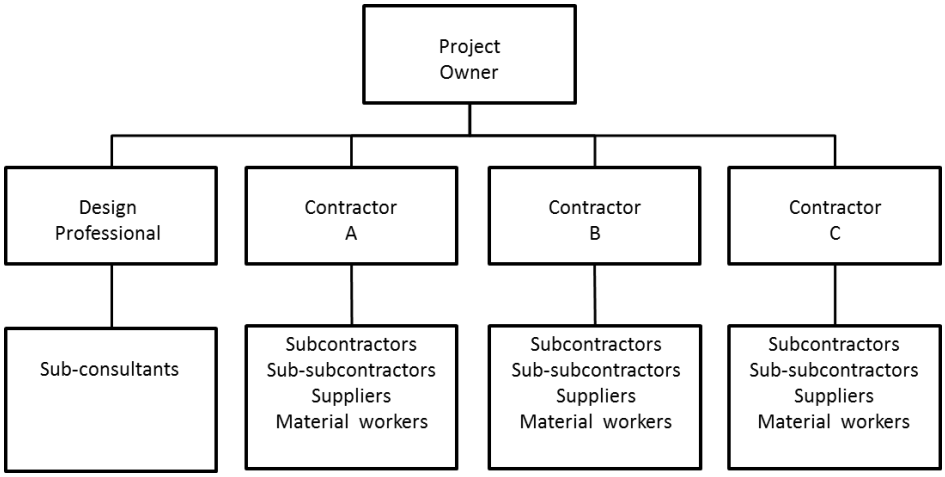


Figure 23.7—Multiple Prime Contractor Organization

While this arrangement allows the owner to get to construction more quickly than the arrangement outlined above, this arrangement puts the owner at risk for coordination problems arising between the independent prime contractors. There is no contractual relationship (that is, no privity of contract) between the independent prime contractors. There is a relationship between each contractor and the owner. Thus, any conflicts, delays, or coordination problems between the multiple prime contractors may become claims to the owner.

The third basic contracting arrangement is the Design-Build Contract. Based upon the discussion of this project delivery method above, the contracting arrangement is shown in Figure 23.8.

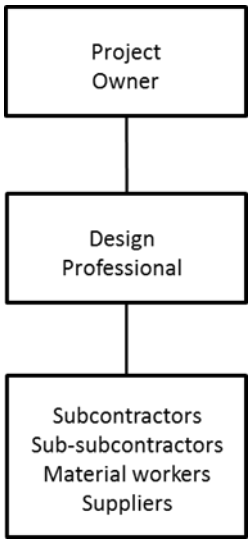


Figure 23.8—Design Build Organization

As noted in the discussion above, the concepts underlying the Design-Build Method are to shorten the project delivery time and to have a single point of responsibility for both design and project execution. A variation of the typical Design-Build Contract is a Turnkey Contract. A Turnkey Contract is a Design-Build Contract; but is, most often, more comprehensive in its scope of work. For example, in a Turnkey Contract, the contractor may purchase the property for the capital project, may finance the project, etc.
The fourth generic contracting arrangement is often referred to as the Agency Construction Management Contract approach. This contractual arrangement is also referred to as Project Management or Program Management Contracting. In this arrangement, the owner still maintains a direct contractual relationship with both the design professional and the contractor, but also retains the services of an independent construction manager to coordinate between the two other entities and oversee the activities of the others. The construction manager is often granted the authority of a limited agent of the owner. That is, they have been vested with contractual authority to take certain actions in the name of the owner and the owner will be bound by such actions. Other actions, however, may be reserved exclusively to the owner and his/her. This contractual arrangement is shown in Figure 23.9.

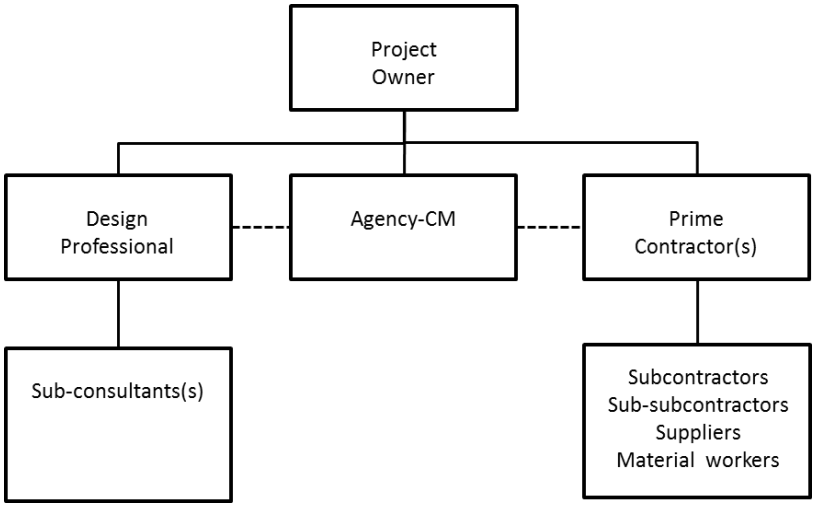


Figure 23.9—Agency Construction Management Organization

The final generic contracting arrangement is most often called the Construction Manager at Risk Model. This contractual model employs the construction manager, project manager or program manager in lieu of a general contractor. Thus, the owner maintains a direct relationship with the design professional and a separate relationship with the construction manager. All trade contracts are then issued by the construction manager. In this manner, the owner provides for multiple prime contracting to speed up the project delivery cycle, while at the same time, sheds some potential liability with respect to the multiple primes. The construction manager holds all trade contracts and the liability for the same. The construction manager typically executes a contract with a firm fixed price—lump sum, or a guaranteed maximum price with the owner. The construction manager is able to take economic advantage of

the difference between their contract value with the owner and the sum total of all contracts with trade contractors. This contractual arrangement is shown in Figure 23.10.

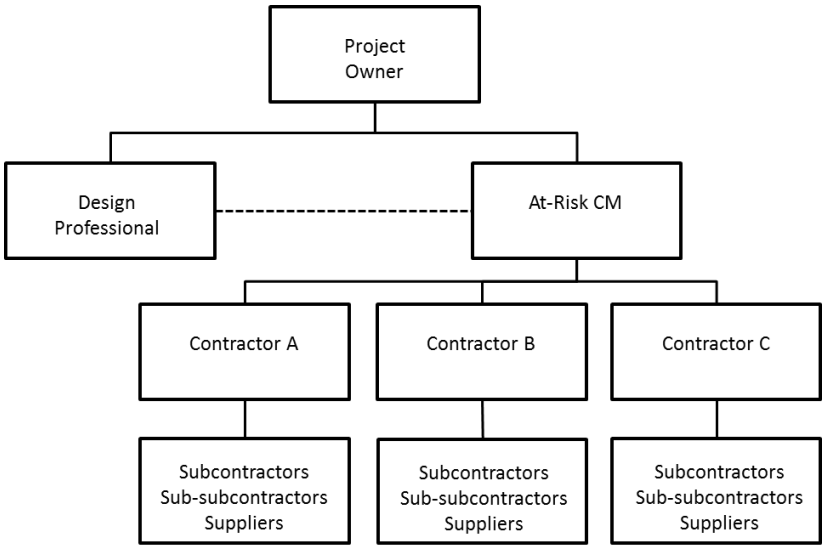


Figure 23.10—Construction At-Risk Organization

There are variations of each of the five generic contracting arrangement models set forth above. However, these five models generically address the majority of typical contracting arrangements.

Contractor Prequalification

As noted earlier, many forms of contract require more than a simple low bid. Owners may find themselves in a position of having to prequalify contractors. This is especially true in situations where owners opt to use a Design-Build Model, a Guaranteed Maximum Price and/or Target Contracts. Prequalification may be problematic for public works owners as many government procurement statutes call for award to the lowest, responsive, responsible bidder, and are deliberately designed to encourage maximum public competition. Where owners have the legal ability to prequalify contractors prior to bidding, it is wise to do so. An owner benefits as there is some degree of assurance that all contractors submitting bids are qualified and capable of successfully performing. The advantage to the contracting community is twofold. First, those bidders who are prequalified can be assured that they are bidding against relatively equal competitors and are not forced to compete against unqualified bidders, who are less likely to understand the work and thus more likely to bid artificially low. The other advantage is to bidders deemed unqualified to bid. While such a decision will obviously be a disappointment, it saves the cost of bidding (which can be very high) and protects unqualified bidders from major mistakes in bidding work they are not truly qualified to perform, thus jeopardizing their financial ability to survive in business.

If an owner chooses to prequalify contractors, an objective, rational system must be established—such that all potential bidders can easily determine whether they are, in fact, qualified. Such objective measures may include the following:

- Past experience on projects of similar size, complexity, technical and schedule requirements.
- Past experience with the design build team proposed on this project.
- Current financial capability.
- Safety ratings on past projects.
- Experienced project team, etc.

Some owners (particularly in the Design-Build arena) have held open competitions to prequalify potential contractors followed by preliminary design competitions among prequalified contractors (with limited remuneration for those invited to the design competition).

Decision to Bid

Perhaps one of the toughest decisions in contracting is the bid/no bid decision that every contractor must make once an opportunity to bid on a project is identified. The contractor needs to carefully examine each potential bidding opportunity. Some of the factors the contractor must consider include:

- **Expertise**—When considering whether to submit a bid, a contractor has to determine whether their organization has the expertise to perform the work. A review of past projects will determine whether the contractor has done anything like this project in the past. If not, some thought should be given to whether they are truly capable of performing the work. In the

alternative, a contractor might consider teaming with another contractor through a joint venture, in order to assure sufficient expertise to perform the work and to spread the risk of the project.

- **Financial Capability**—A contractor considering a bid must look at insurance and bonding requirements, payment provisions, and potential project cash flow scenarios to determine if they have the necessary financial capability to perform the work and successfully complete the project.
- **Bonding Capacity**—Many capital projects require both Performance and Payment Bonds to protect the interests of the owner in the event of contractor failure to complete the work or failure to pay subcontractors and vendors. In such cases, part of the contractor's decision to bid involves their bonding capacity. If a contractor has a bonding capacity of $100 million, and already has $85 million in active projects underway, the contractor may not be able to bid a new $50 million capital project—unless they can successfully convince their surety (bonding company) to increase their bonding coverage limits.
- **Personnel**—A contractor considering a bid must look to their own staffing to determine whether they have sufficient experienced personnel to perform the work during the period of anticipated performance.
- **Equipment**—If a project requires specialized equipment in order to perform the work, a potential bidder must determine whether they have or can acquire the equipment through purchase, lease, or rental arrangements.
- **Specialized Knowledge**—Along with all of the above, a potential bidder should review pre-bid information to determine whether their organization has the requisite skills and knowledge to successfully perform the work under the known terms and conditions.
- **Risk Analysis**—A contractor considering bidding on a potential project should review the general and supplemental, or special, provisions of the contract to determine how project risk is allocated under the contract. For example, if there is not a "No Damages for Delay" clause in the contract, or if the contract has no "Differing Site Condition" clause, then the contractor's risk is increased substantially in the event of delay or they encounter materially different conditions. This being the case, a potential bidder needs to determine if their company can survive if such situations occur during performance of the work.
- **Workload and Other Potential Projects**—Finally, a contractor considering bidding on a project needs to think about their current workload to see how that may impact equipment, key personnel, logistics, bonding, and financial capabilities. Further, a contractor needs to consider what other projects are likely to be bid in the same timeframe that they may want to bid on. A potential problem faced by many contractors is multiple projects being bid at the same time. If a contractor bids on three projects simultaneously, serious consideration must be given to what happens if they are the successful bidder on all three projects. If this event occurs, can they perform?

KEY CONTRACT CLAUSES

In the administration of contracts, all clauses in the contact are important and must be adhered to, to avoid breach of contract situations. However, some clauses are more important than others. Among the more important clauses, in alphabetical order, are the following:

- **Audit**—This clause typically outlines an owner's right to perform reviews (audits) of contractor costs or records. Such clauses ordinarily outline what costs or records are subject to audit, when, and under what circumstances.
- **Changes**—This clause is critical. This is the clause which allows the owner to direct changes to the work—including plans, specifications, time of performance, means, and methods. Absent a changes clause, an owner is precluded from making changes to the work. Of particular importance in this clause, is whether the clause allows the owner to unilaterally direct changes to the work (in which case, if the contractor refuses to comply with the directive, they are in breach of the contract). Alternatively, the clause may require the owner and the contractor to mutually agree on the change (a bilateral change).
- **Contractor Responsibilities**—This clause lays out, in general form, the duties, obligations, and responsibilities of the contractor in performance of the work. This clause assigns specific risks to the contractor, including customarily the risk of adequate labor and equipment to accomplish the work within the required timeframe; the obligation to perform work safely; to perform work in strict accordance with the terms and conditions of the plans and specifications; and to be responsible for the work of subcontractors and suppliers; etc.
- **Delays**—This is, ordinarily, a risk allocation clause with respect to delays in the work. "Excusable Delay" under a contract results in time extensions but no time related damages. That is, a contractor's performance time is extended because of an excusable delay situation, but the contractor is not entitled to collect time extension costs, nor is the owner entitled to impose late completion or liquidated damages for this time. "Compensable Delay," on the other hand, results in both a time extension, as well as time related damages. Customarily, owner caused delay is excusable and compensable to the contractor, while contractor caused delay is the responsibility of the contractor (to either make up the lost time or pay the contractually stipulated late completion damages). Third party caused delay (sometime referred to as force majeure delay) is, most often, excusable and non-compensable to the contractor.
- **Differing Site Conditions or Changed Conditions**—Another risk allocation clause, the Differing Site or Changed Conditions clause, routinely allocates risk of latent site conditions to the owner. The clause normally provides an equitable adjustment to the contract, in the event the contractor encounters a materially different condition at the site during performance of the work.

- **Dispute Resolution**—This clause customarily sets forth the mechanism to resolve disputes during the performance of the work. Most disputes clauses contain some form of a stepped resolution system. For example, the clause may require on site negotiation between project managers, followed by an appeal to project executives, followed by three days of mediation, followed by binding arbitration under a formal set of rules. Often times, the location (jurisdiction) of the disputes resolution forum will be set forth (i.e., arbitration hearings shall be conducted in New York City).
- **Force Majeure**—Some contracts contain a Force Majeure clause, or a clause dealing with delays to the work caused by unforeseeable events beyond the control of both the owner and the contractor. Such clauses often provide lists of examples of force majeure events—acts of God, acts of the government, civil disorder, acts of war, adverse weather, fires, floods, strikes, etc. Other contracts provide for such events in the excusable delay clause.
- **Governing Law**—Many contracts involve parties from differing locations with subcontractors and suppliers from even more locations. Accordingly, contracts often specify which law applies to a dispute, regardless of where the dispute is handled. For example, a contract executed in Mississippi may specify that Texas law applies because the owner is a Texas corporation. The trier of fact (a judge or an arbitrator) is then bound by this contractual stipulation, regardless of where the hearings are conducted.
- **Indemnification**—To indemnify another is to protect them against loss or damage, either by paying for the loss or standing in their place in the event of a legal dispute. An indemnification clause in a contract typically requires a contractor to indemnify the owner against all loss resulting from contractor errors, omissions, accidents, third party property damage, etc.
- **Insurance**—Most contracts have lengthy and complicated clauses requiring owners and contractors to furnish multiple insurance policies prior to commencing work, among which are—builder's risk/all other risk; worker's compensation; automobile, aircraft, and/or marine liability; general liability; bodily injury; broad form property damage; completed operations; personal injury; etc.
- **Late Completion Damages**—Since time is of the essence with most contracts, contracts often contain a clause specifying damages for late completion. In general terms, there are two types of late completion damages—actual and liquidated. Actual damages are those damages an owner actually suffers when a contract is completed late and may include loss of revenue, increased engineering, architectural or inspection services, increased financing costs, etc. Liquidated damages, on the other hand, is a pre-agreed upon amount the contractor will pay the owner in the event the project is completed late as a result of no excusable delay causes—that is, as a result solely to the contractor's fault. Such damages are typically expressed in terms of a daily cost (i.e., $250,000 per day) and need not be proven as actually incurred if the project is completed late.
- **Limitation of Liability**—In order to cap (or limit) a contractor's risk exposure from late completion damages, performance penalties, etc., under a contract, many contracts contain a clause limiting maximum liability to a percentage of the value of the contract. For example, total aggregate damages shall in no case exceed 35 percent of the value of the contract; or limit damages to a maximum dollar value, such that total damages shall not exceed $75.0 million.
- **No Damages for Delay**—Many contracts contain a clause limiting a contractor's recovery for any and all delays to a time extension only, no delay damage costs. While some states have adopted statutes precluding No Damages for Delay clauses in public works contracts, others strictly enforce such clauses provided they are clearly written and the delay involved was within the contemplation of the parties at the time of contract execution (change orders or owner reviews of contractor submittals, for example).
- **Order of Precedence**—Contracts often recognize that the potential for conflicting provisions within a set of contract documents exists. Accordingly, most contracts contain an Order of Precedence clause, intended to provide guidance to both the owner and the contractor in the event of conflicting provisions. Typically, specifications have precedence over drawings, details over general, special provisions over general provisions, and so on and so forth. The legal concept is to provide guidance to people on projects in the event there are two or more conflicting provisions relating to a topic.
- **Owner Responsibilities**—Similar to a contractor responsibility clause, an Owner Responsibilities clause ordinarily sets forth the obligations of the project owner, including adequate project financing; all required and necessary owner furnished permits; appropriate site access; etc.
- **Payments (Compensation/Remuneration)**—This is a key contract clause in terms of project cash flow. This clause sets forth how often the contractor is to be paid, in what manner and what are the conditions precedent to the issuance of payment.
- **Quantity Variations**—Many contracts contain estimated quantities to be installed. In the event as-bid quantity estimates vary substantially (+/- ten percent for example) many contracts (both unit price and lump sum) contain a Quantity Variations clause which allows either the owner or the contractor to request a redetermination of the as-bid unit price on affected portions of the work.
- **Schedules**—A schedule clause typically sets forth the requirements for contractor scheduling including format (bar chart vs. CPM); level of detail; submittal requirements; frequency of schedule updating; damages for failure to submit; delay or time extension analysis requirements; actions to be taken in the event of forecasted late schedule; etc.
- **Suspension of Work**—This suspension clause typically allows a project owner to suspend or stop all or some of the work, with or without cause. Such clauses normally provide for some adjustment to the terms of the contract in such events, including a time extension and payment of delay costs. However, recovery of time and cost may be limited by the terms of

the contract. Often, if the actual cause of the suspension order is something for which the contractor is responsible (i.e., unsafe work conditions, work not in compliance with contract requirements, etc.) no recovery of time or cost is allowed.

- **Termination**—Almost all contracts have a provision allowing the owner to end, in whole or in part, performance of the work prior to project completion. There are, typically, two types of termination—termination for convenience and termination for default. Termination for convenience usually occurs when a project owner decides, for their own reasons, not to complete the project as contracted. Such situations might arise if the owner's needs change, if project financing fails, or if the underlying project economics change substantially. In such a circumstance, the owner may elect to terminate the contractor's performance for the convenience of the owner and pay off the contractor in accordance with the terms of the clause. Termination for default arises only when a contractor is found to be in material breach of the contract. Almost always this clause requires the owner to provide the contractor a cure notice outlining the material breach. (For example, failing to staff the project in such a manner as to assure timely project completion.) Should the contractor fail to remedy the breach in a timely manner, then the owner may terminate the contractor for default. Usually the owner will terminate the contractor from the project and call upon the contractor's financial guarantees to complete the work (i.e., letter of credit or surety bond). Some contracts also provide a contractor the right to terminate their participation in a project under certain carefully proscribed circumstances (such as, failure to make payments, bankruptcy of the owner, suspension of work for more than a defined period of time, etc.). Should the defined situations arise, the contractor is allowed to terminate their own involvement in the project.
- **Time of the Essence/Time of Performance**—If project completion by a time or date certain is important to an owner, then he or she must say so in the contract. As timely project completion is normally important, most contracts contain a clause stating that, "Time is of the essence of this contract." Such a clause must be included to make enforceable a time of performance clause and collection of late completion damages. If absent from the contract, the time of project completion is considered unenforceable. The Time of Performance clause, typically expressed either in work or calendar days after issuance of the Notice to Proceed, sets forth when the work must be completed and the consequences of failure to meet these dates.
- **Warranty**—A warranty clause, which ordinarily continues in existence for some specified period of time after project completion, guarantees the contractor's work after project acceptance. It is not uncommon for warranty clauses to require a warranty for one year after project completion, during which time, if any portion of the project fails or is otherwise found defective, the contractor is obligated to return to the project and make it right or agree to some commercial settlement of the issue.

As noted above, all clauses in a contract are important and must be complied with in order to avoid any allegation of breach of contract. Good contract administration requires knowledge of and compliance with all clauses. Some clauses, especially those identified above, are more important than others. Cost engineers and project controls personnel must become intimately familiar with the terms and conditions of any contract they work on. In reading and interpreting contracts, it is a fundamental rule of law that contracts must "be read as a whole." That is, it is a rare situation for a single clause in a contract to be definitive of an issue. Therefore, people dealing with contracts must read the entire document many times in order to fully understand and appreciate all terms and conditions.

CONCLUSION

Involvement in the administration of contracts is one of the central roles of cost engineers and practicing project controls professionals. Contract administration is a skill, which should be mastered by professionals who practice cost engineering or project controls. Contract administration lies at the heart of what most cost engineers do on a daily basis. Cost engineering is also an integral part of contract administration. The two areas are inextricably intertwined. To be good at one skill, one has to master the other.

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Chapter 24—Strategic Asset Management

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Abstract
Management of strategic assets involves any and all of the unique physical and/or intellectual property that is of long term, ongoing value to the project. Keeping this definition in mind, this article discusses strategic asset management from the perspective of AACE International's concepts of cost engineering. Emphasis is on the responsibilities of the project owner and implementing contractor to manage the strategic assets. The interaction between the owner's strategic asset management and the contractor's project control systems form the links for successful planning and execution of the entire project. Measurements of success are found in: safety, operational efficiency, and resource consumption. Each of these planning factors must be effectively evaluated and followed through the course of development and implementation of the project.

Keywords
Asset Life Cycle, Benchmarking, Capital Asset/Economic Life Cycle, Capital Budget, Product Life Cycle, Project Life Cycle, Enterprise, Enterprise Resource Planning, Front End Loading, Gated Project System, Influence Curve, Project System, Strategic Asset, Strategic Asset Management, Total Cost Management, and Value Improving Practice.

INTRODUCTION

Learning Objectives

The objective of this chapter is to show how the skills and knowledge of cost engineering work from an asset owner's perspective (as opposed to a project team perspective). For the asset owner, the concepts, tools, and resources of cost engineering are applied in an integrated way through the Strategic Asset Management sub-process of Total Cost Management (TCM). This chapter describes Strategic Asset Management and provides examples of how it works in practice. After completing this chapter, the reader should be able to:

- Understand how cost engineering practices can be applied in an integrated way using the Strategic Asset Management process.
- Understand how Strategic Asset Management is applied in different industries and for different asset types.
- Understand typical roles and responsibilities of cost engineers in Strategic Asset Management.

CONCEPTS—TCM AND STRATEGIC ASSET MANAGEMENT

AACE International's Total Cost Management Framework defines TCM as the sum of the practices and processes that an enterprise uses to manage the total life cycle cost investment in its portfolio of strategic assets [1]. The practices are called Cost Engineering and the process through which the practices are applied is called TCM. The TCM Framework further defines a strategic asset as any physical or intellectual property that is of long term or ongoing value to an enterprise. Strategic assets may vary from industrial plants to transportation systems to software programs; essentially anything that an enterprise makes significant investments in can be considered a strategic asset.

Each asset has a life cycle. For example, a building owner evaluates designs, builds, leases, maintains, renovates, and eventually demolishes a building during its life cycle—at each stage of the building's life the owner makes cost investments in it that must be managed. As part of the cost management process, the building owner monitors the operating cost and profitability of the existing building, evaluates the performance of competitor's buildings, assesses alternate building investment opportunities, and initiates, plans and controls new building construction, and/or building maintenance projects. The building owner may manage a large portfolio of operating building assets, as well as construction and maintenance projects in various stages of their life cycles.

In the building example, Strategic Asset Management is the process where the building owner measures the building's operating performance, assesses improvement ideas, and conceives, evaluates, and initiates building investment projects. Project resources and costs are managed through the Project Control process (see the Integration chapter in the text introduction). The Strategic Asset Management and Project Control sub-processes are linked in TCM.

The bridge or link between the owner's Strategic Asset Management and Project Control processes is called its "project system." The project system is a subset of the Strategic Asset Management process that includes the steps for planning asset investments, implementing investment decisions, and then measuring project system and asset performance. In a stages-and-gates or "gated" project system, the performance of planning tasks is done in successively more definitive stages. At the completion of each planning stage, plan deliverables (e.g., scope description, designs, estimates, schedules, etc.) are reviewed by the asset owner at a gate review. If the project is meeting its stated objectives at the review, the asset owner approves sufficient funds to complete the next stage of plan development or to complete the project. If the project is not meeting objectives, it can be cancelled, re-scoped, or redirected.

Business conditions and objectives are in constant flux and projects do not always go as planned. Asset owners use gated project systems to ensure that the project's portfolio is always aligned with current objectives, and that limited resources are invested judiciously. After a project is completed and the asset has been in use for some time, a final gate review is conducted to assess the operational performance of the asset. A project system that results in fast, low-cost projects, but assets that do not function properly, is not a successful project system. Project and asset performance must be measured and assessed together. This last review, or measurement step, closes the link between the Strategic Asset Management and Project Control processes.

Figure 24.1 illustrates the steps above as a process map (i.e., TCM). In Strategic Asset Management on the left half of the TCM process map, the asset owner performs the following steps for each asset in their portfolio, and for their project system.

- **Performance Measurement**—Measurements (e.g., safety, cost, operability, etc.) are taken of how well existing assets and the project system is performing.
- **Performance Assessment**—Performance measurements of assets and the project system are compared to strategic plans. Corrective, mitigating, or improvement actions are taken as needed. Ideas are considered for new or improved assets or project systems.
- **Planning**—Considering the enterprise's objectives and requirements, asset portfolio and project system improvement ideas are conceptualized, evaluated, and converted into plans for investing resources in new or improved assets or project systems.
- **Implementation**—Investment plans and requirements are communicated to and executed by project teams. Project teams request resources as necessary. Project performance is measured and reported, thus continuing the Strategic Asset Management cycle.

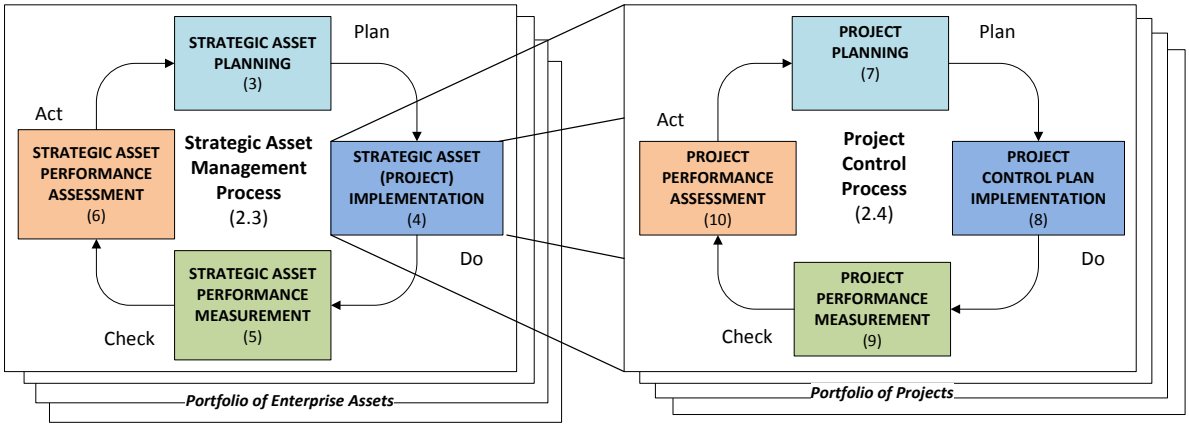


Figure 24.1—The Total Cost Management Process

In the Project Control process on the right of Figure 24.1, strategic asset investments are implemented through the execution of projects or programs. The role of contractors (and others that contribute to the process, but do not own the asset) on project teams increases as projects get larger and require more resources; however, the asset owner, not the project team or contractors, remains accountable for overall TCM process performance. Note the utilization of the PDCA (Plan, Do, Check, Act) Cycle in Figure 24.1

The asset owner's "project system" is the subset of the Strategic Asset Management planning, implementation, and measurement steps, as well as the Project Control performance assessment step; (i.e., the bridge between Strategic Asset Management and Project Control).

Figure 24.2 breaks the Strategic Asset Management process map into more detailed steps that are recognizable as cost engineering practices. The next few paragraphs walk through these steps starting with asset and project system performance measurement on the right hand side of the Figure.

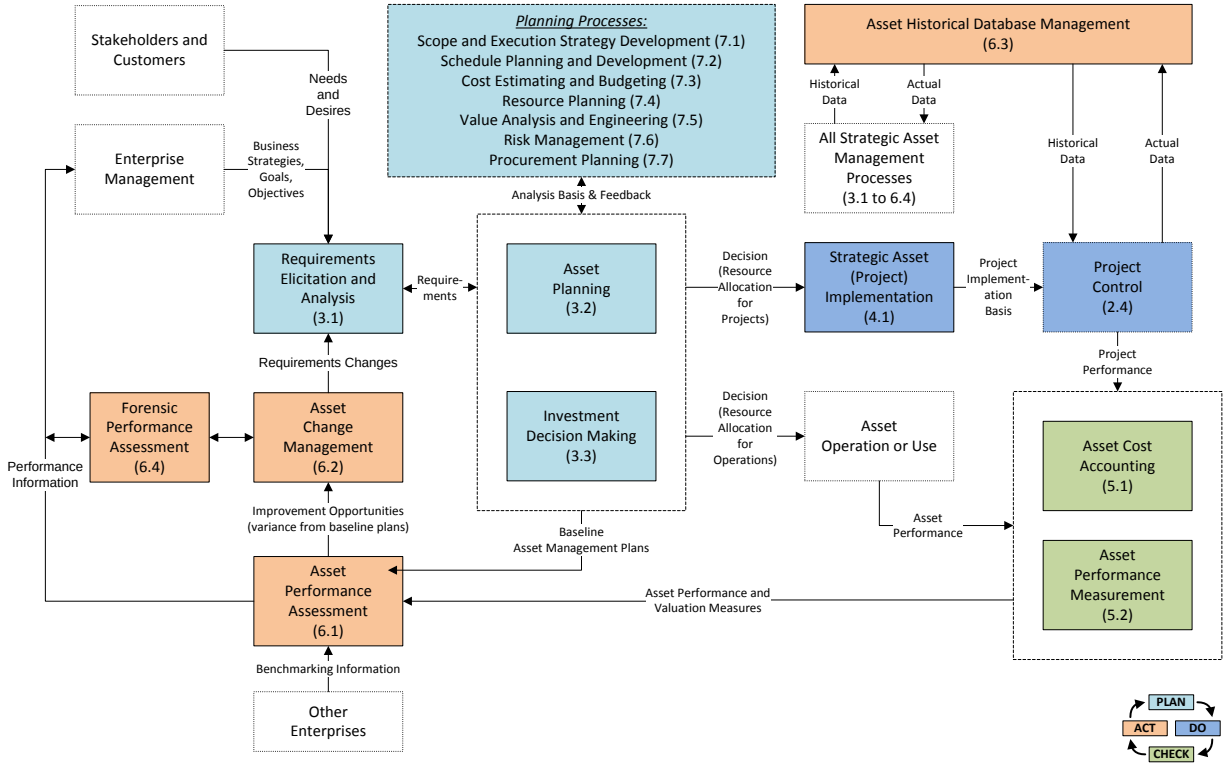


Figure 24.2—The Strategic Asset Management Process

Asset and Project System Performance Measurement

In the first step of Strategic Asset Management the asset owner measures the performance of existing assets and operations, as well as the performance of projects that have been implemented. Measures for the project system may include, but are not limited to, safety, cost, and schedule. Measures for assets in use may include, but are not limited to, safety, operational efficiency, and resource consumption (e.g., materials, labor, energy, etc.).

Existing assets and ongoing projects demand from and return resources to the owner, the most common resource being money. Information about the flow of resources is captured in the accounting element of the performance measurement step. Traditionally, accounting systems have focused on monetary transactions. However, accounting systems have expanded to Enterprise Resource Planning (ERP) systems that, as the name implies, capture information about the input and output of resources for the entire enterprise (e.g., procurement, inventory, etc.).

TCM requires that ERP implementations measure both asset costs (e.g., cost by asset to support depreciation calculations and profitability assessment) and project costs (e.g., costs by activity to support earned value assessment). Unfortunately, many owners' ERP system implementations account for project costs as a type of asset ledger work-in-progress holding account. Using that approach, the ERP system is useless for assessing project system performance; project system information must be captured and assessed using a separate information system.

Fortunately, ERP systems now offer project modules to better meet project management needs. The ERP system is a key interface point between cost engineering and accounting/finance. Working together, owner cost engineers and accountants need to ensure that their ERP systems address both asset and project information needs. Interestingly, ERP software system implementations have become a major class of project themselves in the last decade.

Asset Performance Assessment

Continuing the process to the left side of Figure 24.2, the asset owner evaluates the asset and project system performance measures in comparison to performance plans, as well as internal and external historical performance data. The asset owner looks for variances between measurements and plans for performance aspects—such as safety, quality, reliability, schedule, operability, and of course, cost. The owner investigates any variances found to determine if they are caused by isolated events or systematic problems. In many cases, the immediate user of the asset, or the manager of the project system, identifies the cause of the variance and fixes it through an immediate corrective action. In other cases, the problem requires further assessment.

At the highest level of abstraction, owners assess the total long-term economic return or financial profit from its asset investments and project system performance. Return-on-Investments (ROI) and Return-on-Assets (ROA) are common financial measures. The pressure to improve financial performance is relentless, while the availability of resources to make improvements is limited. Various parts of the enterprise will be competing for those resources. Because the business environment is dynamic, competitive, and uncertain, the Strategic Asset Management performance and requirement assessment steps attempt to balance opportunities and risks against demand and supply for resources in such a way that the enterprise's objectives are met.

The enterprise's objectives are inputs to the Strategic Asset Management requirements assessment step. These objectives are determined in the enterprise's strategy formation process (e.g., what is our mission?; what business should we be in?; etc.). For those with interest in the topic, one source identifies at least ten schools of strategic management thought [2]. Cost engineers should be aware that Strategic Asset Management, as defined in TCM, is related to, but not synonymous with strategic management as a general field of study.

A key requirements assessment tool for the asset owner is benchmarking. This compares the enterprise's asset and project system performance measures to external peer enterprise measures. Benchmarking also identifies internal and external best practices that have been shown to improve performance. Internal benchmarking recognizes that there is diversity in practices and performance within most enterprises. External benchmarking helps ensure that performance is not only on plan, but competitive within their own industry and across other industries.

When asset or project system performance problems cannot be fixed without the investment of significant resources, investment opportunities for improving performance must be identified and evaluated. The effort to identify and develop improvement ideas can itself be a project that consumes considerable resources; therefore, the owner needs to reassess their overall business objectives, requirements, and resource constraints in the strategic requirements assessment step. In this step, constantly changing regulations, industry standards, competitive positions, and market strategies are monitored and evaluated. External benchmarking can help identify competitive positions and strategies.

With their knowledge of both the technical and cost characteristics of the owner's assets and projects, and their analytical skills, cost engineers are often called upon to lead or support performance assessment endeavors, such as benchmarking and profitability evaluations.

Asset Planning

In the performance assessment step, the owner identified assets or project system improvement challenges that required the investment of significant resources for resolution. In the asset planning steps, the owner identifies asset investment and project system options, defines and evaluates them, and decides upon which option(s) to pursue. Every investment decision is made in consideration of strategic objectives and requirements. Once an investment decision is made, owner management communicates the decision to the asset operator and/or the project team, making sure that the scope of the decision and performance objectives for the asset and/or project is clearly understood.

The investment options identification step in the top left center of Figure 24.2 finds ways to improve asset or project system performance. This step is highly creative; it seeks to understand both the nature of the opportunity, as well as the entire range of alternative solutions. A major challenge in this step is to get fresh ideas from both partial and impartial perspectives. Options identification includes a wide variety of continually evolving practices such as benchmarking, cost driver analysis, brainstorming, problem solving, market research, business process analysis, and so forth.

Larger enterprises often have centralized asset planning departments. Common department titles include Strategic Planning, Capital Planning, Facility Planning, or Product Planning. Whether there is a planning department or not, the asset owner's business management creates special teams to tackle specific challenges identified by the assessment step. The planning team thus formed includes a cross-section of personnel with technical, operating, and finance experience. The asset planning effort is generally business-driven (i.e., led by business managers, not technical personnel) because of the need to keep a close eye on enterprise business objectives and strategies at this phase. Owner cost engineers are key participants (though generally not leaders of) the planning effort, because of their skills in key planning practices and their knowledge of asset and project system technology and life cycle costs.

In brainstorming or other option identification sessions, the planning team generates improvement ideas; most are discarded out of hand, while others pass subjective evaluation tests of their potential. For those ideas that look promising, the planning team further develops the scope and definition to a point where their feasibility in terms of potential cost, risk, value, and profitability can be quantitatively analyzed and measured. At some point, the measures reach a level of confidence such that business management can make a decision to discard or continue developing the ideas.

The options analysis and decision steps are in the top center of Figure 24.2. Analysis is an iterative process (i.e., if an idea is still feasible after initial analysis, it is refined and evaluated again and again until it is either discarded or selected for implementation).

Many ideas are going through this process simultaneously. Therefore, a gated project system is desirable so that planning resources are regularly redirected to those concepts that have the most potential.

At some point, the cost of every improvement option is estimated because cost is a component of most decision making criteria (e.g., ROI). The scope definition at early planning stages is minimal, early estimating techniques are stochastic in nature (i.e., parametric analysis, conceptual estimating, factored estimating, etc.). The owner cost engineer, with thorough knowledge of the owner's assets and project system, prepares these estimates using the cost estimating and analysis tools described elsewhere in this publication. Owner cost engineers that initiate and improve on ideas rather than just analyze them (hence the term engineer) are highly valued members of the planning team. The cost engineer also analyzes the options using risk analysis, value engineering, and economic analysis including profitability. These practices all provide quantitative measures upon which owner business management can base its go/no-go decisions.

While go/no-go decisions during the earliest phases of idea generation are usually made subjectively; decisions on options in later stages of definition are usually based on quantitative decision analysis techniques. These techniques include decision tree analysis and other decision-making tools. .

Implementation

Once owner business management makes a decision to implement an asset or project system improvement idea, a project team is formed to implement it. While management is represented on the project team, project responsibility is handed off to a project team manager who often has an operating or technical background depending on the nature of the improvement idea. The planning focus is now on developing the technical scope and execution plans.

At the responsibility hand-off, management conveys formal documentation to the project team of the project's business objectives, conceptual scope, and performance requirements. Objectives and requirements include both the expected conduct of the project (e.g., use the established project system) and the performance of the final asset to be delivered back to the owner. At this point, the project is added to the enterprise's capital budget (or operating budget if costs are not capital). However, the project team is not authorized to spend the full budgeted amount; only enough funds are authorized to carry project definition to the next project system gate review.

The scope of the project at the hand-off is still conceptual in nature. For example, the scope of a chemical plant may include little more than the desired production capacity of the plant, the chemical product specifications, the location for the plant, and the expected date of first production. During implementation, the project team further evaluates and defines the technical scope and project execution plan until they are well enough defined (and the project estimate is considered reliable enough) to ask management for full funding for project execution.

The project definition phase is often called the Front-End Loading (FEL) phase. It is called FEL because the goal is to remove significant uncertainties about the project scope and execution plan during the front-end of the project (i.e., before full funding is authorized by management). Good early definition practices result in more competitive and predictable project outcomes in terms of safety, cost, quality, schedule, and asset performance. With good FEL, late changes in scope are minimized. At the completion of FEL, the project has a detailed budget and schedule that serve as the basis for project control during project execution.

Some project systems call the Asset Planning phase Business FEL, and the Implementation phase Project FEL. The most effective Business and Project FEL systems not only define the scope and reduce the uncertainties of options, they use practices such as value engineering to improve their value (value improving practices or VIPs). The potential to influence the value of an asset diminishes as asset planning and implementation progress. Figure 24.3 illustrates this basic principle known as the Influence Curve.

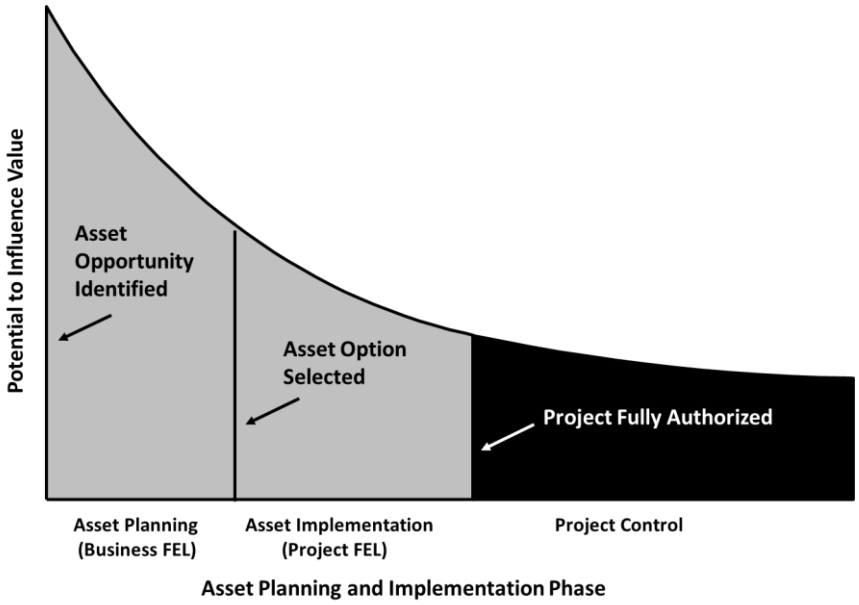


Figure 24.3—The Influence Curve

As projects are implemented and fully authorized (at which point the project team is in full control), the owner asset planning department measures the project portfolio's performance as part of the project system. The owner also continues to measure the performance of its existing assets and operations as well. Asset and project system performance measurement is where we began our discussion of the Strategic Asset Management cycle. Having closed the theoretical loop, some basic examples are provided below of how the theory might work in practice for various types of strategic assets.

APPLICATION—CAPITAL OR FIXED ASSETS

Capital or fixed assets include manufacturing plants and equipment, buildings, roads, and similar items that are not easily moved and have significant useful life spans. Fixed assets are generally created, modified, and retired through a project process rather than a manufacturing process. When determining the profit of an enterprise, the costs of capital assets are generally depreciated rather than being deducted as a current expense. Some items such as commercial airplanes may be a capital asset, but not a fixed asset. The process of creating something as complex as an airplane, is a cross between a project process (i.e., a temporary endeavor with a defined beginning and end) and an ongoing manufacturing process.

As an application example of Strategic Asset Management for capital assets, consider the case of a company that produces photographic film. One of the company's existing assets, at location X, is a machine that coats rolls of specially prepared plastic sheet with photosensitive chemicals and then dries the coating before winding the coated sheet back up for later conversion to a finished product (i.e., rolls of consumer film). This large film-making machine, housed in a light-tight and environmentally-controlled building, continually coats, dries, and winds up thousands of square meters of film per day.

The production planning department of the film manufacturing plant continually monitors film production costs using the company's ERP system. The system tracks the inventory, usage and cost of the rolls of plastic sheet material, the scores of different coating chemicals, the labor of the plant operating personnel, and many other resources. For costs that are difficult to associate with any particular product (such as the cost of planning department personnel), the production planning department uses activity-based costing methods to allocate these costs to specific products. The production planning department includes people with a mix of operations, business (e.g., accounting, finance, etc.), and engineering (e.g., industrial, chemical, etc.) backgrounds. This large plant makes frequent (though individually small) capital investment decisions, so the planning department staff includes a cost engineer.

The consolidated cost and resource measurements of film production for plant X and other company plants are reported through the ERP system to the company's film manufacturing division Strategic Planning Department. In addition to ERP information about the company's internal plant operating costs, the Strategic Planning Department monitors external film technical, business, and cost trends. The internal and external measurements are constantly in flux. The strategic planners observe and assess problems with film production quality and costs that cannot be readily dealt with at the plant level. The planners also observe and assess the market actions of competitors in the film business and they measure the competitor's cost performance through industry benchmarking. Finally, the planners flag division management when their assessment shows that there are significant threats to, or opportunities for, greater profit. Like the plant-level planners, the division-level planners work on a team that has personnel with operations, business, and engineering backgrounds. Cost engineers, depending on their knowledge of a particular class of assets, are part of this team as well.

At any time, the list of opportunities and challenges identified by the strategic planning department is long. There is constant cost pressure because silver prices (a key coating ingredient) and operating labor costs are escalating, while consumer film market prices are dropping. The technology used by competitors is also improving. In this case, through market intelligence and benchmarking, the strategic planners determine that a competitor will probably lower their prices when a film plant they are constructing is completed. In order to match the expected market price, while keeping profits at least level, the planning group's profitability analysis indicates that the unit manufacturing cost component of the product cost must be decreased by about 10 percent within 18 months. Division management agrees with the planning group's assessment and directs the planning team to present them with several feasible improvement options and a decision analysis within 30 days.

The strategic planning team, including cost engineering, holds several days of brainstorming sessions to identify alternative solutions. Most of the alternatives were considered in the past, but the previous analyses are out-of-date and must be revised. Among the ideas considered: make changes in how the film machine at plant X is operated to cut waste and increase uptime; increase the line speed of the film machine at plant X; move some production to plant Y that has some available capacity at lower unit manufacturing costs; build a new production line at either plant X or Y; or build a new film making facility at a very low operating cost location Z. The new plant Z idea is discarded because there is no reasonable way that a complete new facility, with its effects on the entire company's strategies, could be analyzed, designed and built in 18 months. However, the other ideas cannot be discarded out of hand.

At this point, the cost engineer's role becomes significant because each option requires conceptual cost estimating, schedule analysis, risk analysis, value engineering, and ultimately decision analysis; all the tools in the owner cost engineer's skill set. The planning team now starts outlining the conceptual scope of each option and assigns team members to analyze various scope components. Operational specialists consider enterprise and manufacturing resource elements, design engineers consider technical approaches, and cost engineers tie the analysis together with their understanding of design and the cost, schedule, and risk of capital projects.

The cost engineer is prepared for this role. The cost engineering department has an extensive historical database of the cost and schedule of capital projects for film manufacturing [3]. Through benchmarking, they also know what similar projects and facilities cost for their competitors. In addition, they have used the internal and external historical data to develop conceptual estimating, scheduling, and risk-analysis tools. These tools permit the cost engineer to rapidly analyze the numerous technical approaches that the design engineers develop. The cost engineer works closely with the operations specialist and the design engineers to refine approaches and discard unworkable options. The strategic planning team, including the cost engineer, works with the plant planners and engineers to get additional ideas and work out options. However, because coordination with the plant is difficult (some options adversely affect the jobs of plant personnel), it is important that the strategic planning team be very familiar with the plant.

During the course of several weeks of analysis, the team narrows the ideas down to two that meet business objectives: operational changes to cut waste and increase uptime of the machine at plant X, or increase the line speed of the film machine at plant X. In the final week of analysis, the team documents the scope of each option and conducts an early value engineering session for each. The cost engineer prepares a final conceptual estimate and schedule for each and conducts a risk analysis to determine the range of possible cost and schedule outcomes. The team then performs decision tree analysis that weighs the risks and ROI of each option. At the end of 30 days, division management holds a review meeting, and based on the decision analysis, decides to increase the line speed of the film machine at plant X. The division's capital budget is then updated, and funding for further project scope definition is approved.

After the management decision, the division planning team documents the conceptual scope and objectives and holds a kick-off meeting with a newly-formed project team. From there, the project team begins to further define the technical aspects of the accelerated line speed in accordance with the company's project system/FEL process. The cost engineer from the division planning team continues to support the project team. The cost engineer is an important knowledge-bridge from business to technical FEL.

Many design decisions still need to be made; for example the X machine dryer length needs to be extended to accommodate the accelerated line speed and there are a number of equipment layout options for the drying area. The layout options involve building options as well; one option saves on plant real estate by using a multi-story layout with a costly heavy structure, the other option is spaced out on a single floor level that needs only a low cost building shell. Value engineering sessions are again conducted to optimize the function and cost of the process flow, layouts, and building additions. Again, the owner cost engineer has the cost database, estimating, and risk evaluation tools to support these planning efforts.

Eventually, the project team completes the equipment layouts, piping diagrams, instrumentation diagrams, and other project definition deliverables. A project control level cost estimate and a CPM resource-loaded schedule are also prepared to serve as the basis for a management review session, and approval of full project funding. The project execution plan serves as the basis for project control during project execution. At this point, the owner cost engineer's role is as advisor and consultant to the owner project manager. Cost engineers for the project contractors, with owner oversight, take the lead role for day-to-day project control.

APPLICATION—PRODUCTS

Products include manufactured goods and similar items that have a limited useful life span and are not fixed in place. Products are generally created through an ongoing, discrete or continuous manufacturing or production process, rather than a project process. When determining the profit of an enterprise, the costs of creating products are generally deducted as a current expense rather than being depreciated over a number of years as with a capital asset.

As an application example of Strategic Asset Management for products, consider an expansion of the film company case previously described. At the same time that the division planning team was informing management that a ten percent manufacturing cost reduction within 18 months was needed, they also noted that the competitor's new film plant was likely going to be able to make film with better image capture capability. The division management was apprised of the product technical challenges and they direct the planning group to present them with several feasible options and a decision analysis within 30 days that will meet the cost, time, and product improvement objective.

Fortunately, a program had been underway in the research and development laboratories that offered the company two alternative coating formulations that would provide comparable or better capabilities. One formulation adds an additional coating layer, while the other changes the chemistry of an existing coating layer. Furthermore, the additional layer option requires a modification to the company's one-hour film developing machine product used by photo shops. The product improvement is such that the planning department thinks it can support a premium price. As the division planning team enters the asset planning phase, they now have to deal with product decisions, as well as capital asset decisions, and, as they so often are, the capital and product planning are highly intertwined.

The planning team now includes personnel with operations, business, and engineering backgrounds, as well as a representative from the research department, specialists in product planning, and a design engineer from the one-hour developing machine group. Sub-teams are formed to concentrate on the manufacturing cost alternatives, the film product alternatives, and the developing machine alternatives. They all report to a lead planning manager and regularly meet as a group to integrate the assessment of the many alternative branches.

The film product sub-team considers consumer perception of the images produced by the alternative formulations. They also evaluate marketing, sales, distribution, inventory, and other considerations for the new alternative products. The developing machine sub-team considers the implications of modifying the several models of machines in their customer's photo shops. The manufacturing team considers the alternatives discussed previously, but now they also must deal with modifications to the film coating process to address the formulation options.

Eventually the team performs a decision tree analysis that weighs the risks and ROI of each option. At the end of 30 days, division management conducts a review meeting, and based on the decision analysis, decides to increase the line speed of the film machine at plant X and upgrade its coating and drying process to handle the extra coating layer, and to modify the one-hour developing machines as needed. The division's capital budget is then updated and additional funding is granted for further program definition. This decision initiates both capital and product-development projects that will be managed as a program.

After the management decision, the division planning team documents the conceptual scope and objectives and holds a kick-off meeting with the newly-formed program team. Capital and product project teams are formed within the program. The capital project team begins to further define the technical aspects of the accelerated line speed and modified coating process. The film product development team begins to further define the business plans (sales, marketing, inventory, distribution, price, etc.) for the new film. The one-hour developer machine product team begins to further define both the technical aspects of the modified machine design, as well as the business aspects of dealing with its customers in whose shops the machines are used.

As the teams continue with scope development, cost engineering practices are used for capital and product planning—including cost estimating, risk analysis, and value engineering. The basic processes are the same for both capital assets and products, but the details of how they are used differ. Product estimates deal with the assembly process versus the construction process for capital estimates. Product value engineering focuses on product (e.g., the one-hour developer machine) functions that can be combined or eliminated as opposed to plant manufacturing process (e.g., the film dryer) functions for capital project value engineering. The owner cost engineer's major role has traditionally been to support capital planning; however, the more the cost engineer understands both of the products and capital planning aspects, the more valuable their input will be to the program.

APPLICATION SOFTWARE

Software is difficult to classify as either a capital asset or a product. As a manufactured good, it is captured in the form of a plastic disk or a silicon chip, but the value and cost of the software has little relationship to the manufactured good (i.e., the disk or chip). Software may or may not have a limited useful life span depending on its function; some software is quickly outdated while other software is used for decades. Finally, software may or may not be fixed in place depending on what device it is installed in. Software's ambiguous nature (is it an asset or product?) is reflected in wide variations in how companies expense or depreciate software development costs. In any case, many things require software in order to perform their functions, and software is a "strategic asset" of long term or ongoing value to the enterprise holding the software copyright or patent.

As an application example of Strategic Asset Management for software, consider a final extension of the film company case. Previously, the case described how the one-hour developing machine sub-team had to consider the implications of modifying the several models of machines in their customer's shops. Furthermore, this sub-team has to consider both the developer machine operating software and user interface associated with the machine.

The existing operating software is based on coding and a chipset for a now defunct machine model and the software's capabilities for modification and improvement are limited. Also, the planning department has received feedback that shop owners are complaining about the hard-to-understand user interface. So now, the overall planning team includes a one-hour developing machine sub-team design engineer and a software engineer as well. The developing machine sub-team must now consider the implications of modifying the several models of machines in their customer's shops, and the known software limitations and user interface complaints.

Eventually, the overall planning team performs a decision tree and other analyses that weigh the risks and return on net assets of each option. At the end of 30 days, division management conducts a review meeting, and based on the decision analysis, decides to increase the line speed of the film machine at plant X and upgrade its coating and drying process to handle the extra coating layer, and to design and manufacture a new developing machine model using the latest software and user interface (a modified version taken from the prototype of a different, but related model that had been in research and development.)

As the developing machine sub-team continues with scope development, cost engineering practices are used for software development. Software estimates deal with software configuration and coding activities versus the construction or assembly activities for capital and product estimates. Software value engineering focuses on software and interface functions that can be combined or eliminated or developed as object code modules. Again, the more the cost engineer understands the capital, product, and software planning aspects, the more valuable their input will be to the program.

CONCLUSION

This chapter outlines the basics of Strategic Asset Management with its measurement, assessment, planning, and implementation steps. Gated project systems that link the Strategic Asset Management and Project Control are described. Examples were provided of how the asset management steps could be applied to three types of strategic assets: a capital asset, a product, and software. The examples show that the same basic asset management steps and systematic approach apply to all strategic asset types. The examples also show that while the approach is basic, Strategic Asset Management implementation can be complex because the assets in an enterprise's portfolio are often interrelated in myriad ways. More example applications can be described by simply substituting specific descriptors of a given asset type and asset management organization. The Strategic Asset Management process of TCM is universal.

The application example also illustrates the typical roles and responsibilities of cost engineers in Strategic Asset Management teams. The teams feature personnel from all relevant parts of the enterprise, including, business, operations, engineering, product planning, software development, and research and development. While there may not be anyone on the team with the title of cost engineer, someone on the team must practice the cost engineering skills of estimating, planning and scheduling, and value engineering. Few cost engineers will perform each skill for each asset; however, it is important to understand the Strategic Asset Management process so that no matter what your role is, you can communicate and work effectively with the asset management team.

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Chapter 25—Change Management Practical Guide

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Abstract
Projects benefit from the planning and implementation of a change management process. The benefits include early recognition of risks and efficient use of project resources. Change management is the process for identifying, resolving, reporting, and administering changes to a project that affects its scope, cost, or schedule. Change management and control has proven to be an essential part of controlling a project and is critical to the project team's ability to successfully manage the work effort. This chapter emphasizes the use of Total Cost Management (TCM) in the implementation of a change management process. Applying the TCM process map, this chapter provides a practical guide for hands-on implementation of change management

Keywords
Change Order, Change Procedure, Change Proposal, Change Request, Contingency, Deviation Request, Management Reserve, Order of Magnitude Estimate, Pending Change Order, Performance Management Baseline, Project Variance (or Deviation) Notice, Quantity Variation, and Revised or Current Budget.

INTRODUCTION
Most company and client change management procedures specify that all variances and potential changes to scope, cost, or schedule be identified, evaluated and managed on every project. The contract between the parties will identify, often with particularity, the change management process the parties are to follow. All potential changes shall be brought to the responsible person's attention upon identification, as described in this publication, and ultimately to the project manager's attention so that appropriate action can be initiated. The definitions used in this publication represent industry preferred terminology. However, specific project requirements or client preferences may impact the terminology. This publication, written from a contractor's perspective, includes consideration of engineering, procurement, construction, and commissioning scopes of work, and should be tailored for the specific scope of work for each type and size of project. The change management process flow through the contractor first, then to the client. For owner self-managed projects, the change management process could be simplified.

Depending on the size and type of work of the program, responsibilities and specific procedures, may vary. The key is to maintain the intent of as provided in the Total Cost Management (TCM) Framework for Change Management, which will satisfy client and contractor requirements. The TCM process map provides an easy to use practical guide to implement project risk management. TCM is patterned after Total Quality Management (TQM), developed by W. Edwards, Deming, and others. TCM and TQM are integrative philosophies of management for continuously improving the quality of products and processes.

Learning Objectives
After completing this chapter, the reader should be able to:

- Understand basic change management concepts.
- Identify steps in the change management process.
- Conduct a change management review process.
- Implement practical steps in the change management process.
- Understand change management implementation and its impact on the budget and schedule.

Change Management Overview
The TCM definition of change management refers to the process of managing any change to the scope of work and/or any deviation, performance trend, or change to an approved or baseline project control plan. The change management process is used to approve or disapprove changes in the scope and baseline plans, thereby closing the project control cycle loop. The process includes the identification, definition, categorization, recording, tracking, analyzing, disposition (i.e., approval or disapproval for incorporation into approved or baseline project control plans), and reporting of deviations, trends, and changes. An example process map is shown in Figure 25.1.

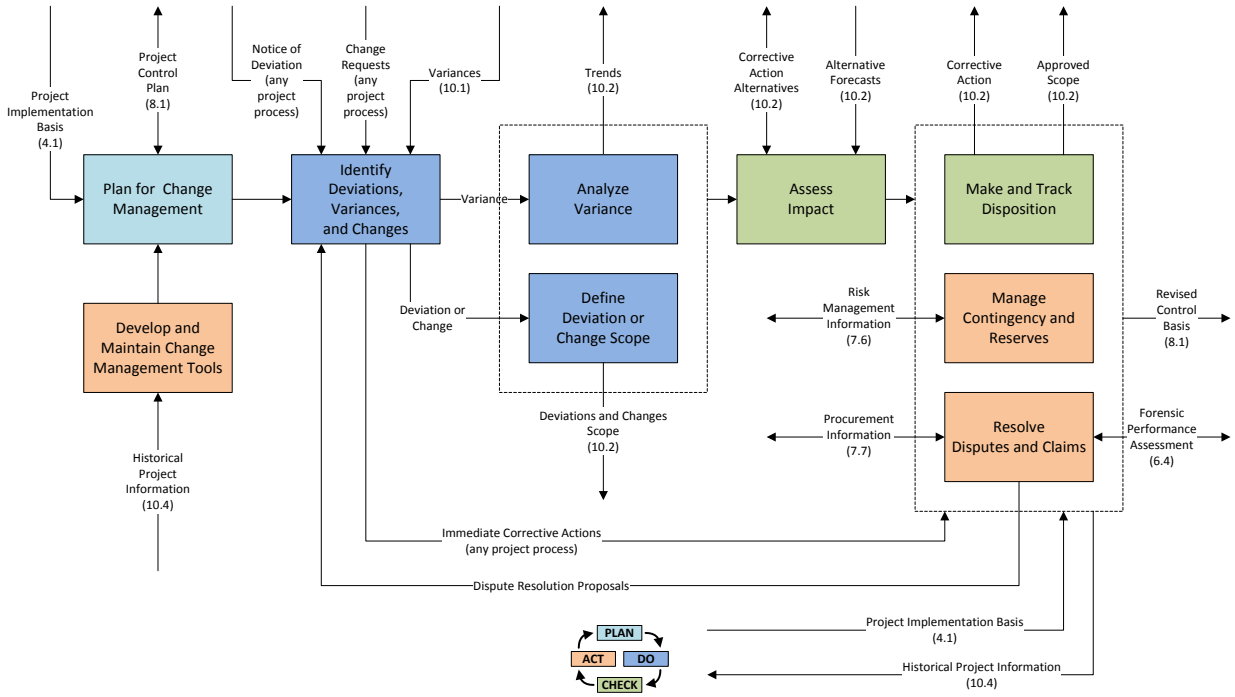


Figure 25.1—TCM Process Map for Change Management

DESCRIPTION OF ACTIVITIES

General

In many cases, contracts require that the contractor and the client agree on, and implement, a detailed project change management procedure within a specific time frame, typically 30 days from notice to proceed. On larger EPC projects, this may take the form of a job bulletin, coordination procedure or be detailed in the Project Execution Plan that is reviewed and approved by both parties. This is a good practice and should be accomplished as quickly as possible on all projects. The contractor needs to follow a practical guide to manage both client and internal change. The attached flow charts illustrate the process for both the internal and client change procedures.

Initiating a Change (Project Variance)

Either the contractor, or an authorized representative the client, may initiate a change. Neither the contractor nor the client's change requests are to be implemented without formal contractor project manager and client approval (See the change management flow chart outlined in Figure 25.2). Each proposed project variance is to be assigned a Project Variance Notice (PVN) number, which is logged by the Project Controls Manager (PCM)/Change Control Manager (CCM). PVNs initiated by the contractor are forwarded to the functional manager, typically the project engineer, procurement manager, or construction manager, for preliminary screening and the identification of all effected disciplines. This is so a complete PVN, including all impacts, can be provided to the contractor project manager.

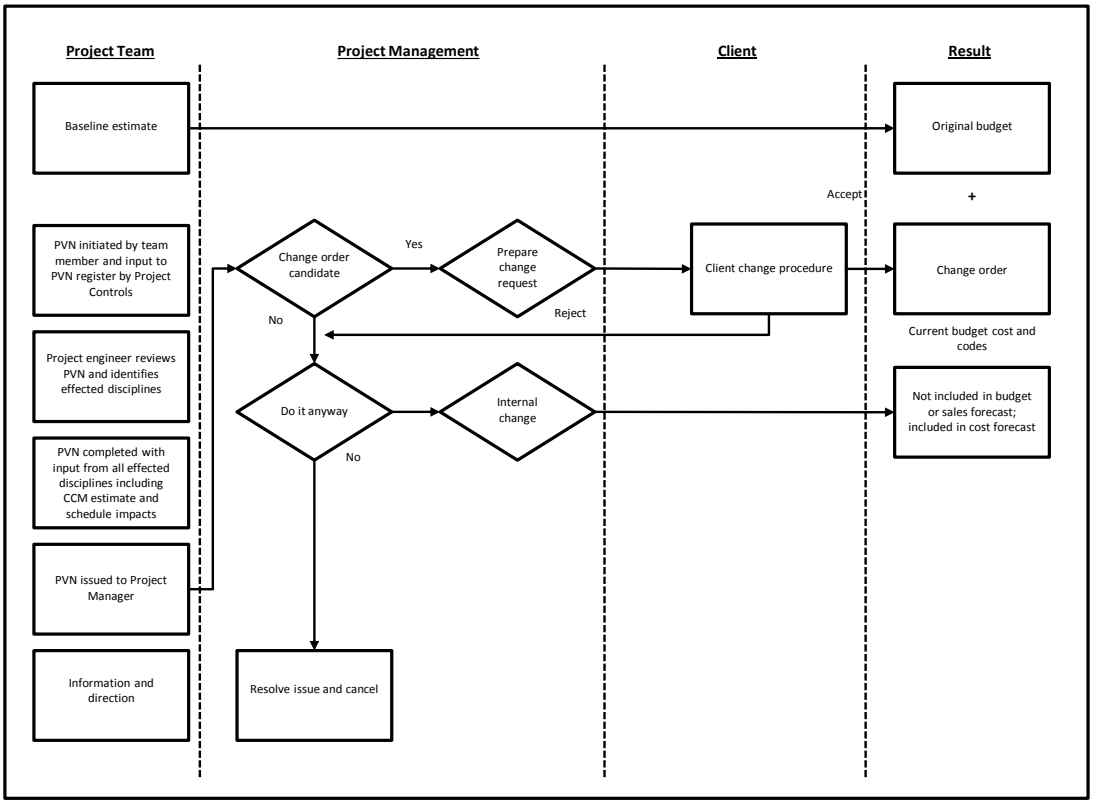


Figure 25.2—Change Management Flow Chart – Contractor's Perspective

Project Variance Notice (PVN)/Change Request

When a team member identifies a potential variance, or is requested to make a change by the client, the team member prepares the PVN Change Request/Change Order form and submits it to project controls for numbering and input into the PVN register. The PVN is passed to the functional manager who identifies all affected disciplines and routes it accordingly. The form is a summary level document and should be supplemented with any additional information needed to understand and justify the variance. As a minimum, the form needs to include:

- **Section 1**—**Title**—Use a concise title to reflect the nature of change.
- **Section 2**—**Change Information**—Add a brief but detailed written description of the proposed change. The description should contain sufficient information for the functional manager to determine all of the affected disciplines and for the project manager to decide whether the change is a contract change order/variation candidate. Indicate the source of the change (i.e., client, vendor, subcontractor, etc.), and provide the basis for the proposed change. This basis can be from source document references, where the change was originally requested. This could be: conference notes, meeting minutes, deviation request, memo/letters, estimate, or revised specification from the client. Leave the change type blank, as this will be completed by the project manager after the review of the PVN. Describe in broad terms the engineering, construction, and schedule impacts in the appropriate blocks. Examples are, "probable increase in concrete and steel quantity"; or, "engineering rework and schedule delay". If an order of magnitude estimate is available, enter it in the "Cost Impact" block. Transmit the original PVN to the PCM/CCM, or the designated project controls person and send a copy to the functional manager.
- **The PCM/CCM**, or project controls person, will assign a sequential number to the PVN, log it into the change register, and coordinate with the project manager for review. Once the project manager evaluates the PVN, the "Change Type" block will be completed.
- **Section 3**—**Approvals**—The client and/or the contractor project manager, as appropriate and depending on the change type and contractual requirements, will check the appropriate authorization block, specify the work that is authorized, the budget or price for the work, and sign the form. The project team is not to proceed with any of the work described in the variance until given approval by the contractor project manager. If the initiator has not received resolution from the contractor project manager in a timely manner, the initiator should follow up with the project manager.

- **Section 4—Distribution**—The PCM/CCM, or project controls person, will check the appropriate blocks and arrange for distribution.

PVN Review Meeting
The contractor project manager will conduct periodic PVN review meetings to keep the project team fully informed of the status of each new and pending variance.

- **Periodic Internal Review**—The periodic internal meeting, usually weekly, will be coordinated and scheduled by the PCM/CCM or project controls person. Attendees will include the contractor project manager, functional managers, PCM, and other selected project team members. The purposes of these meetings are to decide the disposition of new variances, ensure all affected members of the team are fully informed of the impact of each variance, and to review the status of pending variances. The PCM/CCM or project controls person is to prepare an agenda listing the subject variances. Each review team member is to be provided a copy of each new PVN and a variance log prior to the meeting.
- **Special Internal Review**—The contractor project manager will schedule a special review, having the same format as the periodic meeting, for any variance having significant impact which cannot wait for the periodic meeting.
- **Client Review**—When required to support the schedule, the contractor project manager will meet with the client as soon as possible to discuss specific changes. In addition, a periodic client review, usually weekly, is to be coordinated and scheduled by the contractor project manager. Attendees typically include the contractor project manager, project engineer, PCM, CCM/project controls person and selected client personnel. Its purpose is to present new change requests and review the status of pending changes. The PCM will prepare an agenda listing the subject variances and each review team member will be provided a copy of each new change request and a variance log prior to the meeting.

Client Change Procedure
The following procedure is recommended for managing client change in the absence of a contractual provision to the contrary:

1. **Change Request**—Any variance proposed by the client, and any contractor initiated variance that the contractor project manager believes should be a change order, is to be processed into a PVN and change request, if agreed to by the client project manager within three working days of receipt by the contractor project manager. The contractor will prepare an order of magnitude analysis of the scope of the change and its effect on cost and schedule. The cost to prepare a change proposal is included, should the client request one. Should the contractor be unable for any reason to complete the change request within three days, the contractor project manager will advise the client, and reach agreement with the client on a revised submission date. The client will have a reasonable period to approve or reject any change request. The specific amount of time will be dependent on the criticality of the impact to the schedule and/or the contractual requirements. The client may respond as follows:

    - Reject the proposed change request—Contractor will process the rejected change in accordance with the internal change procedure.
    - Approve the proposed change request immediately as a change order.
    - Direct that a more detailed analysis of the change be made and that it be resubmitted as a change proposal.

2. **Change Proposal**—Upon request by the client, the contractor will prepare a change proposal, including any detailed analysis of the scope, price, schedule impact, or other information of the change that the client may direct. The client shall bear the cost of preparing any change proposal, regardless of the disposition of the change. The client shall have a reasonable period of time, consistent with the criticality of the schedule, to approve or reject any change proposal. The client may respond as follows:

    - Reject the change proposal.
    - Approve the change proposal as a change order.

Change Order
When the client approves the change request or a change proposal as a change order, the PCM/CCM or project controls person will complete the necessary forms and obtain contractor and client authorization signatures as specified in the contract or in the project specific change management procedure. If the client provides verbal approval, or if there is any question regarding the approval authority of the client representative providing approval, the contractor project manager is to prepare an acknowledgement in writing to the client's "official" representative with copies to the PCM/CCM or project controls person.

Contractor Internal Change Procedure
The following procedure will be followed for any change to the project execution plan, scope, cost, work hours, or schedule encountered during engineering, procurement, or construction that the contractor project manager views as not being a change order to be sent to the client. This procedure is not dependent upon contract type, and should be followed for all projects:

- **Project Manager's Evaluation**—The contractor project manager may determine that a PVN is not a candidate for a change order to the client. The contractor project manager may either approve or reject the PVN, or indicate that it is an internal change.
- **Reject the PVN**—If the contractor project manager determines that the change in scope should not be implemented, the contractor project manager will reject the PVN accordingly. The contractor project manager will resolve the issue that generated the variance with the variance originator, and the PCM/CCM or project controls person will indicate that it has been rejected.
- **Approve the PVN**—If the contractor project manager gives approval to implement the variance, he/she will mark the PVN as an approved internal change and distribute copies to the project team. This resolution is an authorization for the project team to proceed with the work and/or include any costs, or schedule impacts in the forecast. It does not increase the total budget of the project. If the contractor project manager can provide funding from a management reserve, other project provision or contingency, or via a budget shift, the budget for the variance can be increased.

Disputed Change

The client generally reserves the right to make any change to the project. However, the client may disagree with the contractor on what constitutes a change, or disagree with the contractor on equitable compensation for time or cost to make the change, yet direct the contractor to proceed with the change. In this case, the contractor project manager must perform the following:

- Consult with the contract and legal departments to ensure the project follows the proper course of action.
- Explicitly follow the contractual change procedure and associated change management protocols.
- Dispute the change in writing.
- Receive written instruction from the client defining the course of action.
- Maintain separate and complete records of the time spent; work-hours expended, costs incurred, and any schedule impacts for each disputed change. This is particularly important if the expectation is that the disputed change could result in significant effort, or if the business may be exposed to risk as a result of contract requirements by executing the disputed scope. The project controls person will assist in establishing the best method to capture the costs and schedule impacts for each disputed change. When separate cost codes can be established, they must be recorded on every record of expenditure, including time sheets, purchase orders, or subcontract change notices.

Change Register

The contractor will maintain a change register containing all PVNs, change requests; change proposals, pending change orders, change orders, disputed changes, and cancelled changes. Logs will include the PVN number, the change request, change proposal or change order number (if applicable), the proposed or approved value, and remarks describing the current status of the change, date approved or rejected, and impact on scheduled completion date. The contractor will issue the change register to the client as specified in the contract.

Pending Change Order

Each pending change order, for which the client directs work to proceed in advance of written change order approval, is to be maintained in a change log. The cost of the pending change must be considered in the forecast, but not in the current budget nor in the approved revenue. If the detailed cost distribution for the change is known, it should be tracked separately, in order to facilitate trending and forecasting. When approved by the client, the current budget and revenue is to be updated.

Change Order Identification

Identify each change order consecutively, using a three character numeric block. Note that more than one variance may be included in a single change order. If there is a client change order numbering convention, it should be cross referenced to the contractor character number system.

CONCLUSION

Projects have always involved change. However, they tend to have varying degrees of change management approaches. Management practices have usually been applied in different less formal ways. The effects of this approach have not always been clear. The change management methods outlined in this chapter, if appropriately applied, will more clearly identify project change, the degree of change impact on risk, and the potential consequences. Successful application of these methods will save money and minimize the effects of negative events identified through the change management process.

RECOMMENDED READINGS

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Chapter 26—Overview of Construction Claims & Disputes

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Abstract
Claims (requests for additional time and/or money) are common on construction projects. Claims have grown around the world in the past 50 years, and are now a major construction management issue, often overshadowing on time or on budget performance. The rise of construction claims has been paralleled by the advent of stricter contracts, creation of dispute resolution alternatives, and in many locations, a hardening of owner/employer attitudes toward contractors' pleas for relief from delays, changes/variations in the work, site conditions, etc. No member of the project team, from designers to construction managers, to contractors, to owners/employers, has been left untouched by the rise of claims. Claims, left unresolved on the project, may result in legal disputes involving arbitration or litigation. This chapter provides a short presentation on the basics of how and why claims develop; what it takes to make a claim; what it takes to defend against a claim; how some forms of project delivery methods may affect the number and types of claims on projects; and some suggestions to help avoid, minimize, and resolve claims. This chapter also provides an overview of the various legal dispute resolution venues and forums in the event that claims go unresolved on the project.

Key Words
Acceleration, Causation, Claim, Constructive Acceleration, Constructive Change/Variation, Constructive Suspension of Work, Damages/Quantum, Delay, Delay Cost, Differing Site Conditions, Directed Change/Variation, Direct Cost, Field Office Overhead, Force Majeure, Home Office Overhead, Impact Cost, Indirect Cost, Liability/Entitlement, Lost Productivity, Notice, Suspension of Work, Termination for Convenience, and Termination for Default.

INTRODUCTION

The purpose of this chapter is to discuss changes/variations to contracts and disputes arising under contracts. Practicing cost professionals must learn the fundamentals of contracts and claims in order to fulfill their role properly. Construction claims are inevitable on most projects. As a result, contractors and subcontractors must learn the rule of the road for each type of claim in order to prepare and submit well documented claims. On the other side, owners/employers, cost professionals, and construction managers must learn how to review, analyze, and resolve or defend against claims, in accordance with the terms and conditions of the contract; while following the dispute resolution procedure set forth in the contract.

Construction claims have created an entire industry dedicated to preventing, asserting, and resolving difficulties that occur on construction projects. Claims are a byproduct of our world's economic condition with its fast pace and hyper competitiveness. The number of construction claims has grown in the majority of the world in the past 50 years and are now a major management issue, often overshadowing project execution focus beyond on time or on budget performance. As a result, the rise of construction claims have been paralleled by the advent of stricter contracts, an explosion of dispute resolution alternative methods and in many locations, a hardening of owner/employer attitudes toward contractors' pleas for relief from perceived changes/variations in the work. No member of the project team, from designers to construction managers, to contractors, to owners/employers, has been left untouched by the rise of these claims. While the body of knowledge concerning construction claims is a vast and complicated subject, this short chapter will outline the basics of why and how claims develop; what it takes to make a persuasive claim; what it takes to successfully defend against a claim; and proposes some rules to help avoid, minimize, and resolve construction claims.

Learning Objectives
After completing this chapter, the reader should be able to:

- Understand the typical types and causes of claims that may arise on contracts for capital projects.
- Understand what must be documented in order to successfully prepare, present, review, analyze and resolve or defend against claims.
- Understand how to successfully defend against each type of claim.
- Understand how claims arising on projects may be resolved without the need to complete the project in arbitration or litigation.

CHANGES/VARIATIONS

A frequent occurrence on virtually all contracts is change/variation. All contracts contemplate the probability of changes/variations to the work. It is incumbent upon both the owner/employer and the contractor to establish formal systems to identify changes/variations as soon as they arise and to estimate and negotiate the full scope, time, cost, and impact of the change/variation as quickly as possible. Projects that do not deal adequately with change/variation as it occurs foster end-of-job disputes. On almost all contracts, changed work involves cost and may involve time (as in a delay to the end date of the job). Of equal importance, and perhaps at even greater cost, is the impact of changes/variations, including such things as work being delayed into bad weather periods, increased labor or material costs, and productivity impacts to unchanged work. All such elements of changes/variations should be dealt with as promptly as possible in order to avoid later disputes.

CONTRACTOR CLAIMS

Frequently, disagreements over contract requirements arise, which cannot be resolved through the change/variation order process. Typically, such disputes are referred to under the rubric of a claim. Many design and construction contracts have clauses that address disputes or claims and the notices required in order to properly assert a dispute or claim. The purpose of this chapter is to identify and discuss prime contractor claims asserted against owners/employers, as well as owner/employer claims against contractors.

Subcontractors often file claims against prime contractors; and, if entitlement has been proven, prime contractors are likely to pass through such claims to owners/employers. However, to pass through a subcontractor claim, the prime contractor must determine that the claim meets all the requirements of the prime contract. Accordingly, all passed through subcontractor claims must meet the same tests as are outlined in this chapter.

Claims, like changes/variations, should be addressed promptly and resolved in accordance with the terms of the contract, as soon as possible, and at the lowest possible level in both the owner/employer and the contractor organizations. The longer a claim remains unresolved on a project, the harder and more expensive it will be to resolve later.

Definitions

- **Claim**—The term claim is generally defined in law as a written demand or assertion by one of the contracting parties seeking, as a matter of legal right, payment of additional money, adjustment to the time of performance and/or some other change/variation to the terms of the contract, arising under or related to the contract.
- **Change/Variation Order**—A change/variation order, on the other hand, is a directive from the owner/employer, or their representative, to the contractor directing him or her to perform work differently or perform different work than contracted.

Claims and change/variation orders come together on a project when the owner/employer agrees that the contractor's claim has entitlement. Resolution of such a claim on the project is typically accomplished through negotiation and execution of a change/variation order.

Universe of Claims—Under most contracts, there is a finite number of claim types possible. While the causes of claims are numerous, the clauses of the contract under which relief may be sought are limited. Typically, there are only eleven (11) types of claims in most contracts, which are described as follows:

1. **Owner/Employer Directed Change/Variation**—Claims resulting from directed changes/variations ordinarily involve a dispute over the time, cost and/or impact of an owner/employer directed change/variation. That is, while a change/variation to some aspect of the work has clearly been directed by the project owner/employer, or their representative, there is a dispute concerning the time, cost and/or impact of the changed work. Changes/variations to contract requirements are common during almost all construction projects. Virtually all contracts have a Change/Variations clause—a contract provision allowing the owner/employer to change or vary the scope of work during the project by directive.

   Typically, decisions to make changes/variations to the work are evidenced by the issuance of a written directive (e.g., change/variation order or contract modification). Most contracts authorize the owner/employer to issue changes/variations bilaterally (that is, with full agreement on scope, time, and cost between the parties) or unilaterally (where the owner/employer directs the contractor to proceed with a change/variation even in the absence of full agreement on time, cost, and/or impact). When changes/variations are forward priced and bilaterally executed with a properly worded waiver of future claims clause, there is mutual agreement on the scope, time, and cost of the change/variation before the work is performed. In such a case, there is little likelihood of a claim for damages arising from such an agreement.

   The risk of performing the change/variation is assumed by the contractor. However, when work is performed before the owner/employer and contractor can agree on the amount of additional time and cost, disagreements may arise concerning the fairness of the stipulated time and costs incurred. Such disagreements can turn into claims, or if left unresolved on the project, into legal disputes at the end of the work.

2. **Constructive Changes/Variations**[16]—Unlike directed changes/variations, a constructive change/variation results from an owner/employer action, which has the unintentional effect of requiring the contractor to do more than is required by the contract and results in additional cost or time being incurred. Absent purposeful intent it is usually an accidental or

[16] Constructive is a legal term standing for the proposition that Courts may construe or interpret facts based upon conduct and circumstance and look past the technical legal arguments and contract language to determine intent and impact of one's actions.

unintended change/variation to the work. Owner/employer comments pertaining to a contractor submittal, not intended to be a change/variation, may cause or result in a change/variation to the work and result in time and/or cost impacts. That is, some action or inaction of the owner/employer or the owner's/employer's agents causes the contractor to perform work beyond that which is required by the terms of the contract documents. It should be noted that owners/employers may be held liable for the actions of their representatives (e.g., architects, engineers, construction managers, etc.) even if they do not intend to be so bound. Examples of unintended changes/variations are:

- Comments on shop drawing submittals that require the contractor to perform work not required by the contract.
- Directions communicated in documents such as Construction Change/Variation Directives or Bulletins that are issued by the owner's/employer's consultants that modify the performance of the work in some manner not originally intended but which cause the contractor to incur additional time and/or cost.
- Ambiguous contract requirements where the contractor is required to perform additional work in accordance with the owner's/employer's interpretation.

Constructive changes/variations may also arise from situations such as the following:

- Defective contract documents (e.g., errors, omissions, ambiguities, impossible or impracticable requirements).
- Inspection actions.
- Nondisclosure of relevant information.
- Constructive acceleration.

Typically, to recover for a constructive change/variation, the contractor generally must document that:

- The work performed was not required within the original scope of the contract.
- Appropriate notice of change/variation was given to the owner/employer as required by the contract.
- The change/variation was actually required by the owner/employer (not volunteered by the contractor).
- Additional costs and/or time were actually incurred in performing the changed work.

3. **Differing Site Conditions**—These are classically described as encounters with latent (hidden) physical conditions at the site differing materially from the conditions indicated in the contract documents or conditions normally encountered and reasonably anticipated in work of this nature in this area. Many contracts have a Differing Site Condition or Changed Condition clause that addresses unanticipated or hidden physical conditions at the site that differ from those represented in the contract. Such clauses are often required by law in public works contracts. The purpose of the Differing Site Conditions clause is to transfer the risk of latent site conditions to the owner/employer in an effort to minimize bid contingency costs. In the U.S., most Differing Site Conditions clauses address two types of differing conditions; while some contracts also address a third type:

   - **Type 1**—Subsurface or latent (hidden) physical conditions at the site differing materially from those indicated in the contract documents.
   - **Type 2**—Unknown physical conditions at the site, of an unusual nature, differing materially from those ordinarily encountered in work of the character provided for in the contract.
   - **Type 3**—Material that the contractor believes may be material that is hazardous and/or toxic waste, or that is required to be removed to a specially licensed disposal site in accordance with provisions of existing law.

   Type 1
   Type 1 differing site conditions are physical conditions that differ materially from those shown or indicated in the bid documents. The extent of the Type 1 conditions is measured by the difference between what the contract documents represented at the time of bid and what the contractor actually encounters at the site while performing work.

   Examples of **Type 1** differing site conditions include:

   - Rock where none is shown or anticipated in the subsurface conditions report or other project records.
   - Subsurface water where none is indicated.
   - Buried pipes or utilities where none are depicted.

   Type 2
   Type 2 differing site conditions are conditions that are so unpredictable that the contractor could not have reasonably foreseen their existence at the time of bidding. The extent of the Type 2 conditions is measured by the difference between what the contractor could have reasonably anticipated, based on its experience with work of this nature or what is generally known concerning this type of work in this area and what is actually encountered in the field.

Examples of **Type 2** differing site conditions include:

- Discovery of timber piles beneath concrete plaza.
- Inclusion of considerable debris in previously placed backfill material not discoverable by site visit.
- Soft ground beneath previously undisturbed dry soils.
- Archaeological or paleontological finds.

Type 3
Type 3 differing site conditions generally revolve around unexpected encounters with hazardous or toxic materials, not identified in the contract documents, materials which must by statute or government regulations be transported to and disposed of in specially licensed disposal sites. However, this clause only covers hazardous materials existing on the site prior to bidding and generally does not include any materials brought on site by the contractor or their subcontractors at any tier.

Examples of **Type 3** differing site conditions include:

- Presence of waste oil deposit not mentioned in documents and not discoverable by site inspection in the pre-bid or pre-tender time period.
- Subsurface hazardous or toxic waste materials in an apparently undisturbed area.
- Hazardous materials, such as asbestos, encountered within a building where the building condition survey did not reveal the presence of such materials.

International contracts do not typically break down differing site conditions into these two or three categories. FIDIC's Unforeseeable Physical Conditions clause includes all three conditions, but in a single clause without specific reference to the three types of conditions described above. The resulting outcome of either site conditions clause is the same, assuming the contractor meets the conditions for recovery as set forth in the contract.

Differing site conditions may be either natural or manmade. They may be above or below ground. The contractor must demonstrate that it encountered a material difference; that they provided written notice of the condition to the owner/employer prior to removing or disposing of the material; and that the condition encountered caused it to actually expend additional cost and/or time. The contractor must also establish that it actually and reasonably relied upon the representations in the contract documents concerning the existing site conditions. Although not common, owner/employers may be entitled to a downward price adjustment if the site conditions encountered differ materially from those represented in the contract, with the result that the contractor spends less time or cost in performing the contracted work.

4. **Directed Suspension of Work**—This is an owner/employer directive to stop some work on all or part of the project for a limited period of time. A Suspension of Work clause usually allows the owner/employer to order all or part of the work stopped during the performance of the work, for their own convenience. Accordingly, the clause allows the contractor to recover time and costs associated with such orders to cease work. Some Suspension of Work clauses preclude the recovery of profit on the suspension costs, while other contracts also preclude recovery of avoidable costs. The point here is that while Suspension of Work clauses may provide for an adjustment to the time and cost of the contract they may not provide for the full equitable adjustment common in other clauses as they may have specific cost disallowances. Directives to suspend or cease work by the owner/employer must be provided in writing. The contractor is required to comply with the terms of the directive, so the owner/employer should make certain that the directive is specific. When an owner/employer suspends all or part of the work, they may expect the contractor to file a claim for associated time and costs. However, if the owner/employer stops work because the contractor is performing the work in an obviously unsafe manner, for example, then the suspension order was actually caused by the contractor's actions and resulting extra cost claims are typically invalid. A contractor is not entitled to an adjustment resulting from its own fault or negligence.

5. **Constructive Suspension of Work**—This is an accidental or unintended work stoppage caused by an owner/employer action or inaction which, while not intended to cause a work stoppage, has the effect of doing so. For example, failure by the owner/employer to act on a contractor submittal concerning a piece of equipment which is on the project's critical path may cause a constructive suspension. While the owner/employer may not have intended to stop work, their failure to approve the submittal may unwittingly cause the contractor to not purchase, deliver, and install the equipment in a timely manner. Constructive suspension of work typically is caused by an owner's/employer's acts or omissions that have the effect of unreasonably delaying the contractor's work. These are typically unintended acts that result in an unanticipated delay.

Examples of **Constructive Suspension** actions include:

- Delayed approval of shop drawings.
- Delayed issuance of changes/variations.
- Site or right-of-way unavailability.
- Delayed delivery of owner/employer furnished items.

To recover for a constructive suspension of work, the contractor generally must demonstrate that:

- A delay to the work occurred.
- The delay was caused solely by action or inaction of the owner/employer or its agents.
- The contractor incurred additional time and/or costs as a direct result of the constructive suspension act or lack of action.

Some contracts may contain some variation of a No Damages for Delay clause. Enforceability of such clauses is dependent upon U.S., state, or local law. Prudent owners/employers should check with competent legal counsel prior to inserting such clauses into a contract as many U.S. States have prohibited such clauses by statute. Contractors should review contracts to determine whether they contain such a clause. If so, contractors should check with legal counsel to determine whether the clause is enforceable in the jurisdiction where the work is to be performed or whether an exception may apply.

6. **Force Majeure**—Force majeure events are usually described as unforeseeable events brought about or caused by third parties or acts of God, over which neither the owner/employer nor the contractor could exercise any control, nor which causes a delay to the project.

   Examples of **Force Majeure** events are typically included in the contract and may include:

   - Transportation delays not reasonably foreseeable.
   - Labor disputes not involving the contractor.
   - General labor disputes impacting the project but not specifically related to the worksite.
   - Fires.
   - Floods.
   - Earthquakes.
   - Tsunamis.
   - Acts of war or terrorism.
   - Epidemics.
   - Adverse governmental actions or acts of government in its sovereign capacity.
   - Unusually severe or adverse weather conditions.

   The key to a force majeure claim does not necessarily rest with whether the event is included in the list of events named in the clause but whether the event was unforeseeable; was beyond the control of both the owner/employer and the contractor; could not have been prevented by either party; and caused a delay to the work. Many contracts do not include a separate Force Majeure clause. For example, U.S. government contracts do not contain such a clause, but the acceptability of such delays is included in the Termination for Default clause wherein the clause states that the contractor may not be terminated for default in the event that such situations arise and cause a delay to the project. Regardless of where such language is found in a contract, most contracts deal with force majeure events as excusable, non-compensable delay (discussed further below).

7. **Delays**—Delay is a term of art in that there are numerous ways to define the term. Delay, in many contracts, is generally defined as an impact to the contractually specified completion date or the adjusted contract completion date. That is, delay is any event which causes the project to complete later than planned and beyond the current contract completion date, although the term is often used to describe time impacts to particular activities or groups of activities. Project delay is the most common problem in construction and one of the most difficult to mitigate. Unlike cost impact which can generally remedied by additional costs recovery being granted to the contractor, delay is lost time and lost time cannot be replaced or recovered. The best that can happen is that the time lost as a result of a delay is replaced by time added at the end of the job—most likely in an entirely different phase of work and possibly in a different time of year.

There are, in general terms, seven types of delay on projects—float consumption, inexcusable, excusable, compensable, concurrent, pacing and early completion delays.[17] Most contracts deal with the following four types of delays.

- **Excusable, Non-Compensable Delay**—Delay caused by third parties or incidents beyond the control of both the owner/employer and the contractor. Such delays were discussed in the section on Force Majeure claims, above. Examples typically include acts of God, unusual weather, strikes, fires, floods, acts of government in its sovereign capacity, and so forth. In such situations, the contractor is normally entitled to a time extension and relief from liquidated damages, but no compensation for delay costs.
- **Excusable, Compensable Delay**—Delay caused by the owner/employer or the owner/employer's agents. An example is the issuance of changes/variations that delay the project's end date. The contractor is generally entitled to a time extension and time related compensation as a result of the delay.
- **Inexcusable Delay**—Delay caused solely by the contractor, its subcontractors or suppliers at any tier. Examples include lack of qualified craft labor, or late delivery of contractor-furnished equipment or materials. The contractor is generally not entitled to relief for such a delay and must either make up the lost time or be contractually liable to the owner/employer for late completion or liquidated damages.
- **Concurrent Delay**—This is a situation where two or more delays occur within the same time frame, each of which would have independently impacted the project's critical path. This is sometimes referred to as overlapping delays. The delays may be owner/employer caused, contractor caused, caused by others, or caused by one another. If the concurrent delays are caused by both the owner/employer and the contractor, and to the extent that they actually overlap by at least one day or more, neither party is entitled to damages. That is, the contractor receives a non-compensable time extension but no delay damages. Similarly, the owner/employer will not recover liquidated damages or late completion damages and does not have to pay delay damages to the contractor.

8. **Directed Acceleration**—Directed acceleration is an instruction from the owner/employer to the contractor to complete work earlier than required under the contract or earlier than scheduled/programmed or required by the owner/employer in order to maintain a completion date despite an acknowledged delay. In such situations, a contractor is typically entitled to recover the costs expended in complying or attempting to comply with the directive. The owner/employer may give the contractor a directive to speed up work and shorten the time of performance, or to overcome owner/employer caused delays already experienced (i.e., to buy back delay time). Typically, such a directive is issued in the form of a change/variation. If, on the other hand, the contractor is mitigating their own delays, the contractor's recovery of lost time is often referred to as voluntary acceleration. So long as the contractor is not mitigating its own delays, the net increase in the contractor's costs incurred in complying with this directive (e.g., added equipment or labor, overtime pay, etc.) is usually recoverable by the contractor. In cases of directed acceleration the contractor need not show that the acceleration was actually successful. Simply showing that the acceleration was honestly attempted and costs incurred is sufficient. A claim closely related to acceleration is deceleration, where the owner/employer directs the contractor to slow down or pace the work of others.

9. **Constructive Acceleration**—This is an owner/employer action or failure to act, which inadvertently results in a contractor being required to complete work earlier than required under the contract and causes a cost impact. For example, issuing change/variation orders while refusing to examine delay related issues may bring about constructive acceleration. Constructive acceleration is the unintended shortening of the time of performance for the project. To recover on a claim of constructive acceleration, the contractor must generally show that:

   - Delay occurred which entitles the contractor to a time extension.
   - Notice of delay and a time extension request was properly submitted.
   - No time extension was granted or part of the time extension owed under the contract was denied or, after a reasonable period of time, the owner/employer remained silent on the issue of time extensions and such silence is considered a deemed denial of the time extension requests.
   - The contractor was required or directed to complete on time or threatened with the imposition of late completion damages for failure to do so.
   - The contractor filed a notice of constructive acceleration.
   - The contractor actually accelerated its operation and incurred additional costs as a result of the acceleration.

   Failure of the contractor to factually demonstrate each of these requirements may undermine any recovery for constructive acceleration.

[17] While it is beyond the scope of this chapter to deal with the issue of delay in any depth, it is strongly recommended that all cost professionals become thoroughly familiar with the subject of delay and review every contract carefully to determine how delay is defined, determined, and dealt with under the contract. Cost professionals should review AACE Recommended Practice 29R-03, *Forensic Schedule Analysis*, to familiarize themselves with the various methods of schedule delay analysis.

10. **Termination for Convenience**—This is an action on the part of the owner/employer to end, in whole or in part, the work of the contract prior to completion through no fault of the contractor. Many contracts have a clause allowing the owner/employer to delete all or portions of the project work for the convenience of the owner/employer. If the owner/employer determines not to complete all or portions of the work for whatever reason, a written termination for convenience directive is normally issued to the contractor. Such clauses also typically provide for the contractor's entitlement to compensation in the event such a decision is reached.

11. **Termination for Default**—This is an action to end the work of the contract prior to completion because of a material breach of the contract by the contractor. Failing to mobilize to the site to commence work after the notice to proceed was issued constitutes such a material breach, for example. In the event that the contractor fails to perform in accordance with the contract in a substantial manner, the owner/employer may, under certain circumstances, terminate the contractor's right to continue the work for default. Consistently failing to provide sufficient labor, materials, or equipment; consistently failing to maintain required quality; or consistently refusing to comply with laws and building codes, are situations that may give rise to a termination for default. In such circumstances, owners/employers are typically required to provide the contractor with a cure notice setting forth specifically what conditions must be remedied within a stipulated time period. Should the contractor fail to cure its performance, the owner/employer may elect to terminate the contract for default and call upon the contractor's surety (i.e., bonding company) to fulfill its obligations under the performance bond. The surety often has the option of completing the remaining work itself through a contractor of its choice; paying the owner/employer a negotiated settlement up to the full value of the bond; or do nothing if it believes that the owner's/employer's termination action was wrongful. Default terminations have serious consequences for the contractor and their sureties, and almost always lead to disputes and legal actions.

For each of the above 11 types of claims, there are a myriad of rules and U.S. Case Law.[18] Each particular situation is quite fact intensive and is highly dependent upon the exact wording of the contract clause being relied upon. Additionally, there are normally written notice requirements which must be fulfilled or the contractor may risk losing its legal right to an equitable adjustment. It is, therefore, important for a cost professional to study the field of claims to learn the basics of each type of claim and then examine carefully the terms and conditions of each contract.

Owner/Employer Claims

Owner/employer claim rights are substantially limited in most contracts when compared to contractor claims rights. Owners/employers typically have only three types of claims available to them to assert against contractors. They are the following:

1. **Late Completion Damages**—Assuming the contract contains a time is of the essence clause the owner/employer has the right to assess and collect damages for late completion of the work caused solely by the contractor without any owner/employer involvement. There are generally two types of late completion damages, as follows:

    - **Actual Damages**—In situations such as this, the owner/employer as the claimant would have to keep careful track of all actual damages where incurred, as a result of the contractor's late completion of the work and be able to provide documentation of the claim's damages with a degree of reasonable accounting accuracy. The failure to track and document such claimed damages will waive the owner's/employer's right to collect such damages. Further, in many large capital expenditure EPC projects, the parties agree to mutually waive consequential damages, and as such, the actual damages the owner may seek, are significantly restricted and are usually limited to Liquidated Damages.
    - **Liquidated Damages**—Liquidated damages are stipulated amounts, usually on a daily basis, agreed to at the time the contract is executed. Owners/employers will estimate the damages they are likely to incur if the project is not completed on time and include this daily cost in a Liquidated Damages clause in the contract at the time of bidding. Such damages do not need to be proven if the project is completed late; do not need to be offset by an early completion bonus; nor do they need to reflect actual damages in order to be collected. However, if challenged, the owner/employer must be able to provide documentation that the daily amount included in the contract was a reasonable estimate of the damages, as know at the time of bidding.

2. **False or Fraudulent Claims**—The U.S. Federal government, some 27 States, and a number of large municipalities, have statutes to protect themselves against false claims (e.g., claims based upon knowing misrepresentations or falsified records). Other countries have similar legislation in the form of fraud statutes to protect project owners from false or fraudulent claims. If a claim can be proven to be false or fraudulent as defined in various statutes, recovery may include restitution, civil penalties, costs, punitive damages and attorney's fees. While such laws have been on the books for many years, some public owners/employers now assert such claims as a means for affirmative recovery when contractor claims are filed against them.

[18] See, for example, Robert F. Cushman and James J. Myers, *Construction Law Handbook*, Aspen Law & Business, Wolters Kluwer Company, Frederick, MD, 1999.

3. **Design Deficiency or Standard of Care Claims**—The basis for many contractor claims against owners/employers lies in the deficient performance of the Architect/Engineer (A/E). In turn, some owners/employers seek recovery from the A/E for the damages the contractor may recover from the owner/employer. Many aspects of the A/E's work may be called into question, such as design defects, tardy shop drawing review, untimely response to Requests for Information (RFIs), or inadequate site reports. To recover against the A/E, the owner/employer must typically show that the A/E failed to meet the standard of care in the performance of its duties as a design professional. Likewise, standard of care claims may be filed against contractors for defective construction which decreases the value of the constructed project or which causes the owner to expend money to repair such defects. In most jurisdictions, standard of care (applied to architects, engineers and contractors) is defined as that degree of skill and care ordinarily exercised by other similarly qualified professionals, practicing at the same time and location and under similar circumstances on projects of similar size and complexity. In essence, it is a comparison with the performance of the A/E's peers. The specific technical elements comprising standard of care are not typically defined by statute; rather, they are adjudicated on a case by case basis in arbitration, in court or in an alternative dispute resolution forum.

4. **Consequential Damages**—Consequential damages, sometimes called Special Damages, are damages that arise as a result of the failure of the owner/employer or the contractor to live up to their obligations under the contract. Such damages are not direct; foreseeable damages such as the cost of repairing deficient work or the cost arising from a 30-day delay. Consequential damages typically are indirect and unforeseeable damages resulting from some action or lack of action on the contract and not within the contemplation of the parties at the time the contract was executed. Many construction contracts include Consequential Damages clauses which purport to preclude contractors from recovering such damages as lost profits; impaired bonding capacity; interest on borrowings; and, in some cases, home office overhead costs. However, such clauses do not typically include a list of consequential damages for owners/employers. Many negotiated EPC contracts include a mutual waiver of consequential damages or a cap on consequential damages, sometimes expressed as a set percentage of the value of the contract.

Burden of Proof

The basic equation of a successful claim is summarized as follows:

- **Notice** – Formal, or sometimes informal (e.g. constructive notice), communication from one party that potentially will be seeking time and/or cost relief for an issue that has arisen. Notice gives the parties early warning of that issue and ability to mitigate and/or change direction as to how to deal with the issue. Absent a contractually compliant notice, it will often be argued that the claimant has waived its rights under the contract for relief.
- **Liability/Entitlement**—An event or circumstance has occurred during project performance, which gives rise to a legal right to an adjustment under the contract. For example, a change/variation, delay or differing site condition.
- **Causation**—The event or circumstance causes the contractor to do something or take some action which otherwise would not have occurred were it not for the event. For example some portion of the work is revised and performed differently than originally planned and work has to be re-sequenced as a result.
- **Damages /Quantum**—The work costs more and/or takes longer than planned.

The burden of proof of all four elements of a claim rests squarely on the shoulders of the party making the claim (the claimant). This is true whether the claim is filed by the contractor or filed by the owner/employer. That is, the claimant bears the burden of affirmatively proving all four elements of the claim based upon a preponderance of the evidence. To successfully recover in a claim situation, the claimant must document and prove all four parts before becoming eligible to recover any damages at all. Thus, notice of claim, clear documentation of facts (including both liability and causation) and concise tracking of damages (segregation of both cost and time for the changed work) are necessary prerequisites for a successful claim.

Damages

Damages are the costs or time incurred by a party as a result of one or more of the claims previously discussed. Monetary damages must be proven with some reasonable degree of specificity. Many contracts allow the owner/employer to audit the contractor's books and records to verify claimed costs. Some contracts specifically require the contractor to submit a statement of certification, under penalty of perjury, when submitting a claim concerning the accuracy of the facts of the claim and the damages claimed. Damages associated with construction claims generally fall into these categories.

Direct Costs—Direct costs are those costs incurred as a direct result of the claim situation. They may be described as hard dollar costs or costs incurred in performing extra work. Examples include:

- Labor costs, including fringe benefits and payroll taxes
- Materials
- Equipment
- Subcontractor direct costs
- Mobilization/demobilization costs
- Storage costs

Indirect Costs—Indirect costs are costs not allocable to any specific item of direct work. Such indirect costs may or may not be time related costs; that is, they continue or increase as the project performance period is extended. There are generally two basic types of indirect costs: field office overhead and home office overhead.

Field Office Overhead—Examples of costs in this category include:

- Project management staff
- Superintendents
- Project office
- Temporary utilities and security
- Maintenance and clean-up
- Communications
- Project vehicles
- Laydown and maintenance area costs
- Office equipment

Home Office Overhead—Examples of costs in this category include:

- Corporate management
- Accounting and payroll staff
- Other personnel costs
- Engineering
- Estimating
- Computers and office equipment
- Corporate insurance
- Home office space rent
- Central equipment yard

Home office overhead is often a controversial topic in the claims arena, at least in the U.S. Courts, and alternative dispute resolution forum decisions vary widely as to its acceptability. In some circumstances, estimating formulas (such as the Eichleay Formula in the U.S. and the Hudson or Emden Formulas in the U.K. and other British or former British Commonwealth nations) are used to calculate home office overhead. However, special legal rules and conditions may apply. This is an area of damages where one should consult with competent legal counsel and a claims consultant.

Over the years, some contractors have recovered additional home office overhead costs, either on the theory that extended home office overhead is incurred when a project is delayed, or that unabsorbed home office overhead has occurred when owner/employer-caused delay has impacted the revenue stream of the contract. Extended home office overhead is the concept that the owner/employer owes home office overhead cost for every day of owner/employer caused delay. Unabsorbed home office overhead is the concept that the owner/employer owes home office overhead only if an owner/employer caused delay results in a substantial decrease in the project's cash flow for an unknown duration and there is no other contract work the contractor can pursue to absorb that overhead cost. As most contracts do not specifically address home office overhead as an element of delay damages contractor claims for these types of damages can differ greatly.

Delay Costs—Delay costs are a category of costs that may be incurred by both contractors and owner/employers when a key project completion milestone is delayed. Examples of contractor delay costs include:

- Idle equipment
- Idle personnel
- Additional or extended storage costs
- Escalation costs of labor, materials, and equipment attributable solely to the delay

Owner/employer delay costs may be either liquidated or actual. Liquidated damages are provided for in the contract and specify the amount to be charged to the contractor for each day of contractor caused delay. Liquidated damages should be a reasonable forecast of costs that the owner/employer may incur for late completion. If the daily amount is so high that it is disproportionate to the owner's/employer's probable costs, it may be judged by a trier of fact as an unenforceable penalty. Owner/employer actual damages, unless precluded in a Consequential Damages clause or determined to be a consequential damage by a court or an arbitration panel, may include:

- Lost use of the facility
- Lost rental income
- Lost profits

- Delayed proceeds of the sale of the facility
- Increased or extended financing costs
- Extended general conditions and/or personnel costs
- Increased storage costs
- Holdover penalties
- Extended professional fees

No Damages for Delay and Consequential Damages clauses may complicate the recovery of delay costs by either party. As special legal rules often apply, it is best to consult with competent legal counsel and a claims consultant.

Impact Costs—One of the fastest growing areas of claimed costs is associated with impacts. These are often costs associated with non-critical path delays, which manifest themselves as productivity losses, but may also include impact to unchanged work. Such costs are typically labeled cumulative impact costs or disruption costs. Usually, the contractor asserts that it experienced higher than anticipated costs in performing a specific set of activities as a result of some action or event for which the contractor and their subcontractors are not responsible. The productivity loss might be claimed using:

- Activity total costs (planned vs. actual)
- Measured Mile Analysis (un-impacted productivity vs. impacted productivity)
- Industry standard (actual vs. industry standard)[19]

Impacts that are associated with activities on the critical path may also be part of a related delay claim. If the impact causes a critical path delay and the delay is both excusable and compensable, it is possible that the contractor would be entitled to delay damages as well as the direct cost claims related to the impacts.

Other Contractor Damages—There are other contractor costs that may arise in claims. These costs are not always recoverable as a result of specific contract disallowance clauses or a Consequential Damages clause. Examples include:

- Bond and insurance costs
- Loss of early completion bonus or completed plant performance (“name plate”) bonus
- Lost profits as a result of restricted bonding capacity
- Interest cost
- Legal and consultant fees
- Claim preparation costs

Other Owner/Employer Damages—Owner/employers may also incur other costs as a result of improper performance by contractors. Such costs may include:

- Defective work—repair or replacement costs
- Costs to complete the project
- Re-procurement costs—in the event of termination for default
- Extended warranty costs
- Third-party claim costs
- Legal and consultant fees

The recovery of these owner/employer losses is often governed by the terms of the contract documents. It is important that one read the contract for these points. If appropriate, owners/employers should consider modifying future contracts, after seeking advice from competent legal counsel, to protect their own interests as appropriate.

Claims and Project Delivery Methods

Construction claims arise on all types of project delivery methods. While several project delivery methods purport to reduce the probability of claims, claims are possible whenever the allocation of risk as stated in the contract is inconsistent with the parties’ abilities to control and manage such risks (Refer to Chapter 23, Contracting for Capital Projects, for a more detailed discussion of project delivery methods.) Some currently used alternative project delivery methods include:

- **Unit Price**—Under this delivery method, the engineer provides a detailed estimate of materials and material types as part of the contract documents. Under this system all bidders know the quantities and it is only the cost for the various units that vary. These contracts often become lump sum contracts and changes/variations are priced based on the change/variation in the number of units installed. Typically, such contracts include a Variation of Quantity clause and may even include an Economic Price Adjustment clause to deal with claims revolving around significant changes to certain

[19] Cost professionals should review AACE Recommend Practice 25R-03, *Estimating Lost Labor Productivity in Construction Claims*, to familiarize themselves with the various methods of productivity loss analysis.

quantities (both up and down) or impacted prices of certain quantity prices as a result of external forces (i.e., the price of oil may radically change the cost of asphalt material after award of the contract). Claims may also arise when the performance of the work becomes more difficult, making the original pricing of the units insufficient to cover actual installation/construction costs.

- **Design-Bid-Build**—In many construction projects, claims arise in the context of traditional design-bid-build method projects, where there is a separate design agreement between the designer and owner/employer. Upon completion of the design, competitive bids are received and a construction contract is then awarded. This traditional system, often called lump sum, low bid or hard dollar contracting, has been the primary delivery system in the U.S. public sector since the founding of the nation and therefore has the longest history of claims and legal precedents. However, if properly designed, managed and administered, this system will have few claims.

- **Design-Build**—Design-Build contracts require the Design-Build (D-B) contractor to complete the design based upon the owner's/employers' design program and then to construct it. Some owners/employers perceive that they are free from any contractual risk of claims using design-build project delivery method. While the design-build project delivery system can be an appropriate vehicle for shifting design and schedule/program-related risk from the owner/employer to the design-builder and can be a faster method of project delivery, it is no guarantee of a claims free contracting method. Ambiguities in contract language can and do arise under design-build contracts, leading to claims. Similar to Fast Track Construction (discussed below), sometimes construction begins early with drawings that are not yet final or Issued for Construction, which can cause rework and inefficiencies. While such instances typically fall to the design-builder's responsibility, these instances can create greater cost pressures on the rest of the project, thereby heightening sensitivities to other potential claims. The owner/employer using the design-build method is well advised to retain experienced legal counsel in drafting the contract terms and to favor tried and tested risk mitigation strategies over novel approaches. If this is done, design-build projects can provide greater certainty with respect to cost and schedule/program.

- **Fast-Track Construction**—This delivery system anticipates construction starting on early portions of the project prior to design completion of later parts of the project. Owners/employers using this system believe that the costs associated with an earlier overall completion outweigh the risks associated with starting construction with an incomplete design. Risks include the need to coordinate the interfaces between the multiple contracts and this can lead to claims or disputes. If properly designed, managed and administered claims can be minimized or avoided.

- **Multiple Prime Construction**—This system is usually mandated by U.S. State law or public policy on public contracts.[20] It requires that the project be broken up into several prime contracts (usually general, mechanical, electrical and plumbing) each with a separate contract with the owner/employer but not with the other prime contractors. This system requires that the owner/employer or their agent, typically a construction manager, provide overall coordination among the prime contractors. While potentially saving the owner/employer the profit markups typically contained in a single contract, this method also has considerable risk for the owner/employer as there is no direct contractual relationship between the contractors. Thus, if the general contractor delays or impacts the work of the electrical, mechanical and plumbing contractors, the impacted contractors can only file claims against the owner/employer as there is no privity of contract with these other prime contractors.

- **Construction Management at Risk (CM@R)** —In this system, the CM@R is initially engaged by the owner/employer to perform preconstruction services including estimating, scheduling and constructability reviews. As the design nears completion, the CM@R executes a construction contract for a specific scope and specific price, commonly known as a Guaranteed Maximum Price (GMP or G-max) contract. This method tends to generate fewer claims since the CM@R is more informed as to the scope and risks of the project.

- **Alliance Contracting (a.k.a. Integrated Project Delivery or IPD)** —This relatively new delivery method requires a three way contract between the owner/employer, the contractor and the designer, which establishes the risks and responsibilities of each party. In exchange for greater certainty of cost and liability, each party gives up some of its ability to claim for changes/variations in the project. This method is very specialized and needs unique insurance coverage; generally it is restricted to teams that have worked together frequently.

- **Public Private Partnership**—This is another relatively new delivery system. In this system, a developer works with an owner/employer and provides a wide range of services that could include site selection, programmatic development, and financing, in addition to the traditional design and construction activities. Public entities, especially in cities and historic districts, can stipulate general program parameters and then shop for the developer who will deliver them. The advantage

[20] For example, New York, New Jersey, North Carolina, Pennsylvania and Wisconsin have multiple prime contracting statutes for some or all public works construction. Additionally, some State or local agencies, as a matter of public policy, have adopted multiple prime contracting as a way to keep work local and enhance the local economy.

to the owner/employer is that it can develop needed buildings or infrastructure with minimum upfront cost. In exchange, it often relinquishes detailed project control to the developer.

Each of these systems has advantages and disadvantages and the hoped for successful project outcome will depend upon the selection of the appropriate method and the culture and ability of the owner/employer and other parties to perform in accordance with the risks as allocated. No system is free from the possibility of claims; however, an appropriate delivery method that is executed properly can reduce claims. Regardless of the method selected an owner/employer who does not have adequate in house construction management staff capabilities, should consider the engagement of an Agency Construction Manager (CMa). The CMa will enable the owner/employer to evaluate alternative project delivery methods and represent the owner/employer throughout the design and construction process. The professional CMa service is delivery method neutral and solely represents the owner/employer interests throughout the project as the owner/employer endeavors to achieve timely completion within budget and without claims.

Analysis of a Claim
The following is a multi-phased approach to analyzing a claim:

Phase One—Entitlement and Causation Analysis

- Identify each issue
- Identify and evaluate all relevant contract language
- Establish issue files[21]
- Analyze each issue
- Determine potential for contractual entitlement
- Allocate summary level or order of magnitude costs to specific issues if possible to prioritize further analysis
- Request additional information, if necessary

Phase Two—Delay Analysis

- Obtain baseline and updated project schedules/program, in native electronic format if possible.
- Compare as-planned, schedule/program updates and as-built schedules/programs to determine which activities were delayed and whether concurrent delays occurred.
- Identify periods of delay, disruption or acceleration.
- Associate claim issues with the identified periods.
- Perform a detailed schedule/program analysis.
- Identify the party or parties responsible for the delay(s).

Depending on the size of the project, schedule/program analysis can be very complex and labor intensive. There are a number of forensic schedule/program analysis methods, all with varying degrees of reliability and validity, depending upon the actual circumstances of the delays and the documentation available.[22] It is further recommended that unless the construction management staff has considerable forensic schedule analysis experience, competent professional advice should be sought before conducting complex delay analysis.

Phase Three—Damage Analysis

- Determine direct costs associated with each claim issue that has entitlement.
- Determine potential indirect and impact costs, if any, associated with these issues.
- Determine overhead costs, either from the contract or project records for each claim issue.
- Prepare damage calculations for each issue.

Phase Four—Settlement Negotiations

- Complete the analysis of each issue before commencing negotiations.
- Meet with the other party to negotiate settlement of each issue.
- Use an independent mediator or third-party-neutral as appropriate to facilitate settlement.

[21] Issue files are a compilation of contemporaneous project records and documents that speak to or are related to the issue at hand. In effect, issue files represent the chronology of events related to the issue and contain all known information related thereto. Documentation of Notice, the facts and circumstances that led to the issue being raised as a change/variation claim, mitigations taken, and actual costs and time incurred in addressing the issue, are all critical components of a complete issue file.
[22] Cost professionals should refer to AACE Recommended Practice 29R-03, *Forensic Schedule Analysis*, for further advice.

- Once settlement is achieved, draft and execute the appropriate settlement documents complete with a waiver of further claims concerning each issue settled. Involvement of legal counsel in drafting such language is strongly recommended.

DISPUTE RESOLUTION

Construction contracts traditionally include a Disputes clause. This is the clause that details, to a greater or lesser degree, the process by which all contract disputes will be prosecuted. The clause typically includes the law of the contract (that is, a provision which stipulates what law the contract will be interpreted under); the stages of progressively more formal dispute resolution procedures and forums that will be used (such as mediation, arbitration, litigation) in the event a negotiated settlement is not reached); what venue (location and perhaps even court jurisdiction) the dispute hearings will be held in; and possibly even what language the dispute proceedings will be conducted in if the contract is an international contract. While there are numerous variations on the dispute resolution theme there are four basic methods of resolving disputes on project, as follows.

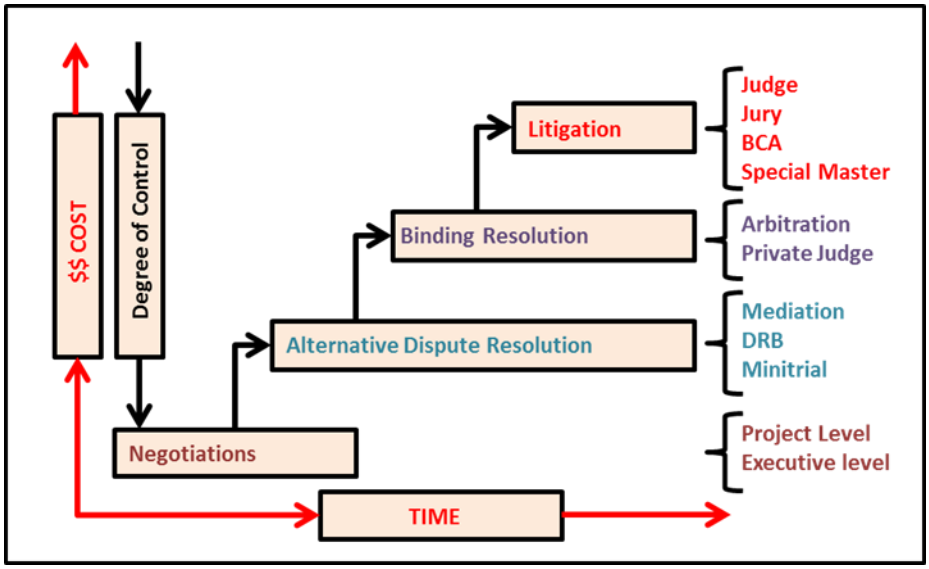
- **Negotiation**—Face to face negotiation may be accomplished in the field between project teams or elevated in the owner's/employer's and the contractor's organizations. In either case, the concept is for the project participants to discuss the disputed issue and mutually arrive at an acceptable business solution to the problem. Typically, negotiations between the parties begin at the project level, such as between foremen and project managers, and then escalate to higher level personnel if resolution is not met. Outsiders are rarely involved in such negotiations; the process and the outcome are entirely under the control of the parties to the dispute who are most familiar with the facts of the situation; and the outcome is confidential.

- **Mediation**—Mediation is a form of a structured negotiation between the parties using the services of an outside, neutral facilitator—the mediator. It is a voluntary submission of the dispute to a process, which is largely controlled by the parties. The mediator's only power is the power of persuasion and the mediator's role is generally to help bring the parties closer together until agreement on a solution can be arrived at. Mediators may list strengths and weaknesses of each party's positions, convey the same as deemed appropriate for facilitating settlement, or assist in identifying potential costs if litigation ensued. Unfortunately, the parties, not the mediator, control both the process and the outcome of the mediation.

- **Arbitration**—Arbitration is a more formalized and legalistic proceeding whereby the dispute is heard by an outside organization typically operating under a national or international set of rules such as the American Arbitration Association (AAA), the Judicial Arbitration and Mediation Service (JAMS), or the International Chamber of Commerce (ICC). There may be a single arbitrator or a panel appointed by one of these organizations. Formal hearings are generally conducted with direct testimony, cross examination, submittal of evidence, qualification, and testimony of experts, etc. At the end of the hearing(s) the arbitrator or arbitration panel rules on the outcome of the dispute. In most arbitration proceedings, the arbitrator's ruling is enforceable at law in a court of competent jurisdiction and may be appealed only for very limited causes. While all of the rules governing a formal court case do not apply in arbitration, most arbitrators are skilled trial attorneys; and, therefore, an arbitration hearing is much more formalized and legalistic than mediation. Additionally, the parties no longer control either the process or the outcome; and, generally, are not free to quit the process of their own volition. Depending upon the size and complexity of the dispute, arbitrations can be lengthy and very expensive.

- **Litigation**—This is a formal lawsuit in a court of competent jurisdiction (frequently stipulated in the contract's Venue clause) pursuant to the terms of the contract and under the rules of the jurisdiction where the lawsuit is filed. Lawsuits are time consuming, lengthy, very expensive, and can be quite disruptive to ongoing business operations. The outcome may rest more on legal technicalities than on fact or circumstance. The parties to such litigation retain little control over the process or the outcome.

- **Alternative Dispute Resolution**—There are numerous other forms of Alternative Dispute Resolution (ADR) available to the parties involved in a dispute, as shown in Figure 26.1. Many are voluntary and need not be mandated by contract. It is almost axiomatic that the sooner the owner/employer and the contractor get an issue to a dispute resolution forum, the less expensive it will be for both sides, almost regardless of the outcome. Likewise, the more involvement actual project participants maintain in the dispute resolution process, the more likely it is that the outcome will be driven by facts and not legal tactics. As Figure 26.1 shows, the further up the staircase the parties move, (1) the more time and cost will be expended; and (2) the less control the parties will have over the outcome of the process.



Source – Navigant Construction Forum™ -- http://www.navigant.com/NCF

Figure 26.1—Dispute Resolution Staircase

CONCLUSION

Claims are inevitable on projects; but, they need not end up in arbitration or litigation. Good, proactive claims management on the part of the cost professionals working for both contractors or owners/employers, should result in the equitable resolution of project claims; provided that, the claims are prepared and documented in accordance with the rules applicable to the type of claim by the contractor and reviewed and analyzed in accordance with the terms and conditions of the contract by the owner/employer and their representatives. In order to accomplish successful resolution of claims, cost professionals working on projects must become knowledgeable of construction claims, remain alert to potential claims on a daily basis, and understand how such potential claim situations must be managed, in order to avoid turning claims into legal disputes. This chapter is just the beginning of learning about good claims management. To be a true cost professional, there is much more to learn.

RECOMMENDED READING

1. Bramble, B. B. & West, J. D. (1999). *Design-Build Claims*, Gaithersburg, MD: Aspen Law & Business.
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SECTION 6 – ECONOMIC ANALYSIS, STATISTICS, PROBABLILITY AND RISK

Chapter 27 – Financial and Cash Flow Analysis
Chapter 28 – Practical Corporate Investment Decision-Making Guide
Chapter 29 – Statistics and Probability
Chapter 30 – Optimization
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Chapter 32 – Risk Management Practical Guide
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Chapter 27—Financial & Cash Flow Analysis

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Abstract
Engineering decision making usually applies economic analysis to assist in the solution of substantial engineering problems where economic aspects dominate over a considerable period of time. Using the time value of money, discount factors, and the principle of equivalence, the economic costs of a problem can be modeled and evaluated. Those who have previously taken a basic financial management or engineering economic course will find this material a refresher of the concepts and basic tools of financial mathematics. The chapter provides a review of the time value of money to assist in the performance of basic cash flow analysis to support economic and financial analyses that inform investment decisions. The use of these financial tools for decision making will be illustrated in subsequent chapters.

Keywords
Cash Flow, Discount Factor, Economic Cost, Equivalence, Future Value, Interest, and Present Value.

INTRODUCTION

Engineering economics is a decision making process based on an evaluation of the monetary consequences of the products, projects, and services that engineers provide. Similarly, financial analysis is the process of evaluating businesses, projects, budgets, and other finance-related entities to determine their suitability for investment. These processes use financial mathematics based on the time value of money. During the life-cycle of a project, the owners and project managers are frequently confronted with the need to select from a variety of alternative activities. This process can be greatly simplified if the monetary valuation for each of the possible activities and/or following decision or course of action is quantified objectively as an economic cost. These costs are typically determined in two ways: (1) the summation of accounting costs; calculated from the sum of the inflows and outflows such as the cash outlays of effort, material, resources, time consumed, utilities consumed, and risks incurred; or, (2) the quantification of risk and opportunity costs calculated from the cost of employing resources for one activity rather than the other. This chapter will review and demonstrate the required financial math calculations to enable the modeling and subsequent analysis of economic cost problems in terms of the time value of money.

Learning Objectives
After completing this chapter, the reader should be able to:

- Calculate simple and compound interest rates, solve interest problems using basic single payments, uniform series, and gradient formulas.
- Calculate present value, future value, and equivalent uniform annual value of cash flow series.
- Calculate a cash flow for an asset investment option from a specified set of cost and revenue forecasts.
- Determine the discount rate of return of a cash flow series.
- Understand (and calculate) the concept of opportunity cost and benefits.

TIME VALUE OF MONEY

Interest—When monetary consequences of a problem occur over a substantial period of time there is usually a cost associated with the use of the money. Since money is a valuable asset, people are willing to pay to have it available for their use. This could be likened to paying rent for the use of money just as one would pay rent for the use of an apartment. With money, the charge for its use is called interest rather than rent.

Simple Interest—The interest due is proportional to the length of time the principal is outstanding and does not accrue or compound on previous interest. Each subsequent interest payment is calculated based on the total principal, ignoring accumulated interest to date, as shown in Example 27.1.

Example 27.1—Simple Interest Calculation

(i = 10%) at end of year (EOY)	
Principal	$1,000
Interest accrued EOY1	100
Interest accrued EOY02	100
Total @EOY2 with simple interest	$1,200

Compound Interest—The interest is considered an increased increment of principal that is capable of earning additional interest with time. Each subsequent interest payment is calculated based on the total principal plus accumulated interest to date, as shown in Example 27.2.

Example 27.2—Compound Interest Calculation

(i = 10%)	
Principal	$1,000
Interest accrued EOY1	100
New principal with accrued interest	$1,100
Interest accrued EOY02	110
Total @EOY2 with compound interest	$1,210

Nominal Interest Rate—The nominal annual interest rate, *r*, will disregard the effects of compounding periods that are more frequent than annually. Common practice is to express interest rates at the nominal rate even though there is compounding such as quarterly, monthly, or daily. The nominal rate, *r*, is not the same as the annual rate, *i*, except in the case of annual compounding.

Effective Interest Rate—If the effective interest rate per period, ϕ, is given it can be converted easily to the effective annual interest rate, *i*, which includes the effects of compounding over *k* periods per year by using Equations 27.1 or 27.2. In general, it can be assumed that the rate given in a problem is annual unless stated otherwise. If compounding is annually, the rate given is the effective rate. If compounding is anything other than annual, the rate given is the nominal rate.

$$i = \left(1 + \frac{r}{k}\right)^{k} - 1$$

Equation 27.1

$$i = (1 + \varphi)^{k} - 1$$

Equation 27.2

Example 27.3—Effective and Annual Interest Calculations

Given a nominal interest rate of 12 percent, find the effective and annual rates for yearly, semi-annual, and monthly compounding.

Solution—The nominal rate, *r*, is given as 12 percent. The effective rate per period can be found by dividing the nominal rate, *r*, by the number of compounding periods per year, *k*. The effective annual rate then can be found through the application of Equation 27.1 or 27.2.

For yearly compounding, the effective annual rate is the same as the nominal rate of 12 percent.
For semi-annual compounding (k = 2):

$$\varphi = \frac{r}{k} = \frac{.12}{2} = .06 \; or \; 6\%$$

$$i = (1 + \varphi)^{k} - 1 = (1 + .06)^{2} - .06\,4 \; or \; 12.4\%$$

For monthly compounding (k=12):

$$\varphi = \frac{r}{k} = \frac{.12}{12} = .01 \; or \; 1\%$$

$$i = (1 + \varphi)^{k} - 1 = (1 + .01)^{12} - 2\,0127 \; or \; 12.7\%$$

Continuous Compounding—Discrete compounding occurs when interest payments are made at the end of finite compounding periods—such as annually, monthly, quarterly, or daily. As the duration of the interest period becomes infinitely short, the number of compounding periods per year becomes infinite and is referred to as continuous compounding. It can be shown that the limit of Equation 27.1 as *k* approaches infinity is the continuous compounding rate:

$$i = e^{r} - 1$$

Equation 27.3

For continuous compounding in Example 27.3:

$$i = e^{r} - 1$$

$$i = e^{.12} - 1 = .128 \; or \; 12.8\%$$

As a result of this relationship, the formulas and factors to be developed for discrete compounding are not directly applicable to problems involving continuous compounding.

Minimum Attractive Rate of Return—The interest rate used in feasibility studies is often called the Minimum Attractive Rate of Return (MARR), the minimum rate of return at which the owner is willing to invest. The selection of a suitable rate of return can be quite complex and could vary from problem to problem and usually involves the analysis and selection of the highest one of the following three values:

- Cost of borrowed money from banks, insurance companies, etc.
- Cost of capital or the composite value for the capital structure of the firm.
- Opportunity cost or the rate-of-return of the best project that is rejected.

When there is a significant risk or uncertainty in a project, a commonly used method is to increase the MARR. This may result in the selection of alternatives with early revenue at the expense of those with equal or greater benefits that are realized later in the project. Care should be taken, since a MARR that is unrealistically low or high will magnify or discount the importance of future costs and benefits, leading to the mistaken conclusion that a project is either too costly or with low risk.

Example 27.4—MARR Selection
A firm is evaluating the feasibility of a design and construction project and needs to know what interest rate should be used in the study. The following data has been compiled:

- Cost of borrowed money, loan A = nine percent
- Investment opportunity, project B = 16 percent
- Cost of capital = 20 percent

Solution—The MARR should be equal to or greater than the highest of the three values. Choose 20 percent since the cost of capital is greater than both the cost of borrowed money and the investment opportunity cost.

Equivalence
An essential concept in engineering economic analysis is that of equivalence: the ability to compare cash flows at different points in time. Equivalence is based on the time value of money, and the cardinal rule is that two cash flows (also called alternatives) only can be compared at a common interest rate[23]. For example, if a person is indifferent to the acceptance of $110 one year from now in lieu of $100 today, we can say that the two sums of money are equivalent at an interest rate of ten percent. If however, the interest rate is something greater than ten percent, the sums are not equivalent and one would prefer the $100 today.

In another example, there are two alternatives. Alternative one has a lower initial cost and higher operating cost, while alternative two has a higher initial cost and lower operating costs. Based on the time value of money, the sums of money at different times cannot be summed up directly. Using equivalence calculations, both streams of money, or cash flows, can be converted to either lump-sum values at any point in time or a series of uniform benefits/costs that will allow for a direct comparison. The conversion of dissimilar cash flows to similar lump-sums at a particular interest rate and a common point in time (or a uniform series of benefits/costs) provides values with a valid basis for comparison.

Discount Factors
Discount factors that have been derived for discrete compounding are summarized in Table 27.1. The algebraic form of the various factors can become quite complex. The functional notation illustrates a standardized notation that simplifies the calculations and permits the use of tabulated factors, as shown in Tables 27.2 and 27.3. The notation is as shown in Equation 27.4:

$$(X/Y,i,n)$$

Equation 27.4

which can be read as, "to find the equivalent amount *X* given amount *Y*, interest rate *i*, and the number of discounting or compounding periods *n*." These factors can be used to simplify cash flows or convert cash flows for comparison purposes.

- **Present Value** (P) —A single lump sum occurring at time zero, the first of *n* time periods.
- **Annual Value** (A) —Annual amount or annuity. A uniform series of end-of-period payments or receipts.
- **Future Value** (F) —A single lump sum value occurring at the end of the last of *n* time periods
- **Gradient Value** (G) —Uniform or arithmetic gradient amount; a constant increase or decrease in funds flow at the end of each period.

[23] Note that this differs from the concept of *monetary equivalents*. The concept presented here only addresses the preference of the decision maker in respect to the time value of money, not the monetary valuation of the "utility" of or preference for a non-cash attribute of an alternative.

Table 27.1—Discount Factors for Discrete Compounding

Factor	Notation	Formula	Application
Single Payment Compound Amount	(F/P, *i, n*)	$(1+i)^n$	Given present value find future value
Single Payment Present Value	(P/F, *i, n*)	$(1+i)^{-n}$	Given future value find present value
Uniform Series Sinking Fund	(A/F, *i, n*)	$\frac{i}{(1+i)^n-1}$	Given future value find annual value
Uniform Series Capital Recovery	(A/P, *i, n*)	$\frac{i(1+i)^n}{(1+i)^n-1}$	Given present value find annual value
Uniform Series Compound Amount	(F/A, *i, n*)	$\frac{(1+i)^n-1}{i}$	Given annual value find future value
Uniform Series Present Value	(P/A, *i, n*)	$\frac{(1+i)^n-1}{i(1+i)^n}$	Given annual value find present value
Arithmetic Gradient Present Value	(P/G, *i, n*)	$\frac{(1+i)^n-1}{i^2(1+i)^n}-\frac{n}{i(1+i)^n}$	Given gradient value find present value
Arithmetic Gradient Uniform Series	(A/G, *i, n*)	$\frac{1}{i}-\frac{n}{(1+i)^n-1}$	Given gradient value find annual value

Table 27.2—Compound Interest Discount Factors at Five Percent

	Single Payment		Uniform Payment Series				Gradient	
n	F/P	P/F	A/F	A/P	F/A	P/A	A/G	P/G
1	1.050	.9524	1.000	1.0500	1.000	0.952	0	0
2	1.102	.9070	.4878	.5378	2.050	1.859	0.488	0.907
3	1.158	.8638	.3172	.3672	3.152	2.723	0.967	2.635
4	1.216	.8227	.2320	.2820	4.310	3.546	1.439	5.103
5	1.276	.7835	.1810	.2310	5.526	4.329	1.902	8.237
6	1.340	.7462	.1470	.1970	6.802	5.076	2.358	11.968
7	1.407	.7107	.1228	.1728	8.142	5.786	2.805	16.232
8	1.477	.6768	.1047	.1547	9.549	6.463	3.244	20.970
9	1.551	.6446	.0907	.1407	11.027	7.108	3.676	26.127
10	1.629	.6139	.0795	.1295	12.578	7.722	4.099	31.652

Table 27.3—Compound Interest Discount Factors at Six Percent

	Single Payment		Uniform Payment Series				Gradient	
n	F/P	P/F	A/F	A/P	F/A	P/A	A/G	P/G
1	1.060	.9434	1.0000	1.0600	1.000	0.943	0	0
2	1.124	.8900	.4854	.5454	2.060	1.833	0.485	0.890
3	1.191	.8396	.3141	.3741	3.184	2.673	0.961	2.569
4	1.262	.7921	.2286	.2886	4.375	3.456	1.427	4.945
5	1.338	.7473	.1774	.2374	5.637	4.212	1.884	7.934
6	1.419	.7050	.1434	.2034	6.975	4.917	2.330	11.459
7	1.504	.6651	.1191	.1791	8.394	5.582	2.768	15.450
8	1.594	.6274	.1010	.1610	9.897	6.210	3.195	19.841
9	1.689	.5919	.0870	.1470	11.491	6.802	3.613	24.577
10	1.791	.5584	.0759	.1359	13.181	7.360	4.022	29.602

Cash Flow

The model used for engineering economic analysis is based on the conversion of an existing cash flow to an equivalent cash flow at a particular interest rate through the application of predefined factors. The cash flow can be represented in tabular or graphical format. The conversion factors are called discount factors and are readily available in either algebraic form or in tables.

Many hand-held calculators and mobile device applications have been programmed with these factors in addition to most computer spreadsheet applications. A thorough understanding of the structure and development of the model and its components will permit the solution of engineering economic problems using any of three approaches: tabular, calculator/mobile device, or computer spreadsheet application.

Cash Flow Diagrams—Cash flow occurs whenever cash or something of monetary value flows from one party to another. It can be either cash flow in, for example cash receipts, or when payments or disbursements are made, which is a cash flow out. This can be shown in a tabular format as in Table 27.4 or in the case of relatively simple problems; a cash flow diagram similar to Figure 27.1(a) can be developed. The cash flow diagram consists of two basic parts: (1) the horizontal or time line, and (2) the vertical arrows or cash flow lines. The horizontal line is divided into periods appropriate for the specific problem under consideration. Each short vertical line represents an interest period based on an end of period convention. Thus, the points in Figure 27.1(b) represent interest periods one through five. These periods could be daily, monthly, quarterly, semi-annually, or annually depending on the problem statement. All transactions will be assumed to occur at the end of the interest period.

The vertical lines represent cash flow and are placed according to the timing and direction of the cash flow. Receipts or income are represented by an upward arrow, while disbursements and expenditures are downward arrows representing cash out. Although the arrows are not usually to scale, a longer line will distinguish a large cash flow from a smaller one. The arrows also will be labeled with the monetary value of the transaction. Other standard labels that might be used on arrows are *P*, *A*, *F*, *G*, and *S*, as shown in Figure 27.1(c).

Table 27.4—Cash Flow Table

Year	Income	Expense
0		$20,000
1	$5,000	$500
2	$5,000	$600
3	$5,000	$700
4	$5,000	$800
5	$5,000	$900

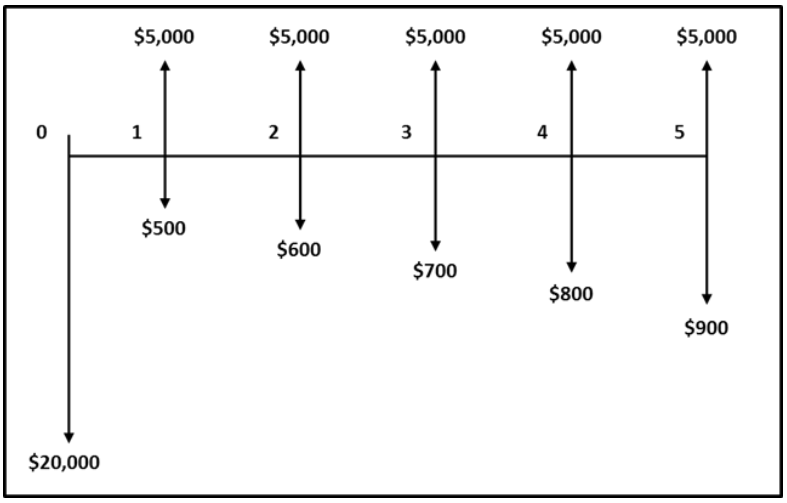


Figure 27.1 (a) —Cash Flow Diagram

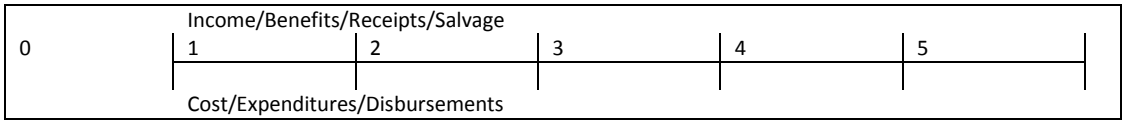


Figure 27.1 (b) —Cash Flow Conventions

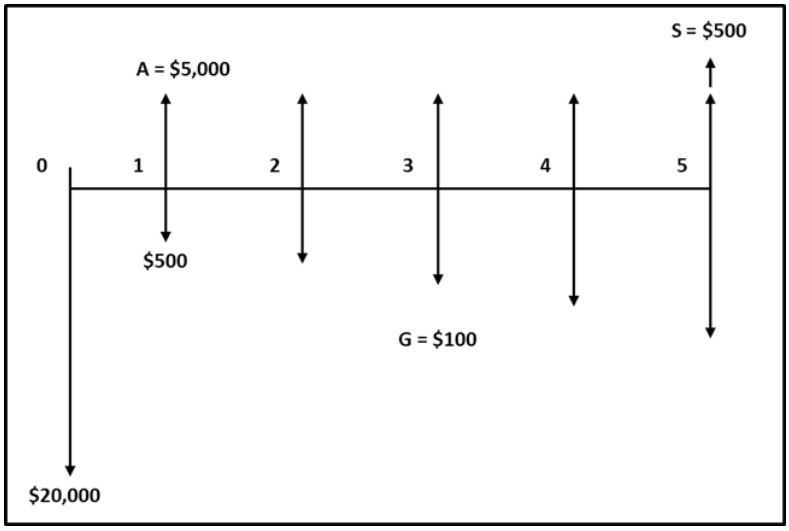


Figure 27.1 (c) —Cash Flow Notation

Salvage Value (S)—In many studies there may be a residual value resulting in income at the end of the useful life of an asset. The resale or salvage value may be associated with the anticipated market value of the asset at that point in time. It is shown as an upward arrow on the cash flow diagram. If the salvage value is low with respect to other cash flow, it is usually omitted. Any significant costs associated with disposal at the end of useful life also can be shown on the diagram as a downward arrow.

CASH FLOW ANALYSIS METHODS

There are two fundamental approaches to the analysis of a given cash flow, equivalent value and rate-of-return. In most types of engineering economic problems, the primary calculation is determining the present value or equivalent uniform annual cost of a cash flow to provide a basis for analysis.

Equivalent Net Value—The equivalent net value method simply converts to one of the basic forms, (i.e., the equivalent present value, or annual value, using discount factors and the required MARR). The "net" is the difference between all costs and all benefits (savings and other gains). Thus, the Net Present Value (NPV) takes into account the time value of money adjusting to expenditures and returns, as they occur over time, so they can be evaluated equally. A negative result means the proposed cash flow is unacceptable because it does not provide the required return-on-investment. Positive results are desirable and indicate an investment that will meet the prescribed criteria. This technique will be illustrated in Examples 27.5 through 27.7.

Example 27.5—Equivalent Net Present Value (NPV)

A contractor is considering the acquisition of a piece of equipment with anticipated financial impact as shown in Table 27.5. If the contractor's MARR is six percent, should the investment be made?

Solution—There are two approaches that can be taken to this problem. The net present value of a cash flow can be computed given a lump-sum future value, a uniform series, or an arithmetic gradient series. Because of the small number of compounding periods, the simple approach would be to compute the present value of the net cash flow for each year based on the MARR of six percent, as shown in Table 27.5. If summation of the net present values results in a positive number, the investment is desirable.

Table 27.5—Net Present Value Analysis for Example 27.5

Year	Expense	Income	Net	Factor	P
0	$38,000	$0	($38,000)	1.000	($38,000)
1	$0	$11,000	$11,000	.9434	$10,377
2	$1,000	$11,000	$10,000	.8900	$8,900
3	$2,000	$11,000	$9,000	.8396	$7,556
4	$3,000	$11,000	$8,000	.7921	$6,337
Total	$44,000	$44,000	$0		($4,830)

Although the income and expense were equal, the application of an interest rate, or the time value of money, resulted in a negative present value of $(4,830); therefore, the investment is not desirable.

A second approach to the problem would be to use a cash flow diagram, as shown in Figure 27.2.

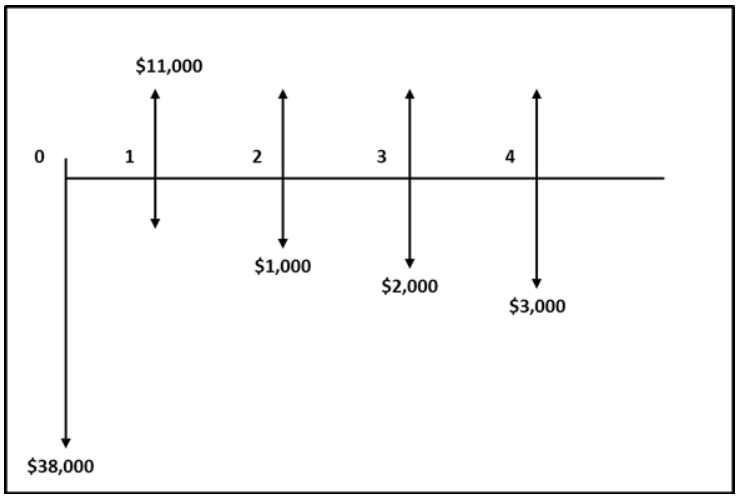


Figure 27.2—Cash Flow Diagram for Example 27.5

From the diagram it can be seen that the cash flow is comprised of a lump-sum expense of $38,000, an arithmetic gradient expense of $1,000, and a uniform series income of $11,000. Application of the discount factors gives the following:

$P = P_0 + P_1 + P_2$

Where:

P_0 = -$38,000

$P_1 = -G\ (P/G,i,n)$ = -$1,000(P/G, 6%, 4) = -$1,000(4.945) = - $4,945

$P_2 = A\ (P/A,i,n)$=$11,000 $(P/A, 6\ \%, 4)$ = $11,000 (3.465) = $38,115

P = -$38,000-$4,945+ $38,115 = -$4,830

Which agrees with the previous solution.

Example 27.6—Equivalent Net Future Value
Using the cash flows of the previous example calculate the net future value at EOY4.

Solution—The future value method uses the end of the planning horizon as a reference point. Typical applications would include investment growth and equipment replacement costs. As before, we will first use a tabular solution based on the net cash flow, as shown in Table 27.6.

Table 27.6—Future Value Analysis for Example 27.5

Year	Net	Factor	F
0	($38,000)	1.262	($47,956)
1	$11,000	1.191	$13,101
2	$10,000	1.124	$11,240
3	$9,000	1.060	$9,540
4	$8,000	1.000	$8,000
Total			($6,075)

As in the previous example, the future value is negative, and the investment would not be considered acceptable. This problem can also be solved using the cash flow diagram of Figure 27.2.

Notice that in Table 27.1 there are no discount factors that convert from *G* to *F*. To solve this problem, the gradient series first can be converted to a present worth and then to a future worth.

$F = F_0 + F_1 + F_2$

Where:

F_0 = -*P (F/P, i, n*)= - $38,000 (*F/P*, 6 %, 4) = -$38,000 (1.262) = -$47,956

F_1 = -*G (P/G, i, n) (F/P, i,n)* = -$1,000 (*P/G*, 6 %, 4) (*F/P*, 6%, 4)
= -$1,000 (4.945) (1.262) = -$6,241

F_2 = *A (F/A,i,n)* = $11,000 (*F/A*, 6 %, 4) = $11,000 (4.375) = $48,125

F = -$47,956 - $6,241 + $48,125 = -$6,072

The slight difference in results can be attributed to rounding errors, but again the future value is negative and the investment would not be acceptable. Note that regardless of the approach, the results are the same and the investment would not be acceptable.

Example 27.7—Equivalent Annual Value
Referring again to the cash flow of example 27.5, calculate the equivalent uniform annual value.

Solution—The basis of this method is the conversion of all cash flows to an Equivalent Uniform Annual Value (EUAV). It is often expressed as either an Equivalent Uniform Annual Cost or Benefit (EUAC or EUAB) to distinguish between costs and benefits. Typical applications for this method are in the calculation of loan payments and determining the cost of capital recovery.

In order to compute an annual value, it is first necessary to convert the lump sum at time zero and the gradient to equivalent uniform series then add them to the existing uniform series.

$A = A_0 + A_1 + A_2$

Where:

A_0 = *P (A/P, i, n)* = -$38,000 (*A/P*, 6%, 4) = -$38,000 (.2886) = -$10,967

A_1 = *G (A/G, i,n)* = -$1,000 (*A/G*, 6%, 4) = -$1,000 (1.427) = -$1,427

A_2 = $11,000

A = -$10,967 - $1,427 + $11,000 = -$1,394

Another approach would be to convert the lump sum, either *P* or *F* as determined in the previous examples through application of the appropriate discount factors.

From Example 27.5,

P = - $4,830 and,
A = *P* (*A/P, i, n*) = -$4,830 (*A/P*, 6%, 4) = -$4,830 (.2886) = -$1,394

Or, using Example 27.6,

F = - $6,075 and,

A = *F* (*A/F, i, n*) = -$6,075 (*A/F*, 6 percent, 4) =-$6,075 (.2286) = -$1,389

If a purely algebraic approach had been taken, these three numbers would have been equal. As before, the difference can be attributed to rounding errors from the tabular factors.

Capitalized Cost—Problems occasionally arise involving extremely long analysis periods. For example, in governmental analysis of permanent structures such as roads, dams, and pipelines, the required maintenance can be spread over an infinite period ($n = \infty$). In these cases the analysis is called capitalized cost or more simply stated, capitalized cost is the present sum of money (P) that would have to be set aside now, at a given interest rate (i), to provide a perpetual uniform cash flow (A). In equation form,

$$P = A/i$$

Equation 27.5

Example 27.8—Capitalized Cost
What is the capitalized cost of a public works project that will cost $15,000,000 now and will require $1,000,000 in annual maintenance? The effective annual interest rate is ten percent.

Solution—The capitalized cost will be equal to the initial cost of $15,000,000 plus the capitalized annual cost as determined from Equation 27.5.

P = $15,000,000 + $1,000,000/.10 = $25,000,000

Benefit-Cost Ratio—This analysis technique is a variation of the equivalent value method often used in municipal project evaluations. It is based on the ratio of benefits (irrespective of the beneficiaries) to costs. An alternative is considered acceptable if the ratio equals or exceeds 1.0 as shown in the following criteria:

$$B/C = PV \text{ of benefits}/ \text{PV of costs} = EUAB/EUAC \geq 1$$

Equation 27.6

Another way of stating this in terms of difference is:

$$PV \text{ of benefits} - PV \text{ of costs} \geq 0 \text{ or } EUAB - EUAC \geq 0$$

Equation 27.7

Example 27.9—Benefit/Cost Analysis
A state department of transportation is evaluating the construction of a bypass highway. The estimated initial cost is $30,000,000 and the capitalized cost of perpetual annual maintenance is $9,000,000. The estimated capitalized value of annual user benefits due to factors such as decreased travel time, fewer accidents, reduced fuel consumption, etc., is $40,000,000. Should the bypass be built?

Solutions—Using Equation 27.7, subtract the present value of costs from the present value of benefits.

PV of benefits - *PV* of costs = ($40,000,000) - ($30,000,000 + $9,000,000) = $1,000,000

Since the difference is greater than zero, the alternative is acceptable and the bypass should be constructed. If the present value of the benefits in this case were less than $39,000,000, the alternative would not be acceptable. The estimation of non-cash flow factors is a critical element of this methodology.

Rate-of-Return—The fundamental concept behind Rate-of-Return (ROR) analysis is that the ROR is the interest rate at which benefits are equivalent to costs. Thus, the cash flow is solved for the unknown value, i. This technique is also known as internal rate of return and can be used with either present value or annual value equivalents.

Example 27.10—Rate-of-Return Analysis
A $10,000 investment returned $2,342 per year over a five-year period. What was the rate of return on this investment?

Solutions—Set the present value of benefits equal to the present value of costs, then algebraically isolate the discount factor and treat it as an unknown.

$2,342 (*P/A, i*, 5) = $10,000

(*P/A, i*, 5) = $10,000/$2,342 = 4.27

Now look at the compound interest factors in the Tables 27.2 and 27.3 for the value of *i* where (*P/A*, *i*, 5) = 4.270; since no tabulated value matches the exact value of 4.270, the next step is finding the values on either side of the desired value of 4.270 and by interpolating to find the rate of return *i*.

From Tables 27.2 and 27.3, the respective P/A factors of five percent and six percent are shown as:

i	(P/A, *i*, 5)
5.0%	4.329
?	4.270
6.0%	4.212

In this example, the rate of return for the investment was found to be five-and-a-half percent by interpolation.

MULTIPLE ALTERNATIVES

Many projects will require a selection from among several mutually-exclusive alternatives. The selection of one alternative will preclude the selection of any other alternative. When comparing alternatives using present value methods, it is necessary to analyze over a common planning horizon. In the event that alternatives do not have equal lives, consideration must be given to the difference. A common technique is to select an analysis period equal to the least common multiple of the alternative lives. Another approach is to select an analysis period and determine the salvage value for each alternative at that point in time. When using the annual value analysis method there is no need to establish equal lives for each alternative. Three simple rules will help identify the preferred alternative when using the net equivalent value methods:

- Compute the net present value (annual value, future value, or capitalized cost) of each alternative at the required MARR.
- Rank the alternatives.
- Select the alternative having the highest net present value (annual value or future value). In the case of problems involving long or infinitely lived analysis periods, the superior alternative will have the lowest capitalized cost.

Example 27.11—Present Value Analysis of Multiple Alternatives

Given the mutually-exclusive alternatives A, B, C, and a minimum attractive rate of return (MARR) of five percent, which one would be chosen?

Year	A	B	C
0	($2,500)	($2,700)	($3,000)
1	$0	$650	$0
2	$0	$650	$350
3	$0	$650	$700
4	$0	$650	$1,050
5	$3,100	$650	$1,400
Total	**$600**	**$550**	$500

Solution—Compute the net present value for each alternative, rank the alternatives, and select the highest value. Notice the values on the total line would suggest that alternative A at $600 would be favored if time value of money is ignored. Now consider the impact of the time value of money.

PV_A = - $2,500 + $3,100 (*P/F*, 5%, 5) = -$2,500 + $3,100 (.7835) = -$71

PV_B = - $2,700 + $650 (*P/A*, 5%, 5) = -$2,700 + $650 (4.329) = $114

PV_C = - $3,000 + $350 (*P/G*, 5%, 5) = -$3,000 + $350 (8.237) = -$117

This analysis provides an interesting result. Alternative A, which had the least initial cost and apparently the most return, is in fact not the preferred alternative based on maximization of the net present value at five percent. Alternative B is preferred in this case.

Incremental Analysis—The ranking method does not apply to rate-of-return or benefit-cost ratio analysis. Assume that the rate-of-return on two alternatives is known. Investment A yields 100 percent and investment B yields 20 percent. Do we select A as the preferred alternative? Given further information we find that investment A was one dollar returning two dollars at the end of year one, and investment B was $10,000, returning $12,000 at the end of year one. The profit for A is one dollar and B is $2,000. The lower Rate of Return (ROR) alternative in this case is probably the better investment if the MARR is 20 percent or less. This illustrates the need for a procedure to evaluate the return on the increment of initial investment if one alternative requires a higher initial investment than the other. This process should also apply to multiple alternatives. By examining the differences between alternatives, we can determine whether or not the differential costs are justified based on the differential benefits.

This technique is based on the paired comparison of alternatives. The following steps should be followed in an incremental rate-of-return analysis:

1. Identify all alternatives. Be sure to consider the option of maintaining the status quo or what is often called "do-nothing."
2. Calculate the rate-of-return for each alternative and discard any alternative with ROR < MARR.
3. Arrange remaining alternatives in ascending order of initial cost.
4. Calculate the rate-of-return on the difference between the first two (lowest initial cost) alternatives (if this ΔROR ≥ MARR, retain the higher cost alternative, otherwise retain the lower cost alternative).
5. Select the retained alternative from the previous step, compare it to the next higher alternative using the calculation of step 4 and calculate the rate-of-return on their difference (if this ΔROR ≥ MARR, retain the higher cost alternative, otherwise retain the lower cost alternative).
6. Repeat this process until all alternatives have been evaluated.

Example 27.12—Rate-of-Return Incremental Analysis
Given the mutually-exclusive alternatives A, B, C and a minimum attractive rate of return (MARR) of five percent, which one should be chosen? Use the incremental rate-of-return method and assume the "do-nothing" alternative is not available.

Year	A	B	C
0	($2,500)	($2,738)	($3,000)
1	$0	$650	$0
2	$0	$650	$350
3	$0	$650	$700
4	$0	$650	$1,050
5	$3,191	$650	$1,400

Solution—The first step is to compute the rate-of-return for each alternative.

For alternative A:
$2,500 = $3,191 (*P/F, i,* 5)
(*P/F, i*, 5) = $2,500/$3,191 = .7835 and from Table 27.2, *i* = 5%

For alternative B:
$2,738 = $650 (*P/A, i,* 5)
(*P/A, i*, 5) = $2,738 /$650 = 4.212 and from Table 27.3, *i* = 6%.

For alternative C:
$3,000 = $350 (*P/G, i*, 5)
(*P/G, i*, 5) = $3,000/$350 = 8.571
From the tables find:

Table 27.2	Table 27.3
(*P/G*, 5%, 5)	(*P/G*, 6%, 5)
8.237	7.934

The P/G factor increases as the interest rate decreases. Since the P/G factor of 8.571 is higher than the P/G factor of 8.237 at a five percent interest rate, the corresponding i for alternative C would be less than five percent. Thus, alternative C should be rejected because it does not achieve the MARR.

The remaining alternatives are already ranked in ascending order of initial cost. The next step is to compute the difference by subtracting the lower initial cost from the higher initial cost as shown below:

Year	A	B	B-A
0	($2,500)	($2,738)	($238)
1	$0	$650	$650
2	$0	$650	$650
3	$0	$650	$650
4	$0	$650	$650
5	$3,191	$650	($2,541)

Next, calculate the rate-of-return by setting present value of benefits equal to present value of costs.

$650 (P/A, i, 4) = $238 + $2,541 (P/F, i, 5)

Since we have two unknown discount factors in this equation, it must be solved by trial and error using the appropriate factors for varying values of *i*. The first try should be the MARR of five percent, which results in present values of:

Year 0 cost:			($238)
Year 5 cost:	($2,541)(0.7835)	=	($1,991)
Total cost		=	($2,229)
Year 1-4 benefits	$650 (3.546)	=	$2,305
Total benefits less cost		=	$76

Since the benefits are greater than the costs, or in other words, the net present value of the increment is greater than zero, the rate-of-return on the increment must be something greater than five percent and we therefore accept the increment and retain the higher cost alternative, B.

Example 27.13 Benefit-Cost Ratio Incremental Analysis
Given the following mutually-exclusive alternatives and a MARR of five percent, which one should be chosen? Use the benefit-cost method and assume the "do-nothing" alternative is not available.

Year	A	B	C
0	($2,500)	($2,738)	($3,000)
1	$0	$650	$0
2	$0	$650	$350
3	$0	$650	$700
4	$0	$650	$1,050
5	$3,191	$650	$1,400

Solution—The first step is to compute the benefit-cost ratio for each alternative.

For alternative A:
B/C = $3191 (P/F, 5%,5)/$2500 = $3191 (.7835)/2500 = 1, which is acceptable.

For alternative B:
B/C = $650 (P/A, 5%,5)/$2738 = $650 (4.329)/$2738 = 1.03, which is also acceptable.

For alternative C:
B/C = $350 (P/G, 5%, 5)/$3000 = $350(8.237) = 0.96, which is not acceptable.

Alternative C is then rejected and we can proceed with the incremental analysis between A and B. These alternatives are already ranked in ascending order, and the difference is computed by subtracting the lower initial cost from the higher initial cost as shown below:

Year	A	B	B-A
0	($2,500)	($2,738)	($238)
1	$0	$650	$650
2	$0	$650	$650
3	$0	$650	$650
4	$0	$650	$650
5	$3,191	$650	($2,541)

Next, calculate the ratio of the present value of benefits and costs.

B/C = $650(P/A, 5%, 4)/($238+$2541(P/F, 5%, 5))
= $650(3.546)/($238+$2541(.7835)) = 1.03

As in the previous example, the ratio on the increment is favorable and we retain alternative B.

Practical Considerations

In private-sector firms, engineering economy problems must also account for taxes. Most firms pay several kinds of taxes such as income, property, sales, and value-added taxes. The computation of these taxes depends on understanding the principles underlying complex and ever-changing tax codes. This includes depreciation, marginal tax rates, tax credits, and deductible interest to name a few [1]. When the consequences of income tax and other tax effects are incorporated into the economic analysis we will have an after-tax cash flow. All of the principles and techniques previously discussed can be applied to an after tax analysis.

Changes in value as a result of escalation and inflation also complicate economic analysis. Most cost and benefits in the real world have prices that follow the same rate of inflation as the economy as a whole. Since all cash flows are inflating at about the same rate, inflation can be ignored. However, escalation, which includes inflation, may affect each cash flow differently (e.g., steel price increases 15 percent as a result of economic conditions while inflation, driven by monetary policy, is only five percent). Inflation is addressed by simply stating everything in constant dollar terms and using a real interest rate [1]. When necessary, inflation or deflation can be modeled as a constant rate of change using the geometric gradient factor. For cases involving differential escalation for various products, different values of the geometric gradient factor can be applied to each product.

Non-Cash Flow Factors

An economic analysis is a quantitative analysis of cash flows that can be measured in currency and evaluated using the time value of money. Real projects have multiple objectives or desired outcomes whose achievement can be characterized as either quantifiable or non-quantifiable. Quantifiable outcomes can usually be converted to cash flow. An example of this would be the time savings for commercial traffic related to a highway project in Example 27.9. Although not initially quantified in money, this benefit could be converted to cash flow in currency by estimating the cost per trip time savings and the number of trips per year. Some non-cash flow factors cannot be *objectively* introduced as numbers. Intangibles are those attributes that are not readily expressed in terms of money and cannot be quantitatively measured as a cash flow. Examples would include stakeholder preferences, utility, goodwill, environmental effects, the consequence of failure, fit with the organizations capability, or simply a "gut feel" about the project. Decision analysis methods that consider multiple attributes as well as uncertainty and risk preference or aversion (i.e., multi-criteria or multi-attribute decision analysis) are increasingly being used as issues such as sustainability become higher priorities.

CONCLUSION

Economic analysis is a technique that assists in the solution of substantial engineering problems where economic aspects dominate over a considerable period of time. Using the time value of money, discount factors, and the principle of equivalence, the economic costs of a problem can be modeled and evaluated. The use of these financial tools for decision making will be illustrated in subsequent chapters.

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Chapter 28—Practical Corporate Investment Decision-Making Guide

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Abstract
This topic is designed to assist the project management professional on how to relate insights gained from economic analysis to investment making requirements and information to company executives. Most companies have economists and cost engineers that provide thorough economic analysis on potential investment opportunities. Often project professionals have difficulty with succintly communicating to executives who may not have enough time to invest in the volume of analytical material that project economists typically produce to defend the project payout. Company executives are immersed in company strategy and are searching for projects to meet the company strategy. Economics and investment strategy are two related, but different subjects.

Therefore, project personnel need to describe their projects to meet the key features that the company executive requires to render a decision to meet corporate strategy. This chapter identifies the key concepts and details necessary to translate project economics into a few basic parameters to assist executives in investment decision making. The chapter is written to provide the structure required for a Project Assessment Document (PAD) with explanation and examples of the technical requirements for each section.

Keywords
Average Annual Rate of Return (AARR), Benchmarking, Business Development, Capital Investment, Cash Flow, Cost Variance, Drivers, Economic Indicators, Operating Expense, Portfolio Management, Project Assessment Document (PAD), Rate of Return (ROR), Return on Investment (ROI), Risks, Schedule, and Strategic Planning.

INTRODUCTION
This topic is designed to assist the general project management professional in communicating basic economic analysis to upper management as they make project investment decisions. The analysis is often detailed comprehensive, and well-understood since most companies have economists and cost engineers. The issue arises when project professionals message appears to be too time consuming, dwells on many unnecessary details; which might all be good; but may not help in decision making. What must be understood is that company executives are immersed in company strategy and are shopping for projects to meet that company strategy. Making a project investment decision, for them, is often compared to buying a car. It is not necessary to review the design specifications, interview the engineers, listen to the supply chain risk scenarios, etc. Rather, the car purchase decision is based on issues such as: history of the company, fuel efficiency, reliability, referrals and staying within a fixed budget. This creates a short list of car manufacturers and models to make our selection. Once all of these items are reviewed, the final selection criteria becomes "when can I have it?" and "what free amenities are included to sweeten the deal?" Since executives review projects in a similar way, project engineers need to spend more of their time communicating how their projects meet the key features that the company executive requires to render a decision to meet corporate strategy, rather than impressing the decision makers with the routine details of the project economic evaluation.

The investment decision makers for a corporation are tasked with selecting projects that meet satisfactory cash flow requirements and return on investments. Capital is a limited resource and there may be several projects competing for selection by business units, project managers, business development professionals, etc. Not all the potential projects can or should be selected for funding. Not only is there a limit to the capital, but there is also a limit to other resources such as staffing, project management, partners, etc. A final decision is often based on understanding whether or not the project goals are worth the financial risk.

A consistent, structured executive review format with key deliverables is required to facilitate the review of projects that are competing for the limited capital funding. A Project Assessment Document (PAD) should be used and required by all proposed projects so that the results can be placed into a universal dashboard for uniform assessment and prioritization. Without a uniform project assessment, the decision makers are faced with irregularities between projects and the lack of reporting transparency may lead to a poor decision to fund and sponsor a project.

The project team's role is to assist the executives with decision making for the corporation. Project teams need to understand that there is a slate of projects and countless permutations to meet the company strategy. There are three important elements that project teams must communicate as they are required by investment decision makers: 1) value to the company, 2) effect on cash flow, and 3) transparency of the potential risk to the company.

Learning Objectives
After completing this chapter, the reader should be able to:

- Understand the importance of a transparent and objective summary for investment decision makers.
- Understand the importance of the lost profit opportunity (LPO) on the economics of a proposed project.
- Understand why the average annual rate of return (AARR) is preferred over a simple return on investment (ROI) and the relationship of AARR is to LPO.
- Identify the basic steps to investment decision making.
- Understand how a company utilizes the portfolio management process to decide which project(s) to fund.
- Identify the seven key elements to an executive summary.
- Understand how the variances to non-controllable and controllable expenses affect the project cost forecast.
- Identify factors and their individual effects on the AARR.
- Understand the need for using a cash flow curve to include risks associated with economic, financial, and political risks to develop a forecast of the project's revenue generation life cycle.
- Present revenue breakdown of the sale price per unit in order to place profit in context of the operating and material costs.
- Present economic indicators in a standard industry format to assess the project with respect to industry benchmarks.
- Understand risks to annual revenue generation and why it is important to understand the risks to cash flow.
- Understand how projects are selected to manage a corporation's cash flow and strategic vision.

How to Use this Chapter
This chapter is formatted to provide the reader with a structured project economic evaluation document that can be used to communicate with investment decision makers. The Roman numbered sections (I – IV) address the investment decision making process holistically. The eight numbered sections (1-8) with recommended graphs and tables illustrate the default structure of the Project Assessment Document (PAD). The Project Assessment Document is structured as a series of progressive approval steps or check list for the decision makers. If all the steps are satisfactory, then the project is a candidate for consideration among the many projects that the corporation must consider to include in the portfolio. If one step creates uncertainty, or does not meet the corporation's need, then the remaining steps do not need to be reviewed. This is because the data and analysis in all subsequent steps build upon each previous step. In each section, this chapter describes the type of content that should be addressed. Therefore, each section in this chapter is much longer than the actual data and explanations that are required for an actual Project Assessment Document (PAD).

Foundational Principles and Concepts Review
In order to provide a comprehensive Project Assessment Document for corporate investment decision making, a few topics need to be explained, including: The Gated Review Process, Lost Profit Opportunity (LPO), Average Annual Rate of Return (AARR), and Portfolio Management Process. There are plenty of papers, books, and educational material available to read and treat each of these as independent subjects. However, it is wise to revisit and understand these subjects as they apply and are interrelated to the economic assessment of a project.

Definition of a Gated Review Process
Project approval processes generally follow a series of steps, or "gates". The Front-End Engineering and Design (FEED) or Front-End Loading (FEL) phase of a project is examined to determine the associated effect on cost and schedule competitiveness. FEL is a term for the gated process first coined by DuPont in the early 1980s [3]. FEL is pre-project planning to develop sufficient strategic information to assess risks to make decisions concerning resources and insure success. The optimal critical success factors that define FEL are determined based upon the project's outcome as assessed by the key performance indicators. FEL is considered a critical success factor for a project. Critical success factors are "statements of how improved business practice must be achieved if an organization is to be able to attain its mission." The key performance indicators are "the means by which an organization can measure the progress being made to ensure that the critical success factors are being achieved." Figure 28.1 is a generic gated process for the purpose of developing projects in a planned development cycle. This also includes consideration for retiring the project once the life cycle has been reached.

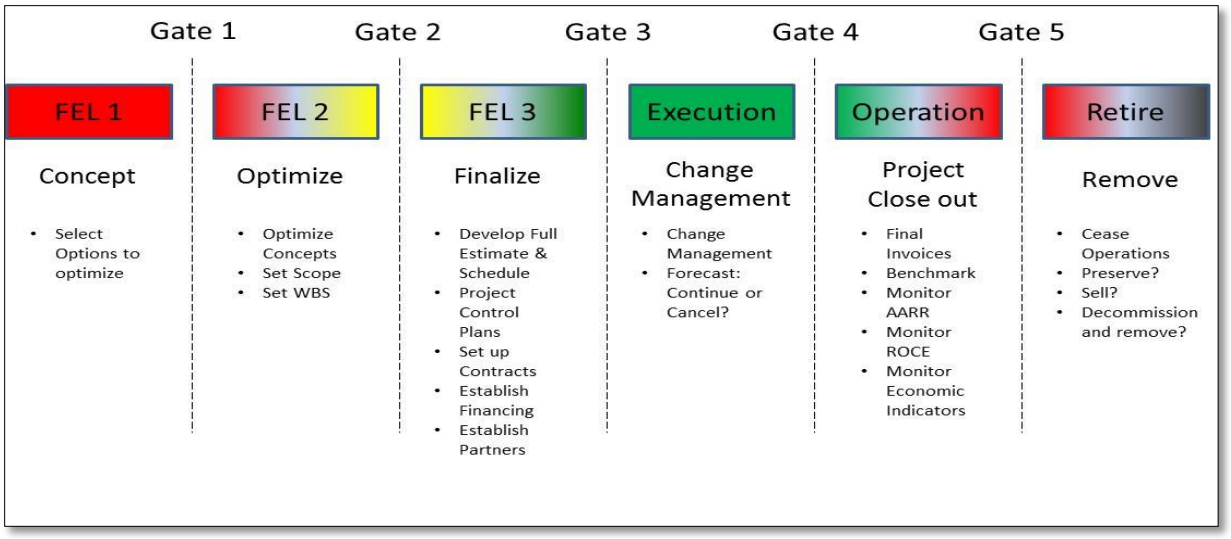


Figure 28.1—Gated Project Review Process

The FEL phase (FEL 1 through FEL 3) is the decision phase where intensively-focused planning and functional design is conducted early on in the project lifecycle. The influence on design and the potential to make changes is relatively high in the FEL phase and the cost to make those changes is relatively low. The FEL phase is intended to develop a distinct set of project and technical deliverables to support an economic analysis that allows investors to make an effective funding decision. The FEL phase is the decision stage, which establishes a contract with investment partners that addresses: revenue generation start date, targeted return on investment, and approaches for being competitive, etc.

The FEL with an appropriate gate process is important to understand in order to bring context to investment decision making. Projects should be assessed in a step-wise or gated process. Each subsequent gate includes information that is more detailed and better understood than the previous gate. However, the increased details of the project should not be included in the PAD. As uncertainties and risks are addressed in each gate, their values will be reflected in the PAD.

The PAD structure, length, and content will remain basically unchanged from FEL 1 through FEL 3. The values from the tables and graphs of each PAD should be placed into an executive dashboard for the corporation to review in the context of the overall portfolio of projects. If a project in FEL 1 becomes unfavorable in FEL 2, then the review of the portfolio dashboard should automatically indicate which other projects should have more attention. The earlier a project can be better defined, the earlier a corporation can make more definitive decision to cancel or continue funding the project.

Lost Profit Opportunity (LPO) [5]

Lost Profit Opportunity is largely discussed when corporations or investors seek damages and need to prove lost profit claims. In this case, a company would be wise to understand the operating point and market conditions where profit is forecasted to become negative to a point that increased sales will never bring the project back to operating in the black. This is why “Black Friday” in the United States is important to retailers. Retailers may operate in “the red” until the holiday season when they start offering deep discounts to increase sales volumes to offset the previous ten months profit loss.

LPO is also used in the context of limiting the effect of a business interruption. As an example of asset utilization, assume product production stops because a broken part. One option is to order a new part from the manufacture. This option means that production will be stopped for three weeks. However, the company has a spare part (asset utilization) at another facility that can be used to make repairs and restart production in one week. If the company uses the spare part from another facility, then the LPO is one week (3-1) claim to the corporation. Other terms that are similar to the concept of Lost Profit Opportunity include:

- Opportunity Loss or Cost
- Economic Profit (or loss)
- Recovery Lost Profits
- Lost Profits and Damages
- Asset Utilization

For companies that have a product with a volume throughput like: assembly lines, chemicals, pharmaceuticals, oil refining, etc. there is an operating point at which they must maintain production to breakeven. This operating point is typically 85 to 90 percent of the design capacity. The business development, being optimists, evaluate projects upon the potential for the project to exceed the base

design specifications. Economists and asset operators are more realistic in their project economic evaluation because they understand the life cycle costs and challenges to keep an asset in production. Therefore, economists and operators view profitability based on meeting 85 to 90 percent of design capacity and the requirements to catch up on lost production when there is an event that stops production for an unspecified amount of time.

One method to quantify LPO is to include how the historic, practical, and realistic operating conditions influence the Average Annual Rate of Return (AARR). Without understanding the Lost Profit Opportunity, the project would be justified solely on optimistic merits. The LPO tempers the project economics by encouraging dialog between the business development professionals with the company economists and accountants early in the very first review gate (FEL 1). In other words, there is a discussion of realism over overly ambitious expectations.

Average Annual Rate of Return (AARR) [5]

This chapter addresses projects that are complex and dynamic. Market conditions, escalation, labor costs, volume sales, maintenance, operation costs, etc., change frequently and a single point calculation like Return on Investment (ROI) is not a recommended critical decision indicator. An average of the results of several ROI calculations over the premised life cycle of a project is often a better economic indicator.

Average Annual Rate of Return (AARR) is the measure (expressed as a percentage) of profitability of an asset over a period of time. AARR is the average yearly profit over the operating lifecycle of a facility and dividing it by the final cost of the project. The final cost of the project is the total cost required to deliver a project to begin operating and generating revenue.

The way to think about AARR in terms of a benchmark is to compare it to the corporate discount rate of a company (business unit or subsidiary). The corporate discount rate should be the rate that must be earned to satisfy the rate of return required by company investors. It includes the cost of capital (money) (if money must be borrowed to finance a project) and should naturally be lower than the company discount rate. The corporate discount rate includes the cost of capital plus any other factor that is added to the cost of capital such as risk, insurance, miscellaneous, etc. (and is therefore a higher value than the cost of capital). Risky investments evaluated with too low of a discount rate make projects look better than they should. Low risk investments with too high of a discount rate appear worse than they are. It is important to use a discount rate that is well documented from the corporation's operating budget.

The AARR for approved projects should be more than the discount rate which sets the Net Present Value (NPV) equal to zero. If the corporate discount rate is 13 percent, then by definition, a project will have a positive Net Present Value if it has an AARR > 13 percent. The 13 percent is a "hurdle rate", higher than the bare cost of capital, which establishes the minimum AARR a project must achieve before investment in the project will be approved. Based on this understanding, an analysis model can be developed to explain the effect of design capacity on AARR. AARR is the measure of the project which needs to exceed the hurdle rate to justify the project. Otherwise, the money should be invested elsewhere. Project professionals must understand this so that they can provide the most accurate picture of how (or whether) the project aligns with investment decision-makers' long-term plans.

To illustrate this point, an analysis was performed on the effects of percent production rate achieved on the AARR. There is a direct linear relationship between the production rate and AARR of the units in this study. Figure 28.2, shows the trend between AARR and percent operability. As the percent operability increases from 70 percent to 100 percent, the AARR also increases by one third, from 10 to 15.

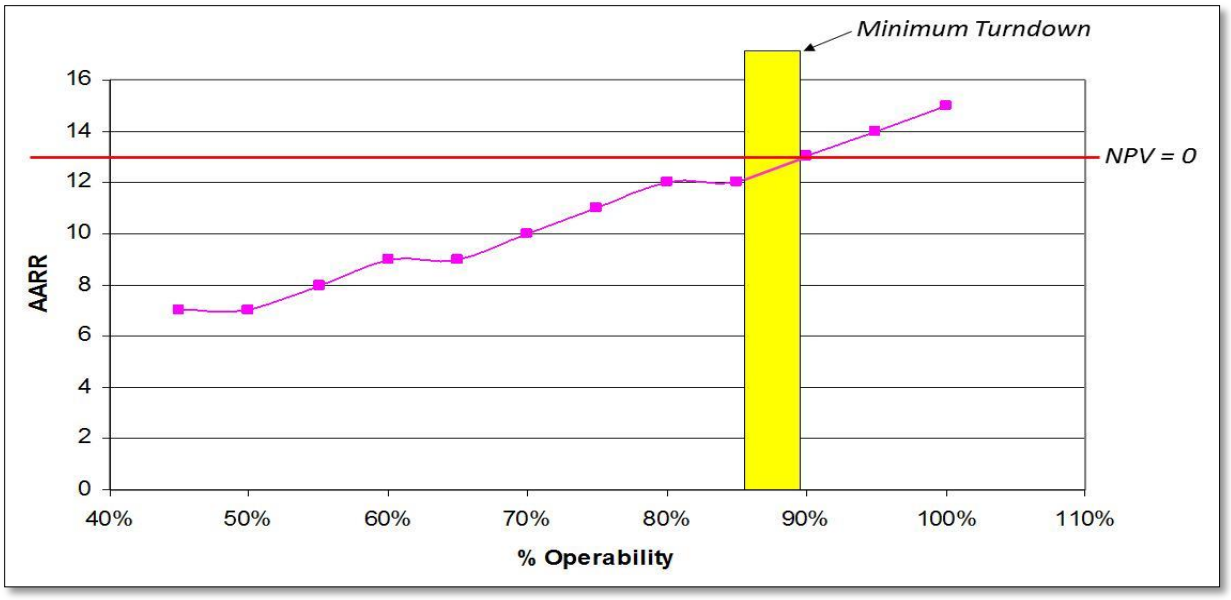


Figure 28.2— Effects of Production on AARR

Knowing that there is a direct relationship between the effects of the production rate in the unit and its ability to return the investment, gives an investor a strong financial incentive to achieve design capacity.

Figure 28.2 illustrates that when operability drops from 100 percent to 95 percent, then AARR drops from 15 percent to 14 percent (one percentage point). Any time this drop of one AARR percentage point occurs, the profit loss over a 20 year operating lifespan can be 15 percent of the project's Total Installed Cost (TIC). (*If a project AARR drops from 15 percent AARR at FEL3 to 10 percent AARR at closeout, the potential profit loss over 20 years can be an astronomical 75 percent of the TIC.*)The profit that could have been generated based on 15 percent AARR is reduced by approximately 40 percent when the AARR drops four percentage points. A company's economics are severely affected if the planned design capacity is not used. Running a facility in a downturn mode is not optimum for economic payout. In some cases, running a facility in a downturn mode, below design capacity, can create other issues including poor yields, increased maintenance and mechanical reliability problems. These issues further diminish the profitability of the project and reduce the company's cash flow.

Comparison of Influences on AARR

Figure 28.3 shows the influences of cost, schedule, and operability on AARR. Each project was analyzed to determine the change of AARR if the project only had a change in one of three areas: cost, schedule, or design throughput. The reason that this is important is because project teams often authorize overtime and longer work weeks in an attempt to achieve the original construction end date when a project falls behind schedule. Given that a moderately longer project schedule has minor influence on the AARR, and cost has a significant influence on AARR, it is generally better to accept a longer project schedule and hold project costs down. Many projects make a decision to trade cost for schedule. Figure 28.3 was part of a study to determine the causes and effects of projects that do not achieve design capacity [5]. Dozens of projects were chosen for the study based on: being the same process, constructed in the same five-year time span, engineered by the same contractor, constructed by very similar construction contractors, and were all based in the United States. However, the operability results varied. The study provided significant insight to LPO.

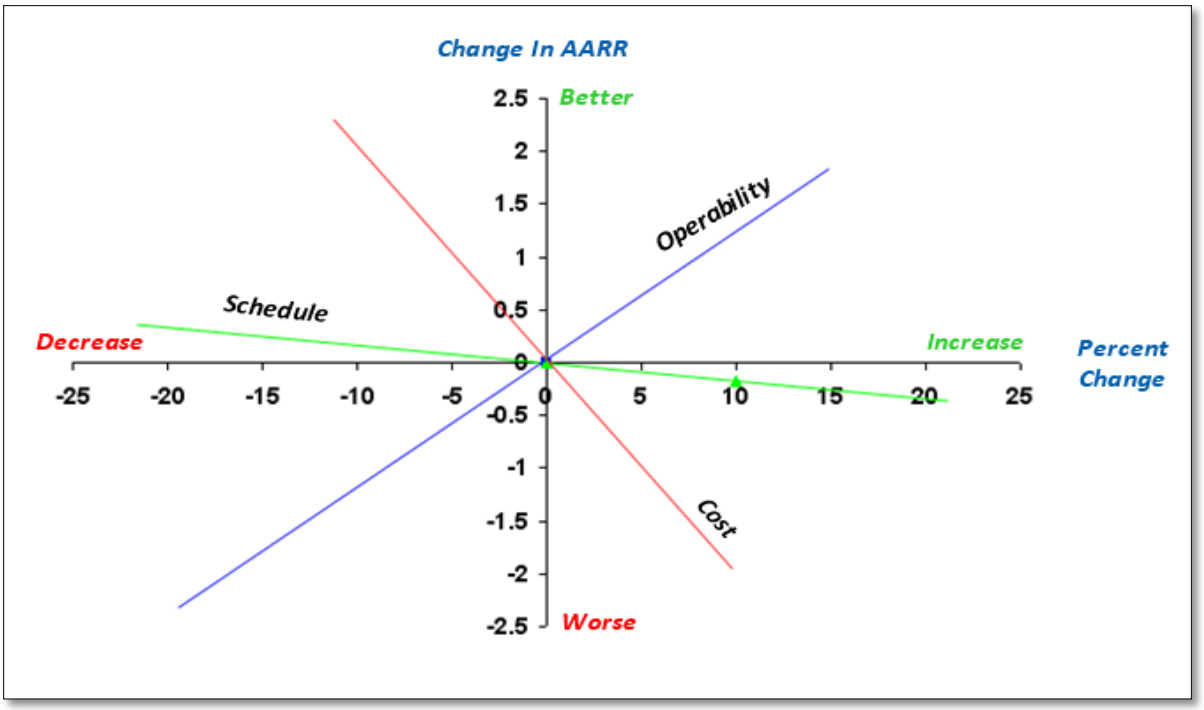


Figure 28.3— Comparison of Influence on AARR

However, trading cost for schedule is a larger detriment on AARR. Since the cost curve is so much steeper than the schedule curve, trading higher costs for a faster schedule is nearly ten times worse for the AARR of the project (in the example in Figure 28.3). Project teams need to understand this so that they can make better economic decision for the corporation. The cash flow of the company is almost always a priority in all economic decision making. Project teams that sacrifice cost for schedule may be adversely affecting the corporation's cash flow.

What is an investment?

Investment is the act of contributing money or capital into an enterprise with the expectation of future profit. While investments may eventually make or lose money, there is always the expectation that the investment make a reasonable return on investment. Investments have a limited decision time-frame to determine the impact of a 20-year life cycle on the company cash flow. Targets must be set to evaluate the level of returns. Risks characteristic to the investments must be determined by creating risk profiles to determine if a risk should be taken, avoided, or mitigated. Also, how much funding will be invested, flexibility in time to monitor investments and knowledge of the changing capital market must be considered.

Investment decision making process is principally the same for any sector and for anyone from a CEO to an individual. The decision process and the extent of robustness can be very simple or require integrated teams of specialists and years to compile. The basic steps to investment decision making [1, 2, 4] can be summarized as:

- Setting strategy and analysis
- Establishing a risk/return on goals
- Searching for investment opportunities
- Forecasting expected costs and returns
- Evaluating forecast returns
- Assessing and adjusting for risk
- Making the decision to act
- Implementing accepted proposals

However, the full life cycle of a project needs to be considered. Once the facility starts requiring maintenance and repairs to keep it functional, these expenses affect the company cash flow and the profitability of the project. Therefore, the three additional steps that need to be included in investment decision making are:

- Auditing operating performance
- Managing the asset

- Terminating the asset

Portfolio Management Process

The strategy of a company rests in the portfolio of projects that range from "think tank" or conceptual ideas to projects that are generating revenue. For companies with more than a single asset, a portfolio must also be established to target the investment strategy and business objective. Figure 28.4, "Portfolio Management Process" [4], is a typical model of the decision engine that constantly analyzes the cash flow of a corporation. Projects are fed into the portfolio management process and do not operate independently. Projects exist for the good of the corporation and not solely for the good of the particular project team or business unit inside the corporation.

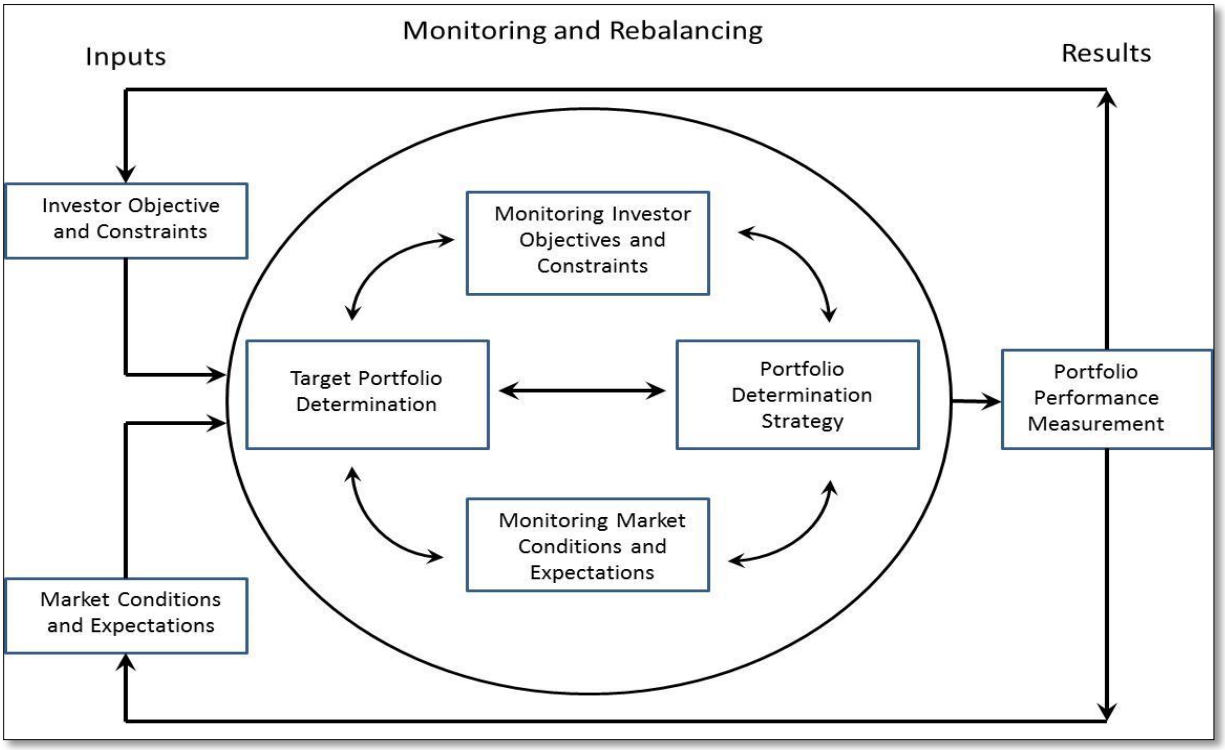


Figure 28.4—Portfolio Management Process

The key to making an investment decision is the effect of the acquisition (investment, project, real estate, or tools) to the current portfolio (small business to large, multi-national corporation). Companies must determine whether the investment is aligned with the current objectives of the investors and supported by the market. Once an investment is made, the process of monitoring and rebalancing the current investment is added to the portfolio of all the investments. Ultimately, the current investment will be retired or terminated sometime in the future.

Now that the basic foundation of the gate process, LPO, and AARR has been laid, the Project Assessment Document (PAD) needs to address the information that is germane to these subjects. The project professionals' job is to present the project analysis to executives in such a way that executives may make the best investment decision possible for the organization. They are also vital to the organization as they continue to monitor and communicate the activity around the investment. The remaining part of this document is the project justification document, or PAD, itself, starting with Section one, "Executive Summary" and ending with Section eight, "Risks to Revenue Generation." Whether the project is an expense or revenue generation project, the Project Assessment Document (PAD) is the same.

EXECUTIVE SUMMARY

Project teams need to understand that there is a risk to over-stating and over-simplifying the project to the decision makers. Therefore, teams must be transparent with the decision makers. Generally, this means that the project justification document needs to identify what the team knows and what the team does not know about the project, what is included and what is excluded from project costs, and quantify the risks in terms of cost, schedule and economic metrics.

Investors need to clearly understand the maturity level of the project scoping documents, and the associated risks to the project and risks to the company. Risks must include an analysis of not only the technical issues to building the project, but also the forecast of

the market and governmental environment that may affect the profit margin. Basically, the investor needs to understand what the project is and why they need to invest.

There are seven key elements to an executive summary:

1. Statement of the problem that the project addresses
2. Project Summary
3. Key Project Drivers
4. Primary Risks and Uncertainties
5. Capital Cost Variance
6. Average Annual Rate of Return Variance
7. Cash Flow Analysis

Statement of the Problem
Investment decision makers (CEO, CFO, Vice Presidents and General Project Managers) need a concise description of the proposed project to evaluate how the project aligns with their strategic investment portfolio. The project teams' first task in the executive summary is to clearly communicate that they understand the corporation's strategy and how the project supports that strategy in a manner that is intuitive to the decision makers.

Project Summary
This section describes the investment opportunity. Diagrams, maps, description of the technology, etc. are included in this section. The descriptions should be fairly high-level, but thorough enough for anyone unfamiliar with the project to understand the business opportunity. While this entire document may be a completely internal company-reviewed exhibit, there are cases when a company seeks outside financing. In this case, where a company elects to seek outside financing or is seeking a business partner, the importance of clarity of the project in this section is heightened. This section should be basically written like a public press release. There should be nothing confidential within the description of the project. The project summary describes what the project is, why it is important, and what the value is to the investor.

The focus of the information about the project should include parameters that are related to the decision making and not include extraneous information such as in-depth process descriptions, history of past funding failures, confidential business planning, etc.

Describe the project in tangible terms such as: how it provides jobs, how it is a novel solution to an existing problem, or how it is an expansion to an existing business sector. Be sure to convey how large the project is in terms of financing, how it affects the economy (local, state, country), how long it will take to construct, and the date investors will realize a return on their investment. If this is a "first-of-its- kind" project, it will take more explanation to convince the investment decision maker why they should invest and how the risk to their investment is mitigated.

Key Project Drivers
Project drivers need to be identified and measurable. There are different types of projects, such as environmental, regulatory, pilot or test projects, and projects that are ready for commercial applications. Every project has its individual drivers and challenges.

In this section of the document, there is a general tendency to use adjectives that are not an adequate substitute for describing drivers. Avoid common mistakes such as these examples:

- "The sheer magnitude of the resource, combined with known technology and low political risk, results *in an opportunity* that warrants considerable attention."
- "This project provides a *solid platform*."
- "Provides an *excellent* training environment."
- "*Evolutionary improvements* will deliver significant future economic upside for the project."
- Or, "There is a potential to add *significant value* from growth increments."

Project drivers need to be tangible. This section is better described as a high-level summary of the current operations, current market needs, changes to regulations, new product lines, new profit opportunities, etc. and how this project is going to address these items. This section is different from Section 1.2 (Project Summary) because this section summarizes the key factors that influence the economic indicators and cash flow for the company. If these project drivers are removed, then the project does not need to be progressed. For example, if the project addresses environmental issues, then a brief summary should address: the environmental issue, what happens if the issue is not resolved, the penalties, the deadline that must be met, the cost of compliance, etc. The summary should also discuss why this project is the best solution. Other competing technologies and solutions should be mentioned to bring clarity as to why this project is best for the investor.

Primary Risks and Uncertainties

Every project has risks and uncertainties. The final cost and schedule will be directly affected when the risks become actual events. The type of risk and magnitude of the risk should be summarized in this section. The investor needs to understand how the risks affect pricing, economics, supply, regulations, and capital.

Capital Cost Variance

As shown in Figure 28.5, a waterfall chart can be used to illustrate six common risks to a project sanction request. The best case scenario is for the project to complete with no additional costs. Worst case would be for each of the events to occur. In the example, the best case is that the project costs 175 million USD and the worst case is 400 million USD. There are two major risk categories to the cost variance: non-controllable and controllable. The base economic indicators and calculations of the project are based on the 175 million sanctioning request. The funding decision for the corporation will include the worst case scenario.

Non-controllable costs for most projects are categorized in Equity Change, Foreign Exchange and Escalation. These are costs that are not completely controlled at the will and direction of the corporation.

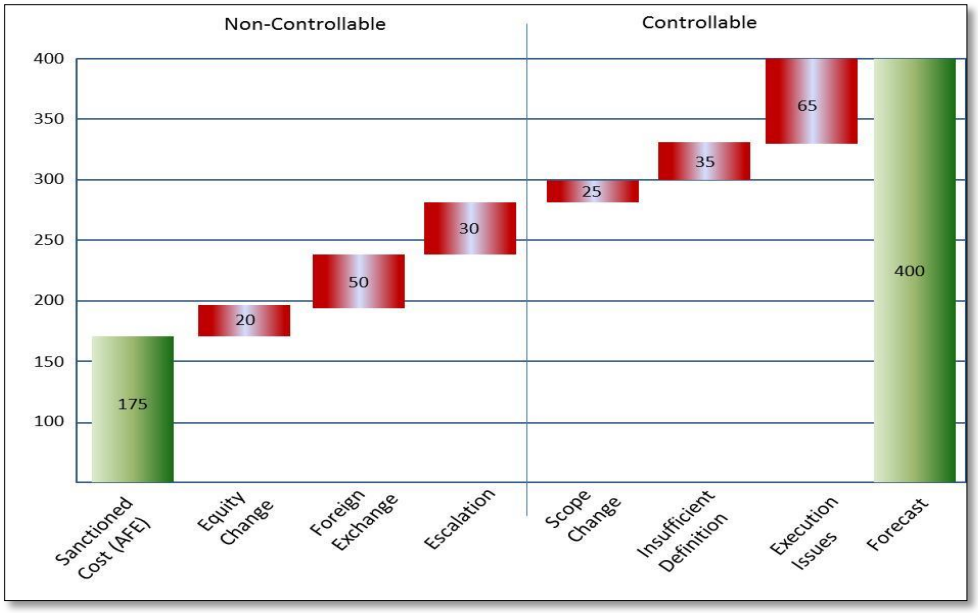


Figure 28.5—Capital Cost Variance

Equity Change is the change to the ownership of the project. For large corporations, these projects often have joint venture partners that share the project from its initial funding to its completion, and they share in its profit. For the small business owner, that partner may be the lending institutions and their requirements to guarantee that the small business owner will have a reasonable chance to pay off the loan. The lending institution enacts requirements that can be burdensome. For larger corporations, the project partnership may vary and change depending on the terms and conditions of which partner is managing the project, funding terms, delayed interest payment, etc. Sometimes partners will trade near-term funding and capitalization agreements for future ownership percentage. This equity change can transfer money from future operations into current project funding costs.

Foreign Exchange (FX) is a function of trade laws and regulations to protect the purchasing, lending, and trade power of a country. Therefore, components are purchased and acquired in part or totally from a foreign country, and that has a direct economic effect on the local economy. For example, the final assembly of electronic assembly may be purchased from a local supplier. However, the cost of the component is affected by the source of the individual components that were shipped from a foreign country.

Escalation is the increase in price of a commodity over a period of time. The change is directly associated with market demand. When a commodity becomes more in demand, and the ability to produce the commodity becomes more difficult, then the commodity will increase (escalate) in cost. For the small business owner, a change in labor cost and availability will have an immediate effect on the cost of services, products, and profitability. Escalation is not a controlled cost, but it can be managed to some extent. For a large corporation, escalation can be managed in part by long range planning of material requirements on future projects. For example, assume the portfolio of projects slated for future years five through ten indicate substantial purchases of steel, pipe, and compressors. The purchasing and contracts departments can then negotiate pricing with the potential suppliers today to lock in pricing agreements for the future. The benefit to the suppliers is that they reduce the risk of unknown future sales with commitments placed today. The risk to the corporation is that if some of the projects are not funded, then any earnest money

committed to the supplier may be forfeited. While the supplier may receive payment for a canceled commitment, the payment will not entirely offset the cost of acquiring, storing, and paying taxes on the inventory held in escrow for the company's future project. However, the corporation did not have to purchase the material and the supplier will probably sell the cancelled order at future market prices, which hopefully have increased (though it is not guaranteed).

Controllable costs generally fall into one of three categories Scope Change, Insufficient Definition, and Execution Issues. Each of these is completely controlled at the will and direction of the corporation.

Scope Change relates to changes in design specifications of the proposed project. It could be that the project premise is to produce a product at the rate of 100,000 units per day, and there are options to which design components can be used. The scope change variance is the total sum of the most likely changes to the design specification. In addition to the base scope, there are changes that impact the design premise which most likely will not be known until the project has completed more engineering. The investment decision makers now have the information to evaluate the risk of this project with similar scope change risks of other projects being considered for funding.

Insufficient Definition is more problematic for projects where a company is making an assessment to enter an emerging market. The emerging market can be a new location where they have little or no experience, a new technology, or a market that is in turmoil because of changing governments, regulations, current events, material shortages, etc. In any case, the project team has been requested to estimate the cost and schedule for a project that the design premise is not as developed and robust. Therefore, if the definition of the project changes, then there is a risk to the cost that should be reported and clearly itemized as to what is and is not included.

Execution Issues address problems and changes experienced during engineering, construction, fabrication, installation, and startup. These issues are unrealized risks, and the cost to address them must be itemized, assigned a risk value, and summarized for the Capital Cost Variance graph. Some execution issues may include: safety incidents, labor dispute, labor shortage, material delay, delay in financing, work stoppages. Execution issues may also include instances where the corporation requests a project to slow down, idle, or work under restrictions on expenditures sometime after the project has been authorized for funding. There are a number of execution issues that may arise, and their cost to the project is significant.

The likelihood of all cost variances occurring must be evaluated by the project team with the decision makers. The ultimate cost of the project will depend on limiting or mitigating potential cost variances from these six common risks. The project that is selected for funding is typically not the one with the higher forecasted cost.

For example, assume there are two projects that have the same funding and risk costs as in Figure 28.5. While the risks have the same magnitude for both projects, it may be that the six risks to Project One have a low likelihood to occur whereas the same risks for Project Two have a moderate to high likelihood of occurring. The forecast to Project Two may be much closer to the worst case value of 400 million USD because the entire cost risk of 225 million USD may actually occur. The forecast to Project One may realize half of the total risk and have a forecast around 287MM. Initially, the answer seems straightforward that Project One should be selected for funding. However, based on strategic business decisions that are confidential to the corporation's executives, Project Two may be selected for funding. It could be that Project Two has higher risks, may cost more, has a lower return on investment, but it starts producing revenue much sooner than Project One. In the portfolio management process, the decision to fund Project Two over Project One may be based on the monitoring and rebalancing of the portfolio to meet the vision (results) the corporation expects.

Average Annual Rate of Return Variance

In addition to understanding absolute cost (and variances) to fund a project, is important to understand the variances to the profitability (AARR) of the project. In the event that the project is an expense project (does not generate revenue), this exercise is still valid.

The next significant waterfall chart is Figure 28.6. It graphically illustrates variance of Average Annual Rate of Return (AARR) on the project's sanctioned cost.

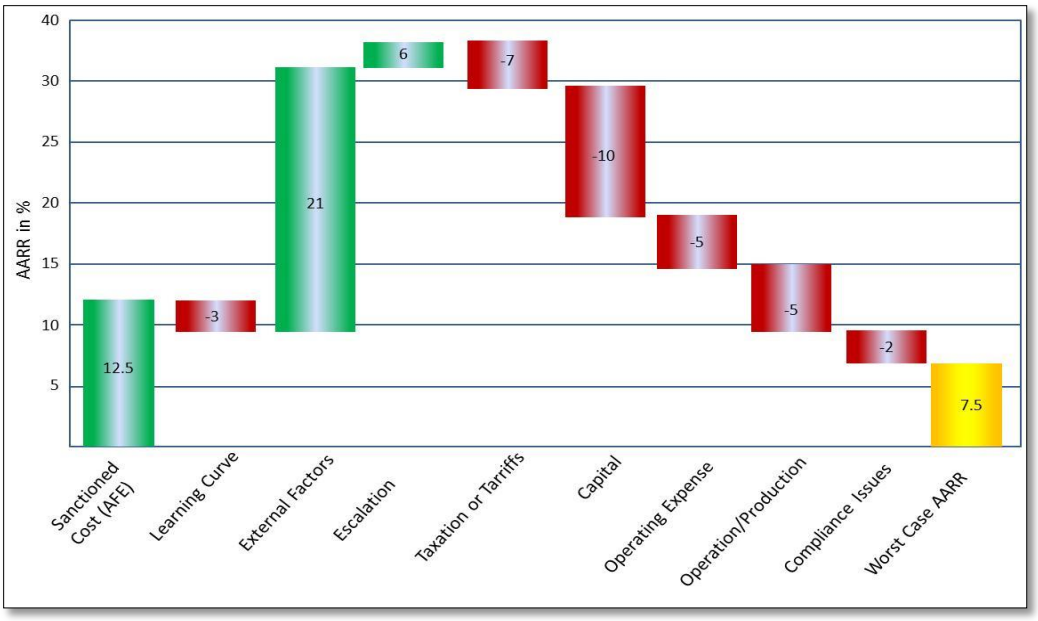


Figure 28.6—Average Annual Rate of Return Variance

The main factors contributing to the change in AARR are identified below.

- **Learning Curves**—Unless this project is a duplicate design of several projects in a series, then no credit should be given to learning curves. In fact, there should be a penalty to the AARR to account for discovery work during design and construction.
- **External Factors**—Changes to the project during design, construction and the first five years of operation should be evaluated. This is an area where economists and business development personnel clash. Business development often proposes that the project will "make more money", but they do not take into consideration all of the things that are constantly changing. The numbers are generally overly optimistic. Whereas accounting and economists evaluate the project more pessimistically because they manage the annual budgets, operating expenses, maintenance expenses, dividends, and the overall financial viability of the company. Accountants and economists run scenarios that affect the revenue such as material shortages, lower product yields, loss of market share, etc.
- **Escalation analysis**—Raw material pricing and resource supply projection for the project. Along with this is the variability of each commodity and scenarios to evaluate best case and worse case scenarios that will affect the AARR in both the short-term and long-term sustainability of the product line.
    - Changes in foreign exchange rate have adversely impacted project economics.
    - Cost escalation on a macro level is generally two to three percent per year. However, escalation for a single commodity, such as steel or labor cost, can spike for one month and extend the payout period for a project several years.
- **Taxation or tariffs**—A hugely complicated field, and companies need to have in-country specialists to understand the effect on the project and negotiate terms and conditions. Tax structures change based on the economic policies of a country and can change significantly during a project.
- **Capital**—From small businesses to corporations, business plans include scenario planning for funding the project. In some cases, a certain percentage of the ownership may be required or significant day-to-day control given to a funding entity such as a lending institution or partner. The effect on the AARR must be calculated and assessed. Basically, it is a matter of quantifying lending requirement in terms of borrowing money.
- **Operating Expense**—Utilities like natural gas, water, electricity, air, etc., are affected by market demand, foreign supply, transportation, labor contracts, and governmental policies and are non-controllable by the company and can be volatile. The pricing in these variable costs have a direct effect on profit margins.
- **Operability or production** is a key factor that is generally set at 90 percent (refer to Figure 28.22 Section VI, Average Annual Rate of Return (AARR)) for the Net Present Value (NPV) breakeven point. Once a product line is on-line and cannot sustain 90 percent of the design capacity for a year, then the payout period is extended. In cases where design throughput cannot be achieved, there should be an execution plan for permanently shutting down the product line to minimize loss profits.

- **Governmental Compliance Costs**—As laws are passed to protect and preserve national or international resources; modifications to a production line may be too costly and render a product line non-profitable. Generally, projects are reviewed based on complying with current laws and standards. However, pending legislation or legislation that may be passed in near term or long term still need to be evaluated. Evaluation of the AARR must include the variance to the AARR if part or all of the foreseeable compliance requirements are enacted.

Like the Capital Cost Variance analysis, the actual AARR is a value based on the likelihood that any or all of the risks categorized may occur. In this example, the business development professionals make a case that the external factors include events that will make this project very successful and nearly offset any other adverse or collection of adverse events. This graph facilitates discussions about what else could affect the AARR. A project's forecasted cost in Figure 28.5, may be reasonable and within discretionary funding limits of the corporation leadership. However, if the project (see Figure 28.6) is required to have a payout that is above an AARR target of 13 percent, then the project is marginal in the base case and the worst case scenario (seven-and-a-half percent) indicates that the project should be further studied or canceled.

Cash Flow Analysis

Cash flow is used to determine a project's rate of return and it is the financial health indicator of a project or company. Cash flow is the flow of money in or out of a business over a finite period of time. It addresses the problems with liquidity and financial risks. Making profit is not the same as liquidity. A project may generate profit but may not have the ability to pay debt. Cash flow has three basic components for a business: operations, financing, and cash flow from investments. Cash flow is important to evaluate over a period of time. Cash flow analysis must be revised on a highly regular basis to reflect changes in the supporting information.

Cash flow = Cash Receipts - Cash Payments

Equation 28.1

Operations = Revenue generated by operations – Materials – Labor

Equation 28.2

Financing = Loan + Cash Received from equity or issue of debt - Loan repayment – Taxes – Dividends - Share repurchase

Equation 28.3

Investments = Sale of Assets + Purchase Capital – Acquisitions

Equation 28.4

Corporations should assess project funding as though they were lenders. This view will expose risks that require evaluation. These categories will be listed with key consideration points for each; however, none will be explained in this chapter. The point of this list is to provide a check list of some common elements that directly affect the cash flow of a project and corporation.

Economic Risks

- Inflation
- Debt service as a percentage of exports of goods and services
- International liquidity ratios
- Foreign trade collection experience
- Current account balance as a percent of goods and services
- Parallel foreign exchange rate market indictors

Financial Risks

- Loan default or unfavorable loan restructuring
- Delayed payment of supplier's credits
- Repudiation of contracts by governments
- Expropriation of private investments

Political Risks

- Economic expectations versus reality
- Economic planning failures
- Political leadership
- External conflict
- Military in politics
- Organized religion in politics
- Law and order tradition
- Racial and nationality tensions
- Political terrorism

- Civil war
- Political party development
- Quality of the bureaucracy

Therefore, evaluating the cash flow of the project and how the project affects the cash flow of the corporation is the key fundamental component to deciding whether or not to authorize a project. Given the list of complexity shown in this section, each may require a subject matter expert and a risk specialist. The investment decision makers are keenly aware of the risks, but the results must be simplified into a cash flow diagram.

It is important to understand cash flow because it makes up the third important view of projects, following Capital Cost Variance and Average Annual Rate of Return Variance. Several cash flow diagrams should be included to depict the profit viability of the asset over its design lifespan. The next two figures depict the same hypothetical project of an 80,000 barrels per day refining facility with an initial cost of 615 million U.S. dollars. In Figure 28.7, the AARR is 21; the project breakeven is in fifth year of operation. The design and economic premise is to run for 20 years.

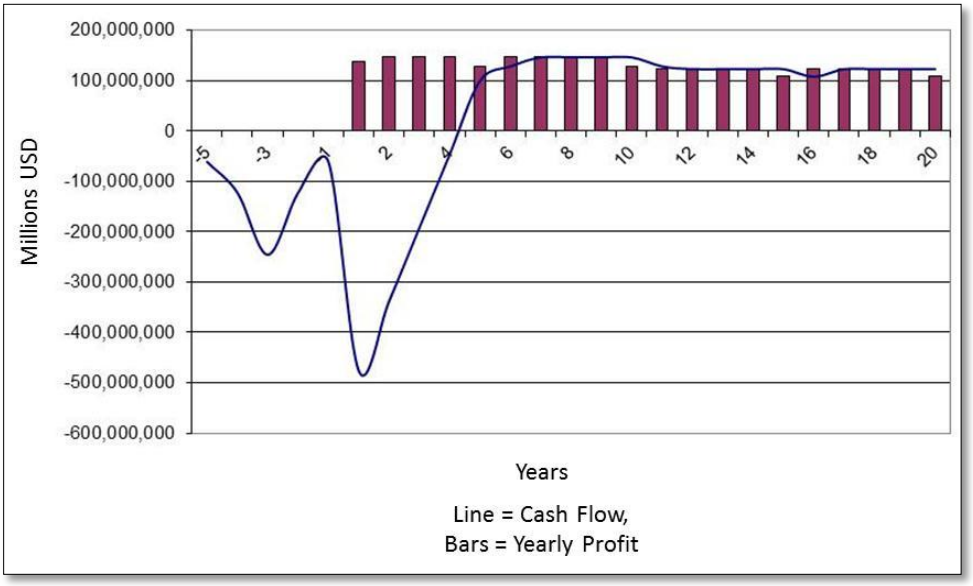


Figure 28.7—Cash Flow: Business Development Model

The importance of evaluating the external factors on a project is highlighted in Figure 28.8. If the cost of the variable costs increase two percent per year, then the facility profit lifespan is shortened to ten years. The AARR on a 20 year evaluation is now nine percent. If the variable costs themselves do not increase two percent per year, but the total of all the other external and internal factors affecting the cost to operate the facility aggregate to two percent per year the effect is the same.

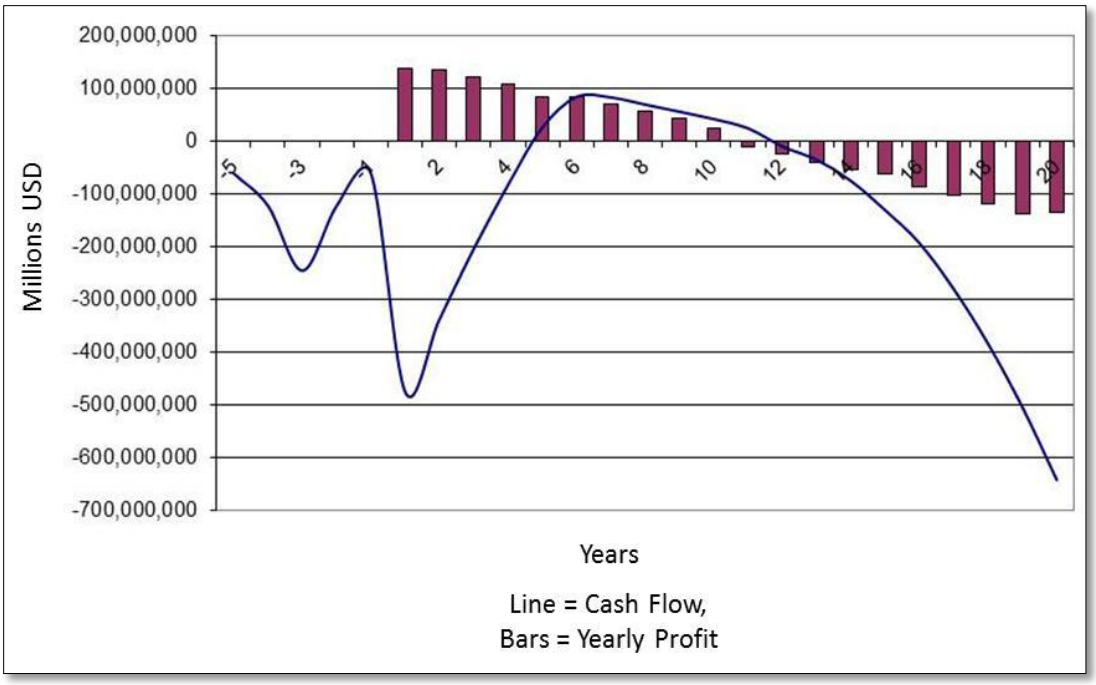


Figure 28.8—Cash Flow: Accounting/Economist's Model

As mentioned before, there is typically a clash between business development and accountants/economists in their evaluation approach to projects. In this scenario, as Figure 28.8 demonstrates, while the AARR drops to nine there are a few other considerations to be made to proceed with the project even though a nine percent AARR falls below an NPV value of 13 percent for a company funding threshold. It could be that this project has benefit if it only operates for ten years. It if is still viable after ten years, then the profits can be diverted to new projects as well as shareholder dividends.

PROJECT DESCRIPTION

Where the executive summary in Section 1 focused on the investment opportunity at a fairly high-level, this section places more emphasis on the major components that comprise the project. These components should be enumerated and followed by a one to two paragraph description.

This is a description of the project and should include the evaluation of the opportunity that is the basis for the project. A subject cannot be evaluated or understood without context. In one or two paragraphs, summarize how the project program is going to be executed. This project may involve other projects (i.e., this project is one project within a program of projects). Provide a summary of how resources will be acquired, basic contracting strategies, an organization chart, and roles and responsibilities. Basically, if the project is funded, describe the plan of action the team has considered to execute the project. Describe to the investor the assurances you can make to guarantee a successful project.

The purpose of the project description and evaluation is to entice potential investor to proceed with reading through the rest of the documentation by highlighting key facts of interest. Descriptions between 100 to 175 words are generally well received. Longer descriptions tend to be too detailed and can possibly derail other sections by introducing conflicting information. Therefore, take care to leave detailed explanations for the sections that are designed to directly address technical issues in more detail.

Project Technology

It is important for the investor to understand how robust the proposed project is. New products and new technology are riskier than projects that are expanding with existing technology, proven infrastructure, established markets, etc. For projects that are proposing funding for new technology and products, brief summaries are necessary to describe the appraisal, pilot, and development programs.

Facilities

The characteristics of the project facilities need to be presented. The main project facility needs to describe the inputs/material source/feed stocks, the process itself, and the characteristics of the product. When a project is an addition or expansion to an existing facility, then the project also needs to describe what considerations the project has made to the existing facility. The considerations should address additional personnel, training, economic development advantages, etc. as well as additional utilities required such as electricity, natural gas, roadways, rail, air, etc. The documentation needs to address considerations required if there

are revisions to direct assets such as assembly lines, operating product lines, and indirect assets/support such as corporate staffing, insurance, legal, and permitting agencies.

PROJECT DRIVERS

Drivers are specific influencing factors to a project's success, and are often based on research from the corporation or project team. The success of a project is based on how different factors influence the project. Some drivers can be directly controlled. However, the drivers that are outside the control of the corporation are the factors that need to be identified, researched, and understood.

Revenue Drivers

The primary revenue drivers for any commercial project are production (rate and volume) and prices. This section should discuss the key premises that define these revenue components and the technical, commercial, and political factors that contribute to uncertainty and risk.

Revenue – Technical Issues

Product Strategy: Generally, a project will not run at full capacity on the first day of start-up. There are many obstacles even to the best projects. Unforeseen events: material supply, start-up, integration with existing facilities, training, product quality, etc., need to be considered and a plan presented with realistic production rates. If the project is producing a product, then the investors will need to know if the project can produce more than its average rated throughput to account for when the project is off-line for maintenance or other reasons. Also, if sales exceed expectations, a trade-off analysis needs to be done between product quality and throughput (delivery rate). If sales do not meet expectations, then there should be a business plan to shut down the product line with an economic recovery plan. An impact analysis should also be done to determine the results on the company portfolio. When sales fail to meet the business plan (or expectations), there are financial instruments and obligations that are still required to be met. Therefore, the cash flow to meet these will have to be pulled from other business lines. Resorting to bankruptcy is a last resort when all other avenues to recover loss of profit and debt payment have been exhausted.

For a project that is selling product in large volumes, the worse-case scenario is that the cost of product equals the cost of the facility, a break-even event, resulting in no profit. When the project is producing a new product, the cost of development must also be part of the AARR. Depending on the product and market sector, costs for development may have already been dispersed in the cost of corporate overhead, tax breaks, or some other form of capital write-off. Writing off the capital cost of development may provide the company an advantage depending on the laws and tax structure of the location where the company is incorporated, operating, headquartered, or governed by tax law. Development costs can sometimes be reported as a capital loss depending on the tax structure. These can be deducted or written off based on a set of depreciation or amortization schedules. This is an area where the project team needs assistance from accountants, lawyers, and tax specialists.

In Figure28.9 the project is designed for a throughput of 20,000 units. However, the market analysis, based on a demographic survey, has determined a forecast of expected orders will lag the production throughput for the first three to four years and exceed production thereafter. The project for years four to seven may be able to exceed the initial facilities design basis through optimization. But as future demand in year four increases, another expansion may be planned in year four with completion slated for year eight, followed by a third project and/or more optimization.

Year	Daily Production	Expected Orders
1	8,000	5,000
2	9,000	8,000
3	15,000	18,000
4	20,000	20,000
5	20,000	25,000
6	25,000	30,000
7	30,000	35,000
8	35,000	40,000
9	45,000	45,000
10	50,000	50,000
11	55,000	55,000
12+	60,000	60,000

Figure 28.9—Project Implementation Strategy

Each project should have an upside and downside analysis. These sections are to provide transparency to the reviewer to indicate issues that have been considered but not part of the economics of the project.

Upside:

The PAD should explain the boundaries and conditions around future market growth expectations. The PAD should contain an itemized list of uncertainties and technology improvements that may provide advantages to project profitability. There may be

technical initiatives that are being pursued within the sector today with the expectation that there will be evolutionary improvements in the technology and possibly a number of breakthroughs. There should be short term achievable improvements to grow market shares and a plan to address expected changes.

Downside:
In the event that technological, construction, design, or other challenges have proven to be highly ambitious, product features may be delayed, or product specifications may have to be changed. This may also limit the market potential, sales, costs, pricing, delivery, and profit. The challenges need to be separated into those that will never be remedied, those that can be easily addressed, and those that will need to be moved to future project/product development phase. In all cases, the effect on cost and schedule and economics should be stated.

Revenue—Commercial Issues
All projects contain a profit margin (whether positive or negative). Profit can range from being perceived as excessive to customary and reasonable. It is important to review profit in the context with the other items that comprise the selling price a product. The profit can be that which the contractor makes for performing the project, or profit can be a percentage of the sale price of the product.

Figure 28.10 is an example of reviewing the profit in the overall context. All revenue breakdown should be in the currency of where the company funds the projects. In the example shown in Figure 28.10, the figures are in U.S. dollars. While the project may be advertised as making excessive profits, excessive is a subjective term. In Figure 28.10, assume the selling price per unit is $121.8 per unit. For every unit sold, $3.20 of profit is generated. The material cost, in this illustration, is considered fairly stable and is not further broken down. However, the operating account has more detail. If taxes increase ten percent, then taxes will increase from $12 to $13.2. The increase of $1.20 in taxes will diminish the profit from $3.20 to $2 for a time. While the selling price can and should be increased to reflect the added taxes, it may not be realistic in all cases.

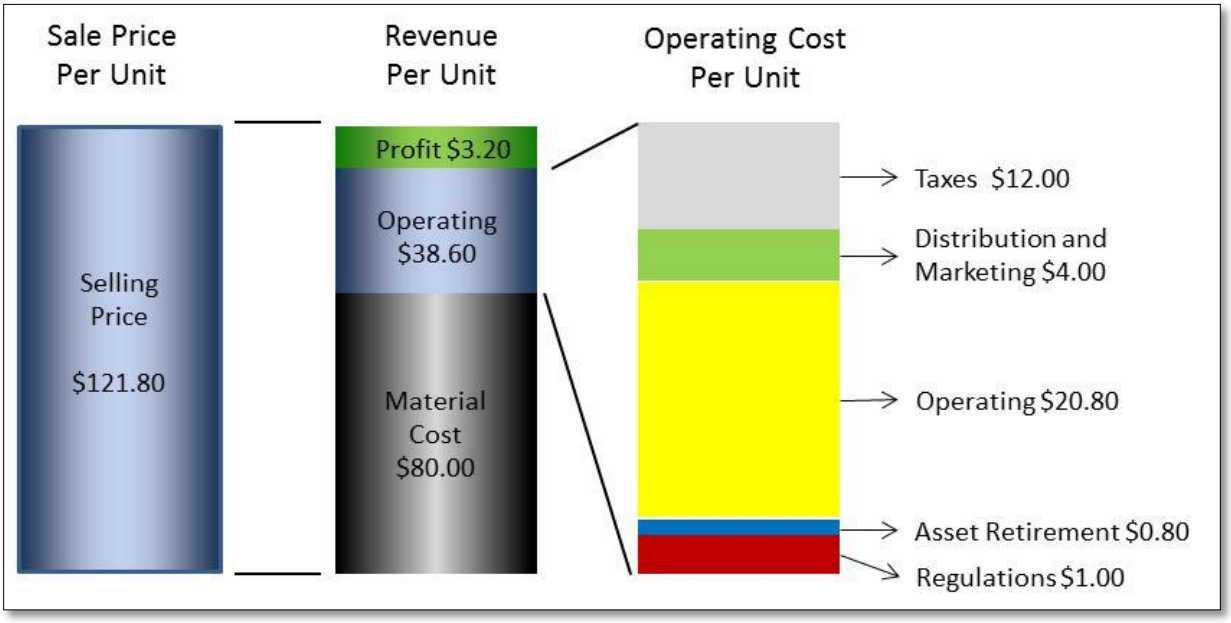


Figure 28.10—Revenue Breakdown

As mentioned in Section 1.6, the variance in operating costs (refer to Figure 28.6) has a significant impact on the AARR of the project. The increase in any of the operating cost elements in this Figure 28.10 directly contributes to the negative cash flow projection (refer to Figure 28.8).

From a commercial view, there needs to be a consideration and strategy to address how to protect the profit margin. Some of the strategy should include: a pricing strategy, market share, customer loyalty, uniqueness of the product, customer demand, and customer tolerance to increased selling price. Long term contracts and hedging of resources in the operating budget need to be tied to long range planning and contingency planning.

There is not a set list of commercial issues to address. The commercial issues will vary depending on the type of project. The core commercial issues need to address what it will take to keep the product viable over the planned life cycle and keep the cash flow positive.

CAPITAL DRIVERS

The capital cost profile in shown in the Figure 28.11. While the project team is focused on the cost of the project they have developed, the ultimate investment decision for the company must consider the total cost of the project along with other cost drivers that are often outside the control of the project team. Other cost drivers may include the cost of land to lease or purchase for the project, development costs, infrastructure costs (this may include costs for negotiated pricing for utilities) and the cost of capitalized interest. Capitalized interest is the interest on the loan from a financial institution or could be negotiated interest paid to a partner.

	Year							
Category	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**Total**
Land Acquisition				10				10
Development Costs	5	15	6	2				28
Sanctioning Capital				20	60	110	40	230
Infrastructure Costs				40	20	30	10	100
Total Capital	5	15	6	72	80	140	50	368
Capitalized Interest				1.5	7	7.3	3	14.8

Figure 28.11—Capital Expenditures

Capital – Technical Issues
The technical design alternatives need to be quantified in terms of capital cost. Typically, there is always a base case for proposed projects followed by a list of priced options. Each option affects the cost, schedule, and profit margin. It could be that a longer schedule may cost more, but the design may render more profit. Longer schedules may be deliberate in order to take advantage of greater market share.

Along with the base project cost and options, there are alternative options to the project or alternative projects that should be discussed. One of the alternatives may be to "do nothing" if the project failed to meet design parameters to selecting a different design concept. Looking at the overall portfolio, there are other proposed projects that may not generate as much profit margin but they provide cash flow early on for the company. Investment decisions should also consider the timing of the cash flow to the entire company.

Capital—Commercial Issues
There are many commercial issues around obtaining funding for a project. When there is a choice as to where to build, incorporate, hire, etc., then commercial issues may determine the construction site, or agreements to funding the project.

The credit risk of a company is a determining factor for the type of loans that may be available. Also, lenders may be competing with the company's creditors for the same capital. Not only do lending institutions provide capital, but lending money to a company may not be in their best interest to their stockholders, shareholders, dividends, and their wealth management programs. Lending institutions evaluate credit risk based on three major risk categories: economic, financial, and political. These are the same risks that require evaluation for projects that a company may decide to self-fund.

Taxation is a tool used by governments to provide services and to regulate commerce. Governments that have taxation authority may include local governments (cities) and ultimately include national-level government. Services provide citizens of a country a range of benefits from the esoteric (funding for the arts, museums, education foundations, etc.) to the practical (security, roads, utilities, healthcare, etc.). Governments regulate commerce through taxation. This is often performed to provide advantages to a business within a country (or local government) by encouraging the use of local materials and labor. Therefore, taxation for a project is not as simple as applying a flat tax on a particular commodity or to the estimate of the total installed cost. In order to optimize profits and payout of a project, businesses must consider how to lower investment costs, and taxes is a significant cost in the economic decision of a project.

If a government initiative is to provide more jobs, then taxes for using local labor may be waived, and taxes on foreign labor brought into the country may be taxed at a higher rate. A foreign business may need to decide what labor is critical to protect its intellectual property and which labor activities can leverage local labor. Intellectual property can include aspects of engineering design to crane operators on large heavy lift cranes. The wages to protect a company's intellectual property (engineering, crane operators, etc.) will be taxed, in this case, but the local labor wages may be taxed at a significantly lower rate. The government grows the economy through lowering unemployment and the business improves the economic payout through lower taxes by providing local jobs.

Taxation can be very complicated. A business should have tax specialists for the location where a project is being considered. The tax specialist will need a detailed estimate, project scope, schedule (some taxes are tied to dates), list of contractors, contractor location to analyze tax liabilities with respect to the project site. Sometimes project elements in a "Trade Free Zone" will not be taxed if the project creates a specified quantity of new permanent jobs and they are employed for a period of ten years or more. In addition, the taxes may have to be held in escrow by the government and will be refunded to the business at the end of ten years. However, the company must provide documentation that the stipulated jobs were created and jobs were filled for ten years. For an existing facility, a government may want the business to demonstrate the specified new jobs did not result in decreasing a similar number of jobs from another location. Sometimes, the escrow money held in escrow may not include interest earned for the ten years it was held, and this affects the project economics. In examples like this, the business will need legal counselors and contract specialists to assist the tax specialist.

The application of a flat tax rate to a project's cost estimate will overstate the total estimated cost of the project. For small projects, the excess estimated tax can be considered part of the overall contingency for the project cost. On large projects, the overstatement of tax could be enough money to fund a small project or several mid-size projects. Overstatement of estimated taxes is capital money that the business takes a financial loss because it cannot be invested or used for business growth. Ultimately, if the project had to be financed through a bank, then not only is their loss on interest rate, but the business had to pay loan interest on estimated taxes that were never paid.

EXPENSE DRIVERS

The primary expense cost components are fixed operating costs, variable operating costs, utilities, and fuel. This section discusses key premises that define these expense components and technical, commercial and political factors that contribute to uncertainty and risk. Explanations for sub-categories in Section 5 (5.1, 5.2, 5.3) are included as a guide to the type of consideration and type of information that is expected to be included. These definitions are not exhaustive, but the identified issues address a wide range of projects from architectural to industrial.

Operating Expense—Technical Issues

There are three subcategories to operating expenses: technical, commercial, and regulatory. Technical issues are the fixed and variable expenses that the business unit has the most control. Commercial issues generally address local contracts for utilities and other local contractual issues. Regulatory issues address laws, restrictions, and requirements by the government of the country of the project site.

Technical issues are the considerations of expenses that may vary because of the difference in exchange rates, hyperinflation, and a better understanding of the operating requirements. Higher fuel prices, increased contract services costs, monitoring (seismic, environmental, safety, etc.), and a better understanding of maintenance requirements heavily influence operating expenses. When operating expenses are lowered through engineering design and controls, then project capital costs are often affected, which may impact the AARR. It is important that the business unit be involved early with the project management to explore the trade-off between operating expenses and capital investment. Ultimately, the corporation has to decide on the balance between operating expenses and capital investment to select a project scope that meets the cash flow requirements that the corporation needs as well as meeting its staying power in the market.

Figure 28.12 is a breakdown of the operating budget in Figure 28.10. The first column, Annual Average Percentage, illustrates the budget percentages on a gross cost basis. If this project is a partnership, then the figures in column two (Avg./Year $US Net) are on the cost to the company or a net-basis. The last column ($Unit) shows the cost of the operating expenses per each unit produced.

Operating Cost Category	Annual Average Percentage (%)	Avg./Year $US Net	$/Unit
Fixed Operating Expense	38%	28.1	8.07
Variable Operating Expense	49%	36.3	10.41
Natural Gas	4%	3.3	0.96
Regulatory Compliance Operating Cost	9%	6.8	1.95
Total	**100%**	**74.5**	**21.4**

Figure 28.12—Operating Expenses

Fixed Expenses: Fixed expenses include labor, maintenance, insurance, and office expense. The basis for these estimated costs is provided from the several years of project development testing. Research into other similar-sized commercial facilities in operation is critical. Maintenance assumptions are based on project development testing, field visits and consultation with industry experts. For "first-of-a-kind" projects, analogous information may need to be developed from the process components that comprise the new project. For example: equipment cost, operating data, and maintenance records from several compressors can be aggregated to estimate the life-cycle cost of a compressor for a project that is "first-of-a-kind." This is an area where subject matter experts are vital in the decision making process of any project whether it is proven technology or new technology.

Variable Expenses
There are several key drivers for the variable expense component: fuel, utilities, chemicals, electrical, maintenance, and raw material costs. Unit cost assumptions need to be developed in the same manner as fixed costs. There needs to be research into costs from other similar operating projects, or subject matter experts need to create scenarios based on analogous data from equipment components.

The cost of utilities is a major cost factor for most projects. Early in the design there must be considerable effort directed toward technology improvements that would lead to lower utility consumption or alternative energy sources. Even when the project is complete and commences operation, lowering utility usage through improving efficiency or lower costs through contract negotiations is an ongoing effort. A recommended value-improving practice is to develop a utility model for the project during the design phase of a project and then train and transfer the model to the business unit once the project is complete.

Operating Expense—Commercial Issues
Projects seldom operate independently and detached from the outside world. The connections to the outside world which are required to operate the project, acquire raw materials, or ship products fall into a category of commercial issues. These interfaces can be materialistic or legal. A utility is a material category and tariffs are a legal category. There may be other commercial categories to consider. Generally, labor is considered part of the operating expenses and addressed in that section of the PAD.

Utilities
Utilities (gas, water, electricity, etc.) are a key driver in economic justification of any project. Even utilities that are produced by the project are likely to be dependent on a utility that is purchased. For example, assume a project has a boiler to produce steam. The boiler is owned and controlled by the corporation; however, the supply of fuel or other utility to operate the boiler is acquired from an outside supplier. In this case, the cost of production is controlled by the negotiated price of the outside supplier. In the case of water supply, a project may be large enough to consider a water recycling system to recover, filter, purify, and reuse water that a process has discharged. The cost of water recycling must be evaluated against the supply limitations of the outside water supplier. The outside water supplier may be governed by permits and legislation that limits how much water can be produced and sold to the project. This requires negotiation of commercial issues with the water supplier and also with local, state, and other governmental bodies. This is an area where having contract specialists and governmental affairs input, is important to the ultimate project funding decision.

Tariffs
A tariff is like a tax; however, a tariff is assessed to influence the trade/flow of goods. When projects are planned, there is generally an announcement of where the project will be located. This announcement also triggers the economic planning boards of the community where the project is planned to be built. Tariffs can be assessed on any factor that the community decides will improve their economy and well-being. There is no standard tariff. It could be that a tariff is assessed as a cost per throughput of the product produced. Or if the roadway infrastructure requires to be improved to the proposed facility, then there may be a tax on the vehicle weight and number of vehicles that enter and leave the facility. There will probably be a list of tariffs/taxes the project will need to start negotiating before the project is considered for approval. Negotiations on tariffs will continue throughout the life of the project. When there is a project has several options to locating the facility, tariffs and taxes may influence the project economics enough to decide the final construction site.

Operating Expense—Regulatory Issues
The conditions imposed by the current regulations should be included into the project design. Governmental licenses and permits need to be renewed on a regular cycle (yearly, five years, ten years, etc.) this will influence future expansion applications beyond the initial project design premise. There is a strong likelihood that new standards may limit the life, operating conditions, or operating cost of the project. Familiarity with the market needs is important; however, familiarity with political trends and legislation can sometimes be far more important. Environmental compliance laws (especially clean air) may become onerous to a point that few projects may be able to meet the current regulations or will be able to remain in operation past ten years. Given what is known about the trend of regulations, the investment decision makers need to know the economic life of the proposed project.

SCHEDULE DRIVERS
The technical, commercial and political factors that contribute to the project schedule and any uncertainty and risk are discussed in this section. Figure 28.13 shows typical project schedule investment milestones. This schedule is different from a project schedule because it contains key milestone dates for funding approvals rather than milestone dates that track the progress of engineering and constructing the project.

Task Name	Start	Finish
Development Planning		
Partner Negotiations		
Internal Sanctioning Approval		
Full Partner Approval		
Project Execution		
Permits Granted		
Startup		
First Production		
Full Production		

Figure 28.13—Schedule of Key Project Drivers

Schedule—Technical Issues
The project must address the technical issues for the risks associated with the project schedule, such as those that may delay project start-up. In the case of a project employing new technology, this is an area that must be fairly robust. Companies sanction projects to support the company cash flow or to comply with regulations that also affect the business operating parameters. Delaying a project not only leads to cost overruns, but a delay to entering a market may adversely affect the market share potential the company has based on the economics. In the case of specialty chemicals like pharmaceuticals, the company could be completely shut out of a potential market because the competition "got there first" and met most of the market demand. In the case of consumer goods, it could be the competition created a loyalty of the consumers who typically do not like a product that has different features. Finally, if the product is a utility, the rates of the competition may have established a contract rate that is so low with the consumers that makes it uneconomical to start the project.

Technical and design issues may be addressed and remedied with enough engineering that results in cost and schedule under-run. While some technical issues for new technology may be daunting, there are still some very common technical issues that are completely controllable and generally always lead to cost and schedule overruns because of lack of planning. These areas are:

- Late Change to Scope or Design
- Force Majeure [6]

The thoroughness and criticality of project reviews and communication cannot be under-emphasized. The success of a project is rooted in the earliest phase of project development: Front-end Loading (FEL). Planning and vetting of scope and the project design very early is a successful mechanism to avoid late design changes.

ECONOMICS SUMMARY
As with any basis document (e.g. Basis of Estimate), a declaring of what "is" and "is not" included needs to be stated. Over the course of developing a project, the economic basis may change. Most importantly, if the completed project is not meeting the economic indicators in the future, then it is important to review the basis of the original economic premises to determine what changed.

Economic Premises
Include a clear and simple economic premise because these items influence the AARR, LPO, and cash flow projections to justify a project. When the project is funded in whole by a single company, the funding is more straightforward than projects with two or more partners. Projects with partners need to clearly identify who is funding the project and carrying the interest. In some cases, the carrying party may have responsibility for 100 percent of the working interest until the project has started production and generating revenue. This means there needs to be clarity around what is meant by "generating revenue". If the project is delayed and there are cost overruns, the partner may decide that generating revenue is based on the original project cost, seek damages, and delay paying any working interest until the project is viable.

Base Case Economic Indicators
The project needs to declare the NPV on both a gross cost basis and net cost basis (when there are partners). The net cost basis provides insight to the shared risks of the project to each partner. The percent ownership should be evaluated to determine that each partner is equitably responsible for shared critical items such as funding, project management, site selection, marketing, etc. Projects need 100 percent commitment by all the partners. Generally, partners with small percentage ownership are not as motivated to bringing the project to a successful completion. Low percentage owners are typically more investors than owners.

AARR should be also declared on a gross and net basis along with the basis of the calculation, such as the cost basis and life of the facility. In the case where the project has several phases, then the capital investment and other economic indicators for the first phase and the final total project cost should be presented. In some cases, substantial research and development costs will need to be included in the project funding. Or, the economics of the project are dependent on introducing the product to gain market share

and then profits will be based on future sales. Therefore, the profitability of the project is clouded. Basic economic indicators should be considered to evaluate the project on a near-term and long-term basis.

The Profitability Index (PI) is the ratio between the present value of the future cash values and the initial investment. The initial profitability index is also called the profit investment ratio. This ratio is important because it quantifies the amount of value the project creates per unit of investment. A PI greater than one means the project should be considered. A company has many considerations to make when selecting projects for funding. One important tool is prioritizing projects by their PI values. Those projects with the largest PI values should be ranked higher in the company's project sanctioning portfolio.

The Return On Capital Employed (ROCE) should be higher than the rate at which the company borrows money. ROCE is the measurement of the performance of a company and is basically the operating income divided by the assets employed (See Equation 28.5). Assets employed includes: equity, fixed, and current assets, gross borrowings plus current assets. ROCE lowers the shareholders earnings if the company has to borrow more money. The basis to calculating ROCE is comprised of three main elements: 1) Profit before interest, tax, and dividends, 2) Total Assets, and 3) Current liabilities.

ROCE = Operating Income / Assets Employed

Equation 28.5

For example, a company that realizes a profit of 40 million USD and has 600 million in assets has a better ROCE (0.0667 = 40/600) than a company that makes the same profit with 800 million USD in assets (ROCE = 0.05 = 40/800). This example assumes that both companies have the same liabilities. ROCE measures how efficient a company is to generating profit out of capital. If the liabilities, taxes, interest, etc. are different, then the ROCE Equations need to address the liabilities, such as in Equation 28.6.

ROCE = Sales / (Assets Employed - liabilities)

Equation 28.6

ROCE reported by many corporations is as complex as the accounting practices of the corporation. As another example, ROCE can accommodate operating income before taxes and interest.

ROCE = (Sales / Assets Employed) X [(Operating Income before taxes and interest)/Sales]

Equation 28.7

Reporting ROCE for the project must match the ROCE reporting practices of the corporation as reported internally and in reports to stockholders.

	Phase 1	Total Project
NPV @ 13% ($ MM)		
PI Ratio @ 13% (Cash)		
AARR (%)		
Cash Breakeven (Years)		
Other Indicators		
ROCE (%) average over project life		
Total Production Throughput		
Phase 1 Capital Investment		
Total Capital Investment		
Unit Measures		
Capex/units produced		
Opex/units produced		
Total Costs/units produced		

Figure 28.14—Economic Indicators

Figure 28.14 provides corporations with benchmarking or other economic indicators. These indicate how well the proposed project compares to other operating facilities (internal or with competitors). These indicators are generally key industry performance metrics that have a direct influence on the investment decision. The Unit Measures are economic indicators of cost performance (efficiency) metrics: Capex/units produced, Opex/units produced, and Total cost/units produced. For example, these may be cost per volume, cost per weight, or cost per each. For these unit cost measures to be meaningful there must be numbers that are intuitive to the decision makers and cause them to take action if the indicators are not within expected ranges for the product being produced.

Risks to Revenue Generation

Cash flow is an extremely important assessment to a corporation. Projects are funded and built to address the corporation cash flow. Therefore, it is important to understand the risks to the cash flow.

A tornado diagram can be used to depict the magnitude of the risks to the project's revenue. It is best to keep the diagram as simple as possible and not overlay different scales and indices. Figure 28.15 shows the variance on annual revenue generated, and the center value for the tornado diagram is the point that is most likely to be met. Several key uncertainties are listed on the left and the diagram is organized from the event with the most volatility at the top and the least volatile on the bottom. Each variance from the center is color coded as to high, medium, and low to illustrate the likelihood that the event may occur. The units across the X-axis should be declared. This chart could depict the cost variances of the risks (Y-axis) in terms of profit, cost of the project, forecast cost of the project, etc.

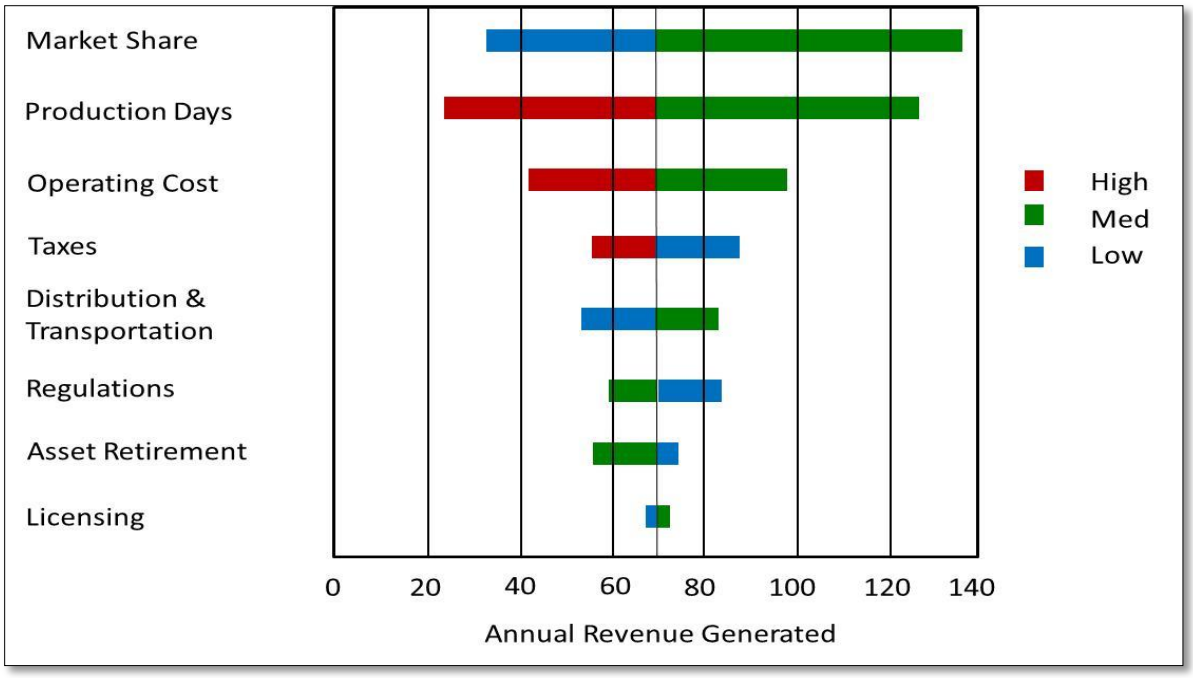


Figure 28.15—Risks to Annual Revenue Generated

Given that Market Share and Production Days have nearly the same volatility in Figure 28.15, the risk that should be scrutinized the most is Production Days. There is a limited amount of resources that can be funded to manage the risks to a project. In this example, it may be a better decision to take action to monitor and control the problems that cause production to decline. The risk to production appears to be significant, and the corporation will have more direct control of the project's production than trying to control or heavily influence the market.

Too often project members overlay different indices on the bars. For example, market share may have numbers indicating quantities of units produced, production days may overlay the number of days the project is off-line, the Operating Costs may depict the effect on the AARR to the project. Overlaying another set of indices is not recommended. Develop a tornado diagram for every concept. Include only the tornado diagram that is the most significant to the decision making process. Nuances of secondary effects on the project economics are distracting. If the tornado diagram does not affect the value of the project based on three significant figures, then the diagram should not be included.

CONCLUSION

Capital is a limited resource, and there are numerous projects that are competing for funding. Investment decision makers review many projects and options to evaluate what is best for the company. A consistent structured format and key deliverables, such as the Project Assessment Document (PAD) outlined in this chapter, facilitate the review process. There are three important elements required by investment decision makers:

1. **Value to the Company**—The point of undertaking a thorough and structured project assessment is to determine if a company's market and operating condition will be better or worse than it is currently operating. A project may have a great profit margin, but if it is not aligned with the core values (strategic vision) of the company, then it may become hard

to sustain. Business lines that are difficult to sustain and that pull a company away from its strategic vision will erode the cash flow of the company over time.

2. **Effect on Cash Flow**—Cash flow analysis is generally more important than profit analysis. Business development professionals focus on the profitability of new concepts. However, cash flow analysis is the final determination by economists, lenders, stockholders, etc. to determine the viability of company. Projects must support and improve the cash flow of a company. Not all projects generate profit, but they can improve the cash flow of a company.

3. **Transparency of Risk to the Company**—Projects with more assessment transparency will assist investment decision makers with the information they need to make the critical calls that best serve the company's sustainability in the marketplace. A transparent, clear, and realistic accounting of risks is vital to assessing a project. It is far better to fund a well-defined (but less profitable) project than to fund (gamble on) a highly profitable project with uncertain risks.

Projects are part of the corporation's portfolio management process. Having a structured review format with key deliverables will facilitate the review of the project and how it fits within the portfolio. The project team's role is to assist the executive with the decision making for the corporation and understand they are only one project in a slate of projects and permutations that the investment decision makers must evaluate for the future of the corporation's strategic vision.

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Chapter 29—Statistics & Probability

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Abstract
There are two distinctive methods to describe a set of data, namely, qualitative and quantitative data types. Numerical methods used for describing quantitative data include mean, median, mode, range, variance, and standard deviation, etc. Cost professionals combine these methods with probability analyses to perform conceptual estimates, assess project risks, and support decision–making process. These tools are also applied to manufacturing quality controls.

This chapter serves as a refresher of the fundamental aspects of statistics and probability. Examples are used to describe data types, illustrate concepts of random variables, and explain relative frequency and probability distribution. Readers will learn both normal and standard normal distributions and their applications. An introduction of basic regression analysis concept and its application in cost management is also covered.

Keywords
Frequency Distribution, Mean, Median, Mode, Normal Distribution, Percentile, Probability Distribution, Random Variables, Range, Regression Analysis, Standard Deviation, and Variance.

BASIC CONCEPTS

- Introduction
- Describing data
    - Frequency distribution
    - Central tendency: mean, median, and mode
    - Variability: range, variance, standard deviation
    - Percentile, z-score
- Selected key probability distributions
    - Random variables
    - Discrete and continuous probability distributions
    - Normal distribution and standard normal distribution
- Regression analysis

INTRODUCTION
Statistics is the field of study where data are collected for the purpose of drawing conclusions and making inferences. Descriptive statistics is the summarization and description of data. Inferential statistics is the estimation, prediction, and/or generalization about the population based on the data from a sample.

Four elements are essential to inferential statistical problems:

1. **Population** is the collection of all elements from which statistical inferences are to be developed. The size of the population is usually denoted by *N*. Very often the population is so large that a complete census is out of the question. Sometimes, not even a small population can be examined entirely because it may be destructive or prohibitively expensive to obtain the data. Under these situations, we draw inferences based upon a part of the population (called a sample*).*
2. **Sample** is a subset of data randomly selected from a population; the size of a sample is usually denoted by *n*.
3. **Statistical inference** is an estimation, prediction, or generalization about the population based on the information from the sample.
4. **Reliability** is the measurement of the "goodness" of the inference.

The first two elements are the primary focus of this chapter. Numerical characteristics of a population are called parameters of the population. The corresponding numerical characteristics calculated from a sample are called sample statistics.

Learning Objectives
After completing this chapter, the reader should be able to:

- Understand basic definitions and terminologies in probability and statistics.
- Apply statistical techniques in decision making.
- Understand the basic concept of regression analysis.

DESCRIBING DATA
In general, data can be classified as either Qualitative or Quantitative.

Qualitative Data
Qualitative data can be categorized or summarized.

Example A:
As of August 30, 2011, the total membership of AACE International is 7,653 [1]. This can be classified according to member types:

Regular and Associates Members	7,149
Student Members	124
Emeritus Members	97
Honorary Life Members	19
Prepaid Life Members	235
Retired Members	29
Total	**7,653**

Or, according to geographical region distribution:

Region	Predominant Areas	Members
1	Canada	1,199
2	Northeast USA	1,170
3	Eastern USA	878
4	Central USA	882
5	Mid Central USA	935
6	Western USA	1,082
7	Middle East, Africa, Europe	1,160
8	Far East, South America	347
Total		**7,653**

Quantitative Data
The description of quantitative data is more complex. It can be described graphically or numerically. Some graphic methods for describing quantitative data are:

- Frequency distribution; relative frequency (f/n)
- Stem and leaf plot
- Histogram

Numerical methods for describing quantitative data include the following:

- Measures of Location (Central Tendency)
  - Mean (Average)
  - Median
  - Mode
- Measures of Dispersion
  - Range
  - Variance
  - Standard Deviation
- Relative Standing
  - Percentile
  - Z-score

Example B:
Many companies invest on training their employees. The following average training hours for every employee are selected from the Jan. 20, 2003 "The 100 Best Companies to Work For" from *Fortune* magazine. To simplify the example, only the Top 50 company data are used. Numbers are rounded to the nearest five for convenience.

145	50	35	50	25	160	30	40	40	20
40	20	30	95	40	50	30	50	20	35
70	35	40	45	30	45	70	45	40	140
50	70	20	30	40	40	30	60	40	50
60	55	50	35	40	30	45	75	20	40

The data will first be divided into smaller equal intervals (classes*).* The number of observations that fall into each class, the frequency, is then counted. These classes cover the entire spectrum of data but should not overlap. There may be open-ended intervals when the first class contains no lower limit or the last data class contains no upper limit. The number of classes depends on the number of observations in the data set. In practice, a frequency distribution usually has between five and 20 classes.

The Stem and Leaf Plot is developed by first determining the stem and then adding leaves. Using Example B, the stem is formed by the tens digit and the leaves are the ones digit. Note that the stem values are placed to the left of the vertical line and the leaves on the right.

Example: 45 is shown as:

Stem	**Leaf**
4	5

Stem	**Leaf**	**f** **frequency**	**f/n** **relative frequency**
2	5, 0, 0, 0, 0, 0	6	6/50
3	5, 0, 0, 0, 5, 5, 0, 0, 0, 5, 0	11	11/50
4	0, 0, 0, 0, 0, 5, 5, 5, 0, 0, 0, 0, 0, 5, 0	15	15/50
5	0, 0, 0, 0, 0, 0, 5, 0	8	8/50
6	0, 0	2	2/50
7	0,0,0,5	4	4/50
8		0	0
9+	95, 145, 160, 140	4	4/50
		50	50/50

Figure 29.1—Stem, Leaf, Frequency and Relative Frequency Distributions of Example B

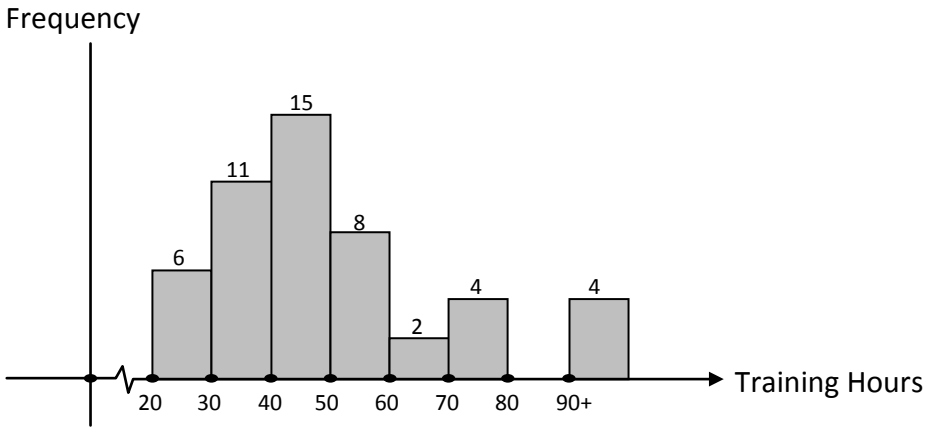


Figure 29.2—Histogram of Training Hours

Graphical description provides a general idea of how the data are distributed. To understand data characteristics better, it is necessary to analyze data numerically. In Example B, the sum of the training hours from the above 50 companies is:

145 + 50 + 35 + 50 +…. + 20 + 40 = 2,445 hours

Mean (average): the sum of measurements divided by the number of measurements
Population mean is denoted by ***μ = sum of all numbers in population, N***
Sample mean is denoted by $\overline{x}$ ***= sum of all numbers in sample, n***
The mean of Example B is: **2,445/50 = 48.9 hours**

Median*:** the middle number, when the data observations are arranged in either ascending or descending order. If the number ***n of measurements is even, the median is the average of the two middle measurements in the ranking.
The median of Example B is: **40 hours**

For symmetric data set, the mean equals to the median.
If the median is less than the mean, the data set is skewed to the right.
If the median is greater than the mean, the data set is skewed to the left.

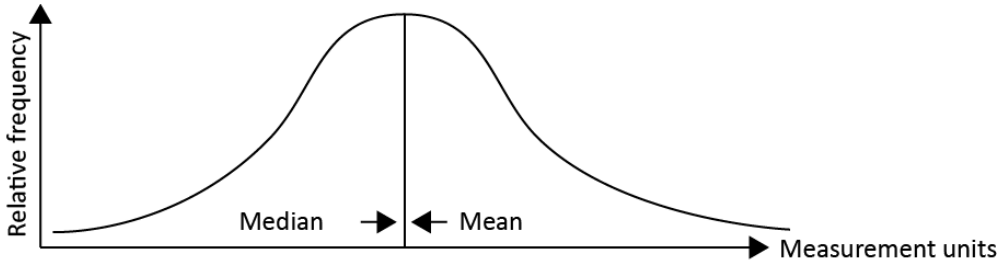


Figure 29.3—Symmetry

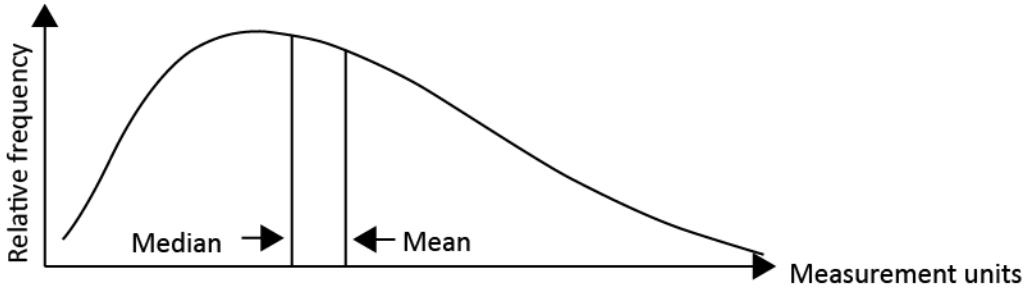


Figure 29.4—Rightward Skewness

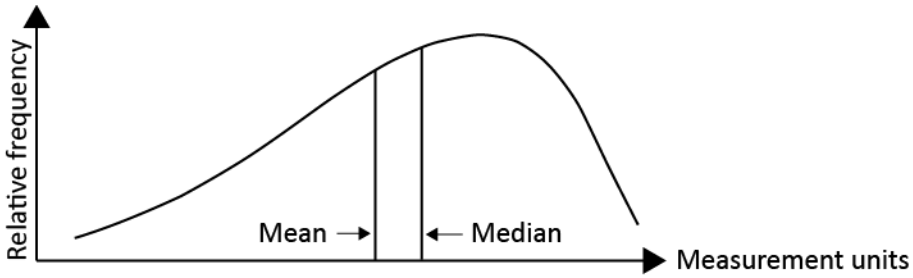


Figure 29.5—Leftward Skewness

***Mode*:** the measurement that occurs most often in the data set
The mode of Example B is: **40 hours**

If the observations have two modes, the data set is said to have a bi-modal distribution.
When the data set is multi-modal, the mode(s) is no longer a viable measure of the central tendency.
In a large data set, the modal class is the class containing the largest frequency. The simplest way to define the mode will then be the mid-point of the modal class.

Comparison of the Mean, Median, and Mode
The mean is the most commonly used measure of central location; however, it is affected by extreme values. For example, the high incomes of a few employees will influence the mean income of a small company. Under such situation, the median maybe a better measure of central tendency.

The median is of most value in describing large data sets. It is often used in reporting salaries, ages, sale prices, and test scores.

The mode is frequently applied in marketing. For example, the modal men's shirt neck size, and sleeve length, and shoe size, etc.

Numerical Measures of Variability
Measures of central tendency do not describe the spread of the data set, which may be of greater interest to the decision-maker.

The simplest measure of the variability of a data set is its **range**.

***Range*:** The difference between the largest and the smallest values of the data set.
*The range of Example B is: **160 - 20 = 140 hours***

The range only uses the two extreme values and ignores the rest of the data set. One instinctive attempt to measure the dispersion would be to find the deviation of each value from the mean and then calculate the average of these deviations. One will find that this value is always zero, an answer which is no accident. The alternative might be to calculate the average absolute deviation. However, this measure is rarely used because it is difficult to handle algebraically and does not have the unique mathematical properties possessed by the *variance*.

***Variance*:** The average of the squared deviations from the mean.

The population variance is denoted by $\sigma^2 = \frac{\Sigma(x-\mu)^2}{N} = \frac{\Sigma x^2 - N\mu^2}{N}$

The sample variance is denoted by $s^2 = \frac{\sum\left(x-\bar{x}\right)^2}{n-1} = \frac{\Sigma x^2 - n\bar{x}^2}{n-1}$

Note that the divisor (n-1) *is used instead of the more obvious* n. *This will make the sample variance,* $\mathbf{s^2}$*, a better estimate of the population variance,* σ^2*. The explanation for this practice is beyond the scope of this chapter but can be found in many statistics textbooks. Calculating sample variance, the divisor (n-1) should be used. When n is a large number, the difference between using n and (n-1) as the divisor is minimal. This simply implies that the bigger the sample size, the better the sample values can represent the population parameters.*

The variance of Example B is: $S^2 = \frac{162825 - 50(48.9)^2}{50-1}$ = 882.95 hours

The variance has a squared unit and is in a much larger scale than that of the original data. To offset these, the square root of this value is adopted leading to the popular use of standard deviation.

***Standard Deviation*:** The positive square root of the variance.
The population standard deviation is denoted by σ
The sample standard deviation is denoted by s

The sample standard deviation *s* of Example B is $\sqrt{882.95}$ = 29.71 hours

The standard deviation may also be approximated by: range/6. Some may prefer to use range/4.

Measures of Relative Standing
Another measure of interest is the description measurement of the relative location of a particular observation within a data set.

***pth percentile*:** In any data set, the pth percentile is the number with exactly p percent of the measurements fall below it and (100-p) percent fall above it when the data are arranged in ascending or descending order.

The *first (lower) quartile:* the 25th percentile
The *third (upper) quartile:* the 75th percentile

The *second (middle) quartile* is the median.

Using the same Example B, the 80th percentile is 60 hours.

Another measure of relative standing is the famous **z-score.**

A ***z-score*** is the number of standard deviations a point is above or below the mean of a set of data.

The **population z-score** for a measurement x is $z = (x - \mu)/\sigma$

The **sample z-score** for a measurement x is $z = (x - \bar{X})/s$

RANDOM VARIABLES AND KEY PROBABILITY DISTRIBUTIONS
Analyzing frequency distribution for every decision-making situation would be very time consuming. Fortunately, many physical events that appear to be unrelated have the same underlying characteristics and can be explained by the same laws of probability. The mathematical model used to represent frequency distributions is called a probability distribution. Understanding the concept and knowing which distribution to use in a particular situation will save considerable time and effort in the decision-making process. We will start from the concept of a random variable.

Random Variable: A variable whose numerical value is determined by the outcome of a random experiment.

A *random experiment* is the type of experiment that may produce different results in spite of all efforts to keep the conditions of performance constant. If a random variable can take on only countable number of values, then we call it a discrete random variable. For example: the number of sales made by a salesperson in a given day.

Random variables that can assume any value within some interval or intervals are called continuous random variables. For example, the length of time an employee is late for work.

Probability Distribution: A table listing all possible values that a random variable can take on together with the associated probabilities.

Notice that the random variable itself will be denoted by X, while the lower-case *x* denotes a particular value of X.

The symbol *p(x)* = Pr(*X=x)* means the probability that the experiment yields the value *x.*

In a discrete probability distribution, each *p(x)≥0* for all values of *x* and *∑p(x)* = 1.

Example C: Two coins are tossed. Let X be the number of heads appeared.

Probability distribution of X

X	0	1	2
p(x)	1/4	2/4	1/4

The probabilities *p(x)* can be interpreted as long-range relative frequencies. For instance, if two coins were flipped many, many times, we could anticipate obtaining two tails (*x=0)* about one-fourth of the time, one head and one tail (*x=1)* one-half of the time and two heads one-fourth of the time. Therefore, the probability distribution for a random variable X, *p(x)*, is a theoretical model for the relative frequency distribution, *f / n*, of a population.

Like the frequency distribution, the mean and standard deviation of a probability distribution need to be calculated to describe the central location and spread of the probability distribution.

The predicted long-range average of a discrete random variable X, often called the expected value (or mean) of X, is defined by

$$\mu = E(x) = \sum xp(x)$$

The population variance is defined as

$$\sigma^2 = E(x-\mu)^2 = \sum(x-\mu)^2 p(x)$$

The standard deviation σ is the square root of the variance.

In the above two-coin example, the expected value of X is:

$$\mu = \sum xp(x) = 0*p(x=0) + 1*p(x=1) + 2*p(x=2) = 0*(1/4) + 1*(1/2) + 2*(1/4) = 1$$

It is very important to remember that the expected value is not a number we "expect" to get at a given experiment. What this expected value tells is that if we were to toss two fair coins many, many times, carefully record the number of heads appeared in each toss, and at the end calculate the average number of heads, then this average would be one.

The variance and the standard deviation of X is respectively,

$$\sigma^2 = (0-1)^2 (1/4) + (1-1)^2 (1/2) + (2-1)^2 (1/4) = 1/2$$

$$\sigma = \sqrt{\frac{1}{2}}$$

Another example of expected value is as follows:

Suppose you work for an insurance company and sell individual ten-year $100,000 term life insurance coverage at an annual premium of $240. Actuarial table shows that the probability of death during the next year for a person of your customer's age, sex, health, etc., is .001. What is the expected gain to the company for a policy of this type?

Gain(X)	**Event**	**Probability**
$240	customer lives	.999
$240–$100,000	customer dies	.001

If the customer lives, the company keeps the $240 premium. If the customer dies, the company must pay $100,000 and will have a net gain of $(240-100,000). The expected gain is therefore:

$$\mu = E(X) = (240)(.999) + (240-100,000)(.001) = 240(.999+.001)-100,000(.001) = \$140$$

Please note that for each policy sold, the insurance company is taking a risk of either gaining $240 or losing $99,760. However, if the company were to sell a very large number of such insurance policies to customers possessing the characteristics described above, the company would on the average net $140 per policy written.

Discrete Random Variables

There are several theoretical discrete probability distributions that have extensive applications in decision-making. One will be introduced in this section.

Binomial Distribution

Many decisions are of the either/or variety. A company bidding for a contract may either get the contract or the bid will be unsuccessful. The responses to a public opinion poll may be either "favor" or "oppose". Many experiments (situations) have only two possible alternatives, such as Yes/No, Pass/Fail, or Acceptable/Defective.

Consider a series of experiments which have the following properties:

- The experiment is performed *n* times under identical conditions.
- The result of each experiment can be classified into one of two categories, say, success (S) and failure (F).
- The probability of a success, denoted by *p*, is the same for each experiment. The probability of a failure is denoted by ***q.*** Note that *q=1-p.*
- Each experiment is independent of all the others.
- The binomial random variable *X* is the number of successes in ***n*** experiments.

Probability of *x* successes in *n* experiments:

$$p(x) = \binom{n}{x} p^x q^{n-x} \qquad x=0,1,2,\ldots,n$$

The name binomial arises from the fact that the probabilities *p(x)*, *x*=0,1,2,…,n, are terms of the binomial expansion, $(q+p)^n$.

Mean, Variance, and Standard deviation for a Binomial Random Variable*:*

Mean: $\mu = np$
Variance: $\sigma^2 = npq$
Standard deviation: $\sigma = \sqrt{npq}$

Several extensive tables of the binomial distributions for some values of *p* and *n* have been published. Either one of the cumulated probabilities $Pr(X \leq x)$ or $Pr(X \geq x)$ is listed.

Example D: Suppose your company ships electrical fuses in lots, each lot containing 10,000 fuses. Your quality control plan requires that you will randomly sample twenty-five fuses from each lot and accept the lot if the number of defective fuses, x, is less than two. If $x \geq 2$ you will reject the lot and will conduct a complete re-inspection. What is the probability of accepting a lot (x=0, 1) if the actual fraction defectives in the lot is (a) 0.1? (b) 0.01?

Solution:
n = 25
(a) If p = 0.1; q = 1 - p=1 - 0.1=0.9

Pr (Accepting the lot) = Pr(x<2) = Pr ($x \leq 1$) = Pr(x = 0) + Pr(x = 1)

$$= \binom{25}{0} (.1)^0 (.9)^{25} + \binom{25}{1} (.1)^1 (.9)^{24}$$

= .27121

(b) If p = .01; q = .99

$$Pr(x \leq 1) = \binom{25}{0} (.01)^0 (.99)^{25} + \binom{25}{1} (.01)^1 (.99)^{24} = .77782 + .19642$$

= .97424

The two measurements indicate that, under the proposed quality control plan, there is 73 percent chance that you will reject the lot if in fact ten percent (p=0.1) of the fuses manufactured are defective. If only one percent of the fuses are defective, the chance of re-inspection is very small (less than three percent).

Continuous Random Variables

The probability distribution for a continuous random variable is often denoted by *f(x)* and is variously called a probability density function**.** The primary difference between probabilities for discrete and continuous random variables is that while probabilities for a discrete random variable are defined for specific values of the variable, the probabilities of a continuous random variable are defined for a range of values of the variable. The graphic form of *f(x)* is a smooth curve and the area under the curve corresponds to probabilities for *x.* For example, the area *A* beneath the curve between the two points *a* and *b*, is the probability Pr(*a<x<b).*

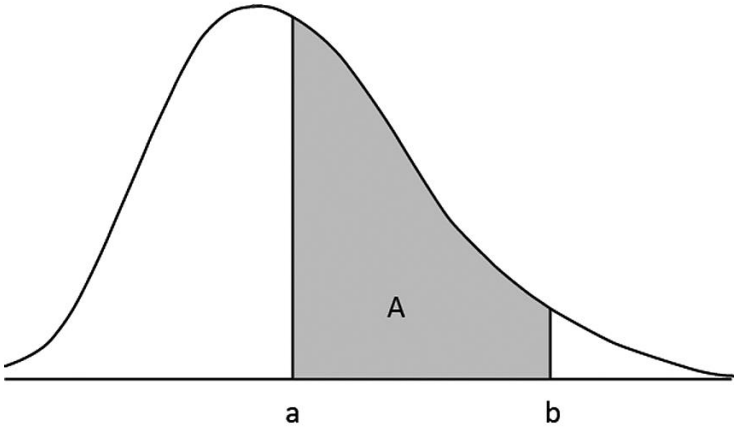


Figure 29.6—Continuous Distribution

Since there is no area over a point, the probability associated with any particular value of x, say, *x=a,* is equal to zero. Hence, Pr(*a≤x≤b)*=Pr*(a<x<b).* In other words, the probability is the same regardless of whether the endpoints of the interval are included. The total area under the curve, which is the total probability for *x,* equals to one.

The areas under most probability density functions are obtained by the use of calculus or other numerical methods. This is often a difficult procedure. However, as with commonly used discrete probability distributions, there are tables that exist for finding probabilities under commonly used continuous probability distributions.

Similar to the requirements for a discrete probability distribution, we require:

$f(x) \geq 0$ and $\int f(x)dx = 1$ for all x

The Normal Distribution

The most important continuous distribution in statistical decision making is the normal distribution. It is important for the following two reasons:

- As odd as it may seem, many observed variables are normally distributed, or approximately so.
- Many of the procedures used in statistical inference require the assumption that a population is normal.

Probability distribution for a normal random variable X is:

$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \qquad \text{for } -\infty < x < +\infty$$

Where

μ = mean of the random variable X
σ = standard deviation of X
e = 2.71828…
π = 3.14159…

Cost professionals may not use the above probability distribution function on a regular basis; however, the formula is shown as a refresher of the known normal distribution.

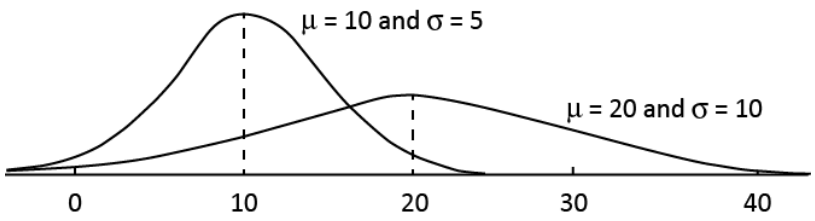


Figure 29.7—Two Normal Distributions

The graph of a normal distribution is called a normal curve and it has the following characteristics:

- It is bell-shaped and is symmetrical about the mean. The mean, median, and mode are all equal. Probability density decreases symmetrically as *x* values move from the mean in either direction. Since the total area (probability) under the curve is one, the area on each side of the mean is half.

- The curve approaches but never touches the horizontal axis. However, when the value of X is more than three standard deviations from the mean, the curve approaches the axis so closely that the extended area under the curve is negligible.

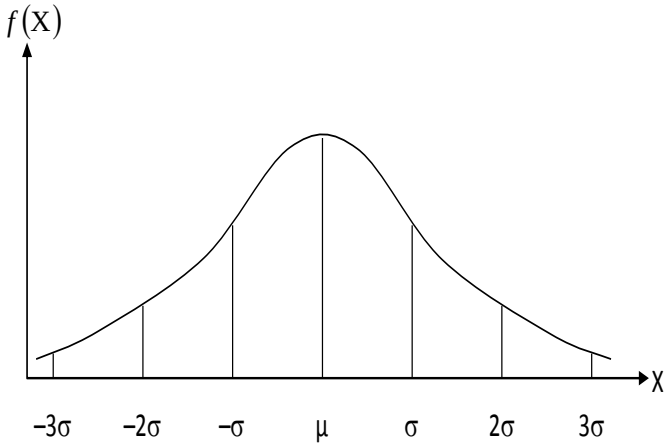


Figure 29.8—A Normal Probability Distribution

The Standard Normal Distribution
If X is a normally distributed random variable with mean μ and standard deviation σ**,** the random variable *Z,* defined by:

$$Z=(X-\mu)/\sigma$$

is a new normally distributed variable with mean zero and standard deviation of one. The probability distribution of *Z* is called the ***Standard Normal Distribution.*** Notice that ***Z*** gives the number of standard deviations that a value of *X* lies above or below the mean. By using the Z-score, all normal distributions can be transformed to Standard Normal Distribution. We can say that X in a normal distribution $N(\mu,\sigma^2)$, is equivalent to the new variable $Z=(x-\mu)/\sigma$ in the standard normal distribution $N(0,1)$.

The standard normal distribution table that gives the area under the standard normal curve is available. Some tables give the area between the mean zero and any particular value of *Z, Pr(0<Z<z),* where *0<z<3.59*. Remember that the area represents the probability that a value of *Z* will lie between zero and the given value and it must always be positive. Further, since the normal curve is symmetric, the area between *–Z* and zero is the same as the area between zero and *Z;* the area between *–Z* and *+Z* is twice the area between zero and *Z.*

Table 29.1—Frequently Quoted Z-Value and Probabilities

Z	Pr(-z<Z<z)	Pr(Z<-z) or Pr(Z>z)
1.00	**.683**	**.158**
1.282	.80	.10
1.645	.90	.05
1.96	.95	.025
2.00	**.954**	**.023**
2.326	.98	.01
2.576	.99	.005
3.00	**.997**	**.0015**

Example E: The actual amount of coffee ground which a filling machine puts into six-ounce jars varies from jar to jar and it may be assumed as a normal random variable with a standard deviation of 0.04 ounce. If the jar contains less than six ounces, it is considered unacceptable. Determine the mean fill of the machine so that only one percent of the jars will be unacceptable.

Solution: Let x be the amount of coffee in the jar. We are given $\sigma = 0.04$

We are asked to find the average fill, μ, such that Pr($x<6$)=.01

$$Px(x<6) = Pr\left(\frac{x-\mu}{\sigma} < \frac{6-\mu}{\sigma}\right) = Pr\left(z < \frac{6-\mu}{.04}\right) = .01$$

From the above table we find that:

Pr($z < -2.326$) = .01, therefore:

$$\frac{6-\mu}{.04} = -2.326 \qquad \mu = 6.093$$

If the average fill is set at 6.093 ounces, only one percent of the jars will contain less than six ounces.

REGRESSION ANALYSIS

Regression analysis is an advanced subject in the field of statistics. In this introductory "Statistics and Probability" chapter, only basic concept of regression analysis is covered. Readers are recommended to refer to other statistics textbooks and user manuals of regression analysis software packages for more details. The real benefit of performing regression analysis is in the ability of predicting or estimating a dependent variable from given independent variables using a probability model developed from a group of known sampled data. For example, experienced estimators might use a parametric estimating method during the early stage of project development to support screening and feasibility study. A parametric model (equation) could be developed to represent cost relationships that could provide a logical and predictable correlation between the physical or functional characteristics of a plant (or process system) and the installed cost. In this example, the installed cost, y, is a dependent variable. Its value depends on the independent variables x_1, x_2, …,x_k, etc (physical or functional characteristics).

Regression analysis is a tool to estimate the parameters of the adopted probability model. Several models could be developed from a given set of observed (historical) data. The analysis provides an approximation to illustrate how closely the model is able to predict or estimate the dependent variable, y, with respect to the true value based on given independent variables. Since the model is an approximation, the value predicted from the model would generally deviate from the true value by a random error. Errors tend to be either positive or negative values. The sum of all errors is zero. However, the sum of squared error (SSE) is a positive number providing a more meaningful indicator to determine how closely the proposed model is capable of predicting dependent variable value against the true value. In a regression analysis, the model that yields the minimum SSE is chosen as the best fit. This is known as the "Least Square Approach' to fit a model.

Linear Regression
A typical regression modeling uses a five-step procedure [2]:

Step 1 – Hypothesize the deterministic component of the probability model.
Step 2 – Use sample data to estimate unknown parameters in the model.
Step 3 – Specify the probability distribution of the random error term, and estimate any unknown parameters of this distribution.
Step 4 – Statistically check the usefulness of the model.
Step 5 – When the model is deemed satisfactory, use it for prediction or estimation.

The simplest regression analysis uses a linear (straight line) model to approximate (simulate) the relationship between two variables and the least squares method of fitting a linear regression model developed from the sample data. The resultant model is then used to predict the dependent variable, *y*, from a given independent variable *X*. The general form of probability models consists of:

$$y = \text{Deterministic component} + \text{Random error}$$

Equation 29.1

where y is the dependent variable to be predicted. The mean value of the random error is assumed to be zero. This is equivalent to assuming that the mean (or expected value) of *y*, E(y), equals the deterministic component of the model, i.e., E(y) = Deterministic component

For a linear probabilistic model the above general form (Equation 29.1) is expressed as:

$$y = \alpha + \beta x + \varepsilon$$

Equation 29.2

Where:

y = Dependent variable (variable to be modeled)
x = Independent variable (variable used as a predictor of y)
ε (epsilon) = Random error component due to the deviation from the true value
α = y-intercept of the line, i.e., point at which the line intercepts or cuts through the y- axis
β = Slope of the line, i.e., amount of increase (or decrease) in the deterministic component of y for every unit change in x.

Fitting this model (Equation 29.2) to a set of sample data is an example of linear regression analysis because of using a straight line to fit data points.

The deterministic component of Equation 29.2 is used as a prediction equation to obtain the predictive value of y, i.e., $\hat{y}_i$:

$$\hat{y}_i = \alpha + \beta x_i$$

Equation 29.3

For a given data point, such as the point (x_i, y_i), the observed value of y is y_i. $(y_i - \hat{y}_i)$ represents a deviation of the ith value of y from the predicted value. The sum of squares of the deviations of the y-value about the predicted values for all the n data points is:

$$SSE = \Sigma[(y_i - \hat{y}_i)^2]$$

Equation 29.4

Estimators of α and β.are used in Equation 29.2. Determining estimators of α and β can be found in any regression analysis textbook. The quantities α and β that make the SSE a minimum are called the least squares estimators of the population parameters α and β, and the predicted Equation 29.3 is called the least square line in the linear regression model.

Multiple Regression
In the real world, the dependent variable, y, is influenced by more than one variable and could also include a non-linear relationship. This requires modeling the expected value (mean) of y as a function of two or more independent variables leading to a multiple regression analysis. The five-step modeling procedure used in the linear modeling is also applicable to multiple regression. The linear probability model (Equation 29.2) is expanded to:

$$y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \ldots. + \beta_k x_k + \varepsilon$$

Equation 29.5

The deterministic component of Equation 29.5 is used as a prediction equation to obtain the predictive value of y, i.e., $\hat{y}_i$:

$$\hat{y}_i = \alpha + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + ... + \beta_k x_{ki}$$

Equation 29.6

A measure of how well a model fits a set of data is defined as multiple coefficient of determination, R^2:

$$R^2 = 1 - \Sigma(y - \hat{y})^2 / \Sigma(y - \bar{y})^2$$

Equation 29.7

where $\hat{y}$ is the predicted value of y from the model; $\bar{y}$ is the mean value of y.

R^2 represents the fraction of the sample variation of the y-values that is explained by the least squares of prediction Equation 29.6. Thus, $R^2 = 0$ implies a complete lack of fit of the model to the data and $R^2 = 1$ implies a perfect fit with the model passing through every data point. In general, the larger the value of R^2, the better the model fits the data.

Cost professionals may not use above regression analysis equations on the regular basis; however these equations are included as fundamentals to understand regression analysis software packages. Readers are further recommended to review the "parametric method" section of Chapter 9 (Cost Estimating, authored by Larry Dysert) to learn how regression analysis is applied in predicting the installed cost of a new induced draft cooling tower based on six recent similar installations.

CONCLUSION

This chapter covers some of the basic concepts of statistics and probability. It begins with the common understanding of two data types, namely; qualitative and quantitative data. An example of employee training hours is used to illustrate the concepts of frequency, relative frequency, and histogram. Numerical methods using mean, median, and mode are applied to determine data location (central tendency). Additional practical tools such as range, variance, and standard deviation are applied to measure data dispersion. The experiment of tossing two coins is used to introduce the concept of discrete random variable and probability distribution. Example D demonstrates the use of binomial distribution for quality control purpose in a fuse manufacturing company. After learning the well- known normal distribution, the standard normal distribution Z-score is presented for wider practical applications. The calculation of average coffee fill on a manufacturing process is an example of using the standard normal distribution Z-score table. Basic concepts of linear and multiple regression analyses are included as an introduction for further studies of their practical applications in cost estimating. A five-step regression modeling procedure is also presented to provide a fundamental knowledge useful for cost professionals.

Basic concepts of probability are used frequently by cost professionals. P80 or P50 reflects the chosen contingency amount matching the expectation of having 80 percent or 50 percent probability of not overrunning the approved budget. In the risk management plan, risk elements are identified by their potential impact and the probability of occurrence in the risk identification process. Before performing the Monte Carlo simulation, appropriate probability distribution functions must be determined. When proposing a GMP (Guaranteed Maximum Price) contract, a contractor must thoroughly evaluate the possible range of final project cost and the probability of occurrence while maintaining a competitive bid. Using parametric estimate method to generate early estimates in process industry requires knowledge of statistics. Understanding the fundamental concepts of probability and statistics will enhance the capability of cost professionals.

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RECOMMENDED READING

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Chapter 30—Optimization

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Abstract
Optimization is the process of determining the best performance for a system. The first step in the optimization process is the creation of a valid model of the system to be optimized. There are numerous types of cost engineering models and the object of the optimization process is to determine the value of the independent parameters for the best performance of the system. The optimization can focus on maximization or minimization. There are numerous techniques for optimization and the ones presented are Linear Programming, Monte Carlo Simulation, and Sensitivity Analysis. The techniques presented are illustrated with example problems using Microsoft Excel®

Keywords
Linear Programming, Model, Monte Carlo Simulation, Optimization, and Sensitivity Analysis.

INTRODUCTION
Optimization can be defined as the process of determining the best or most effective result, using a quantitative measurement system that uses variables such as currency, time, or quantity of materials. The measurement variable most frequently used in financial analysis, engineering economics, cost engineering or cost estimating tends to be currency, such as US Dollars, Euros, Yen, Won, Rupees, or Pounds Sterling. The optimization may occur in terms of net cash flows, profits, costs, benefit/cost ratio, etc. Other measurement units may be used, such as units of production or production time, and optimization may occur in manufacturing in terms of maximizing production units, minimizing cost, or minimizing production time or in construction in terms of minimizing cost and schedule.

There are a wide variety of optimization techniques such as linear programming, dynamic programming, geometric programming, queuing theory, statistical analysis, risk analysis, Monte Carlo simulation, artificial neural networks, sensitivity analysis, search techniques, and etc. The first step in optimization is the creating of a valid model for optimization. The techniques presented in this chapter for solving and evaluating models are linear programming, Monte Carlo simulation, and sensitivity analysis. Example problems are presented to illustrate the techniques, using the spreadsheet program Excel. It is interesting to note that both linear programming and Monte Carlo simulation were formalized in the 1940s from the intensive research efforts of complex systems during World War II.

Learning Objectives
After completing this chapter, the reader should be able to:

- Understand how to develop a valid model of the system to be optimized.
- Understand linear programming and solving a simple linear programming problem.
- Perform a simply Monte Carlo simulation.
- Apply sensitivity analysis to support decision-making.

Modeling
There are a wide variety of engineering models. Some of the various types are classified as static or dynamic, deterministic or probabilistic, iconic including scale models, experimental models and prototype models, analogue models, and symbolic models including theoretical and empirical models based upon mathematical relationships [3]. Linear programming and Monte Carlo models are of the symbolic type as they are based upon theoretical and/or empirical relationships and use mathematical representations of these relationships to find solutions for the models [3].

Models are simplified representations of reality by simple or complex systems used to optimize the performance of the real system. The models are used where it is impossible, impractical, or cost prohibitive to create experimental outcomes to directly measure the outcomes to optimize the performance of the actual complex system. The models may be small scale actual systems, such as a pilot plant, but most models are mathematical models of the system using empirical or theoretical relationships and these are the types which are emphasized in this chapter. The model is a representation of the desired system as completely accurate model representations are often impossible. The system being represented may be the operation of a blast furnace, the operation of an aircraft or automobile, the operation of an oil refinery, the construction of a structure, the cost estimate for a manufactured product, etc.

Mathematical relationships are developed to represent the behavior of the complex system. These relationships may be based upon the laws of physics, chemistry, economics, mathematics, biology, sociology, or other science in mathematical form. The equations may be from existing laws, either theoretical or empirical, which have existed or developed from the analysis of data from the performance of the system. Regression techniques have been frequently used to develop equations to predict the performance of system based on data obtained from the system being modeled or similar systems. Most of the regression techniques result in linear

relationships, but many of the complex systems have non-linear relationships. More advanced techniques, such as artificial neural networks, can also be used to develop complex models with non-linear relationships, but the relationships developed in neural networks are nearly impossible to explain in a logical manner. The models are calibrated with actual data to simulate performance and the accuracy of the model can be estimated by comparing the actual results and the model results. When the desired level of accuracy is obtained, then the model can be used to predict responses and to optimize the performance of the actual complex system.

Two major modeling techniques used are linear programming and Monte Carlo simulation. These are explored in the following sections. These techniques are used in a wide variety of situations; such as determining the charge/batch mixes for foundries, steel companies and food companies, determining the best location for distribution centers to minimize shipping costs, determining the best scheduling of activities for projects, the evaluation of contingencies and risk for projects, and predicting ranges for product costs. The performance of the model is frequently evaluated by sensitivity analysis to determine the importance of the control variables upon the system output. Sensitivity analysis will be discussed in the last section.

Linear Programming

Linear programming is an efficient mathematical method for determining an optimal strategy for optimizing a linear objective function subject to a set of linear constraints. Before the development of linear programming, linear equations were solved by the simultaneous linear equation method, but the possibility for solutions with negative values was a serious problem. It was the first method that could be applied to large, complex problems and was developed during the 1940s with the increased use of computers to solve mathematical problems. George B. Dantzig is frequently credited as being the "Father" of linear programming and 1947 is considered the year in which it started [1]. Linear Programming was first known as "programming in a linear structure" and the term "Linear Programming" was suggested by T.C. Koopman to George B. Dantzig in 1951 [1].

The goal of linear programming is to determine the level for all the activities, sometimes referred to as variables, of the system which:

- **Restrict the activities** to be non-negative (that is only positive or zero activities are permitted and this prevents the problem of negative resources which was obtained when simultaneous equations were used for solutions). Activities are frequently resources which are consumed to produce a product or achieve a schedule. The amount (in pounds) of coke to produce a ton of pig iron in a blast furnace would be considered as an activity.
- **Balance the constraint Equations** (if the constraint equations cannot be balanced, the result is an "infeasible" solution). The constraint equations represent the restrictions placed upon the resources such as a maximum amount of a resource available or a minimum amount to be in the product. For example, in the production of a ton of pig iron, the maximum amount of sulfur allowed in the product is one pound.
- **Optimize the Objective Function** (either minimize or maximize; such as minimize cost or maximize profit) to a single point. Only one objective will be considered. Typical objective functions are to minimize the cost or time to produce the product; or complete the project; or maximize the amount of production, the cash flows, or the profit of a project.

Mathematically, the linear programming problem can be formulated as:

Optimize (Maximize or Minimize) the objective function represented by Z where

$$Z = \sum_{i=1}^{n} C_i X_i \qquad i = 1, n$$

Equation 30.1

Where:

Z = objective function
C_i=cost of ith item
X_i=activity (amount) of ith item

Subject to the linear constraints:

$$\sum_{j=1}^{m} \sum_{i=1}^{n} A_{i,j} X_{i,j} = B_j \quad j=1, m \text{ and } i=1, n$$

Equation 30.2

Where:

$A_{i,j}$ = coefficient of ith item in j^{th} constraint
$X_{i,j}$ = activity (amount) of ith item in j^{th} constraint
B_j = constraint (resource) limit of jth constraint

The technique will be illustrated by a graphical example and then by using the Microsoft Excel program. The following example problem is about a housing developer determining which type of houses to build.

Example Problem: Developer Decision

The RCC Development Company wants to maximize its profits in a new housing development. The RCC Company has two types of homes, a three-bedroom model and a four-bedroom model. The Company has ten lots available in the new development, Mountaineer Estates, and the profit on a three-bedroom home is $30,000 and on the four-bedroom home is $40,000. The maximum funding from the bank for the project is $2,400,000; the cost to construct a three-bedroom home is $200,000; and the four-bedroom costs $300,000. How many homes of each type should RCC Development Company construct?

The problem can be formulated in thousand dollar units as:

Objective Function $30\ X_1 + 40\ X_2 = Z$ (maximum profit in thousand dollar units)

Constraints $X_1 + X_2 \leq 10$ (lots available for development)

$200\ X_1 + 300\ X_2 \leq 2{,}400$ (available funds from bank in thousand dollar units)

Where

X_1 = number of 3-bedroom homes to be built

X_2 = number of 4-bedroom homes to be built

Z = total profit from development project in thousand dollar units

The constraints are set as less than inequalities, as all of the lots do not need to be used nor do all of the funds need to be borrowed from the bank. The dollars are thousand dollar units to simplify the problem. The problem and the solution are shown graphically in Figure 30.1 The two constraints and the X1 and X2 axes form the boundary for the solution area and the objective function increases to a maximum and the vertex at X1 = 6 and X2 = 4. The Z lines indicate how the objective function is increasing and for this problem only integer values of X1 and X2 would be meaningful. The objective function increases from Z = 120 to Z=240, and finally to the maximum value of Z = 340. The houses to be built are six three-bedroom homes (X1) and four four-bedroom homes (X2).

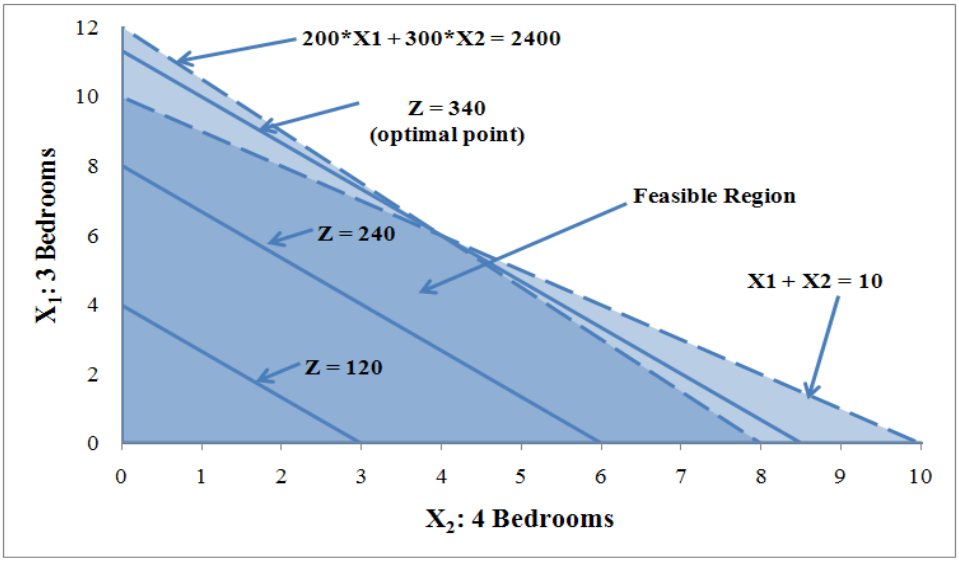


Figure 30.1—Graphical Solution of Developer Decision Problem

There are many software programs for solving linear programming problems and one should use the software one has confidence in. The Microsoft Excel software uses the add-in "Solver" to obtain a solution to the problem. This software is not very user friendly and better software packages exist, but it is readily available to many users; and with practice, one can set-up and solve simple problems with minimal difficulty. The data for this particular problem would appear as indicated in Figure 30.2. Entering the constraints is somewhat difficult, but instructions from an Excel© reference book or operations research textbook are very helpful [4, 6].

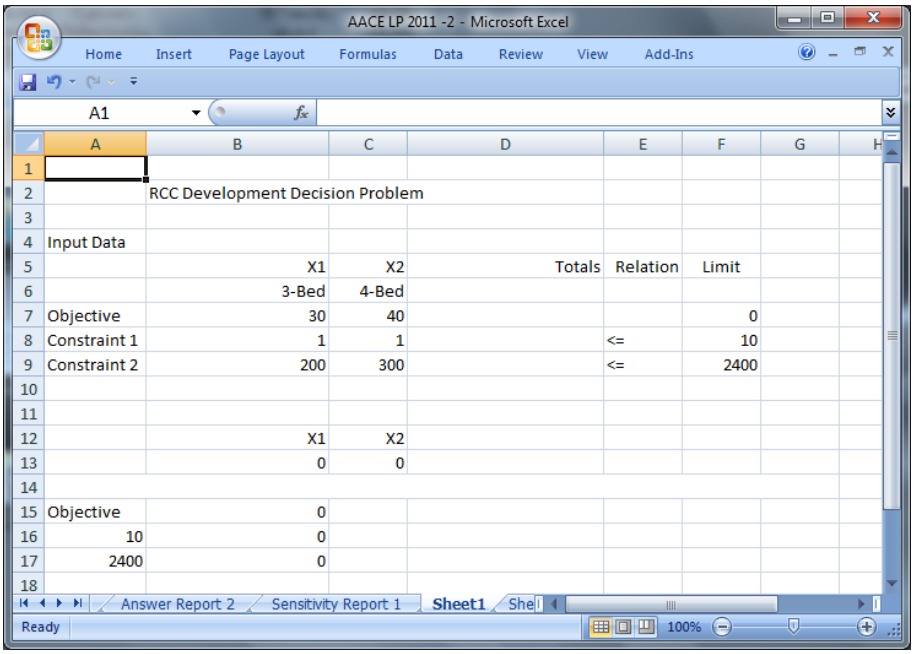


Figure 30.2—Input to Microsoft Excel© Solver Program for RCC Development Decision Model

The model requires that the initial values be selected as zero, and thus the first screen to the model will appear as indicated in Figure 30.2. Figure 30.3 is a screen shot of the solver input variables based upon the input of Figure 30.2. The solution to the program is shown in Figure 30.4 and the solution values are beneath the X_1 and X_2 values, where $X_1 = 6$ and $X_2 = 4$. Column D indicates that all the resources were used; that is all ten lots were used and the borrowing limit of $2,400,000 (2,400 thousand dollar units) was reached. The profit was $340,000 (340 in Figure 30.4 and in Figure 30.5 as units are thousand dollar units for profit). Figure 30.5 is the Solver Answer Report for the Development Decision Model. The initial values in Figure 30.5 are the initial solution values obtained by the model.

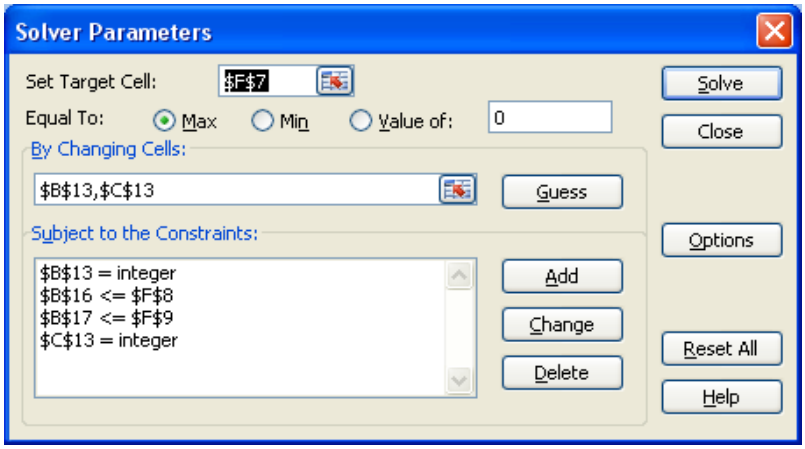


Figure 30.3—Excel© Screen Shot of Solver Input Parameters for Linear Programming Problem

The equations for the excel program indicated in Figure 30.3 for objective function and the two constraints in cells $F$7 and the constraints $B$16, and $B$17 were:

$F$7 = $B$7*$B$13+$C$7*$C$13
B 16 = $B$8*$B$13+$C$8*$C$13
$B$17 = $B$9*$B$13+$C$9*$C$13

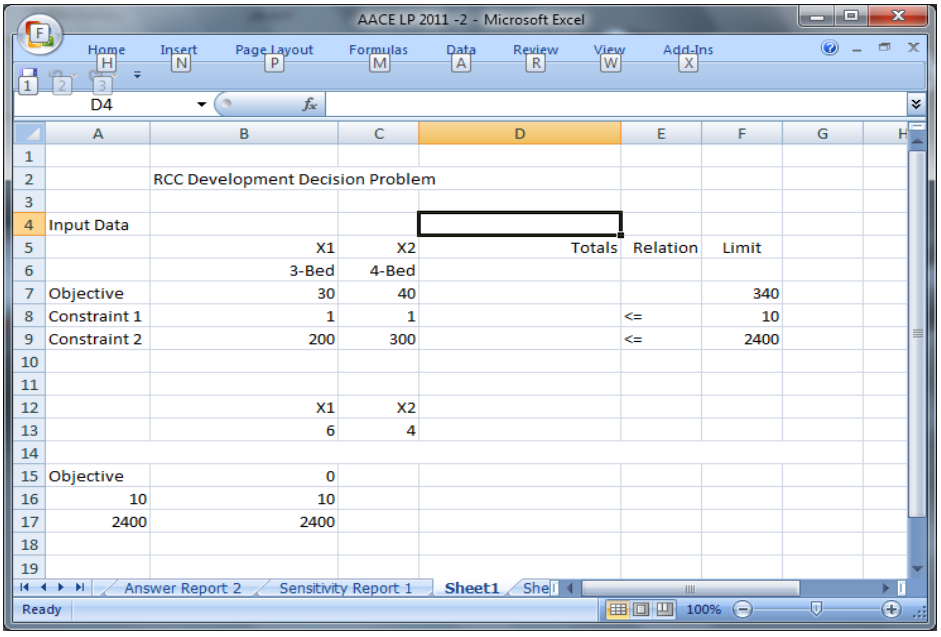


Figure 30.4—Output of Microsoft Excel© Solver Program for RCC Development Decision Model

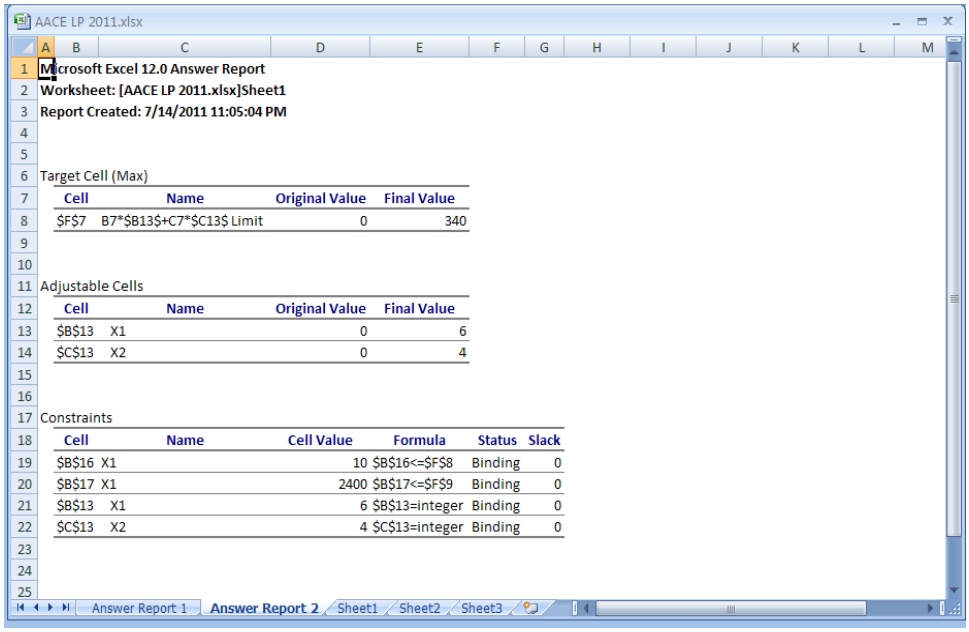


Figure 30.5—Screen Shot of Excel© Solver Answer Report for Linear Programming

Problem—This was an unusual problem in that the answers were integers for the houses even though integers were not forced. However, for the solution shown in Figure 30.4, the variables were made integers in Figure 30.3 to illustrate how to obtain integer solutions, as integer values are often required since one would not desire an unfinished house. A sensitivity analysis is presented for this problem in the last section of this chapter.

Monte Carlo Simulation

Monte Carlo Simulation is a technique which uses the generation of a series of random numbers to simulate a population followed by statistical analysis to make predictions [2,6,7]. Stanislaw Ulam is often given credit for inventing the Monte Carlo Method during the early 1940s [7]. Nicholas Metropolis is given credit for naming this new methodology after the casinos in Monte Carlo [7]. The use of statistical sampling for solving quantitative problems had been used before, but Ulam, Metropolis, and John von Neuman developed computer algorithms for solving problems and started the development of a formal methodology for statistical sampling and simulation.

Monte Carlo Simulation is a sampling method to simulate the process of sampling from the actual population being studied to initiate real life or make a prediction. The model uses inputs randomly generated from probability distributions to simulate the process of sampling the actual distribution. Monte Carlo Simulation methods are very useful in studying complex systems with significant uncertainty in the inputs, such as in estimating costs. It is used in investment analysis to evaluate the risk in making investments.

There are numerous commercial computer software programs available to perform Monte Carlo Simulations such Crystal Ball, @Risk, Risk Solver, Arena [2], and GoldSim. Microsoft Excel can be used to develop Monte Carlo Simulations, but the commercial software programs are more commonly used for complex problems as they have been designed with an emphasis for simulation.

To illustrate how the simulation works, a simple example will be presented to estimate the cost of a construction project. The first phase of the project is to obtain the necessary permits and it is expected with a 30 percent chance to cost $100,000, (under favorable conditions with a probability of 0.30); or a 70 percent chance to cost $250,000, (under unfavorable conditions with a probability of 0.70). The second phase of the project is the construction, and if the weather is good (60 percent chance); the estimate is $2,000,000, (under favorable conditions with a probability of 0.60); and if the weather is poor (40 percent chance) the estimate is $2,500,000, (under unfavorable conditions with a probability of 0.40). The third phase is that the economy is entering uncertain times and inflation may occur. There is a 0.20 probability that inflation will not occur (20 percent chance), a 0.50 probability that inflation will increase (50 percent chance) the cost by $200,000; and a 0.30 probability that inflation will increase (30 percent chance), the cost by $300,000. Monte Carlo Simulation will be used to estimate the cost of the project. Random numbers are selected to represent the probabilities of the three phases to estimate the causes given in Table 30.1.

Table 30.1—Random Number Assignment for Three Phases

Phase 1		Phase 2		Phase 3	
Permit Conditions	**Random #s**	**Weather**	**Random #s**	**Inflation**	**Random #s**
Favorable	00-29	Good	00-59	No Inflation	00-19
Unfavorable	30-99	Poor	60-99	$200,000	20-69
				$300,000	70-99

A sample simulation for ten estimates will be made. Random numbers are from the tables of Fisher and Yates as published by Olsen [5]. Random numbers can also be obtained by using the Excel program. Random numbers are one of the keys to the success of the simulation and the values obtained are presented in Table 30.2.

Table 30.2—Random Numbers for the Three Phases and the Total Cost

Phase 1		Phase 2		Phase 3		Total
RN	**Cost**	**RN**	**Cost**	**RN**	**Cost**	**Cost**
57	$250,000	41	$2,000,000	34	$200,000	$2,450,000
35	$250,000	10	$2,000,000	72	$300,000	$2,550,000
27	$100,000	76	$2,500,000	57	$200,000	$2,800,000
33	$250,000	47	$2,000,000	59	$200,000	$2,450,000
72	$250,000	91	$2,500,000	13	$0	$2,750,000
24	$100,000	44	$2,000,000	82	$300,000	$2,400,000
63	$250,000	04	$2,000,000	43	$200,000	$2,450,000
94	$250,000	95	$2,500,000	80	$300,000	$3,050,000
09	$100,000	49	$2,000,000	46	$200,000	$2,300,000
53	$250,000	66	$2,500,000	15	$0	$2,750,000

The average of the ten simulated values gives an estimate for the total cost of $2,595,000. This expected value as given by:

Expected Value = 0.3*100,000 + 0.7*250,000 + 0.6*2,000,000 + 0.4*2,500,000 + 0.2*0 + 0.5*200,000 + 0.3*300,000
= $2,595,000

In this case, the simulated average and expected values were equal and this is unusual for such a few number of simulation runs. Usually, the simulation must be run hundreds or thousands of times to have the average become close to the expected value in complex simulations. The standard deviation is:

$232,678 from this set of data, and the one tail 80 percent confidence limit (Z = 0.84) would be:
X = 2,595,000 + 0.84*232,678 = $2,790,450.

Thus, there is a 20 percent risk that the project will cost more than $ 2,790,450, and in the simulation of ten runs; two or 20 percent were above the $2,790,450. The simulation would need to be repeated many more times to have a better estimate of the standard

deviation. The expected value, or mean, of the simulation value should approach the actual distribution mean, and it is unusual that it occurs in such as few number of simulation runs. The typical number of simulation runs is over 1,000 to obtain results with highly reliable estimates in complex systems, but since this system is very simple only a few runs (at least 100) were necessary to obtain a good estimate of the mean.

Table 30.3 contains the results for a simulation model run on Excel for 30 simulations. The variation in the three components is large, but the variation in the sum of the components is smaller. However, the simulation indicates there is considerable variability in the totals, and although the average of the runs (2,636,667) approaches the expected value (2,595,000) the variation in the data ranges from the low of 2,100,000 to the maximum of 3,050,000 in the simulation of 30 runs. The wide variation is a result of the large variation in the three individual components and this indicates that this would be a rather risky project.

One problem with simulation models is that when one runs the simulation with random numbers, the solutions change each time a simulation is run, as the new random numbers will typically result in a new solution. This can be prevented by using the same random numbers and can be done to make certain that the model is operating in the manner designed. However, the major purpose of simulation model is not only to indicate a solution, but also to indicate the variability in the solution. Projects with high variability typically are risky projects.

Sensitivity Analysis

Sensitivity analysis is a technique used to quantify the variation in the solution with respect to the variables and the constants (including any coefficients) used in the formulation of the problem. Sensitivity is one of the components calculated in the risk analysis of projects. Sensitivity analysis is done to investigate the impact of changes in variables and assumptions made for constants used upon the solution obtained, often referred to as the base case. The purpose is to determine which variables and constants have the most influence upon the solution. Most investigations involve sensitivity of the variables as the constants should be constant, but in model simplification some variables may have been assumed as constants and may not truly be constants. The sensitivity can be considered as the change in the objective function of the linear program with respect to incremental changes in the variable under consideration.

The solution to the original LP problem was $X_1 = 6$, $X_2 = 4$, and the objective function was $340,000, when the limit on the number of houses was ten and the limit on the bank funding was $2,400,000. When the limit on the number of houses was increased to 11, while the limit on bank funding remained at $2,400,000, the solution increased the number of three-bed-room houses (X_1) increased to X_1=9, the number of four-bedroom houses (X_2) decreased to X_2=2, and the objective function increased to $350,000; which is an increase of $10,000 over the base case. The mix of the types of houses changed drastically with X_1 increasing from six to nine and X_2 decreasing from four to two. The overall sensitivity would be that for an increase of one unit in the number of houses (a ten percent increase), the increase in the profits was $10,000 (an increase of about three percent). If on the other hand, the limit on the number of houses remained at ten and the amount of bank funding increased by $100,000, the profits increased to $350,000; which is the same amount as the increase in one house. Thus, the profits increased by $10,000 for an increase of funding of $100,000; or, the profits increased by approximately three percent when the funding increased by 4.2 percent. The mix of houses changed from $X_1 = 6$ to X_1=5 and from X_2=4 to $X_2 = 5$, that is the regular houses decreased by one and the expensive houses increased by one.

Permit Conditions			Weather Conditions			Inflation				
Condition	Amount	Probability	Condition	Amount	Probability	Condition	Amount	Probability		
Favorable	100,000	0.3	Good	2,000,000	0.6	None	0	0.2		
Unfavorable	250,000	0.7	Poor	2,500,000	0.4	Low	200,000	0.5		
						High	300,000	0.3		
	lookup2=b21:c22			lookup3=e21:f22			lookup4=h20:i22			
							data4=h25:h74			
	Mean	2595000								
							Cutoffs	Demand		
	Cutoffs	Demand		Cutoffs	Demand		0	0		
	0	100000		0	2000000		0.2	200000		
	0.3	250000		0.6	2500000		0.7	30000		
Trial	demand	rand		amt	rand		amt	rand	Total	
1	100000	0.14140655		2500000	0.7593221		200000	0.5850894	2800000	
2	250000	0.66046088		2500000	0.8236304		200000	0.3791443	2950000	
3	250000	0.50051557		2500000	0.8297739		300000	0.7817911	3050000	
4	100000	0.16405752		2000000	0.1534758		0	0.185037	2100000	
5	100000	0.20481405		2500000	0.8943791		300000	0.97822	2900000	
6	250000	0.38214979		2000000	0.2033503		200000	0.5345992	2450000	
7	250000	0.7100959		2000000	0.3330346		200000	0.4676282	2450000	
8	100000	0.27953081		2000000	0.322062		200000	0.444297	2300000	
9	250000	0.75650669		2000000	0.241411		300000	0.7556957	2550000	
10	250000	0.79563426		2500000	0.9680335		200000	0.4256442	2950000	
11	250000	0.78769973		2000000	0.0467364		300000	0.9703615	2550000	
12	250000	0.95126277		2500000	0.971632		200000	0.5153695	2950000	
13	250000	0.40015555		2000000	0.1135574		300000	0.7651685	2550000	
14	100000	0.29388283		2000000	0.3107765		200000	0.4233019	2300000	
15	250000	0.55998712		2000000	0.3509953		0	0.1975502	2250000	
16	250000	0.98080718		2500000	0.6108644		200000	0.651959	2950000	
17	250000	0.35122691		2000000	0.2791357		200000	0.5522567	2450000	
18	100000	0.0250237		2500000	0.8547806		200000	0.5274861	2800000	
19	250000	0.85884208		2000000	0.5315708		200000	0.4505917	2450000	
20	100000	0.22391566		2000000	0.5984562		0	0.1075174	2100000	
21	250000	0.61676004		2000000	0.3823608		0	0.0189501	2250000	
22	250000	0.84917527		2000000	0.0173575		0	0.0660999	2250000	
23	250000	0.34461364		2000000	0.4317812		200000	0.2717682	2450000	
24	250000	0.72704419		2500000	0.8718396		300000	0.9940627	3050000	
25	250000	0.34050043		2500000	0.817564		0	0.0029333	2750000	
26	250000	0.39753482		2500000	0.7124942		0	0.1802709	2750000	
27	250000	0.89255024		2500000	0.7093213		200000	0.4976182	2950000	
28	250000	0.89345291		2500000	0.8965713		200000	0.3534848	2950000	
29	250000	0.36583252		2500000	0.7118432		200000	0.3296438	2950000	
30	100000	0.12493935		2500000	0.7880307		300000	0.7724249	2950000	
	Average	Expected Value		Average	Expected Value		Average	Expected Value	Average	Expected Value
	210000	2050000		2250000	2200000		176666.667	190000	2636666.67	2595000
Fraction of time		Simulation	Actual		Simulation	Actual		Simulation	Actual	
	100000	0.26666667	0.3	2000000	0.5	0.6	0	0.2333333	0.2	
	250000	0.73333333	0.7	2500000	0.5	0.4	200000	0.5333333	0.5	
							300000	0.2333333	0.3	

Table 30.3—Simulation Example with 30 Runs

If one decreased the number of houses from ten to nine, while the bank funding remained at $2,400,000, the objective function was reduced to $330,000, a decrease of $10,000. The mix of houses changed from $X_1 = 6$ to $X_1 = 3$ and X_2 from four to six; therefore, when the number of houses decreased by one unit, the profit decreased by $10,000. This is the same amount by which the profit increased when the number of houses increased by one, so the sensitivity of profit to the number of houses is $10,000 per unit house for one house.

Similarly, if the bank funding is reduced by $100,000 to $2,300,000 (2,300) and the number of houses to be built remains at ten, the objective function is reduced to $330,000, the profit decreased by $10,000. The mix of houses changed from $X_1 = 6$ to $X_1=7$ and X_2 from four to three. Thus, when the bank funding decreased by $100,000, the profit decreased by $10,000. The overall sensitivity would be that for a decrease of $100,000 in the bank funding (that is ten $1,000 units or a decrease of about four percent), the decrease in the profits was $10,000 (a decrease of about three percent). The increase in profits was the same ratio for the increase in bank funding; that is a $100,000 increase in bank funding led to an increase of profits of $10,000.

The sensitivity results of the Excel Solver Program are given in Figure 30.6 and are the same as a change in constraint one having a Lagrange Multiplier of ten, which implies that the objective function will change by ten (that is $10,000 per unit since they are thousand dollar units) for each unit of change in the number of houses. Similarly, for a unit change in the bank funding of $100,000,

or 100 of the $1,000 units, the objective function changes in the Lagrange Multiplier of 0.1 which is 0.1 times $100,000 or $10,000 (or 0.1 x 100 = 10 of the $1,000 units or $10,000). The sensitivity analysis requires that the variables must not be specified as integers and the integer constraints were removed to obtain the sensitivity report. For this particular problem, the solution did not change.

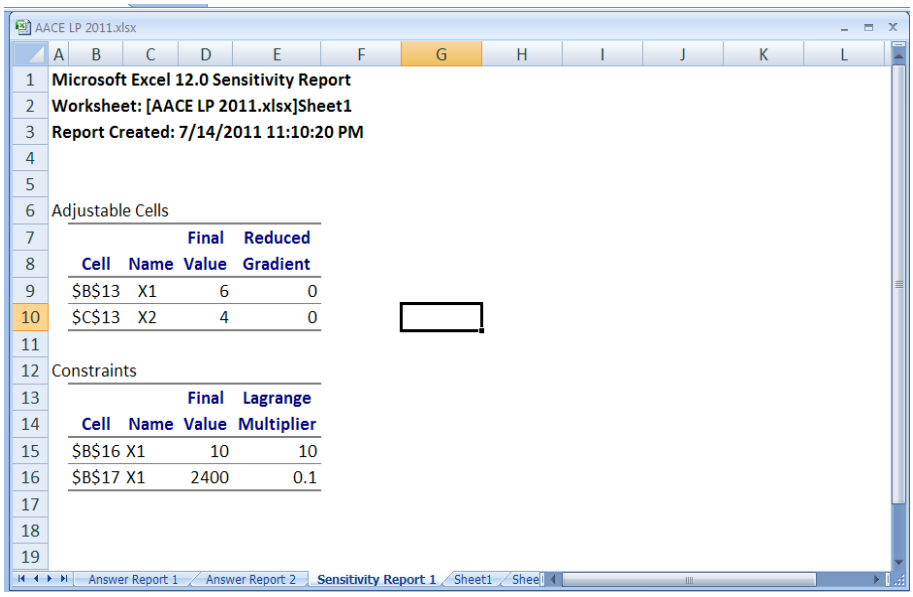


Figure 30.6—Screen Shot of Sensitivity Results from Excel© Solver

The examples are rather simplistic, but they give an indication of sensitivity. The sensitivity presented is with respect to the objective function, and not to the specific type of houses being built. As the amount of bank funding increased, the number of expensive houses increased and the number of regular houses decreased. On the other hand, as the amount of bank funding was constant and the number of houses increased from ten to 11, the number of three-bedroom houses increased by three and the number of four-bedroom houses decreased by two for each unit increase in the number of houses. The funding for three of the three-bedroom houses is the same as the funding for two of the four-bedroom houses, but the profit from three-bedroom houses was $90,000, and the loss in profits for two four-bedroom houses was $80,000; thus, a net gain of $10,000 occurred.

An example of a non-linear change in sensitivity is in the evaluation of the Present Value of a project. If an investment of $10,000 is made and the investment returns are $3,000 per year for five years, what is the Present Value of the project considering required returns of six, eight, and ten percent? The Present Values are:

PV(i%)	*= -$10,000 + $3,000(P/A,i,n=5)*	*= -$10,000 + $3,000 *([(1+i)^n – 1] / [i(1+i)^n])*
PV(6%)	= -$ 10,000 + $ 3,000(4.212)	= + $2,636
PV(8%)	= -$ 10,000 + $ 3,000(3.993)	= + $1,979
PV(10%)	= -$ 10,000 + $ 3,000(3.791)	= + $1,373

The increment between eight and six percent is $657 and the increment between ten and eight percent is $606, or a difference of 7.8 percent. Thus, the sensitivity is changing non-linearly as the return rate is changing by a constant amount (two percent).

Sensitivity analysis is important as one must know which variables have the greatest impact upon the system output. The variables with the greatest sensitivity should be monitored more carefully than variables which have low sensitivity. Variables with large ranges and high sensitivity have the potential of high impact upon the system performance and should be monitored carefully to optimize system performance.

CONCLUSION

Optimization is an exciting area which consists of a variety of techniques to improve performance by improving the utilization of resources. Three of the most commonly used techniques—linear programming, Monte Carlo Simulation, and sensitivity analysis—have been presented and illustrated with examples. Optimization techniques are used by companies, both large and small, to improve their performance to be competitive in the global market place.

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Chapter 31—Risk Management Fundamentals

Allen C. Hamilton, CCP

Abstract
Risk Management Fundamentals is a basic overview of key elements. It is intended to give the reader definitions and risk management reference materials for further study. The benefits and fundamental approach will be covered. The key steps of risk management will be briefly covered including planning, assessment, treatment, and control. Risk management techniques will be covered including quantitative risk analyses—simulation, sensitive analysis, and decision trees. Finally, risk management software programs will be discussed.

Keywords
Contingency, Risk, Risk Analysis, and Risk Management.

INTRODUCTION
The purpose of this chapter is to give a general overview of risk management and examples of how to apply the tools of risk management.

Learning Objectives
After completing this chapter, the reader should be able to:

- Understand risk management.
- Identify techniques for identifying risk, assessing risk occurrence and impact, and analyzing risk.
- Understand risk treatment.
- Perform quantitative risk analyses (i.e. simulation, sensitivity analysis, and decision trees).
- Understand risk analysis software.
- Describe the use of contingency.

Skills and Knowledge of Cost Engineering and Total Cost Management Alignment
The subject of risk management is covered in the respective sections of two key AACE International publications: *Skills and Knowledge of Cost Engineering* and the *Total Cost Management (TCM) Framework*. This chapter embraces both publications to provide a consistent fundamental knowledge of risk management.

Skills and Knowledge of Cost Engineering
- Supporting Skills and Knowledge
- Process and Functional Skills and Knowledge
    - Planning
    - Risk Management

Total Cost Management Framework [4]
Total Cost Management (TCM) is the effective application of professional and technical expertise to plan and control resources, costs, profitability, and risk. Simply stated, TCM is a systematic approach to managing cost throughout the life cycle of any enterprise, program, facility, project, product or service. The *TCM Framework* is a representation of that "systematic approach".
- Introduction to TCM
- Strategic Asset Management Process
- Project Control Process
    - Chapter 7—Project Control Planning
    - Sub chapter 7.6—Risk Management

The development of the *Skills and Knowledge of Cost Engineering* is a product of the AACE Education Board and its development is guided by and coordinated with the Technical Board and the *TCM Framework*. In addition, this chapter was also coordinated with the Decision and Risk Management Professional (DRMP) certification committee.

Definitions
There are a variety of definitions associated with risk and risk management. The following samples are selected from dictionaries and professional organizations:

Dictionary
Risk—the possibility of suffering harm or loss (American Heritage Dictionary, Houghton Mifflin Co.)

Risk—uncertainty of an event which if occurred would result in a negative or positive effect on the project (Project Management Institute) [6]

AACE Recommended Practice 10S-90, *Cost Engineering Terminology*, 2009 [1]
Risk—(1) An ambiguous term that can mean any of the following: a) All uncertainty (threats + opportunities); or b) Undesirable outcomes (uncertainty = risks + opportunities); or c) The net impact or effect of uncertainty (threats – opportunities). The convention used should be clearly stated to avoid misunderstanding. (2) Probability of an undesirable outcome. (3) In total cost management, an uncertain event or condition that could affect a project objective or business goal.

Professional Practice Guide—Risk
Risk, 2nd Edition, AACE International, Professional Practice Guide [3]
AACE published the second edition to the Risk Professional Practice Guide in 2007. The guide contains about 300 references, including Cost Engineering articles and papers presented at AACE Annual Meetings.

TCM 7.6.1 Risk Management Description, 2011 [4]
Risk: An uncertain event or condition that could affect a project objective or business goal.

Risk management is the process of identifying risk factors (risk assessment), analyzing and quantifying the properties of those factors (risk analysis), treating the impact of the factors on planned asset or project performance and developing a risk management plan (risk treatment), and implementing the risk management plan (risk control) [4]. The goal of risk management is to increase the probability that a planned asset or project outcome will occur without decreasing the value of the asset or project. Risk management presumes that deviations from plans may result in unintended results (positive or negative) that should be identified and managed.

Project Management Institute, Guide to the Project Management BOK, PMBOK 2000 [6]
Risk Management is the systematic process of identifying, analyzing, and responding to project risk.

The industry has a wide variety of definitions for risk. There is also a difference in the terms applied to the steps in the management of risk. The above group of definitions attempts to provide some guidance of these definitions. In general we follow those definitions and risk management steps provided in two references the Total Cost Management Framework [2] and AACE Recommended Practice 10S-90, *Cost Engineering Terminology* [1].

Why Understand Risk Management?
Risk management is an important tool in the project management tool kit. Risk is an important topic in project management and there are a large number of resources and many papers in the academic and popular literature. Many projects deal with uncertainty and the results of many different aspects of project performance and delivery. For example, is there anyone who can say for certain what the project cost will be and when it will be completed? Risk management allows us to plan for project risks and how to manage them.

In June 1815, Napoleon met Wellington and Blucher at the Battle of Waterloo. Napoleon was defeated and it brought to an end of his military and political career. He was exiled to St. Helena and died in 1821. At Waterloo it has been calculated Napoleon had an 18 percent chance of winning based on a statistical calculation of leadership, morale, troop size, intelligence, and technology between the two opposing armies [7]. Would you engage in an undertaking that had an 18 percent chance of success? Perhaps you would also ask, "If successful, what would be the benefit?"

Benefits of Risk Management
All organizations need to deal with risk. There are many examples of failed companies that were unable to deal with risk. The bankruptcy of Enron Inc. in 2001 is an example. Enron's participation in and use of risky (illegal) accounting procedures is one of the primary causes of their bankruptcy. Arthur Anderson, their accounting firm, failed to recognize the risk and consequences of these procedures on both Enron and themselves. The bankruptcy of Lehman Brothers in 2008 is also an example of a company that failed to deal properly with risk. Lehman borrowed significant amounts of money, was highly leveraged in housing-related assets and vulnerable to market changes [5].

One of the greatest risks to projects is the lack of adequate and appropriate planning. An important part of project planning is risk management. Risk management in all its forms, both simple and complex, will potentially have a positive effect on the implementation of projects. Risk management will also assist projects where the risk was not identified or anticipated. By using risk management, projects will develop techniques and methods to deal with both anticipated and unanticipated events.

In his book, Against the Gods, Peter Bernstein describes the history and development of risk management and says in the introduction, "By showing the world how to understand risk, measure it, and weigh its consequences, they converted risk-taking into

one of the prime catalysts that drives modern Western society" [2]. Much of his book is devoted to the history and development of risk as it related to society, macroeconomic events, and investments. However, the application of the theories is just as relevant to a project as to a society.

Fundamental Approach to Risk Management
When we perform risk management we should keep in mind the following:

- You do not need an advanced degree in mathematical theory to perform risk management.
- Risk management can be applied in different ways suitable to project needs from easy/simple to complex.
- One of the more important elements of risk management is to establish a process of dealing with risks.

The approach to risk management is similar to the classic approach to management and project management.

Plan → Implement → Monitor → Control

Risk management usually follows a similar sequence. Risk Management is a systematic and iterative process comprising four steps:

1. Plan—to establish Risk Management objectives.
2. Assess—to identify and analyze risk.
3. Treat—by planning and implementing risk responses.
4. Control—by monitoring, communicating and enhancing Risk Management effectiveness.

Ref: TCM 7.6.1 [4]



Figure 31.1—Risk Management Steps

A BIT OF HISTORY

Counting, Gambling, Insurance and Investments
Risk management, as we know it, has developed as a result of an outgrowth of the study of mathematics, specifically statistical analysis. The development of the knowledge and science of risk management was used to satisfy the needs of counting, gambling, insuring and investing. It all began with the adoption of the Hindu-Arabic numbering system in place of the Roman system. In the 9th century, we had the rules for adding, subtracting, multiplying, and dividing; and the 15th/16th centuries brought us algebra, statistics, and probabilities. With probabilities, we had the beginnings of risk management using probabilistic analysis.

During World War II, as Moscow was being bombed, a well-known Soviet professor of statistics declined the use of the local air raid shelter and said, "There are seven million people in Moscow. Why would I expect them to hit me?" One night he showed up at the shelter and a woman asked him why he was there. He said, "There are seven million people in Moscow and one elephant (in the zoo). Last night they got the elephant" [2].

Much of the historical progress of risk management has been helped by the desire to quantify the risks of investments. Today there is a lot of information available on the degree to which stocks, bonds, and mutual funds move and behave. Alpha, Beta, and R

squared are commonly used to analyze the behavior of investments. Even Shakespeare in the *Merchant of Venice* had something to say about investments:

> My ventures are not in one bottom trusted,
> Nor to one place; nor is my whole estate
> Upon the fortune of this present year;
> Therefore, my merchandise make me not sad.
>
> (Antonio, Act I, Scene 1)

Finally, in the 18th/19th century, we saw the development of averages, means, normal distributions, correlation, and the symmetrical distribution around the mean. All of which leads us to the beginning of the 20th century and the essence of risk management which is to:

- Maximize areas where we have some control.
- Minimize areas where we have no control.

RISK MANAGEMENT STEPS

Risk Planning
Prior to beginning risk management, it is appropriate to review the risk management plan as outlined in the project plan, project execution plan, or project coordination procedure. If nothing has been done in this regard, it should be started. Risk planning in its broadest scope should establish the approach, form, content, and results of risk management. Depending on the project, this can range from one to dozens of pages of content with any number of attachments for forms and checklists. It is important to establish what is to be covered by risk management and what is not going to be covered. A mission statement and scope of work description are essential in this regard. The scope of risk management could include the intention to include external, internal, strategic, and tactical risks. Many risk management efforts focus on risks which the project management team can influence; these are commonly called internal risks.

Risk planning should also establish criteria for the identification of risk items, assessment criteria scales for occurrence and impact, analysis approaches, and general treatment strategies. In addition, responsibilities of risk treatment and follow-up should be identified.

Risk Assessment
Risk assessment is the next major step in risk management. The objective of risk assessment is to establish the relative importance and impact of the risk item. Assessment allows the team to establish a prioritization of the risk items. Each risk item is assessed a probability of occurrence and a potential impact. The assessment scale that has shown to work best is a high/medium/low scale. Other more refined scales, such as one to five or one to ten, can be used. The author uses the high/medium/low because individuals and teams can come to a quicker decision on assessment with a more compact scale.

Risk Identification
Risk identification is the most common step in risk management. Everyone touches on risk issues every day. Think about it. Driving, writing that critical memo, and the choice of answers to an important client. Many of our actions are weighed with the calculation of which choice to make based on the most positive outcome. In risk management on projects this is reduced to a more objective level. Risks on projects are the identification of activities which can have a negative or positive impact on outcomes or performance. The new aspect of the view of risk management is to widen the previous approach to risk management and include both negative (risks) and positive items (opportunities).

The first step in risk identification is to put together a list of project risks. There are many techniques to putting together a risk identification list:

- Authored by one person.
- Authored by a team with surveys and interviews.
- Circulate a chart where risk items can be added by each person.
- Free form list put together by a group at a meeting, e.g., brainstorming session.
- List developed by a group facilitated by a third party.

The author's experience is that the use of a small focused group is a productive approach to assembling a risk identification list. The interaction among and between members of the group has been shown to be very productive. Discussion about risk items leads to identifying new items, which may not have been developed in an isolated setting. In risk identification, it is better to be inclusive and ensure capturing more risks, rather than trying to manage the risk item; editing at too early a stage. Creativity is also helpful in ensuring a more complete identification of risks. The inclusion of people outside the project may also be helpful in facilitating some

fresh views of the issues. The editing of the list happens later, rather than during identification. Identification items may be classified into groups depending on how the project is set up. It is desirable to have items which would be clearly identified as the responsibility of an organizational group or individual.

Elements of risk identification can include the following:

- Statement of the problem, issue, or risk
- Short description of the risk
- Identification of the fundamental risk
- Provide a basis for categorizing the risk

Internal and External Risks

There are two general classifications of risk, external and internal. The classifications differentiate between the ability of the company or project team to control the risk. External risks are risks the company or project team cannot control. Examples of external risks include "Force Majeure"; risks such as hurricanes, price changes due to market forces, and major labor strikes. The company or project cannot control the occurrence of external risk but can mitigate the impact if the risk occurs. Internal risks are risks the company or project team *can* control. Examples of internal risks are ineffectual contractor as a result of a flawed selection process, design errors as a result of using untrained personnel, and cost overruns as a result of excessive project changes. The company or project can control the occurrence of internal risks and can mitigate the impact if the risk occurs. See Table 31.4 for a list of external and internal risk categories and risk items.

Table 31.1—Risk Identification and Assessment Chart [8]

Risk Identification and Assessment		
Describe Risk	Probability of Occurrence (High, Medium, Low)	Potential Impact (High, Medium, Low)

We now have the risk items in groups, if appropriate, and the assessments would be made from the list. The probability of occurrence is usually straight forward. But to provide some criteria on occurrence:

- **High**—very likely the risk item will occur e.g. >75 percent
- **Medium**—the risk item may or may not occur e.g. 25-75 percent
- **Low**—unlikely the risk item will occur e.g. <25 percent

The percentages are only illustrative of an approach. The occurrence assessment is almost always subjective and cannot usually be calculated. However, the percentages indicated would serve to distribute items across all three categories. Establishing the occurrence criteria is up to the team and may have been outlined in the risk planning documents. Assessing the probability of occurrence is best done in a group where the assessments can be discussed and challenged.

The next assessment is potential impact. Impact usually means financial impact although other impacts can be used such as days of delay or other important project parameter. The criteria for high, medium, and/or low should be established rather than gauging for each item. The impact assessment may either be subjective or calculated. Most teams provide a group subjective impact assessment and if necessary will later do further investigation or calculation to refine the initial assessment. The risk plan should identify the impact criteria. Potential criteria for impact:

- **High**—greater than five percent of project cost
- **Medium**—from 0.5 to five percent of project cost
- **Low**—up to 0.5 percent of project cost

The cost impacts above are examples and the project team will need to establish the cost criteria most useful to the project. For example, the high threshold of five percent was chosen because many projects will have to be reviewed or re-appropriated when the cost of the project increases ten percent. A single impact of five percent would be a significant cost increase and the potential of re-appropriation would be likely. Projects will need to establish the threshold at which impacts are judged significant. In a very cost sensitive environment, the thresholds would be made lower.

RISK TREATMENT

Analysis

The first part of analysis is the review of the risk items. The review confirms the occurrence and impact assessments where completed and were consistent with the established criteria for each. The items can now be grouped if they were not already done so. There may be reasons to consider the combining or splitting of risk items. Responsibility and control of the risk treatment are

good reasons to consider changing items. Analysis is a good time to refine the cost impact assessments on items where the scope is unclear. Investigation of cost assessments may be helpful later.

The main objective of analysis is the selection of risk items for treatment. Many techniques are available for this purpose. It is possible to make a selection from the list and proceed. It is also possible to perform a calculation to give each risk item a score. However, the most common method and one that is easy to understand and visualize is the occurrence/impact matrix. The matrix uses the high/medium/low assessments for each risk item to establish the criteria for treatment. The matrix is usually used as follows:

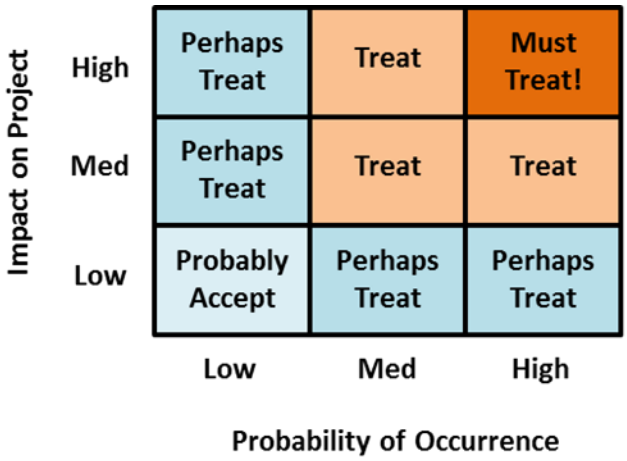


Figure 31.2—Risk Treatment Chart

The risk planning document will determine the criteria for occurrence and impact. It should also determine the criteria for selection of risk items for treatment. This is commonly determined by how items show up on the matrix. For example, a risk item that had a high impact and a high occurrence would be a high-high. As Figure 31.2 shows, this item would be reviewed for treatment. It is up to the project to determine the item approach according to the matrix. Some projects leaders may believe it is appropriate to treat all items which have a high impact on the project, or to probably accept all items which have a low impact on the project.

RISK CONTROL

Treatment

Once a risk item is selected for treatment during analysis, a risk treatment technique is chosen. It is assumed the risk items selected will be acted upon or placed in a priority list that will be acted upon depending on resources. Some items may require a review of different techniques to treat using the optimal solution. Methods of handling risk include:

- **Avoidance**—Structuring the project so that the risk is avoided, examples include project relocation, delay, restructure, and redesign.
- **Prevention**—Structure the project so that the risk is prevented. Prevention may reduce the risk factors, so that the risk does not occur; or, if it does, the severity is reduced. Examples include safety and security measures, and redesign. Hazardous Operations Design Reviews (HAZOPS) are good examples of risk prevention in engineering and design offices. Safety inspections are also good examples of prevention activities.
- **Reduction**—The occurrence of a loss may justify actions which will prevent or reduce reoccurrence.
- **Transfer**—A common method of risk treatment is to transfer the risk to an organization that is more competent or willing to assume it. The transfer is usually accomplished by contract. All owners, contractors, and subcontractors have a collective share of the total project risk. There are two parts to the assumption of these risks, the company is technically qualified and financially prepared to accept the consequences. Project claims are a good example of the misunderstanding of who is at risk and who accepts the consequences.
- **Hedging**—A specialized part of transfer where the risk of price fluctuations is assumed by a speculator through the purchasing and selling of futures contracts. It is assumed the commodity futures contracts are covered by an organized exchange, such as the Chicago Board of Trade. Examples include crude oil, fuel oil, and foreign currency. Purchase orders for equipment or material in foreign currency can be fixed by the purchase of currency futures.
- **Insurance**—There may be some activities where insurance is appropriate.

Most treatment actions will be avoidance, prevention, and reduction. Many projects will have established the approach to project execution in the areas of transfer, hedging, and insurance, by the time the project team is engaged in risk management. If risk

management begins early in project development, transfer and hedging will be important considerations. A sample risk treatment chart is shown in Table 31.2:

Table 31.2—Risk Treatment [9]

Risk Treatment					
Describe Action	Cost to Treat (H,M,L)	Probability of Success (H,M,L)	Do it? (Yes/No)	Who Leads?	By When?

After the treatment action is selected, the cost to treat and the probability of success are judged. Again, a high/medium/low category is illustrated, but the calculation can be expanded and made more precise. The cost to treat can be relatively straight forward to estimate or calculate. A high/medium/low category is easy to assign and easy to understand. In risk treatment, we are looking for low costs to treat and high probability of success. In other words, getting the most money for what we spend on treatment. The ranking of risk items in terms of cost of impact and probability of occurrence can be combined with the ranking of treatment. For example, the highest ranking treatment items would have:

- High impact on the project
- High probability of occurrence
- Low cost to treat
- High probability of success

The next decision in risk management is to decide if the treatment action will be taken. The priority listing may be helpful in this regard. The decision to do it can also depend on the availability of resources. Ranking treatment items also helps in deciding to what extent the project is able to execute its treatment actions. In some situations, one might prefer a yes or no for taking treatment action, but there is no reason that the use of high/medium/low category would not be beneficial.

When the risk items are described, it is helpful to describe items that, if treated, will be the responsibility of a group, department, or individual. Risk management is facilitated by reducing treatment items to clearly defined individuals. It is also facilitated by describing the risk item so it is the sole or primary responsibility of one group. Risk items may span many groups, but if possible it is helpful to divide or reduce items to more limited groups. Finally, it should be identified when the treatment item needs to be completed, especially if the action is time dependent.

Follow-up

It goes without saying that any activity is not effective if it is not monitored and carried out. This includes risk management. Taking the time to have meetings and establish the risk management plan is wasted if the work is completed, put in a notebook, and ignored until one of the risk items occurs without the treatment being performed. Regular monitoring and updating of risk management progress should be part of a regular cycle of project activities. Monthly or quarterly reviews and status reporting of risk treatment is recommended.

QUANTITATIVE RISK ANALYSIS

Up to this point, we have used both a qualitative and a quantitative approach to risk management. The emphasis has been to provide an approach that can be readily used with limited use of quantitative tools. Some of these analytical tools can be of use to refine and focus the qualitative conclusions. Quantitative risk analysis is the application of mathematical techniques and models to numerically establish the probability of risk and the consequences of risk. The most common of these techniques are simulation, sensitivity analysis, and decision tree analysis.

Simulation

A simulation is the development of a model of the uncertainties of project, in terms of cost or time, and the effect on the project. The effect is usually expressed as a curve of the outcomes verses the probability. The general approach to simulation using the Monte Carlo method is:

1. Develop a model
    a. Cost estimate ➡ work breakdown structure
    b. Schedule ➡ network diagram
2. Select the group for analysis
    a. Cost variables ➡ summaries,
    b. Schedule ➡ activities, high level summaries, management schedule
3. Identify uncertainty
    a. Select the probability distributions and provide inputs, i.e. high, likely, low, etc.
    b. Identify the form of the results
4. Analyze the model with simulation
5. Generate reports and analyze information

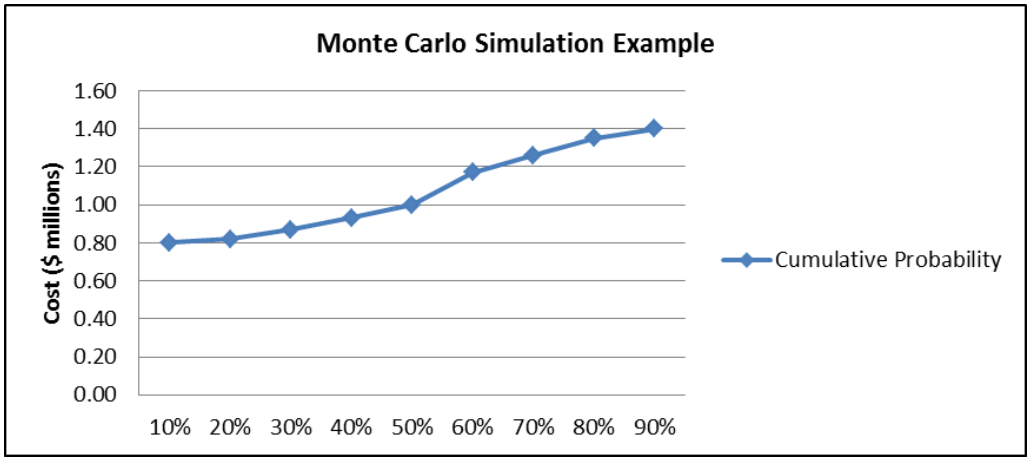


Figure 31.3—Monte Carlo Simulation Example

The curve, shown in the simulation output curve in Figure 31.3, indicates the project cost would vary from $0.8 million with a ten percent probability to $1.4 million with a 90 percent probability. This information is also used to select or derive contingencies.

Sensitivity Analysis

Sensitivity analysis is the substitution of variables in a risk model to test the effects of these changes. This is easily accomplished with the wide availability of electronic tools such as spreadsheets. Some spreadsheets have tools that also facilitate sensitivity analysis. Add-on tools are also available from vendors. This is also sometimes referred to as "what if" analysis. For example, in cost items one can substitute various quantities and unit costs to test the "sensitivity" of risk items to changes. Concern would be raised if small changes result in large effects on the costs.

Table 31.3—Sensitivity Analysis

Base	1,000 units	$1 per unit labor	$1 per unit material	Total $2,000
Sensitivity 1 **Additional 200 units**	1,200 units +20%	$1 per unit labor	$1 per unit material	Total $2,400 +20%
Sensitivity 2 **Labor at $1.50**	1,000 units	$1.50 per unit labor +50%	$1 per unit material	Total $2,500 +25%

Decision Tree Analysis

Decision analysis methods are techniques to evaluate and compare the probabilities and financial results of investment choices. Decision tree analysis is a diagram which shows the costs, probability, and financial consequences of taking certain decisions with mutually exclusive options. Table 31.3 shows the decision points and the costs for taking each decision. Decision trees are easy to set up and calculate. They are most useful when the decision costs and probabilities are established by knowledgeable individuals. Performing the calculations on all paths of the decision yields all of the expected monetary values. The value with the lowest cost or greatest savings should be given the highest priority. A sample of a simple decision tree is shown in Figure 31.4:

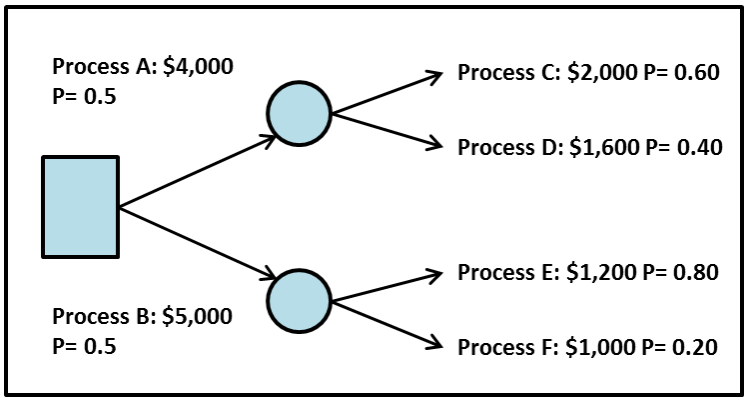


Figure 31.4—Decision Tree Analysis

In the example, a process is being analyzed which has a primary process, A or B, and a secondary process, C or D and E or F. The probability of each of the pairs of processes equals 1.00, A+B, C+D, and E + F. The expected monetary value of the individual process equals the value times the probability. The process and sub process monetary values are added to arrive at value for a case. Therefore:

AC = $4,000 (0.5) + $2,000 (0.6) = $3,200
AD = $4,000 (0.5) + $1,600 (0.4) = $2,640
BE = $5,000 (0.5) + $1,200 (0.8) = $3,460
BF = $5,000 (0.5) + $1,000 (0.2) = $2,700

The lowest cost case is process AD and should be selected. However, case BF is $2,700, and is within a few percentage points of the cost of AD. Sensitivity analysis may be helpful to refining the judgment.

RISK MANAGEMENT SOFTWARE

Cost Risk Analysis
If you are looking for risk management software, there are a handful of major recognized companies and many other companies. The basic question in cost risk analysis is whether to do it yourself or get assistance. All of the software packages listed can be used by individuals or project teams. If your project budget can support it and you have recognized that you have risks that will benefit from one of these packages, getting outside help is a better approach.

Risk Analysis Software Directory
Risk World at: www.riskworld.com/software/
The directory includes about 100 companies that provide risk related software and services. However, many have specialties that may not apply to your situation.

Cost Analysis Software Companies
AACE International is a non-profit professional organization. It does not endorse products or services provided by for-profit companies. Any company names referred in this chapter is solely for education purpose and does not constitute an endorsement by AACE International.

Three companies that provide cost risk analysis software that is sold to the public and are supported by both the companies and user groups are:

- Crystal Ball, Oracle Corp.
- @Risk, Palisade Corp.
- Risk Solver, Frontline Systems Inc.

The risk in quoting software companies is the dynamics of the industry where there are many companies that may not be there by the time you read this chapter. Crystal Ball and @Risk have been available for a number of years. Each has features that may appeal to certain users. Risk Solver is newer and has certain higher level features that may appeal to more advanced users. It can be said that all have a user interface, base set of probability distributions, and displays of results from which to make decisions [8]. A useful review of the three software programs was made by Sam Sugiyama in the 2008 issue of Foresight [8].

All of the simulation programs are Excel add-ins. The programs integrate with Excel and the look and feel is the same with additional functions from the simulation programs. The inputs for the model are probability distributions with additional values. The outputs of the programs are graphs and tables to show the most likely outcomes, the probability that these will fall within a range and the dependence of outcomes to inputs. Each program features a wide array of statistical and analysis tools to facilitate the Monte Carlo Simulation, along with other decision support tools such as optimizing and forecasting.

Schedule Risk Analysis
The schedule risk analysis programs below integrate with their companion scheduling programs. The risk programs can only be used with specific scheduling programs, so review the application of the add-in with the scheduling software. For illustration purpose, only two examples for use with MS Project and Primavera are shown. Analysis follows the classic risk analysis approach of developing a model (network schedule), identify uncertainty (ranges and probability distribution functions), and analyze the model with simulation. The schedule risk analysis programs have the additional functionality of performing this analysis on the schedule activity resources such as people, work hours and costs. Most analysis is on the results of the project completion date or project duration.

- Project ➡ @Risk for Project, Palisade Corp.
- Primavera ➡ Oracle Corp.

RELATED RISK SUBJECTS

Contingency—An amount added to an estimate to allow for items, conditions, or events for which the state, occurrence, or effect is uncertain and that experience shows will likely result, in aggregate, in additional costs. Typically, estimates using statistical analysis or judgment, based on past asset or project experience. Contingency usually excludes:

- Major scope changes such as changes in end product specification, capacities, building sizes, and location of the asset or project.
- Extraordinary events such as major strikes and natural disasters.
- Management reserves.
- Escalation and currency effects.

Some of the items, conditions, or events for which the state, occurrence, and/or effect is uncertain include, but are not limited to, planning and estimating errors and omissions, minor price fluctuations (other than general escalation), design developments and changes within the scope, and variations in market and environmental conditions. Contingency is generally included in most estimates, and is expected to be expended [1].

To reflect the view of contingency with respect to risk management, contingency is the amount of resources added to the project, cost or schedule, which satisfies the risk approach to the business and the project.

Contingency, cost and schedule, is one of the most frequent outputs of risk management. Used appropriately, and resulting from risk analysis, it is a powerful project control tool. Contingency also allows management to select the risk level appropriate to the companies approach to its business. If the business is risky and the opportunities large, the desire to assume risk may be greater than businesses that are risk adverse and more conservative.

The use of risk management techniques and risk analysis allows projects to proceed on the basis of both a qualitative and quantitative assessment of project variability. As explained in the simulation example of Figure 31.3, the project cost can vary from $0.8 to $1.4 million between ten percent and 90 percent probability. If the business environment requires greater risk, then the project cost that will be chosen will have a greater chance of variability. Specifically, the cost may have a good chance of being exceeded.

Table 31.4a—Risk Management—Risk Identification Categories: External Risks
The company cannot control the occurrence of these risks, but the company can mitigate the impact of the risk if it occurs.

TYPE OF RISK	EXAMPLES	
Natural Hazards	• Storm • Flood • Earthquake	• Tornado • Fire
New or Revised Government Regulations	• Environmental issues • Design standards • Product specifications • Facility sitting issues	• Product pricing • Taxes • Import/export issues • Supply of raw materials
Market Conditions	• Customer/User demand • Availability/cost of raw materials • State of overall economy	• Action by competition • Salvage value after useful life
Acts of Deliberate Intent	• Vandalism • Sabotage	• Terrorism • Labor strikes
Indirect Effects Occurring as a Result of the Project	• Environmental impacts • Social impacts	
Contractor/Vendor Performance	• Shortage of qualified contractors/vendors • Quality of work • Bankruptcy	• Contract claims • Schedule slippage
Financial	• Obtaining financing • Currency fluctuations • Inflation	• Taxes • Cash flow problems
Legal	• Lawsuits • Contract problems • Process license disputes	• Patent rights • Obtaining permits
Technical	• New technology is unproven • Technology becomes obsolete	• New technology makes project too complex

Table 31.4b–Risk Management—Risk Identification Categories: Internal Risks
The company can control the occurrence of these risks, and the company can mitigate the impact of the risk if it occurs.

TYPE OF RISK	EXAMPLES	
Management Problem	• Loss of control • Unresolved conflict • Lack of policies and procedures • Inadequate planning • Inadequate personnel training • Changing priorities	• Management turnover • Goals unclear, unrealistic, or not aligned • Poor definition of roles and responsibilities • Not understanding the complexity of large projects
Schedule Delays as a Result of:	• Management problems • Unrealistic schedule • Regulatory approvals • Labor shortages • Labor productivity • Unforeseen site conditions • Lack of scope definition • Project changes (scope, design, field, methods of execution, etc.) • Schedule not coordinated between design, procurement, construction , and startup	• Poor contractor/vendor • Accidents or sabotage • Work stoppages • Late deliveries • Material shortages • Lack of access • Inadequate planning and scheduling procedures • Difficulties with startup or turnover • Design and/or field rework
Cost Overruns as a Results of:	• Management problems • Schedule delays • Inappropriate procurement or contracting strategy • Contractor/Vendor claims • Inexperienced workforce	• Labor negotiations • Errors in cost estimate • Unrealistic cost budget • Inadequate cost control procedures
Technical/Quality Problems	• Shortage of trained and experienced design personnel • New technology • Design cannot be built • Inappropriate design standards	• Facility proves unreliable • Facility too expensive to operate and/or maintain • Facility cannot perform to specifications

CONCLUSION

Projects have always had risks, the preparation, and response to the risk has varied. It is clear that many project failures would have been viewed differently by people who approve funding if the degree of risk had been clearly identified. This chapter had provided some guidance on basics of risk management. It is clear that projects that are adequately prepared to perform appropriate risk management techniques will be able to respond and treat risk events and will be better able to meet project objectives.

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RECOMMENDED READING

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Chapter 32—Risk Management Practical Guide

Allen C. Hamilton, CCP

Abstract
The chapter is a continuation and expansion of risk management fundamentals. The risk management sections of planning, assessment, treatment, and control will be covered in more detail. Risk assessment will be covered with respect to identification, analysis, and treatment thresholds. This practical guide chapter will give a detailed explanation of conducting a risk management workshop and a Monte Carlo Simulation (MCS) workshop. The risk workshop section will provide guidance on how to perform the workshop, where to have it, and who should participate. In addition, the chapter will also give an explanation of conducting a Monte Carlo Simulation workshop. The benefits and drawback of MCS will be given and explained. The MCS section will also provide guidance as to how, who, and what to expect. The preparation, meeting location, and the analysis/results will be covered.

Keywords
Contingency, Risk, Risk Management Process Map, Risk Treatment Matrix, and Risk Workshop.

INTRODUCTION
This Risk Management Practical Guide will build and expand on the principles outlined in Risk Management Fundamentals, Chapter 31. Risk management has played an increasing role in the development of projects. Risk management is a way to more clearly identify and assess the alternatives to potential events in a project. This chapter is a suggested approach to implementing the risk management process.

The chapter will provide a practical guide on how to do risk management. Of particular importance is the planning for and conducting risk management workshops, the use of Monte Carlo Simulation (MCS) workshops, and associated interviews. Contingency determination (i.e., using MCS) needs to fully address the potential impacts of project risks identified in the risk management plan. A well-managed risk management closure will benefit future projects with a higher degree of success.

This chapter follows sections of the *Total Cost Management Framework*. A key approach to project implementation is adapting to the Total Cost Management (TCM) Framework. This chapter uses the TCM process map to provide an easy-to-use practical guide to implement project risk management. TCM is patterned after Total Quality Management (TQM), developed by W. Edwards Deming and others [6]. TCM and TQM are integrative philosophies of management for continuously improving the quality of products and processes.

Learning Objectives
After completing this chapter, the reader should be able to:

- Understand risk management.
- Identify the detailed steps of identifying risk, assessing risk occurrence and impact, and analyzing risk.
- Understand how to conduct a risk management workshop.
- Identify the key elements of conducting a Monte Carlo Simulation workshop.
- Describe the practical steps of risk treatment and risk control.
- Understand risk management close out and benefits to future projects.

Risk Management Overview
The *TCM Framework* defines risk as “an uncertain event or condition that could affect a project objective or business goal.”

The key steps in risk management are shown in Figure 32.1.

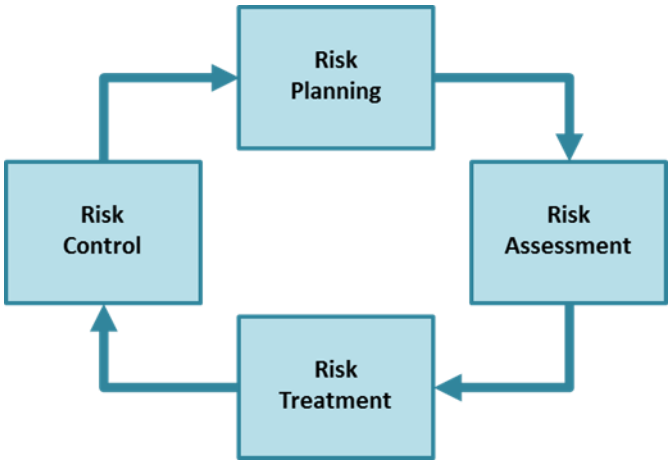


Figure 32.1—Risk Management Cycle

- **Plan**—Identify the key elements of the project risk management plan including objectives, roles and responsibilities, level and frequency of risk analysis, risk register updates, and reporting.
- **Assess**—Identify the threats/opportunities and analyze them to determine potential impact to project outcomes and determine appropriate treatment priorities.
- **Treat**—Plan and implement the treatment of the identified risks.
- **Control**—Monitor the implementation of risk treatment actions, report on status, and adjust actions according to results.

A detailed TCM process map for risk management is shown in Figure 32.2, to facilitate the subsequent discussion.

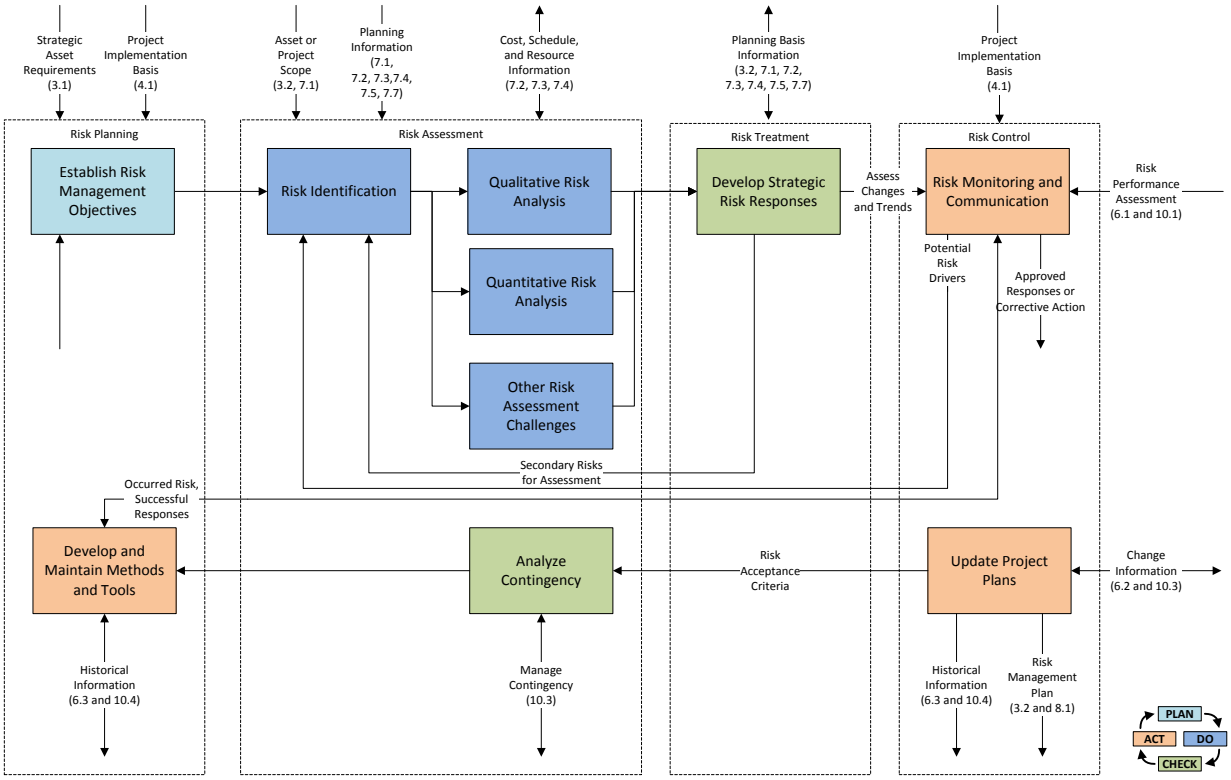


Figure 32.2— TCM Process Map for Risk Management

RISK PLANNING

Risk planning should establish the approach, form, content, and expected results of risk management. To be effective in risk management, it is important to establish a plan that is appropriate to the project. The best risk management plans are derived from the project execution plans. A plan written during project development is the best approach. The outline of the risk management implementation should be in the project plan at the start of basic engineering. The final version of the risk management plan should be reviewed and approved early in detailed engineering. The first workshops should take place during project development, typically prior to the start of detailed design. This will allow the project team the greatest flexibility in the treatment of identified project risks. The recognition of cost vs. time influence diagrams (the effectiveness of influencing project cost is at the front end) is well known in the construction industry. To summarize, the earlier a project can recognize and implement risk management treatment the more effective it will be.

The risk management plan will support the business goals and project objectives. Business goals define success and objectives and how the team will achieve these goals by setting specific objectives. The risk management plan will establish objectives which are supportive of the project. The risk management objectives set the standard for performance against the variables that a project team can control.

Considerations should be made to identify the formation of the risk management team. In addition, the overall interaction between risk management and the project teams should be clarified. Roles and responsibilities among and between the project and risk management members should be set, reviewed, and approved. If the risk management plan is not reviewed, approved, and supported, the plan could sit on a book shelf until a significant negative event occurs. Risk management is a continuous process, but with distinct phases. The lead responsible party for each phase should be identified.

After the risk management organizational structure is set up, the methods and tools for identification, analysis, treatment, and control should be evaluated. Methods will vary from simple and easy to complex and difficult. Project size, complexity, and organizational procedures will have a strong effect on the choices. Smaller, less complex projects may emphasize manual methods and tools such as word processing and spreadsheets. When project size or complexity increases, more robust methods and tools, such as a data base and proprietary software may be needed. A flow chart of the risk management actions is useful; a comprehensive version is shown in Figure 32.2 and is part of the *TCM Framework*, Subchapter 7.6, Risk Management.

A risk item is an identifiable and definable occurrence that may have a negative or positive effect on the project. Criteria for the development of risk items should be assembled prior to assessment. Assessment criteria guidelines should be reviewed and approved prior to the start of assessment-identification. Off- the-shelf software is available for assessment (risk register) and analysis (qualitative and quantitative). The quantitative analysis of contingency is performed by software applications. Examples of these applications include Crystal Ball (Oracle), @Risk (Palisade Corp.), Primavera Risk Analysis (Oracle), Acumen RISK (Deltek), and Risky Project (Intaver). *AACE International does not endorse specific software products*.

Planning for the development, timing, implementation, and maintenance of risk management is an essential step to success. The project derives greater benefit the earlier risk management starts in the project. The schedule for risk management can range from a list of milestones, a bar chart, or a network schedule. The choice for the scheduling tool to support risk management should be appropriate to the project size and needs.

RISK ASSESSEMENT

Risk planning will establish the form, content, and output of risk management. Risk assessment is the second step in the process. Assessment is in two parts, identification, and analysis. The steps for assessment are:

- The risk management lead notifies the participants and schedules the meetings and interviews.
- Meetings and interviews to identify threats and opportunities are conducted, based on the risk management criteria formulated in planning, approved by management, and distributed to participants (i.e., risk identification).
- Threat or opportunity items and any additional needed resources are quantified and evaluated by the appropriate subject matter experts. The results of analysis are prepared to support the treatment phase (i.e., risk analysis).

Risk Assessment—Identification

The identification of threats and opportunities is an important step to ensure a suitable and manageable population of risk items. The planning phase will have generated guidelines for the participants in considering risk items. Guidelines should stress the identification of risk items that can impact project goals and objectives. In addition risk items should have the following logical sequence:

Threat or Opportunity ➡ Causes ➡ Effects (Positive and Negative)

A number of factors will determine the decision to use meetings and or interviews. A combination may produce the best results. Meetings are usually more appropriate for the development of the risk identification. It is important to have an open atmosphere. The encouragement of brainstorming during the meeting will also facilitate a successful session. It is more effective to include a

broader scope of risk items, instead of missing a critical risk item. Small meetings of participants with cross sectional experience and capabilities are preferred. Large meetings present challenges to coordination and efficiency. Meetings of ten or more people need a strong lead that can control the group, encourage suitable interchange of ideas, and move the meeting forward. A member of the risk management team should act as the secretary. The meeting documentation should be available to the participants during the meeting. The goal of risk identification is to propose the risk items, briefly discuss the suitability, and clarify the item for analysis. Extensive and prolonged discussion on a single item is inappropriate. Better to leave it on the list and move forward to the next risk item. Risk interviews pledging confidentiality to the interviewee may uncover risks that are not raised at open workshops. Workshops often revert to the status quo and people may not discuss risks that are unpopular, embarrassing, or difficult to manage.

The risk identification stage will need a register that has been formed in risk planning. In addition, tools such as spreadsheets, data bases or risk management software will need to be used. Suggested risk register information:

- Risk item identification number or code, unique to each risk
- Category or Work Breakdown Structure (WBS) coding
- Title
- Description (short)
- Justification for inclusion, (i.e., description of potential effect)
    - o Issue ➡ risk ➡ effect
- Accountable group or individual (owner)
- Assumptions
- Category

Risk identification categorization is useful to organize the risk items into similar groups for clarity of interaction and treatment. Sample categories are as follows:

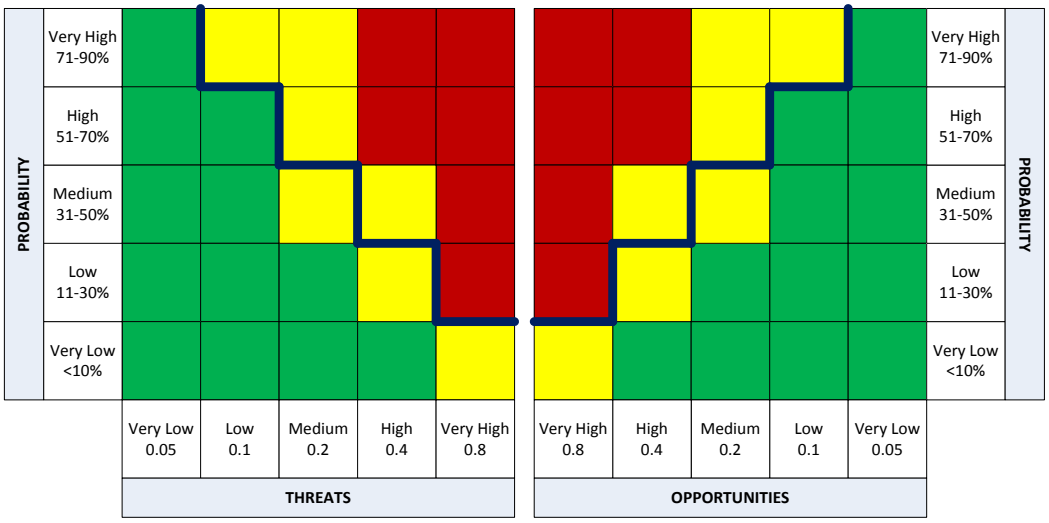
- Owner Company
- Contract
- Contractor
- Technical Requirements
- Project Management Team
- Cost and Schedule Risks
- Risk Assessment—Analysis

During risk identification it is recommended to perform as much analysis as can be accomplished in few minutes. Rules of thumb, judgment calls and “ball park” estimates are suitable for a starting point that can be refined later. This later analysis can be performed by a risk management team member or one of the project team staff, such as an estimator or scheduler. Risk analysis is in two broad areas, qualitative and quantitative.

1. **Qualitative Risk Analysis**
   Qualitative risk analysis is the assignment of subjective values to risk management items in order to provide guidance on the treatment of the item. The values are usually probability of occurrence and magnitude of threat/opportunity for the project being considered. The assignment of these values is a way to establish the threshold of taking action (treatment) on a risk item. Risk management of small projects may benefit from a three point scale as described in the *TCM Framework*, Chapter 31, Risk Management Fundamentals. The scale described in this chapter and recommended in TCM 7.6 is a five point scale (very low, low, medium, high, very high) which would provide increased differentiation of probability and impacts for the risk items. The values for each of the scores should be established during risk planning and provided to the participants. The preliminary score for each risk item should be assigned during the risk identification meetings. The scores of the threats/opportunities are most commonly based on time or money, but can occasionally take other values depending on the category.

As an example of the criteria for analysis, a threat or opportunity with a potential impact of $100,000, or more than one percent of total project budget, is a potential example of a very high score. However, the scores should be appropriate to the project size. Any schedule risk item which potentially affects the project completion date is a candidate for a high score. The threshold for risk treatment should be identified in risk planning. The threshold in Figure 32.3 is the dark line and any risk item with a matrix score above the threshold line would be a candidate for treatment. Some risk management exercises also try to assess the ease of implementing the treatment. Easy implementation may justify lower score and high probability events. The criteria should also address minimum scores for any risk item to be considered.

Risk assessment scores should receive appropriate analysis after the conclusion of the identification meetings if there is a question of accuracy. The cost and schedule impact of implementing the proposed treatment may also influence the degree of follow-up analysis. All of the scores will need to be documented in order to establish baseline scores for reference during control step.



Legend:

Red zone highlights major risks that require immediate action.

Yellow zone highlights moderate risks that require heightened action.

Green zone highlights minor risks that are "business as usual".

Risk threshold line

Figure 32.3—Risk Assessment and Treatment Threshold Matrix

2. **Quantitative Risk Analysis**
The path to establishing a value for a risk item can involve definitive methods where scope and values are explicit. Quantitative techniques using probabilistic methods are expected value, range estimating, and parametric modeling. These methods require an experienced practitioner and appropriate support. Expected value uses three point estimates (high, likely, and low) and the probability for each to generate a value. Range estimating is similar with three point values used in the analysis. In addition, range estimating is effective when there are limited inputs into the overall cost or schedule. Monte Carlo Simulation is commonly used with expected value and range estimating and should incorporate specific risk events that have been identified. Parametric modeling combines risk to impact relationships. An advantage of these methods is that sensitivity analysis can be performed to better identify the risk drivers. These methods generate numerical tables and values associated with the risks and may be helpful in making decisions on treatment. It is important to properly document the assumptions and results so all will understand the impacts.

RISK WORKSHOPS PLANNING

Planning a risk management workshop should be consistent with the overall planning both for the project and risk management. The workshop planner should clearly understand the intended approach to take for the workshop, the form of the workshop, and the expected results. The expected results should support the project objectives. Key considerations for the workshop planning are the project size and complexity, resources allocated to risk management, and the level of support from management. Smaller, less complex projects may justify a half day workshop conducted by a staff member who performs all follow-up on a part-time basis. Large, complex projects may justify a third party moderator to conduct the workshop over a multi-day period, who issues a comprehensive report to be used by the risk management lead. It is assumed that all key risk assessment criteria have been identified and approved. The decision to include both threats and opportunities should have been made and risk item coding outlined. The analysis criteria should be confirmed and any basis for estimating threats and opportunities should be communicated.

Follow up interviews should be planned as appropriate to delve into any details regarding the identified risks, or the potential impacts associated with them, or to discuss any issues that could not be discussed in the workshop setting.

Workshop Moderator Internal vs. External

A key decision about the workshop is the choice between an internal and external moderator. The advantage of internal moderators is that they can be mobilized quickly, they are knowledgeable about the project, they know the project staff, and they know the political and organizational structure of the project. If the internal moderator is also the risk management lead, it is a natural extension of his or her duties and may be important for establishing teamwork within the project. After the workshop is completed,

the internal moderator can use the meeting to support implementation. Internal moderators can also provide more nuanced follow-up and resolution of issues.

External moderators have distinct advantages that justify their consideration. Big projects should consider having external moderators. They are independent and hopefully objective in their approach and do not have project history to influence their work. They are also experts in this type of meeting and should be competent in managing the meeting to a successful conclusion. They can perform the role of a broker if there are issues or conflicts in the project team. They can be asked to perform the workshop and periodic follow-up. Often risk management is performed and may suffer from a lack of follow-up until an issue becomes a critical problem.

Workshop Setting

The best location for the workshop is somewhere where the atmosphere will support the focus on the workshop goals. An optimum location is an offsite conference room which is a short drive from the project site. All food should be available in the conference room; including breakfast, lunch, and coffee breaks. The key is to make the experience of the workshop as supportive as possible. When the workshop takes place onsite, or in the existing plant, the potential for interruptions and distractions increase dramatically. When the participants can walk out of conference room and down the hall to their office, it is too easy for someone to disappear. One strategy is to have the project manager open the meeting, stress the importance of the workshop, remind everyone how important it is to stay in the meeting, and emphasize that everyone should contribute freely to the discussion.

Selection of the Participants

The workshop works best with the minimum number of people required to identify the issues, briefly define the risk items and make informed assessments, but should be large enough to represent the various project stakeholders and each functional group. The planning phase should identify the anticipated length of the meeting and the need to subdivide the sessions if the potential risk items are numerous and complex. The most fundamental split for example could be design/engineering and construction, but may also include procurement, permitting/regulatory, commissioning, and start-up functions. The invited participants should be able to identify the risk item, describe it; and, if possible, assess it in one to five minutes. The group should have reasonable familiarity with each other, responsibilities and job functions.

Start of the Meeting and Developing Issues

The planning has been completed, the site secured, and the participants are present; it is time to start the workshop. First, establish meeting ground rules, such as the process of proposing risk items, who will record and keep the register, and time limits on risk item deliberations. It is important to stress that the importance of a free form for anyone to propose risk items for consideration and their discussion. The risk management planning and workshop planning will already have established the form and criteria for the risk management register. Each risk item will need to have one or more responsible parties to retain the ownership. The importance of assigning an individual for each risk item is that the person will be responsible for implementing the risk management response and resolution. Another important individual is the person who will provide the trigger for the decision to implement risk control.

Risk Item Coding

If the risk register is anticipated to include 20 or more risk items, it is important that risk items be coded to pre-established criteria. Coding helps to organize and communicate the risk items. The register should be set up to identify the primary impact area/phase/group of each risk item and any significant secondary impact. Risk items coding frequently follows the work breakdown structure and the project cost coding. The coding should facilitate reporting on the status of the risk item population. These codes should be established prior to the start of the workshop, to facilitate the use of the codes during the workshop.

Initial Assessment: Identification and Qualitative Assessment

Once a risk item is proposed it should have a brief title. The fewest key words (preferably no more than six) to describe the risk items is best, but they need to be unique. Next, a brief description should be noted and entered into the register. A couple of sentences should be sufficient. Remember the group needs to identify, define; and, if possible, assess in one to five minutes. The risk item proposer should be able to give the title and description in one minute. Next, the proposer should briefly define the effect of the risk if it occurs, (i.e., how it will impact the project objectives). Follow-up discussion should expand the initial description. All risk item proposals go into the register. It is more important to include than to spend a lot of time figuring what to exclude. The exclusions are easy to identify after the meeting is completed. Finally, the group needs to agree on the impact and possibility of occurrence. The probability of occurrence may not be easy to evaluate, but the participants will need to assess a level on the scale (very low, low, medium, high, and very high). The impact, including cost and schedule, may be difficult to estimate, but a holding value will need to be entered and confirmed later.

Risk Item Communications, Reporting and Control

The workshop should also confirm the risk management planning with respect to the flow of information and communications, as risk item treatments are implemented. The monthly report is a common area for the reporting of the status of risk items. The register should be available to all participants and updates provided in a commonly accessible area.

CONTINGENCY

Contingency is an amount added to an estimate or schedule finish date to allow for items, conditions, or events for which the state, occurrence, or effect is uncertain. Experience shows these occurrences will likely result, in aggregate, in additional costs and time.

Contingency usually excludes:

- **Major scope changes** such as changes in end product specification, capacities, building sizes, and location of the asset or project.
- **Extraordinary events** such as major strikes and natural disasters.
- **Management reserves.**
- **Escalation** and **currency** effects.

In TCM, establishing contingency is considered part of the risk management process. The risk impacts from treatment that are incorporated into the project plan are assumed to be included in the project budget. Contingency is then established to address what is called residual risk. When the estimate and schedule is completed, the impact of all opportunities and threats are assessed against the estimate and schedule. Many projects treat risk management and establishing contingency as parallel activities and do not integrate the assessed impacts from identified risks and associated risk management actions into setting the level of contingency for the project. The Project and Risk Management Plan will need to explicitly show how the two will be addressed in the estimate.

According to John Hollmann [4], the three common methods to calculate contingency on projects are:

1. **Expert Judgment**—This is perhaps the most widely used. It is the application of assessments based on personal and past company experience with projects. It is frequently used in companies where there is an expert who has demonstrated competence with making these judgments. It is also used in companies where there is a close collaboration with project professionals that have objective and realistic evaluations of project risks. James D. Whiteside indicates that judgments are affected by experience, intuition, collective reasoning, emotions and personalities [7].
2. **Predetermined Guidelines**—These are frequently used by larger owner companies that have project support groups. The guidelines are specific criteria for design, engineering and information by estimate class. The criteria can also allow for maturity of development, such as assumed, preliminary, semi-detailed, and definitive. Large sophisticated project organizations collect project historical cost data and use the analysis to guide the setting of contingency. The AACE Recommended Practice 18R-97, *Cost Estimate Classification System: As Applied in Engineering, Procurement, and Construction for the Process Industries*, is an excellent reference for such contingency levels although actual results often exceed these ranges for both cost and schedule [1].
3. **Monte Carlo Simulation**—This is used by project organizations that have a reasonable degree of experience with computational tools and methods. MCS uses a statistical analysis of the cost estimate and schedule associated with the project risks to determine an amount of money or time to add to achieve a given outcome. Expected values and range estimating are used with MCS to generate a table or curve with associated probabilities for the potential cost and schedule outcomes. In the probabilistic approach the choice of a probability from the range of potential outcomes is important [4].

To determine the amount of contingency that is to be applied to the project, the tolerance for overall risk has to be established. Risk tolerance means the company will be willing to assume risk for an expected reward (profit or return). Risk aversion means the company wants to mitigate, transfer, or avoid risk to ensure an expected outcome. When statistical analysis has been performed, the organization's level of risk tolerance will be reflected in the chosen probability. This is expressed as shown in Table 32.1.

Table 32.1—Probability vs. Tolerance and Aversion

Unit	Probability	Contingency Amount	Reward	Risk Tolerance	Risk Aversion
P10	10%	Low	High	High	Low
P50	50%	Medium	Medium	Medium	Medium
P90	90%	High	Low	Low	High

(See **TCM Framework**, *Section 10.3*, *Change Management*, for a discussion on the treatment of changes and their relationship to contingency).

Monte Carlo Simulation (MCS)

Monte Carlo Methods (Simulation) are a broad class of computational algorithms that rely on repeated random sampling to obtain numerical results. Monte Carlo methods are used to model phenomena with significant uncertainty in inputs, such as the calculation of risk in business [8].

Montel Carlo Simulation (MCS) is a method for iteratively evaluating a deterministic model using sets of random numbers as inputs. It is typically used when model is complex, nonlinear, or involves more than just a couple of uncertain parameters [2, 7].

Many organizations perform MCS a limited number of times for a single project. It is most commonly used during the project phase prior to receiving full funding. The appropriation estimate and schedule is the basis to use for this analysis.

Introduction to MCS
MCS was first identified in Buffon's needle problem experiments in the 1930s, and then Enrico Fermini commenced the initial implementations, but was most commonly applied by Stanislaw Ulam in 1946, at the Los Alamos National Laboratory. It is widely used in business, including the capital investment community. Since it is a method that uses computational algorithms and requires a basic understanding of statistics, it is usually used by project control specialists. It is a method that is easily subject to misuse if not properly implemented. If properly implemented, it can provide a robust analysis of the impact of threats, opportunities, and uncertainty on the project cost and schedule outcomes.

Benefits and Drawbacks
MCS depends on the subdivision of the total estimate into cost items or the schedule into activities that are assessed. Typically, the cost items or schedule activities are at some summary level, as the detailed estimate and schedule may contain thousands of lines of data. These assessments reflect the potential variability of the cost and schedule outcomes. Everyone knows that the baseline estimate and schedule will be different from the actual results. Most project groups are optimistic that the project will meet its objective. The reality is that many are wrong. The only group that has shown to be realistic is successful lump sum bidding organizations. Organized and disciplined cost and schedule risk analysis can lead to more understandable and realistic bids.

Achieving the benefits of MCS requires project teams with broad competences who are prepared to make decisions with knowledge of the degree of risk present. MCS can be used to fully develop the degree that project risks may affect project outcomes, including threats and opportunities. The most important benefit is to numerically quantify a range of project cost and schedule outcomes and the probability associated with each. MCS also gives the project team an opportunity to review the estimate and schedule in a group setting. The MCS meeting can also expand the participants' understanding of concern with respect to the estimate from other members.

The drawback of MCS is that it may misrepresent the full range of potential outcomes and the risk drivers if the models do not accurately reflect the correlation of inputs, dependencies between inputs, or have narrow ranges of inputs [4]. The Hollmann experience includes MCS failures that result from all of these drawbacks.

- **No input correlation**—If one had all the time and resources, we could correlate the range assessments to the estimate line items. Few organizations have such time and resources. However, this should not stop any organization from collecting project performance data and using them on future estimate and risk assessments.
- **Input ranges are narrow or similar**—Project staff members have often insisted that high and low ranges for many cost items or variables are +10 percent and -10 percent. Narrow ranges are one of the problems with MCS. Narrow ranges generate narrow results. Few projects have the characteristics of narrow ranges. The best ranges are ones that have occurred in the past, or by judgment or intuition are possible to occur.

Preparation
The MCS moderator will need to make the following preparations:

- Prepare an outline of the meeting to be sent to all participants and stake holders.
- Prepare schedule of meeting sessions.
- Identify the participants and the sessions that they are scheduled to attend.
- Include the summary of the estimate line items and schedule activities to be assessed.
- Explain the importance of input ranges and assessments.
- Ask for comments and suggestions; also allow time to incorporate these if appropriate.
- Identify the report format, if it is to be distributed; some reports are considered confidential.

Table 32.2—MCS Participants Matrix by Type of Estimate

Participant	Early Estimate FEL-1	Business Case Estimate FEL-2	Appropriation Estimate FEL-3
Facilitator	X	X	X
Chief Estimator	X	X	X
Lead Estimator			X
Project Mgr.	X	X	X
Business Mgr.	X	X	Part Time
Project Engineer	X	X	X
Process Engineer	X	X	X
Project Control Mgr.		X	X
Constructor Mgr.		X	X
Procurement Mgr.		X	X
Plant Engineering Mgr.		X	Part Time
Discipline Engineers			X

Conducting MCS Meetings [3]
Conducting the MCS meeting is similar to the risk management meeting. Therefore, an offsite meeting location is best. Full participation is enhanced by providing all food in the meeting room. The agenda should be presented and should be what the participants were sent in the pre-meeting package. The agenda should include time allocations for all sections. The format and criteria for the selection of inputs should be discussed. It is assumed the participation schedule will have been agreed to and sent to all members. A member should be identified who will record the meeting discussion and MCS inputs.

Stress the importance of input ranges that reflect worst case outcomes. Accept all range proposals and reasons. Discourage criticism of proposals and reasons. Experts should lead or guide discussion of high and low ranges. Allocate time for discussion, with some link to estimate line item importance and potential impact. Ensure that minor cost or schedule impact items are settled quickly. Avoid prolonged discussions: propose a compromise, defer to subject matter experts, or defer a decision subject to further research or analysis. The meeting recorder should be diligent in noting any follow-up actions and the parties responsible for each follow-up item. The moderator will indicate what information the participants will receive after the meeting.

Analysis and Results [3]
After the conclusion of the MCS meeting, an analysis will be performed and a report issued. First, the moderator or Monte Carlo technician will input the assessment values. These assessments are the high, low and most probable values. The Monte Carlo programs are relatively easy to learn. One of the problems with the ease of using the programs is the degree to which they could be misused or manipulated. The second step in the analysis is to program the relationships and interdependencies of the line items. Examples of parameters that have strong relationships are wage rates, work hours, and labor productivity. The initial analysis is performed and the output is reviewed for anomalies, inappropriate ranges and data distortions. Based on the initial analysis, it is possible that some estimate line item or schedule activity assessments may need to be adjusted. It may be necessary to obtain advice and support from the subject matter expert to adjust key data inputs.

Once the review is completed and the MCS responsible party is satisfied, the report is finalized. The key document in this regard is the expected cost by probability.

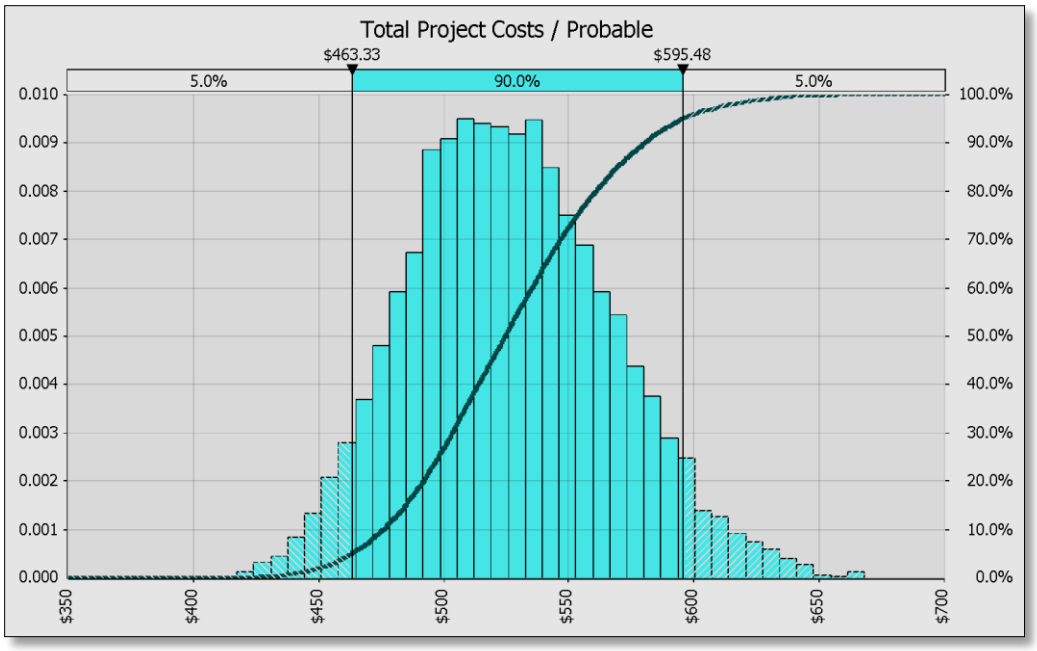


Figure 32.4—Probability of Cost Underrun

Figure 32.4 is a simplified version of this report. The key parameter for MCS is the selection of the probability. The selection of a probability is dependent on the maturity of the estimate and the risk tolerance, or aversion of the organization, as discussed in an earlier section. A 10 percent probability (P10) would have a low chance of being achieved, but some organizations may have a tolerance for this risk. A 90 percent probability (P90) would have a high chance of being achieved and some risk adverse organization may prefer this expected cost. Many organizations recognize that projects overrun and under run and choose a 50 percent (P50).

The report is written and issued. A number of organizations consider the report as confidential because of the sensitivity of the information. Usually, the probability and the contingency amount or percentage is approved by the appropriate authority.

Follow-up

Key follow-up activities include tracking and documenting project performance. If the project is still in the planning stage, a comparison between the current estimate and the MCS results will indicate the possible need to make changes in the next project planning phase. If a project is completed, a normalized version of the actual should be compared to the authorization estimate. Many organizations will benefit from a history of MCS. Dale E. Clarke in his technical article explains the benefits of performing MCS on projects over a ten year period [3].

RISK TREATMENT

Risk assessment would have determined the risk items and the priorities for treating. From risk analysis, the primary risk items to be treated would be identified. The risk management team and other members would develop recommendations and decisions on how to treat those risks. Risks should be assigned risk owners who will be responsible for overseeing the implementation of the risk treatment action. All risk items should have owners, which should be reflected in a risk register. It is best if the owner is an individual rather than an organization. One of the reasons to choose a risk owner is because the risk item will relate directly or indirectly to his or her job function. Risk management control will measure results and it is easier to follow-up with individuals rather than organizational units.

The following three bullet lists are adopted from the *TCM Framework*, Section 7.6.2.3, Risk Treatment [5]. Key actions performed during the risk treatment phase include:

- **Evaluating** all appropriate response strategies.
- **Selecting** an appropriate risk response plan strategy (or combination of strategies).
- **Developing** action items in support of the selected response.
- **Validating** proposed actions with assigned actionees, including dates for implementation.
- **Ascertaining** post-response targets and gains.
- **Ascertaining** response plan resource requirements.
- **Updating** project schedule or budget if the anticipated treatment value gain is positive.

- **Identifying** any secondary threats or opportunities that may arise from the response.

Response strategies for threats, listed in order of preference are:

- **Avoid**—Involves eliminating either the probability or impact. This can be done by either clarifying requirements, acquiring expertise, or changing the project management plan.
- **Reduce**—Involves mitigating key drivers to reduce probability and/or impact.
- **Transfer**—Involves transferring the threat to a competent third party who is better able to control. This could be through insurance or contractual transfer (via indemnity, exclusion, or hold harmless clauses).
- **Accept**—No proactive action is to be taken because cost either outweighs benefit or there is already an adequate contingency provision that will protect the project objectives.

Response strategies for opportunities, listed in order of preference are:

- **Exploit**—Involves taking steps that guarantee the opportunity will arise (e.g. changing specification, scope, or supplier).
- **Share**—Involves sharing the risk with a third party better able to manage it (perhaps by applying a pain/gain formula).
- **Enhance**—Involves increasing the probability and/or impact of key risk drivers.
- **Accept**—No proactive action is to be taken because cost either outweighs benefit or there is already an adequate contingency provision that will protect the project objectives.

The documentation for each risk item will include the risk action, strategy, and plan for implementing, if the risk is triggered. In addition, an indication of possible change management classification should be outlined.

The Risk Treatment step includes the actual implementation of the risk treatments for the risks. As previously discussed, addressing project risks needs to be prioritized, based on the probability of occurrence and the magnitude of impact. Other considerations in developing priorities may include the timing of the risk, (i.e., a risk associated with an earlier phase of the project may need to be treated before one that impacts a late phase of the project).

RISK CONTROL

Risk control is a vital step in the risk management process cycle. Monitoring and reporting on the risk management process is critical for project and organizational management to understand the effectiveness of the risk treatments and any changes in the risk profiles. Adjusting the risk management plans as needed to react to the increasing, decreasing, or new risks during the project is important component of control. The responsibility for risk control must be identified in the risk management plan. For example: Will risk control be part of project cost control, a separate group, or a hybrid? Most projects incorporate the risk management function into project control.

Monitoring will begin during risk assessment as risks are identified and a register is developed. Most monitoring will be performed by the risk management lead and the risk item owner. The risk management owner should provide updates on the risks and associated treatments to the person or group responsible for risk control.

A triggered risk will result in reviewing the original risk in the register. The review will include confirming the original risk definition, classification, and analysis. Next, the strategy for risk treatment will be confirmed and validated if necessary. The project plan including the risk management plan will determine if the change management procedure or the separate risk management procedure will apply. The change and risk management procedures should be coordinated and the outcome of risk management be reported. It is assumed that the change management procedure will be the primary tool for risk action. The appropriate documentation will need to be completed, reviewed, and approved. The control procedures will also have to be clear regarding the risk treatment expenses and benefits. The risk treatment may result in incurring expenses in order to minimize a threat or obtain a benefit (opportunity). The procedures will also need to be clear as to the handling of the risk in respect to contingency. Risk control may result in a variety of responses including:

- **Act on Risk**—The original risk occurs and is acted upon consistent with the original plan.
- **Modify Risk**—Risk assumptions, analysis, and treatment strategies may need to be modified.
- **New Risk**—Events or developments will highlight new risks that need to be assessed.

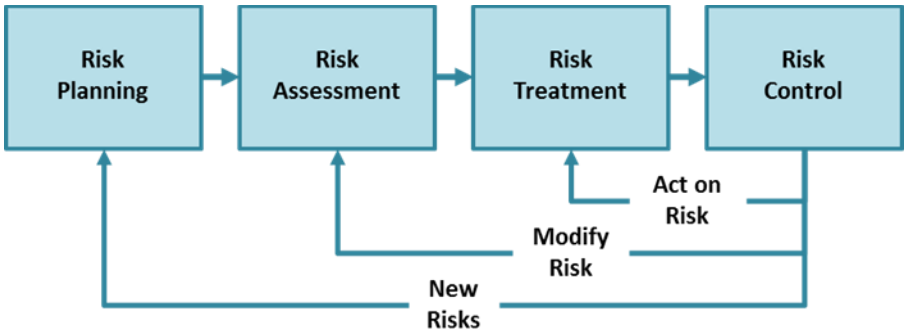


Figure 32.5—Risk Control

The risk items that are activated will need to progress through a procedure that recognizes the balances among responsible parties in the project. The risk management team will need to reconcile any differences with the risk owners. Conclusions as to treatment may differ among parties. Sufficient data and analysis will help to minimize subjective analysis. The procedures and process are more important than one individual making decision on risk treatment actions. Risk management committees may also help to facilitate consensus as to risk treatment actions.

Periodic reporting is helpful to measure progress and results in risk management. It is also helpful to provide a comprehensive approach which will provide sufficient details to ensure compliance with the risk management objectives. These details include quantification of total risk counts and the outcomes of the process are recommended. Reporting savings and opportunities should also be part of the report.

RISK MANAGEMENT CLOSURE

One of the important steps in projects implementing risk management is closure activities. Total Cost Management is a continuously improving process where the results of the activity are summarized and made available for the next project. Successes, changes, and failures are all important to be captured so the next project can benefit from the experience and failures are not repeated. The transfer of the knowledge should take place prior to the departure and reassignment of key personnel. Key steps in risk management closure are:

Collect and Debrief ➡ Evaluate and Document ➡ Archive

- **Collect and Debrief**—All appropriate records and documentation should be collected. Reports, analysis, and a comprehensive risk register or data base should be preserved. The key participants should be debriefed. Keys information from the debriefing should include observations on risk management effectiveness, success, failure, and any activities to make risk management better on the next project.
- **Evaluate and Document**—The plans, actual results and observations of the participants should be summarized in a form usable for future projects and risk practitioners. The evaluations and conclusions may be presented to the participants for a final review. It is important the results be usable by other projects.
- **Archive**—If key learnings of risk management cannot be found, the results will be of no use to other projects. A central archiving system that is accessible to future projects is essential.

CONCLUSION

Projects have always had risks, the preparation and response to the risk has varied. It is clear that many project failures would have been viewed differently by people who approve funding if the degree of risk had been clearly identified. This chapter has provided some guidance on the practical approach to risk management and the conducting of the risk management and Monte Carlo Simulation workshops. It is clear that projects that are adequately prepared to perform appropriate risk management techniques will be able to respond and treat risk events and will be better able to meet project objectives.

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Chapter 33—Total Cost Management Overview

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Abstract
The TCM Framework is AACE International's landmark technical publication that describes the concepts of total cost management. These principles are expressed within the framework as a series of process map guides that illustrate the highly integrated matrix of cost engineering and other business and management practices that support the lifecycle of asset investments. "Simply stated, TCM is a systematic approach to managing cost throughout the lifecycle of an enterprise, program, facility, project, product, or service."

Keywords
Ideation, Plan-Do-Check-Act (PDCA) Management Cycle, Process Map, Strategic Asset, Strategic Asset Management (SAM), Total Cost Management (TCM), TCM Framework.

INTRODUCTION
The constitution of AACE International provides the following definition for Total Cost Management:

> "Total Cost Management is the effective application of professional and technical expertise to plan and control resources, costs, profitability, and risks. Simply stated, it is a systematic approach to managing cost throughout the lifecycle of an enterprise, program, facility, project, product, or service. This is accomplished through the application of cost engineering and cost management principles, proven methodologies, and the latest technology in support of the management process"[1].

Total Cost Management has been described as the "sum of the practices and processes that an enterprise uses to manage the total lifecycle cost investment in its portfolio of strategic assets" [2]. Strategic assets are physical or intellectual property that is of long-term or lasting value to an enterprise. Examples of assets include a building, an oil refinery, a software application, or a stage production. Assets are economic resources generally expected to provide a positive economic benefit to its owner (i.e., an asset has an economic value that can be bought or sold). Assets are created through the investments of money, time, and human and physical resources. Projects are the temporary endeavor (a set of activities with a defined beginning and end) to create, modify, terminate, or otherwise affect the state of an asset. Thus, projects are how the decision to invest in an asset is put into effect.

As an example, an oil company may plan, build, maintain, expand, and finally demolish an oil refinery during its lifecycle. At each of these various phases, investments are required. Initially the oil company begins a project to plan, design, and construct the original oil refinery (or asset). During operation of the refinery, the oil company monitors operating costs and profitability; and may evaluate alternative investment opportunities to maintain, expand, or modify the facility asset that may result in additional projects. Eventually, the refinery has served its useful life, and another project (and investment) may be required to close and demolish the refinery. All of these activities and stages within the asset's lifecycle fall within the scope of total cost management, which identifies the practices and processes to manage the total resource investment during the lifecycle of the asset.

It is important to recognize that in Total Cost Management, cost "goes beyond the traditional monetary definition to include any investment of resources in the enterprise's assets" [2]. Everything invested in assets and projects is a cost; which includes time, resources, and obviously money. Total Cost Management is a comprehensive approach to managing the total resource investment in assets.

Learning Objectives
After completing this chapter, readers should be able to:

- Understand the project lifecycle.
- Understand how TCM is applied throughout the lifecycle of an asset.
- Utilize the process map guides in the *TCM Framework*.
- Understand the Plan, Do, Check, Act (PDCA) management cycle.
- Understand the Strategic Asset Management Process and associate process map.
- Understand the Project Control Process and associate process map.

AACE International's Total Cost Management Framework
The *Total Cost Management Framework: An Integrated Approach to Portfolio, Program, and Project Management* (TCM Framework) is AACE International's landmark technical publication that describes the concept of total cost management. It expresses the principles of total cost management in a series of process maps used to illustrate the highly integrated matrix of cost engineering and other business and management practices that support the lifecycle of asset investments.

The TCM Framework is not intended to be a strict and rigid set of rules. The process maps identify the steps involved in managing asset investments and projects, but are intended to be flexible. It provides a structure that emphasizes the integration of all of the various cost engineering and other business practices used in Total Cost Management. The skills and knowledge of the various individual cost engineering disciplines are described elsewhere in this book; Total Cost Management illustrates the integration of all of these individual skill areas into a comprehensive management process. The overall process is meant to be both industry and asset generic—it can apply to any enterprise, program, facility, project, product, or service.

Total Cost Management is a Process

Processes are the foundation of quality-driven business management. A key element of achieving quality is to develop and implement predictable and repeatable processes to achieve organizational success. AACE's *TCM Framework* identifies a comprehensive set of processes related to Total Cost Management.

The foundation of the TCM process model is the Plan-Do-Check-Act (PDCA) Management Cycle, often known as the Deming or Shewhart Cycle. It is a generally accepted, quality-driven, continuous improvement model that is very applicable to general project management principles. Figure 33.1 illustrates the PDCA process steps:

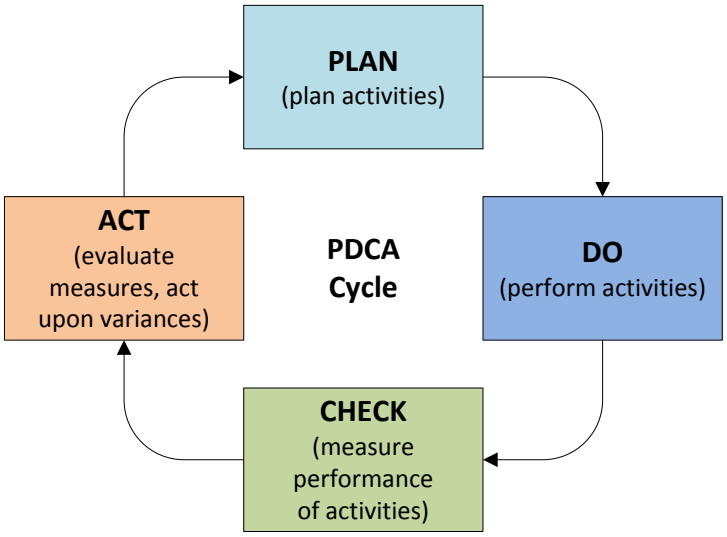


Figure 33.1—PDCA Cycle

In the PDCA Cycle, the term check is generally synonymous with measure; and the term assess has been substituted with the word act, as in "to take corrective action."

The PDCA process is cyclical; thus, promoting continuous improvement. Every individual process within total cost management is designed to employ feedback and improvement. Another key element to Total Cost Management is that every individual process relates back to identified objectives and requirements. AACE's Total Cost Management process is divided into two aspects: 1) the Strategic Asset Management Process, and 2) the Project Control Process.

Strategic Asset Management refers to the macro process of managing the total life cycle cost investment of resources in an enterprise's complete portfolio of strategic assets. At any particular point in time, the portfolio may contain many different assets each of which may be at a different stage in their lifecycle. Although investments are made in an asset through the performance of a project or program, strategic asset management is not concerned with day-to-day project tasks; it focuses instead on initiating and managing the overall portfolio of projects in a way that addresses the strategic objectives of the enterprise.

Project Control is the recursive process cycle that is nested within the "do" step of the Strategic Asset Management process cycle. A project is the temporary endeavor an enterprise undertakes to create, modify, maintain, or retire an asset. At completion of the project, a usable or operational asset is returned to the enterprise's asset portfolio.

Figure 33.2 illustrates how the PDCA model is applied in a hierarchical and recursive manner in the highest level TCM Process Map. The basic PDCA process is applied for each asset and group or portfolio of assets. It is then again applied for each project being performed to create, modify, maintain, or retire those assets.

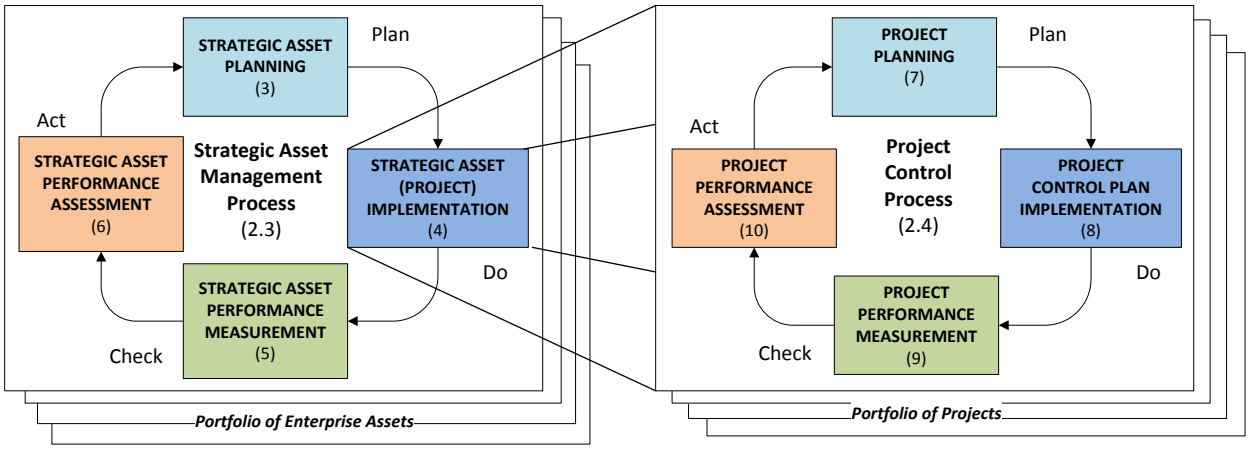


Figure 33.2—High Level TCM Process Map

Note that the numbers assigned to the boxes in the process maps refer to specific sections of the *TCM Framework*.

The PDCA steps in the strategic asset management process, as identified in Figure 33.2 involve:

- **Plan—**Establish plans for the investment of resources in assets, based upon the goals and objectives of the enterprise.
- **Do—**Implement plans; initiate projects as needed (to the project control process steps).
- **Check—**Measure asset and project performance.
- **Act—**Evaluate asset performance against the plan; identifying corrective mitigating, or improving asset performance initiatives as may be determined (leading back to strategic asset planning).

The PDCA steps in the project control process as identified in Figure 33.2 involve:

- **Plan—**Establish plans for the specific project (identified in strategic asset management planning).
- **Do—**Implement plans and project execution.
- **Check—**Measure project performance.
- **Act—**Evaluate project performance against the plan; identifying corrective, mitigating, or improving project performance initiatives as may be determined (updating project plans).

The cyclical nature of the PDCA control steps are applied iteratively throughout the lifecycle of the asset or project. Whether focusing on an asset or a project, each has various stages (or phases) that occur during their lifecycle. Each stage or phase has a grouping of processes and sub-processes that may result in an intermediate deliverable or progress milestone. Every stage, process, and sub-process may use the concept of the PDCA Cycle.

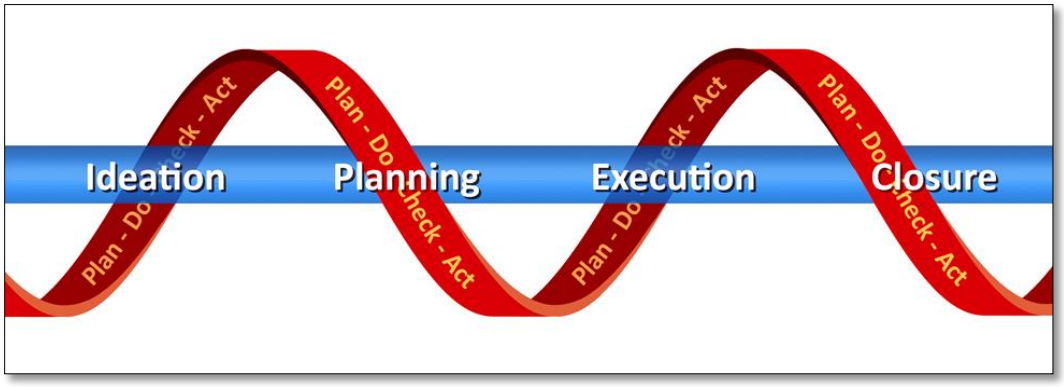


Figure 33.3—Applying the PDCA Concept to the Project Lifecycle

Figure 33.3 illustrates a project lifecycle with PDCA shown as a spiral. The axis represents the life cycle phases of a project from ideation through closure. The spiral attempts to show that the PDCA process is employed continually to achieve various milestones

or deliverables at each phase of the project life cycle. The asset life cycle can be represented in the same way by substituting the asset life cycle phases along the axis.

The lifecycle of a strategic asset can be summarized by the following five stages:

- **Ideation—**Determine an opportunity for a new asset or an improvement to an existing asset; research, evaluate, define, and develop potential solutions that address the opportunity; and select the optimal solution.
- **Creation—**Create or otherwise implement the asset solution, typically through execution of a project or program.
- **Operation—**Deploy the new or improved asset into service, operation, or other use.
- **Modification—**Modify, improve or otherwise change the asset, typically through execution of a project or program.
- **Termination**—Decommission, retire, demolish, or otherwise terminate the asset from the enterprise's portfolio.

Between the beginning and the end of the asset lifecycle, many of the stages and processes are iterative and sometimes overlapping. An asset is typically modified many times over its lifecycle, with ongoing ideation leading to changes and improvements that are incorporated generally through execution of a project.

Within the asset lifecycle, projects are the temporary endeavors for the ideation, creation, modification, and termination of an asset. Operation of the asset itself is not considered as a project, but during operation of the asset many projects may be undertaken to maintain, modify, relocate, enhance or otherwise improve the utility of the asset.

The lifecycle of a project can be summarized by the following four phases:

- **Ideation—**Given the project objectives, assess alternative concepts and solutions for performing the project; establish the project requirements and project performance goals.
- **Planning—**Develop project plans to address the selected project concept and performance goals.
- **Execution—**Implement the project plans, and execute the project to meet the project objectives and performance goals.
- **Closure—**Review, test, validate, and turn over the completed asset to the customer; document project learnings for future use in ideation and planning.

Generally, each phase of the project yields one or more deliverables (e.g., a requirements document, project plan, design document, model, etc.) that become resources or inputs to the next phase.

The phases are recursive, meaning that each phase may be treated as a project in itself that produces a final deliverable but not the final asset. For example, the ideation phase itself has a lifecycle that may include planning for ideation, executing the ideation process, and closure of the ideation phase, that results in completion of the project requirements document. At this recursive level, the closure of a phase typically results in one or more defined project deliverables and the achievement of a project milestone or decision point (often called a decision gate). If the deliverables do not pass a quality review for that phase (often called a gate review), then the deliverable is recycled for further development.

Generally, before beginning project execution, all required planning deliverables (including the project cost estimate and schedule) are thoroughly reviewed, and if not found acceptable then the project may be stopped. Often, projects may be stopped at any time during ideation or planning if it becomes clear that the project objectives cannot be met within the cost or schedule constraints identified in the ideation phase.

The Strategic Asset Management Process

The strategic asset management process is typically a business-led process. It is concerned with managing the long-term economic return or profitability of the complete asset portfolio of the enterprise. Business management is continually evaluating the asset portfolio to decide whether a new asset should be developed, whether existing assets should be modified in any way (expanded or changed), or whether an existing asset should be retired or decommissioned. Generally, any particular enterprise is limited in the funds available for investments in assets (for creating, maintaining, or modifying assets); and therefore, the strategic asset management process is focused on "selecting the right project."

Compounding strategic asset management is that there are often internal business units within the overall enterprise, each of which may be in competition for the limited funds of the enterprise. Also, business environments are at times highly dynamic and uncertain. Therefore, the strategic asset management process attempts to first identify, and then balance business opportunities and risks against demand and supply for resources in such a way that maximizes business objectives and overall profitability.

Figure 33.4 illustrates a more detailed process map for the Strategic Asset Management process.

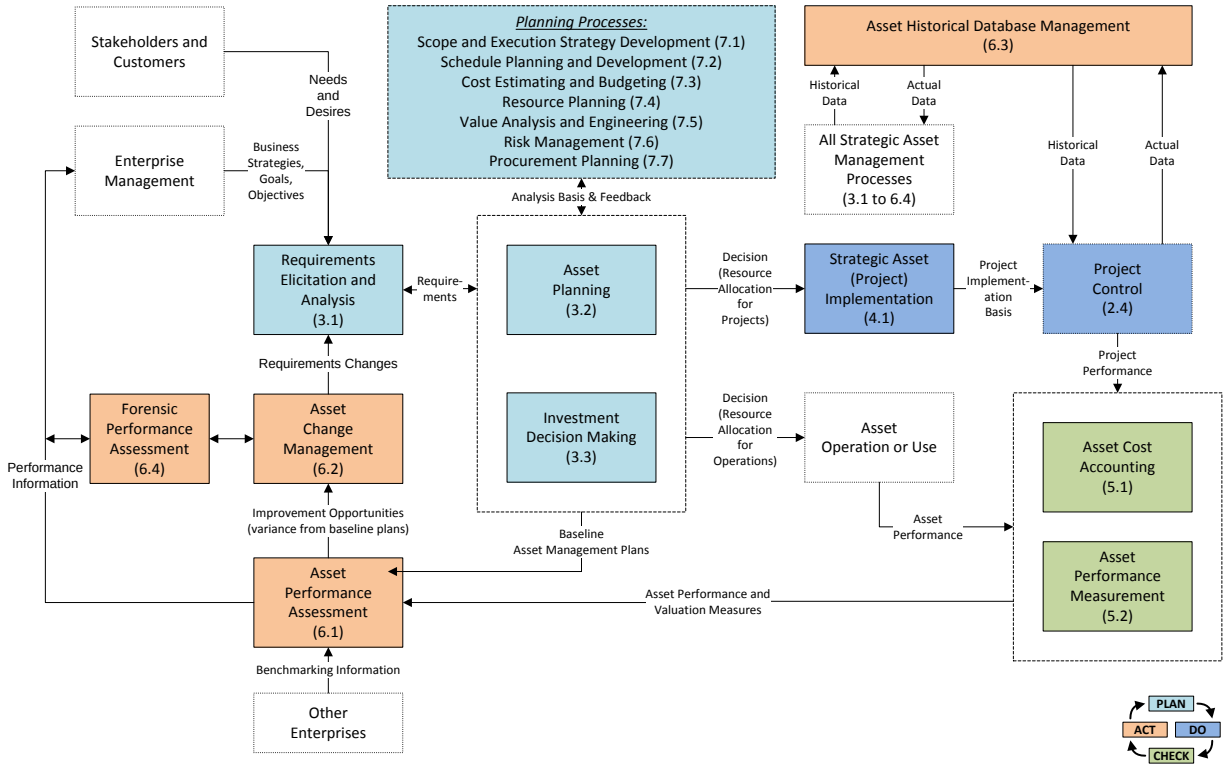


Figure 33.4—Strategic Asset Process Map

The Strategic Asset Process Map further develops the hierarchy of the PDCA cycle into additional process steps. In the complete TCM Framework, each of the numbers associated with a process box in Figure 33.4 has an even more detailed process map that identifies sub-processes.

The strategic asset management process begins with assessing the strategies, goals, objectives, and resource requirements that are established by enterprise management. Then the needs and desires of both customers and stakeholders are elicited, analyzed, and translated into asset performance requirements for the enterprise (Process Step 3.1). As an example, the asset performance requirements for a hotel owner may identify the need for additional capacity in luxury hotel rooms (either through a new hotel or expansion of an existing hotel), or may indicate a new geographical area to develop a luxury hotel.

Considering the performance of existing assets (Process Steps 6.1 and 6.4), and the updated asset requirements from the enterprise, various asset investment options are identified and developed (Process Step 3.2), and then the options are evaluated and a decision is made to select the preferred option (Process Step 3.3). In the example of the hotel owner, the decision may be made to add an additional tower to an existing luxury hotel.

The planning processes (identified as Process Steps 7.1 to 7.6) provide support to the asset planning and investment decision making steps. These steps assist in developing the conceptual scope, budget, and schedule for the options under consideration, and identify potential risks to aid in assessment of the investment decision. This helps to further define and establish the goals and objectives for the selected option as it moves forward as a potential project to develop the new asset.

Project implementation (Process Step 4.1) is the transition point from strategic asset management to the project control process. The asset investment plans and requirements are communicated to and then executed by project teams. Project implementation focuses on steps that help ensure that good investment decisions become good projects.

After the asset has been created or modified (or possibly retired) by the execution of the project, asset performance is measured, including cost accounting measurements (Process Step 5.1), and non-cost performance measurements such as quality (Process Step 5.2). In the case of the hotel example after the new hotel tower is built and placed into operation, hotel operating costs, occupancy rates, and customer surveys may be measured.

Asset performance assessment (Process Step 6.1) uses techniques for determining if the profitability, cost of quality, and other parameters vary from established plans and benchmarks. Also, adverse or positive trends or changes in overall asset performance are evaluated. In the hotel example, this assessment helps determine if the correct asset alternative was selected by examining the impact of the new hotel tower on overall performance of the portfolio of hotels (assets); and determining if the new portfolio is meeting overall objectives of the enterprise.

Benchmarking and other means may be used to identify additional potential improvement opportunities for new or existing asset performance. If overall asset performance is meeting objectives, the strategic asset management process continues as is. If there are performance deviations noted in the assessments, then action should be taken to correct or improve the asset performance trend. Using the hotel example, if occupancy rates in the new luxury tower are lower than expected, then a planned third tower may be cancelled or modified to be a mid-quality tower instead of a luxury tower.

If asset performance corrections or improvements will affect portfolio investment plans; or if changes to stakeholder needs, requirements, or resource availability occur, then these changes are managed using a change management process (Process Step 6.2). Finally, asset and project performance, history, and lessons learned are captured in a historical database for use in future asset management (Process Step 6.3).

The Project Control Process

The Project Control Process is a technical-led process. It is a process for controlling the investment of resources in an asset during project execution. Project Control is the recursive process cycle nested within the "do" step (Process Step 4.1) of the Strategic Asset Management Process Cycle. While Strategic Asset Management is always an on-going activity for the enterprise, a project is a temporary endeavor with a defined beginning and end that the enterprise undertakes to create, modify, maintain, or retire an asset. Projects are how asset investment decisions are put into effect. During the life of a project, various resources (material, labor, time) are invested in the asset by the project team. Ultimately, a usable or operational asset is returned to the enterprise's asset portfolio at the completion of the project. Project control is focused on delivering an asset that meets all of the business objectives (budget, cost, performance, operability, safety, etc.) identified by strategic asset planning; it is about "doing the project right."

Figure 33.5 illustrates the TCM process map for the Project Control process.

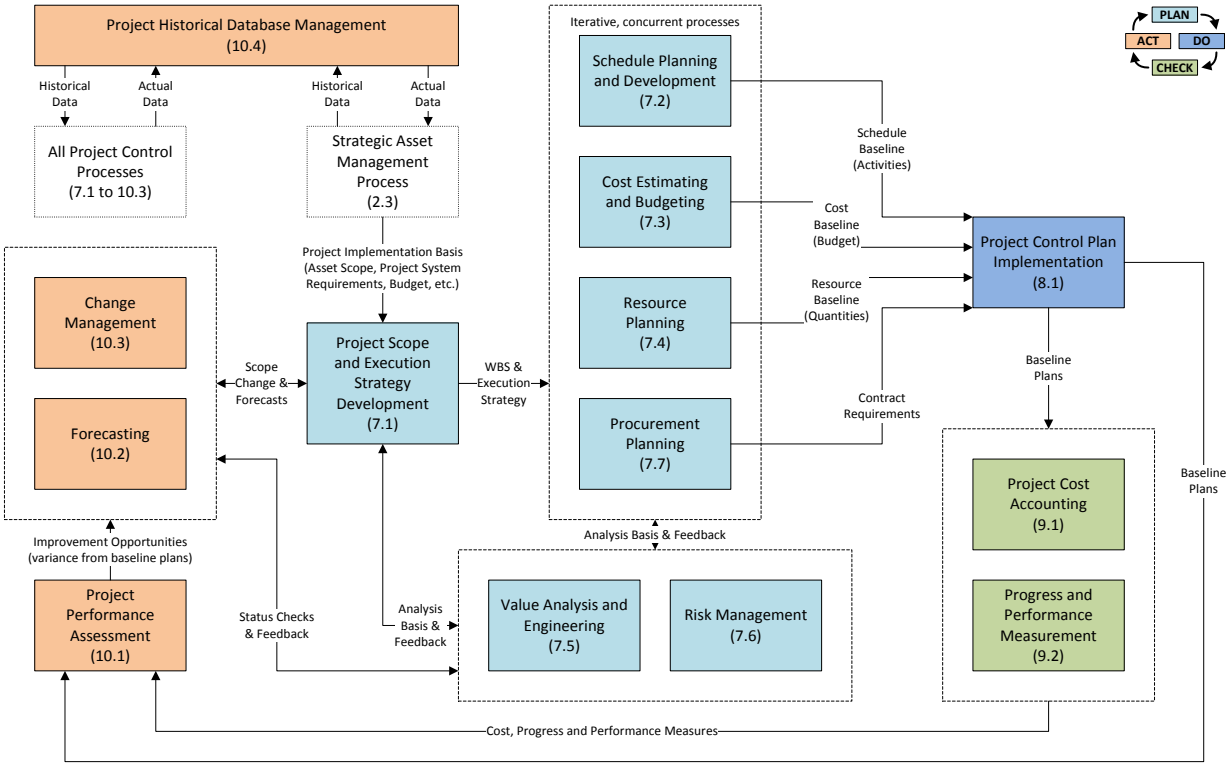


Figure 33.5—Project Control Process Map

Similar to the Strategic Asset Process Map, the Project Control Process Map further develops the hierarchy of the PDCA cycle into additional process steps. In the complete TCM Framework, each of the numbers associated with a process box in Figure 33.5 has an even more detailed process map that identifies sub-processes.

The project control process begins with assessing the strategic asset requirements, and translating those into a set of project performance requirements, including identifying the complete project scope and project execution strategy (Process Step 7.1). Continuing to use the hotel example and from a project control perspective, this would involve developing the scope description to understand overall capacity, quality, and location for the hotel; as well as developing the execution strategy (e.g., design, bid, build versus a design-build contracting strategy). Initial constraints on budget and schedule may also be established.

As the technical teams are developing the specific scope for the project (architectural and engineering development), the project control process is developing the integrated project plans for cost, schedule, and resource planning and procurement (Process Steps 7.2, 7.3, 7.4, and 7.7). The plans should result in time-phased baselines that against which project performance will be measured. During planning, value engineering (Process Step 7.5) ensures that the project scope and plans balance the functional importance of scope requirements versus cost. Risk Analysis (Process Step 7.6) ensures that the project plans, and subsequent project decisions, address uncertainty.

The project plans will be communicating to and implemented by the performing parties (the collective project team including contractors and supporting third parties) in Process Step 8.1. As the project progresses, performance measurements include accounting for cost expenditures and commitments; as well as physical progressing to identify and compare the resources used to the amount of work completed (Process Steps 9.1 and 9.2).

Performance assessment uses evaluative techniques to determine if the project progress and expenditure of resources (including time) vary from the baseline plans, which includes identifying both positive and negative trends or changing performance (Process Step 10.1). Based on the performance assessment, forecasts for cost and schedule are determined and compared to baseline plans (Process Step 10.2). If there are performance deviations from plans, and particularly if the cost or schedule forecasts exceed the planned budget and/or schedule, then action should be taken to correct or improve project performance. Corrective actions, as well as any other changes to initial project plans, are addressed and managed through a change management process (Process Step 10.3). The results from performance assessment and change management often result in changes to the control baselines, and the entire project control process becomes an iterative cycle throughout project execution. Finally, project performance, history, and lessons learned are captured in a historical database for use in future asset management and project control (Process Step 10.4).

The project control process is performed iteratively as the project moves through the various phases of the project lifecycle, from conceptual design through detailed design and final project execution (construction in the case of building a hotel), and any final start-up or project commissioning activities. Typically, at the beginning of each phase, a new control baseline is established based on updated scope information and the translation of the updated scope into cost and schedule goals. For example, new cost estimates and schedules are prepared during each project phase; and they generally become more detailed as project scope is further developed. However, the project control process is also continually performed iteratively within each phase as performance deviations occur and change management results in changes to plans within each phase.

Total Cost Management is an Integrated Process

The various chapters in *Skills and Knowledge of Cost Engineering* focus primarily on individual functions, such as cost estimating or scheduling. However, TCM attempts to illustrate the integration of all the various skills and knowledge areas that are required into processes that support overall management of both strategic assets, as well as the individual projects undertaken to create and develop those assets. For example, a schedule is developed in Process Step 7.2, schedule progress is measured in Process Step 9.2, the schedule is assessed in Process Step 10.1, and updated and maintained in Process Steps 10.2 and 10.3.

It is important for an individual scheduler to understand the particular skills of scheduling; but equally important to understand how those skills are applied in an integrated manner with all of the other disciplines required for effective management of projects and assets. The Total Cost Management Framework: An Integrated Approach to Portfolio, Program, and Project Management and Skills and Knowledge of Cost Engineering combine to provide a collaborative education to any cost engineering practitioner.

REFERENCES

1. AACE Constitution and Bylaws as adopted June 2, 1956. Last amended and approved by the Membership, March 15, 2012, and approved by the Board of Directors, April 8, 2012. Morgantown, WV: AACE International.
2. Hollman, J. K. & Quern, W. (2001). The Total Cost Management (TCM) Framework; *Cost Engineering; 43*(3).

Chapter 34—The International System of Units (SI)

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Abstract
The Système International d'Unites (The international system of units), also known as (SI), was adopted as a global measurement unit system in 1960. The SI uses seven base units accompanied by twenty two derived units; each with special names and symbols. The base and derived units support each other and are coherent, absolute and unique. Key benefits of using SI include ease of interchangeability, reduction in number of manufacturing parts and expansion of global markets. Estimating, contracting and project controls will be greatly enhanced when all parties use the SI units to eliminate potential errors and facilitate communication.

Keywords: Base Units, Conversion Factors, Derived Units, Hard Conversion, Metric Conversion, SI, Significant Digits, and Soft Conversion.

INTRODUCTION
As cost conscious managers, we are well aware that practical action by corporate bodies must be justified by monetary consideration. The Système International d'Unites (SI) changeover is a deliberate practical move that must ultimately yield a net gain. Keeping track of costs and benefits is, therefore, an important part of "metric conversion" activity.

Learning Objectives
After completing this chapter, readers should be able to:

- Identify the seven base SI units of measurement.
- Understand the unit conversions from SI to non-SI units and vice versa.
- Understand the rules for SI style and usage to be able to properly express units.

Objective
AACE International has adopted a policy of using international units of measure (SI) in all of its publications, in compliance with the United States Metric Conversion Act of 1975; which was enacted to increase the use of the metric system. This chapter is intended to help readers better understand the concept and the proper application of the SI system.

Background
The measuring system customarily used in the U.S. is basically the same as the one brought by the colonists from England. This system had origins in a variety of cultures; such as the Babylonian sexagesimal counting system (360^0 circle), Egyptian (common fractions, i.e., repeated halving), Roman (base number 12 as in dozen, hours, months), Celts, and Vikings (base number 20). The Chinese were the first to introduce the decimal notation (ten fingers and the abacus), which later appeared in Arabia in the 15th Century.

Through colonization and dominance of world commerce, the English system of weights and measures spread to many parts of the world, including the American colonies. In 1585, Simon Stevin introduced decimal fractions and predicted that eventually a common measuring system would be based on factors of ten. Stevin's idea was realized in 1795, when the French Convention Nationale passed a law establishing "le système métrique decimal;" which is the basis, practically unchanged, of the metric system today.

The British Association for the Advancement of Science introduced the Centimeter/Gram/Second (CGS) system in 1873, which was adopted by 35 nations, but not by Britain. In 1866, the metric system was made legal in the U.S., mainly because engineers found the CGS units too small for engineering use. Engineers preferred the MKS (Meter-Kilogram-Second) set of base units which became the European metric system; however, this system had a number of conversion factors which made it almost as unwieldy as the British system. In 1901, Giovanni Giorgi proposed that by recognizing an electric unit such as the ampere as a base unit, the system could be made coherent, i.e., without conversion factors between units (see Table 34.1).

Giorgi's MKSA system was adopted by the Conférence Générale des Poids et Mesures (CGPM); and in 1960, the CGPM renamed it le Système International d'Unites (SI). This involved declaring 29 old units obsolete and replacing them with new coherent units.

In the U.S., President Gerald Ford submitted the Metric Conversion Act of 1975, leading to the creation of the USA Metric Board for a voluntary implementation of the new system.

Table 34.1— Base Units and Derived Units with Special Names

Quantity	**Base Unit**	**Symbol**
length	meter	m
mass	kilogram	kg
time	second	s
electric current	ampere	A
thermodynamic temperature	kelvin	K
amount of substance	mole	mol
luminous intensity	candela	cd
Quantity	**Derived Unit**	**Symbol**
land surveyor's area	hectare	ha
liquid volume	liter	L
plane angle	radian	rad
solid angle	steradian	sr
frequency	hertz	Hz
force	newton	N
pressure, stress	pascal	Pa
energy, work, heat quantity	joule	J
power	watt	W
quantity of electricity	coulomb	C
electric potential, emf	volt	V
electric capacitance	farad	F
electric resistance	ohm	Ω
electric conductance	siemens	S
magnetic flux	weber	Wb
flux density, magnetic induction	tesla	T
inductance	henry	H
luminous flux	lumen	lm
illuminance	lux	lx
radioactive activity	becquerel	Bq
absorbed radioactive dose	gray	Gy
catalytic activity	katal	kat

The SI originally consisted of seven fundamental base units, two supplementary units, and several derived units with special names. The supplementary units were changed to derived units in October 1995, resulting in only two major categories (Table 34.1).

SI also uses prefixes that relate to base units in intervals of one thousand (Table 34.2). In the lower range, for popular use, the intervals are ten (centi-, hecto-, kilo-). Thanks to those prefixes, most measurements can be expressed with numerical values of two to four digits. The relationship to the base unit is immediately evident from the prefix. There is no need for any more than one basic unit for any given quantity. Length is always expressed in meter (m) with a prefix if applicable [mm, cm, km (verbal emphasis on the prefix, kilometer, not kilometer!). This eliminates different names for different lengths such as miles, fathoms, chains, feet, knots, etc. with many conversion factors. SI quantities relate by simply moving the decimal point.

Table 34.2—Prefixes for Multiples and Submultiples of SI Units

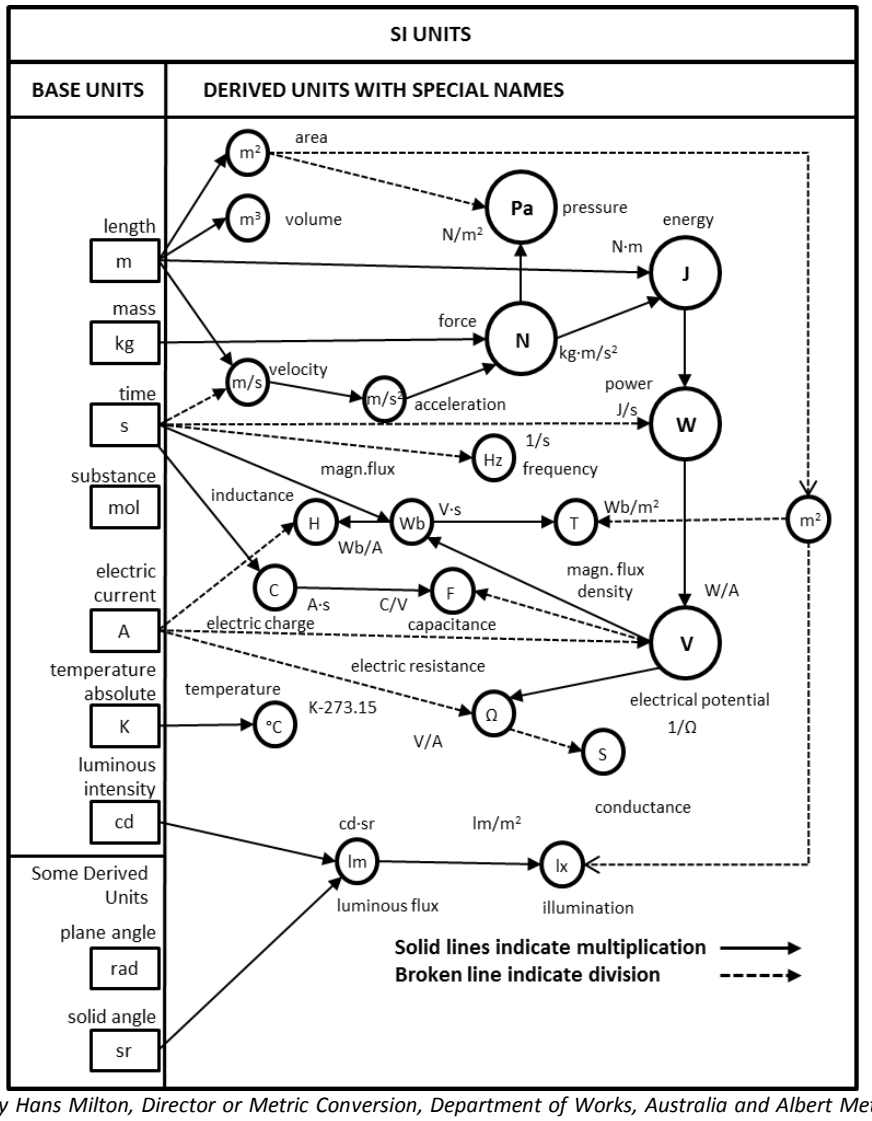
Prefix	**Symbol**	**Factor**
tera-	T	10^{12} = 1 000 000 000 000
giga-	G	10^{9} = 1 000 000 000
mega-	M	10^{6} = 1 000 000
kilo-	k	10^{3} = 1 000
hecto-	h	10^{2} = 100
deca-	da	10^{1} = 10
deci-	d	10^{-1} = 0.1
centi	c	10^{-2} = 0.01
milli-	m	10^{-3} = 0.0001
micro-	µ	10^{-6} = 0.000 001
nano-	n	10^{-9} = 0.000 000 001
pico-	p	10^{-12} = 0.000 000 000 001

SI IS COHERENT, ABSOLUTE, AND UNIQUE

SI is coherent
There are no factors relating different units; they are all related to each other by unity: One Newton is the force required to give a mass of one kilogram an acceleration of one meter per second-squared.

SI is absolute
SI expressions are unqualified. Force is always expressed in Newton, there is no other name for force; mass is kg and pressure is Pascal. The imperial system used pound for force (psi) and also for mass (density). Ounces could denote weight or volume. U.S. gallons are different from Imperial gallons etc.

SI is unique
The same units are used whether we do thermal, mechanical or electrical calculations. This is a tremendous advantage because all engineering disciplines speak the same language. One unit of power, the watt, is used now where nineteen units were used previously (Btu, calories, HP, etc.). The Pascal is replacing some 30 different units including obsolete old metric units.



Based on a design by Hans Milton, Director or Metric Conversion, Department of Works, Australia and Albert Mettler, Secretary of the Canadian Metric Association. 1973. Reproduced by permission of the Canadian Metric Association.

Figure 34.1—SI is coherent

If one Newton force is exerted through a length of one meter, it produces the energy of one joule. If this takes place in one second, the power produced is one watt. Furthermore, if one Newton force is distributed over an area of one square meter, it produces a

pressure of one Pascal. We find the same coherence with the other base units. For example, the derived units for ampere are volt, ohm, siemens, coulomb, henry, weber, tesla, and farad. (Figure 34.1)

METRIC CONVERSION

The advantage of the SI over the customary and the old "metric" system is obvious but not necessarily appreciated by the general public in its day-to-day living. Many individuals still seem to be deprived of facts and may not really understand the system. It is not the learning of the new, but the unlearning of the old which appears so painful to many.

Soft Conversion

We can make it easy for ourselves by using computers and conversion tables that instantly give us SI measurements. However, instant "converters" do not convert conventional units into SI units; instead they translate the existing into the "metric" equivalent of the existing. By doing this, we are only using the old system with new numbers.

This subjective approach is called "soft" conversion. What we are doing when we "soft" convert is adding another expression to the proliferation of expressions already in existence for each quantity of measurement. During the transition period of conversion we cannot avoid changing existing physical quantities into their SI equivalents.

Significant Digits

With the ease of computerized calculations there is a temptation of using an overabundance of insignificant digits. A 20 foot hydro pole is not 6096 mm long. A seven ounce jar of pickled eggs is not 198 g. The liquid is probably expressed in fluid ounces and should have been converted into 200 cm^3 or 200 mL.

Hard Conversion:

Our customary system worked in the past, but it is standing on its own. The SI is a different improved system which also stands on its own. The two systems are incompatible. When a speed limit sign on our highways show 60 and our speedometer shows 60 km/h without any reference to miles, we have hard converted.

Britain's double bed size has been 54 inches wide and 75 inches long (137 cm × 191 cm). Britain's Bedding Federation changed this into a hard converted 150 cm by 200 cm bed. This is a true changeover, taking advantage of the opportunity to increase the size of the bed at the same time. This holds true with standard door openings and other modular building components. With hard conversion we have the opportunity to update, houseclean, rationalize, remove duplication, and otherwise improve upon the standards we use.

Conversion Factors

Table 34.3 shows conversions from old units to SI units, such as converting gal/hr into L/s.

For example:

1.051 503 E-03 means 1.051 503 x 10^{-3} or 1.051 503/1000 = 0.001 051 503.
1 gal/hr. = 0.0001 051 503 L/s

To convert from 1 L/s to gal/hr., we divide
1/0.001 051 503 = 9.510 223 E+02.

Converting 5000 gal/hr = 5000 x 0.001 051 503 = 5.26 L/s.

Here we have used the prefix d (deci), where dm^3 (L) is 1/1000 of a cubic meter (m^3). Prefixes move the decimal point back and forth.

Table 34.3—Multiplication Table to Convert to SI

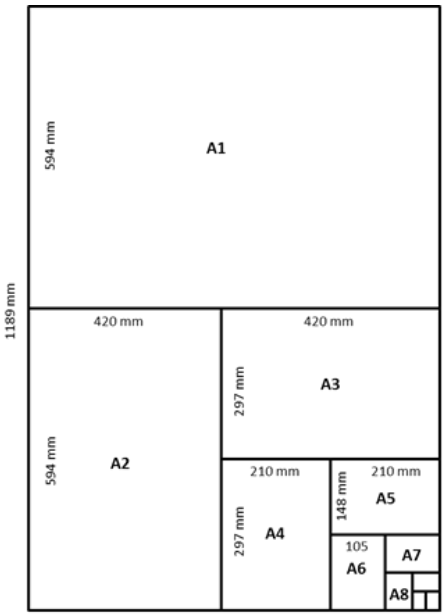
Convert From	To	Multiply By	Replacing Other Old Units
Length and Area			
inch	millimeter (mm)	2.54 E+01	rod, league, pole, chain, fathom, mil, printer's pica, point, angstrom, caliber, section, barn, township
yard	meter (m)	9.144 000 E-01	
foot	meter (m)	3.048 000 E-01	
mile	meter (m)	1.609 347 E+03	
acre	meter^2 (m^2)	4.046 873 E+03	
ft^2	meter^2 (m^2)	9.290 304 E-02	
Please note, that one acre (10 square chains) is 43 560 ft^2 or a square of 208.710 327 feet each side, which is not a whole number. The acre (approx. 63m x 64m) may be compared to the derived unit hectare (ha), which is 100m x 100m or approx. 2.5 acres.			
Volume and Flow			
gallon (US dry)	meter^3 (m^3)	4.404 884 E-03	barrel oil, board foot, bushel, cup, gill (US), gill (UK), pint, cubic yard, quart, cubic inch per minute
gallon (US liquid)	meter^3 (m^3)	3.785 412 E-03	
gallon (Canada)	meter^3 (m^3)	4.546 090 E-03	
register ton	meter^3 (m^3)	2.832 685 E+00	
ounce (US fluid)	meter^3 (m^3)	2.957 353 E-05	
ounce (Canada)	meter^3 (m^3)	2.841 307 E-05	
gallon (US)/min	meter^3/second	6.309 020 E-05	
ft^3/second	meter^3/second	2.831 685 E-02	
Mass			
pound (lb. avoirdupois)	kilogram (kg)	4.535 924 E-01	grain, ounce (avoirdupois), slug, ounce (troy), stone, pennyweight, cwt
pound (troy)	kilogram (kg)	3.732 417 E-01	
ton (long)	kilogram (kg)	1.016 047 E+03	
ton (short)	kilogram (kg)	9.071 847 E+02	
Kilogram (kg) mass is the constant property of a body. In popular speech, "weight" is used to mean mass at rest. This mass can change its weight due to acceleration to a point when it becomes "weightless." We only weigh one-sixth as much on the moon as on earth, but our mass is still the same.			
Force (Weight)			
pound-force (lbf)	newton (N)	4.448 222 E+00	Dyne, kip, thrust
kilopond	newton (N)	9.806 650 E+00	
poundal	newton (N)	1.382 550 E-01	
Pressure or stress			
psi (lb/inch^2)	pascal (Pa)	6.894 757 E+03	cm Hg, cm H_2O, dyne/cm^2, foot of H_2O (39.2°F), lbf/ft^2, torr, kip/in^2, poundal/ft^2, etc. (approx. 30 units)
atm (standard)	pascal (Pa)	1.013 250 E+05	
bar	pascal (Pa)	1.000 000 E+05	
inch Hg (60°F)	pascal (Pa)	3.376 85 E+03	
inch H_2O (60°F)	pascal (Pa)	2.488 4 E+02	
Viscosity			
slug/ft·s	Pa·s	4.788 026 E+01	Poise, stoke, lb/ft·s, rhe, etc.
poundal·s/ft^2	Pa·s	1.488 164 E+00	
ft^2/s	m^2/s	9.290 304 E-02	

Table 34.3—Multiplication Table to Convert to SI (cont.)

Convert From	To	Multiply By	Replacing Other Old Units
Energy, work			
Btu (international)	joule (J)	1.055 056 E+03	Various types of Btus and Calories, er, ft·lbf, therm, Watthour, hp-hour (mech.)
calorie (international)	joule (J)	4.186 800 E+00	
electronvolt	joule (J)	2.602 19 E-19	
kW·h	joule (J)	3.600 000 E+06	
Power			
horsepower (hp)	kilowatt (kW)	7.456 999 E-01	$hp_{electric}$, therm/h, ftlbf/s, $hp_{mechanical}$, brake hp
Btu/h	watt (W)	2.930 711 E-01	
Other units			
temperature (°F)	celsius (°C)	(5/9)(F-32)	centigrade, réaumur
Kelvin (°K) is used for absolute temperature (-273°C)			
frequency (rpm)	Hertz (Hz) = 1/s	1.666 667 E-02	cycles per second
Fuel consumption			
mass (ton/hr)	kg/s	2.519 958 E+04	(see example below)
volume (gal/hr)	dm^3/s or L/s	1.051 503 E-03	
automobile	L/100km	no conversion	
*USA uses fuel **efficiency**, i.e. miles per gallon. Conversion to L/100km is non-linear.*			
time (hr., min., sec.)	h, s	no conversion	just spelling of symbol

Metric Conversion is an Investment

The SI changeover has a continuous benefit, similar to the invention of the wheel. Even if we let the costs lie where they fall, benefits will accrue when conversion is complete. Therefore, conversion to SI is an investment, not an expense. The SI implementation is only a part of a bigger change, the standardization of industrial products and processes.



In a law passed in 1798, France defined the sizes of official documents requiring stamp duty. The original size A0 =1 m2, halved four times, leads to the size A4. Following this method for cutting paper, results in no paper trimming waste.

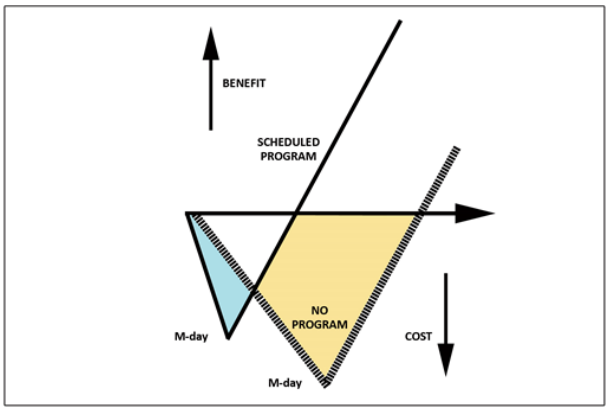
Today, the ISO—A4 size is mandatory in over 45 countries worldwide. The Canadian government mandated A4 sizes in 1975, but ran into opposition by the paper industry and had some implementation problems.

Figure 34.2—International Paper Sizes

It will be difficult for the general public to abandon a system that worked fairly well and to learn a new "language." It will be a slow transition process to a full SI practice.

The fact that most of the world is already using ISO standards will probably have an adverse impact on American competitiveness. It will likely be the exporting industry and technical societies who push for a USA metric conversion.

Standardization includes a rationalization of products (reduction in the total number of sizes and ranges), which is where a great deal of the benefits of conversion will occur. The benefits of rationalization remain long after the expenses of conversion are paid.



The cost/benefit chart (Figure 34.3; *1986 AACE Transactions)* expresses an idea, not a specific duration for a planned M-day. In Canada, it was about 10 to 12 years.

Fred Jelen, who wrote the book on *Cost Optimization Engineering,* commented on this cost/benefit idea when he said, "*The only decision to be made is how to make it most economically. The real issue is not so much the cost but rather determining the optimum time at which to make the conversion."*

It has taken us 200 years to implement a worldwide uniform measuring system, but a completion date has not yet been established in the United States.

Figure 34.3—Cost/Benefit Chart

RULES FOR SI STYLE AND USAGE

1. Symbols and prefixes are the same in all languages.
2. Symbols are written in lower case, except when the unit is derived from a proper name, i.e. m = meter; W = watt. The exception is the non-SI but commonly used unit for cm^3, the liter = L. The capital letter is to avoid confusion between "l(el)" and "1(one)".
3. Symbols and prefixes are printed in upright type. There is no spacing between prefixes and units, e.g.

   km cm ng
   not *km cm ng*
   and not *k m c m n g*
4. Symbols are not pluralized, e.g. 100 g or 50 km not 100 gs or 50 kms
5. There is no period after a symbol except at the end of a sentence, e.g.
   m (not m.), mL (not mL.)
6. A sentence does not start with a symbol and prefix: The symbol for kilogram is kg (not: kg is the symbol for...).
7. Preference is given to decimal notation:
   Use 3.25 % rather than 3¼ %; 0.75 km rather than ¾ km.
   But "I walked 2 km in half an hour" (not in 0.5 h)
8. For values less than one (1), a zero (0) is used, i.e. 0.56 (not **.**56)
9. There is a space between the last digit of a number and the first letter of the symbol except for ^{o}C, e.g. 36 kg not (36kg); or 22°C not (22 ^{o}C)
10. The multiplication symbol is used instead of a dot or period: 5 × 7 or 5*7 not (5 ·7 or 5.7)
11. Compound prefixes must never be used: 3 mg (milligram) not 3 μkg (microkilogram)
12. Spaces are used instead of commas to put large numbers in reading blocks of three,
    e.g., 23 456 789.24 not 23,456,789.24
13. Only one unit is used to designate quantities: 5.36 m not (5 m, 36 cm) or 3.7 kg not (3 kg, 700 g)
14. An oblique stroke (/) with no spaces is used with symbols rather than the word "per": km/h not (km / h) and not (km per h). When writing units in full, then kilometer per hour, not (kilometer/hour)
15. ISO standard time units start with the largest unit, e.g.:

Year	Month	Day	Hour	Minute	Second					
2011	07	06	18	21	08	or	2011 -	07 -	06 -	18 : 21 : 08

    Europe has changed from DDMMYY to the ISO standard of YYMMDD. The USA still uses MMDDYY, which can be very confusing and legally costly in international trade. If we have a commitment with China, for example, (they use ISO dating) for 11 - 03 - 10 (ISO date), we will probably make payments earlier than necessary. (Nov. 3, 2010 instead of March 10, 2011).
16. SI units are pronounced on the first syllable:
    centimeter, kilometer, Celsius, micrometer (μ), megahertz; never ki**lo**meter.
    Instruments are pronounced micrometer (μ),, thermometer, speedometer.
    (The spelling in Canada is different:, e.g. kilomet**re** vs. the instrument thermomet**er**).
17. Numeral and symbol should not be on separate lines, nor should a number be hyphenated.
18. The choice of the appropriate multiple of an SI unit is governed by the application, preferably having values between 0.1 and 1000. 3.94 mm instead of 0.003 94 m

19. The symbol is placed behind the numeral:
    15.7 km or 350 mL; (not km 15.7 or mL 350)
    (Prefixes may also apply to dollars, e.g. 250 k$ and 5 M$)
20. Temperature will be expressed in "degrees Celsius" or ^{o}C without space after the numeral, not (centigrade or degrees C), e.g. 40°C and –40°C; and not (+ 40 ^{o}C, neither minus 40°C or 40°C minus).
    Please note, that there are no warmer or cooler temperatures, they can only be higher or lower. Furthermore, 20°C is not "half as cold" as 40°C
21. Number and symbol should not be separated by an adjective.
    Write: You get "200 km free" with your car rental.
    Do not write: You get "200 free km" with your car rental"
22. If a presentation in dual values is unavoidable, give preference to the SI value: 20 kg (44lbs.); not 44 lbs. (20 kg).
23. Former units of area and volume will be changed as follows:
    Square meter (formerly sq.m) now becomes m^{2}, pronounced square meter, not meter square. Similarly, cubic centimeter is now cm^{3}, not cu.cm. or c.c. (It is acceptable to use mL instead of the SI unit cm^{3} in case of liquids and gases).
24. In order to not confuse quantity symbols in engineering with SI symbols, the ISO recommends to use italic type lettering where practical, e.g.,
    Quantities:
    m = mass; *d* = depth or diameter; *M* = bending moment; *A* = area, *a* = acceleration.
    SI symbols:
    m = meter; d = deci-; M = mega-; A = ampere
    Combined:
    Gravity = g and *Force = F = m* (kg) × *a* (m/sec^{2}).

CONCLUSION

The International System of Units (SI) has become a standard measurement practice. Adopting SI has economic benefits of maximizing the use of standard parts, ease of parts interchangeability and procurement, maintaining consistent estimate database, enhancing contract management and communication. To increase access to global markets, familiarity of SI and conversion to non-SI units and vice versa, is essential.

RECOMMENDED READING

This chapter is based on information in the public domain and established international standards. In addition:

1. AACE International. *Cost Engineers' Notebook*. Morgantown, WV: AACE International.
2. AACE International (1980). *Cost Engineering, 22*(6).
3. AACE International (1986). *Cost Engineering, 28*(11), pp. 26.
4. Former Canadian Metric Association (CMA), *Metric Fact Sheets* (out of print), Scarborough, ON.
5. Heinze, K. (1996). Appendix B, Cost Management of Capital Projects. New York, NY Marcel Dekker.

Sources of SI Information

1. American National Metric Council (ANMC), Publications Department, 4330 East West Highway, Suite 1117, Bethesda, MD, 20814 - 4408.
2. American National Standards Institute (ANSI), ANSI/IEEE Std. 268-82, 11 W 42nd St., New York, NY 10036.
3. American Society of Civil Engineers (ASCE), Metric Units in Engineering - Going SI, Revised Edition 1995, 345 East 47th Street, New York NY 10017 - 2398.
4. American Society for Testing and Materials (ASTM), ASTM E380, Standard Practice For Use Of The International System Of Units and Units In Building Design And Construction, 1916 Race St., Philadelphia, PA 19103.
5. Canadian Standards Association (CSA), Metric Practice Guide, CSA Z234, 178 Rexdale Blvd., Rexdale, ON, Canada M9W 1R3.
6. National Bureau of Standards (NBS), Various Publications, U.S. Department of Commerce, Gaithersburg, MD 20899.
7. National Research Council of Canada, Manual on Metric Building Practice, Otttawa, ON Canada K1A 0R6.
8. R.S. Means Co. Building Construction Cost Data, Metric Addition, R.S. Means Co., Box 800, Kingston, MA 02364.
9. U.S. Metric Association, 10245 Andasol Ave., Northridge, CA 91325-1504.



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